Table 54.5 Criteria for observation at home

1. Head CT scan not indicated, or CT scan normal if indicated[22]
2. initial GCS $\geq$ 14
3. no high risk criteria
4. no moderate risk criteria except loss of consciousness
5. patient is now neurologically intact (amnesia for the event is acceptable)
6. there is a responsible, sober adult that can observe the patient
7. patient has reasonable access to return to the hospital E/R if needed
8. no “complicating” circumstances (e.g. no suspicion of domestic violence, including child abuse)

Table 54.6 Sample discharge instructions for head injuries

Seek medical attention for any of the following:
1. a change in level of consciousness (including difficulty in awakening)
2. abnormal behavior
3. increased headache
4. slurred speech
5. weakness or loss of feeling in an arm or leg
6. persistent vomiting
7. enlargement of one or both pupils (the black round part in the middle of the eye) that does not get smaller when a bright light is shined on it
8. seizures (convulsions or fits)
9. significant increase in swelling at injury site
Do not take sedatives or pain medication stronger than acetaminophen (paracetamol in some countries) for 48 hours. Do not take aspirin or other anti-inflammatory medications because of interference with platelet function and theoretical increased risk of bleeding

Table 54.7 Findings with moderate risk of ICI

1. history of change or loss of consciousness on or after injury
2. progressive H/A
3. EtOH or drug intoxication
4. posttraumatic seizure
5. unreliable or inadequate history
6. age < 2 yr (unless trivial injury)
7. vomiting
8. posttraumatic amnesia
9. signs of basilar skull fracture
10. multiple trauma
11. serious facial injury
12. possible skull penetration or depressed fracture
13. suspected child abuse
14. significant subgaleal swelling[21]

b) in-hospital observation to rule-out neurologic deterioration if patient does not meet criteria in Table 54.5 (including cases where CT scan is not done).

Managing patients with in-hospital observation and only getting a CT scan in cases of deterioration (GCS score ≤ 13) is as sensitive as CT in detecting intracranial hematomas,[23,24,25,26,27] but is less cost effective than routinely performing an early CT scan and discharging patients who have a normal CT and no other indication for hospitalization[23]

Category 3. High risk for intracranial injury

Criteria

Possible findings are shown in Table 54.8.

Management recommendations

1. admit to hospital
2. STAT unenhanced head CT scan
3. if there are focal findings on neurologic examination

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Table 54.8 Findings with high risk of ICI

• depressed level of consciousness not clearly due to EtOH, drugs, metabolic abnormalities, postictal, etc.
• focal neurological findings
• decreasing level of consciousness
• penetrating skull injury or depressed fracture

a) notify operating room to be on standby
b) if CT scan or MRI is not available, consider emergency burr holes (p.836)
4. determine if intracranial monitor (p.858) is indicated
5. SXR usually not recommended: a fracture is rarely surprising, and a SXR is inadequate for assessing for intracranial injury. A SXR is possibly useful for localizing a radio-opaque penetrating foreign body (knife blade, bullet…) for the O.R.

Other risk factors

Occipital vs. frontal fractures

Patients with occipital fractures may be at higher risk of significant intracranial injury (ICI). May be related to the fact that in forward trauma, one may protect oneself with the outstretched arms. Furthermore, the facial bones and air sinuses exert an impact absorbing effect. In 210 patients with facial fractures,[28] the highest incidence of ICI was seen in those with upper facial fractures. Those with mandibular and midfacial region fractures (without upper facial involvement) had a lower likelihood of ICI, and those with mandibular region trauma only were least likely to have ICI.

54.5 Radiographic evaluation

54.5.1 CT scans in trauma

General information

An unenhanced (i.e. non-contrast) CT scan of the head usually suffices for patients seen in the emergency department presenting after trauma or with a new neurologic deficit. Enhanced CT or MRI may be appropriate after the unenhanced CT, but are not usually required emergently (exceptions include: significant brain edema due to suspected neoplasm that is not demonstrated without contrast).

The main emergent conditions to rule out (and brief descriptions):

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1. blood (hemorrhages or hematomas):
   a) extra-axial blood: surgical lesions are usually ≥ 1 cm maximal thickness
      • epidural hematoma (EDH) (p.892): usually biconvex and often due to arterial bleeding. May cross dural barriers (unlike SDH) such as falx, tentorium
      • subdural hematoma (SDH) (p.891): usually crescentic, usually due to venous bleeding. May cover larger surface area than EDH (dural adherence to inner table limits extension of EDH). Chronology of SDH: acute = high density, subacute ≈ isodense, chronic ≈ low density
   b) subarachnoid blood (SAH): Trauma is the most common cause of SAH. Unlike aneurysmal SAH where blood is thickest near the circle of Willis, traumatic SAH (tSAH) usually appears as high density spread thinly over convexity and filling sulci or basal cisterns. However, when the history of trauma is not clear, an arteriogram may be indicated to R/O a ruptured aneurysm (that might have precipitated the trauma in some cases)
   c) intracerebral hemorrhage (ICH): increased density in brain parenchyma
   d) hemorrhagic contusion (p.891): often "fluffy" inhomogeneous high-density areas within brain parenchyma, usually adjacent to bony prominences (frontal and occipital poles, sphenoid wing). Typically less well defined than primary ICH
   e) intraventricular hemorrhage (p.1192): present in ≈ 10% of severe head injuries.[29] Associated with poor outcome; may be a marker for severe injury rather than the cause of the poor outcome. Use of intraventricular rt-PA has been reported for treatment[30]
2. hydrocephalus: enlarged ventricles may sometimes develop following trauma
3. cerebral swelling: obliteration of basal cisterns (p.921), compression of ventricles and sulci…
4. evidence of cerebral anoxia: loss of gray-white interface, signs of swelling
5. skull fractures:
   a) basal skull fractures (including temporal bone fracture)
   b) orbital blow-out fracture
   c) calvarial fracture (CT may miss some linear nondisplaced skull fractures)

- linear vs. stellate
- open vs. closed
- diastatic (separation of sutures)
- depressed vs. nondepressed: CT helps assess need for surgery

6. ischemic infarction: findings are usually minimal or subtle if < 24 hrs since stroke
7. pneumocephalus: may indicate skull fracture (basal or open convexity)
8. shift of midline structures (due to extra- or intra-axial hematomas or asymmetric cerebral edema): shift can cause altered levels of consciousness (p.921)

Indications for initial brain CT

1. presence of any moderate[31] or high risk criteria (Table 54.7 and Table 54.8) which include: GCS ≤ 14, unresponsiveness, focal deficit, amnesia for injury, altered mental status (including those that are significantly inebriated), deteriorating neuro status, signs of basal or calvarial skull fracture
2. assessment prior to general anesthesia for other procedures (during which neurologic exam cannot be followed in order to detect delayed deterioration)

Follow-up CT

Routine follow-up CT (when there is no indication for urgent follow-up CT, see below):
1. many facilities perform a repeat head CT at 24 hours for patients who are clinically stable but had findings on initial head CT of: traumatic SAH, small SDH or EDH, intraparenchymal contusions
2. for patients with severe head injuries:
   a) for stable patients, follow-up CTs are usually obtained between day 3 to 5, (some recommend at 24 hrs also) and again between day 10 to 14
   b) some recommend routine follow-up CT several hours after the "time zero" CT (i.e. initial CT done within hours of the trauma) to rule-out delayed EDH (p.894), SDH (p.898), or traumatic contusions (p.891) [32]
3. for patients with mild to moderate head injuries:
   a) for those with an abnormal initial CT, the CT scan is usually repeated prior to discharge
   b) stable patients with mild head injury and normal initial CT do not require follow-up CT

Urgent follow-up CT: performed for neurological deterioration (loss of 2 or more points on the GCS, development of hemiparesis or new pupillary asymmetry), persistent vomiting, worsening H/A, seizures or unexplained rise in intracranial pressure (ICP) in patients with an ICP monitor.

54.5.2 Spine films

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1. cervical spine: must be cleared radiographically from the cranio-cervical junction down through and including the C7-T1 junction. Spinal injury precautions (cervical collar…) are continued until the C-spine is cleared. The steps in obtaining adequate films are outlined in Spine injuries, Radiographic evaluation and initial C-spine immobilization (p.952)
2. thoracic and lumbosacral LS-spine films should be obtained based on physical findings and on mechanism of injury; see Spine injuries, Radiographic evaluation and initial C-spine immobilization (p.952)

54.5.3 Skull x-rays

A skull fracture increases the probability of a surgical intracranial injury (ICI) (in a comatose patient it is a 20-fold increase, in a conscious patient it is a 400-fold increase[33,34]). However, significant ICI can occur with a normal skull x-ray (SXR) SXR (SXR was normal in 75% of minor head injury patients found to have intracranial lesions on CT, attesting to the insensitivity of SXRs[23]). SXRs affect management of only 0.4–2% of patients in most reports.[20]

A SXR may be helpful in the following:
1. in patients with moderate risk for intracranial injury (Table 54.7) by detecting an unsuspected depressed skull fracture (however, most of these patients will get a CT scan, which obviates the need for SXR)
2. if a CT scan cannot be obtained, a SXR may identify significant findings such as pineal shift, pneumocephalus, air-fluid levels in the air sinuses, skull fracture (depressed or linear)… (however, sensitivity for detecting ICI is very low)
3. with penetrating injuries: helps in visualization of some metallic objects

54.5.4 MRI scans in trauma

Usually not appropriate for acute head injuries. This is due to longer acquisition time, less access to patient during study, increased difficulty in supporting patient (requires special non-magnetic ventilators, cannot use most IV pumps…) and MRI is less sensitive than CT for detecting acute blood.[35] There were no surgical lesions demonstrated on MRI that were not evident on CT in one study.[36] There may be some additional benefit in combining CT with an MRI performed directly in the emergency department.[37]

MRI may be helpful later after the patient is stabilized, e.g. to evaluate brainstem injuries, small white matter changes,[38] e.g. punctate hemorrhages in the corpus callosum seen in diffuse axonal injury (p.848)… Spinal MRI is indicated in patients with spinal cord injuries.

Rapid sequence MRI may be useful for follow-up in pediatrics to minimize radiation exposure.

54.5.5 Arteriogram in trauma

Cerebral arteriogram (p.911): useful with non missile penetrating trauma.

54.6 Admitting orders for minor or moderate head injury

54.6.1 General information

Traditionally, mild head injury has been defined as GCS ≥ 13. However, the increased frequency of both surgical lesions and CT scan abnormalities in patients with GCS = 13 suggests that they would be better classified with the moderate rather than mild head injuries.[22] See Indications for CT and admission criteria for TBI (p.830) for admitting criteria.

54.6.2 Admitting orders for minor head injury (GCS ≥ 14)

1. activity: BR with HOB elevated 30–45°
2. neuro checks q 2 hrs (q 1 hr if more concerned; consider ICU for these patients). Contact physician for neurologic deterioration
3. NPO until alert; then clear liquids, advance as tolerated
4. isotonic IVF (e.g. NS + 20 mEq KCl/L) run at maintenance (p.870): ≈ 100 cc/hr for average size adult (peds: 2000 cc/m^2/d). Note: the concept of "running the patient dry" is considered obsolete
5. mild analgesics: acetaminophen (PO, or PR if NPO), codeine if necessary
6. anti-emetic: give infrequently to avoid excessive sedation, avoid phenothiazine anti-emetics (which lower the seizure threshold); e.g. use trimethobenzamide (Tigan®) 200 mg IM q 8 hrs PRN for adults

54.6.3 Admitting orders for moderate head injury (GCS 9–13)

1. orders as for minor head injury (see above) except patient is kept NPO in case surgical intervention is needed (including ICP monitor)
2. for GCS = 9–12 admit to ICU. For GCS = 13, admit to ICU if CT shows any significant abnormality (hemorrhagic contusions unless very small, rim subdural…)
3. patients with normal or near-normal CTs should improve within hours. Any patient who fails to reach a GCS of 14–15 within 12 hrs should have a repeat CT at that time[31]

54.7 Patients with associated severe systemic injuries

54.7.1 Intraabdominal injuries

Diagnostic peritoneal lavage (DPL) looking for bloody fluid or FAST (focused abdominal sonogram for trauma) are often used by trauma surgeons to assess for intra-abdominal hemorrhage. If negative and the patient is hemodynamically stable, the patient should be taken for cranial CT (with DPL – if the initial fluid is not bloody, the remainder of the lavage fluid may be collected for quantitative analysis as the head CT is being done).

Patients with grossly positive DPL or positive FAST and/or hemodynamic instability may need to be rushed to the O.R. for emergent laparotomy by trauma surgeons without benefit of cerebral CT. Neurosurgical management is difficult in these patients, and must be individualized. These guidelines are offered:

 CAUTION: many patients with severe trauma may be in DIC (either due to systemic injuries, or directly related to severe head injury possibly because the brain is rich in thromboplastin[39]). Operating on patients in DIC is usually disastrous (p.167). At the least, check a PT/INR/PTT

1. if GCS > 8 (which implies at least localizing)
   a) operative neurosurgical intervention is probably not required
   b) utilize good neuroanesthesia techniques (elevate head of bed, judicious administration of IV fluids, avoiding prophylactic hyperventilation…)
   c) obtain a head CT scan immediately post-op
2. if patient has focal neurologic deficit, an exploratory burr-hole should be placed in the O.R. simultaneously with the treatment of other injuries. Placement is guided by the pre-op deficit (p.836)
3. if there is severe head injury (GCS ≤ 8) without localizing signs, or if initial burr hole is negative, or if there is no pre-op neuro exam, then
   a) measure the ICP: insert a ventriculostomy catheter (if the lateral ventricle cannot be entered after 3 passes, it may be completely compressed or it may be displaced, and an intraparenchymal fiber-optic monitor or subarachnoid bolt should be used)
      - normal ICP: unlikely that a surgical lesion exists. Manage ICP medically and, if a IVC was inserted, with CSF drainage
      - elevated ICP (≥ 20 mm Hg): inject 3–4 cc of air into ventricles through IVC, then obtain portable intraoperative AP skull x-ray (intra-operative pneumoencephalogram) to determine if there is any midline shift. If there is mass effect with ≥ 5 mm of midline shift is explored[40] with burr-hole(s) on the side opposite the direction of shift. If no mass effect, intracranial hypertension is managed medically and with CSF drainage
   b) routine use of exploratory burr holes for children with GCS = 3 has been found not to be justified[41]

54.7.2 Fat embolism syndrome

General information

Most often seen after a long bone fracture (usually femoral, but may include clavicular, tibial, and even isolated skull fracture). Although almost all patients have pulmonary fat emboli at autopsy, the syndrome is usually mild or subclinical, only ≈ 10–20% of cases are severe, and the fulminant form leading to multiple organ failure is rare. Clinical findings usually appear within 12–72 hrs of injury, and do not always include the complete classic clinical triad of:

- acute respiratory failure (including hypoxemia, tachypnea, dyspnea) with diffuse pulmonary infiltrates (usually seen as bilateral fluffy infiltrates). May be the only manifestation of fat emboli in up to 75% of cases
- global neurologic dysfunction: may include confusion (PaO_2 usually not low enough to account for these changes[42]), lethargy, seizures
- petechial rash: seen ≈ 24–72 hrs after the fracture, usually over thorax

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Other possible findings include:

- pyrexia
- retinal fat emboli

There is no specific test for fat embolism syndrome (FES). The following have been proposed, but have poor sensitivity and specificity: fat globules in the urine (positive in ≈ one–third[43]) and serum, serum lipase activity. In cases of unexplained neurologic or pulmonary abnormalities, it may be possible to diagnose FES if on bronchoalveolar lavage[44] > 5% of cells in the washings staining for neutral fat with red oil 0. Nonspecific tests include ABG (findings: hypoxemia, hypocarbia from hyperventilation, respiratory alkalosis).

Treatment

Pulmonary support with oxygen, and mechanical ventilation if necessary including use of PEEP. The use of steroids is controversial. Ethyl alcohol (to decrease serum lipase activity) and heparin have not been shown to be of benefit. Early operative fixation of long bone fractures may reduce the incidence of FES.[45]

Outcome

Usually related more to the underlying injuries. Although FES itself is usually compatible with good recovery, 10%mortality is usually quoted.

54.7.3 Indirect optic nerve injury

General information

≈ 5% of head trauma patients manifest an associated injury to some portion of the visual system. Approximately 0.5–1.5%of head trauma patients will sustain indirect injury (as opposed to penetrating trauma) to the optic nerve, most often from an ipsilateral blow to the head (usually frontal, occasionally temporal, rarely occipital).[19] The optic nerve may be divided into 4 segments: intraocular (1 mm in length), intraorbital (25–30 mm), intracanalicular (10 mm), and intracranial (10 mm). The intracanalicular segment is the most common one damaged with closed head injuries. Funduscopic abnormalities visible on initial exam indicates anterior injuries (injury to the intraocular segment (optic disc) or the 10–15 mm of the intraorbital segment immediately behind the globe where the central retinal artery is contained within the optic nerve), whereas posterior injuries (occurring posterior to this but anterior to the chiasm) takes 4–8 weeks to show signs of disc pallor and loss of the retinal nerve fiber layer.

Treatment

See reference.[19]

No prospective study has been carried out. Optic nerve decompression has been advocated for indirect optic nerve injury, however, the results are not clearly better than expectant management with the exception that documented delayed visual loss appears to be a strong indication for surgery. Transethmoidal is the accepted route, and is usually done within 1–3 weeks from the trauma.[46] The use of "megadose steroids" may be appropriate as an adjunct to diagnosis and treatment.

54.7.4 Post-traumatic hypopituitarism

Trauma is a rare cause of hypopituitarism. It may follow closed head injury (with or without basilar skull fracture) or penetrating trauma.[47] In 20 cases in the literature[48] all had deficient growth hormone and gonadotropin, 95%had corticotropin deficiency, 85%had reduced TSH, 63%had elevated PRL. Only 40%had transient or permanent DI.

54.8 Exploratory burr holes

54.8.1 General information

In a trauma patient, the clinical triad of altered mental status, unilateral pupillary dilatation with loss of light reflex, and contralateral hemiparesis is most often due to upper brainstem compression by uncal transtentorial herniation which, in the majority of trauma cases, is due to an extraaxial intracranial hematoma. Furthermore, the prognosis of patients with traumatic herniation is poor. Outcome may possibly be improved slightly by increasing the rapidity with which decompression is undertaken, however, an upper limit of salvageability is probably still only ≈ 20% satisfactory outcome.

Burr holes are primarily a diagnostic tool, as bleeding cannot be controlled and most acute hematomas are too congealed to be removed through a burr hole. However, if the burr hole is positive, it is possible that modest decompression may be performed, and then the definitive craniotomy can be undertaken incorporating the burr hole(s).

With widespread availability of quickly accessible CT scanning, exploratory burr holes are infrequently indicated.

54.8.2 Indications

1. clinical criteria: based on deteriorating neurologic exam. Indications in E/R (rare): patient dying of rapid transtentorial herniation (see below) or brainstem compression that does not improve or stabilize with mannitol and hyperventilation.[49]
   a) indicators of transtentorial herniation/brainstem compression:
      - sudden drop in Glasgow Coma Scale (GCS) score
      - one pupil fixes and dilates

- paralysis or decerebration develops (usually contralateral to blown pupil)

b) recommended situations where criteria should be applied:
- neurologically stable patient undergoes witnessed deterioration as described above
- awake patient undergoes same process in transport, and changes are well documented by competent medical or paramedical personnel

2. other criteria
   a) some patients needing emergent surgery for systemic injuries (e.g. positive peritoneal lavage + hemodynamic instability) where there is not time for a brain CT (p. 834)

54.8.3 Management

Controversial. The following should serve only as guidelines:

1. if patient fits the above criteria (emergent operation for systemic injuries or deterioration with failure to improve with mannitol and hyperventilation), and CT scan cannot be performed and interpreted immediately, then treatment should not wait for CT scan
   a) in general, if the O.R. can be immediately available, burr holes are preferably done there (equipped to handle craniotomy, better lighting and sterility, dedicated scrub nurse…) especially in older patients (> 30 yrs) not involved in MVAs (below). This may more rapidly diagnose and treat extraaxial hematomas in herniating patients, although no difference in outcome has been proven
   b) if delay in getting to the O.R. is foreseen, emergency burr holes in the E/R should be performed
2. placement of burrhole(s) as outlined under Technique below

54.8.4 Technique

Position

Shoulder roll, head turned with side to be explored up. Three pin skull-fixation used if concern about possible aneurysm or AVM (to allow for retractors and increased stability) or if additional stability is desired (e.g. with unstable cervical fractures), otherwise a horse-shoe head-holder suffices and saves time and makes it easier to turn the head to access to the other side if needed.

Choice of side for initial burr hole

Start with a temporal burr hole (see below) on the side:

1. ipsilateral to a blown pupil. This will be on the correct side in > 85% of epidurals[50] and other extra-axial mass lesions[51]
2. if both pupils are dilated, use the side of the first dilating pupil (if known)
3. if pupils are equal, or it is not known which side dilated first, place on side of obvious external trauma
4. if no localizing clues, place hole on left side (to evaluate and decompress the dominant hemisphere)

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Approach

Burr holes are placed along a path that can be connected to form a “trauma flap” if a craniotomy becomes necessary (Fig. 54.2). The “trauma flap” is so-called because it provides wide access to most of the cerebral convexity permitting complete evacuation of acute blood clot and control of most bleeding.

First outline the trauma flap with a skin marker:

1. start at the zygomatic arch < 1 cm anterior to the tragus (spares the branch of the facial nerve to the frontalis muscle and spares the anterior branch of the superficial temporal artery (STA))
2. proceed superiorly and then curve posteriorly at the level of top of the pinna
3. 4–6 cm behind the pinna it is taken superiorly
4. 1–2 cm ipsilateral to the midline (sagittal suture) curve anteriorly to end behind the hairline

Burr hole locations

1. first (temporal) burrhole: over middle cranial fossa (# 1 in Fig. 54.2) just superior to the zygomatic arch. Provides access to middle fossa (the most common site of epidural hematoma) and usually allows access to most convexity subdural hematomas, as well as proximity to middle meningeal artery in region of pterion

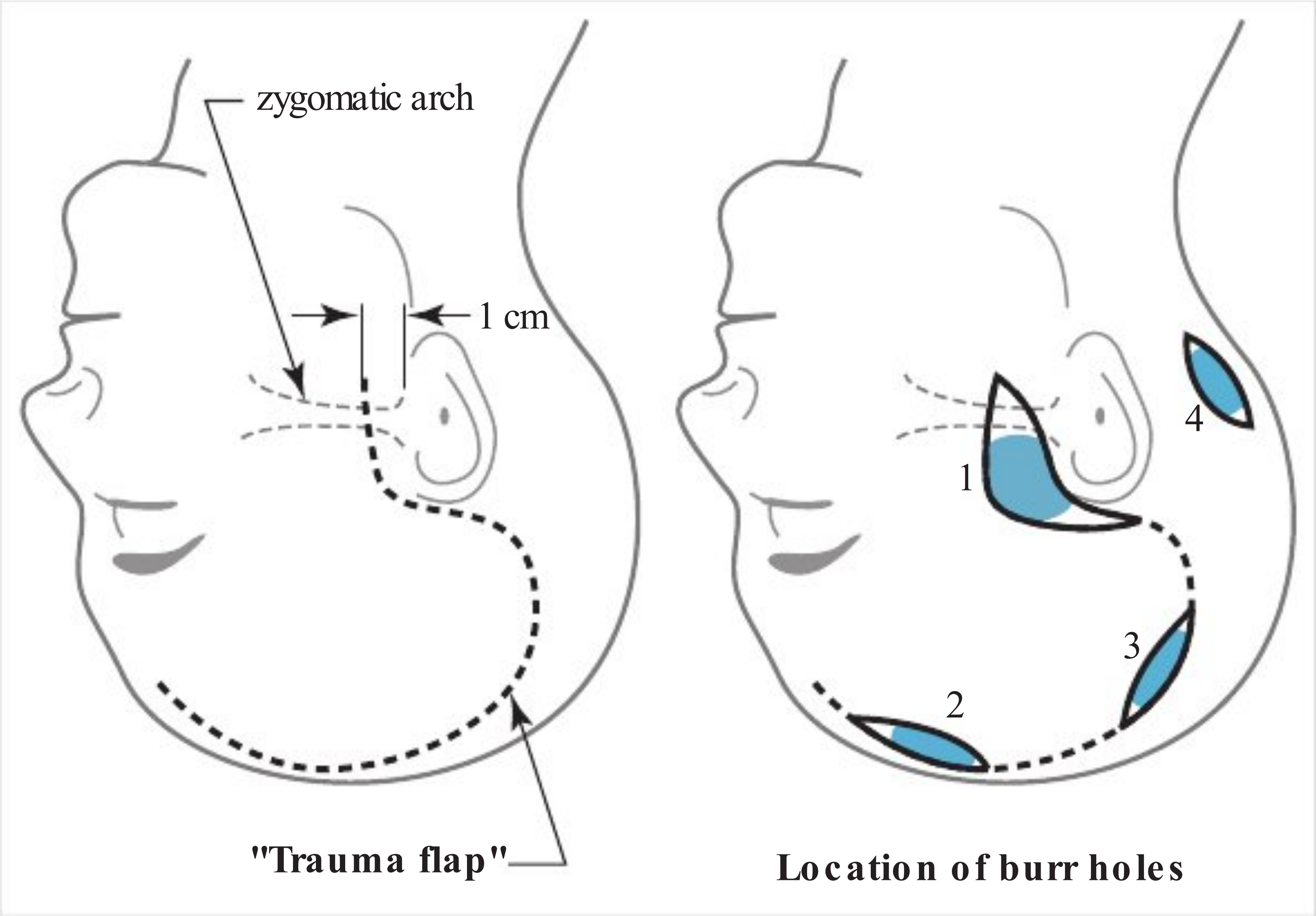

Fig. 54.2 Technique to convert burrhole(s) into trauma flap (adapted[51,52])

2. if no epidural hematoma, the dura is opened if it has bluish discoloration (suggests subdural hematoma(SDH)) or if there is a strong suspicion of a mass lesion on that side
3. if completely negative, usually perform temporal burr hole on contralateral side
4. if negative, further burr holes should be undertaken if a CT cannot now be done
5. proceed to ipsilateral frontal burr hole (#2 in Fig. 54.2)
6. subsequent burr holes may be placed at parietal region (#3 in Fig. 54.2) and lastly in posterior fossa (#4 in Fig. 54.2)

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Literature

In 100 trauma patients undergoing transtentorial herniation or brainstem compression as outlined above,[51] exploratory burr holes (bilateral temporal, frontal and parietal, done in the O.R.) were positive in 56%. Lower rates in younger patients (<30 yrs) and those in MVAs (as opposed to falls or assaults). SDH was the most common extraaxial mass lesion (alone and unilateral in 70%, bilateral in 11%, and in combination with EDH or ICH in >9%).

When burr holes were positive, the first burr hole was on the correct side 86% of the time when placed as suggested above. Six patients had significant extraaxial hematomas missed with exploratory burr holes (mostly due to incomplete burr hole exploration). Only 3 patients had the above neurologic findings as a result of intraparenchymal hematomas.

Outcome

Mean follow-up: 11 mos (range: 1–37). 70 of the 100 patients died. No morbidity or mortality was directly attributable to the burr holes. Four patients with good outcome and 4 with moderate disability had positive burr holes.

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55 Concussion, High Altitude Cerebral Edema, Cerebrovascular Injuries

55.1 Concussion

55.1.1 General information

Key concepts

- A traumatic biomechanically induced complex pathophysiologic process affecting the brain with no identifiable structural abnormalities on imaging studies
- Concussion is a subset of mild TBI (mTBI) and is therefore not equivalent to mTBI
- Indicators of concussion: post-traumatic alterations in any of: orientation, balance, speed of reaction, and/or impaired verbal learning and memory in a patient with GCS 13–15[1]
- Does not require loss of consciousness (LOC) or even a direct blow to the head
- Grading scales have been abandoned in favor of experienced assessment assisted by various "sideline" tools

Concussion occurs in a subset of patient with mild traumatic brain injury (mTBI; Fig. 54.1). It is considered "mild" because it is usually not life-threatening by itself. While most victims recover completely, effects of concussion can be serious, and in some instances, may be lifelong.

Much of the discussion in this chapter relates to concussion in sports which, is the largest source of data on the subject, and generalization to other types of trauma must be done circumspectly.

There has been a move away from grading scales for concussion, and the current recommendation is for the diagnosis to be determined in the judgement of an experienced examiner with the assistance of various assessment tools, ideally with the availability of pre-injury baseline metrics for comparison.

Concussion can occur without a direct blow to the head, e.g. with violent shaking of the torso and head. Concussion symptoms can present soon after an insult or in a delayed fashion.

The subject may not be aware that they have sustained a concussion.

55.1.2 Epidemiology

Incidence: 1.6–3.8 million concussions occur per year in the United States from sports and recreational activities. It is estimated that 50%of concussions go unreported.[2]

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55.1.3 Concussion genetics

There is no clear evidence to support a genetic predisposition to concussion. Apolipoprotein E4, Apo E G-219 T promoter and tau exon 6 have been studied in small retrospective and prospective trials without definitive association.[2,3]

55.1.4 Concussion – definition

There is no universally accepted definition for concussion.[4] Of the many contemporary definitions,[2,3,4,5,6,7] most key elements are contained in the Concussion in Sport Group 2012 consensus definition[3] summarized below. However, opinions differ e.g. whether there is any long-term effect of concussion or if that necessitates a different diagnosis.

Definition: Concussion is a complex pathophysiological process affecting the brain resulting in alteration of brain function, that is induced by nonpenetrating biomechanical forces, without identifiable abnormality in standard structural imaging.

The Concussion in Sport Group[3] elaborates on this definition as follows:

- Results in a graded set of neurological symptoms that may or may not involve loss of consciousness (LOC).
- Symptom onset is usually rapid, short-lived and resolves spontaneously. Manifestations may include transient deficits in balance, coordination, memory/cognition, strength, or alertness.

- May result in neuropathological changes, but the acute clinical symptoms largely reflect a functional disturbance rather than a structural injury.
- Resolution of the clinical and cognitive features typically follows a sequential course.
- Typically associated with grossly normal standard structural neuroimaging studies.

55.1.5 Concussion versus mTBI

- Concussion and mTBI are not interchangeable. Concussion may be thought as a subcategory of mTBI on the less severe end of the brain injury spectrum, though with similar clinical symptoms[2,5,6,8] (see Fig. 57.2).
- A major difference between the two is that mTBI may demonstrate abnormal structural imaging (such as cerebral hemorrhage/contusion) and concussion, by definition, must have normal imaging studies. mTBI is part of an injury severity spectrum primarily based on GCS score. TBI is evaluated 6 hours after injury and differentiated into mild, moderate and severe; see Grading (p.824). Concussion is evaluated directly after the insult and based on a clinical diagnosis aided by a multitude of standardized assessment tools. To include concussion under the full spectrum of traumatic brain injury then it must fall at the low end of mTBI and overlap with the subset of "minimal" injury. Most mTBIs with negative imaging can be considered concussions but the majority of sports concussions cannot be classified as mTBI.[5,8]

55.1.6 Risk factors for concussion

- History of previous concussion increases risk for further concussion
- Being involved in an accident: bicyclist, pedestrian or motor vehicle collision
- Combat soldier
- Victim of physical abuse
- Falling (especially pediatrics or elderly)
- Males are diagnosed with sports-related concussion more than females (due to increased number of male participation in sports studied) but females have a higher risk overall when compared to males who play in the same sport. (i.e. soccer and basketball)[7]
- Participating in sports with high risk of concussion:
  - American football
  - Australian rugby
  - Ice hockey
  - Boxing
  - *Soccer is the highest risk for females
- (For contrast, sports with the lowest risk of concussion: baseball, softball, volleyball & gymnastics)
- BMI > 27 kg/m^2 and less than 3 hours of training per week increase risk of sports-related concussion[7]

55.1.7 Diagnosis

Triggers

Findings suggestive of concussion are listed in Table 55.1. The diagnosis of concussion should be considered when any of these findings occur following trauma. In pre-verbal children, findings may include those in Table 55.2.

General diagnostic information

Clinical evaluation

No physiologic measure has been identified that can detect the underlying changes that lead to the manifestations of concussion. Therefore the diagnosis relies on: self-reporting of abnormal function (symptoms), observed physiologic abnormalities (signs) including assessment of cognitive dysfunction,[11] sometimes with the assistance of imaging tests to rule out a structural substrate.

A clinical diagnosis of concussion is made if there are abnormal findings in balance, coordination, memory/cognition, strength, reaction speed or alertness after a traumatic insult to the head. Findings include confusion, amnesia, headache, drowsiness or LOC (LOC is not a requirement for diagnosing concussion,[6] patients themselves may be unaware whether or not they experienced LOC[4]). Frequent neurobehavioral features of concussion are shown in Table 55.1. In children who may not be able to verbalize their symptoms, evidence of concussion may include findings in Table 55.2. Positive imaging findings would necessitate a more severe diagnosis such as cerebral contusion

Table 55.1 Possible findings in concussion[2,9,10]

Physical	Cognitive	Emotional	Sleep
• vacant stare or befuddled expression • dazed or stunned • headache or pressure sensation in the head • nausea • vomiting • fatigue • "seeing stars" • photophobia • phonophobia • ringing in the ears (tinnitus) • delayed verbal & motor responses: • difficulty focusing attention • inability to perform normal activities • speech alterations: slurred or incoherent, disjointed or incomprehensible statements • incoordination stumbling • any period of LOC, paralytic coma, unresponsiveness to stimuli	• feeling like being in a fog • slow to answer questions or follow instructions • easy distractibility • disorientation (e.g. walking in the wrong direction) • unaware of date, time or place • memory deficits: amnesia for the event • repeatedly asking same question that has been answered	• exaggerated emotionality: inappropriate crying • distraught appearance • irritability • nervousness	• drowsiness • insomnia • hypersomnia • difficulty falling asleep or staying asleep

Table 55.2 Concussion findings in children

Listlessness and easy fatigability, change in sleeping patterns
Irritability
Appearing dazed
Balance impairment
Excessive crying
Change in eating habits
Loss of interest in favorite toys

Approach

- Take a concussion specific symptom survey including inquiries about: H/A, N/V, light sensitivity, tinnitus, feeling like being in a fog, sleep disturbances
- History of diagnoses that might have an impact on the assessment or on a current concussion
  - History of prior concussions
  - H/A history
  - ADD/HD
  - Learning disabilities
  - Medications (prescribed and other) that might affect alertness or cognition
- Perform a good general neurological exam
- Include a concussion specific neuro exam
  - Check orientation
  - Assess for amnesia and impaired verbal memory
  - Balance: Romberg test (look for significant sway or breaking stance), single leg stance
  - Eye movements: optokinetic nystagmus (OKN), smooth pursuit
  - Simultaneous task performance: e.g. snap fingers while walking
- Include assessment aides ("sideline tools") as appropriate (see below)

Assessment aids

- There is no single validated assessment tool for diagnosis of concussion.[4] It is primarily a clinical diagnosis that is ideally made by certified healthcare providers who are familiar with the patient based on a detailed history and physical examination and a continuum of evaluation from the sideline to the clinic (diagnosis is ideally within made within 24 hours of injury)[2,3,4,5,6,7]

- Diagnosis may be aided by concussion assessment tools such as the SCAT3, ImPACT™.
  No test has shown high validity on independent testing, and no test should be used as the sole method of diagnosing concussion or for determining suitability for return to play. Athletes have also learned to "game" some baseline tests to avoid removal from play after possible concussion
- SCAT3 (Sports Concussion Assessment Tool – 3rd Edition)[12]: Derived from the 2012 Zurich Conference.[3] The SCAT has become the most commonly used standardized tool for sideline assessment of sport concussion. The sensitivity and specificity of concussion assessment tools change over the course of a concussion so a tool designed for sideline use (i.e. SCAT3) is not appropriate for office use.
  - SCAT3™ is a trademarked tool developed by the Concussion in Sports Group for use only by medical professionals for assessing sports-related concussion.
  - It can be found at http://bjsm.bmj.com/content/47/5/259.full.pdf.
  - To be used in athletes of 13 years or older (for 12 and younger, use Child SCAT3[13])
  - Is a multimodal assessment tool with 8 sections that includes self-reported symptoms and evaluation of functional domains such as cognition, memory, balance, gait and motor skills
  - Takes 8–10 min to administer
  - A "normal" SCAT3 does not rule out concussion
  - It has not been validated
- Other types of sports concussion assessment tools (many can be viewed on YouTube):
  - Neurocognitive testing (may take up to 20 minutes to administer)
  - SAC (Standardized Assessment of Concussion)[14]: a neurocognitive test that includes tests of immediate memory, delayed recall, serial 7's, digit span
  - ImPACT™ (Immediate Post-Concussion Assessment and Cognitive Testing): a widely used commercially produced computer test (https://www.impacttest.com). Independent validation studies have yielded conflicting results and results can diverge from observations[15]
  - PCSS (Post-Concussive Symptom Scale)
  - CSI (Concussion Symptom Inventory)
  - BESS (Balance Error Scoring System): the subject stands in each of various standardized positions for 20 seconds each, and the number of errors are recorded (breaking stance, opening eyes, taking hands off hip…).
  - SOT (Sensory Organization Test)
  - "Concussion Quick Check" app for mobile devices produced by the AAN
  - King-Devick eye movement testing: only takes 2–3 minutes to administer. On printed cards or tablet computer (http://kingdevicktest.com/for-concussions/).
- Formal neuropsychological testing: it is recommended that this be reserved for patients with prolonged cognitive symptoms
- Concussion serum biomarkers: no moiety has been identified that can reliably diagnose concussion on serum or saliva testing. Neuron-specific enolase, S100, and cleaved tau protein have been studied for prognostication after mTBI and concussion. S100 has demonstrated only a 33.3%sensitivity for postconcussive symptoms and 93%sensitivity for an Extended Glasgow Outcome Scale < 5 at 1 month. Another study involving pediatric patients with mTBI showed no difference in levels of neuron-specific enolase or S100B in asymptomatic and symptomatic children. A prospective study found no significant correlation between cleaved tau protein and postconcussive syndrome in patients with mTBI[16]

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On-site/sideline evaluation

Any individual suspected of having a concussion (displaying ANY findings in Table 55.1) should be removed from the activity (for athletes, stopped from playing) and assessed by a licensed healthcare provider trained in the evaluation and management of concussions with attention to excluding a cervical spine injury.[2,3] If no provider is available, return to the activity is not permitted and urgent referral to a physician should be arranged.

After ruling out emergency issues, the provider should perform a concussion assessment (may employ standardized tools such as SCAT3™ or other methodologies).

The patient should not be left alone, and serial evaluations for signs of deterioration should be made over the following few hours.

For return to play guidelines, see below.

55.1.8 Indications for imaging or other diagnostic testing

Imaging in concussion is typically used to rule out more serious traumatic injuries.

Indications[4] for CT or MRI imaging:

- Adults with or without LOC or amnesia
  - Focal neurologic deficit
  - GCS < 15
  - Severe headache
  - Coagulopathy
  - Vomiting
  - Age > 65 years old
  - Seizures
- Peds
  - LOC > 60 secs
  - Evidence of skull fracture
  - Focal neurologic deficit

Other imaging studies:

- Diffusion Tensor Imaging (DTI): used to quantify white matter tract integrity throughout the brain with 4 types of analysis methods – voxel analysis, region of interest (ROI) analysis, histogram analysis, and tractography. There is no strong consensus regarding the best method for utilizing DTI for diagnosis or prognosis in the individual patient but multiple studies have shown group differences in DTI parameters between mTBI and control patients.[8]
- Functional MRI (fMRI): consists of 2 types (task-based fMRI and resting state fMRI) and is based on the blood oxygen level dependent (BOLD) effect, in which specialized MRI sequences measure/detect regions of increased oxygen rich blood flow to areas of upregulated neuronal activity. Both task-based and resting state fMRI modalities have shown group differences between mTBI and control patients (specifically in frontal lobe dysfunction) but further studies need to be completed on both a single time point and longitudinal basis before these techniques can be widely adopted for individual diagnosis and therapeutic guidance.[8]
- Imaging studies that are currently used primarily in concussion research: positron emission tomography (PET), single photon emission CT (CT-SPECT), MR-spectroscopy (MRS).

Quantitative EEG (QEEG) is another research tool for concussion that assesses brain activity, patterns of cortical activation and neuronal networks. The concept is that post-concussion studies are compared to baseline. Currently undergoing proof of concept evaluation.

55.1.9 Acute pathophysiology

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Biomechanical force results in unregulated ionic (K^+ efflux, Na^+/Ca^{2+} influx) flux and unrestricted hyperacute glutamate release from sublethal mechanoporation of lipid membranes at the cellular level. This triggers voltage/ligand gated ion channels causing a cortical spreading depression-like state that is thought to be the substrate behind immediate postconcussive symptoms. Subsequently, ATP-dependent ionic pumps are extensively upregulated to restore cellular homeostasis causing widespread intracellular energy reserve depletion and an increase in ADP. Cells then pass into a state of impaired metabolism (energy crisis) that can last up to 7–10 days and may be associated with alterations in CBF. This impaired metabolic state is associated with vulnerability to repeat injury as well as behavioral and spatial learning impairments. Cells also undergo cytoskeletal damage, axonal

Table 55.3 Physiologic perturbations and their proposed corresponding symptomology[17]

Pertubation	Symptom
Ionic influx →	Migraine headache, photophobia, phonophobia
Energy Crisis →	Vulnerability to second injury
Axonal injury →	Impaired cognition, slowed processing, slowed reaction time
Impaired neurotransmission →	Impaired cognition, slowed processing, slowed reaction time
Protease activation, altered cytoskeletal proteins, cell death →	chronic atrophy, persistent impairments

dysfunction, and altered neurotransmission with the as yet unproven impression that each of these pathologic processes correlate with a separate symptomology.[17]

55.1.10 Post concussion syndrome (PCS)

Occurs in 10%–15% of concussed individuals. As with most concussion related pathologies, there are multiple definitions of PCS. An amalgam of some definitions is as follows: Patients having ≥ 3 symptoms including headache, fatigue, dizziness, irritability, difficulty concentrating, memory difficulty, insomnia, and intolerance to stress, emotion, or alcohol, and symptoms must begin within 4 weeks of injury and remain for ≥ 1 month after onset of symptoms.[16,18] In one retrospective study, the following conclusions were reached[18]:

- > 80% of PCS patients had at least 1 previous concussion
- average number of previous concussions was 3.4
- median duration of PCS was 6 months
- 50% of patients were < 18 years of age
- LOC does not increase the risk for PCS

55.1.11 Prevention of concussion

- The AAN guidelines conclude that protective headgear in rugby is "highly probable" to decrease the incidence of concussion.[7] However, the AMSSM (American Medical Society for Sports Medicine) hold that there is no clear evidence that soft or hard helmets reduce the severity or incidence of concussion (in football, lacrosse, hockey, soccer, and rugby).[2,3] Biomechanical studies have shown helmets reduced impact forces on the brain but this has not translated into concussion prevention.[3]
- There is insufficient data to determine whether one type of football helmet protects better than another in preventing concussions[3,7]
- No significant evidence that a mouthpiece protects against concussion[3,7]

55.1.12 Management of concussion and post-concussion syndrome

Return to Play (RTP)

- No system of return to play (RTP) guidelines has been rigorously tested and proven to be scientifically sound
- After sustaining a concussion, athletes should not return to play the same day.[2,3,4,5,6,7] Prohibited by some state laws.
- a symptomatic player should not return to competition.
- If there is any uncertainty: "When in doubt, sit them out"
- Evaluation should proceed in a stepwise fashion. A player needs to be completely asymptomatic both at rest and with provocative exercise before full clearance is given.[3] There is no standardized RTP protocol. Each player's progression should be individualized.[2] Generally, the athlete's level of activity should be gradually increased over 24 hour increments from light aerobic activity to full contact practice. The athlete is evaluated after each progression. If postconcussive symptoms occur then the player is dropped back to the previous asymptomatic level and then allowed another attempt at progression after a 24-hour rest period. 80–90% of concussions resolve within 7–10 days. This recovery time may be longer for children or adolescents.[3]
- The CDC endorses a graded 5-step return to play for student athletes[19] as shown in Table 55.4. The athlete should move to the next step only if they have no new symptoms. If symptoms return or new ones develop, then medical attention should be sought and, after clearance the student can return to the previous step.

Contraindications for return to play are shown in Table 55.5.

Management of post-concussive syndrome

An extremely complicated topic, partly because of potential for litigation and the fact that symptoms are often vague and nonspecific and there may be no objective findings to corroborate subjective symptoms.

Most symptoms from concussion resolve within 7–10 days and do not require treatment. The most common exception to this is post-traumatic headache, the most common subtype being acute post-traumatic migraine.

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Table 55.4 5-step return to play progression

Step	Description
Baseline	Athlete is back to regular school activities without symptoms
1	Light aerobic activity: only to increase heart rate for 5–10 minutes. No weight lifting
2	Moderate activity: increase heart rate with body or head movement. May include moderate intensity weight training (less time and intensity than their typical routine)
3	Heavy, non-contact activity: may include running, high-intensity stationary biking, regular weight training, non-contact sports-specific drills
4	Practice & full contact: in controlled practice
5	Competition

Table 55.5 Cerebral contraindications for return to contact sports

1. persistent postconcussion symptoms
2. permanent CNS sequelae from head injury (e.g. organic dementia, hemiplegia, homonymous hemianopsia)
3. hydrocephalus
4. spontaneous SAH from any cause
5. symptomatic (neurologic or pain producing) abnormalities about the foramen magnum (e.g. Chiari malformation)

Typical symptoms include: H/A, dizziness, insomnia, exercise intolerance, depression, irritability, anxiety, memory loss, difficulty concentrating, fatigue, light or noise hypersensitivity.

Patients with protracted symptoms may require more directed treatment.

- Psychological and neuropsychological involvement is often employed.
- Pharmacologic treatment: there are no evidence based studies of the utility of medications for post-concussive symptoms (aside form H/A).
- Intractable headaches: occurs in ≈ 15%of concussions
  - Expert neurology consultation is usually required for difficult to control headaches
  - The first line drugs are OTC medications
  - Triptans are usually employed for nonresponders
  - Third line drugs includ Ketorolac or DHE-45 (dihydroergotamine)
  - Steroids may be beneficial for some
  - Avoid: narcotics, butalbital/caffeine preparations (Fioricet, Esgic…), beta blockers and calcium channel blockers

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55.1.13 Second impact syndrome (SIS)

A rare condition described primarily in athletes who sustain a second head injury while still symptomatic from an earlier one. Classically, the athlete walks off the field under their own power after the second injury, only to deteriorate to coma within 1–5 minutes and then, due to vascular engorgement, develops malignant cerebral edema that is refractory to all treatment and progresses to herniation. Mortality: 50–100%

A syndrome compatible with SIS was first described by Schneider[20] in 1973, and was later dubbed the “second impact syndrome of catastrophic head injury” in 1984.[21] Although it is contended that SIS is rare (if it exists at all) and may be overdiagnosed,[22] its apparent predilection for teens and children still warrants extra precaution following concussion.

55.1.14 Chronic traumatic encephalopathy (CTE)

There is limited evidence-based research involving the pathophysiology and natural history of CTE. Thought to be a distinct neurodegenerative disease (tauopathy) associated with repetitive brain trauma, not limited to athletes with reported concussions, and can only be diagnosed postmortem with a pathology-confirmed analysis. Small studies have shown that there is a variable age of onset with variable behavioral, mood, and cognitive deficits present at the time of death (92%symptomatic at time of death).[10,16]

See section about CTE (p.924) for further details.

55.2 Other TBI definitions

55.2.1 Contusion

A contusion is a TBI with CT findings that may include:

- low attenuation areas: representing associated edema
- high attenuation areas (AKA “hemorrhagic contusions”): usually produce less mass effect than their apparent size. Most common in areas where sudden deceleration of the head causes the brain to impact on bony prominences (e.g. temporal, frontal and occipital poles). These areas may progress (or “blossom” in neuroradiological jargon) to frank parenchymal hemorrhages. Surgical decompression may sometimes be considered if herniation threatens (p.891).

55.2.2 Contrecoup injury

(French: “counter blow”) in addition to the potential injury to the brain directly under the point of impact, the force imparted to the head may cause the brain to be thrust against the skull directly opposite the blow. May result in contusions typically in locations described above.

55.2.3 Other definitions

Posttraumatic brain swelling

This term encompasses two distinct processes:

1. increased cerebral blood volume: may result from loss of cerebral vascular autoregulation (p.856). This hyperemia may sometimes occur with extreme rapidity, in which case it has sometimes been referred to as diffuse or “malignant cerebral edema”[23] which carries close to 100% mortality and may be more common in children. Management consists of aggressive measures to maintain ICP < 20 mm Hg and CPP > 60 mm Hg.[24] CPP ≥ 70 mm Hg is generally recommended, see ICP treatment threshold (p.866).
2. true cerebral edema: classically at autopsy these brains “weep fluid”.[25] Both vasogenic and cytotoxic cerebral edema (p.90) can occur within hours of head injury[25,26] and occasionally may be treated with decompressive craniectomy (p.891).

Diffuse axonal injury (DAI) (AKA diffuse axonal shearing)

A primary lesion of rotational acceleration/deceleration head injury.[27] In its severe form, hemorrhagic foci occur in the corpus callosum and dorsolateral rostral brain stem with microscopic evidence of diffuse injury to axons (axonal retraction balls, microglial stars, and degeneration of white matter fiber tracts). Often cited as the cause of loss of consciousness in patients rendered immediately comatose following head injury in the absence of a space occupying lesion on CT[28] (although DAI may also be present with subdural[29] or epidural hematomas[30]).

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May be diagnosed clinically when loss of consciousness (coma) lasts > 6 hours in absence of evidence of intracranial mass or ischemia. May be graded as shown in Table 55.6.

55.3 High-altitude cerebral edema

Acute high-altitude sickness (AHAS) is a systemic disorder that affects individuals usually within 6–48 hrs after ascent to high altitudes. Acute mountain sickness (AMS) is the most common form of AHAS, with symptoms of nausea, headache, anorexia, dyspnea, insomnia and fatigue[31]

Table 55.6 Grading DAI

DAI grade	Description
mild	coma > 6–24 hrs, followed by mild-to-moderate memory impairment, mild-to-moderate disabilities
moderate	coma > 24 hrs, followed by confusion & long-lasting amnesia. Mild-to-severe memory, behavioral and cognitive deficits
severe	coma lasting months with flexor and extensor posturing. Cognitive, memory, speech, sensorimotor and personality deficits. Dysautonomia may occur

and is often assessed using the Lake Louise system.[32] The incidence is ≈ 25% at 7,000 feet, and ≈ 50% at 15,000 feet. Other symptoms of AHAS include edema of feet and hands, and pulmonary edema (HAPE = high altitude pulmonary edema). Ocular findings include retinal hemorrhages,[33] nerve fiber layer infarction, papilledema and vitreous hemorrhage.[34] Cerebral edema (HACE = high altitude cerebral edema), usually associated with pulmonary edema, may occur in severe cases of AHAS. Symptoms of HACE include: severe headache, mental dysfunction (hallucinations, inappropriate behavior, reduced mental status), and neurologic abnormalities (ataxia, paralysis, cerebellar findings).

The unproven “tight-fit” hypothesis postulated that individuals with less compliant CSF systems (smaller ventricles and CSF spaces) were more vulnerable to AMS.[35] A small study of 10 volunteers[36] analyzing CT scans before ascent and symptoms showed a trend that supports the hypothesis.

Prevention: gradual ascent, 2–4 day acclimatization at intermediate altitudes (especially to include sleeping at these levels), avoidance of alcohol or hypnotics.

Treatment of cerebral edema: immediate descent and oxygen (6–12 L/min by NC or face-mask) are recommended. Dexamethasone 8 mg PO or IV followed by 4 mg q 6 hrs may help temporize.

55.4 Traumatic cervical artery dissections

55.4.1 General information

Cervical arterial dissections are a subset of cervical cerebrovascular injuries as shown below.

Cervical cerebrovascular injuries:

- penetrating injury (p.1017)
- traumatic dissection: the subject of this chapter
  - due to blunt trauma
  - due to stretching: e.g. from neck hyperextension or therapeutic spinal manipulation
  - iatrogenic: dissection caused by intimal tear from angiography catheters
- traumatic compression or occlusion
  - kinking from malalignment: e.g. with cervical fracture-dislocation
  - compression by bone fragments: e.g. by fractures through foramen transversarium

This chapter deals with traumatic cervical artery dissections. There is significant overlap with spontaneous cerebrovascular arterial dissections (p.1322), however, features that are more pertinent to post-traumatic dissections are covered here.

Optimal screening, diagnostic and treatment methods are controversial. A 13% mortality rate is considered low. Nearly one-third of patients are not treatable.

55.4.2 Epidemiology

Incidence: 1–2% of blunt trauma patients[37] (among those who stayed > 24 hrs in a trauma hospital the incidence was 2.4%[37]).

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55.4.3 Risk factors

Traumatic risk factors for blunt cerebrovascular injury (BCVI) are shown in Table 55.7. Risk factors not directly related to the type of trauma include fibromuscular dysplasia, where dissections may follow minor injuries because of increased susceptibility. BCVI can occur even in the absence of identifiable risk factors.[37]

55.4.4 Presentation

Signs and symptoms of BCVI are shown in Table 55.8.

55.4.5 Evaluation of patients with risk factors or signs/symptoms of BCVI

The following is an adaptation of the guidelines of the Western Trauma Association flow chart[39] (including footnotes!) into an outline format. Their recommendations are based on observational studies and expert opinion (no Class I data was available).

NB: CTA on scanners with ≥ 16 detectors (16-slice multidetector CT angiography, (16MD-CTA) have an accuracy near 99%[40] & equivalent predictive value to cerebral angiogram. MRA[41,42] and

Table 55.7 Traumatic risk factors for BCVI[38,39]

• high energy transfer mechanism associated with: ◦ displaced mid face fracture: LeForte fracture type II or III (p.887) ◦ basilar skull fracture involving carotid canal
• TBI consistent with DAI and GCS < 6
• cervical vertebral body or transverse foramen fracture, subluxation, or ligamentous injury at any level
• any fracture involving C1–3
• near hanging with anoxic brain injury
• clothesline-type injury or seat belt abrasion with significant cervical swelling, pain, or mental status changes

Table 55.8 Signs & symptoms of BCVI[39]

• arterial hemorrhage from neck/nose/mouth (? go to O.R.)
• cervical bruit in pt. < 50 yrs old
• expanding cervical hematoma
• focal neurologic deficit: TIA, Horner's syndrome, hemiparesis, VBI
• neurologic deficit inconsistent with head CT
• stroke on CT or MRI

ultrasound[43,44] are not considered adequate for BCVI screening. If unavailable, then catheter angiography should be employed.

1. 16MD-CTA should be obtained as follows:
   a) emergently in patients with signs/symptoms of BCVI (Table 55.8)
   b) asymptomatic patients with risk factors (Table 55.7) for BCVI:
      - if the presence of BCVI would alter therapy (e.g. no contraindication to heparin) then MDCTA should be done within 12 hours if possible
      - if heparin is contraindicated due to associated injuries, timing of MDCTA is determined by patient stability
2. if the MDCTA is equivocal, or if it is negative but clinical suspicion remains high: a catheter arteriogram should be done (otherwise, if negative: stop)
3. grading: if the MDCTA or the arteriogram shows positive findings (p.1324):
   a) the injury is graded using the scale shown in Table 55.9 [45] (sometimes referred to as the "Denver grading scale")
   b) proceed with grade-based management (see below)

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55.4.6 Management of documented BCVI

Antiplatelet therapy is as effective in preventing stroke as anticoagulation for cerebrovascular dissection.[46,47]

Grade specific therapy

- Grade I & II
  - Most resolve on their own
  - Even though there might be a slight benefit of heparin over aspirin for low grade injuries, due to the low overall risk the general trend is to treat these with aspirin
- Grade III
  - Anticoagulate with heparin. Rationale: heparin and aspirin are roughly equivalent for Grade III, however, most will need to be restudied in 7-10 days
  - Repeat angiogram or 16MD-CTA 7-10 days post-injury. See below for subsequent management
- Grade IV: endovascular occlusion to prevent embolization
- Grade V: highly lethal injury
  - accessible lesions should be considered for urgent surgical repair (anecdotal)
  - inaccessible lesions (the majority): incomplete transection may be amenable to endovascular stenting with concurrent antithrombotics; complete transections should be ligated (or occluded endovascularly)

Table 55.9 BCVI grading scale,[45] "Denver grading scale"

Grade	Description
I	luminal irregularity with <25% stenosis
II	≥ 25% luminal stenosis or intraluminal thrombus or raised intimal flap
III	pseudoaneurysm
IV	occlusion
V	transection with free extravasation

For Grade III repeat MDCTA or angiography 7–10 days post injury to assess healing.[48] Results:
- lesion healed: discontinue anticoagulation
- non-healed lesions:
  - consider endovascular stenting "with caution" for severe luminal narrowing or expanding pseudoaneurysm (controversial: results have been mixed – favorable[46] and unfavorable[49])
  - transition from heparin to aspirin (75–150 mg/d) alone
  - repeat MDCTA or angiography 3 months post injury (rationale: most heal with canalization in 6 wks). Results:
    - healed lesion: consider discontinuing aspirin
    - non-healed: optimal drug and duration is not known. Recommendation[39]: lifelong antiplatelet therapy with either aspirin or clopidogrel. Dual therapy is used for acute coronary syndromes and following angioplasty (± stenting) but is not recommended in patients who have had a stroke or TIA[50]

Heparinization:

When anticoagulation is employed, perform a baseline PTT and then begin heparin drip 15 U/kg/hr IV. Repeat PTT after 6 hours, and titrate to PTT = 40–50 seconds.

Trauma contra-indications to anticoagulation: patients that are actively bleeding, have potential for bleeding, or in whom the consequences of bleeding are severe. Specific examples include: liver and spleen injuries, major pelvic fractures, and intracranial hemorrhage.

Dissection-related anticoagulation risks include: extension of the medial hemorrhage (with possible SAH), and intracerebral hemorrhage (conversion of pale infarct to hemorrhagic).

55.4.7 Carotid artery blunt injuries

General information

See general information related to cerebral arterial dissections and spontaneous dissections (p. 1322). For evaluation and management, see above.

This section considers blunt (i.e. nonpenetrating) specifically related to ICA dissection. Neck hyperextension with lateral rotation is a common mechanism of injury, and is thought to stretch the ICA over the transverse processes of the upper cervical spine. In posttraumatic dissection, ischemic symptoms are the most common.[51]

Etiologies:
1. following MVAs: the most common etiology
2. attempted strangulation[52]
3. spinal manipulation therapy: VA dissections are more common than ICA

Most carotid dissections start ≈ 2 cm distal to the ICA origin.

Clinical

The risk of stroke with various ICA dissection grades is shown in Table 55.10. The risk of stroke increases with increasing grade for ICA injuries. This does not hold true for VA injuries.

Grade I injuries: 70% heal with or without heparin. 25% will persist. 4–12% will progress to more severe grade. Data suggests that anticoagulation reduces the risk of progression.[38]

Grade II: ≈ 70% progress to more severe grade even with heparin therapy.

Grade III & IV: most persist.

Table 55.10 Risk of stroke with ICA dissection

Grade[a]	Description	Stroke risk
I	stenosis < 25%	3%
II	stenosis > 25%	11%
III	pseudoaneurysm	44%
IV	occlusion	uniformly lethal

[a]for grading, see Table 55.9

Table 55.11 Time to presentation after non-penetrating trauma

Time	%
0–1 hours	6–10% of cases
1–24 hours	57–73%
after 24 hours	17–35%

Initially, there may be no neurologic sequelae, however, progressive thrombosis, intramural hemorrhage or embolic phenomenon may develop in a delayed fashion. The distribution of time delays following trauma to time of presentation are shown in Table 55.11 (the majority are evident within the 1st 24 hours).

Management

See Management of documented BCVI (p. 849).

Outcome

Natural history is not well known. Many patients with minor symptoms may not present and presumably do well. In one series, 75% of patients returned to normal, 16% had a minor deficit, and 8% had a major deficit or died.[53]

55.4.8 Vertebral artery blunt injuries

General information

See anatomy of vertebral artery segments (p. 80).

Blunt vertebral artery injury (BVI) is very rare, being found in 0.5–0.7% of patients with blunt injuries with aggressive screening.[54] It may produce vertebrobasilar insufficiency (VBI) or posterior circulation stroke. Fractures through the foramen transversarium, facet fracture-dislocation, or vertebral subluxation are frequently identified in patients with BVI[38,55,56] (overall incidence increases to 6% in the presence of cervical fracture or ligamentous injury[54]).

Etiologies

While motor vehicle accidents are the most common mechanism of injury, any trauma that can injure the C-spine can cause BVI (diving accidents, spinal manipulation…).

1. automobile accidents
2. spinal manipulation therapy (SMT): including chiropractic[57] or similar, which comprise 11 of 15 case reports reviewed by Caplan et al.[58] VA dissections were independently associated with SMT within 30 days in multivariate analysis (odds ratio = 6.62, 95% CI 1.4 to 30)[59]
3. sudden head turning
4. direct blows to the back of the neck[58]

Stroke from BVI

The Denver grade of the dissection in BVI does not correlate with risk of stroke or mortality (as it does with ICA dissection).[60] Unlike with carotid injuries, there is rarely a premonitory "warning" TIA. Time from injury to stroke: mean 4 days (range: 8 hours -12 days).

Evaluation

When BVI is identified, it is critical to assess the status of the contralateral VA.

Practice guideline: Vertebral artery blunt injuries

Evaluation

Level I[61]

- Patients meeting the "Denver Screening Criteria" (symptoms shown in Table 55.8 or risk factors shown in Table 55.7) should undergo 16MD-CTA to screen for BVI

Level III[61]

- Catheter angiogram is recommended in select patients after blunt cervical trauma if 16MD-CTA is not available especially if concurrent endovascular intervention is a consideration.
- MRI is recommended for BVI after blunt cervical trauma in patients with incomplete SCI or vertebral subluxation injuries.

Treatment

Practice guideline: Vertebral artery blunt injuries

Treatment

Level III[61]

- No specific guidelines were made among treatment options (anticoagulation, antiplatelet therapy, or no treatment)
- The role of endovascular therapy for BVI has not been defined

Strokes were more frequent in patients with BVI who were not treated initially with IV heparin despite an asymptomatic BVI.[38] However, based on historical controls, it is not clear if either screening or treatment improves overall outcome.[54]

Recommendations: treat all BVI with aspirin. Restudy chronic occlusion in 3 months.

Treatment options include endovascular stenting when amenable. This can restore near-normal flow, but long-term results are lacking.[62] Also, stenting requires ≥≈3 months of antiplatelet therapy which is contraindicated in some situations.

Outcome

Overall mortality with unilateral BVI ranges from 8-18%[60] which is lower than with ICA dissections (17-40%). Bilateral VA dissection appears highly fatal.

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56 Neuromonitoring

56.1 General information

This section considers neuromonitoring instrumentation that can be done primarily at the patient's bedside, and therefore does not include CT perfusion studies, PET scans…. The bulk of neuromonitoring literature deals with intracranial pressure (ICP). Other parameters that can be monitored include: jugular venous oxygen monitoring (p.865), regional CBF (p.866), brain tissue oxygen tension (p.865), and brain metabolites (pyruvate, lactate, glucose…) (p.866).

The role of adjunctive monitoring is currently unknown. Unanswered questions include: should neuromonitoring be disease specific (e.g. is SAH different from TBI), which monitors provide additional unique information, what are the critical values of the monitored entity, and what interventions should be undertaken to correct abnormalities?

56.2 Intracranial pressure (ICP)

56.2.1 Background

Intracranial pressure (ICP) is discussed in this section on trauma because of the close relationship between elevated ICP and brain damage from head injury. However, factors involved in diagnosing and treating intracranial hypertension (IC-HTN) also may pertain (with modifications) to brain tumors, dural venous thrombosis, etc.

56.2.2 Cerebral perfusion pressure (CPP) and cerebral autoregulation

Secondary brain injury (i.e. following the initial trauma) is attributable in part to cerebral ischemia; see Secondary injury (p.824). The critical parameter for brain function and survival is not actually ICP, rather it is adequate cerebral blood flow (CBF) to meet $CMRO_2$ demands; see discussion of CBF & $CMRO_2$ (p.1264). CBF is difficult to quantitate, and can only be measured continuously at the bedside with specialized equipment and difficulty.[1] However, CBF depends on cerebral perfusion pressure (CPP), which is related to ICP (which is more easily measured) as shown in Eq (56.1).

$$\begin{aligned}&\{\text{cerebral perfusion pressure}\} = \{\text{mean arterial pressure}^{*}\} - \{\text{intracranial pressure}\}\\&\text{or, expressed in symbols}\\&CPP = MAP^{*} - ICP\end{aligned} \tag{56.1}$$

*note: the actual pressure of interest is the mean carotid pressure (MCP) which may be approximated as the MAP with the transducer zeroed ≈ at the level of the foramen of Monro.[2]

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As ICP becomes elevated, CPP is reduced at any given MAP. Normal adult CPP is > 50 mm Hg. Cerebral autoregulation is a mechanism whereby over a wide range, large changes in systemic BP produce only small changes in CBF. Due to autoregulation, CPP would have to drop below 40 in a normal brain before CBF would be impaired.

In the head injured patient, older recommendations were to maintain CPP ≥ 70 mm Hg due to increased cerebral vascular resistance.[3] However, recent evidence suggests that elevated ICP (≥ 20 mm Hg) may be more detrimental than changes in CPP (as long as CPP is > 60 mm Hg[4])[5] (higher levels of CPP were not protective against significant ICP elevations[5]).

56.2.3 ICP principles

The following are approximations to help simplify understanding ICP (these are only models, and as such are not entirely accurate):

1. normal intracranial constituents (and approximate volumes):
   a) brain parenchyma (which also contains extracellular fluid): 1400 ml
   b) cerebral blood volume (CBV): 150 ml
   c) cerebrospinal fluid (CSF): 150 ml
2. these volumes are contained in an inelastic, completely closed container (the skull)

3. pressure is distributed evenly throughout the intracranial cavity (in reality, pressure gradients exist[6,7])
4. the modified Monro-Kellie doctrine[8] states that the sum of the intracranial volumes (CBV, brain, CSF, and other constituents (e.g. tumor, hematoma…)) is constant, and that an increase in any one of these must be offset by an equal decrease in another. The mechanism: there is a pressure equilibrium in the skull. If the pressure from one intracranial constituent increases (as when that component increases in volume), it causes the pressure inside the skull (ICP) to increase. When this increased ICP exceeds the pressure required to force one of the other constituents out through the foramen magnum (FM) (the only true effective opening in the intact skull) that other component will decrease in size via that route until a new equilibrium is established. The cranio-spinal axis can buffer small increases in volume with no change or only a slight increase in ICP. If the expansion continues, then the new equilibrium will be at a higher ICP. The result:
   a) at pressures slightly above normal, if there is no obstruction to CSF flow (obstructive hydrocephalus), CSF can be displaced from the ventricles and subarachnoid spaces and exit the intracranial compartment via the FM
   b) intravenous blood can also be displaced through the FM via the IJVs
   c) as pressure continues to rise, arterial blood is displaced and CPP decreases, eventually producing diffuse cerebral ischemia. At pressures equal to mean arterial pressure, arterial blood will be unable to enter the skull through the FM, producing complete cessation of blood flow to the brain, with resultant massive infarction
   d) increased brain edema, or an expanding mass (e.g. hematoma) can push brain parenchyma downward into the foramen magnum (cerebral herniation) although brain tissue cannot actually exit the skull

56.2.4 Normal ICP

The normal range of ICP varies with age. Values for pediatrics are not well established. Guidelines are shown in Table 56.1.

56.2.5 Intracranial hypertension (IC-HTN)

General information

Traumatic IC-HTN may be due any of the following (alone or in various combinations):
1. cerebral edema
2. hyperemia: the normal response to head injury.[10] Possibly due to vasomotor paralysis (loss of cerebral autoregulation). May be more significant than edema in raising ICP (p.901) [11]
3. traumatically induced masses
   a) epidural hematoma
   b) subdural hematoma
   c) intraparenchymal hemorrhage (hemorrhagic contusion)
   d) foreign body (e.g. bullet)
   e) depressed skull fracture
4. hydrocephalus due to obstruction of CSF absorption or circulation
5. hypoventilation (causing hypercarbia → vasodilatation)
6. systemic hypertension (HTN)
7. venous sinus thrombosis
8. increased muscle tone and valsalva maneuver as a result of agitation or posturing → increased intrathoracic pressure → increased jugular venous pressure → reduced venous outflow from head
9. sustained posttraumatic seizures (status epilepticus)

Table 56.1 Normal ICP

Age group	Normal range (mm Hg)
adults and older children[a]	<10–15
young children	3–7
term infants[b]	1.5–6

[a]the age of transition from "young" to "older" child is not precisely defined
[b]may be subatmospheric in newborns[9]

A secondary increase in ICP is sometimes observed 3–10 days following the trauma, and may be associated with a worse prognosis.[12] Possible causes include:

1. delayed hematoma formation
   a) delayed epidural hematoma (p.894)
   b) delayed acute subdural hematoma (p.898)
   c) delayed traumatic intracerebral hemorrhage[13] (or hemorrhagic contusions) with perilesional edema: usually in older patients, may cause sudden deterioration. May become severe enough to require evacuation (p.892)
2. cerebral vasospasm[14]
3. severe adult respiratory distress syndrome (ARDS) with hypoventilation
4. delayed edema formation: more common in pediatric patients
5. hyponatremia

Clinical presentation – Cushing's triad

The classic clinical presentation of IC-HTN (regardless of cause) is Cushing's triad which is shown in Table 56.2. However, the full triad is only seen in ≈ 33% of cases of IC-HTN.

Patients with significant ICP elevation due to trauma, brain masses (tumor) or hydrocephalus (but paradoxically not with pseudotumor cerebri) will usually be obtunded.

CT scan and elevated ICP

Whereas CT findings may be correlated with a risk of IC-HTN, no combination of CT findings has been shown to allow accurate estimates of actual ICP. 60% of patients with closed head injury and an abnormal CT will have IC-HTN.[15] (Note: "abnormal" CT: demonstrates hematomas (EDH, SDH or ICH), contusions,[15] compression of basal cisterns (p.921), herniation or swelling.[16,17])

Only 13% of patients with a normal CT scan will have IC-HTN.[15] However, patients with a normal CT AND 2 or more risk factors identified in Table 56.3 will have ≈ 60% risk of IC-HTN. If only 1 or none are present, ICP will be increased in only 4%

56.2.6 ICP monitoring

Indications for ICP monitoring

Practice guideline: Indications for ICP monitoring

For salvageable patients with severe traumatic brain injury (GCS ≤ 8 after cardiopulmonary resuscitation)

Level II[17]: with an abnormal admitting brain CT(note: abnormal" CT: demonstrates hematomas (EDH, SDH or ICH), contusions,[15] compression of basal cisterns (p.921), herniation or swelling[16,17])

Level III[17]: with a normal admitting brain CT, but with ≥ 2 of the risk factors for IC-HTN in Table 56.3

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1. neurologic criteria: see Practice guideline: Indications for ICP monitoring (p.858)
   a) some centers monitor patients who don't follow commands. Rationale: patients who follow commands (GCS ≥ 9) are at low risk for IC-HTN, and one can follow sequential neurologic exams in these patients and institute further evaluation or treatment based on neurologic deterioration
   b) some centers monitor patients who don't localize, and follow neuro exam on others

Table 56.2 Cushing's triad with elevated ICP

1. hypertension
2. bradycardia
3. respiratory irregularity

Table 56.3 Risk factors for IC-HTN with a normal CT

- age > 40 yrs
- SBP < 90 mm Hg
- decerebrate or decorticate posturing on motor exam (unilateral or bilateral)

2. multiple systems injured with altered level of consciousness (especially where therapies for other injuries may have deleterious effects on ICP, e.g. high levels of PEEP or the need for large volumes of IV fluids or the need for heavy sedation)
3. with traumatic intracranial mass (EDH, SDH, depressed skull fracture…)
   a) a physician may choose to monitor ICP in some of these patients[16,18]
   b) post-op, subsequent to removal of the mass
4. non-traumatic indications for ICP monitoring:
   a) some centers monitor ICP in patients with acute fulminant liver failure with an INR > 1.5 and Grade III of IV coma. A recent study shows that a subarachnoid bolt may be inserted after administration of factor VII 40 mcg/kg IV over 1–2 minutes (the bolt is inserted as soon as possible (usually within 15 minutes and no more than 2 hours after administration)) without significant risk of hemorrhage. All patients were treated with hypothermia; other ICP treatment measures were used for refractory IC-HTN

Contraindications (relative)

1. "awake" patient: monitor usually not necessary, can follow neuro exam
2. coagulopathy (including DIC): frequently seen in severe head injury. If an ICP monitor is essential, take steps to correct coagulopathy (FFP, platelets…) and consider subarachnoid bolt or epidural monitor (an IVC or intraparenchymal monitor is contraindicated). See recommended range of PT or INR (p. 157).

Duration of monitoring

D/C monitor when ICP normal × 48–72 hrs after withdrawal of ICP therapy. Caution: IC-HTN may have delayed onset (often starts on day 2–3, and day 9–11 is a common second peak especially in peds). Also see delayed deterioration (p. 824). Avoid a false sense of security imparted by a normal early ICP.

Complications of ICP monitors

General information

Table 56.4 for a summary of complication rates for various types of monitors.[3]

1. infection: see below
2. hemorrhage[3]: overall incidence is 1.4% for all devices (see Table 56.4 for breakdown). The Angioma Alliance[21] definition of hemorrhage: acute or subacute symptoms (any of: headache, seizure, impaired consciousness, or new/worsened focal neurological deficit referable to the anatomic location of the CM) accompanied by radiological, pathological, surgical, or rarely only cerebrospinal fluid evidence of recent extra- or intralesional hemorrhage. This definition does not include either an increase in CM diameter without other evidence of recent hemorrhage, nor the presence of a hemosiderin halo. Risk of significant hematoma requiring surgical evacuation is ≈ 0.5–2.5%[15,22,23]
3. malfunction or obstruction: with fluid coupled devices, higher rates of obstruction occur at ICPs > 50 mm Hg
4. malposition: 3% of IVCs require operative repositioning

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Table 56.4 Complication rates with various types of ICP monitors

Monitor type	Bacterial colonization[a]	Hemorrhage	Malfunction or obstruction
IVC	ave: 10–17% range[19,20]: 0–40%	1.1%	6.3%
subarachnoid bolt	ave: 5% range: 0–10%	0	16%
subdural	ave: 4% range: 1–10%	0	10.5%
parenchymal	ave: 14% (two reports, 12% & 17%)	2.8%	9–40%

[a]some studies report this as infection, but do not distinguish between clinically significant infection and colonization of ICP monitor

Infection with ICP monitors

Colonization of the monitoring device is much more common than clinically significant infection (ventriculitis or meningitis). See Table 56.4 for colonization rates. Fever, leukocytosis and CSF pleocytosis have low predictive value (CSF cultures are more helpful). Range of reported infection rates: 1–27%[24]

Practice guideline: Infection prophylaxis with ICP monitors

Level III[25]: neither prophylactic antibiotics nor routine ventricular catheter exchange is recommended to reduce infection

Identified risk factors for infection include[20,24,26,27]:
1. intracerebral, subarachnoid or intraventricular hemorrhage
2. ICP > 20 mm Hg
3. duration of monitoring: contradictory results in literature. One prospective study in 1984 found an increased risk with monitor duration > 5 days (infection risk reaches 42% by day # 11).[22,26] Another found no correlation with monitoring duration.[28] A retrospective analysis[20] found a non-linear increase of risk during the first 10–12 days after which the rate diminished rapidly
4. neurosurgical operation: including operations for depressed skull fracture
5. irrigation of system
6. leakage around IVCs
7. open skull fractures (including basilar skull fractures with CSF leak)
8. other infections: septicemia, pneumonia

Factors not associated with increased incidence of infection:
1. insertion of IVC in neuro intensive care unit (instead of O.R.)
2. previous IVC
3. drainage of CSF
4. use of steroids

Treatment of infection

Removal of device if at all possible (if continued ICP monitoring is required consideration may be given to inserting a monitor at another site) and appropriate antibiotics.

Types of monitors

1. intraventricular catheter (IVC): AKA external ventricular drainage (EVD), connected to an external pressure transducer via fluid-filled tubing. The standard by which others are judged (also below; note: other options for IVCs utilize transducers tipped with fiberoptic or strain gauge devices which are located within the intraventricular catheter; in this discussion, "IVC" does not refer to this type)
   a) advantages:
      - most accurate (can be recalibrated to minimize measurement drift)[29]
      - lower cost
      - in addition to measuring pressure, allows therapeutic CSF drainage (may help reduce ICP directly, and may drain particulate matter, e.g. blood breakdown products after SAH, that could occlude arachnoid granulations[30])
   b) disadvantages
      - may be difficult to insert into compressed or displaced ventricles
      - obstruction of the fluid column (e.g. by blood clot, or by coaptation of the ependymal lining on the catheter as the ventricle collapses with drainage) may cause inaccuracy
      - some effort is required to check and maintain function, e.g. IVC problems (p. 862) and IVC trouble shooting (p. 863)
      - transducer must be consistently maintained at a fixed reference point relative to patient's head (must be moved as HOB is raised/lowered)
2. intraparenchymal monitor (e.g. Camino labs or Honeywell/Phillips[31,32]): similar to IVC but more expensive. Some are subject to measurement drift,[33,34] others may not be[35]
3. less accurate monitors
   a) subarachnoid screw (bolt): risk of infection 1%, rises after 3 days. At high ICPs (often when needed most) surface of brain may occlude lumen → false readings (usually lower than actual, may still show ≈ normal waveform)

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b) subdural: may utilize a fluid coupled catheter (e.g. Cordis Cup catheter), fiberoptic tipped catheter, or strain gauge tipped catheter
c) epidural: may utilize a fluid coupled catheter, or fiberoptic tipped catheter (e.g. Ladd fiberoptic). Accuracy is questionable
d) in infants, one can utilizing an open anterior fontanelle (AF):
   - fontanometry[36]: probably not very accurate
   - applanation principle: may be used in suitable circumstances (viz.: if the fontanelle is concave with the infant upright, and convex when flat or head down) to estimate the ICP within 1 cm H_2O.[9] The infant is placed supine, and the AF is visualized and palpated while the head is raised and lowered. When the AF is flat, the ICP equals atmospheric pressure, and ICP can be estimated in cm H_2O as the distance from the AF to the point where the venous pressure is 0 (for a recumbent infant, the midpoint of the clavicle usually suffices). If the AF is not concave with the infant erect, then this method cannot be used because either the ICP exceeds the distance from the AF to the venous zero point, or the scalp may be too thick

Conversion factors: between mm Hg and cm H_2O are shown in Eq (56.2) and Eq (56.3) (the density of mercury is 13.6 times that of water, and CSF is fairly close to water).

$$1 \text{ mm Hg (torr)} = 1.36 \text{ cm } H_2O \tag{56.2}$$

$$1 \text{ cm } H_2O = 0.735 \text{ mmHg (torr)} \tag{56.3}$$

Intraventricular catheter (IVC)

Insertion technique

For technique to place catheter in frontal horn, see Kocher's point (p.1512). The right side is usually used unless specific reasons to use the left are present (e.g. blood clot in right lateral ventricle which might occlude IVC).

Set-up

Fig. 56.1 shows a typical external ventricular drainage (EVD) system/ ventriculostomy ICP monitor. Not every system will have the same components (some may have less and some may have more). Note that the effect of having an opening on the top of the drip chamber (through an air-filter) is the same as having the drip nozzle open to air, and therefore as long as this filter is not wet or plugged the pressure in the IVC is regulated by the height of the nozzle (as read on the pressure scale; note that the "0" is level with the nozzle).

The external auditory canal (EAC) is often used as a convenient external landmark for "0" (approximates the level of the foramen of Monro). In Fig. 56.1 the drip chamber is illustrated at 8 cm above the EAC.

Normal functioning of the IVC system

The system should be checked for proper functioning at least every 2–4 hours, and any time there is a change in: ICP (increase or decrease), neuro exam, or CSF output (for systems open to drainage).

1. check for presence of good waveform with respiratory variations and transmitted pulse pressures
2. IVCs: to check for patency, open the system to drain and lower the drip chamber below level of head and observe for 2–3 drops of CSF (normally do not allow more than this to drain)
3. for systems open to drainage:
   a) volume of CSF in drip chamber should be indicated every hour with a mark on a piece of tape on the drip chamber, and the volume should increase with time unless ICP is less than the height of the drip chamber (in practice, under these circumstances the system would usually not be left open to drainage).
   NB: the maximum expected output from a ventriculostomy would be ≈ 450–700 ml per day in a situation where none of the produced CSF is absorbed by the patient. This is not commonly encountered. A typical amount of drainage would be ≈ 75 ml every 8 hrs
   b) drip chamber should be emptied into drainage bag regularly (e.g. q 4 or 8 hours) and any time the chamber begins to get full (record volume)

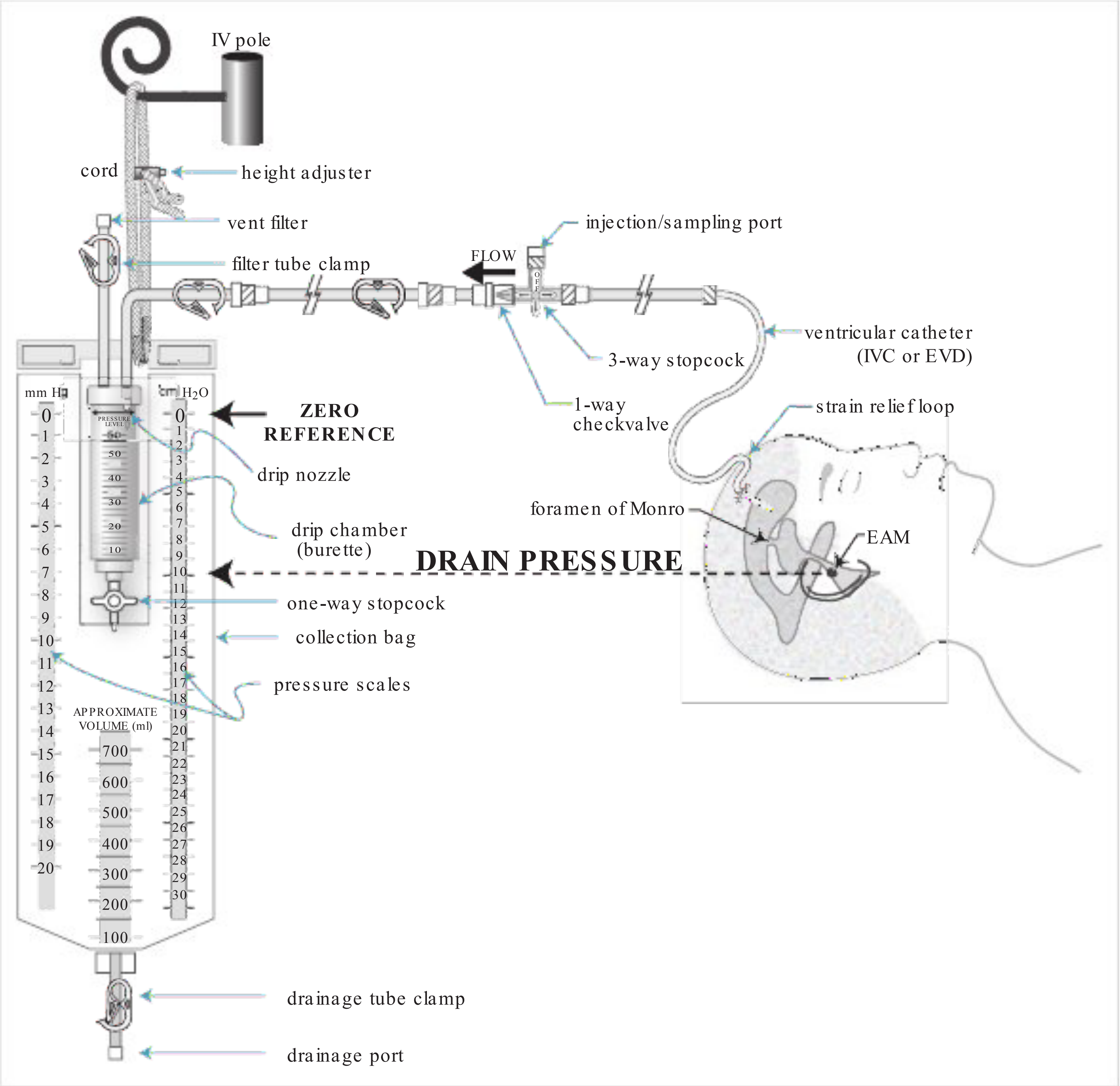

Fig. 56.1 Medtronic® ventricular drainage system/ICP monitor

4. in cases where there is a question whether the monitor is actually reflecting ICP, lowering the HOB towards 0° should increase ICP. Gentle pressure on both jugular veins simultaneously should also cause a gradual rise in ICP over 5–15 seconds that should drop back down to baseline when the pressure is released

IVC problems

The following represents some of the error or pitfalls that commonly occur with external ventricular drainage. Some also apply to ICP monitoring in general.

1. air filter on drip chamber gets wet (prevents air from passing through filter)
   a) result: fluid cannot drain freely into drip chamber (the pressure is no longer regulated by the height of the drip nozzle)
      - if the outflow from the drip chamber is clamped, then no flow at all is possible
      - if the clamp on the drip chamber outlet is open, then the pressure is actually regulated by the height of the nozzle in the collection bag and not the nozzle in the drip chamber
   b) solution: if a fresh filter is available, then replace the wet one. Otherwise one must improvise (with the risk of exposing the system to contamination): e.g. replace the wet filter with a filter from an IV set, or with a sterile gauze taped over the opening
2. air filter on collection bag gets wet: this will make it difficult to empty the drip chamber into the bag

a) this is not usually an urgent problem unless the drip chamber is full and the collection bag is distended tensely with air
b) the filter will dry out with time and will usually start to work again
c) if it is necessary to empty the drip chamber before the filter is dry, then using sterile technique insert a needle into the bag drainage port and decompress the bag of fluid and air
3. improper connections: a pressurized irrigation bag with or without heparinized solution should never be connected to an ICP monitor
4. changing position of head of bed: must move drip chamber up or down to keep it level with the same external landmarks (e.g. level of auditory canal):
 a) when open to drainage, this will assure the correct pressure will be maintained
 b) when opened to pressure transducer, will maintain correct zero
5. when open to drain, pressure reading from transducer is not meaningful: the pressure cannot exceed the height of the drip chamber in this situation (because at that point, fluid will drain off), and the opening to the “atmosphere” in the drip chamber will dampen the waveform
6. drip chamber falls to floor:
 a) overdrainage, possible seizures and/or subdural hematoma formation
 b) solution: securely tape chamber to pole, bed-rail…, check position regularly

IVC troubleshooting

See also IVC problems above.

▶ IVC no longer works:
1. manifestation of problem:
 a) dampening or loss of normal waveform
 b) no fluid drains into drip chamber (applies only when catheter has been opened to drain)
2. possible causes:
 a) occlusion of catheter proximal to transducer
  • slide clamp closed or stopcock closed
  • catheter occluded by brain particles, blood cells, protein
 b) IVC pulled out of ventricle
  • test: temporarily lower drip nozzle and watch for 2–3 drops CSF
  • solution:
 c) verify all clamps are open
 d) flush no more than 1.5 ml of non-bacteriostatic saline (AKA preservative-free saline) with very gentle pressure into ventricular catheter (NB: in elevated ICP the compliance of the brain is abnormally low and small volumes can cause large pressure changes)
  • if no return then brain or clot is probably plugging catheter. If it is known that the ventricles are ≈ completely collapsed then the IVC may be OK and CSF should still drain over time. Otherwise this is a non-functioning catheter, and if a monitor/drain is still indicated then a new catheter may need to be inserted (CT may be considered first if the status of the ventricles is not known). If catheter is clotted by intraventricular hemorrhage, rt-PA may sometimes be used (p.1344) [37]

▶ ICP waveform dampened:
1. possible causes:
 a) occlusion of catheter proximal to transducer: see above
 b) IVC pulled out of ventricle: no fluid will drain
 c) air in system:
  • solution: allow CSF to drain and expel air
  • caution: do not allow excessive amount of CSF to drain (may allow obstruction of catheter, subdural formation…). Do not inject fluid to flush air into brain
 d) following decompressive craniectomy: due to the fact that the monitor is no longer in a closed space, this is a normal finding in this setting

Types of ICP waveforms

Normal waveforms

The normal ICP waveform (as occurs with normal blood pressure and in the absence of IC-HTN) as illustrated in ▶ Fig. 56.2 is rarely seen since ICP is usually monitored only when it is elevated. The origin of the variations seen in the normal tracing is somewhat in dispute. One explanation describes these two types of waveforms[38]:

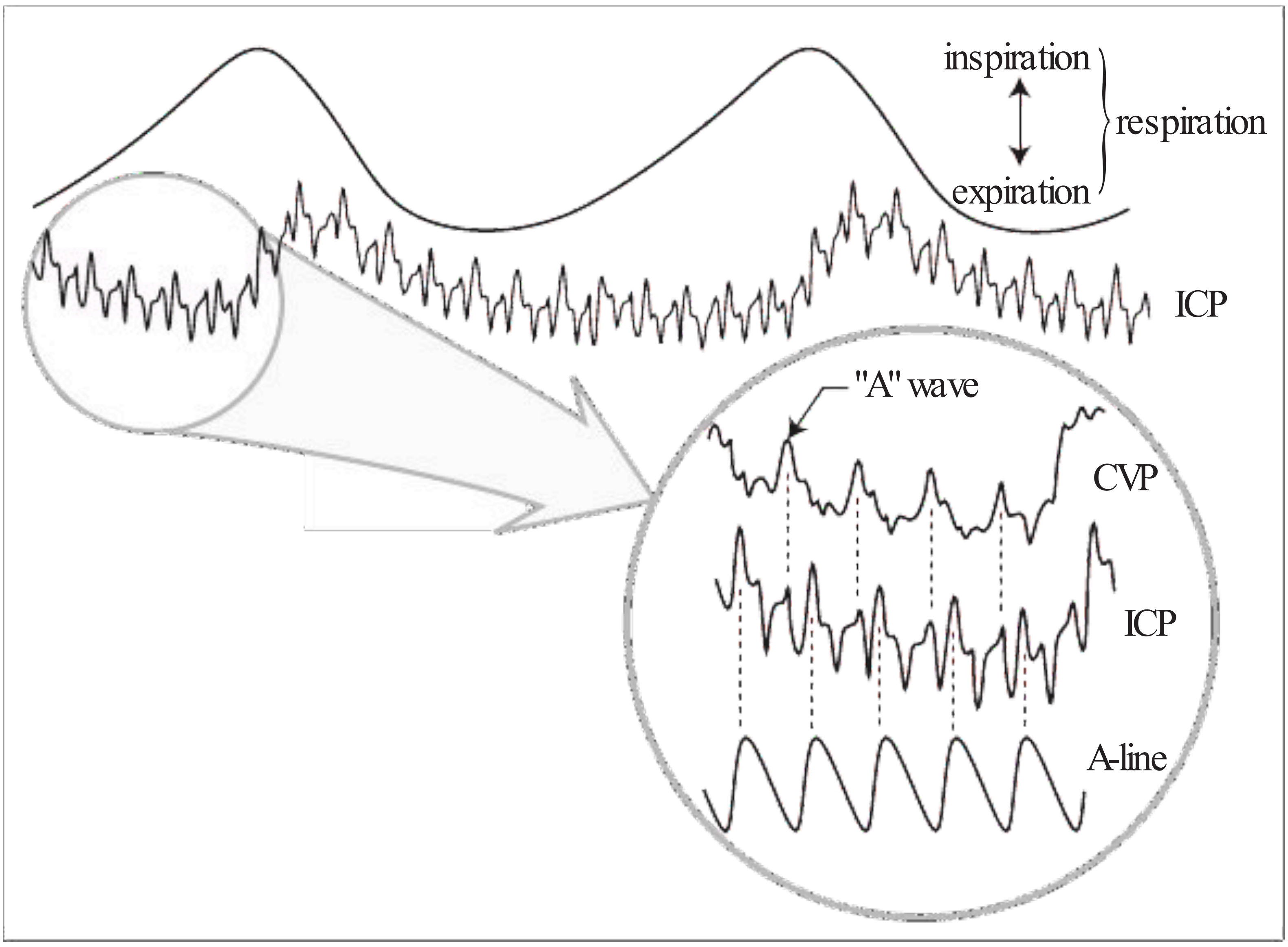

Fig. 56.2 Normal ICP waveform

1. small pulsations transmitted from the systemic blood pressure to the intracranial cavity
   a) large (1–2 mm Hg) peak corresponding to the arterial systolic pressure wave, with a small dicrotic notch
   b) this peak is followed by smaller and less distinct peaks
   c) followed by a peak corresponding to the central venous "A" wave from the right atrium
2. blood pressure pulsations are superimposed on slower respiratory variations. During expiration, the pressure in the superior vena cava increases which reduces venous outflow from the cranium causing an elevation in ICP. This may be reversed in a mechanically ventilated patients, and is opposite to that in the lumbar subarachnoid space which follows the pressure in the inferior vena cava

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Pathological waveforms

As ICP rises and cerebral compliance decreases, the venous components disappear and the arterial pulses become more pronounced. In right atrial cardiac insufficiency, the CVP rises and the ICP waveform takes on a more "venous" or rounded appearance and the venous "A" wave begins to predominate.

A number of "pressure waves" that are more or less pathologic have been described. Currently, this classification is not considered to be of great clinical utility, with more emphasis being placed on recognizing and successfully treating elevations of ICP. Plateau waves will rarely be seen because they are usually aborted at the onset by instituting treatments outlined herein (p.866). A brief description of some of these waveforms is included here for general information[39]:

1. Lundberg A waves AKA plateau waves (of Lundberg, ▯ Fig. 56.3): ICP elevations ≥ 50 mm Hg for 5–20 minutes. Usually accompanied by a simultaneous increase in MAP (it is debated whether the latter is cause or effect)
2. Lundberg B waves AKA pressure pulses: amplitude of 10–20 mm Hg is lower than A waves. Variation with types of periodic breathing. Last 30 secs – 2 mins
3. Lundberg C waves: frequency of 4–8/min. Low amplitude C waves (AKA Traube-Hering waves) may sometimes be seen in the normal ICP waveform. High amplitude C-waves may be pre-terminal, and may sometimes be seen on top of plateau waves

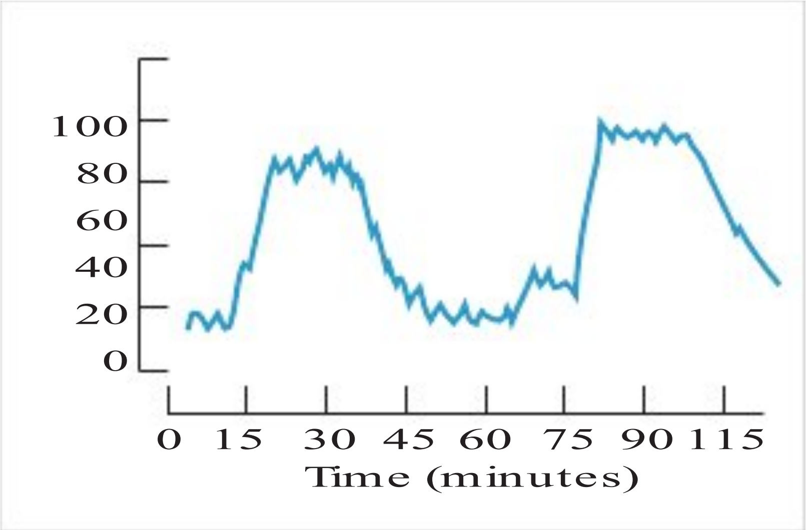

Fig. 56.3 Plateau waves (Lundberg A waves)

56.3 Adjuncts to ICP monitoring

56.3.1 Jugular venous oxygen monitoring

Indications for SjVOs or $pBtO_2$ monitoring include the need for augmented hyperventilation (PCO_2 = 20-25) to control ICP. Parameters related to oxygen content of the blood in the jugular veins are global in nature and are insensitive to focal pathology. Requires retrograde placement of catheter near to the origin of the internal jugular vein at the base of the skull. Parameters that can be measured:

1. jugular venous oxygen saturation ($SjVO_2$): measured continuously with special fiberoptic catheter. Normal $SjVO_2$: ≥60% Desaturations to <50%suggest ischemia. Multiple desaturations (<50%), or sustained (≥10 minutes) or profound desaturation episodes are associated with poor outcome.[40,41] Sustained desaturations should prompt an evaluation for correctable etiologies: kinking of jugular vein, anemia, increased ICP, poor catheter position, CPP <60 mm Hg, vasospasm, surgical lesion, $PaCO_2$ <28 mm Hg. High $SjVO_2$ >75%may indicate hyperemia or infarcted tissue and is also associated with poor outcome[42]
2. jugular vein oxygen content (CVO_2). Requires intermittent sampling of blood
3. arterial-jugular venous oxygen content difference ($AVdO_2$)[43]: $AVdO_2$ >9 ml/dl (vol%) probably indicates global cerebral ischemia,[44,45] while values <4 ml/dl indicate cerebral hyperemia[46] ("luxury perfusion" in excess of the brain's metabolic requirement[45])

56.3.2 Brain tissue oxygen tension monitoring ($pBtO_2$)

Indications for SjVOs or $pBtO_2$ monitoring include the need for augmented hyperventilation (PCO_2 = 20-25) to control ICP. Monitored e.g. with Licox® probe. The likelihood of death increases with longer times of brain tissue oxygen tension ($p_{Bt}O_2$) <15 mm Hg or even a brief drop of $pBtO_2$ <6.[47] Initial $pBtO_2$ <10 mm Hg for >30 minutes correlates with increased risk of death or bad outcome.[48] Also, see Practice guideline: Brain oxygen monitoring (p.867).

Probe placement:

1. TBI: assumed to be a diffuse process, often placed on least injured side
2. SAH: placed in vascular distributions at greatest risk of vasospasm
   a) ACA (with ACA or a-comm aneurysm): standard frontal placement (≈ 2–3 cm off midline on appropriate side)
   b) MCA (with ICA or MCA aneurysm): 4.5–5.5 cm off midline
   c) ACA-MCA watershed area: 3 cm lateral to midline
3. ICH: usually placed near the site of the hemorrhage

Effect of $pBtO_2$ monitoring/intervention on outcome: no randomized studies

1. in TBI[49]: goal was to maintain $pBtO_2$ >25 mm Hg. Adding $pBtO_2$ monitoring resulted in improved outcome. May have been result of increased attentiveness ("Hawthorne effect")
2. in SAH[50]: a moving correlation coefficient (ORx) between CPP and $pBtO_2$ was used to label high ORx as disturbed autoregulation, and this value on post SAH days 5 & 6 had predictive value for delayed infarction

Management suggestions for $pBtO_2$ <15–20 mmHg:

1. consider jugular venous O_2 saturation monitor or lactate microdialysis monitor for confirmation

2. consider CBF study to determine generalizability of $pBtO_2$ monitor reading
3. treatment: proceed to each tier as needed
   a) tier 1
      - keep body temperature < 37.5 C
      - increase CPP to > 60 mm Hg (use fluids preferentially to pressors until CVP > 8 cm H_2O, then use pressors)
   b) tier 2
      - increase FiO_2 to 60%
      - increase $paCO_2$ to 45–50 mm Hg
      - transfuse PRBCs until Hgb > 10 g/dl
   c) tier 3
      - increase FiO_2 to 100%
      - consider increasing PEEP to increase PaO_2 if FiO_2 is at 100%
      - decrease ICP to < 10 mm Hg (drain CSF, mannitol, sedation…)

56.3.3 Bedside monitoring of regional CBF (rCBF)

Thermal diffusion flowmetry permits continuous rCBF monitoring by assessing thermal convection due to tissue blood flow. The probe tip is inserted into the white matter of the brain. Commercially available systems include Hemedex® monitoring system (Codman) utilizing the QFLOW 500® probe which is not MRI compatible.

Probe placement: issues similar to those discussed for $pBtO_2$ (see above).

Readout:

1. K value (thermal conduction): range for white matter is 4.9–5.8 mW/cm-°C (the monitor suppresses CBF readings if the K value is outside this range)
   a) K<4.9: the probe tip is probably out of the brain tissue or white matter – the probe should be advanced 1–2 mm
   b) K>5.8 the tip is probably too deep, near a blood vessel, or in the ventricle or epidural or subdural space – the probe should be retracted 1–2 mm
2. CBF
   a) normal white matter: 18–25 ml/100g-min
      - white matter CBF < 15: may indicate vasospasm or ischemia
      - white matter CBF < 10: may indicate infarction
   b) normal gray matter: 67–80 ml/100g-min

Observational data: in a small study of SAH (n=5) and TBI (n=3)[51] there was good correlation between rCBF and $pBtO_2$ 91% of the time. Monitoring was not possible 36% of the time due to patient fever (wherein the system prevents monitoring).

56.3.4 Cerebral microdialysis

Compounds assayed include: lactate, pyruvate, lactate/pyruvate ratio, glucose, glutamate, urea and electrolytes including K^+ & calcium. Some observational data:

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1. lactate levels increase during episodes of $SjVO_2$ desaturation[52]
2. decreased extracellular glucose was associated with increased mortality[53]

56.4 Treatment measures for elevated ICP

56.4.1 General information

This section presents a general protocol for treating documented (or sometimes clinically suspected) intracranial hypertension (IC-HTN). Guidelines promulgated by the Brain Trauma Foundation[3,54,55,56] are generally followed. Unless otherwise stated, guidelines are for adult patients (≥ 18 years age).

56.4.2 Treatment thresholds

Intracranial pressure treatment thresholds

The optimal ICP at which to begin treatment is not known. Various cutoff values are used at different centers above which treatment measures for intracranial hypertension (IC-HTN) are initiated.

Although 15, 20 and 25 have been quoted, the Brain Trauma Foundation guideline is ICP > 20 mm Hg[57] as shown in Practice guideline: ICP treatment threshold (p. 867). Caution: patients can herniate even at ICP < 20[58] (depends on location of intracranial mass).

Rationale: There is high mortality and worse outcome[5] among patients with ICP persistently > 20 compared to 20% in those where ICP could be kept < 20.[59] Better control may be possible by treating early rather than waiting and trying to control higher ICPs or when plateau waves occur.[60]

Practice guideline: ICP treatment threshold

Level II[57]: treatment for IC-HTN should be initiated for ICP > 20 mm Hg

Level III[57]: the need for treatment should be based on ICP in combination with clinical examination & brain CT findings

Cerebral perfusion pressure (CPP)

The optimal value for CPP has yet to be determined. The threshold for ischemia is in the range of CPP < 50–60 mm Hg. Because of deleterious systemic effects, paradigms of maintaining CPP > 70 mm Hg have been superseded. Practice guideline: Cerebral perfusion pressure issues (p. 867) outlines current recommendations regarding CPP.

Practice guideline: Cerebral perfusion pressure issues

Level II[61]: avoid aggressive use of fluids and pressors to maintain CPP > 70 mm Hg (because of risk of adult respiratory distress syndrome (ARDS))

Level III[61]: avoid CPP < 50 mm Hg

Level III[61]: ancillary monitoring of CBF, oxygenation or metabolism assists CPP management

Brain oxygenation parameters

Suggestions for treatment thresholds are shown in Practice guideline: Brain oxygen monitoring (p. 867). It remains to be determined which interventions are useful to achieve this, and whether this improves outcome.

Practice guideline: Brain oxygen monitoring

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Level III[62]: jugular venous O_2 saturation < 50% or brain tissue oxygen tension ($pBtO_2$) < 15 mm Hg are treatment thresholds

56.4.3 ICP management protocol: Quick reference summary

Table 56.5 summarizes a protocol for control of IC-HTN (below for details).

Dosages are given for an average adult, unless specified as mg/kg. Treatment may be initiated prior to insertion of a monitor if there is acute neurologic deterioration or clinical signs of IC-HTN, but continued treatment requires documentation of persistent IC-HTN.

For persistent IC-HTN consider “second tier” therapies (p. 871).

Temporary measures which may be used to quickly treat an acute ICP crisis are shown in Table 56.6.

Table 56.5 Summary of measures to control IC-HTN[a] Goals: keep ICP < 20 mm Hg, and CPP ≥ 50 mm Hg[57,61]

Step	Rationale/Remedy
GENERAL MEASURES (should be utilized routinely)	
elevate HOB to 30–45°	↓ ICP by enhancing venous outflow, but also reduces mean carotid pressure → no net change in CBF
keep neck straight, avoid neck constrictions (tight trach tape, tight cervical collar…)	constriction of jugular venous outflow causes ↑ ICP
avoid arterial hypotension (SBP < 90 mm Hg)	• Hypotension reduces CBF • □: normalize intravascular volume, use pressors if needed
control hypertension if present	• □: nicardipine if not tachycardic • □: beta-blocker if tachycardic (labetalol, esmolol…) • □ avoid overtreatment → hypotension
avoid hypoxia (PaO_2 < 60 mm Hg or O_2 sat < 90%)	hypoxia may cause further ischemic brain injury □: maintain airway and adequate oxygenation
ventilate to normocarbia ($PaCO_2$ = 35–40 mm Hg)	□ avoid prophylactic hyperventilation (p. 872)
light sedation: e.g. codeine 30–60 mg IM q 4 hrs PRN	(same as heavy sedation, see below)
controversial: prophylactic hypothermia. If used, hold at target temp > 48 hrs	Hypothermia →↓ $CMRO_2$ – efficacy not rigorously proven (p. 872)
unenhanced head CT scan for ICP problems[b]	rule out surgical condition
SPECIFIC MEASURES FOR IC-HTN	
proceed to successive steps if documented IC-HTN persists – each step is ADDED to the previous measure)	
heavy sedation (e.g. fentanyl 1–2 ml or MSO4 2–4 mg IV q 1 hr) and/or paralysis (e.g. vecuronium 8–10 mg IV)	reduces elevated sympathetic tone and HTN induced by movement, tensing abdominal musculature…
drain 3–5 ml CSF if IVC present	reduces intracranial volume
hyperventilate to $PaCO_2$ = 30–35 mm Hg ("blows off" CO_2)	CO_2 is a potent vasodilator Hyperventilation → ↓ $PaCO_2$ → ↓ CBV → ↓ ICP □ hyperventilation also → ↓ CBF
mannitol 0.25–1 gm/kg, then 0.25 mg/kg q 6 hrs, increase dose if IC-HTN persists & serum osmol ≤ 320 (NB: skip this step if hypovolemia or hypotension)	Mannitol → initially↑ plasma volume & ↑ serum tonicity which draws fluid out of brain → ↓ intracranial volume, may also improve rheologic properties of blood. □ Mannitol is an osmotic diuretic, and eventually → ↓ plasma volume
if there is "osmotic room" (i.e. serum osmol<320) bolus with 10–20 ml of 23.4% hypertonic saline (HS)	some patients refractory to mannitol will respond to HS
Augmented hyperventilation to ↓ $PaCO_2$ to 25–30 mm Hg	Due to risk of cerebral ischemia from ↓ CBF, monitor $SjVO_2$ (p. 865) or CBF if possible
If IC-HTN persists, consider unenhanced head CT[b] & EEG[c]. Proceed to "second tier" therapy (p. 871)	

[a]see text for details (p. 870). As IC-HTN subsides, carefully withdraw treatment
[b]if IC-HTN persists, and especially for a sudden unexplained rise in ICP or loss of previously controlled ICP, give strong consideration to repeating cranial CT to rule out a surgical condition, i.e. "clot" (SDH, EDH, or ICH) or hydrocephalus
[c]EEG to rule-out subclinical status epilepticus which is a rare cause of sustained IC-HTN

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Table 56.6 Measures to treat an acute ICP crisis[a]	
Step	Rationale
Verify basics: check airway, neck position…(see general measures in Table 56.5). For resistant or sudden IC-HTN, consider STAT unenhanced head CT	
be sure patient is sedated and paralyzed (Table 56.5)	(Table 56.5)
drain 3–5 ml CSF if IVC present	↓ intracranial volume
mannitol[b] 1 gm/kg IV bolus or 10–20 ml of 23.4% saline	↑ plasma volume → ↑ CBF → ↓ ICP, also ↑ serum osmolality → ↓ extracellular brain water
hyperventilate with Ambu® bag (do not reduce $PaCO_2$ < 25 mm Hg)	"blow off" (reduce) $PaCO_2$ → ↓ CBV → ↓ ICP. CAUTION: due to reduced CBF, use for no more than several minutes (p. 872)
pentobarbital[c] 100 mg slow IV or thiopental 2.5 mg/kg IV over 10 minutes	sedates, ↓ ICP, treats seizures, may be neuroprotective; also myocardial depressant → ↓ MAP

[a]for measures to treat ICP that is trending up over a longer period, see Table 56.5 or information starting
[b]skip this step and go to hyperventilation if hypotensive, volume depleted, or if serum osmolality > 320 mOsm/L
[c]the availability of pentobarbital in the U.S. has been reduced, and other sedatives may need to be substituted (p. 876)

56.4.4 ICP management protocol details

Goals of therapy

1. keep ICP ≤ 20 mm Hg (prevents "plateau waves" from compromising cerebral blood-flow (CBF) and causing cerebral ischemia and/or brain death[30])
2. keep CPP ≥ 50 mm Hg.[61] The primary goal is to control ICP, simultaneously, CPP should supported by maintaining adequate MAP[63] (no study shows any deleterious effect on ICP, morbidity or mortality as a result of normalizing intravascular volume or inducing systemic hypertension to achieve the desired CPP).

Surgical treatment

1. Traumatic Intracranial masses should be treated as indicated. See surgical indications for subdural (p. 895), epidural (p. 893) or intraparenchymal (p. 891) hematoma or posterior fossa mass lesions (p. 905)
2. patients with hemorrhagic contusions ("pulped brain") showing progressive deterioration may benefit from surgical excision of portions of the contused brain tissue especially if not eloquent brain (p. 871)
3. decompressive craniectomy may be considered for IC-HTN that cannot be controlled medically

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General care

Major goals

1. avoid hypoxia (pO_2 < 60 mm Hg)
2. avoid hypotension (SBP ≤ 90 mm Hg): 67% positive-predictive value (PPV) for poor outcome (79% PPV when combined with hypoxia)[64]

Details of general treatment measures

1. prophylaxis against steroid ulcers (if steroids are used) and Cushing's (stress) ulcers (seen in severe head injury and in increased ICP, accompanied by hypergastrinemia)[65,66,67,68,69] for all patients including peds; see Prophylaxis for stress ulcers (p. 129).
   a) elevating gastric pH: titrated antacid and/or H2 antagonist (e.g. ranitidine 50 mg IV q 8 hrs) or proton pump inhibitor. See discussion of potential increased mortality as a result of increased gastric pH (p. 129)
   b) sucralfate

2. aggressive control of fever (fever is a potent stimulus to increase CBF, and may also increase plateau waves)[30]
3. arterial line for BP monitoring and frequent ABGs
4. CVP or PA line if high doses of mannitol are needed (goal: keep patient euvolemic)
5. IV fluids
   a) choice of fluids:
      - isolated head injury: IVF of choice is isotonic (e.g. NS + 20 mEq KCl/L)
      - avoid hypotonic solutions (e.g. lactated ringers) which may impair cerebral compliance[70]
   b) fluid volume:
      - provide adequate fluid resuscitation to avoid hypotension
      - normalization of intravascular fluid volume is not detrimental to ICP
      - although fluid restriction reduces the amount of mannitol needed to control ICP,[71] the concept of "running patients dry" is obsolete[72]
      - if mannitol is required, patient should be maintained at euvolemia
      - also exercise caution in restricting fluids following SAH; see Cerebral salt wasting (p.118)
      - if injuries to other systems are present (e.g. perforated viscus), they may dictate fluid management
   c) pressors (e.g. dopamine) are preferable to IV fluid boluses in head injury

Measures to lower ICP

General measures that should be routine

1. positioning:
   a) elevate HOB 30–45° (see below)
   b) keep head midline (to prevent kinking jugular veins)
2. light sedation: codeine 30–60 mg IM q 4 hrs PRN, or lorazepam (Ativan®) 1–2 mg IV q 4–6 hrs PRN
3. avoid hypotension (SBP < 90 mm Hg): normalize intravascular volume, support with pressors if needed
4. control HTN; in ICH, aim for patient's baseline, see Initial management of ICH (p.1339)
5. prevent hyperglycemia: (aggravates cerebral edema) usually present in head injury,[73,74] may be exacerbated by steroids
6. intubation: for GCS ≤ 8 or respiratory distress. Give IV lidocaine first (below) and antibiotics (p.827)
7. avoid hyperventilation: keep $PaCO_2$ at the low end of eucapnia (35 mm Hg)
8. prophylactic hypothermia: non-statistically significant trend suggests reduced mortality.[75] Maintain target temperature for > 48 hours

Measures to use for documented IC-HTN

First, check General measures that should be routine above. Proceed to each step if IC-HTN persists.

1. heavy sedation and/or paralysis when necessary (also assists treatment of HTN) e.g. when patient is agitated, or to blunt the elevation of ICP that occurs with certain maneuvers such as moving the patient to CT table. Caution: with heavy sedation or paralysis, the ability to follow the neurologic exam is lost (follow ICPs)
   a) for heavy sedation (intubation recommended to avoid respiratory depression → elevation of $PaCO_2$ → ↑ ICP): e.g. one of the following:
      - MSO4: □ 2–4 mg/hr IV drip
      - fentanyl: □ 1–2 ml IV q 1 hr (or 2–5 mcg/kg/hr IV drip)
      - sufentanil: □ 10–30 mcg test dose, then 0.05 -2 mcg/kg/hr IV drip
      - midazolam (Versed®): □ 2 mg test dose, then 2–4 mg/hr IV drip
      - propofol drip (p.106): 0.5 mg/kg test dose, then 20–75 mcg/kg/min IV drip □ avoid high-dose propofol (do not exceed 83 mcg/kg/min)
      - "low dose" pentobarbital (adult: 100 mg IV q 4 hrs; peds: 2–5 mg/kg IV q 4 hrs)
   b) paralysis (intubation mandatory): e.g. vecuronium 8–10 mg IV q 2–3 hrs
2. CSF drainage (when IVC is being utilized to measure ICP): 3–5 ml of CSF should be drained with the drip chamber at ≤ 10 cm above EAC. Works immediately by removal of CSF (reducing intracranial volume) and possibly by allowing edema fluid to drain into ventricles[76] (latter point is controversial)
3. "osmotic therapy" when there is evidence of IC-HTN:
   a) mannitol (also see below) 0.25–1 gm/kg bolus (over < 20 mins) followed by 0.25 gm/kg IVP (over 20 min) q 6 hrs PRN ICP > 20. Recent literature suggests that 1.4 gm/kg initial dose is more effective. May "alternate" with:

furosemide (Lasix®) (also see below): adult 10–20 mg IV q 6 hrs PRN ICP > 20. Peds: 1 mg/kg, 6 mg max IV q 6 hrs PRN ICP > 20

b) keep patient euvolemic to slightly hypervolemic
c) if IC-HTN persists and serum osmolarity is < 320 mOsm/L, increase mannitol up to 1 gm/kg, and shorten the dosing interval
d) if ICP remains refractory to mannitol, consider hypertonic saline, either continuous 3%saline infusion or as bolus of 10–20 ml of 23.4%saline
(D/C after ≈ 72 hours to avoid rebound edema)
e) hold osmotic therapy if serum osmolarity is ≥ 320 mOsm/L (higher tonicity may have no advantage and risks renal dysfunction; see below) or SBP < 100

4. hyperventilation (HPV) to $PaCO_2$ = 30–35 mm Hg (for details, see below)
   a) do not use prophylactically
   b) avoid aggressive HPV ($PaCO_2 \leq 25$ mm Hg) at all times
   c) use only for
      - short periods for acute neurologic deterioration
      - or chronically for documented IC-HTN unresponsive to sedation, paralytics, CSF drainage and osmotic therapy
   d) avoid HPV during the first 24 hrs after injury if possible
5. steroids: the routine use of glucocorticoids is not recommended for treatment of patients with head injuries (see below)

"Second tier" therapy for persistent IC-HTN

If IC-HTN remains refractory to the above measures, and especially if there is loss of previously controlled ICP, strong consideration should be given to repeating a head CT to rule out a surgical condition before proceeding with "second tier" therapies which are either effective but with significant risks (e.g. high-dose barbiturates), or are unproven in terms of benefit on outcome. Also consider an EEG to rule-out subclinical status epilepticus (seizures that are not clinically evident); see treatment measures for status epilepticus (p.471); some medications are effective for both seizures and IC-HTN, e.g. pentobarbital, propofol…

1. high dose barbiturate therapy (p.875): initiate if ICP remains > 20–25 mm Hg
2. hyperventilate to $PaCO_2$ = 25–30 mm Hg. Monitoring $SjVO_2$, $AVdO_2$, and/or CBF is recommended (see below)
3. hypothermia[77,78]: patients must be monitored for a drop in cardiac index, thrombocytopenia, elevated creatinine clearance, and pancreatitis. Avoid shivering which raises ICP[78]
4. decompressive surgery:
   a) decompressive craniectomy removal of portion of calvaria.[79] Controversial (may enhance cerebral edema formation[80]). Craniectomy decreased ICP to < 20 mm Hg in 85%[81] regardless of pupillary response to light, timing of craniectomy, brain shift and age. Outcomes were improved when IC-HTN responded.[81,82,17] Further randomized trials are indicated. Early decompressive craniectomy may be considered in patients undergoing emergent surgery (for fracture, EDH, SDH…).[83] Flap must be at least 12 cm in diameter, and duraplasty is mandatory. Also, see Hemicraniectomy for malignant MCA territory infarction (p.1303)
   b) removal of large areas of contused hemorrhagic brain (makes room immediately; removes region of disrupted BBB). If contused, consider temporal tip lobectomy (no more than 4–5 cm on dominant side, 6–7 cm on non-dominant) (total temporal lobectomy[84] is probably too aggressive) or frontal lobectomy. Has not shown great therapeutic promise
5. lumbar drainage: showing some promise. Watch for "cerebral sag"
6. hypertensive therapy

Adjunctive measures

1. lidocaine: 1.5 mg/kg IVP (watch for hypotension, reduce dose if necessary) at least one minute before endotracheal intubation or suctioning. Blunts the rise in ICP as well as tachycardia and systemic HTN (based on patients with brain tumors undergoing intubation under light barbiturate-nitrous oxide anesthesia; extrapolation to trauma patients is unproven)[85]
2. high frequency (jet) ventilation: consider if high levels of positive end-expiratory pressure (PEEP) are required[86] (NB: patients with reduced lung compliance, e.g. pulmonary edema, transmit more of PEEP through lungs to thoracic vessels and may raise ICP). PEEP ≤ 10 cm H_2O does not cause clinically significant increases in ICP.[87] Higher levels of PEEP > 15–20 are not recommended. Also, rapid elimination of PEEP may cause a sudden increase in circulating blood volume which may exacerbate cerebral edema and also elevate ICP

Details of some measures employed in treating increased ICP

Elevating head of bed (HOB)

Seemingly simple, but there is still some controversy. Early data obtained from dog studies indicated that keeping the HOB at 30–45° optimized the trade-off between the following two factors as the HOB is elevated: reducing ICP (by enhancing venous outflow and by promoting displacement of CSF from the intracranial compartment to the spinal compartment) and reducing the arterial pressure (and thus CPP) at the level of the carotid arteries. Some studies showed a deleterious effect from elevating the HOB and were used to justify nursing these patients with HOB flat.[2]

Recent data[88] indicate that although mean carotid pressure (MCP) is reduced, the ICP is also reduced and the CBF is unaffected by elevating the HOB to 30°. The onset of action of raising the HOB is immediate.

Prophylactic hypothermia

Practice guideline: Prophylactic hypothermia

Level III[75]: prophylactic hypothermia:
- improves the chances of having a moderate to good outcome – 4–5 on the Glasgow Outcome Score Table 88.5 – at the end of the follow-up period when target temperatures of 32–35° C (91.4–95° F) were used (note: no clear relationship was found for cooling duration or rewarming rate)
- showed a non-significant trend suggesting that it lowers mortality when the target temperature is maintained for > 48 hrs (note: the actual target temperature and rewarming rate did not influence mortality)

Hyperventilation

Intraarterial carbon dioxide ($PaCO_2$) is the most potent cerebrovascular vasodilator, the effect of which is probably mediated by changes in pH caused by the rapid diffusion of CO_2 across the BBB.[89] Hyperventilation (HPV) lowers ICP by reducing $PaCO_2$ which causes cerebral vasoconstriction, thus reducing the cerebral (intracranial) blood volume (CBV).[90] Of concern, vasoconstriction also lowers cerebral blood flow (CBF) which could produce focal ischemia in areas with preserved cerebral autoregulation as a result of shunting.[91,92] However, ischemia does not necessarily follow as the O_2 extraction fraction (OEF) may also increase, up to a point.[93]

Practice guideline: Hyperventilation for ICP management[a]

Level I[94]: in the absence of IC-HTN, chronic prolonged hyperventilation (HPV) ($PaCO_2 \le 25$ mm Hg) should be avoided

Level II[95]: prophylactic hyperventilation ($PaCO_2 \le 25$ mm Hg) is not recommended

Level III
- HPV may be necessary for brief periods when there is acute neurologic deterioration, or for longer periods if there is IC-HTN refractory to sedation, paralysis, CSF drainage and osmotic diuretics[94]
- HPV should be avoided ≤ 24 hrs after head injury[95]
- if HPV is used, jugular venous oxygen saturation ($SjVO_2$) (p. 865) or $PbrO_2$ (p. 865) should be measured to monitor brain O_2 delivery[95]

[a]See also Practice guideline: Early/prophylactic hyperventilation (p. 827).

Hyperventilation (HPV), is to be used in moderation only in specific situations[3] (see below). Prophylactic HPV may actually be associated with a worse outcome[96] (note: prophylactic HPV implies cases where there are no clinical signs of IC-HTN and where IC-HTN unresponsive to other measures has not been documented by ICP monitoring). When indicated, use HPV only to $PaCO_2$ = 30–35 mm Hg (see Caveats for hyperventilation below). CBF in severe head trauma patients is already about half of normal during the first 24 hrs after injury (typically < 30 cc/100 g/min during the first 8 hours, and may be < 20 during the first 4 hours in patients with the worst injuries).[97,98,99,100] In one study,

Table 56.7 Summary of recommendations for $PaCO_2$ following head trauma (see text for details)

$PaCO_2$ (mm Hg)	Description
35–40	normocarbia. Use routinely
30–35	hyperventilation. Do not use prophylactically. Use only as follows: briefly for clinical evidence of IC-HTN (neurologic deterioration) or chronically for documented IC-HTN unresponsive to other measures
25–30	augmented hyperventilation. A second tier treatment. Use only when other methods fail to control IC-HTN. Additional monitoring recommended to R/O cerebral ischemia
<25	aggressive hyperventilation. No documented benefit. Significant potential for ischemia

the use of HPV to $PaCO_2 = 30$ mm Hg within 8–14 hrs of severe head injury did not impair global cerebral metabolism,[93] but focal changes were not studied. Hyperventilation to $PaCO_2 < 30$ mm Hg further reduces CBF, but does not consistently reduce ICP and may cause loss of cerebral autoregulation.[45] If carefully monitored, there may be occasion to use this. There are no studies showing any improvement in outcome with aggressive HPV ($PaCO_2 \leq 25$ mm Hg) which can cause diffuse cerebral ischemia.[45] A summary of the ranges of $PaCO_2$ and the recommendations is shown in Table 56.7.

Reducing $PaCO_2$ from 35 to 29 mm Hg lowers ICP 25–30% in most patients. Onset of action: ≤30 seconds. Peak effect at ≈ 8 mins. Duration of effect is occasionally as short as 15–20 mins. Effect may be blunted by 1 hour (based on patients with intracranial tumors), after which it is difficult to return to normocarbia without rebound elevation of ICP.[101,102] Thus, HPV must be weaned slowly.[30]

Indications for hyperventilation (HPV)

1. HPV for brief periods (minutes) at the following times
   a) prior to insertion of ICP monitor: if there are clinical signs of IC-HTN (Table 54.2)
   b) after insertion of a monitor: if there is a sudden increase in ICP and/or acute neurologic deterioration, HPV may be used while evaluating patient for a treatable condition (e.g. delayed intracranial hematoma)
2. HPV for longer periods: when there is documented IC-HTN unresponsive to sedation, paralytics, CSF drainage (when available) and osmotic diuretics
3. HPV may be appropriate for IC-HTN resulting primarily from hyperemia (p.901)

Caveats for hyperventilation

1. avoid during the first 5 days after head injury if possible (especially first 24 hrs)
2. do not use prophylactically (i.e. without appropriate indications, see above)
3. if documented IC-HTN is unresponsive to other measures, hyperventilate only to $PaCO_2 = 30$–35 mm Hg
4. if prolonged HPV to $PaCO_2$ of 25–30 mm Hg is deemed necessary, consider monitoring $SjVO_2$, $AVdO_2$, or CBF to rule-out cerebral ischemia (p.865)
5. do not reduce $PaCO_2 < 25$ mm Hg (except for very brief periods of a few minutes)

Mannitol

Practice guideline: Mannitol in severe traumatic brain injury

Level II[103,104]
mannitol is effective for control of IC-HTN after severe TBI(note: current information did not allow recommendations regarding hypertonic saline to be made[104]

- intermittent boluses may be more effective than continuous infusion
- effective doses range from 0.25–1 gm/kg body weight
- avoid hypotension (SBP < 90 mm Hg) which may result from the diuretic effect of mannitol which can lead to ↓ circulating fluid volume

Level III[103]
- indications: signs of transtentorial herniation or progressive neurological deterioration not attributable to systemic pathology
- euvolemia should be maintained (hypovolemia should be avoided) by fluid replacement. An indwelling urinary catheter is essential
- serum osmolarity should be kept < 320 mOsm when there is concern about renal failure

No controlled clinical trial has been conducted to show the benefits of mannitol over placebo.[3] The exact mechanism(s) by which mannitol provides its beneficial effects is still controversial, but probably includes some combination of the following

1. lowering ICP
   a) immediate plasma expansion[105,106,107]: reduces the hematocrit and blood viscosity (improved rheology) which increases CBF and O_2 delivery. This reduces ICP within a few minutes, and is most marked in patients with CPP < 70 mm Hg
   b) osmotic effect: increased serum tonicity draws edema fluid from cerebral parenchyma. Takes 15–30 minutes until gradients are established.[105] Effect lasts 1.5–6 hrs, depending on the clinical condition[108,109,3]
2. supports the microcirculation by improving blood rheology (see above)
3. possible free radical scavenging[110]

With bolus administration, onset of ICP lowering effect occurs in 1–5 minutes; peaks at 20–60 minutes. When urgent reduction of ICP is needed, an initial dose of 1 gm/kg should be given over 30 minutes. When long-term reduction of ICP is intended, the infusion time should be lengthened to 60 minutes[111] and the dose reduced (e.g. 0.25–0.5 gm/kg q 6 hrs). A large previous dose reduces the effectiveness of subsequent doses[71]; thus it is desirable to use the smallest *effective* dose (small frequent doses may be preferable, e.g. 0.25 mg/kg q 2–3 hrs; also results in fewer peaks as mannitol "troughs" are smoothed out). Titrating to ICP (instead of dosing at regular intervals) results in less mannitol being given.[71,112] The effectiveness of mannitol may be synergistically enhanced when combined with the use of loop acting diuretics (e.g. furosemide, see below),[113] and alternating these medications has been suggested.[71]

Cautions with mannitol

1. mannitol opens the BBB, and mannitol that has crossed the BBB may draw fluid into the CNS (this may be minimized by repeated bolus administration vs. continuous infusion[106,114]) which can aggravate vasogenic cerebral edema.[115] Thus, when it is time to D/C mannitol, it should be tapered to prevent ICP rebound[111]
2. caution: corticosteroids + phenytoin + mannitol may cause hyperosmolar nonketotic state with high mortality[30]
3. excessively vigorous bolus administration may → HTN and if autoregulation is defective → increased CBF which may promote herniation rather than prevent it[116]
4. high doses of mannitol carries the risk of acute renal failure (acute tubular necrosis), especially in the following[10,117]: serum osmolarity > 320 mOsm/L, use of other potentially nephrotoxic drugs, sepsis, pre-existing renal disease
5. large doses prevents diagnosing DI by use of urinary osmols or SG (p.121)
6. because it may further increase CBF,[118] the use of mannitol may be deleterious when IC-HTN is due to hyperemia (p.901)

Furosemide

The use of furosemide (Lasix®) has been advocated, but little data exists to support this.[3] Loop acting diuretics may reduce ICP[119] by reducing cerebral edema[120] (possibly by increasing serum tonicity), and may also slow the production of CSF.[121] They also act synergistically with mannitol[122] (above).

□: 10–20 mg IV q 6 hrs, may be alternated with mannitol such that the patient receives one or the other q 3 hrs. Hold if serum osmolarity > 320 mOsm/L.

Hypertonic saline (HS)

May reduce ICP in patients refractory to mannitol,[123,124] although no improvement in outcome over mannitol has been demonstrated.[124,125] Potentially deleterious effect on stroke penumbra in animal studies. Studies[126,127] are not adequate to make recommendations regarding use.[104]

□: Continuous infusion: 3% saline at 25–50 ml/hr may be given through a peripheral IV. Bolus: 10–20 ml of 7.5–23.4% saline must be given through a central line. HS should be discontinued after ≈ 72 hours to avoid rebound edema.[124] Hold if serum osmolarity > 320 mOsm/L.

Steroids

Practice guideline: Glucocorticoids in severe head injury

Level I[128]: the use of glucocorticoids (steroids) is not recommended for improving outcome or reducing ICP in patients with severe TBI (except in patients with known depletion of endogenous adrenal hormones[129,130]). High-dose methylprednisolone is associated with increased mortality and is contraindicated[128]

Although glucocorticoids reduce vasogenic cerebral edema (e.g. surrounding brain tumors) and may be effective in lowering ICP in pseudotumor cerebri, they have little effect on cytotoxic cerebral edema which is the more prevalent derangement following trauma; see Cerebral edema (p.90).

Significant side effects may occur with steroids[131] including coagulopathies, hyperglycemia[132] with its undesirable effect on cerebral edema – see Possible deleterious side effects of steroids (p.594) – and increased incidence of infection (due to immunosuppression). High-dose methylprednisolone is associated with increased mortality.[133]

Non-glucocorticoid steroids (e.g. 21-aminosteroids, AKA lazaroids, including tirilazad[134,135]) and the synthetic glucocorticoid triamcinolone[136] have also failed to show overall benefit.

High-dose barbiturate therapy

Practice guideline: Barbiturates in severe head injury

Level II[137]: □ prophylactic use of barbiturates for burst suppression EEG is not recommended

Level II[137]: high-dose barbiturates are recommended for IC-HTN refractory to maximal medical and surgical ICP lowering therapy. Patients should be hemodynamically stable before and during treatment

Theoretical benefits of barbiturates in head injury derive from vasoconstriction in normal areas (shunting blood to ischemic brain tissue), decreased metabolic demand for O_2 ($CMRO_2$) with accompanying reduction of CBF, free radical scavenging, reduced intracellular calcium, and lysosomal stabilization.[138] There is little question that barbiturates lower ICP, even when other treatments have failed,[139] but regarding outcome, studies have shown both benefits[140,141] and lack of same.[142,143] A subgroup of patients with preserved vasoreactivity may benefit from the use of barbiturates[144]; and when reserved for use in patients who failed to respond sufficiently to other measures, barbiturates have been shown to lower ICP.[145] Patients that do respond have a lower mortality (33%) than those in whom ICP control could not be accomplished (75%).[141]

The limiting factor for therapy is usually hypotension due to barbiturate induced reduction of sympathetic tone[146 (p 354)] (causing peripheral vasodilatation) and direct mild myocardial depression. Hypotension occurs in ≈ 50% of patients in spite of adequate blood volume and use of dopamine.[147]

NB: the ability to follow the neurologic exam is lost with high-dose barbiturates, and one must follow ICP.

"Barbiturate coma" vs. high-dose therapy: If barbiturates are given until there is burst suppression on EEG, this is considered true "barbiturate coma." This results in near maximal reductions in $CMRO_2$ and CBF.[3] However, most regimens should technically be called "high dose intravenous therapy" since they simply try to establish target serum barbiturate levels (e.g. 3–4 mg% for pentobarbital), even though there is poor correlation between serum level, therapeutic benefit, and systemic complications.[3]

Adjunctive measures to administration of high-dose barbiturates:

1. consider a Swan-Ganz (PA) catheter placed during the first hour of loading dose
2. high-dose barbiturates often causes paralytic ileus: therefore NG tube to suction & IV hyperalimentation are usually needed

Indications

The use of barbiturates should be reserved for situations where the ICP cannot be controlled by the previously outlined measures,[141] as there is evidence that prophylactic barbiturates do not favorably alter outcome, and are associated with significant side effects, mostly hypotension,[147] that can cause neurologic deterioration.

Choice of agents

A number of agents have been studied, however, there is inadequate data to recommend one drug over another. The most information is available on pentobarbital (see below). Alternative agents which have not been as well studied: thiopental (see below), phenobarbital (p.451) & propofol (p.877).

Drug info: Pentobarbital (Nembutal®)

Pentobarbital has a fast onset (full effects within ≈ 15 minutes), short duration of action (3–4 hrs), and a half–life of 15–48 hrs.

Protocols for pentobarbital therapy in adults

There are many protocols. A simple one from a randomized clinical trial[145]:

1. loading dose:
   a) pentobarbital 10 mg/kg IV over 30 minutes
   b) then 5 mg/kg q 1 hr × 3 doses
2. maintenance: 1 mg/kg/hr

A more elaborate protocol:

1. loading dose: pentobarbital 10 mg/kg/hr IV over 4 hrs as follows:
   a) FIRST HOUR: 2.5 mg/kg slow IVP q 15 min × 4 doses (total: 10 mg/kg in first hr), follow BP closely
   b) next 3 hours: 10 mg/kg/hr continuous infusion (put 2500 mg in 250 ml of appropriate IVF, run at K ml/hr × 3 hrs (K = patient’s weight in kg))
2. maintenance: 1.5 mg/kg/hr infusion (put 250 mg in 250 ml IVF and run at 1.5 × K ml/hr)
3. check serum pentobarbital level 1 hr after loading dose completed; usually 3.5–5.0 mg%
4. check serum pentobarbital level q day thereafter
5. if level ever > 5 mg% and ICP acceptable, reduce dose
6. baseline brain stem auditory evoked response (BAER) early in treatment. May be omitted on clinical grounds. Repeat BAER if pentobarbital level ever > 6 mg%. Reduce dose if BAER deteriorates (NB: hemotympanum may interfere with BAER)
7. goal: ICP < 24 mm Hg and pentobarbital level 3–5 mg%. Consider discontinuing pentobarbital due to ineffectiveness if ICP still > 24 with adequate drug levels × 24 hrs
8. if ICP < 20 mm Hg, continue treatment × 48 hrs, then taper dose. Backtrack if ICP rises

Neuro function takes ≈ 2 days off pentobarbital to return (Table 56.8). If is desired to perform a brain death exam, the pentobarbital level needs to be ≈ ≤ 10 mcg/ml before the exam is valid.

Table 56.8 CNS effects of various pentobarbital levels[a]

Degree of CNS depression	mg%	mcg/ml
level for valid brain death exam	≤ 1	≤ 10
sedated, relaxed, easily aroused	0.05–0.3	0.5–3
heavy sedation, difficult to arouse, respiratory depression	2	20
“coma” level (burst suppression occurs in most patients)	5	50

[a]levels reported are for intolerant patients; there is significant variability between patients and tolerant patients may not be sedated even at levels as high as 100 mcg/ml

Drug info: Thiopental (Pentothal®)

May be useful when a rapidly acting barbiturate is needed (e.g. intra-op) or when large doses of pentobarbital are not available. One of many protocols follows (note: thiopental has not been as well studied for this indication, but is theoretically similar to pentobarbital[148,149]):

1. loading dose: thiopental 5 mg/kg (range: 3–5) IV over 10 minutes → transient burst suppression (< 10 minutes) and blood thiopental levels of 10–30 mcg/ml. Higher doses (≈ 35 mg/kg) have been used in the absence of hypothermia to produce longer duration burst suppression for cardiopulmonary bypass
2. follow with continuous infusion of 5 mg/kg/hr (range: 3–5) for 24 hours
3. may need to rebolus with 2.5 mg/kg as needed for ICP control
4. after 24 hours, fat stores become saturated, reduce infusion to 2.5 mg/kg/hr
5. titrate to control ICP or use EEG to monitor for electrocerebral silence
6. “therapeutic” serum level: 6–8.5 mg/dl

Drug info: Propofol (Diprivan®)

Level II[137]: propofol may control ICP after several hours of dosing, but it does not improve mortality or 6 month outcome. □ Caution: high-dose propofol (total dose > 100 mg/kg for > 48 hrs) can cause significant morbidity (see propofol infusion syndrome).

□: 0.5 mg/kg test dose, then 20–75 mcg/kg/min infusion. Increase by 5–10 mcg/kg/min q 5–10 minutes PRN ICP control (do not exceed 83 mcg/kg/min = 5 mg/kg/hr).

Side effects: include Propofol Infusion Syndrome (p. 133). Use with caution at doses > 5 mg/kg/hr or at any dose for > 48 hrs.

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57 Skull Fractures

57.1 General information

Classified as either closed (simple fracture) or open (compound fracture).

Diastatic fractures extend into and separate sutures. More common in young children.[1]

57.2 Linear skull fractures over the convexity

90%of pediatric skull fractures are linear and involve the calvaria.

Table 57.1 shows some differentiating features to distinguish linear skull fractures. See also Indications for CT and admission criteria for TBI (p.830).

By themselves, linear skull fractures over the convexity rarely require surgical intervention.

57.3 Depressed skull fractures

For special considerations in pediatrics, see Depressed skull fractures (p.915) in pediatrics section.

57.3.1 Indications for surgery

See Practice guideline: Surgical management of depressed skull fractures (p.882). Some additional observations regarding surgery to elevate a depressed skull fracture in an adult:

1. consider surgery for depressed skull fractures with deficit referable to underlying brain
2. more conservative treatment is recommended for fractures overlying a major dural venous sinus (note: exception: depressed fractures overlying and depressing one of the dural sinuses may be dangerous to elevate, and if the patient is neurologically intact, and no indication for operation (e.g. CSF leak mandates surgery) may be best managed conservatively).

Practice guideline: Surgical management of depressed skull fractures

Indications for surgery

Level III[2]:

1. open (compound) fractures
   a) surgery for fractures depressed > thickness of calvaria and those not meeting criteria for nonsurgical management listed below
   b) nonsurgical management may be considered if
      - there is no evidence (clinical or CT) of dural penetration (CSF leak, intradural pneumocephalus on CT...)
      - and no significant intracranial hematoma
      - and depression is < 1 cm
      - and no frontal sinus involvement
      - and no wound infection or gross contamination
      - and no gross cosmetic deformity
2. closed (simple) depressed fractures: may be managed surgically or nonsurgically

Timing of surgery

Level III[2]: early surgery to reduce risk of infection

Surgical methods

Level III[2]:

1. elevation and debridement are recommended
2. option: if there is no evidence of wound infection, primary bone replacement
3. antibiotics should be used for all compound depressed fractures

Table 57.1 Differentiating linear skull fractures from normal plain film findings

Feature	Linear skull fracture	Vessel groove	Suture line
density	dark black	grey	grey
course	straight	curving	follows course of known suture lines
branching	usually none	often branching	joins other suture lines
width	very thin	thicker than fracture	jagged, wide

There is no evidence that elevating a depressed skull fracture will reduce the subsequent development of posttraumatic seizures,[3] which are probably more related to the initial brain injury.

57.3.2 Surgical treatment for depressed skull fractures

General information

Booking the case: Craniotomy for depressed skull fracture

Also see defaults & disclaimers (p. 27).
1. position: (depends on location of the fracture)
2. post-op: ICU
3. blood: type & screen (for severe fractures: type and cross 2 U PRBC)
4. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery in the area of the skull fracture to bone fragments that may have been displaced, to repair the covering of the brain, to remove any foreign material that can be identified and any permanently damaged brain tissue (i.e. dead brain tissue), remove any blood clot and stop any bleeding identified, possible placement of intracranial pressure monitor. If a large opening has to be left in the skull, it may require surgery to correct in a number of months (3 or more)
   b) alternatives: nonsurgical management
   c) complications – usual craniotomy complications (p. 28) – plus any permanent brain injury that has already occurred is not likely to recover, seizures may occur (with or without the surgery), hydrocephalus, infection (including delayed infection/abscess)

Technical considerations of surgery

Surgical goals (modified[4])
1. debridement of skin edges
2. elevation of bone fragments
3. repair of dural laceration
4. debridement of devitalized brain
5. reconstruction of the skull
6. skin closure

Techniques
1. with open (compound) contaminated fractures, it may be necessary to excise depressed bone. In these cases or when air sinuses are involved, to minimize the risk of infecting the flap, some surgeons follow the patient for 6–12 months to rule out infection before performing a cosmetic cranioplasty. There has been no documented increase in infection with replacement of bone fragments; soaking the fragments in povidone-iodine has been recommended[4]
2. elevating the bone may be facilitated by drilling burr holes around the periphery and either using rongeurs or craniotome to excise the depressed portion
3. in cases where laceration of a major dural sinus is suspected and surgery is mandated, adequate preparation must be made for dural sinus repair[5]; NB: the SSS is often to the right of the sagittal suture (p. 61)

a) prepare for massive blood loss
b) have small Fogarty catheter ready to temporarily occlude sinus
c) have dural shunt ready (Kapp-Gielchinsky shunt, if available, has an inflatable balloon at both ends)
d) prep out saphenous vein area for vein graft
e) bone fragments that may have lacerated sinus should be removed last

57.4 Basal skull fractures

57.4.1 General information

Most basal (AKA basilar) skull fractures (BSF) are extensions of fractures through the cranial vault.

Severe basilar skull fractures may produce shearing injuries to the pituitary gland.

BSF, especially those involving the clivus, may be associated with traumatic aneurysms. This rarely occurs in pediatrics.[6]

57.4.2 Some specific fracture types

Temporal bone fractures

General information

Although often mixed, there are two basic types of temporal bone fractures:
- longitudinal fracture: more common (70–90%). Usually through petro-squamosal suture, parallel to and through EAC. Can often be diagnosed on otoscopic inspection of the EAC. Usually passes between cochlea and semicircular canals (SCC) sparing the VII and VIII nerves, but may disrupt the ossicular chain
- transverse fracture: perpendicular to EAC. Often passes through cochlea and may place stretch on geniculate ganglion, resulting in VIII and VII nerve deficits respectively

Posttraumatic facial palsy

Posttraumatic unilateral peripheral facial nerve palsy may be associated with transverse petrous bone fractures as noted above.

Management

Management is often complicated by multiplicity of injuries (including head injury requiring endotracheal intubation) making it difficult to determine the time of onset of facial palsy. Guidelines:
1. regardless of time of onset:
   a) steroids (glucocorticoids) are often utilized (efficacy unproven)
   b) consultation with ENT physician is usually indicated
2. immediate onset of unilateral peripheral facial palsy: facial EMG (AKA electroneuronography[7] or ENOG) takes at least 72 hrs to become abnormal. These cases are often followed and are possible candidates for surgical VII nerve decompression if no improvement occurs with steroids (timing of surgery is controversial, but is usually not done emergently)
3. delayed onset of unilateral peripheral facial palsy: follow serial ENOGs, if continued nerve deterioration occurs while on steroids, and activity on ENOG drops to less than 10% of the contralateral side, surgical decompression may be considered (controversial, thought to improve recovery from ≈ 40% to ≈ 75% of cases)

Clival fractures

See reference.[8]

3 categories (75% are longitudinal or transverse):
1. longitudinal: may be associated with injuries of vertebrobasilar vessels including:
   a) dissection or occlusion: may cause brain stem infarction
   b) traumatic aneurysms
2. transverse: may be associated with injuries to the anterior circulation
3. oblique

Clival fractures are highly lethal. May be associated with:
1. cranial nerve deficits: especially III through VI; bitemporal hemianopsia
2. CSF leak

3. diabetes insipidus
4. delayed development of traumatic aneurysms[9]

Occipital condyle fractures

These are considered in the section on Spine fractures (p.966).

57.4.3 Radiographic diagnosis

BSF appear as linear lucencies through the skull base.

CT scan with multiplanar projections is the most sensitive means for directly demonstrating BSF.

Plain skull x-rays and clinical criteria (see below) may also be able to make the diagnosis.

Indirect radiographic findings (on CT or plain films) that suggest BSF include: pneumocephalus (diagnostic of BSF in the absence of an open fracture of the cranial vault), air/fluid level within or opacification of air sinus with fluid (suggestive).

57.4.4 Clinical diagnosis

Some of these signs may take several hours to develop. Signs include:
1. CSF otorrhea or rhinorrhea
2. hemotympanum or laceration of external auditory canal
3. postauricular ecchymoses (Battle's sign)
4. periorbital ecchymoses (raccoon's eyes) in the absence of direct orbital trauma, especially if bilateral
5. cranial nerve injury:
   a) VII and/or VIII: usually associated with temporal bone fracture
   b) olfactory nerve (Cr. N. I) injury: often occurs with anterior fossa BSF and results in anosmia, this fracture may extend to the optic canal and cause injury to the optic nerve (Cr. N. II)
   c) VI injury: can occur with fractures through the clivus (see below)

57.4.5 Management

NG tubes

Caution: cases have been reported with BSF where an NG tube has been passed intracranially through the fracture[10,11,12] and is associated with fatal outcome in 64% of cases. Possible mechanisms include: a cribriform plate that is thin (congenitally or due to chronic sinusitis) or fractured (due to a frontal basal skull fracture or a comminuted fracture through the skull base).

Suggested contraindications to blind placement of an NG tube include: trauma with possible basal skull fracture, ongoing or history of previous CSF rhinorrhea, meningitis with chronic sinusitis.

Prophylactic antibiotics

The routine use of prophylactic antibiotics is controversial. This remains true even in the presence of a CSF fistula; see CSF fistula (cranial) (p.384). However, most ENT physicians recommend treating fractures through the nasal sinuses as open contaminated fractures, and they use broad spectrum antibiotics (e.g. ciprofloxacin) for 7–10 days.

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Treatment of the BSF

Most do not require treatment by themselves. However, conditions that may be associated with BSF which may require specific management include:
1. "traumatic aneurysms" (p.1227)[13]
2. posttraumatic carotid-cavernous fistula (p.1256)
3. CSF fistula: operative treatment may be required for persistent CSF rhinorrhea; see CSF fistula (cranial) (p.384)
4. meningitis or cerebral abscess: may occur with BSF into air sinuses (frontal or mastoid) even in the absence of an identifiable CSF leak. May even occur many years after the BSF was sustained; see Post craniospinal trauma meningitis / post-traumatic meningitis (p.318)
5. cosmetic deformities
6. posttraumatic facial palsy (below)

57.5 Craniofacial fractures

57.5.1 Frontal sinus fractures

General information

Frontal sinus fractures account for 5–15%of facial fractures.

In the presence of a frontal sinus fracture, intracranial air (pneumocephalus) on CT even without a clinically evident CSF leak, must be presumed to be due to dural laceration (although it could also be due to a basal skull fracture, below).

Anesthesia of the forehead may occur due to supratrochlear and/or supraorbital nerve involvement.

The risks of posterior wall fractures are not immediate, but may be delayed (some even by months or years) and include:

1. brain abscess
2. CSF leak with risk of meningitis
3. cyst or mucocele formation: injured frontal sinus mucosa has a higher predilection for mucocele formation than other sinuses.[14] Mucoceles may also develop as a result of frontonasal duct obstruction due to fracture or chronic inflammation. Mucoceles are prone to infection (mucopyocele) which can erode bone and expose dura with risk of infection

Anatomic considerations of the frontal sinus

The frontal sinus begins to appear around age 2 yrs, and becomes radiographically visible by age 8 as it extends above the superior orbital rim.[15] The sinus is lined with respiratory epithelium, the mucous secretion of which drains through the frontonasal duct medially and inferiorly into the middle nasal meatus.

Surgical considerations

Indications

Linear fractures of the anterior wall of the frontal sinus are treated expectantly.

Indications for exploration of posterior wall fractures is controversial.[16] Some argue that a few mm of displacement, or that CSF fistula that resolves may not require exploration. Others vehemently disagree.

Technique

In the presence of a traumatic forehead laceration, the frontal sinus may be exposed through judicious incorporation of the laceration in a forehead incision. Without such a laceration, either a bicoronal (souttar) skin incision or a butterfly incision (through the lower part of the eyebrows, crossing the midline near the glabella) is used.

In the presence of pneumocephalus, if no obvious dural laceration is found the dural undersurface of the frontal lobes should be checked for leaks. Extradural inspection and repair is rarely indicated; the act of lifting the dura off the floor of the frontal fossa in the region of the ethmoid sinuses often creates lacerations.[17] Intradural repair is accomplished using a graft (fascia lata is most desirable; periosteum is thinner but is often acceptable) which is held in place with sutures and must extend all the way back to the ridge of the sphenoid wing (fibrin glue may be a helpful adjunct).

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A periosteal flap is placed across the floor of the frontal fossa to help isolate the dura from the frontal sinus and to prevent CSF fistula.

Dealing with frontal sinus

 Simple packing of the sinus (with bone wax, Gelfoam®, muscle or fat) increases the possibility of infection or mucocele formation.

The rear wall of the sinus is removed (so-called cranialization of the frontal sinus). The sinus is then exenterated (mucosa is stripped from sinus wall down to the nasofrontal duct, the mucosa is inverted over itself in the region of the duct and is packed down into the duct, temporalis muscle plugs are then packed into the frontonasal ducts[16]), then the bony wall of the sinus is drilled with a diamond burr to remove tiny remnants of mucosa found in the surface of bone that may proliferate and form a mucocele.[14] If there is any remnant of sinus, it may then be packed with abdominal fat that fills all corners of the cavity. Post-op risks related to frontal sinus injury include: infection, mucocele formation and CSF leak.

57.5.2 LeFort fractures

Complex fractures through inherently weak “cleavage planes” resulting in an unstable segment (“floating face”). Shown in Fig. 57.1 (usually occur as variants of this basic scheme).

- LeFort I: transverse AKA transmaxillary fracture. Fracture line crosses pterygoid plate and maxilla just above the apices of the upper teeth. May enter maxillary sinus(es)
- LeFort II: pyramidal. Fracture extends upward across inferior orbital rim and orbital floor to medial orbital wall, then across nasofrontal suture. Often from downward blow to the nasal area
- LeFort III: craniofacial dislocation. Involves zygomatic arches, zygomaticofrontal suture, nasofrontal suture, pterygoid plates, and orbital floors (separating maxilla from cranium). Requires significant force, therefore often associated with other injuries, including brain injuries

57.6 Pneumocephalus

57.6.1 General information

AKA (intra)cranial aerocele, AKA pneumatocele, is defined as the presence of intracranial gas. It is critical to distinguished this from tension pneumocephalus which is gas under pressure (see below). The gas may be located in any of the following compartments: epidural, subdural, subarachnoid, intraparenchymal, intraventricular.

57.6.2 Etiologies of pneumocephalus

Anything that can cause a CSF leak can produce associated pneumocephalus (p.386).

1. skull defects
   a) post neurosurgical procedure
      - craniotomy: risk is higher when patient is operated with surgery in the sitting position[18]
      - shunt insertion[19,20]
      - burr-hole drainage of chronic subdural hematoma[21,22]: incidence is probably < 2.5%[22] although higher rates have been reported
   b) posttraumatic
      - fracture through air sinus (frontal, ethmoid…): including basal skull fracture
      - open fracture over convexity (usually with dural laceration)
   c) congenital skull defects: including defect in tegmen tympani[23]
   d) neoplasm (osteoma,[24] epidermoid,[25] pituitary tumor): usually caused by tumor erosion through floor of sella into sphenoid sinus
2. infection
   a) with gas-producing organisms
   b) mastoiditis

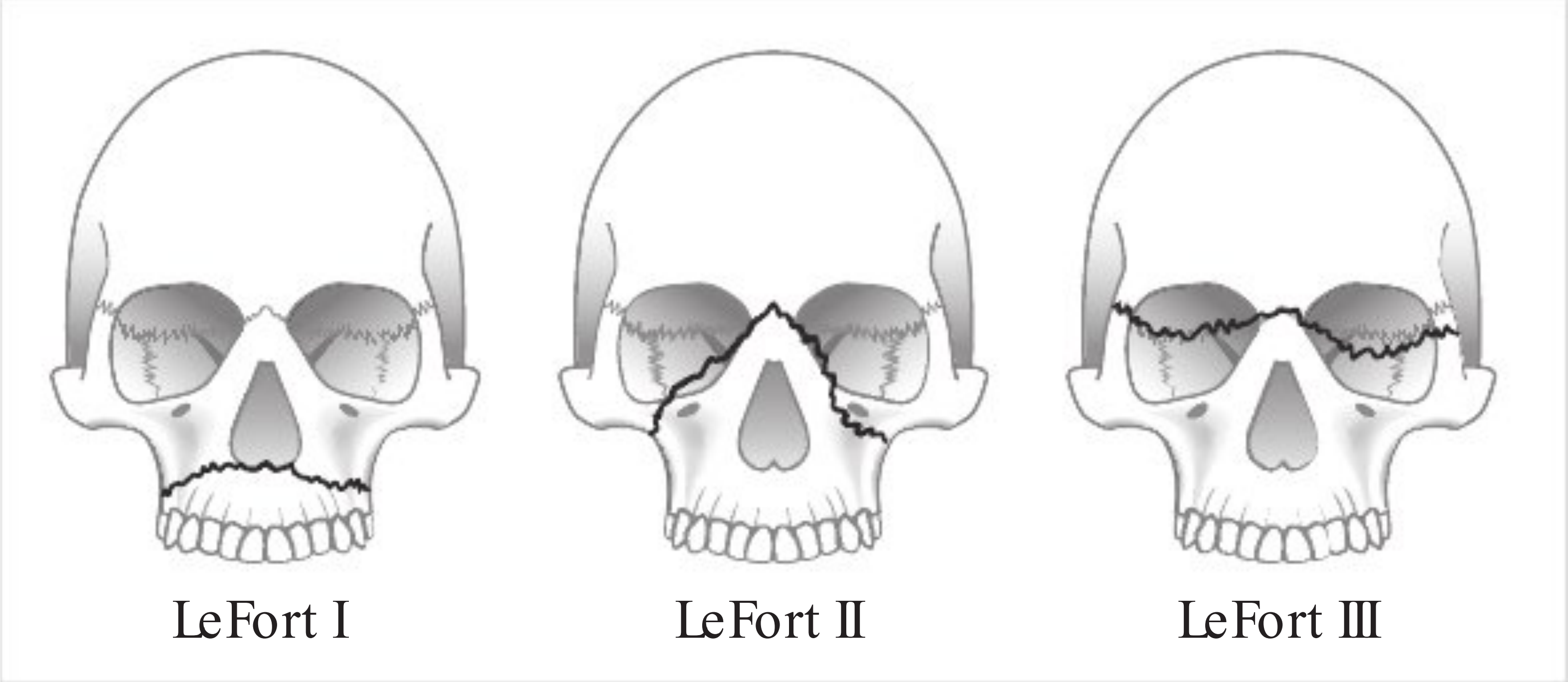

Fig. 57.1 LeFort fractures

3. post invasive procedure:
   a) lumbar puncture
   b) ventriculostomy
   c) spinal anesthesia[26]
4. spinal trauma (LP could be included here as well)
5. barotrauma[27]: e.g. with scuba diving (possibly through a defect in the tegmen tympani)
6. may be potentiated by a CSF drainage device in the presence of a CSF leak[28]

57.6.3 Presentation

H/A in 38%, N/V, seizures, dizziness, and obtundation.[29] An intracranial succussion splash is a rare (occurring in ≈ 7%) but pathognomonic finding. Tension pneumocephalus may additionally cause signs and symptoms just as any mass (may cause focal deficit or increased ICP).

57.6.4 Differential diagnosis (things that can mimic pneumocephalus)

Although intracranial low-density on CT may be associated with epidermoid, lipoma, or CSF, nothing is as intensely black as air. This can often be better appreciated on bone-windows than on soft-tissue windows.

57.6.5 Tension pneumocephalus

Intracranial gas can develop elevated pressure in the following settings:
1. when nitrous oxide anesthesia is not discontinued prior to closure of the dura[30]; see nitrous oxide, N_2O (p.105)
2. when a "ball-valve" effect occurs due to an opening to the intracranial compartment with soft tissue (e.g. brain) that may permit air to enter but prevent exit of air or CSF
3. when trapped room temperature air expands with warming to body temperature: a modest increase of only ≈ 4% results from this effect[31]
4. in the presence of continued production by gas-producing organisms

57.6.6 Diagnosis

Pneumocephalus is most easily diagnosed on CT[32] which can detect quantities of air as low as 0.5 ml. Air appears dark black (darker than CSF) and has a Hounsfield coefficient of –1000. One characteristic finding with bilateral pneumocephalus is the Mt. Fuji sign in which the two frontal poles appear peaked and are surrounded by and separated by air, resembling the silhouette of the twin peaks of Mt. Fuji[22] (see Fig. 57.2). Intracranial gas may also be evident on plain skull x-rays.

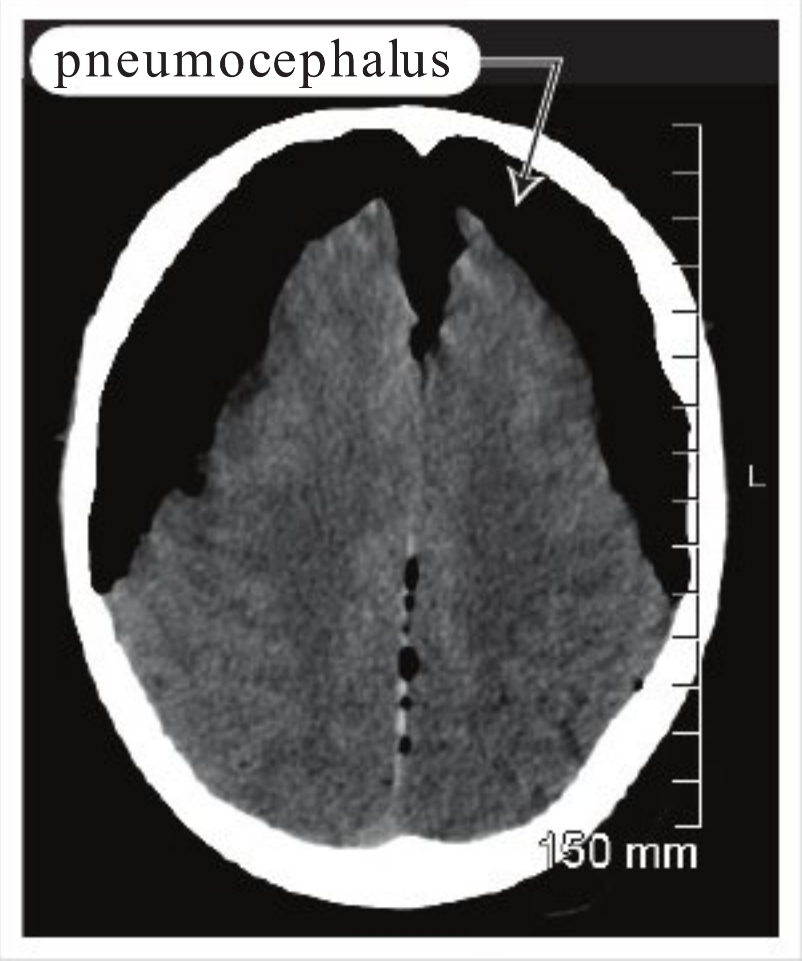

Fig. 57.2 Mt. Fuji sign with bilateral pneumocephalus. Axial noncontrast CT scan

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Since simple pneumocephalus usually does not require treatment, it is critical to differentiate it from tension pneumocephalus, which may need to be evacuated if symptomatic. It may be quite difficult to distinguish the two; brain that has been compressed e.g. by a chronic subdural hematoma may not expand immediately post-op and the "gas gap" may mimic the appearance of gas under pressure.

57.6.7 Treatment

When pneumocephalus is due to gas-producing organisms, treatment of the primary infection is initiated and the pneumocephalus is usually followed.

Treatment of non-infectious simple pneumocephalus depends on the whether or not the presence of a CSF leak is suspected. If there is no leak the gas will be resorbed with time, and if the mass effect is not severe it may simply be followed. If a CSF leak is suspected, management is as with any CSF fistula, see CSF fistula (cranial) (p.384).

Treatment of significant or symptomatic post-op pneumocephalus by breathing 100% O_2 via a nonrebreather mask increases the rate of resorption[33] (100% FiO_2 can be tolerated for 24–48 hours without serious pulmonary toxicity[34]).

Tension pneumocephalus producing significant symptoms must be evacuated. The urgency is similar to that of an intracranial hematoma. Dramatic and rapid improvement may occur with the release of gas under pressure. Options include placement of new twist drill or burr holes, or insertion of a spinal needle through a pre-existing burr hole (e.g. following a craniotomy).

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58 Traumatic Hemorrhagic Conditions

58.1 Posttraumatic parenchymal injuries

58.1.1 Cerebral edema

Surgical decompression is occasionally an option; see Practice guideline: Posttraumatic cerebral edema (p.891).

Practice guideline: Posttraumatic cerebral edema

Indications and timing for surgery

Level III[1]: bifrontal decompressive craniectomy within 48 hrs of injury is a treatment option for patients with diffuse, medically refractory posttraumatic cerebral edema and associated IC-HTN

58.1.2 Diffuse injuries

Patients with severe diffuse injuries occasionally may be considered for decompressive craniectomy; see Practice guideline: Diffuse injuries (p.891).

Practice guideline: Diffuse injuries

Indications for surgery

Level III[1]: decompressive craniectomy is an option for patients with refractory IC-HTN and diffuse parenchymal injury with clinical and radiographic evidence for impending transtentorial herniation

58.2 Hemorrhagic contusion

58.2.1 General information

AKA traumatic intracerebral hemorrhage (TICH). The definition is not uniformly agreed upon. Often considered as high density areas on CT (some exclude areas < 1 cm diameter[2]). TICH usually produce much less mass effect than their apparent size. Most commonly occur in areas where sudden deceleration of the head causes the brain to impact on bony prominences (e.g. temporal, frontal and occipital poles) in coup or contrecoup fashion.

TICH often enlarge and/or coalesce with time as seen on serial CTs. They also may appear in a delayed fashion (below). Surrounding low density may represent associated cerebral edema. CT scans months later often show surprisingly minimal or no encephalomalacia.

58.2.2 Treatment

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Practice guideline: Surgical management of TICH

- Level III[1]: Indications for surgical evacuation for TICH:
  - progressive neurological deterioration referable to the TICH, medically refractory IC-HTN, or signs of mass effect on CT
  - or TICH volume > 50 cm^3 cc or ml
  - or GCS = 6–8 with frontal or temporal TICH volume > 20 cm^3 with midline shift (MLS) ≥ 5 mm (p. 921) and/or compressed basal cisterns on CT (p. 921)
- nonoperative management with intensive monitoring and serial imaging: may be used for TICH without neurologic compromise and no significant mass effect on CT and controlled ICP

58.2.3 Delayed traumatic intracerebral hemorrhage (DTICH)

TICH demonstrated in patients on imaging that was not evident on initial admitting CT scan.

Incidence of DTICH in patients with GCS ≤ 8: ≈ 10%[3,4] (reported incidence varies with resolution of CT scanner,[5] timing of scan, and definition). Most DTICH occur within 72 hrs of the trauma.[4] Some patients seem to be doing well and then present with an apoplectic event (although DTICH accounted only for 12% of patients who "talk and deteriorate"[6]).

Factors that contribute to formation of DTICH include local or systemic coagulopathy, hemorrhage into an area of necrotic brain softening, coalescence of extravasated microhematomas.[7]

Treatment is the same as for TICH (see above).

Outcome for patients with DTICH described in the literature is generally poor, with a mortality ranging from 50–75%.[7]

58.3 Epidural hematoma

58.3.1 General information

Incidence of epidural hematoma (EDH): 1% of head trauma admissions (which is ≈ 50% the incidence of acute subdurals). Ratio of male:female = 4:1. Usually occurs in young adults, and is rare before age 2 yrs or after age 60 (perhaps because the dura is more adherent to the inner table in these groups).

Dogma was that a temporoparietal skull fracture disrupts the middle meningeal artery as it exits its bony groove to enter the skull at the pterion, causing arterial bleeding that gradually dissects the dura from the inner table resulting in a delayed deterioration. Alternate hypothesis: dissection of the dura from the inner table occurs first, followed by bleeding into the space thus created.

Source of bleeding: 85% = arterial bleeding (the middle meningeal artery is the most common source of middle fossa EDHs). Many of the remainder of cases are due to bleeding from middle meningeal vein or dural sinus.

70% occur laterally over the hemispheres with their epicenter at the pterion, the rest occur in the frontal, occipital, and posterior fossa (5–10% each).

58.3.2 Presentation with EDH

"Textbook" presentation (< 10%–27% have this classic presentation[8]):

- brief posttraumatic loss of consciousness (LOC): from initial impact
- followed by a "lucid interval" for several hours
- then, obtundation, contralateral hemiparesis, ipsilateral pupillary dilatation as a result of mass effect from hematoma

Deterioration usually occurs over a few hours, but may take days and rarely, weeks (longer intervals may be associated with venous bleeding).

Other presenting findings: H/A, vomiting, seizure (may be unilateral), hemi-hyperreflexia + unilateral Babinski sign, and elevated CSF pressure (LP is seldom used any longer). Bradycardia is usually a late finding. In peds, EDH should be suspected if there is a 10% drop in hematocrit after admission.

Contralateral hemiparesis is not uniformly seen, especially with EDH in locations other than laterally over the hemisphere. Shift of the brain stem away from the mass may produce compression of the opposite cerebral peduncle on tentorial notch which can produce ipsilateral hemiparesis (so called Kernohan's phenomenon or Kernohan's notch phenomenon),[9] a false localizing sign.

60% of patients with EDH have a dilated pupil, 85% of which are ipsilateral.

No initial loss of consciousness occurs in 60%. No lucid interval in 20%. NB: a lucid interval may also be seen in other conditions (including subdural hematoma).

58.3.3 Differential diagnosis

- subdural hematoma
- a posttraumatic disorder described by Denny-Brown consisting of a "lucid interval" followed by bradycardia, brief periods of restlessness and vomiting, without intracranial hypertension or mass. Children especially may have H/A, and may become drowsy and confused. Theory: a form of vagal syncope. CT must be done to rule-out EDH.

58.3.4 Evaluation

Plain skull x-rays

Usually not helpful. No fracture is identified in 40% of EDH. In these cases the patient's age was almost always < 30 yrs.

CT scan in EDH

"Classic" CT appearance occurs in 84% of cases: high density biconvex (lenticular) shape adjacent to the skull. In 11% the side against the skull is convex and that along the brain is straight, and in 5% it is crescent shaped (resembling subdural hematoma).[10] An EDH may cross the falx (distinct from SDH which is limited to one side of the falx) but is usually limited by skull sutures. EDH usually has uniformly density, sharply defined edges on multiple cuts, high attenuation (undiluted blood), contiguous with inner table, usually confined to small segment of calvaria. Mass effect is frequent. Occasionally, an epidural may be isodense with brain and may not show up unless IV contrast is given.[10] Mottling of density has been described as a finding in hyperacute EDH.[11]

58.3.5 Mortality with EDH

Overall: 20–55% (higher rates in older series). Optimal diagnosis and treatment within few hours results in 5–10% estimated mortality (12% in a recent CT era series[12]). Mortality without lucid interval double that with. Bilateral Babinski's or decerebration pre-op → worse prognosis. Death is usually due to respiratory arrest from uncal herniation causing injury to the midbrain.

20% of patients with EDH on CT also have ASDH at autopsy or operation. Mortality with both lesions concurrently is higher, reported range: 25–90%.

58.3.6 Treatment of EDH

Medical

CT may detect small EDHs and can be used to follow them. However, in most cases, EDH is a surgical condition (below).

Nonsurgical management may be attempted in the following:

Small (≤ 1 cm maximal thickness) subacute or chronic EDH,[13] with minimal neurological signs/symptoms (e.g. slight lethargy, H/A) and no evidence of herniation. Although medical management of p-fossa EDHs has been reported, these are more dangerous and surgery is recommended.

In 50% of cases there will be a slight transient increase in size between days 5–16, and some patients required emergency craniotomy when for signs of herniation occurred.[14]

Management

Management includes: admit, observe (in monitored bed if possible). Optional: steroids for several days, then taper. Follow-up CT: in 1 wk if clinically stable. Repeat in 1–3 mos if patient becomes asymptomatic (to document resolution). Prompt surgery if signs of local mass effect, signs of herniation (increasing drowsiness, pupil changes, hemiparesis…) or cardiorespiratory abnormalities.

Surgical

Surgical indications and timing

See also more details (p. 893). EDH in pediatric patients is riskier than adults since there is less room for clot. The threshold for surgery in pediatrics should be very low.

Practice guideline: Surgical management of EDH

Indications for surgery

Level III[15]:

1. EDH volume > 30 cm^3 should be evacuated regardless of GCS
2. EDH with the all of the following characteristics can be managed nonsurgically with serial CT scans and close neurological observation in a neurosurgical center:
   a) volume < 30 cm^3
   b) and thickness < 15 mm

c) and with midline shift (MLS) < 5 mm (p. 921)
d) and GCS > 8
e) and no focal neurologic deficit

Timing of surgery

Level III[15]: it is strongly recommended that patients with an acute EDH and GCS < 9 and anisocoria undergo surgical evacuation ASAP

(Note: Volume of a lens = 1.6 to 2 × r^2t = 0.4 to 0.5 × d^2t ≈ (A × B × T)/2 as in an ellipsoid, 1/2 the products of the height times the AP diameter and the thickness T. For a 1.5 cm thick EDH to be < 30 cc it would have to have a diameter (not radius) < 6.3-7 cm. For a 1 cm thick EDH to be < 30 cc, it would have to have a diameter < 7.7 – 8.6 cm.)

Booking the case: Craniotomy for acute EDH/SDH

Also see defaults & disclaimers (p. 27).
1. position: (depends on location of bleed, usually supine)
2. blood: type & screen (for severe SDH: T & C 2 U PRBC)
3. post-op: ICU
4. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the skull to remove blood clot, stop any bleeding identified, possible placement of intracranial pressure monitor
   b) alternatives: nonsurgical management
   c) complications: usual craniotomy complications (p. 28) plus further bleeding which may cause problems (especially in patients taking blood thinners, antiplatelet drugs including aspirin, or those with coagulation abnormalities or previous bleeds) and may require further surgery, any permanent brain injury that has already occurred is not likely to recover, hydrocephalus

Surgical technical issues

Evacuation is performed in the O.R. unless the patient herniates in E/R and access to OR is not within acceptable timeframe. Objectives:
1. clot removal: lowers ICP and eliminates focal mass effect. Blood is usually thick coagulum, thus exposure must provide access to most of clot. Craniotomy permits more complete evacuation of hematoma than e.g. burr holes[15]
2. hemostasis: coagulate bleeding soft tissue (dural veins & arteries). Apply bone wax to intra-diploic bleeders (e.g. middle meningeal artery). Also requires large exposure
3. prevent reaccumulation: (some bleeding may recur, and dura is now detached from inner table) place dural tack-up sutures to edges of craniotomy and use central "tenting" suture

58.3.7 Special cases of epidural hematome

Delayed epidural hematoma (DEDH)

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Definition: an EDH that is not present on the initial CT scan, but is found on subsequent CT. Comprise 9–10% of all EDHs in several series.[16,17]

Theoretical risk factors for DEDH include the following (NB: many of these risk factors may be incurred after the patient is admitted following a negative initial CT):
1. lowering ICP either medically (e.g. osmotic diuretics) and/or surgically (e.g. evacuating contralateral hematoma) which reduces tamponading effect
2. rapidly correcting shock (hemodynamic "surge" may cause DEDH)[18]
3. coagulopathies

Observation agrees with what one would predict based on the above in that DEDH tend to occur in patients with severe head injury and associated systemic injuries. However, DEDH have been reported in mild head injury (GCS > 12) infrequently.[19] Presence of a skull fracture has been identified as a common feature of DEDH.[19]

Key to diagnosis: high index of suspicion. Avoid a false sense of security imparted by an initial "nonsurgical" CT. 6 of 7 patients in one series improved or remained unchanged neurologically despite enlarging EDH (most eventually deteriorate). 1 of 5 with an ICP monitor did not have a heralding increase in ICP. May develop once an intracranial lesion is surgically treated, as occurred in 5 of 7 patients within 24 hrs of evacuation of another EDH. 6 of 7 patients had known skull fractures in the region where the delayed EDH developed,[17] but none of 3 had a skull fracture in another report.[18]

Posterior fossa epidural hematoma

Comprise ≈ 5% of EDH.[20,21] More common in 1st two decades of life. Although as many as 84% have occipital skull fractures, only ≈ 3% of children with occipital skull fractures develop p-fossa EDH. The source of bleeding is usually not found, but there is a high incidence of tears of the dural sinuses. Cerebellar signs are surprisingly lacking or subtle in most. See surgical indications (p.905). Overall mortality is ≈ 26% (mortality was higher in patients with an associated intracranial lesion).

58.4 Acute subdural hematoma

58.4.1 General information

The magnitude of impact damage, as opposed to secondary damage (p.824), is usually much higher in acute subdural hematoma (ASDH) than in epidural hematomas, which generally makes this lesion much more lethal. There is often associated underlying brain injury, which may be less common with EDH. Symptoms may be due to compression of the underlying brain with midline shift, in addition to parenchymal brain injury and possibly cerebral edema.[22,23]

Two common causes of traumatic ASDH:

1. accumulation around parenchymal laceration (usually frontal or temporal lobe). There is usually severe underlying primary brain injury. Often no “lucid interval.” Focal signs usually occur later and are less prominent than with EDH
2. surface or bridging vessel torn from cerebral acceleration-deceleration during violent head motion. With this etiology, primary brain damage may be less severe, a lucid interval may occur with later rapid deterioration

ASDH may also occur in patients receiving anticoagulation therapy,[24,25] usually with, but sometimes without, a history of trauma (the trauma may be minor). Receiving anticoagulation therapy increases the risk of ASDH 7-fold in males and 26-fold in females.[24]

58.4.2 CT scan in ASDH

Crescentic mass of increased density adjacent to inner table. Edema is often present.

Locations:

- Usually over convexity
- Interhemispheric
- Layering on tentorium
- in p-fossa

Changes with time on CT (see Table 58.1): isodense after ≈ 2 wks, only clues may be obliteration of sulci and lateralizing shift, the latter may be absent if bilateral. Subsequently becomes hypodense to brain (p.898). Membrane formation begins by about 4 days after injury.[26]

Table 58.1 ASDH density changes on CT with time

Category	Time frame	Density on CT
acute	1 to 3 days	hyperdense
subacute	4 days to 2 or 3 wks	≈ isodense
chronic	usually > 3 wks and < 3–4 mos	hypodense (approaching density of CSF)
	after about 1–2 months	may become lenticular shaped (similar to epidural hematoma) with density > CSF, < fresh blood

Differences from EDH: SDH is more diffuse, less uniform, usually concave over brain surface, often less dense (from mixing with CSF), and bridging subdural veins (from brain surface to the skull) may be seen (cortical vein sign).

58.4.3 Treatment

Indications for surgery

Level III surgical indications are shown in Practice guideline: Surgical management of ASDH (p.896). Other factors that should be considered:

1. presence of anticoagulants or platelet inhibitors: patients in good neurologic condition may be better served by reversing these agents prior to operating (to increase the safety of surgery)
2. location of hematoma: in general, a SDH high over the convexity is less threatening than a temporal/parietal SDH of the same volume that also has MLS
3. patient's baseline level of function, DNR status…
4. while the guidelines suggest evacuating SDH < 10 mm thick in some circumstances, clots that are smaller than this may not be causing problems but may simply be an epiphenomenon

Practice guideline: Surgical management of ASDH

Indications for surgery

Level III[27]:

1. ASDH with thickness > 10 mm or midline shift (MLS) > 5 mm (on CT) should be evacuated regardless of GCS
2. ASDH with thickness < 10 mm and MLS < 5 mm (see text regarding the evacuation of ASDH < 10 mm thick) should undergo surgical evacuation if:
   a) GCS drops by ≥ 2 points from injury to admission
   b) and/or the pupils are asymmetric or fixed and dilated
   c) and/or ICP is > 20 mm Hg
3. monitor ICP in all patients with ASDH and GCS < 9

Timing of surgery

Level III[27]: ASDH meeting surgical criteria should be evacuated ASAP (for issues regarding timing of surgery, see text)

Surgical methods

Level III[27]: ASDH meeting the above criteria for surgery should be evacuated via craniotomy with or without bone flap removal and duraplasty (a large craniotomy flap is often required to evacuate the thick coagulum and to gain access to possible bleeding sites).

Timing of surgery

Timing of surgery for ASDH is a matter of controversy. As a general principle, when surgery for ASDH is indicated it should be done as soon as possible.

"Four hour rule"

This "rule" was based on a 1981 series of 82 patients with ASDH,[28] which held that:

1. patients operated within 4 hrs of injury had 30% mortality, compared to 90% mortality if surgery was delayed > 4 hrs
2. functional survival (Glasgow Outcome Scale ≥ 4, see Table 88.5) rate of 65% could be achieved with surgery within 4 hrs
3. other factors related to outcome in this series included:
   a) post-op ICP: 79% of patients with functional recovery had post-op ICPs that didn't exceed 20 mm Hg, whereas only 30% of patients who died had ICP < 20 mm Hg
   b) initial neuro exam
   c) age was not a factor in this study (ASDH tend to occur in older patients than EDH)

However, a subsequent study of 101 patients with ASDH found a delay to surgery (delays > 4 hours from the injury) showed a nonstatistically-significant trend where mortality increased from 59% to 69% and functional survival decreased (Glasgow Outcome Scale ≤ 4, see Table 88.5) from 26% to 16%[29]

Booking the case: Acute subdural hematoma

Same as for acute epidural hematoma (p. 894).

Technical considerations

One may start with a small linear dural opening to effect clot removal and enlarge it as needed and only if brain swelling seems controllable. The actual bleeding site is often not identified at the time of surgery.

58.4.4 Morbidity and mortality with ASDH

Mortality Range: 50–90% (a significant percentage of this mortality is from the underlying brain injury, and not the ASDH itself).

Mortality is traditionally thought to be higher in aged patients (60%), and is 90–100% in patients on anticoagulants.[25]

In a series of 101 patients with ASDH, functional recovery was 19%[29] Postoperative seizures occurred in 9%, and did not correlate with outcome. The following variables were identified as strongly influencing outcome:

- mechanism of injury: the worst outcome was with motorcycle accidents, with 100% mortality in unhelmeted patients, 33% in helmeted
- age: correlated with outcome only > 65 yrs age, with 82% mortality and 5% functional survival in this group (other series had similar results[30])
- neurologic condition on admission: the ratio of mortality to functional survival rate related to the admission Glasgow Coma Scale (GCS) is shown in Table 58.2
- postoperative ICP: patients with peak ICPs < 20 mm Hg had 40% mortality, and no patient with ICP > 45 had a functional survival

Of all the above factors, only the time to surgery and postoperative ICP can be directly influenced by the treating neurosurgeon.

58.4.5 Special cases of acute subdural hematoma

Interhemispheric subdural hematoma

General information

Subdural hematoma along the falx between the two cerebral hemispheres (older term: interhemispheric scissure).

May occur in children,[31] possibly associated with child abuse.[32]

In adults, a consequence of: head trauma in 79–91%, ruptured aneurysm[33] in ≈ 12%, surgery in the vicinity of the corpus callosum, and rarely spontaneously.[34]

Incidence is unknown. Spontaneous cases should be investigated for possible underlying aneurysm. Occasionally may be bilateral, sometimes may be delayed (see below)

Table 58.2 Outcome as related to admission GCS

Admission GCS	Mortality	Functional survival
3	90%	5%
4	76%	10%
5	62%	18%
6 & 7	51%	44%

Most often are asymptomatic, or may present with the so-called “falx syndrome” – paresis or focal seizures contralateral to the hematoma. Other presentations: gait ataxia, dementia, language disturbance, oculomotor palsies.

Treatment

Controversial. Small asymptomatic cases may be managed expectantly. Surgery should be considered for progressive neurological deterioration with larger lesions. Approached through a parasagittal craniotomy. Surgery for these lesions can be treacherous – there is risk of venous infarction and one often finds they are dealing with a superior sagittal sinus injury.

Outcome

Reported mortality: 25–42% Mortality is higher in the presence of altered levels of consciousness. Mortality rate may actually be lower (24%) than with all-comers.[34] This is significantly lower than SDH in other sites (see above).

Delayed acute subdural hematoma (DASDH)

DASDHs have received less attention than delayed epidural or intraparenchymal hematomas. Incidence is ≈ 0.5% of operatively treated ASDHs.[7]

Definition: ASDH not present on an initial CT (or MRI) that shows up on a subsequent study. Indications for treatment are the same as for ASDH. Neurologically stable patients with a small DASDH and medically controllable ICP are managed expectantly.

Infantile acute subdural hematoma

General information

Infantile acute subdural hematoma (IASDH) is often considered as a special case of SDH. Roughly defined as an acute SDH in an infant due to minor head trauma without initial loss of consciousness or cerebral contusion,[35] possibly due to rupture of a bridging vein. The most common trauma is a fall backwards from sitting or standing. The infants will often cry immediately and then (usually within minutes to 1 hour) develop a generalized seizure. Patients are usually < 2 yrs old (most are 6–12 mos, the age when they first begin to pull themselves up or walk).[36]

These clots are rarely pure blood, and are often mixed with fluid. 75% are bilateral or have contralateral subdural fluid collections. It is speculated that IASDH may represent acute bleeding into a preexisting fluid collection.[36]

Skull fractures are rare. In one series, retinal and preretinal hemorrhages were seen in all 26 patients.[35]

Treatment

Treatment is guided by clinical condition and size of hematoma. Minimally symptomatic cases (vomiting, irritability, no altered level of consciousness and no motor disturbance) with liquefied hematoma may be treated with percutaneous subdural tap, which may be repeated several times as needed. Chronically persistent cases may require a subduroperitoneal shunt.

More symptomatic cases with high density clot on CT require craniotomy. A subdural membrane similar to those seen in adult chronic SDH is not unusual.[36] Caution: these patients are at risk of developing intraoperative hypovolemic shock.

Outcome

8% morbidity and mortality rate in one series.[35] Much better prognosis than ASDH of all ages probably because of the absence of cerebral contusion in IASDH.

58.5 Chronic subdural hematoma

58.5.1 General information

Originally termed “pachymeningitis hemorrhagica interna” by Virchow[37] in 1857. Chronic subdural hematomas (CSDH) generally occur in the elderly, with the average age being ≈ 63 yrs; exception: subdural collections of infancy (p.903). Head trauma is identified in < 50% (sometimes rather trivial trauma can produce these lesions). Other risk factors: alcohol abuse, seizures, CSF shunts, coagulopathies (including therapeutic anticoagulation[25]), and patients at risk for falls (e.g. with hemiplegia from previous stroke). CSDHs are bilateral in ≈ 20–25% of cases.[38,39]

Hematoma thickness tends to be larger in older patients due to a decrease in brain weight and increase in subdural space with age.[40]

Classically CSDHs contains dark “motor oil” fluid which does not clot.[41] When the subdural fluid is clear (CSF), the collection is termed a subdural hygroma (p. 902).

58.5.2 Pathophysiology

Many CSDH probably start out as acute subdurals. Blood within the subdural space evokes an inflammatory response. Within days, fibroblasts invade the clot, and form neomembranes on the inner (cortical) and outer (dural) surface. This is followed by ingrowth of neocapillaries, enzymatic fibrinolysis, and liquefaction of blood clot. Fibrin degradation products are reincorporated into new clots and inhibit hemostasis. The course of CSDH is determined by the balance of plasma effusion and/or rebleeding from the neomembranes on the one hand and reabsorption of fluid on the other.[42,43]

58.5.3 Presentation

Patients may present with minor symptoms of headache, confusion, language difficulties (e.g. word-finding difficulties or speech arrest, usually with dominant hemisphere lesions), or TIA-like symptoms (p. 1398). Or, they may develop varying degrees of coma, hemiplegia, or seizures (focal, or less often generalized). Often, the diagnosis may be unexpected prior to imaging.

58.5.4 Treatment

Overall management

1. seizure prophylaxis: used by some. It may be safe to discontinue after a week or so if there are no seizure. If late seizure occurs with or without prior use of AEDs, longer-term therapy is required
2. coagulopathies (including iatrogenic anticoagulation) should be reversed
3. surgical evacuation of hematoma indications as follows
   a) symptomatic lesions: including focal deficit, mental status changes…
   b) or subdurals with maximum thickness greater than ≈ 1 cm
   c) or progressive increase in size on serial imaging (CT or MRI scans)

Surgical considerations

Booking the case: Craniotomy: for chronic subdural

Also see defaults & disclaimers (p. 27).
1. position: (usually supine), horseshoe headrest
2. post-op: ICU
3. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the skull to remove blood clot, stop any bleeding identified, placement a drainage tube to allow further fluid to drain after surgery for a day or so
   b) alternatives: nonsurgical management
   c) complications: usual craniotomy complications (p. 28) plus further bleeding which may cause problems (especially in patients taking blood thinners, antiplatelet drugs including aspirin, or those with coagulation abnormalities or previous bleeds) and may require further surgery, hydrocephalus

Surgical options

There is not uniform agreement on the best method to treat CSDHs. For details of techniques (burr holes, whether or not to use subdural drain…) see below.
1. placing two burr holes, and irrigating through and through with tepid saline until the fluid runs clear
2. single “large” burr hole with irrigation and aspiration: see below

3. single burr hole drainage with placement of a subdural drain, maintained for 24–48 hrs (removed when output becomes negligible)
4. twist drill craniostomy: see below (note that small "twist drill" drainage without subdural drain has higher recurrence rate than e.g. burr holes)
5. formal craniotomy with excision of subdural membrane (may be necessary in cases which persistently recur after above procedures, possibly due to seepage from the subdural membrane). Still a safe and valid technique.[44] No attempt should be made to remove the deep membrane adherent to the surface of brain

Techniques that promote continued drainage after the immediate procedure and that may thus reduce residual fluid and prevent reaccumulation:
1. use of a subdural drain: (see below)
2. using a generous burr hole under the temporalis muscle: (see below)
3. bed-rest restriction with the head of the bed flat (1 pillow is permitted) with mild overhydration for 24–48 hours post-op (or if a drain is used, until 24–48 hours after it is removed). May promote expansion of the brain and expulsion of residual subdural fluid. Allowing patients to sit up to 30–40° immediately post-op was associated with higher radiographic recurrence rate (2.3%for those kept flat, vs. 19%for those who sat up) but usually did not require reoperation[45]
4. some advocate continuous lumbar subarachnoid infusion when the brain fails to expand, however there are possible complications[46]

Twist drill craniostomy for chronic subdurals

This method is thought to decompress the brain more slowly and avoids the presumed rapid pressure shifts that occurs following other methods, which may be associated with complications such as intraparenchymal (intracerebral) hemorrhage. May even be performed at the bedside under local anesthesia.

A 0.5 cm incision is made in the scalp in the rostral portion of the hematoma, and then a twist drill hole is placed at a 45° angle to the skull, aimed in the direction of the longitudinal axis of the collection. If the drill does not penetrate the dura, this is done with an 18 Ga. spinal needle. A ventricular catheter is inserted into the subdural space, and is drained to a standard ventriculostomy drainage bag maintained 20 cm below the level of the craniostomy site[47,48,49] (below). The patient is kept flat in bed (see above). Serial CTs assess the adequacy of drainage. The catheter is removed when at least ≈ 20%of the collection is drained and when the patient shows signs of improvement, which occurs within a range of 1–7 days (mean of 2.1 days). Some include a low pressure shunt valve in the system to prevent reflux of fluid or air.

Burr holes for chronic subdural hematomas

To prevent recurrence, the use of small burr holes (without a subdural drain) is not recommended. A generous (>2.5 cm diameter – it is recommended that one actually measure this) subtemporal craniectomy should be performed, and bipolar coagulation is used to shrink the edges of the dura and subdural membrane back to the full width of the bony opening (do not try to separate these two layers as this may promote bleeding). This allows continued drainage of fluid into the temporalis muscle where it may be resorbed. A piece of Gelfoam® may be placed over the opening to help prevent fresh blood from oozing into the opening.

Subdural drain

58

Use of a subdural drain is associated with a decrease in need for repeat surgery from 19%to 10%[50] If a subdural drain is used, a closed drainage system is recommended. Difficulties may occur with ventriculostomy catheters because the holes are small and are restricted to the tip region (so-designed to keep choroid plexus from plugging the catheter when inserted into the ventricles when used as intended as a CSF shunt), especially with thick "oily" fluid (on the positive side, slow drainage may be desirable). The drainage bag is maintained ≈ 50–80 cm below the level of the head.[49,51] An alternative is a small Jackson-Pratt® drain using "thumb-print" indentation of the suction bulb which provides good drainage with a self-contained one-way valve (however, there may be a risk of excessive negative pressure with overcompression of the bulb).

Post-op, the patient is kept flat (see above). Prophylactic antibiotics may be given until ≈ 24–48 hrs following removal of the drain, at which time the HOB is gradually elevated. CT scan prior to removal of the drain (or shortly after removal) may be helpful to establish a baseline for later comparison in the event of deterioration.

There is a case report of administration of urokinase through a subdural drain to treat reaccumulation of clot following evacuation.[52]

58.5.5 Outcome

General information

There is clinical improvement when the subdural pressure is reduced to close to zero, which usually occurs after ≈ 20% of the collection is removed.[49]

Patients who have high subdural fluid pressure tend to have more rapid brain expansion and clinical improvement than patients with low pressures.[51]

Residual subdural fluid collections after treatment are common, but clinical improvement does not require complete resolution of the fluid collection on CT. CTs showed persistent fluid in 78% of cases on post-op day 10, and in 15% after 40 days,[51] and may take up to 6 months for complete resolution. Recommendation: do not treat persistent fluid collections evident on CT (especially before ≈ 20 days post-op) unless it increases in size on CT or if the patient shows no recovery or deteriorates.

76% of 114 patients were successfully treated with a single drainage procedure using a twist drill craniostomy with subdural ventricular catheter, and 90% with one or two procedures.[47] These statistics are slightly better than twist drill craniostomy with aspiration alone (i.e. no drain).

Complications of surgical treatment

Although these collections often appear innocuous, severe complications may occur, including:

1. seizures (including intractable status epilepticus)
2. intracerebral hemorrhage (ICH): occurs in 0.7–5%[53] Very devastating in this setting: one–third of these patients die and one third are severely disabled (also, see below)
3. failure of the brain to re-expand and/or reaccumulation of the subdural fluid
4. tension pneumocephalus
5. subdural empyema: may also occur with untreated subdurals[54]

In 60% of patients ≥ age 75 yrs (and in no patients < 75 yrs), rapid decompression is associated with hyperemia in the cortex immediately beneath the hematoma, which may be related to the complications of ICH or seizures.[53] All complications are more common in elderly or debilitated patients.

Overall mortality with surgical treatment for CSDH is 0–8%[53] In a series of 104 patients treated mostly with craniostomy,[55] mortality was ≈ 4%, all of which occurred in patients > 60 yrs old and were due to accompanying disease. Another large personal series reported 0.5% mortality.[56] Worsening of neurologic status following drainage occurs in ≈ 4%[55]

58.6 Spontaneous subdural hematoma

58.6.1 General information

Occasionally patients with no identifiable trauma will present with severe H/A with or without associated findings (nausea, seizures, lethargy, focal findings including possible ipsilateral hemiparesis[57]...) and CT or MRI discloses a subdural hematoma that may be acute, subacute or chronic in appearance. The onset of symptoms is often sudden.[57]

58.6.2 Risk factors

58

Risk factors identified in a review of 21 cases in the literature[58] include:

1. hypertension: present in 7 cases
2. vascular abnormalities: arteriovenous malformation (AVM), aneurysm[59]
3. neoplasm
4. infection: including meningitis, tuberculosis
5. substance abuse: alcoholism, cocaine[60]
6. hypovitaminosis: especially vitamin C deficiency[37]
7. coagulopathies, including:
   a) iatrogenic (anticoagulation e.g. with warfarin)
   b) Ginkgo biloba (GB) extract: EGb761 and LI1379. Contains ginkgolides (especially Type B) which are inhibitors of platelet activating factor (PAF) at high concentrations,[61] also cause

vasodilation and decreased blood viscosity. There have been case reports showing temporal relationship of hemorrhage to intake of GB,[62] especially at higher doses over long periods of time. However, no consistent alteration was demonstrable in 29 measurable coagulation/clotting variables after 7 days[63] (bleeding time was mildly prolonged in some case reports[62,64]). Some individuals may possibly be more susceptible to the supplement, and there may be as-yet uncharacterized interactions with other entities (such as alcohol, aspirin…) but studies so far have been unrevealing[65]

c) factor XIII deficiency (protransglutaminase).[66,67] In peds: history may include report of bleeding from umbilical cord at birth. Check factor XIII levels as coagulation parameters may be normal or only slightly elevated

8. seemingly innocuous insults (e.g. bending over) or injuries resulting in no direct trauma to the head (e.g. whiplash injuries)
9. intracranial hypotension: spontaneous, after epidural anesthesia, lumbar puncture, or VP shunt[68,69]

58.6.3 Etiology

The bleeding site was determined in 14 of the 21 cases, and was arterial in each, typically involving a cortical branch of the MCA in the area of the sylvian fissure[58] where there is a large number of branches to a wide cortical area.

Possible mechanisms for arterial rupture in idiopathic acute subdural hematoma (ASDH) include tears occur secondary to sudden head movements or trivial head trauma of the following[70,71]:

1. small artery at perpendicular branch point off a cortical artery
2. small artery connecting the dura and cortex
3. adhesions between cortical artery and dura

58.6.4 Treatment

As for traumatic SDH. If symptomatic and/or >≈ 1 cm thick, surgical evacuation is the treatment of choice. For subacute to chronic subdurals, burr-hole evacuation is usually adequate (see above). For acute SDH, a craniotomy is usually required, and should expose the sylvian fissure to identify bleeding point(s). Microsurgical repair of arterial wall has been described.[71]

58.7 Traumatic subdural hygroma

58.7.1 General information

From the Greek hygros meaning wet. AKA traumatic subdural effusion, AKA hydroma. Excess fluid in the subdural space (may be clear, blood tinged, or xanthochromic and under variable pressure) is almost always associated with head trauma, especially alcohol-related falls or assaults.[72] Skull fractures were found in 39%of cases. Distinct from chronic subdural hematoma, which is usually associated with underlying cerebral contusion, and usually contains darker clots or brownish fluid ("motor oil" fluid), and may show membrane formation adjacent to inner surface of dura (hygromas lack membranes).

"Simple hygroma" refers to a hygroma without significant accompanying conditions. "Complex hygroma" refers to hygromas with associated significant subdural hematoma, epidural hematoma, or intracerebral hemorrhage.

58

58.7.2 Pathogenesis

Mechanism of formation of hygroma is probably a tear in the arachnoid membrane with resultant CSF leakage into the subdural compartment. Hygroma fluid contains pre-albumin, which is also found in CSF but not in subdural hematomas. The most likely locations of arachnoid tears are in the sylvian fissure or the chiasmatic cistern. Another possible mechanism is post-meningitis effusion (especially influenza meningitis).

May be under high pressure. May increase in size (possibly due to a flap-valve mechanism) and exert mass effect, with the possibility of significant morbidity. Cerebral atrophy was present in 19% of patients with simple hygromas.

58.7.3 Presentation

Table 58.3 shows clinical findings of subdural hygromas. Many present without focal findings. Complex hygromas usually present more acutely and require more urgent treatment.

Table 58.3 Major clinical features of traumatic subdural hygromas[72]

Type of hygroma	Simple	Complex	Total
number of patients	66	14	80
spontaneous eye opening	74%	57%	71%
disorientation or stupor	65%	57%	64%
mental status change without focal signs	52%	50%	51%
neurological plateau with deficit or delayed deterioration	42%	7%	36%
seizures (usually generalized)	36%	43%	38%
hemiparesis	32%	21%	30%
neck stiffness	26%	14%	24%
anisocoria (maintained light reflex)	15%	7%	14%
headache	14%	14%	14%
alert (no mental status change)	8%	0%	6%
hemiplegia	6%	14%	8%
comatose (responsive to pain only)	3%	43%	10%

58.7.4 Imaging

On CT, the density of the fluid is similar to that of CSF.

Signal characteristics on MRI follow those of CSF.

58.7.5 Treatment

Asymptomatic hygromas do not require treatment. Recurrence following simple burr-hole drainage is common. Many surgeons maintain a subdural drain for 24–48 hrs post-op. Recurrent cases may require either a craniotomy to locate the site of CSF leak (may be very difficult), or a subdural-peritoneal shunt may be placed.

58.7.6 Outcome

Outcome may be more related to accompanying injuries than to the hygroma itself.

5 of 9 patients with complex hygromas and subdural hematoma died. For simple hygromas, morbidity was 20% (12% for decreased mental status without focal findings, 32% if hemiparesis/plegia was present).

58.8 Extraaxial fluid collections in children

58.8.1 Differential diagnosis

1. benign subdural collection in infants (see below)
2. chronic symptomatic extraaxial fluid collections or effusions (see below)
3. cerebral atrophy: should not contain xanthochromic fluid with elevated protein
4. “external hydrocephalus”: ventricles often enlarged, fluid is CSF (p.400)
5. normal variant of enlarged subarachnoid spaces and interhemispheric fissure
6. acute subdural hematoma: high density (fresh blood) on CT (occasionally these will appear as low density collections in children with low hematocrits). Will usually be unilateral (the others above are usually bilateral). These lesions may occur as birth injuries, and typically present with seizures, pallor, tense fontanelle, poor respirations, hypotension, and retinal hemorrhages
7. “craniocerebral disproportion” (head too large for the brain)[73]: extracerebral spaces enlarged up to 1.5 cm in thickness and filled with CSF-like fluid (possibly CSF), ventricles at upper limits of normal, deep sulci, widened interhemispheric fissure, normal intracranial pressure. Patients are developmentally normal. May be the same as benign extra-axial fluid of infancy (see below). Making this diagnosis with certainty is difficult in first few months of life

58.8.2 Benign subdural collections of infancy

General information

Benign subdural collections (or effusions) of infancy,[74,75] are perhaps better characterized by the term benign extra-axial fluid collections of infancy, since it is difficult to distinguish whether they are subdural or subarachnoid.[76] They appear on CT as peripheral hypodensities over the frontal lobes in infants. Imaging may also show dilatation of the interhemispheric fissure, cortical sulci,[77] and sylvian fissure. Ventricles are usually normal or slightly enlarged, with no evidence of transependymal absorption. Brain size is normal. Transillumination is increased over both frontal regions. The fluid is usually clear yellow (xanthochromic) with high protein content. The etiology of these is unclear, some cases may be due to perinatal trauma. They are more common in term infants than preemies. Must be differentiated from external hydrocephalus (p.400).

Presentation

Mean age of presentation is ≈ 4 months.[76]

May show: signs of elevated intracranial pressure (tense or large fontanelle, accelerated head growth crossing percentile curves), developmental delay usually as a result of poor head control due to the large size (Carolan et al. feel that developmental delay without macrocrania runs counter to the concept of “benign” collections[76]), frontal bossing, jitteriness. The poor head control may lead to positional flattening. Other symptoms, such as seizures (possibly focal) are indicative of symptomatic collections (see below). Large collections in the absence of macrocrania are more suggestive of cerebral atrophy.

Treatment

Most cases gradually resolve spontaneously, often within 8–9 months. A single subdural tap (p.1504) for diagnostic purposes (to differentiate from cortical atrophy and to rule out infection) may be done, and may accelerate the rate of disappearance. Repeat physical exams with OFC measurements should be done at ≈ 3–6 month intervals. Head growth usually parallels or approaches normal curves by ≈ 1–2 yrs age, and by 30–36 months orbital-frontal head circumference (OFC) approaches normal percentiles for height and weight. They usually catch up developmentally as OFCs normalize.

58.8.3 Symptomatic chronic extraaxial fluid collections in children

General information

Variously classified as hematomas (chronic subdural hematoma), effusions, or hygromas, with differing definitions associated with each. Since the appearance on imaging and the treatment is similar, Litofsky et al. proposed that they all be classified as extraaxial fluid collections.[78] The difference between these lesions and “benign” subdural effusions (see above) may simply be the degree of clinical manifestation.

Etiologies

The following etiologies were listed in a series of 103 cases[78]:
1. 36%were thought to be the result of trauma (22 were victims of child abuse)
2. 22%followed bacterial meningitis (post-infectious)
3. 19 occurred after placement or revision of a shunt (p.425)
4. no cause could be identified in 17 patients

Other causes include[73]:
1. tumors: extracerebral or intracerebral
2. post-asphyxia with hypoxic brain damage and cerebral atrophy
3. defects of hemostasis: vitamin K deficiency…

Signs and symptoms

Symptoms include: seizure (26%), large head (22%), vomiting (20%), irritability (13%), lethargy (13%), headache (older children), poor feeding, respiratory arrest…

Signs include: full fontanelle (30%), macrocrania (25%), fever (17%), lethargy (13%), hemiparesis (12%), retinal hemorrhages, coma, papilledema, developmental delay…

Evaluation

CT/MRI usually shows ventricular compression and obliteration of the cerebral sulci, unlike with benign subdural collections. The "cortical vein sign" (p.401) helps distinguish this from external hydrocephalus.

Treatment

Options include:

1. observation: follow-up with serial OFC measurements, ultrasound and CT/MRI
2. serial percutaneous subdural taps (p.1504): some patients require as many as 16 taps.[79] Some series show good results and others show low success rate[80,81]
3. burr hole drainage: may include long-term external drainage. Simple burr hole drainage may not be effective with severe craniocephalic disproportion as the brain will not expand to obliterate the extra-axial space
4. subdural-peritoneal shunt: unilateral shunt is usually adequate even for bilateral effusions[78,81,82] (recent recommendations: no study is required to demonstrate communication between the 2 sides[78,83]). An extremely low pressure system should be utilized. The general practice is to remove the shunt after 2–3 months of drainage (once the collections are obliterated) to reduce the risk of associated mineralization of the dura and arachnoid and possible risk of seizures (these shunts are easily removed at this time, but may be more difficult to remove at a later date) [84]

Other recommendations:

At least one percutaneous tap should be performed to rule-out infection.

Many authors recommend observation for the patient with no symptoms or with only enlarging head and developmental delay.

58.9 Traumatic posterior fossa mass lesions

Less than 3% of head injuries involve traumatic mass lesions of the posterior fossa.[85] Epidural hematomas constitute the majority of these (p.894). Other entities (subdural hematoma, intraparenchymal hematoma[86]) comprise the small remainder. See Practice guideline: Surgical management of traumatic posterior fossa mass lesions (p.905) for surgical management recommendations. Any of these can cause hydrocephalus.[85]

Practice guideline: Surgical management of traumatic posterior fossa mass lesions

Indications for surgery

Level III[87]: symptomatic posterior fossa mass lesions or those with mass effect on CT should be surgically removed. Note: mass effect on CT: defined as dislocation, compression or obliteration of the 4th ventricle; compression or loss of basal cisterns (p.921) or the presence of obstructive hydrocephalus

- asymptomatic lesions without mass effect on CT may be managed with close observation and serial imaging

Timing of surgery

Level III[87]: p-fossa mass lesions meeting surgical criteria should be evacuated ASAP due to the potential for rapid deterioration

Surgical methods

Level III[87]: suboccipital craniectomy is the recommended procedure

Most parenchymal hemorrhages managed nonsurgically were < 3 cm diameter.

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59 Gunshot Wounds and Non-Missile Penetrating Brain Injuries

59.1 Gunshot wounds to the head

59.1.1 General information

Gunshot wounds to the head (GSWH) account for the majority of penetrating brain injuries, and comprise ≈ 35% of deaths from brain injury in persons < 45 yrs old. GSWH are the most lethal type of head injury, ≈ two–thirds die at the scene, and GSWH ultimately are the proximal cause of death in > 90% of victims.[1]

59.1.2 Primary injury

Primary injury from GSWH results from a number of factors including:
1. injury to soft tissue
   a) direct scalp and/or facial injuries
   b) soft tissue and bacteria may be dragged intracranially, the devitalized tissue may also then support growth of the bacteria
   c) pressure waves of gas combustion may cause injury if the weapon is close
2. comminuted fracture of bone: may injure subjacent vascular and/or cortical tissue (depressed skull fracture). May act as secondary missiles
3. cerebral injuries from missile
   a) direct injury to brain tissue in path of bullet, exacerbated by
      - fragmentation of bullet
      - ricochet off bone
      - deviations of the bullet from a straight path as it travels: tumbling (forward rotation – pitch), yaw (rotation about vertical axis), rotation (spin), nutation
      - deformation of bullet at impact: e.g. mushrooming
   b) injury to tissue by shock waves, cavitation
4. coup + contrecoup injury from missile impact on head (may cause injuries distant from bullet path)

Because of the complexities of ballistics (some of which are described above) there is often more damage distally than at the entry site even though the bullet slows (losing kinetic energy).

Extent of primary injury is related to impact velocity:
- impact velocity > 100 m/s: causes explosive intracranial injury that is uniformly fatal (NB: impact velocity is less than muzzle velocity)
- non-bullet missiles (e.g. grenade fragments) are considered low velocity
- low muzzle velocity bullets (≈ < 250 m/s): as with most handguns. Tissue injury is caused primarily by laceration and maceration along a path slightly wider than missile diameter
- high muzzle velocity bullets (≈ 600–750 m/s): from military weapons and hunting rifles. Causes additional damage by shock waves and temporary cavitation (tissue pushed away from the missile causes a conical cavity of injury that may exceed bullet diameter many-fold, and causes low-pressure region which may draw surface debris into the wound)

59.1.3 Secondary injury

Cerebral edema occurs similar to closed head injury. ICP may rise rapidly within minutes (higher ICPs result from higher impact velocities). Cardiac output may also fall initially. Together, ↑ ICP and ↓ MAP adversely effect cerebral perfusion pressure.

Other common complicating factors include: DIC, intracranial hemorrhage from lacerated blood vessels.

59.1.4 Late complications

Late complications include:
1. cerebral abscess: migration of bullet may be a tip-off (see below). Usually associated with retained contaminated material (bullet, bone, skin…) but may also result from persistent communication with nasal sinuses

2. traumatic aneurysm[2]
3. seizures
4. fragment migration
   a) migration of a bullet: often indicates abscess[3] or, less commonly, a hematoma cavity. May also migrate within the ventricles
   b) intraventricular fragments may migrate and cause obstructive hydrocephalus[4]
5. lead toxicity: more of an issue with bullet in disc space (p.1017)

59.1.5 Evaluation

Physical exam

Exam should describe visible entrance and exit wounds. In through-and-through missile wounds of the skull, the entrance wound is typically smaller than the exit wound due to bullet mushrooming. Entrance wounds may be especially small with direct contact of the muzzle to the head. At surgery or autopsy, the entrance wound will typically show beveling of the inner table, whereas exit wounds have a beveled outer table.

Imaging

AP and lateral skull x-rays

This is one situation where skull x-rays still may provide useful information, as they are less susceptible to artifact from the bullet than the CT scan. Helps to localize metal and bone fragments, and to help identify entrance/exit sites (omit if time not available)

Head CT scan without contrast

The main assessment tool. Demonstrates location of bone and metal. Delineates bullet trajectory: assesses if bullet passed through ventricles and how many quadrants of the hemisphere have been traversed. Shows amount of blood in brain and assesses intracranial hematomas (epidural, subdural or intraparenchymal).

Angiography in GSWH

Rarely performed emergently. When done, usually performed on ≈ day 2–3.

Indications for angiography[5]:

- unexpected delayed hemorrhage
- a trajectory that would likely involve named vessels in a salvageable patient
- large intraparenchymal hemorrhages in a salvageable patient

59.1.6 Management

Initial management

General measures

1. CPR as required; endotracheal intubation if stuporous or airway compromised
2. additional injuries (e.g. chest wounds) identified and treated appropriately
3. usual precautions taken for spine injury
4. fluids as needed to replace estimated blood loss which may be variable: exercise restraint to avoid excessive hydration (to minimize cerebral edema)
5. pressors to support MAP during and after fluid resuscitation

Treatment specific to the injury

Neurological assessment as rapidly as possible and as thoroughly as time permits.

The Glasgow Coma Scale is still the most widely used grading system and allows better comparison between series than specialized scales for GSWHs.

Decision by experienced neurosurgeon regarding the ultimate treatment of the patient will determine appropriate steps to be taken. Patients with little CNS function (in the absence of shock) are unlikely to benefit from craniotomy. Supportive measures are indicated in most cases (for possibility of organ donation, opportunity for family to adjust to situation, and requirements for observation period to determine actual brain death).

In patients considered for further treatment, rapid deterioration at any point with signs of herniation requires immediate surgical intervention. As time permits, the following should be undertaken:

1. initial steps
   a) control bleeding from scalp and associated wounds (hemostats on scalp vessels)
   b) shave scalp to identify entrance/exit sites, and to save time in the O.R.
2. medical treatment (similar to closed head injury)
   a) assume ICP is elevated:
      - elevate HOB 30–45° with head midline (avoids kinking jugular veins)
      - mannitol (1 gm/kg bolus) as blood pressure tolerates
      - hyperventilate to $PaCO_2$ = 30–35 mm Hg if indications are met (p.873)
      - steroids: (unproven efficacy) 10 mg dexamethasone IVP
   b) prophylaxis against GI ulcers: H2 antagonist (e.g. ranitidine 50 mg IVPB q 8 hrs) or proton pump inhibitor, NG tube to suction
   c) begin anticonvulsants (does not reduce incidence of late seizures)
   d) antibiotics: generally used although no controlled study demonstrates efficacy in preventing meningitis or abscess. Most organisms are sensitive to penicillinase resistant agents, e.g. nafcillin, recommended for ≈ 5 days
   e) tetanus toxoid administration

Surgical treatment

Indications for surgery are controversial. Some authors suggest that better outcome might occur with more aggressive management, and that poor outcome may be a self-fulfilling prophecy.[6] Patients with minimal neurologic function, e.g. fixed pupils, decorticate or decerebrate posturing… (when not in shock and with good oxygenation) should not be operated upon, because the chance of meaningful recovery is close to zero. Patients with less severe injuries should be considered for urgent operation.

Goals of surgery

1. debridement of devitalized tissue: less tissue is injured in civilian GSWH, but elevated ICP post-op may imply more vigorous debridement was needed, especially of non-eloquent brain (e.g. temporal tips)
2. evacuation of hematomas: subdural, intraparenchymal…
3. removal of accessible bone fragments
4. retrieval of bullet fragment for forensic purposes (note: everyone who handles the fragments may be subpoenaed to testify as to the “chain of evidence”). Large intact fragments should be sought as they tend to migrate (note: risk of infection and seizures due to retained bullet fragments is not high in civilian GSWH, therefore only accessible fragments should be sought and removed)
5. obtaining hemostasis
6. watertight dural closure (usually requires graft)
7. separation of intracranial compartment from air sinuses traversed by bullet
8. identification of entry and exit wounds for forensic purposes; see Evaluation (p.909)

Surgical technique

Some key points of surgical technique[7 (p 2098–104)]:
- positioning and draping should make both entry and exit wounds accessible
- devitalized tissue around the entry & exit wounds should be excised
- fractured bone should be excised by a circumferential craniectomy (craniotomy may be used in some civilian GSWH, the entry site within the craniotomy should be rongeured or drilled back to clean bone)
- air sinuses that are traversed should have the mucosa exenterated, and are then packed with muscle, and covered with a graft (e.g. periosteum or fascia lata) to separate them from intracranial compartment
- the dura is opened in a stellate fashion
- pulped brain is removed from within using suction and bipolar in an enlarging cone until healthy tissue is encountered (further injury to deep midline structures should be avoided, here, stay within bullet tract)
- contralateral fragments with no exit wound should only be removed if accessible
- intraventricular fragments can present significant risk. Ventriculoscopy (if available) may be well suited for removing these
- dural closure should be watertight; grafts of pericranium, temporalis fascia, or fascia lata grafts may be used; avoid dura substitutes

- cranioplasty should be delayed 6–12 months to reduce risk of infection
- a post-op CSF fistula that persists >2 weeks should be repaired

ICP monitoring

ICP is often elevated after surgical debridement[6] and monitoring may be warranted.

Outcome

Prognostic factors:

1. level of consciousness is the most important prognostic factor: ≈ 94% of patients who are comatose with inappropriate or absent response to noxious stimulus on admission die, and half the survivors are severely disabled[8]
2. as initially espoused by Cushing, the path of the bullet is also an important prognosticator. Especially poor prognosis is associated with:
   a) bullets that cross the midline
   b) bullets that pass through the geographic center of the brain
   c) bullets that enter or traverse the ventricles
   d) the more lobes traversed by the bullet
3. hematomas seen on CT are poor prognostic findings
4. suicide attempts are more likely to be fatal

59.2 Non-missile penetrating trauma

59.2.1 General information

This section deals with penetrating injuries to the brain (and to some extent to the spinal cord) excluding missile injuries, i.e. gunshot wounds (p.908). Includes trauma from: knives, arrows, lawn darts… Injury to neural tissue tends to be more limited than with missiles because many of the associated injurious aspects of the missile are absent (p.908).

59.2.2 Arrow injuries

As a result of the lower velocity (e.g. 58 m/s) compared to firearms and the sharp tip, injury is usually limited to tissue directly incised by the arrowhead.[9]

59.2.3 Cases with foreign body still embedded

In penetrating trauma, it is usually not appropriate to remove any protruding part of the foreign body until the patient is in the operating room, unless it cannot be avoided. If possible, it is helpful to have another identical object for comparison in planning extrication of the embedded object.[10] To minimize extending the trauma to the CNS, the protruding object should be stabilized in some way during transportation and evaluation. Intraoperatively, devices such as the Greenberg retractor may be used to stabilize the object during preparation and the initial approach.

59.2.4 Indications for pre-op angiography

1. object passes in region of large named artery
2. object passes near dural sinuses
3. visible evidence of arterial bleeding: angiography is not appropriate if hemorrhage cannot be controlled

59.2.5 Surgical techniques

It is impossible to give details to cover every situation. Some guidelines:

1. empiric antibiotic coverage is appropriate; see Meningitis post craniospinal trauma (p.318). Take cultures from the wound and the foreign body to guide later antibiotic therapy
2. optimal control can usually be gained by performing a craniotomy up to and if possible around the object, such that removing the bone flap will not disturb the object. The last remnants of bone may then be removed with a rongeur

3. if at all possible open the dura before removing the object, since removal with the dura closed does not allow adequate control of any bleeding from the brain
4. removal of the object ideally should follow the entry trajectory if possible
5. although gunshot wounds are not sterile as once thought, they are probably less contaminated than penetrating wounds. One should debride any easily accessible impacted bone and other extracranial tissue and material along the track

59.2.6 Post-op care

1. a course of antibiotics are usually appropriate since infection is common
2. consider a post-op arteriogram to rule-out traumatic aneurysm

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60 Pediatric Head Injury

60.1 General information

75% of children hospitalized for trauma have a head injury. Although most pediatric head injuries are mild and involve only evaluation or brief hospital stays, CNS injuries are the most common cause of pediatric traumatic death.[1] The overall mortality for all pediatric head injuries requiring hospitalization has been reported between 10–13%[2] whereas the mortality associated with severe pediatric head injury presenting with decerebrate posturing has been reported as high as 71%[3]

Differences between adult and pediatric head injury:

1. epidemiology:
   a) children often have milder injuries than adults
   b) lower chance of a surgical lesion in a comatose child than in an adult[4]
2. types of injury: injuries peculiar to pediatrics
   a) birth injuries: skull fractures, cephalhematoma (see below), subdural or epidural hematomas, brachial plexus injuries (p.550)
   b) perambulator/walker injuries
   c) child abuse (see below): shaken baby syndrome…
   d) injuries from skateboarding, scooters…
   e) lawn darts
   f) cephalhematoma: see below
   g) leptomeningeal cysts, AKA "growing skull fractures" (p.915)
3. response to injury
   a) responses to head injury of older adolescent are very similar to adults
   b) "malignant cerebral edema": acute onset of severe cerebral swelling (probably due to hyperemia[5,6]) following some head injuries, especially in young children (may not be as common as previously thought[7])
   c) posttraumatic seizures: more likely to occur within the 1st 24 hrs in children than in adults (p.462) [8]

60.2 Management

60.2.1 Imaging studies

Indications for CT are shown below.

When not contraindicated, rapid sequence MRI may be used (avoids additional radiation in the growing child), especially for follow-up imaging.

Practice guideline: Imaging in minor pediatric head injury

- Recommendations[9]: CT scan for children with neurologic or cognitive dysfunction, or suspicion of a depressed or basilar skull fracture.
- Recommendations[9]: when a CT scan is not done in a child ≤ 1 year age meeting the above criteria (e.g. because of sedation concerns), a skull film may be considered.

Based mostly on prospective trials (not randomized) or large case series.

Definitions: pediatrics = ages 1 month – 17 years of age. Minor head injury: GCS ≥ 13 (excludes: suspicion or proof of child abuse, patients requiring hospitalization for other reasons).

≈ 22% of those with a history of loss of consciousness (LOC) > 5 mins have a brain injury, whereas 92% without LOC > 5 mins will have no brain injury.[9]

60.2.2 Home observation

Practice guideline: Home observation in minor pediatric head injury

Recommendations[9]: a child with GCS = 14–15 and normal CT scan can be considered for home observation if neurologically stable (these patients are at near zero risk of having an occult brain injury).

Based mostly on prospective trials (not randomized) or large case series.

Definitions: pediatrics = ages 1 month – 17 years of age. Minor head injury: GCS ≥ 13 (excludes: suspicion or proof of child abuse, patients requiring hospitalization for other reasons).

60.3 Outcome

As a group, children fare better than adults with head injury.[10] However, very young children do not do as well as the school-age child.[11]

All aspects of neuropsychological dysfunction following head injury may not always be related to the trauma, as children who get injured may have pre-existing problems that increase their propensity to get hurt[12] (this is controversial[13]).

60.4 Cephalhematoma

60.4.1 General information

Accumulation of blood under the scalp. Occurs almost exclusively in children.

Two types:

1. subgaleal hematoma: may occur without bony trauma, or may be associated with linear nondisplaced skull fracture (especially in age < 1 yr). Bleeding into loose connective tissue separates galea from periosteum. May cross sutures. Usually starts as a small localized hematoma, and may become huge (with significant loss of circulating blood volume in age < 1 year, transfusion may be necessary). Inexperienced clinicians may suspect CSF collection under the scalp which does not occur. Usually presents as a soft, fluctuant mass. These do not calcify
2. subperiosteal hematoma (some refer to this as cephalhematoma): most commonly seen in the newborn (associated with parturition, may also be associated with neonatal scalp monitor[14,15]). Bleeding elevates periosteum, extent is limited by sutures. Firmer and less ballotable than subgaleal hematoma[16 (p 312)]; scalp moves freely over the mass. 80%reabsorb, usually within 2–3 weeks. Occasionally may calcify

Infants may develop jaundice (hyperbilirubinemia) as blood is resorbed, occasionally as late as 10 days after onset.

60.4.2 Treatment

Treatment beyond analgesics is almost never required, and most usually resolve within 2–4 weeks. Avoid the temptation of percutaneously aspirating these as the risk of infection exceeds the risk of following them expectantly, and in the newborn removal of the blood may make them anemic. Follow serial hemoglobin and hematocrit in large lesions. If a subperiosteal hematoma persists > 6 weeks, obtain a skull film. If the lesion is calcified, surgical removal may be indicated for cosmetic reasons (although with most of these the skull will return to normal contour in 3–6 months.[16 (p 315)]

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60.5 Skull fractures in pediatric patients

60.5.1 General information

This section deals with some special concerns of skull fractures in pediatrics. Also see Child abuse (p.916).

60.5.2 Posttraumatic leptomeningeal cysts (growing skull fractures)

General information

Posttraumatic leptomeningeal cysts (PTLMC) (sometimes just traumatic leptomeningeal cysts) AKA growing skull fractures are not to be confused with arachnoid cysts (AKA leptomeningeal cysts, which are not posttraumatic). PTLMC consists of a fracture line that widens with time. Although usually asymptomatic, the cyst may cause mass effect with neurologic deficit.

PTLMCs were first described in 1816,[17] and are very rare, occurring in 0.05–0.6% of skull fractures.[18,19] Usually requires both a widely separated fracture AND a dural tear. Mean age at injury: < 1 year, over 90% occur before age 3 years[20] (formation may require the presence of a rapidly growing brain[21]) although rare adult cases have been described[22,23,17] (a total of 5 cases in the literature as of 1998[17]). PTLMC rarely occur > 6 mos out from the injury. Some children may develop a skull fracture that seems to grow during the initial few weeks, that is not accompanied by a subgaleal mass, and that heal spontaneously within several months; the term "pseudogrowing fracture" has been suggested for these.[24]

Presentation

Most often presents as scalp mass (usually subgaleal), although there are reports of presentation with head pain alone.[22]

Diagnosis

Radiographic findings: progressive widening of fracture and scalloping (or saucering) of edges.

Screening for development of PTLMC

If early growth of a fracture line with no subgaleal mass is noted, repeat skull films in 1–2 months before operating (to rule-out pseudogrowing fracture). In young patients with separated skull fractures (the width of the initial fracture is rarely mentioned), consider obtaining follow-up skull film 6–12 mos post-trauma. However, since most PTLMCs are brought to medical attention when the palpable mass is noticed, routine follow-up x-rays may not be cost-effective.

Treatment

Treatment of true PTLMC is surgical, with dural closure mandatory. Since the dural defect is usually larger than the bony defect, it may be advantageous to perform a craniotomy around the fracture site, repair the dural defect, and replace the bone.[23] Pseudogrowing fractures should be followed with x-rays and operated only if expansion persists beyond several months or if a subgaleal mass is present.

60.5.3 Depressed skull fractures in pediatrics

See reference.[25]

General information

Most common in frontal and parietal bones. One third are closed, and these tend to occur in younger children (3.4 ± 4.2 yrs, vs. 8.0 ± 4.5 yrs for compound fractures) as a result of the thinner, more deformable skull. Open fractures tended to occur with MVAs, closed fractures tended to follow accidents at home. Dural lacerations are more common in compound fractures.

Simple depressed skull fractures

There was no difference in outcome (seizures, neurologic dysfunction or cosmetic appearance) in surgical vs. nonsurgical treatment in 111 patients < 16 yrs age. In the younger child, remodelling of the skull as a result of brain growth tends to smooth out the deformity.

Indications for surgery for pediatric simple depressed skull fracture:

1. definite evidence of dural penetration
2. persistent cosmetic defect in the older child after the swelling has subsided
3. ± focal neurologic deficit related to the fracture (this group has a higher incidence of dural laceration, although it is usually trivial)

"Ping-pong ball" fractures

See reference.[26]

A green-stick type of fracture →caving in of a focal area of the skull as in a crushed area of a ping-pong ball. Usually seen only in the newborn due to the plasticity of the skull.

Indications for surgery

No treatment is necessary when these occur in the temporoparietal region in the absence of underlying brain injury as the deformity will usually correct as the skull grows.

- radiographic evidence of intraparenchymal bone fragments
- associated neurologic deficit (rare)
- signs of increased intracranial pressure
- signs of CSF leak deep to the galea
- situations where the patient will have difficulty getting long-term follow-up

Technique

Frontally located lesions may be corrected for cosmesis by a small linear incision behind the hairline, opening the cranium adjacent to the depression, and pushing it back out e.g. with a Penfield #3 dissector.

60.6 Nonaccidental trauma (NAT)

60.6.1 General information

AKA Child abuse. At least 10% of children < 10 yrs age that are brought to E/R with alleged accidents are victims of child abuse.[27] The incidence of accidental head trauma of significant consequence below age 3 is low, whereas this is the age group in which battering is highest.[28]

There are no findings that are pathognomonic for child abuse. Factors which raise the index of suspicion include:

1. retinal hemorrhage (see below)
2. bilateral chronic subdural hematomas in a child < 2 yrs age (p.904)
3. skull fractures that are multiple (see below) or those that associated with intracranial injury
4. significant neurological injury with minimal signs of external trauma

60.6.2 Shaken baby syndrome

Vigorous shaking of a child produces violent whiplash-like angular acceleration-decelerations of the head (the infant head is relatively large in proportion to the body, and the neck muscles are comparatively weak)[29] which may lead to significant brain injury. Some researchers believe that shaking alone may be inadequate to produce the severe injuries seen, and that impact is often also involved.[30]

Characteristic findings include retinal hemorrhages (see below), subdural hematomas (bilateral in 80%) and/or subarachnoid hemorrhage (SAH). There are usually few or no external signs of trauma (including cases with impact, although findings may be apparent at autopsy). In some cases there may be finger marks on the chest, multiple rib fractures and/or pulmonary compression ± parenchymal lung hemorrhage. Deaths in these cases are almost all due to uncontrollable intracranial hypertension. There may also be injury to the cervicomedullary junction.[31]

60.6.3 Retinal hemorrhage (RH) in child abuse

"In a traumatized child with multiple injuries and an inconsistent history, the presence of RH is pathognomonic of battering".[28] However, RH may also occur in the absence of any evidence of child abuse. 16/26 battered children < 3 yrs age had RH on funduscopy, whereas 1/32 non-battered traumatized children with head injury had RH (the single false positive: traumatic parturition, where the incidence of RH is 15–30%).

Differential diagnosis of etiologies of retinal hemorrhage:

1. child abuse (including "shaken baby syndrome", see above)
2. benign subdural effusion in infants (p.904)
3. acute high altitude sickness (p.848)
4. acute increase in ICP: e.g. with a severe seizure (may be similar to Purtscher's retinopathy – see below)

5. Purtscher's retinopathy[32]: loss of vision following major trauma (chest crush injuries, airbag deployment[33]...), pancreatitis, childbirth or renal failure, among others. Posterior pole ischemia with cotton-wool exudates and hemorrhages around the optic disc due to microemboli of possibly fat, air, fibrin clots, complement-mediated aggregates or platelet clumps. No known treatment

60.6.4 Skull fractures in child abuse

A series comparing 39 cases of skull fracture from documented child abuse to 95 cases of probable accidental injury[27] showed the following:

1. the parietal bone was the most common site of fracture in both groups (≈ 90%)
2. depression of skull fractures was frequently missed clinically due to overlying hematoma
3. clinical features in patients with skull fractures did not reliably differentiate child abuse from trauma (retinal hemorrhages (RH) were seen in 1 child abuse and 1 accidental trauma patient: note that RH is more common in "shaken child" syndrome which is not commonly associated with skull fractures)
4. 3 characteristics more frequently seen after child abuse than after other trauma:
   a) multiple fractures
   b) bilateral fractures
   c) fractures that cross sutures

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60

61 Head Injury: Long-Term Management, Complications, Outcome

61.1 Airway management

Practice guideline: Timing of tracheostomy

Level II[1]: early tracheostomy reduces the number of days of mechanical ventilation but does not affect mortality or incidence of pneumonia

Practice guideline: Timing of extubation

Level III[1]: early extubation for patients meeting extubation criteria does not increase the risk of pneumonia

61.2 Deep-vein thrombosis (DVT) prophylaxis

Also see further details about thromboembolism (p.167) in neurosurgical patients. The risk of developing DVT is ≈ 20% in untreated severe TBI.[2] See also Practice guideline: DVT prophylaxis in severe TBI (p.918).

Practice guideline: DVT prophylaxis in severe TBI

Level III[3]:
- unless contraindicated, graduated compression stockings or intermittent compression boots are recommended until patients are ambulatory
- low molecular weight heparin (LMWH) (p. 164) or low-dose unfractionated heparin in conjunction with mechanical measures lowers the DVT risk, but a trend suggests they increase the risk of expansion of intracranial hemorrhage (note: there is insufficient evidence: to support use of one pharmacologic agent over another, or to define the optimal dose or timing of agents[3]

61.3 Nutrition in the head-injured patient

61.3.1 Summary of recommendations (see text for details)

Practice guideline: Nutrition

Level II[4]: full caloric replacement should be attained by post-trauma day 7

Σ

1. by post-trauma day 7, replace the following (enterally or parenterally):
   a) non-paralyzed patients: 140% of predicted basal energy expenditure (BEE)
   b) paralyzed patients: 100% of predicted BEE
2. provide ≥ 15% of calories as protein
3. nutritional replacement should begin within 72 hrs of head injury in order to achieve goal #1 by day 7
4. the enteral route is preferred (IV hyperalimentation is preferred if higher nitrogen intake is desired or if there is decreased gastric emptying)

61.3.2 Caloric requirements

Rested comatose patients with isolated head injury have a metabolic expenditure that is 140% of normal for that patient (range: 120–250%).[5,6,7,8] Paralysis with muscle blocker or barbiturate coma reduced this excess expenditure in most patients to ≈ 100–120% of normal, but some remained elevated by 20–30%[9] Energy requirements rise during the first 2 weeks after injury, but it is not known for how long this elevation persists. Mortality is reduced in patients who receive full caloric replacement by day 7 after trauma[10] (a beneficial effect with an earlier goal of replacement by 3 days post-trauma was not found[11]). Since it generally takes 2–3 days to get nutritional replacement up to speed whether the enteral or parenteral route is utilized,[8] it is recommended that nutritional supplementation begin within 72 hrs of head injury.

61.3.3 Enteral vs. IV hyperalimentation

Caloric replacement that can be achieved is similar between enteral or parenteral routes.[12] The enteral route is preferred because of reduced risk of hyperglycemia, infection and cost.[13] IV hyperalimentation may be utilized if higher nitrogen intake is desired or if there is decreased gastric emptying. No significant difference in serum albumin, weight loss, nitrogen balance, or final outcome was found between enteral and parenteral nutrition.[12]

Estimates of basal energy expenditure (BEE) can be obtained from the Harris-Benedict equation,[14] shown in Eq (61.1), (61.2) and (61.3), where W is weight in kg, H is height in cm, and A is age in years.

$$\text{Males: } BEE = 66.47 + 13.75 \times W + 5.0 \times H - 6.76 \times A \tag{61.1}$$

$$\text{Females: } BEE = 65.51 + 9.56 \times W + 1.85 \times H - 4.68 \times A \tag{61.2}$$

$$\text{Infants: } BEE = 22.1 + 31.05 \times W + 1.16 \times H \tag{61.3}$$

61.3.4 Enteral nutrition

Isotonic solutions (such as Isocal® or Osmolyte®) should be used at full strength starting at 30 ml/hr. Check gastric residuals q 4 hrs and hold feedings if residuals exceed ≈ 125 ml in an adult. Increase the rate by ≈ 15–25 ml/hr every 12–24 hrs as tolerated until the desired rate is achieved.[15] Dilution is not recommended (may slow gastric emptying), but if it is desired, dilute with normal saline to reduce free water intake.

Cautions:

NG tube feeding may interfere with absorption of phenytoin; see phenytoin (PHT, Dilantin®) (p.446). Reduced gastric emptying may be seen following head-injury[16] (NB: some may have temporarily elevated emptying) as well as in pentobarbital coma, patients may need IV hyperalimentation until the enteric route is usable. Others have described better tolerance of enteral feedings using jejunal administration.[17]

61.3.5 Nitrogen balance

A normal subject fed a protein-free diet for 3 days will excrete 85 mg of nitrogen/kg/d. These loses increase with injury. The rise in urinary N is due primarily to an increase in urea (comprises 80–90% of urinary N). This is thought to represent an increase in mobilization and breakdown of amino acids, which are felt to originate mainly from skeletal muscle.[18] Some of this represents a primary reaction to injury in which certain vital organs seem to be maintained at the expense of less active organs, and a significantly higher nitrogen balance cannot be achieved by increasing the amount of calories supplied as protein beyond a certain level.[12,15] Catabolism of protein yields 4 kcal/g (compared to 1 kcal/g for carbohydrates and 9 kcal/g for fat), and in the non-injured adult normally supplies only ≈ 10% of energy needs.[19]

As an estimate, for each gram of N excreted (mostly in the urine, however, some is also lost in the feces), 6.25 gm of protein have been catabolized. It is recommended that at least 15% of calories be supplied as protein. The percent of calories consumed (PCC) derived from protein can be calculated from Eq (61.4), where N is nitrogen in grams, and BEE is the basal energy expenditure[5] (see Eq (61.1), (61.2) and (61.3)).

$$\text{PCC (from protein)} = \frac{\text{N (gm N)} \times \frac{6.25 \text{ gm protein}}{\text{gm N}} \times \frac{4.0 \text{ kcal}}{\text{gm protein}}}{\text{BEE}} \times 100 \qquad (61.4)$$

Thus, to supply PCC (protein) = 15% once the BEE is known, use Eq (61.5). Some enteral formulations include Magnacal® (PCC = 14%) and TraumaCal® (PCC = 22%).

$$\text{N (gm N)} = 0.006 \times \text{BEE} \qquad (61.5)$$

61.4 Posttraumatic hydrocephalus

61.4.1 General information

Hydrocephalus was found in 40% of 61 patients with severe head injury (GCS = 3–8) and in 27% of 34 patients with moderate head injury (GCS = 9–13).[20] Hydrocephalus developed by 4 weeks after injury in 58% and by 2 months in 70%[20] There was no statistically significant relationship between posttraumatic hydrocephalus and age, the presence of SAH, or type of lesion (focal or diffuse). Posttraumatic hydrocephalus was associated with worse outcome.[20]

Hydrocephalus after traumatic subarachnoid hemorrhage

Incidence of clinically symptomatic hydrocephalus within 3 months of traumatic subarachnoid hemorrhage (tSAH) is ≈ 12%[21] In this series of 301 tSAH patients, multivariate analysis showed the risk of developing hydrocephalus increased with age, intraventricular hemorrhage, blood thickness ≥ 5 mm, and diffuse distribution of blood (vs. focal distribution). There was no correlation with gender, admission GCS score, basal location of tSAH, or use of decompressive craniectomy.[21] NB: this is potentially confusing, univariate analysis shows the risk of hydrocephalus increases with increasing severity of TBI.

61.4.2 Differentiating true hydrocephalus from hydrocephalus ex vacuo

Delayed ventricular enlargement months to years after TBI may instead be due to atrophy (hydrocephalus ex vacuo) secondary to diffuse axonal injury, and may not represent true hydrocephalus. It may not be possible to accurately differentiate these two conditions, and the decision to shunt may therefore be difficult (similar to the dilemma in patients with NPH vs. atrophy).

61.4.3 Indications for surgical treatment

Factors favoring hydrocephalus, for which shunt should be considered:
1. elevated pressure on 1 or more LPs
2. papilledema on funduscopic exam
3. symptoms of headache/pressure
4. findings of "transependymal absorption" on CT or T2WI MRI (p.406)
5. ± patient's whose neurologic recovery seems worse than expected
6. provocative tests have been recommended (p.255) [22]

Patients with enlarged ventricles who are asymptomatic and are doing well following their head injury should be managed expectantly.

61.5 Outcome from head trauma

61.5.1 Age

In general, the degree of recovery from closed head injury is better in infants and young children than in adults. In adults, decerebrate posturing or flaccidity with loss of pupillary or oculovestibular reflex is associated with a poor outcome in most cases, these findings are not as ominous in pediatrics.

61.5.2 Outcome prognosticators

General information

The frequency of poor outcome from closed head injury is increased with persistent ICP > 20 mm Hg after hyperventilation, increasing age, impaired or absent pupillary light response or eye movement, hypotension (SBP < 90), hypercarbia, hypoxemia, or anemia.[23] This is probably due at least in part to the fact that some of these are markers for significant injury to other body systems. One of the most important predictors for poor outcome is the presence of a mass lesion requiring surgical removal.[24] High ICP during the first 24 hrs is also a poor prognosticator.

Obliteration of basal cisterns on CT

The status of the basal cisterns (BCs) is evaluated on axial CT scan at the level of the midbrain (Fig. 61.1) where they are divided into 3 limbs[25] (1 posterior limb = quadrigeminal cistern, 2 lateral limbs = posterior portion of the ambient cisterns). Note: "basal cisterns" in the trauma literature are a subset of the perimesencephalic cisterns (p. 1232). Possible findings:

1. open: all 3 limbs open
2. partially closed: 1 or 2 limbs obliterated
3. completely closed: 3 limbs obliterated

Compression or absence of the BCs carries a threefold risk of increased ICP, and the status of the BCs correlates with outcome.[25]

In a study of 218 patients with GCS ≤ 8, the BCs were classified on initial CT (within 48 hrs of admission) as: absent, compressed, normal, or not visualized (quality of CT too poor to tell).[26] The relationship of the BCs to outcome is shown in Table 61.1.

18 patients had a shift of brain structures > 15 mm associated with absent BCs, all of them died. The status of the BCs were more important within each GOS score than across scores. Also, see Table 58.3 for further information on CT.

Midline shift (MLS)

The presence of MLS correlates with a worse outcome. For the purpose of standardizing measurements in trauma, MLS is defined at the level of the foramen of Monro[25] as shown in Fig. 61.2, and is calculated using Eq (61.6).

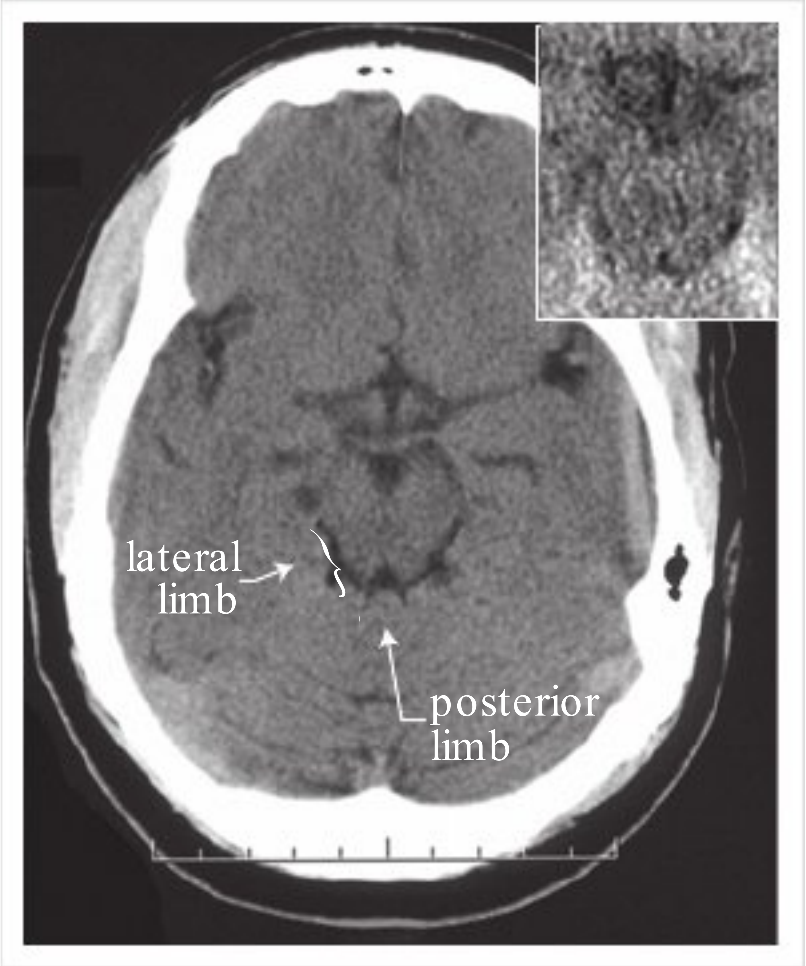

Fig. 61.1 Basal cisterns CT demonstrating open basal cisterns (inset: example of ≈ complete obliteration of BCs)

Table 61.1 Correlation of GOS[a] with basal cisterns

Basal cisterns	Outcome[a]				
	Mortality	Vegetative	Severe disability	Moderate disability	Good
	(GOS 1)	(GOS 2)	(GOS 3)	(GOS 4)	(GOS 5)
normal	22%	6%	16%	21%	35%
compressed	39%	7%	18%	17%	19%
absent	77%	2%	6%	4%	11%
not-visualized	68%	0%	11%	9%	12%

[a]GOS = Glasgow outcome scale, see Table 88.4

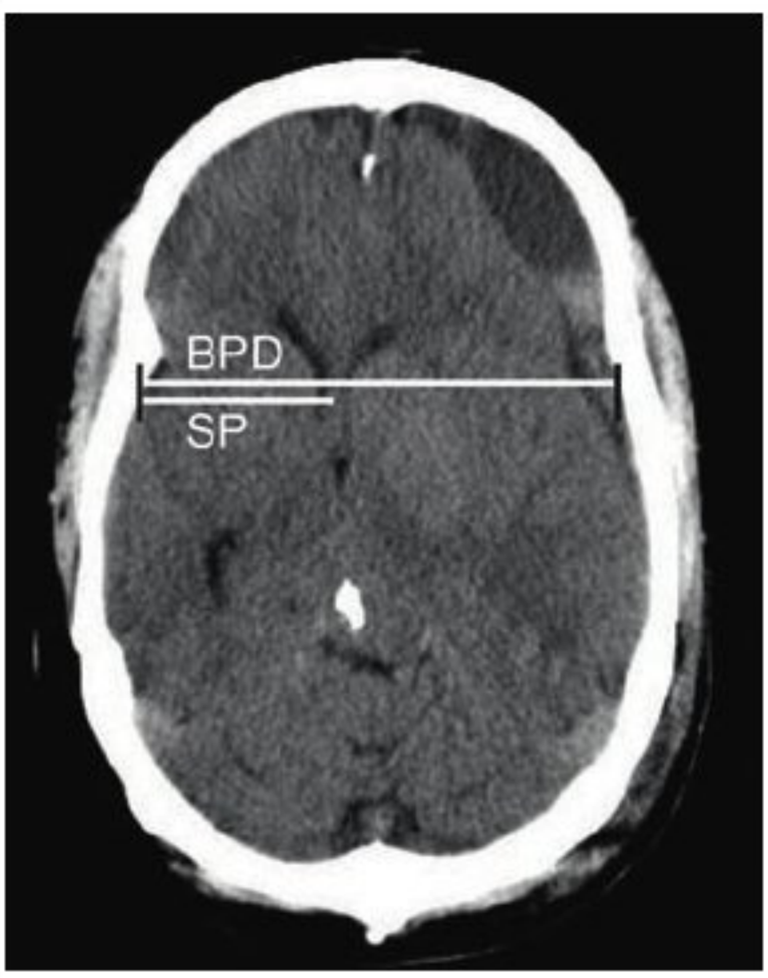

Fig. 61.2 Measurement of midline shift (axial non-contrast CT of a brain with a left-sided acute-on-chronic subdural hematoma)

$$\text{midline shift (MLS)} = \frac{\text{BPD}}{2} - \text{SP} \qquad (61.6)$$

where the midline is found by dividing the biparietal diameter (BPD) (the width of intracranial compartment at this location) by 2, and subtracting SP (the distance from the inner table to the septum pellucidum on the side of the shift). Measurements may be inaccurate if the vertical axis of the patient's head is not parallel to the long axis of the CT scanner.

Midline shift may be associated with altered levels of consciousness (p. 298).

Apolipoprotein E (apoE) ε4 allele

The presence of this genotype portends a worse prognosis following traumatic brain injury.[27] Furthermore, the incidence of severe brain injury in individuals with the apoE-4 allele greatly exceeds the rate of the allele in the general population.[28] This allele is also a risk factor for Alzheimer's disease (see below) as well as for chronic traumatic encephalopathy (p. 924).

61.6 Late complications from traumatic brain injury

61.6.1 General information

Long term complications include:
1. posttraumatic seizures (p.462)
2. communicating hydrocephalus: incidence ≈ 3.9%of severe head injuries
3. posttraumatic syndrome (or postconcussive syndrome): see below
4. hypogonadotropic hypogonadism (p.836) [29]
5. chronic traumatic encephalopathy (p.924)
6. Alzheimer's disease (AD): head injury (especially if severe) promotes the deposition of amyloid proteins, especially in individuals possessing the apolipoprotein E (apoE) ε4 allele,[28] which may be related to the development of AD[30,31,32]

61.6.2 Postconcussive syndrome

General information

Variously defined collection of symptoms (see below) that is usually considered as a possible sequelae to minor head trauma (although some of these features can certainly be seen following more serious head trauma). Loss of consciousness is not a prerequisite to the development of the syndrome.

Controversy exists over the relative contribution of actual organic dysfunction vs. psychological factors (including conversion reaction, secondary gain which may be for attention, financial reward, drug seeking…). Furthermore, the presence of some of these symptoms can undoubtedly lead to the development of others (e.g. headache can cause difficulty concentrating and thus poor job performance and thence depression).

Presentation

A paradox has been noted by clinicians that the complaints following minor head injury seem out of proportion when considered in the context of the frequency of complaints after serious head injury. It has also been noted that patients with early post-traumatic complaints generally improve with time, whereas the late development of symptoms is often associated with a more protracted and fulminant course.

Symptoms commonly considered part of this syndrome include the following (with headache, dizziness and memory difficulties being the most frequent):
1. somatic
   a) headache
   b) dizziness or light-headedness
   c) visual disturbances: blurring is a common complaint
   d) anosmia
   e) hearing difficulties: tinnitus, reduced auditory acuity
   f) balance difficulties
2. cognitive
   a) difficulty concentrating
   b) dementia: more common with multiple brain injuries than with a single concussion (p.924)
      - loss of intellectual ability
      - memory problems: usually impairs short-term memory more than long-term
   c) impaired judgement
3. psychosocial
   a) emotional difficulties: including depression, mood swings (emotional lability), euphoria/giddiness, easy irritability, lack of motivation, abulia
   b) personality changes
   c) loss of libido
   d) disruption of sleep/wake cycles, insomnia
   e) easy fatigability
   f) intolerance to light (photophobia) and/or loud (or even moderate) noise
   g) increased rate of job loss and divorce (may be related to any of above)

Virtually any symptom can be ascribed to the condition. Other symptoms that may be described by patients which are generally not included in the definition:
1. fainting (vaso-vagal episodes): may need to rule out posttraumatic seizures, as well as other causes of syncope

2. altered sense of taste
3. dystonia[33]

Treatment

Treatment for symptoms attributed to this syndrome tends to be more supportive and reassuring than anything else. Often times these patients obtain treatment from primary care physicians, neurologists, physiatrists, and/or psychiatrists/psychologists. Neurosurgical involvement in the continuing care for these patients is usually at the discretion of the individual physician based on his or her practice patterns. Recovery follows a highly variable course.

Early follow-up care (consisting mainly of reassurance, providing information and neuropsychological assessment and counselling) was found to reduce post-concussion symptoms at 6 months in patients with post-traumatic amnesia lasting ≥ 1 hour or those who required hospitalization, but had no benefit in those not requiring hospitalization or having amnesia < 1 hr.[34]

Some symptoms may need to be evaluated for possible correctable late complications (seizures, hydrocephalus, CSF leak…). Alves and Jane[35] perform a head CT, MRI, BAER and neuropsychological battery if symptoms after minor head injury persist > 3 months. An EEG may be appropriate in cases where there is a question of seizures. If all studies are negative, "the authors tell the patient (and the lawyer) that there is no objective evidence for disease and that psychiatric evaluation is warranted." Non-correctable abnormalities on these studies prompt reassurance that significant symptoms should subside by 1 year, and that no specific treatment, other than psychological counselling, is helpful.

61.6.3 Chronic traumatic encephalopathy

General information

Often described in retired boxers, chronic traumatic encephalopathy (CTE) encompasses a spectrum of symptoms that range from mild to a severe form AKA dementia pugilistica,[36] or punch drunk syndrome (among others). Symptoms involve motor, cognitive and psychiatric systems. CTE is distinct from post-traumatic dementia (which may follow a single closed head injury) or from post-traumatic Alzheimer's syndrome. Although generally accepted, not all authorities agree that repeated concussions have any long-term sequelae.[37]

There are some similarities with Alzheimer's disease (AD), including the presence of neurofibrillary tangles having similar microscopic characteristics (the main difference is that they tend to be more superficial in CTE than in AD[38]) and the development of amyloid angiopathy with the attendant risk of intracerebral hemorrhage.[39] EEG changes occur in one-third to one-half of professional boxers (diffuse slowing or low-voltage records).

Neuropathology

Findings include:
1. cerebral and cerebellar atrophy
2. neurofibrillary degeneration of cortical and subcortical areas
3. deposition of β-amyloid protein
   a) forming diffuse amyloid plaques
   b) in a subset of CTE patients this involves the vessel walls giving rise to cerebral amyloid angiopathy

Clinical

Clinical features of CTE are shown in Table 61.2 [36] and include[36]
1. cognitive: mental slowing and memory deficits (dementia)
2. personality changes: explosive behavior, morbid jealousy, pathological intoxication with alcohol, and paranoia
3. motor: cerebellar dysfunction, symptoms of Parkinson's disease, pyramidal tract dysfunction

Grading scales have been devised to rank patients as having probable, possible, and improbable CTE.

The chronic brain injury scale (CBIS) assesses involvement of motor, cognitive, and psychological axes as shown in Table 61.3.

Table 61.2 CTE of boxing[a]

Motor	Cognitive	Psychiatric
Early (≈ 57%)		
dysarthria tremors mild incoordination especially non-dominant hand	decreased complex attention	emotional lability euphoria/hypomania irritability, suspiciousness ease of aggression & talkativeness
Middle (≈ 17%)		
parkinsonism increased dysarthria, tremors, and incoordination	slowed mental speed mild deficits in memory, attention & executive ability	magnified personality decreased spontaneity paranoid, jealous inappropriate violent outbursts
Late (<3%)		
pyramidal signs prominent parkinsonism prominent dysarthria, tremors & ataxia	prominent slowness of thought/speech amnesia attention deficits executive dysfunction	cheerful/silly decreased insight paranoid, psychotic disinhibited, violent possible Klüver-Bucy

[a]in professional boxers with ≥20 bouts

Table 61.3 Chronic brain injury scale

Grade involvement of each of the following axes separately:	Scoring for each axis:
• motor • cognitive • psychological	• 0 = none • 1 = mild • 2 = moderate • 3 = severe
Sum total points	Severity
0	normal
1 – 2	mild
3 – 4	moderate
>4	severe

Risk factors for dementia pugilistica in boxing:

See reference.[36]

- risk increases with length of boxing career, especially > 10 yrs
- age at retirement: risk goes up after age 28 yrs
- number of bouts: especially ≥ 20 (more important than the number of knock-outs)
- boxing style: increased risk among poorer performers, those known as sluggers rather than “scientific” boxers, those known to be hard to knock out or known to take a punch and keep going
- age at examination: long latency causes increased prevalence with age
- and possibly, the number of head blows
- risk increases in patients with the apolipoprotein E (apo E) ε4 allele (as in Alzheimer’s disease) as shown in Table 61.4
- professional boxers (more risk than amateurs)

61

Table 61.4 Odds ratio for developing Alzheimer's disease

Head injury	Apo E ε4 allele	Odds ratio
–	–	1
–	+	2
+	–	1
+	+	10

Neuro-imaging

The most common finding is cerebral atrophy. A cavum septum pellucidum (CSP) is observed in 13% of boxers.[40] CSP in this setting probably represents an acquired condition[41] and correlates with cerebral atrophy.

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Part XV

Spine Trauma

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XV

62 General Information, Neurologic Assessment, Whiplash and Sports-Related Injuries, Pediatric Spine Injuries

62.1 Introduction

20% of patients with a major spine injury will have a second spinal injury at another level, which may be noncontiguous. These patients often have simultaneous but unrelated injuries (e.g. chest trauma, TBI…). Injuries directly associated with spinal cord injuries include arterial dissections (carotid and/or vertebral arteries).

62.2 Terminology

62.2.1 Spinal stability

Many definitions have been proposed. A conceptual definition of clinical stability from White and Panjabi[1]: the ability of the spine under physiologic loads to limit displacement so as to prevent injury or irritation of the spinal cord and nerve roots (including cauda equina) and, to prevent incapacitating deformity or pain due to structural changes.

Biomechanical stability refers to the ability of the spine ex vivo to resist forces.

Predicting spinal stability is often difficult, and to this end various models have been developed, none of which is perfect. See models of stability for cervical spine injuries (p.987), and thoracolumbar fractures (p.1002).

62.2.2 Level of injury

There is disagreement over what should be defined as "the level" of a spinal cord injury. Some define the "level" of a spinal cord injury as the lowest level of completely normal function (thus a patient would be termed a C5 quadriplegic even with minor C6 motor function). However, most sources define the "level" as the most caudal segment with motor function that is at least 3 out of 5 and if pain and temperature sensation is present.

62.2.3 Completeness of lesion

Categorization is important for treatment decisions and prognostication.

Incomplete lesion

Definition: any residual motor or sensory function more than 3 segments below the level of the injury.[2] Look for signs of preserved long-tract function.

Signs of incomplete lesion:

1. sensation (including position sense) or voluntary movement in the LEs in the presence of a cervical or thoracic spinal cord injury
2. "sacral sparing": preserved sensation around the anus, voluntary rectal sphincter contraction, or voluntary toe flexion
3. an injury does not qualify as incomplete with preserved sacral reflexes alone (e.g. bulbocavernosus)

Types of incomplete lesion:

1. central cord syndrome (p.944)
2. Brown-Séquard syndrome (cord hemisection) (p.947)
3. anterior cord syndrome (p.946)
4. posterior cord syndrome (p.947): rare

Complete lesion

No preservation of any motor and/or sensory function more than 3 segments below the level of the injury in the absence of spinal shock. About 3% of patients with complete injuries on initial exam will develop some recovery within 24 hours. Recovery is essentially zero if the spinal cord injury remains complete beyond 72 hours.

Spinal shock

This term is often used in two completely different senses:

1. hypotension (shock) that follows spinal cord injury (SBP usually ≈ 80 mm Hg). See Hypotension (p. 950) for treatment. Caused by multiple factors:
   a) interruption of sympathetics: implies spinal cord injury above T1
      - loss of vascular tone (vasoconstrictors) below level of injury
      - leaves parasympathetics relatively unopposed causing bradycardia
   b) loss of muscle tone due to skeletal muscle paralysis below level of injury results in venous pooling and thus a relative hypovolemia
   c) blood loss from associated wounds → true hypovolemia
2. transient loss of all neurologic function (including segmental and polysynaptic reflex activity and autonomic function) below the level of the SCI[3,4] → flaccid paralysis and areflexia
   a) Duration: may abate in as little as 72 hours, but typically persists 1–2 weeks, occasionally several months
   b) Accompanied by loss of the bulbocavernosus reflex
   c) Spinal cord reflexes immediately above the injury may also be depressed on the basis of the Schiff-Sherrington phenomenon
   d) When spinal shock resolves, there will be spasticity below the level of the lesion and return of the bulbocavernosus reflex
   e) A poor prognostic sign

62.3 Whiplash-associated disorders

62.3.1 General information

"Whiplash" was initially a lay term, which is currently defined as a traumatic injury to the soft tissue structures in the region of the cervical spine (including: cervical muscles, ligaments, intervertebral discs, facet joints…) due to hyperflexion, hyperextension, or rotational injury to the neck in the absence of fractures, dislocations, or intervertebral disc herniation.[5] It is the most common non-fatal automobile injury.[6] Symptoms may start immediately, but more commonly are delayed several hours or days. In addition to symptoms related to the cervical spine, common associated complaints include headaches, cognitive impairment, and low back pain.

62.3.2 Clinical grading

A proposed clinical classification system of WAD is shown in Table 62.1.[7]

62.3.3 Evaluation and treatment

A consensus[8] regarding diagnosis and management of these injuries is shown in Table 62.2 and Table 62.3. Keep in mind that conditions such as occipital neuralgia may occasionally follow whiplash type injuries and should be treated appropriately (Table 62.3).

Table 62.1 Clinical grading of WAD severity

Grade		Description
	0	no complaints, no signs[a]
Whiplash	1	neck pain or stiffness or tenderness, no signs
Whiplash	2	above symptoms with reduced range of motion or point tenderness
Whiplash	3	above symptoms with weakness, sensory deficit, or absent deep tendon reflexes
	4	above symptoms with fracture or dislocation[a]

[a]the definition of whiplash excludes these patients[5]

Table 62.2 Evaluation of WAD

Grade 1 patients with normal mental status and physical exam do not require plain radiographs on presentation
Grade 2 & 3 patients: C-spine x-rays, possibly with flexion-extension views. Special imaging studies (MRI, CT, myelography…) are not indicated
Grade 3 & 4: these patients should be managed as suspected spinal cord injury; see Initial management of spinal cord injury (p. 949), and sections that follow

Table 62.3 Treatment of WAD[8a]

Whiplash is usually a benign condition requiring little treatment and usually resolves in days to a few weeks in most cases.

Recommendation	Grade		
	1	2	3
Range of motion exercises	should be started immediately for all		
Encourage early return to regular activities	immediately	ASAP	
Cervical collars and rest[b]	no	not for > 72 hrs	not for > 96 hrs
Passive modality therapies: heat, ice, massage, TENS, ultrasound, relaxation techniques, acupuncture, and work alteration	no	optional if symptoms last > 3 wks	
Medications: optional use of NSAIDs and non-narcotic analgesics? (recommended for ≤ 3 wks)	no	yes	yes. Limited narcotics may also occasionally be needed
Surgery	no	no	only for progressive neurologic deficit or persisting arm pain

Not recommended: cervical pillows and soft collars, bed rest, spray and stretch exercises, muscle relaxant medication, TENS, reflexology, magnetic necklaces, herbal remedies, homeopathy, OTC medications (except NSAIDs, see above), and intra-articular, intrathecal, or trigger point steroid injections

[a]excluding patients with fractures, dislocations, or spinal cord injuries
[b]soft foam collars are generally discouraged; if they are to be used, the narrow part should be placed in front to avoid neck extension[5]

62.3.4 Outcome

In a study of 117 patients < 56 years of age having WAD due to automobile accidents (excluding those with cervical fractures, dislocations, or injuries elsewhere in the body) conducted in Switzerland[9] (where all medical costs were paid by the state and there was no opportunity for litigation and no compensation for pain and suffering, although there was the possibility of permanent disability), the recovery rate was as shown in Table 62.4. Of the 21 patients with continued symptoms at 2 yrs, only 5 were restricted with respect to work (3 reduced to part-time work, 2 on disability). Patients with persistent symptoms were older, had more varied complaints on initial exam, had a more rotated or inclined head position at the time of impact, had a higher incidence of pretraumatic headaches, and had a higher incidence of certain pre-existing findings (such as radiologic evidence of cervical osteoarthritis). The amount of damage to the automobile and the speed of the cars had little relationship to the degree of injury, and outcome was not influenced by gender, vocation, or psychological factors.

Table 62.4 Recovery of patients with WAD

Time (mos)	Percent recovered
3	56%
6	70%
12	76%
24	82%

62.4 Pediatric spine injuries

62.4.1 General information

Spinal cord injury is fairly uncommon in children, with the ratio of head injuries to spinal cord injuries being ≈ 30:1 in pediatrics. Only ≈ 5% of spinal cord injuries occur in children. Due to ligamentous laxity together with a high head to body weight ratio, immaturity of paraspinal muscles and the underdeveloped uncinate processes, these tend to involve ligamentous rather than bony injuries, see SCIWORA (p.999). There is also the potential for physeal (growth plate) separation in young children, which may have good potential to heal. The cervical spine is the most vulnerable segment (with subaxial injuries being fairly uncommon), with 42% of injuries occurring here, 31% thoracic, and 27% lumbar. The fatality rate is higher with pediatric spine injuries than with adults (opposite to the situation with head injury), with the cause of death more often related to other severe injuries than to the spinal injury.[10]

62.4.2 Evaluation

Practice guidelines for diagnostic workup is shown below (see Practice guideline: Evaluation of pediatric C-spine injuries (p.934)).

62.4.3 Pediatric cervical spine injuries and mimics

General information

See pediatric C-spine anatomy (p.214). In the age group ≤ 9 yrs, 67% of cervical spine injuries occur in the upper 3 segments of the cervical spine (occiput-C2).[11]

Synchondroses

Normal synchondroses (p.215) may be mistaken for fractures, especially the dentocentral synchondrosis of the atlas (p.214) which may be mistaken for an odontoid fracture. Conversely, actual fractures may occur through synchondroses.[12,13] Recommended treatment for fractures through synchondroses: the tendency for synchondroses to fuse suggests that emergency reduction followed by external immobilization be attempted. Internal immobilization/fusion should be reserved for persistent instability.[13]

Pseudospread of the atlas

See reference.[14]

Pseudospread of the atlas (defined as > 2 mm total overlap of the two C1 lateral masses on C2 on AP open-mouth view) is present in most children 3 mos to 4 yrs age. Prevalence is 91–100% during the second year of life. Youngest example at 3 mos, oldest at 5.75 yrs. Normal total offset is typically 2 mm during the first year, 4 mm during the second, 6 mm during the third, and decreasing thereafter. The maximum is 8 mm. Trauma is not a contributing factor.

Pseudospread is probably a result of disproportionate growth of the atlas on the axis. This could be misdiagnosed as a Jefferson fracture (p.971), which rarely occurs prior to the teen-ages (owing to lower weight of children, more flexible necks, increased plasticity of skull, and shock absorbing synchondroses of C1).

Neck rotation can also sometimes simulate the appearance of a Jefferson fracture.

When suspicion of fracture is high: CT scan through C1 can resolve the issue of whether or not there is a fracture.

Pseudosubluxation

Either anterior displacement of C2 (axis) on C3 and/or significant angulation at this level. Seen in children (up to age 10 yrs) on lateral C-spine x-ray after trauma. Up to age 10 yrs, flexion and extension are centered at C2–3; this moves down to C4–5 or C5–6 after age 10. C2 normally moves forward on C3 up to 2–3 mm in peds.[15] When the head is flexed, displacement is expected; may be exacerbated by spasm.[16] Does not represent pathological instability. Fractures and dislocations are unusual in children, and when they do occur, they resemble those in adults.

10 cases reported between ages 4–6 yrs[17]: pain was not uncommon. In each case, either the head or neck was flexed (sometimes minimally); the pseudosubluxation corrected when x-ray was repeated with head in true neutral position.

Recommendation: treat patient for soft-tissue injury and not for subluxation.

Treatment

Practice guideline: Evaluation of pediatric C-spine injuries

Level I[18]
- Use CT to assess the condyle-C1 interval (CCI) for pediatric patients with potential atlanto-occipital dislocation (AOD)

Level II[18]:
- Do not perform C-spine imaging in children > 3 years age with trauma who are:
  - alert
  - neurologically intact
  - without posterior midline cervical tenderness (with no distracting pain)
  - not hypotensive without explanation
  - not intoxicated
- Do not perform C-spine imaging in children < 3 years age with trauma who meet all of the following conditions:
  - have a GCS > 13
  - are neurologically intact
  - have no cervical midline tenderness (without distracting injury)
  - are not intoxicated
  - do not have unexplained hypotension
  - were not in a motor vehicle collision, a fall > 10 feet, or non-accidental trauma (NAT) as the known or suspected mechanism of injury
- Obtain cervical spine x-rays or high-resolution cervical CT in pediatric trauma victims who do not meet either set of criteria above
- Obtain 3-position CT with C1-2 motion analysis to confirm and classify the diagnosis for children suspected of having atlantoaxial rotatory fixation (AARF)

Level III[18]:
- children < 8 yrs age: when restrained, immobilize with thoracic elevation or an occipital recess (allows more neutral alignment due to the relatively large head)
- children < 7 yrs age with injuries of the C2 synchondrosis (p. 215): closed reduction and halo immobilization
- patients with AARF:
  - acute AARF (< 4 weeks duration) that does not reduce spontaneously: reduction with manipulation or halter traction
  - chronic AARF (> 4 weeks duration): reduction with halter or tong/halo traction
  - recurrent or irreducible AARF: internal fixation and fusion
- for isolated cervical spine ligamentous injuries and unstable or irreducible fractures of dislocations with associated deformity: consider primary operative treatment
- for cervical spine injuries that fail non-operative management: operative treatment

Level III[19]:
- children < 8 yrs age: immobilize with thoracic elevation or an occipital recess (allows more neutral alignment due to the relatively large head)

- children < 7 yrs age with injuries of the C2 dentocentral synchondrosis (p. 215): closed reduction and halo immobilization
- consider: primary operative treatment for isolated C-spine ligamentous injuries with associated deformity

62.5 Cervical bracing

62.5.1 Soft collars

Soft (sponge rubber) collar: does not immobilize the cervical spine to any significant degree. Its function is primarily to remind the patient to reduce neck movements.

62.5.2 Rigid cervical collars

Inadequate for stabilizing upper and mid-cervical spine and for preventing rotation.

Common rigid collars:

- Miami J collar & Aspen collar: have removable pads
- Philadelphia collar: no removable pads. Feels hotter to wear

62.5.3 Poster braces

Distinguished from cervicothoracic orthoses (see below) by the lack of straps under the axilla. Includes the four poster brace. Generally good for preventing flexion at midcervical levels.

62.5.4 Cervicothoracic orthoses

Cervicothoracic orthoses (CTO) incorporate some form of body vest to immobilize the cervical spine. The following are presented in increasing degree of immobilization.

Guilford brace: essentially a ring around the occiput and chin connected by two posts to anterior and posterior thoracic pads.

SOMI brace: acronym for Sternal Occipital Mandibular Immobilizer. Good for bracing against flexion (especially upper cervical spine). Inadequate for hyper-extension type injuries because of weak occipital support. Has special forehead attachment to allow patient to eat comfortably without mandibular support.

“Yale brace”: a sort of extended Philadelphia collar. The most effective CTO for bracing against flexion-extension and rotation. Major shortcoming is poor prevention of lateral bending (only ≈ 50% reduced).

62.5.5 Halo-vest brace

Can immobilize the upper or lower cervical spine, not very good for mid-cervical spine (due to snaking of the midcervical spine). Unable to provide adequate distraction support following vertebral body resection when patient assumes upright position (i.e. it is not a portable cervical traction device). Overall reduction of flexion/extension as well as lateral bending is ≈ 90–95%, rotation is reduced by 98% See placement (p. 958).

62.6 Follow-up schedule

After initial management (surgical or nonsurgical) of cervical spine problems (stable or unstable) the follow-up schedule shown in Table 62.5 is suggested to permit recognition of problems in time for treatment[1] (start with 3 weeks and keep doubling the interval to 1 year).

62.7 Sports-related cervical spine injuries

62.7.1 General information

Any of the spine injuries described in this book can be sports-related. This section considers some injuries peculiar to sports.

Table 62.5 Sample follow-up cervical spine clinic visit schedule

Time post-op	Agenda
7–10 d	(for post-op patients only) wound check, D/C sutures/staples if used
4-6 weeks	AP & lateral C-spine x-ray in brace
10-12 weeks	• AP & lateral C-spine x-rays with flexion/extension views out of brace • if x-rays look good and patient is doing well, begin weaning brace
6 months	• AP & lateral C-spine x-rays with flexion/extension views • some surgeons release patients at this time if they are doing well
1 year (optional)	• AP & lateral C-spine x-rays with flexion/extension views • release patient if they are doing well

Table 62.6 Sports-related spinal cord injuries

Type	Description
I	permanent SCI
II	transient SCI without radiographic abnormality
III	radiologic abnormality without neurologic deficit

Bailes et al.[20] classified sports-related spinal cord injuries (SCI) as shown in Table 62.6. Type I injures may be complete or may have features of any of the incomplete SCI syndromes (often in mixed or partial forms). Type II injuries include spinal concussion, spinal neuropraxia (see below), and the burning hands syndrome (see below), all in the absence of radiographic abnormalities and all with complete resolution of symptoms. Patients should be carefully evaluated, and return to competition should not be allowed in the presence of neurologic deficit, radiographically demonstrated injury, certain congenital C-spine abnormalities, and possibly for "repeat offenders" (p.937). Type III injuries are the most common. Unstable injuries should be treated appropriately (p.997).

62.7.2 Football-related cervical spine injuries

General information

Football players with suspected C-spine injury should not have their helmet removed in the field (p.949).

Terminology

The following terms probably originated as locker-room jargon for various cervical spine-related injuries usually sustained in playing football. Medical definitions have subsequently been retro-fitted to them. As a result, the precise definitions may not be uniformly agreed upon. Although the semantics may differ, it is more important from a diagnostic and therapeutic standpoint to distinguish nerve root injuries, brachial plexus injuries, and spinal cord injuries.

1. cervical cord neuropraxia[21] (CCN): sensory changes which may involve numbness, tingling or burning. May or may not be associated with motor symptoms of weakness or complete paralysis. Typically lasts < 15 mins (although may persist up to 48 hrs), involves all 4 extremities in 80% of cases. Narrowing of the sagittal diameter of the cervical spinal canal is felt to be a contributory factor. With resumption of contact activities, recurrence rate is ≈ 56%, with higher risks of recurrence among those with narrower canal diameters. Evaluation should include cervical MRI. Torg[21] feels that uncomplicated cases of CCN (no spinal instability and no MRI evidence of cord defect or edema) have a low risk of permanent injury and does not recommend activity restrictions
2. "stinger" or "burner": distinct from the burning hands syndrome. Unilateral. burning dysesthetic pain radiating down one arm from the shoulder, sometimes associated with weakness involving the C5 or C6 nerve roots. Usually follows a tackle. May result from downward traction on the upper trunk of the brachial plexus (when the shoulder is forcefully depressed with the neck

flexed to the contralateral side) or by direct nerve root compression in the neural foramina (not a SCI)
3. burning hands syndrome[22]: similar to a stinger, but bilateral. Probably represents a SCI; possibly a mild variant of a central cord syndrome (p.944)
4. other neurologic injuries include: vascular injury to carotid or vertebral arteries. Usually related to intimal dissection (p.1018) following a direct blow to the neck or by extreme movements. Symptoms are those of a TIA or stroke

Spear tackler's spine

Rule changes in 1976 banned spearing (the practice of using the football helmet as a battering ram to tackle an opponent) and resulted in a reduction of the number of football-related occurrences of cervical spine fractures and quadriplegia.[23]

Four characteristics of spear tackler's spine:
1. cervical spinal stenosis
2. loss of normal cervical lordosis: as a result, the stress of axial loading is more likely to be imparted to the vertebral bodies, rather than being absorbed by the cervical musculature and ligaments, increasing the risk of burst fractures and quadriplegia
3. evidence of pre-existing traumatic abnormalities
4. documented spear-tackler's technique

Suggested management:

The athlete is removed from competition until the cervical lordosis returns and the player learns to use other tackling techniques. This tackling technique has been banned since 1976.

62.7.3 Return to play and pre-participation guidelines

Return to play (RTP) and pre-participation evaluation guidelines related to the cervical spine are shown in Table 62.7 (modified[24]). These are just guidelines, and do not insure safety. Clinical judgement must always be employed.

Table 62.7 C-spine-related contraindications for participation in contact sports[a]

Condition[b]			C.I.[c]
Congenital[d]			
1.	odontoid abnormalities (serious injury may result from atlanto-axial instability)		
	a.	complete aplasia (rare)	absolute
	b.	hypoplasia (seen in conjunction with achondroplasia and spondyloepiphyseal dysplasia)	absolute
	c.	os odontoideum (probably of traumatic origin)	absolute
2.	atlanto-occipital fusion (partial or complete fusion of atlas to occiput): sudden onset of symptoms & sudden death have been reported		absolute
3.	Klippel-Feil anomaly (congenital fusion of 2 or more cervical vertebrae)[e]		
	a.	Type I: mass fusion of C-spine to upper T-spine	absolute
	b.	Type II: fusion of only 1 or 2 interspaces	
		• associated with limited ROM, occipitocervical anomalies, instability, disc disease or degenerative changes	absolute
		• associated with full ROM and none of the above	none
Acquired			
1.	cervical spinal stenosis[f]		
	a.	asymptomatic	none
	b.	with one episode of cord neuropraxia	relative

Table 62.7 continued

Condition[b]			C.I.[c]
	c.	cord neuropraxia + MRI evidence of cord defect or edema	absolute
	d.	cord neuropraxia + ligamentous instability, symptoms or neurologic findings > 36 hrs, or multiple episodes	absolute
2.	spear tackler's spine (see text)		absolute
3.	spina bifida occulta: rare, incidental x-ray finding		none
Post-traumatic upper cervical spine			
1.	atlantoaxial instability (ADI > 3 mm adults, > 4 mm peds)		absolute
2.	atlantoaxial rotatory fixation (may be associated with disruption of transverse ligament)		absolute
3.	fractures		
	a.	healed, pain-free, full ROM, & no neurologic findings with any of the following fractures: nondisplaced Jefferson fracture; odontoid fracture; or lateral mass fracture of axis	none
	b.	all others	absolute
4.	post-surgical atlantoaxial fusion		absolute
Post-traumatic subaxial cervical spine			
1.	ligamentous injuries: > 3.5 mm subluxation, or > 11° angulation on flexion-extension views		absolute
2.	fractures		
	a.	healed, stable fractures listed here with normal exam: VB compression fracture without posterior involvement; spinous process fractures	none
	b.	VB fractures with sagittal component or posterior bony or ligamentous involvement	absolute
	c.	comminuted fracture with displacement into spinal canal	absolute
	d.	lateral mass fracture producing facet incongruity	absolute
3.	intervertebral disc injury		
	a.	healed herniated disc treated conservatively	none
	b.	S/P ACDF with solid fusion, no symptoms, normal exam and full pain-free ROM	none
	c.	chronic herniated disc with pain, neuro findings or ↓ ROM, or acute herniated disc	absolute
4.	S/P fusion		
	a.	stable one-level fusion	none
	b.	stable two-level fusion	relative
	c.	fusion > 2 levels	absolute

[a]organized contact sports includes[24]: boxing, football, ice hockey, lacrosse, rugby & wrestling
[b] see also cranial-related (and craniocervical) conditions (p. 1151) (e.g. Chiari I malformation…)
[c]C.I. = contraindications, classified as absolute, relative (i.e. uncertain) or none
[d]congenital abnormalities may have particular relevance to Special Olympics
[e]NB: Klippel-Feil may be associated with abnormalities in other organ systems (e.g. cardiac) which may impact on participation in contact sports (p. 271)
[f] Pavlov ratio (p. 1088) has a low positive predictive value for injuries in contact sports and is therefore not a useful screening test (i.e. an asymptomatic Pavlov ratio < 0.8 is not a contraindication to participation)

62.8 Neurological assessment

62.8.1 General information

Evaluation of the level of the lesion requires familiarity with the following concepts about the relationship between the bony spinal canal and the spinal cord and nerves (Fig. 62.1).

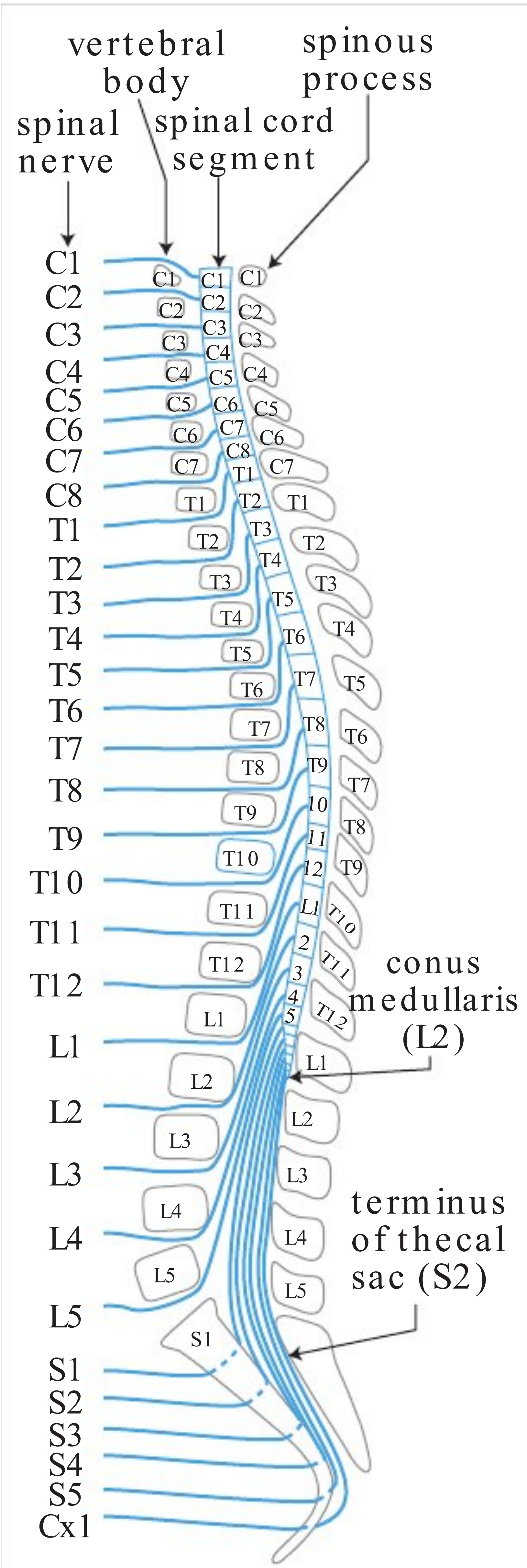

Fig. 62.1 Relationship between spinal cord, nerve roots, and bony spine

1. since there are 8 pairs of cervical nerves and only 7 cervical vertebra
   a) cervical nerves 1 through 8 exit above the pedicles of their like-numbered vertebra
   b) thoracic, lumbar and sacral nerves exit below the pedicles of their like-numbered vertebra
2. due to disproportionately greater growth of the spinal column than the spinal cord during development, the following relationships of the spinal cord to the vertebral column exist:
   a) to determine which segment of the cord underlies a given vertebra:
      • from T2 through T10: add 2 to the number of the spinous process
      • for T11, T12 and L1, remember that these overlie the 11 lowest spinal segments (L1 through L5, S1 through S5, and Coxygeal-1)
   b) the conus medullaris in the adult lies at about L1 or L2 of the spine

62.8.2 Motor level assessment

General information

The following tables are for rapid assessment (see Table 29.5 and Table 29.7 for detailed tables of motor innervation).

ASIA (American Spinal Injury Association) motor scoring system

A system[25,26] that may be rapidly applied to grade 10 key motor segments using the MRC Grading Scale (Table 29.2) from 0–5 on the left and the right, for a total score of 100 possible points (see Table 62.8). NB: most muscles receive innervation from two adjacent spinal levels, the levels listed in Table 62.8 are the lower of the two. The standard considers a segment intact if the motor grade is fair (≥3). For additional information, see www.asia-spinalinjury.org.

Table 62.8 Key muscles for motor level classification (EXTREMITIES)

RIGHT grade	Segment	Muscle	Action to test	LEFT grade
0–5	C5	biceps	flex elbow	0–5
0–5	C6	wrist extensors	cock up wrist	0–5
0–5	C7	triceps	extend elbow	0–5
0–5	C8	flexor digitorum profundus	flex middle distal phalanx	0–5
0–5	T1	hand intrinsics	abduct little finger	0–5
0–5	L2	iliopsoas	flex hip	0–5
0–5	L3	quadriceps	straighten knee	0–5
0–5	L4	tibialis anterior	dorsiflex foot	0–5
0–5	L5	EHL	dorsiflex big toe	0–5
0–5	S1	gastrocnemius	plantarflex foot	0–5
50	← TOTAL POSSIBLE POINTS →			50
GRAND TOTAL: 100				

Table 62.9 Axial muscle evaluation[27]

Level	Muscle	Action to test
C4	diaphragm	tidal volume (TV), FEV1, and vital capacity (VC)
T2–9 T9–10 T11–12	intercostals upper abdominals lower abdominals	use sensory level, abdominal reflexes, & Beevor's sign

More detailed motor evaluation

Table 62.10 Skeletal muscles and their major spinal innervation (major contributing segment is shown in boldface)

Segment	Muscle	Action to test	Reflex
C1–4	neck muscles		
C3, 4, 5	diaphragm	inspiration, TV, FEV1, VC	
C5, 6	deltoid	abduct arm > 90°	
C5, 6	biceps	elbow flexion	biceps
C6, 7	extensor carpi radialis	wrist extension	supinator
C7, 8	triceps, extensor digitorum	elbow and finger extension	triceps
C8, T1	flexor digitorum profundus	grasp (flex distal phalanges)	
C8, T1	hand intrinsics	abduct little finger, adduct thumb	
T2–9	intercostals[a]		
T9,10	upper abdominals[a]	Beevor's sign[b]	abdominal cutaneous reflex[c]
T11,12	lower abdominals[a]		
L2, 3	iliopsoas, adductors	hip flexion	cremasteric reflex[d]
L3, 4	quadriceps	knee extension	infrapatellar (knee jerk)
L4, 5	medial hamstrings, tibialis anterior	ankle dorsiflexion	medial hamstrings
L5, S1	lateral hamstrings, posterior tibialis, peroneals	knee flexion	
L5, S1	extensor digitorum, EHL	great toe extension	
S1, 2	gastrocs, soleus	ankle plantarflexion	achilles (ankle jerk)
S2, 3	flex digitorum, flex hallucis		
S2, 3, 4	bladder, lower bowel, anal sphincter	clamp down during rectal exam	anal cutaneous reflex[e], bulbocavernosus & priapism

[a]also use sensory level to help evaluate these segments
[b]Beevor's sign: used to assess abdominal musculature for level of lesion. Patient lifts head off of bed by flexing neck; if lower abdominal muscles (below ≈ T9) are weaker than upper abdominal musculature, then umbilicus moves cephalad. Not helpful if both upper and lower abdominals are weak
[c]the abdominal cutaneous reflex: scratching one quadrant of abdomen with sharp object causes contraction of underlying abdominal musculature, causing umbilicus to migrate toward that quadrant. Upper abdominal reflex: T8–9. Lower abdominal reflex: T10–12. This is a cortical reflex (i.e. reflex loop ascends to cortex, and then descends to abdominal muscles). The presence of this response indicates an incomplete lesion for cord injuries above the lower thoracic level
[d]cremasteric reflex: L1–2 superficial reflex
[e]anal-cutaneous reflex: AKA anal wink. Normal reflex: mild noxious stimulus (e.g. pinprick) applied to skin in region of anus results in involuntary anal contraction. bulbocavernosus (BC) reflex: see section 62.8.5

Table 62.11 Key sensory landmarks

Level	Dermatome
C2	occipital protuberance
C3	supraclavicular fossa
C4	top of acromioclavicular joint
C5	Lateral side of antecubital fossa
C6	thumb, dorsal surface, proximal phalanx
C7	middle finger, dorsal surface, proximal phalanx
C8	little finger, dorsal surface, proximal phalanx
T1	medial (ulnar) side of antecubital fossa
T2	apex of axilla
T3	third intercostal space (IS)
T4	fourth IS (nipple line)
T5	fifth IS (midway between T6 & T8)
T6	sixth IS (xiphoid process)
T7	seventh IS (midway between T6 & T8)
T8	eighth IS (midway between T6 & T10)
T9	ninth IS (midway between T8 & T10)
T10	tenth IS (umbilicus)
T11	eleventh IS (midway between T10 & T12)
T12	inguinal ligament at mid-point
L1	half the distance between T12 & L2
L2	mid-anterior thigh
L3	medial femoral condyle
L4	medial malleolus
L5	dorsum of foot at 3rd MTP (metatarsal phalangeal) joint
S1	lateral heel
S2	popliteal fossa in the mid-line
S3	ischial tuberosity
S4–5	perianal area (taken as 1 level)

62.8.3 Sensory level assessment (dermatomes and sensory nerves)

ASIA standards[25]

28 key points identified in Table 62.11 are scored separately for pinprick and light touch on the left & right side using the grading scale shown in Table 62.12, for a maximum possible total of 112 points for pinprick (left & right) and 112 points for light touch (left & right).

Table 62.12 Sensory grading scale

Grade	Description
0	absent
1	impaired (partial or altered appreciation)
2	normal
NT	not testable

NB: regarding the "C4 cape" AKA "bib" region across the upper chest and back: sensory segments "jump" from C4 to T2 with the intervening levels distributed exclusively on the UEs (Fig. 1.14). The location of this transition is not constant from person to person.

62.8.4 Rectal exam

1. external anal sphincter is tested by insertion of the examiner's gloved finger
   a) perceived sensation is recorded as present or absent. Any sensation felt by the patient indicates that the injury is sensory incomplete
   b) record resting sphincter tone and any voluntary sphincter contraction
2. bulbocavernosus (BC) reflex (p.941); see also below: Absence suggests the presence of spinal shock, and it may not be possible to declare a suprasacral SCI as complete because there might be spinal shock which could transiently suppress spinal cord function

62.8.5 Bulbocavernosus (BC) reflex

A polysynaptic spinal cord mediated reflex relayed via S2-S4 nerve roots. Contraction of anal sphincter in response to squeezing the glans penis in males, or to tugging on the Foley catheter in either sex is a normal response (must be differentiated from the movement of the Foley catheter balloon).

Loss of reflex can occur with:

1. spinal shock: the BC reflex may be lost with spinal shock as can occur with suprasacral injuries. Reportedly, the return of the BC reflex may be the earliest clinical indicator that spinal shock has subsided.
2. injuries involving the cauda equina or conus medullaris

Presence of BC reflex used to be taken as an indication of an incomplete injury, but its presence alone is no longer considered to have a good prognosis for recovery.

62.8.6 Additional sensory exam

The following elements are considered optional but it is recommended that they be graded as absent, impaired or normal:

1. position sense: test index finger and great toe on both sides
2. awareness of deep pressure/deep pain

62.8.7 ASIA impairment scale

The ASIA impairment scale*[25] is shown in Table 62.13 (a modified Frankel Neurological Performance scale[28]).

* NB: this scale indicates the completeness of spinal cord injury and is distinct from the other ASIA grading scales; see also motor and sensory scoring (p.940).

62.9 Spinal cord injuries

62.9.1 Complete spinal cord injuries

See definition of complete vs. incomplete spinal cord injury (p.930).

In addition to loss of voluntary movement, sphincter control and sensation below the level of the injury, there may be priapism. Hypotension and bradycardia (p.931) (spinal shock) may also present.

Class	Description
A	Complete: no motor or sensory function preserved
B	Incomplete: sensory but no motor function preserved below the neurologic level (includes sacral segments S4–5)
C	Incomplete: motor function preserved below the neurologic level (more than half of key muscles below the neurologic level have a muscle strength grade <3)[a]
D	Incomplete: motor function preserved below the neurologic level (more than half of key muscles below the neurologic level have a muscle strength grade ≥ 3)
E	Normal: Sensory & motor function normal

Table 62.13 ASIA impairment scale

[a]for muscle strength grading see Table 29.2

62.9.2 Bulbar-cervical dissociation

Occurs as a result of spinal cord injury at or above ≈ C3 (includes SCI from atlanto-occipital and atlantoaxial dislocation). Bulbar-cervical dissociation produces immediate pulmonary and, often, cardiac arrest. Death results if CPR is not instituted within minutes. Patients are usually quadriplegic and ventilator dependent (phrenic nerve stimulation may eventually allow independence from ventilator).

62.9.3 Incomplete spinal cord injuries

Central cord syndrome

General information

Key concepts

- disproportionately greater motor deficit in the upper extremities than lower
- usually results from hyperextension injury in the presence of osteophytic spurs
- surgery is often employed for ongoing compression, usually on a non-emergency basis except for rare cases of progressive deterioration

Originally described by Schneider et al.[29] in 1954. Central cord syndrome (CCS) is the most common type of incomplete spinal cord injury syndrome. Usually seen following acute hyperextension injury in an older patient with pre-existing acquired stenosis as a result of bony hypertrophy (anterior spurs) and infolding of redundant ligamentum flavum (posteriorly), sometimes superimposed on congenital spinal stenosis. Translational movement of one vertebra on another may also contribute. A blow to the upper face or forehead is often disclosed on history, or is suggested on exam (e.g. lacerations or abrasions to face and/or forehead). This often occurs in relation to a motor vehicle accident or to a forward fall, often while intoxicated. Younger patients may also sustain CCS in sporting injuries; see burning hands syndrome (p.1421). CCS may occur with or without cervical fracture or dislocation.[30] CCS may be associated with acute traumatic cervical disc herniation. CCS may also occur in rheumatoid arthritis.

Pathomechanics

Theory: the centermost region of the spinal cord is a vascular watershed zone which renders it more susceptible to injury from edema. Long tract fibers passing through the cervical spinal cord are somatotopically organized such that cervical fibers are located more medially than the fibers serving the lower extremities (Fig. 1.13).

Presentation

See reference.[29]

The clinical syndrome is somewhat similar to that seen in syringomyelia.

1. motor: weakness of upper extremities with lesser effect on lower extremities
2. sensory: varying degrees of disturbance below level of lesion may occur
3. myelopathic findings: sphincter dysfunction (usually urinary retention)

Hyperpathia to noxious and non-noxious stimuli is also common, especially in the proximal portions of the upper extremities, and is often delayed in onset and extremely distressing to the patient.[31] Lhermitte's sign occurs in ≈ 7% of cases.

Natural history

There is often an initial phase of improvement (characteristically: LEs recover first, bladder function next, UE strength then returns with finger movements last; sensory recovery has no pattern) followed by a plateau phase and then late deterioration.[32] 90% of patients are able to walk with assistance within 5 days.[33] Recovery is usually incomplete, and the amount of recovery is related to the severity of the injury and patient age.[34]

If CCS results from hematomyelia with cord destruction (instead of cord contusion), then there may be extension (upward or downward).

Evaluation

Findings: young patients tend to have disc protrusion, subluxation, dislocation or fractures.[33] Older patients tend to have multi-segmental canal narrowing due to osteophytic bars, discs, and inbuckling of ligamentum flavum.[33]

C-spine x-rays: may demonstrate congenital narrowing, superimposed osteophytic spurs, traumatic fracture/dislocation. Occasionally, AP narrowing alone without spurs may be seen.[30] Plain x-rays will fail to demonstrate canal narrowing due to: thickening or inbuckling of ligamentum flavum, hypertrophy of facet joints, and poorly calcified spurs.[30]

Cervical CT scan: also helpful in diagnosing fractures and osteophytic spurs. Not as good as MRI for assessing status of discs, spinal cord and nerves.

MRI: discloses compromise of anterior spinal canal by discs or osteophytes (when combined with plain C-spine x-rays, it increases the ability to differentiate osteophyte from traumatic disc herniation). Also good for evaluating ligamentum flavum. T2WI may show spinal cord edema acutely,[35] and can detect hematomyelia. MRI is poor for identifying fractures.

Treatment

The indications, timing and best treatment method for CCS remains controversial.

Practice guideline: Acute traumatic central cord injuries (ATCCS)

Level III[34,36]

- ICU management of patients with acute traumatic central cord syndrome, especially for those with severe neurologic deficits (because of possible cardiac, pulmonary & BP disturbances)
- medical management to include the following: cardiac, hemodynamic and respiratory monitoring and maintainence of MAP 85–90 mm Hg (use BP augmentationif necessary) for the 1st week after injury to improve spinal cord perfusion
- early reduction of fracture-dislocation injuries
- surgical decompression of the compressed spinal cord, particularly if the compression is focal and anterior. Unresolved: the role of surgery in ATCCS with long segment cord compression or with spinal stenosis without bony injury[36] (see text for details)

Indications for surgery

1. continued compression of the spinal cord[37 (p 1010)] that correlates with the level of deficit with any of the following:
   a) persistent significant motor deficit following a varying period of recovery (below)
   b) deterioration of function
   c) continued significant dysesthetic pain
2. instability of the spine

Improvement has been shown in short and long-term follow-up with subacute decompression of the offending lesion.[33] Nonsurgical treatment results in a longer period of pain and weakness in many cases.

Timing of surgery: A perennial point of controversy. Classic teaching was that early surgery for this condition is contraindicated because this may worsen the deficit. In the absence of spinal instability, traditional management consisted of bed rest in a soft collar for ≈ 3–4 weeks, with consideration for surgery after this time, or else gradual mobilization in the same collar for an additional 6 weeks. However, the basis for this recommendation was at least in part derived from an early report of only 8 patients with CCS, 2 of which underwent surgery, with 1 being worse post-op (the operation consisted of laminectomy, opening the dura, sectioning the dentate ligament, and manipulation of the spinal cord in order to inspect the anterior spinal canal).[29] It is presently felt that there is no solid evidence that early decompressive surgery (without cord manipulation) is actually harmful, but there is also no evidence that it is helpful, either. There may be good justification for early surgery in the rare patient who is improving and then deteriorates,[38] however, great restraint must be used in avoiding what would be an inappropriate operation in many patients.[39] Surgery may improve the rate and degree of recovery in selected patients.[40] Surgery has been recommended for patients with gross spinal instability or for patients with significant persistent cord compression (e.g. by osteophytic spurs) who fail to progress consistently after an initial period of improvement,[35] often within 2–3 weeks following the trauma. Better results occur with decompression within the first few weeks or months rather than very late (e.g. ≥ 1–2 years).[37 (p 1010)]

Σ

There is no role for surgery without ongoing compression or instability. The rare patient with ongoing compression undergoing documented progressive deterioration should be decompressed ASAP. Patients that are improving should be followed and decompression can be done electively for ongoing compression. There is controversy regarding timing of surgery for stable CCS and ongoing compression: while Class I or II data are lacking, there seems to be a trend to decompress these patients as soon as they are medically stable without an arbitrary waiting period.

Technical considerations: The most rapid procedure to decompress the cord is often a multi-level laminectomy. This is frequently accompanied by dorsal migration of the spinal cord which may be seen on MRI.[32] With myelopathy, fused patients fare better than those that are just decompressed without fusion. Fusion may be accomplished posteriorly (e.g. with lateral mass screws and rods) at the time of decompression, or anteriorly (e.g. multi-level discectomy, or corpectomy with strut graft and anterior cervical plating) at the same sitting as the laminectomy or staged at a later date.

Prognosis

In patients with cord contusion without hematomyelia, ≈ 50% will recover enough LE strength and sensation to ambulate independently, although typically with significant spasticity. Recovery of UE function is usually not as good, and fine motor control is usually poor. Bowel and bladder control often recovers, however bladder spasticity is common. Elderly patients with this condition generally do not fare as well as younger patients, with or without surgical treatment (only 41% over age 50 become ambulatory, versus 97% for younger patients[41]).

Anterior cord syndrome

General information

AKA anterior spinal artery syndrome. Cord infarction in the territory supplied by the anterior spinal artery. Some say this is more common than central cord syndrome.

May result from occlusion of the anterior spinal artery, or from anterior cord compression, e.g. by dislocated bone fragment, or by traumatic herniated disc.

Presentation

1. paraplegia, or (if higher than ≈ C7) quadriplegia
2. dissociated sensory loss below lesion:
   a) loss of pain and temperature sensation (spinothalamic tract lesion)
   b) preserved two-point discrimination, joint position sense, deep pressure sensation (posterior column function)[42]

Evaluation

It is vital to differentiate a non-surgical condition (e.g. anterior spinal artery occlusion) from a surgical one (e.g. anterior bone fragment). This requires one or more of: myelography, CT, or MRI.

Treatment

Surgical intervention is indicated for patients with evidence of cord compression (e.g. by large central disc herniation) or for spinal instability (ligamentous or bony).

Prognosis

The worst prognosis of the incomplete injuries. Only ≈ 10–20% recover functional motor control. Sensation may return enough to help prevent injuries (burns, decubitus ulcers…).

Brown-Séquard syndrome

General information

Spinal cord hemisection. First described in 1849 by Brown-Sequard.[43]

Etiologies

Usually a result of penetrating trauma, it is seen in 2–4% of traumatic spinal cord injuries.[44] Also may occur with radiation myelopathy, cord compression by spinal epidural hematoma, large cervical disc herniation[45,46,47] (rare), spinal cord tumors, spinal AVMs, cervical spondylosis, and spinal cord herniation (p. 1150).

Presentation

Classical findings (rarely found in this pure form):

1. ipsilateral findings:
   a) motor paralysis (due to corticospinal tract lesion) below lesion
   b) loss of posterior column function (proprioception & vibratory sense)
2. contralateral findings: dissociated sensory loss
   a) loss of pain and temperature sensation inferior to lesion beginning 1–2 segments below (spinothalamic tract lesion)
   b) preserved light (crude) touch due to redundant ipsilateral and contralateral paths (anterior spinothalamic tracts)

Prognosis

This syndrome has the best prognosis of any of the incomplete spinal cord injuries. ≈ 90% of patients with this condition will regain the ability to ambulate independently as well as anal and urinary sphincter control.

Posterior cord syndrome

AKA contusio cervicalis posterior. Relatively rare. Produces pain and paresthesias (often with a burning quality) in the neck, upper arms, and torso. There may be mild paresis of the UEs. Long tract findings are minimal.

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63 Management of Spinal Cord Injury

63.1 General information

The major causes of death in spinal cord injury (SCI) are aspiration and shock.[1] Initial survey under ATLS protocol: assessment of airway takes precedence, then breathing, then circulation & control of hemorrhage (“ABC’s”). This is followed by a brief neurologic exam.

NB: other injuries (e.g. abdominal injuries) may be masked below the level of SCI.

Any of the following patients should be treated as having a SCI until proven otherwise:

1. all victims of significant trauma
2. trauma patients with loss of consciousness
3. minor trauma victims with complaints referable to the spine (neck or back pain or tenderness) or spinal cord (numbness or tingling in an extremity, weakness, paralysis)
4. associated findings suggestive of SCI include
   a) abdominal breathing
   b) priapism (autonomic dysfunction)

Trauma patients are triaged as follows:

1. no history of significant trauma, completely alert, oriented and free of drug or alcohol intoxication with no complaints referable to the spine: most may be cleared clinically without the need for C-spine x-rays; see Radiographic evaluation (p.952)
2. significant trauma, but no strong evidence of spine or spinal cord injury: the emphasis here is in ruling-out a bony lesion and preventing injury
3. patients with neurologic deficit: the emphasis here is to define the skeletal injury and to take steps to prevent further cord injury and loss of function and minimize or reverse the present deficit. The pros and cons of the high-dose methylprednisolone protocol (p.951) should be weighed if a neurologic deficit is identified

63.2 Management in the field

1. Spine immobilization prior to and during extrication from vehicle and transport to prevent active or passive movements of the spine.
   a) For possible C-spine injuries in football players, see Table 63.1 for the National Athletic Trainers’ Association (NATA) guidelines for helmet removal. When CPR is necessary it takes precedence. Caution with intubation (see below)
   b) place patient on back-board

Table 63.1 NATA helmet removal guidelines[a]
NB: do not remove the helmet in the field.
• most injuries can be visualized with the helmet in place • neurological exam can be done with the helmet in place • the patient may be immobilized on a spine board with the helmet in place • the facemask can be removed with special tools to access the airway • hyperextension must be avoided following removal of the helmet and shoulder pads
In a controlled setting (usually after x-rays) the helmet and shoulder-pads are removed together as a unit to avoid neck flexion or extension
Possible indications for removal of helmet
• face mask cannot be removed in a reasonable amount of time • airway cannot be established even with face mask removed • life threatening hemorrhage under the helmet that can be controlled only by removal • helmet & strap do not hold head securely so that immobilizing the helmet does not adequately immobilize the spine (e.g. poor fitting or damaged helmet) • helmet prevents immobilization for transportation in an appropriate position • certain situations where the patient is unstable (M.D. decision)
[a]for more details, see http://www.nata.org

c) sandbags on both sides of the head with a 3 inch strip of adhesive tape from one side of the back-board to the other across the forehead immobilizes the spine as well as a rigid orthosis[2] but allows movement of the jaw and access to the airway
d) a rigid cervical collar (e.g. Philadelphia collar) may be used to supplement
2. maintain blood pressure, see below under Hypotension (p.950)
a) pressors treat the underlying problem (SCI is essentially a traumatic sympathectomy). Dopamine is the agent of choice, and is preferred over fluids (except as necessary to replace losses); see Cardiovascular agents for shock (p.127) for pressors. Avoid phenylephrine (see below)
b) fluids as necessary to replace losses
c) military anti-shock trousers (MAST): immobilizes lower spine, compensates for lost muscle tone in cord injuries (prevents venous pooling)
3. maintain oxygenation (adequate FIO_2 and adequate ventilation)
a) if no indication for intubation: use NC or face mask
b) intubation: may be required for airway compromise or for hypopnea. In SCI, hypopnea may be due to: paralyzed intercostal muscles, paralysis of diaphragm (phrenic nerve = C3, 4 & 5). Hypopnea may also be due to depressed LOC in TBI
c) caution with intubation with uncleared C-spine
- use chin lift (not jaw thrust) without neck extension
- nasotracheal intubation may avoid movement of C-spine but patient must have spontaneous respirations
- avoided tracheostomy or cricothyroidotomy if possible (may compromise later anterior cervical spine surgical approaches)
4. brief motor exam to identify possible deficits (also to document delayed deterioration); ask patient to:
a) move arms
b) move hands
c) move legs
d) move toes

63.3 Management in the hospital

63.3.1 Stabilization and initial evaluation

1. immobilization: maintain backboard/head-strap (see above) to facilitate transfers to CT table, etc. Log-roll patient to turn. Once studies are completed, remove patient from backboard ASAP (early removal from board reduces risk of decubitus ulcers)
2. hypotension (spinal shock): maintain SBP ≥ 90 mm Hg. Spinal cord injuries cause hypotension by a combination of factors (p.931) which may further injure spinal cord[3] or other organ systems
a) pressors if necessary: dopamine is agent of choice (avoid phenylephrine: non-inotropic and possible reflex increase in vagal tone → bradycardia)
b) careful hydration (abnormal hemodynamics → propensity to pulmonary edema)
c) atropine for bradycardia associated with hypotension
3. oxygenation (see above)
4. NG tube to suction: prevents vomiting and aspiration, and decompresses abdomen which can interfere with respirations if distended (paralytic ileus is common, and usually lasts several days)
5. indwelling (Foley) urinary catheter: for I's & O's and to prevent distension from urinary retention
6. DVT prophylaxis: see below
7. temperature regulation: vasomotor paralysis may produce poikilothermy (loss of temperature control), this should be treated as needed with cooling blankets
8. electrolytes: hypovolemia and hypotension cause increased plasma aldosterone which may lead to hypokalemia
9. more detailed neuro evaluation (p.939). Patients may be stratified using the ASIA impairment scale (Table 62.13)
a) focused history: key questions should center on:
- mechanism of injury (hyperflexion, extension, axial loading…)
- history suggestive of loss of consciousness
- history of weakness in the arms or legs following the trauma
- occurrence of numbness or tingling at any time following the injury
b) palpation of the spine for point tenderness, a “step-off”, or widened interspinous space

c) motor level assessment
   - skeletal muscle exam (can localize dermatome)
   - rectal exam for voluntary anal sphincter contraction

d) sensory level assessment
   - sensation to pinprick (tests spinothalamic tract, can localize dermatome): be sure to test sensation in face also (spinal trigeminal tract can sometimes descend as low as ≈ C4)
   - light (crude) touch: tests anterior cord (anterior spinothalamic tract)
   - proprioception/joint position sense (tests posterior columns)

e) evaluation of reflexes
   - muscle stretch reflexes: usually absent initially in cord injury
   - abdominal cutaneous reflexes
   - cremasteric reflex
   - sacral: bulbocavernosus (p.941), anal-cutaneous reflex

f) examine for signs of autonomic dysfunction
   - altered patterns of perspiration (abdominal skin may have low coefficient of friction above lesion, and may seem rough below due to lack of perspiration)
   - bowel or bladder incontinence
   - priapism: persistent penile erection

10. radiographic evaluation: see below
11. medical management specific to spinal cord injury:
    a) methylprednisolone (see below)
    b) experimental/investigational drugs: none of these agents shown to have unequivocal benefit in man: naloxone, DMSO, Lazaroid®. Tirilazad mesylate (Freedox®) was less beneficial than methylprednisolone[4]

63.3.2 General information

Practice guideline: Assessment of SCI in the hospital

Clinical Assessment

Level III[5]: the ASIA international standards for neurological and functional assessment of spinal cord injury (SCI) is recommended (p.940)

Functional outcome assessment

Level II[5]: the Functional Impairment Measure™ (FIM™) is recommended (see Table 88.7)

Level III[5]: the modified Barthel index is recommended (Table 88.6)

Practice guideline: In-hospital critical care management of SCI

Level III[6]: monitor patients with acute SCI (especially those with severe cervical level injuries) in an ICU or similar monitored setting

Level III[6]: cardiac, hemodynamic & respiratory monitoring after acute SCI is recommended

Level III[7]: hypotension (SBP < 90 mm Hg) should be avoided or corrected ASAP

Level III[7]: maintain MAP at 85–90 mm Hg for the first 7 days after SCI to improve spinal cord perfusion

63.3.3 Methylprednisolone

Practice guideline: Methylprednisolone in SCI

Level I[8]

- Methylprednisolone (MP) for the treatment of acute SCI is not recommended.
- GM-1 ganglioside (Sygen) for the treatment of acute SCI is not recommended.

MP is not FDA approved for use in treating acute SCI. There is no Class I or II evidence of benefit supporting this use of MP. Class III data that had been used to advocate its use, but the benefits were likely due to random chance and/or selection bias.[8] Conversely, there is Class I, II and III level evidence that high dose steroids are associated with harmful side effects and even death.[8] Use of high-dose MP among spine surgeons has shown a steady decline,[9] however, it was still used by as many as 56% of respondents to one survey.[9]

63.3.4 Hypothermia for spinal cord injury

The position statement of the joint sections of the AANS and the CNS is that there is not enough evidence to recommend for or against local or systemic hypothermia for acute SCI, and that it should be noted that systemic hypothermia is associated with medical complications in TBI.[10]

63.3.5 Deep-vein thrombosis in spinal cord injuries

General information

Also see Thromboembolism in neurosurgery (p. 167). Incidence of DVT may be as high as 100% when 125I-fibrinogen is used.[11] Overall mortality from DVT is 9% in SCI patients.

Practice guideline: DVT in patients with cervical SCI

Prophylaxis

Level I[12]:

- prophylactic treatment of venous thromboembolism (VTE) in patients with severe motor deficits due to SCI. Choices include:
  - LMW heparin, rotating beds, adjusted dose heparin, or some combination of these measures
  - or, low-dose heparin + pneumatic compression stockings or electrical stimulation

Level II[12]:

- early administration of VTE prophylaxis (within 72 hours)
- treat for 3 months
  - low-dose heparin should not be used alone
  - oral anticoagulation should not be used alone

Level III[12]:

- vena cava interruption filters should not be used for routine prophylaxis; they may be used for select patients who fail anticoagulation or are not candidates for anticoagulation

Diagnosis

Level III[12]:

- duplex doppler ultrasound, impedance plethysmography, venography, and the clinical examination are recommended as diagnostic tests for DVT in patients with SCI

Prophylaxis

A study of 75 patients found titrating dose of SQ heparin q 12 hrs to a PTT of 1.5 times control resulted in lower incidence of thromboembolic events (DVT, PE) than “mini-dose” heparin (5000 U SQ q 12 hrs) (7% vs. 31%).[13] Heparin can cause thrombosis, thrombocytopenia and chronic therapy may produce osteoporosis; see heparin (p. 164).

63.4 Radiographic evaluation and initial C-spine immobilization

63.4.1 Clinical criteria to rule-out cervical spine instability

There is almost no chance of a significant occult cervical spine injury[14,15] in a trauma patient who met all of the criteria in the Practice Guideline below. (Note: Although reports of bony or ligamentous abnormalities have be described as possibly occurring in these patients, there has been no report of a patient who had neurologic injury as a result of these abnormalities.)

Practice guideline: Radiographic evaluation in awake, asymptomatic trauma patients

Level I[16] & Level II[17,18]: radiographic studies are not indicated in patients who meet all of the following (these are basically the NEXUS criteria[19]):

- no mental status changes (and no evidence of alcohol or drugs). Note: altered mental status can include GCS ≤ 14; disorientation to person, place, time or events; inability to remember 3 objects at 5 minutes; delayed response to external stimuli. Evidence of alcohol or drugs includes information from the history, physical findings (slurred speech, ataxia, odor of alcohol on the breath) or positive blood or urine tests
- no neck pain or posterior midline tenderness (and no distracting pain)
- no focal neurologic deficit (on motor or sensory exam)
- do not have significant associated injuries that detract/distract from their evaluation

Cervical immobilization may be discontinued without cervical spine imaging in these patients.

The Canadian C-Spine Rule (CCR) was found to be more sensitive & specific,[20] but the EAST has not embraced it as of this writing.[17]

63.4.2 Cervical immobilization

General information

Cervical collars should be removed as soon as it can be determined that it is safe to do so. The benefits from early collar removal include: reduction of skin breakdown,[21] fewer days of mechanical ventilation,[22] shorter ICU stays,[22] reduction of ICP.[23,24]

Guidelines

Guidelines for “clearing“ the cervical spine and removing the cervical collar are shown in Practice guideline: Cervical immobilization in trauma patients (p.953).

Practice guideline: Cervical immobilization in trauma patients

A cervical collar is not needed in trauma patients who meet these criteria

- Asymptomatic patients as in Practice guideline: Radiographic evaluation in asymptomatic trauma patients (p.953): patients who are alert, without neurologic deficit or distracting injury who have no neck pain or tenderness and full ROM of the cervical spine (Level II[17])
- Penetrating brain trauma: unless the trajectory suggests direct cervical spine injury (Level III[17])
- Level III[25] & Level III[17]: Patients who are awake with neck pain or tenderness and normal cervical CT scan after either (these tests are performed in the absence of an identifiable fracture or obviously unstable dislocation to rule-out ligamentous or other soft-tissue injury that might be occult and unstable)
  - normal & adequate dynamic flexion-extension C-spine x-rays
  - or a normal cervical MRI is obtained. Note: AANS/CNS guidelines from 2002 recommended getting the MRI within 48 hours.[25] MRI is usually employed in this setting when the patient is unable to co-operate for flex-ext x-rays; see MRI findings and issues related to timing (p.957), etc.

In obtunded patients with normal cervical CT scan and gross movement of all 4 extremities

- □ flexion-extension C-spine x-rays should not be performed (Level II[17])
- options:
  - maintain cervical collar until a clinical exam can be performed[17]

- remove the collar on the basis of the normal CT scan alone[17] (the incidence of ligamentous injury with negative CT is < 5%, and the incidence of clinically significant injury is unknown but is much < 1%[17])
- obtain cervical MRI (AANS/CNS guidelines from 2002 recommended getting the MRI within 48 hours[25]) :
  - Level III[17]: the risk and benefit of cervical MRI in addition to CT is unclear, and must be individualized
  - Level II[17]: If the MRI is normal, the collar may be safely removed

63.4.3 Minimum radiographic evaluation

General information

There is controversy regarding what constitutes a minimum radiographic evaluation of the cervical spine in multiple trauma patient. No imaging modality is 100% accurate.

Asymptomatic patients – meeting criteria outlined in Practice guideline: Radiographic evaluation in awake, asymptomatic trauma patients (p.953) – may be considered to have a stable cervical spine and no radiographic studies of the cervical spine are indicated.[17,25] Factors associated with increased risk of failing to recognize spinal injuries include: decreased level of consciousness (due to injury or drugs/alcohol), multiple injuries, technically inadequate x-rays (p.1019).[26]

Primary imaging recommendations

Practice guideline: Radiographic imaging in trauma patients who are obtunded or unevaluable

Includes unresponsive patients or unreliable exam (altered mental status, distracting pain or injuries)

- Level I[18]
  - high-quality computed tomography (CT) imaging is the modality of choice
    - if high-quality CT imaging is available, routine 3-view cervical spine x-rays are not recommended
  - if high-quality CT imaging is not available, 3-view cervical spine x-rays (AP, lateral and open-mouth odontoid view) are recommended. Supplement with CT when available if needed to further define areas that are suspicious or poorly visualized on plain x-rays
- Level II[18]
  - if high-quality CT imaging is normal but the index of suspicion is high, further management should fall to physicians trained in the diagnosis and treatment of spine injuries
- Level III[18]
  - if high-quality CT imaging is normal, options include:
    - continue cervical immobilization until asymptomatic
    - obtain cervical MRI within 48 hours of injury and, if normal, D/C cervical immobilization*
    - D/C cervical immobilization at the discretion of the treating physician
      - the routine use of dynamic imaging (flexion-extension) is of marginal benefit and is not recommended in this situation

* limited and conflicting Class II & III medical evidence

CT scan, while extremely sensitive for bony injuries, is not adequate for assessing soft tissues (e.g. traumatic disc herniation, spinal cord contusion…) or ligamentous injuries (may require flexion-extension x-rays (see below) and/or MRI).

When CT scan is not appropriate/available as the initial radiographic exam

When CT cannot be done, the following guidelines are offered:

See X-rays, C-Spine (p.212) for normal vs. abnormal findings. Table 63.2 lists some indicators that should alert the reviewer that there may be significant C-spine trauma (they do not indicate definite instability by themselves).

1. cervical spine: must be cleared radiographically from the cranio-cervical junction down through and including the C7-T1 junction (incidence of pathology at C7-T1 junction may be as high as 9%[27]):
   a) lateral portable C-spine x-ray while in rigid collar: this study by itself will miss ≈ 15% of injuries[28]
   b) if all 7 cervical vertebra AND the C7-T1 junction are adequately visualized and are normal, and if the patient has no neck pain or tenderness and is neurologically intact (neurologically intact implies patient is alert, not drugged/intoxicated, & able to report pain reliably), then remove the cervical collar and complete the remainder of the cervical spine series (AP and open-mouth odontoid (OMO) view). Lateral, AP, and OMO views together detect essentially all unstable fractures in neurologically intact patients[29] (although the AP view rarely provides unique information[30]). In a severely injured patient, limitation to an AP and lateral view usually suffices for the acute (but not complete) evaluation[31]
   c) if the above studies are normal, but there is neck pain, tenderness or neurologic findings (there may be a spinal cord injury even with normal plain films), or if the patient is unable to reliably verbalize neck pain or cannot be examined for neurologic deficit, then further studies are indicated, which may include any of the following:
      - oblique views (some authors include oblique views in a "minimal" evaluation,[31] others do not[29]): demonstrates the neural foramina – may be blocked with a unilateral locked facet (p.992) -, shows a different projection of the uncinate processes than the AP view, and helps assess the integrity of the articular masses and lamina (the lamina should align like shingles on a roof)[31]
      - flexion-extension views: see below
      - CT scan: helpful in identifying bony injuries, especially in areas difficult to visualize on plain radiographs. However, CT cannot exclude significant soft-tissue or ligamentous injury[32]
      - MRI: utility is limited to specific situation (p.957) and the accuracy has not been determined
      - polytomograms: becoming less available
      - pillar view: devised to demonstrate the cervical articular masses en face (reserved for cases of suspected of having articular mass fracture)[33]: the head is rotated to one side (requires that the upper cervical spine injury has been excluded by previous radiographs), the x-ray tube is off centered 2 cm from midline in the opposite direction and the beam is angled 25° caudad, centered at the superior margin of the thyroid cartilage

Table 63.2 Radiographic signs of C-spine trauma (modified[34])

Soft tissues
• retropharyngeal space > 7 mm, or retrotracheal space > 14 mm (adult) or 22 mm (peds), see Table 12.2 for details) • displaced prevertebral fat stripe • tracheal deviation & laryngeal dislocation
Vertebral alignment
• loss of lordosis • acute kyphotic angulation • torticollis • widened interspinous space (flaring) • axial rotation of vertebra • discontinuity in contour lines (p.212)
Abnormal joints
• ADI: > 3 mm (adult) or > 4 mm (peds) (see Table 12.1 for details) • narrowed or widened disc space • widening of apophyseal joints

d) if subluxation is present at any level and is ≤ 3.5 mm and the patient is neurologically intact (neurologically intact implies patient is alert, not drugged/intoxicated, and able to report pain reliably), then obtain flexion-extension films (see below)
   - if no pathologic movement, may discontinue cervical collar
   - even if no instability is demonstrated, may need delayed films once pain and muscle spasms have resolved to reveal instability

e) if lower C-spine (and/or cervical-thoracic junction) are not well visualized
   - repeat lateral C-spine x-ray with caudal traction on the arms (if not contraindicated based on other injuries, e.g. to shoulders)
   - if still not visualized, then obtain a "swimmer's" (Twining) view: the x-ray tube is positioned above the shoulder furthest from the film, and aimed towards the axilla closest to the film with the tube angled 10–15° toward the head while the arm is elevated above the head
   - if still not visualized: CT scan through non-visualized levels (CT is poor for evaluating alignment and for fractures in the horizontal plane, thin cuts with reconstructions ameliorates this shortcoming)

f) see questions regarding stability of the subaxial spine (p.987)

g) patients with C-spine fractures or dislocations should have daily C-spine x-rays during initial traction or immobilization

2. thoracic and lumbosacral LS-spine: AP and lateral x-rays for all trauma patients who:
   a) were thrown from a vehicle, or fell ≥ 6 feet to the ground
   b) complain of back pain
   c) are unconscious
   d) are unable to reliably describe back pain or have altered mental status preventing adequate exam (including inability to verbalize regarding back pain/tenderness)
   e) have an unknown mechanism of injury, or other injuries that cast suspicion of spine injury
3. reminder: when abnormalities of questionable vintage are identified, a bone scan may be helpful to distinguish an old injury from an acute one (less useful in the elderly; in an adult, a bone scan will become "hot" within 24–48 hrs of injury, and will remain hot for up to a year; in the elderly, the scan may not become hot for 2–3 weeks and can remain so for over a year)
4. if a bony abnormality is identified or if there is a level of neurologic deficit ascribable to a specific spinal level, either a CT or MRI scan through that area should be done if possible

Flexion-extension cervical spine x-rays

Purpose: to disclose occult ligamentous instability.

Rationale: It is possible to have a purely ligamentous injury involving the posterior ligamentous complex without any bony fracture (p.991). Lateral flexion-extension views help detect these injuries, and also evaluate other injuries (e.g. compression fracture) for stability. For patients with limited flexion due to paraspinal muscle spasm (sometimes resulting from pain), a rigid collar should be prescribed, and if the pain persists 2–3 weeks later[35] the flexion-extension films should be repeated.

Options: a cervical MRI done within 48–72 hours of the trauma (may be more sensitive with STIR sequences or equivalent) may identify ligamentous or other soft-tissue injury, especially in patients who cannot cooperate for flexion-extension x-rays.

 Contraindications

- the patient must be cooperative and free of mental impairment (i.e. no head injury, street or prescription drugs, alcohol…)
- there should not be any subluxation > 3.5 mm at any level on cross-table C-spine x-rays, which is a marker for possible instability (p.991)
- patient must be neurologically intact (if there is any degree of spinal cord injury, proceed instead first with imaging studies, e.g. MRI)
- F/E x-rays are no longer recommended in obtunded patients due to a low yield, poor cost-effectiveness, and they may be dangerous[17]

Technique

The patient should be sitting, and is instructed to flex the head slowly, and to stop if it becomes painful. Serial x-rays are taken at 5–10° increments (or followed under fluoro with spot films at the end of movement), and if normal, the patient may be encouraged to flex further. This is repeated until evidence of instability is seen, or the patient cannot flex further because of pain or limitation of motion. The process is then repeated for extension.

Findings

Normal flexion-extension views demonstrate slight anterior subluxation distributed over all cervical levels with preservation of the normal contour lines (Fig. 12.1). Abnormal findings include: "flaring" of the spinous processes, see exaggerated widening (p.214).

Emergent MRI (or myelogram)

General information

Indications for emergent MRI in spinal cord injury (SCI) are listed below.

When an MRI cannot be performed, a myelogram is required (employing intrathecal contrast with CT to follow) Caution: cervical myelogram in patients with cervical spine injuries usually requires C1–2 puncture to achieve adequate dye concentration in the cervical region without dangerous extension of the neck or tilting of the patient as required when dye is injected via LP. Furthermore, pressure shifts from LP exacerbates deficit in 14%of cases with complete block.[36]

Indications

1. incomplete SCI (to check for R/O soft tissue compressing cord) with normal alignment: to check for soft tissue compressing cord
2. neurologic deterioration (worsening deficit or rising level) including after closed reduction
3. neurologic deficit not explained by radiographic findings, including:
   a) fracture level different from level of deficit
   b) no bony injury identified: further imaging is done to R/O soft tissue compression (disc herniation, hematoma...) that would require surgery
   c) always keep in mind the possibility of arterial dissection in this setting (p.1322)

MRI (non-emergent)

General information

MRI may be used to identify potentially unstable occult ligamentous or soft tissue injury. Note: abnormal signal on MRI is not always associated with instability on flexion-extension x-rays.[37] It has been recommended that this MRI should be done within 48 hours[25] or 72 hrs[38] of injury. MRI is not reliable for identifying osseous injury.

Indications for non-emergent MRI (modified):

See reference[39]

1. inconclusive cervical spine radiography, including questionable fractures
2. significant midline paraspinal tenderness and patient unable to have flexion-extension x-rays
3. obtunded or comatose patients

T2WI and STIR are the most helpful sequences. Significant abnormal findings:

1. ventral signal abnormalities with prevertebral swelling
2. dorsal signal abnormalities. Abnormal signal limited to the interspinous is probably not as unstable as when it extends into the ligamentum flavum.[39] These patients were treated with rigid collars or Minerva jackets for 1–3 months, and one that was felt to be very unstable underwent fusion
3. disc disruption indicated by abnormal signal intensity within the disc, increased disc height, or frank disc protrusions

63.5 Traction/reduction of cervical spine injuries

63.5.1 General information

Purpose

To reduce fracture-dislocations, maintain normal alignment and/or immobilize the cervical spine to prevent further spinal cord injury. Reduction decompresses the spinal cord and roots, and may facilitate bone healing.

Practice guidelines

Practice guideline: Initial closed reduction in fracture/dislocation cervical SCI

Level III[40,41]

- early closed reduction of C-spine fracture/dislocation injuries with craniocervical traction to restore anatomic alignment in awake patients
- □ not recommended: closed reduction in patients with an additional rostral injury
- patients with C-spine fracture-dislocation who cannot be examined during attempted closed reduction, or before open posterior reduction, should undergo cervical MRI before attempted reduction (see note below). The presence of a significant herniated disc in this setting is a relative indication for anterior decompression (e.g. by an anterior cervical discectomy and fusion – ACDF) before reduction
- cervical MRI is also recommended for patients who fail attempts at closed reduction (see note below).

Controversies

1. the rapidity with which reduction should be done[1]
2. whether MRI should be done prior to attempted closed reduction (prereduction MRI (p.958), will show disrupted or herniated discs in 33–50%of patients with facet subluxation. These findings do not seem to significantly influence outcome after closed-reduction in awake patients; □the usefulness of prereduction MRI in this setting is uncertain)
   a) in intact patients, to R/O a condition that might cause worsening of neurologic condition with reduction (e.g. traumatic disc herniation) – must be balanced against risks of transferring patients to MRI
   b) in patients with neurologic deficit (complete or partial SCI)

□ Contraindications
1. atlantooccipital dislocation (p.963): traction may worsen deficit. If immobilization with tongs/halo is desired, use no more than ≈ 4 lbs
2. types IIA or III hangman’s fracture (p.973)
3. skull defect/fracture at anticipated pin site: may necessitate alternate pin site
4. use with caution in pediatric age group (do not use if age ≤ 3 yrs)
5. very elderly patients
6. demineralized skull: some elderly patients, osteogenesis imperfecta…
7. patients with an additional rostral injury
8. patients with movement disorders: constant motion may cause pin erosion through the skull

63.5.2 Application of tongs or halo ring

General information

Supplies: gloves, local anesthetic (typically 1% lidocaine with epinephrine), betadine ointment. Optional equipment: razor or hair clipper, scalpel.

Choice of device: a number of cranial “tongs” are available. Crutchfield tongs require predrilling holes in the skull. Gardner-Wells tongs are the most common tongs in use. If, after the acute stabilization, the later use of halo-vest immobilization is anticipated, a halo ring may be used for the initial cervical traction, and then converted to vest traction at the appropriate time (e.g. post-fusion).

Preparation: placed with patient supine on a gurney or bed. Option: shave hair around proposed pin sites (see below). Betadine skin prep, then infiltrate local anesthetic. Option: incise skin with scalpel (prevents pins from driving in surface contaminants).

Gardner-Wells tongs

Pin sites: the pins are placed in the temporal ridge (above the temporalis muscle), 2–3 fingerbreadths (3–4 cm) above pinna. Place directly above external acoustic meatus for neutral position traction; 2–3 cm posterior for flexion (e.g. for locked facets); 2–3 cm anterior for extension.

One pin has a central spring-loaded force-indicator. Tighten pins until the indicator protrudes 1 mm beyond the flat surface. Retighten the pins daily until indicator protrudes 1 mm for 1 or 2 days only, then stop.

Halo ring

Supplies (in addition to above): optional paddle AKA “spoon” to support the head beyond the edge of the bed, traction adapter (called a “traction bail“ from the circular handle on a baille, the old French word for bucket). Read all of this (including pointers) before starting

1. ring size: choose an appropriately sized ring that leaves a ≈ 1–2 cm gap between the scalp and the ring all the way around
2. ring position: generally placed at or just below the widest portion of the skull (the “equator”), but the front should be ≈ 1 cm above the orbital rim and the back should be ≈ 1 cm above the pinna.[42] The ring is usually stabilized with temporary pins that have plastic discs where they contact the skull
3. pin sites: choose the threaded holes in the ring that place the pins as perpendicular to the skull as possible as follows
   a) anterior pins: above the lateral two–thirds of the orbit
   b) posterior pins: just behind the ears
   c) in pediatrics, additional pins may be placed to further distribute the load on the thinner skull
4. pin insertion: the pins are gradually brought close to the scalp which is then anesthetized with local anesthetic. Pins are then sequentially tightened, starting with any pin then going to the kitty-corner pin, then a third pin and finally its opposite. Most halos provide some type of torque wrench to permit approximately 8 in-lb of torque for most adults; 2–5 in-lb for peds
5. placement pointers
   a) the cervical collar is left in place until traction/immobilization is established
   b) try to place the halo as level from left to right as possible. While a skewed placement can be compensated for when attaching the vest, it looks bad
   c) prior to penetrating the forehead skin for anterior pins, have the patient close their eyes and hold them closed as the pins are advanced (this avoids “pinning the eyes open”)
   d) avoid placing pins in the temporalis muscle or the temporal squamosa
   e) do not place pins above the medial third of the orbit to avoid the supraorbital and supratrochlear nerves, and to reduce theorist of penetrating the relatively thin anterior wall of the frontal sinus

Application of traction

For traction, transfer to a bed with ortho headboard with the tongs or halo ring in place. Tie a rope to tongs/halo and feed through a pulley at the head of bed. Slight flexion or extension is achieved by changing the height of the pulley relative to the patient’s long axis.

X-rays: lateral C-spine x-rays immediately after application of traction and at regular intervals and after every change in weights and every move from bed. Check alignment and rule-out overdistraction at any level and atlanto-occipital dislocation; BDI should be ≤ 12 mm (p.963).

Weight: if there is no malalignment and traction is being used just to stabilize the injury and to compensate for ligamentous instability, use 5 lbs for the upper C-spine or 10 lbs for lower levels. See information on reducing locked facets (p.992). May remove cervical collar once patient is in traction with adequate reduction or stabilization.

Post-placement care

Pin tightening: pins are re-torqued in 24 hours. Some authors do one additional tightening the day after that. Avoid further tightenings which can penetrate the skull

Pin care: clean (e.g. half strength hydrogen peroxide), then apply povidone-iodine ointment. Frequency: in hospital: q shift. At home following discharge: twice daily.

Alternatively, simple cleaning with soap and water twice daily is acceptable.

Application of halo vest

For vest placement (i.e. patients not remaining in traction) once the halo ring is placed (see above) it needs to be attached to the vest by posts. The mechanism varies between manufacturers. If possible, have the patient in a cotton T-shirt prior to placing the vest (this may require cutting the neck opening to accommodate the ring).

The vest should be snug, but too tight so as to restrict respirations. Shoulder straps should be contacting the shoulders (the vest will tend to ride up when the patient is sitting). Most vests come with a wrench that is taped to the vest for emergency removal e.g. for cardiopulmonary resuscitation.

Reduction of locked facets

See Practice guideline: Initial closed reduction in fracture/dislocation cervical SCI (p.958) for background information and Reduction of locked facets (p.992) for technique.

Complications

1. skull penetration by pins. May be due to:
   a) pins torqued too tightly
   b) pins placed over thin bone: temporal squamosa or over frontal sinus
   c) elderly patients, pediatric patients, or those with an osteoporotic skull
   d) invasion of bone with tumor: e.g. multiple myeloma
   e) fracture at pin site
2. reduction of cervical dislocations may be associated with neurologic deterioration which is usually due to retropulsed disc[43] and requires immediate investigation with MRI or myelogram/CT
3. overdistraction from excessive weight (especially with upper cervical spine injuries), may also endanger supporting tissues
4. caution with C1-C3 injury, especially with posterior element fracture (traction may pull fragments in towards canal)
5. infection:
   a) osteomyelitis in pin sites: risk is reduced with good pin care
   b) subdural empyema (p.327): rare[44,45]

63.6 Indications for emergency decompressive surgery

63.6.1 Cautions and contraindications

Caution: laminectomy in the face of acute spinal cord injury has been associated with neurologic deterioration in some cases. When emergency decompression is indicated, it is usually combined with a stabilization procedure.

Contraindications to emergent operation

- complete spinal cord injury ≥ 24 hrs (no motor or sensory function below level of lesion) in the absence of spinal shock (i.e. the deficit is attributable to a complete spinal cord injury and not a temporary condition due to spinal shock). Bulbocavernosus reflex is generally used as a guide to the presence of spinal shock (see Bulbocavernosus reflex)
- medically unstable patient
- central cord syndrome (p.944): controversial

63.6.2 Modified recommendations of Schneider

See reference.[46]

In patients with complete spinal cord lesions, no study has demonstrated improvement in neurologic outcome with either open decompression or closed reduction.[47] In general, surgery is reserved for incomplete lesions – possibly excluding central cord syndrome (p.944) – with extrinsic compression, who, following maximal possible reduction of subluxation show:

1. progression of neurologic signs
2. complete subarachnoid block by Queckenstedt test or radiographically (on myelography or MRI)
3. compression of spinal cord (on CT/myelogram, CT, or MRI) e.g. by bone fragments or soft tissue elements (e.g. traumatic disc herniation)
4. necessity for decompression of a vital cervical root
5. compound fracture or penetrating trauma of the spine
6. acute anterior spinal cord syndrome (p.946)
7. non-reducible fracture-dislocations from locked facets causing spinal cord compression

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64 Occiptoatlantoaxial Injuries (Occiput to C2)

64.1 Atlanto-occipital dislocation

64.1.1 General information

64

See Occiptoatlantoaxial-complex anatomy (p. 68) for relevant anatomy.

Atlanto-occipital dislocation (AOD) AKA craniocervical junction dislocation. Disruption of the stability of the craniocervical junction (which results from ligamentous injuries). Probably under-diagnosed, may be present in ≈ 1% of patients with "cervical spine injuries"[1] (definition of cervical spine injuries not specified), found in 8–19% of fatal cervical spine injury autopsies.[2,3] More than twice as common in pediatrics as adults, possibly owing to the flatter (i.e. less cupped) condyles in peds, the higher ratio of cranium to body weight, and increased ligamentous laxity. Patients usually either have minimal neurological deficit or exhibit bulbar-cervical dissociation (BCD) (p. 944). Some may exhibit cruciate paralysis (p. 1419). Most mortality results from anoxia due to respiratory arrest as a result of BCD.

Classification

See reference.[4]

For illustration, see Fig. 64.1.

- Type I. Anterior dislocation of occiput relative to the atlas
- Type II. Longitudinal dislocation (distraction)
- Type III. Posterior dislocation of occiput

Combinations (e.g. anterior-distracted AOD[5]) may also occur.

Practice guideline: Diagnosis of atlanto-occipital dislocation

Level I[6]
- In pediatric patients, CT to assess the condyle-C1 interval (CCI) is recommended to diagnose AOD

Level III[6]
- In pediatric patients, CCI measured on CT has the highest sensitivity and specificity for AOD. The utility in adults has not been reported
- A lateral cervical spine x-ray is recommended to diagnose AOD. If it is desired to employ a radiologic method of measurement, the BAI-BDI method is recommended (see Table 64.1). Prevertebral soft tissue swelling in the upper cervical spine on an otherwise nondiagnostic x-ray should be followed with a cervical CT to rule out AOD

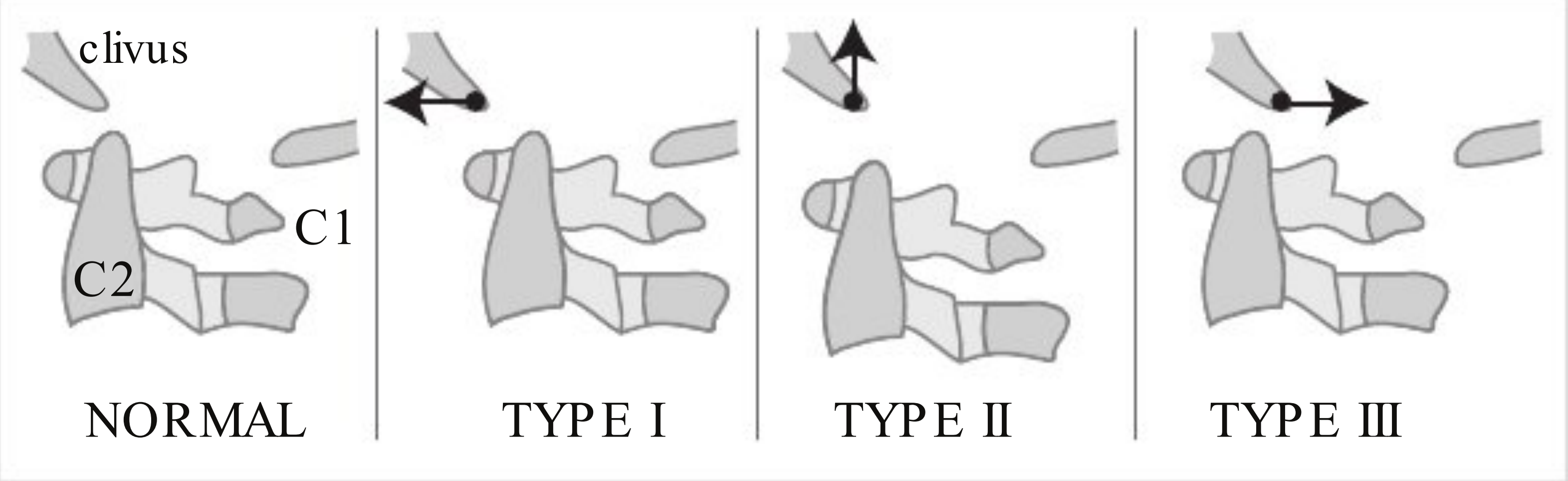


Fig. 64.1 Classification of atlanto-occipital dislocation

Table 64.1 Radiographic evaluation of atlanto-occipital dislocation (AOD)

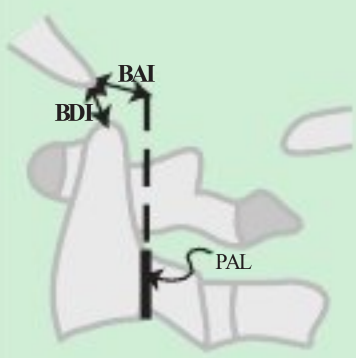
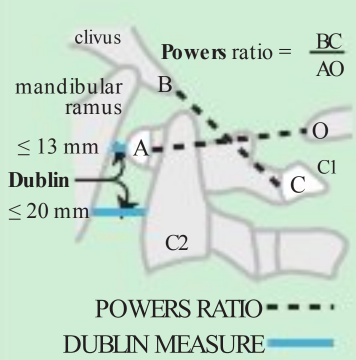
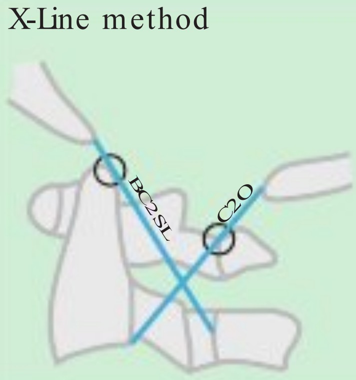
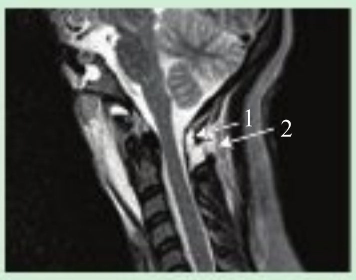
Method	Comments	Normal Values: Plain X-ray	Normal Values: CT
 BAI-BDI method[8a] Both BAI & BDI should be measured in adults.	BAI[9] (basion-axial interval) = distance from basion (inferior tip of the clivus) to rostral extension of posterior axial line (PAL) (the posterior cortical margin of the body of C2). AKA Harris line. Better for anterior or posterior AOD	Adults: -4 ≤ BAI ≤ 12 mm. Normal: BAI & BDI each ≤ 12 mm	May be used,[10] but was not reliably reproducible on CT[11]
		Peds: 0–12 mm (BAI should never be negative)	
	BDI (basion-dental interval) = distance from basion to the closest point on the tip of the dens. Better for distracted AOD	Adult: ≤ 12 mm (range: 2–15 mm) (mean: 7.5 ±þ4.3)	Adult: < 8.5 mm (range: 1.4–9 mm)[11]
		Peds: unreliable in age < 13 yrs because of variable age of ossification and fusion of the odontoid tip (os)	Peds[12b]: < 10.5 mm (95th percentile). With os[c]: < 9.5 mm. Without os[c]: < 11.5 mm
Atlanto-occipital interval (AOI) 	AKA condylar gap.[13] Distance between occipital condyle and superior articular surface of C1 measured on lateral x-ray or sagittal CT reconstructions thru O-C1 junction. Pang[14] averaged the interval between the condyle and C1 at 4 equidistant points on sagittal and 4 on coronal images (8 points total)	Adult[d]: ≤ 2 mm[13]	Adult:: < 1.4 mm (95th percentile) (based on single measurement)[11]
		Peds: ≤ 5 mm (for all of 5 equally spaced measurements[15])	Peds: < 2.5 mm (single measure),[12] or < 4.0 mm (average of 8 measurements in 2 planes)[16]
Powers' ratio & Dublin measure 	Powers' ratio: cannot be used with fractures of C1 or foramen magnum. Only for anterior AOD (see text). Requires identification of 4 reference points: B = basion, A = anterior arch of C1, C = posterior arch of C1, O = opisthion[e]	Adult: < 1 (range: 0.5–1.2) (95th percentile = 0.6–0.9) (see text for details)	Same as plain x-ray[11]
		Peds: < 0.9[f]	
	Dublin measure[17] 25% sensitive[18]	Mandible to anterior atlas: ≤ 13 mm. Posterior mandible to dens: ≤ 20 mm	
X-Line method 	AKA occipital-axial lines method.[18] Requires identification of 6 reference points and 2 lines (sensitivity 75%).[18] Uses a 6 ft target-film distance in a sitting patient[19] which is not always practical in the E/R[5] • C2O line: from the posteroinferior corner of axis body to the opisthionΔ. Should intersect tangentially with the highest point on the C1 spinolaminar line • BC2SL line: from the basion to a point midway on the C2 spinolaminar line. Should intersect tangentially with the posterosuperior dens		

Table 64.1 continued

Method	Comments	Normal Values	
		Plain X-ray	CT
MRI 	Abnormal MRI findings include: abnormal high signal on T2WI in the occipitoatlantal joints or in the posterior occipitoatlantal (O-C1) ligaments. Very sensitive (≈ 100%) but not specific for unstable AOD. The figure at left shows abnormal signal in the posterior O-C1 ligaments (arrow 1) and in the ligamentum flavum and soft tissues (arrow 2).		

[a]original study of lateral x-ray in a supine patient with a target-film distance of 40 in. (1 m). The sensitivity of the BAI-BDI method for AOD is good when all landmarks can be identified, but still may only be ≈ 75%[10]
[b]for this study, peds is defined up to the age of 10 years, by age ≈ 8–10 years the C-Spine reaches adult proportions (not necessarily size)
[c]os = ossiculum terminale (p.981)
[d]the articular process of C1 is often obscured by the tip of the mastoid process on plain films
[e]the opisthion cannot be identified in ≈ 56% of lateral C-spine x-rays[9]
[f]could not be measured in many peds cases often due to lack of ossification (usually of posterior C1 arch)

64.1.2 Clinical presentation

1. may be neurologically intact, therefore must be ruled-out in any major trauma
2. bulbar-cervical dissociation (p.944)
3. may have lower cranial nerve deficits (as well as VI palsies) ± cervical cord injury
4. worsening neurologic deficit with the application of cervical traction: check lateral C-spine films immediately after applying traction (p.959)

64.1.3 Radiographic evaluation

Numerous methodologies have been devised to radiographically diagnose AOD. Most utilize surrogate markers for the end-point of interest: viz. instability of the occipital-cervical junction. None are completely reliable.[7] Measurements on CT scans are more accurate than plain radiographs (landmarks are easier to identify, no magnification or rotation error) – however, normal values differ from plain radiographs. Some methods are shown in Table 64.1. The BAI-BDI method and the AOI method are recommended.

A helpful rule-of-thumb: the inferior tip of the clivus should point directly to tip of dens (this may be obscured on x-ray).

Additional CT clues: there may be blood in the basal cisterns (an indirect sign). On thin-cut axial CT, there may be one or more slices showing no bone at all due to the gap between the occiput and C1.

64.1.4 Technical suggestions for radiographic evaluation

1. X-rays: verify that the film is a true lateral (e.g. check alignment of the two mandibular rami as well as of the posterior clinoids)
2. CT: sensitivity, specificity, and positive/negative predictive values of most of the these measures improves when sagittal CT reconstructions are used instead of plain radiographs[20] (relevant landmarks could be identified in > 99% of CTs, vs. 39–84% on x-ray)

Powers' ratio[1]: distance BC (basion to posterior arch of atlas) is divided by distance AO (opisthion to anterior arch of atlas), see Table 64.1. Interpretation is shown in Table 64.2.

Cannot be used with any fracture involving the atlas or the foramen magnum, or with congenital anatomic abnormalities. Applies only to anterior AOD (i.e. not for posterior or distracted AOD).

64.1.5 Management

Initial management

If AOD is suspected, immediately immobilize the neck with halo orthosis or with sandbags. Do not apply cervical traction in an attempt to reduce AOD because there is a 10% risk of neurologic deterioration with the use of traction in AOD.

Table 64.2 Powers' ratio

Ratio BC/AO	Interpretation	Comment
< 0.9	normal	1 standard deviation below the lowest case of AOD
≥ 0.9 and < 1	"grey zone" (indeterminate)	included 7% of normals and no cases of AOD
≥ 1	AOD	encompassed all AOD cases

Table 64.3 Grading & management of AOD[10]

Grade	Definition	Management
I	no abnormal CT criteria[a] with only moderately abnormal MRI (high signal in posterior ligaments or occipitoatlantal joints)	external orthosis (halo or collar)
II	≥ 1 abnormal CT criteria[a] or grossly abnormal MRI findings in occipitoatlantal joints, tectorial membrane, or alar or cruciate ligaments	surgical stabilization

[a]CT criteria used: Power's ratio, BAI-BDI, X-line

Subsequent management

Controversial whether operative fusion vs. prolonged immobilization (4–12 months) with halo brace is required. However, posterior occipitocervical fusion is usually recommended (p.1138).

Practice guideline: Treatment of atlanto-occipital dislocation

Level III[6]
- internal fixation & arthrodesis (fusion) using one of a variety of methods
- CAUTION: traction is not recommended in the management of AOD

Horn et al.[10] suggest that patients be grouped and then managed as shown in Table 64.3.

In infants: reduce in the OR and fuse (usually with transarticular screws).

64.1.6 Prognosis

The most important predictor of outcome is the severity of neurologic injuries at the time of presentation.[10] Among AOD patients who survived the initial injury, those with severe TBI and brainstem dysfunction or complete bulbar-cervical dissociation all had poor outcome.[10] Those with incomplete SCI or nonsevere TBI may improve.

64.2 Occipital condyle fractures

64.2.1 General information

Key concepts

- uncommon (0.4% of trauma patients)
- may present with lower cranial nerve deficits which may be delayed in onset (e.g. hypoglossal nerve palsy), mono-, para-, or quadriparesis or plegia
- W/U: CT scan with reconstructions (rarely detected on plain x-rays)

- Tx: usually treated with rigid collar. Indications for occipitocervical fusion or halo immobilization: craniocervical misalignment (occipital-C1 interval > 2.0 mm)

Occipital condyle fractures (OCF) were first described in 1817 by Bell.[21]

Rare. Incidence: 0.4%(in a series of 24,745 consecutive trauma patients surviving to the E/R[22]).

64.2.2 Diagnosis

Clinical suspicion of occipital condyle fracture (OCF) should be raised by the presence of ≥ 1 of the following[23]:

- blunt trauma with high energy
- craniocervical injuries
- altered consciousness
- occipital pain or tenderness
- impaired cervical movement
- lower cranial nerve palsies
- retropharyngeal soft-tissue swelling

Practice guideline: Diagnosis of occipital condyle fractures

Level II[24]

- CT to establish the diagnosis of occipital condyle fracture (OCF)

Level III[24]

- MRI to assess the integrity of the ligaments of the craniocervical complex

64.2.3 Classification

A widely used classification system is that of Anderson & Montesano[25] as shown in Table 64.4.

Maserati et al.[22] classified patients simply on the basis of whether craniocervical misalignment was present or absent on CT with reconstructions (they defined craniocervical misalignment as an occipital condyle-C1 interval > 2.0 mm). They felt other classification systems were superfluous as they did not affect outcome in their retrospective review (see Treatment below).

64.2.4 Treatment

Controversial. Lower cranial nerve deficits often develop in untreated cases of OCF, and may resolve or improve with external immobilization. Anderson & Montesano Types I & II have been treated with or without external immobilization (cervical collar or, occasionally, halo) without obvious difference. External immobilization × 6–8 weeks is suggested for Type III fractures because of the higher risk of delayed deficits.

Table 64.4 Anderson & Montesano classification of occipital condyle fractures

Type	Description
I	comminuted from impact: may occur from axial loading
II	extension of linear basilar skull fracture[26]
III	avulsion of condyle fragment (traction injury): may occur during rotation, lateral bending, or a combination of mechanisms. Considered unstable by many

Practice guideline: Treatment of occipital condyle fractures

Level III[24]:
- For OCF with associated atlanto-occipital ligamentous injury or evidence of instability: halo vest immobilization or occipitocervical stabilization (fusion).
- For bilateral OCFs, consider halo vest instead of a collar to provide increased immobilization.
- External cervical immobilization for all other OCFs.

64.2.5 Outcome

In a retrospective review of 100 patients with OCF,[22] 3 patients underwent occipitocervical fusion (p.1474) for craniocervical misalignment (2) or unrelated C1–2 fracture (1). The remainder (without craniocervical misalignment) were treated with a rigid collar and delayed clinical & radiographic follow-up. None of their unoperated patients had neurologic deficit, and none developed delayed instability, malalignment, or neurologic deficit (regardless of their classification on the other systems in use).

64.3 Atlantoaxial subluxation/dislocation

64.3.1 General information

Lower morbidity and mortality than atlantooccipital dislocation.[27] See Occiptoatlantoaxial-complex anatomy (p.68) for relevant anatomy.

Types of atlantoaxial subluxation:
1. rotatory: (see below) usually seen in children after a fall or minor trauma
2. anterior: more ominous (see below)
3. posterior: rare. Usually from erosion of odontoid. Unstable. Requires fusion

64.3.2 Atlantoaxial rotatory subluxation

General information

Key concepts

- typically seen in children
- associations: trauma, RA, respiratory tract infections in peds (Grisel syndrome)
- often present with cock-robin head position (tilt, rotation, sl. flexion)
- classification: Fielding & Hawkins (Table 64.5)
- Tx: early traction often successful. Treat infection in Grisel syndrome. Subluxation unreducible in traction may need transoral release then posterior fusion

Table 64.5 Fielding & Hawkins classification of rotatory atlantoaxial subluxation

Type	Description		AD(mm)	Comment
	TAL[a]	facet injury		
I	intact	bilateral	≤3	dens acts as pivot
II	injured	unilateral	3.1–5	intact joint acts as pivot
III	injured	bilateral	>5	rare. Very unstable
IV	incompetence of the odontoid with posterior displacement			rare. Very unstable

[a]TAL = transverse atlantal ligament, AD = anterior displacement of C1 on C2

Rotational deformity at the atlanto-axial junction is usually of short duration and easily corrected. Rarely, the atlantoaxial joint locks in rotation (AKA atlantoaxial rotatory fixation[28]). Usually seen in children. May occur spontaneously (with rheumatoid arthritis[29] or with congenital dens anomalies), following major or minor trauma (including neck manipulation or even with neck rotation while yawning[28]), or with an infection of the head or neck including upper respiratory tract (known as Grisel syndrome[30]: inflammation may cause mechanical and chemical injury to the facet capsules and/or transverse atlantal ligament (TAL)).

The vertebral arteries (VA) may be compromised in excessive rotation, especially if it is combine with anterior displacement.

Mechanism of subluxation

The dislocation may be at the occipito-atlantal and/or the atlanto-axial articulations.[31] The mechanism of the irreducibility is poorly understood. With an intact TAL, rotation occurs without anterior displacement. If the TAL is incompetent as a result of trauma or infection, there may also be anterior displacement with more potential for neurologic injury. Posterior displacement occurs only rarely.[28]

Classification

The Fielding and Hawkins classification[28] is shown in Table 64.5.

Clinical findings

Patients are usually young. Neurologic deficit is rare. Findings may include: neck pain, headache, torticollis – characteristic “cock robin” head position with ≈ 20° lateral tilt to one side, 20° rotation to the other, and slight (≈ 10°) flexion, see DDx (p. 1390) -, reduced range of motion, and facial flattening.[28] Although the patient cannot reduce the dislocation, they can increase it with head rotation towards the subluxed joint with potential injury to the high cervical cord.

Brainstem and cerebellar infarction and even death may occur with compromise of circulation through the VAs.[32]

Radiographic evaluation

X-rays : Findings (may be confusing) include:

- pathognomonic finding on AP C-spine x-ray in severe cases: frontal projection of C2 with simultaneous oblique projection of C1.[33 (p 124)] In less severe cases, the C1 lateral mass that is forward appears larger and closer to the midline than the other
- asymmetry of the atlantoaxial joint that is not correctable with head rotation, which may be demonstrated by persistence of asymmetry on open mouth odontoid views with the head in neutral position and then rotated 10–15° to each side
- the spinous process of the axis is tilted in one direction and rotated to the other (may occur in torticollis of any etiology)

CT scan: Demonstrates rotation of the atlas.[31]

MRI: May assess the competence of the transverse ligament.

Treatment

Grisel’s syndrome

Appropriate antibiotics for causative pathogen with traction (see below) and then immobilization for the subluxation as follows[30]: Fielding (Table 64.5) Type I: soft collar, Type II: Philadelphia collar or SOMI, Type III or IV: halo. After 6–8 weeks of immobilization, check stability with flexion-extension x-rays. Surgical fusion for residual instability

Traction

If treated within the first few months[34] the subluxation can usually be reduced with gentle traction (in children start with 7–8 lbs and gradually increase up to 15 lbs over several days, in adults start with 15 lbs and gradually increase up to 20). If the subluxation is present > 1 month, traction is less successful. Active left-right neck rotation is encouraged in traction.

If reducible, immobilization in traction or halo is maintained × 3 months[28] (range: 6–12 weeks).

Surgical fusion

Subluxation that cannot be reduced or that recurs following immobilization should be treated by surgical arthrodesis after 2–3 weeks of traction to obtain maximal reduction. The usual fusion is C1 to C2 (p. 1479) unless other fractures or conditions are present.[28] Fusion may be performed even if the rotation between C1 & C2 is not completely reduced. For irreducible fixation, a staged procedure can be done with anterior transoral release of the atlantoaxial complex (the exposure is taken laterally to expose the atlantoaxial joints which must be done carefully to avoid injury to the VAs, soft tissue is carefully removed from the joints and the atlantodental interval, no attempt at reduction was made at the time of this 1st stage) followed by gradual skull traction and then a second stage posterior C1–2 fusion[34]

64.3.3 Anterior atlantoaxial subluxation (AAS)

See reference.[27]

General information

One third of patients with AAS have neurologic deficit or die. For relevant anatomy, see Occiptoatlantoaxial-complex anatomy (p. 68).

Subluxation may be due to:

1. disruption (rupture) of the transverse (atlantal) ligament (TAL): the atlantodental interval (ADI) (see below) will be increased
   a) attachment points of the TAL may be weakened in rheumatoid arthritis (p. 1479)
   b) trauma: may cause anatomic or functional ligament disruption (see below)
2. incompetence of the odontoid process: ADI will be normal
   a) odontoid fracture
   b) congenital hypoplasia, e.g. Morquio syndrome (p. 1151)

Presentation

Neck pain is common. There are no specific patterns to the pain that is characteristic.

“V” shaped pre-dens space

See reference.[35]

Widening of the upper space between the anterior arch of C1 and the odontoid seen on lateral C-spine flexion x-ray. It is not known if this increased mobility represents elongation or laxity of the transverse ligament and/or the posterior ligamentous complex. This may also be a normal finding in flexion in peds.

True subluxation will result in malalignment between C1 and C2. The key differentiating feature is whether the ADI is increased or normal, as indicated above.

Evaluation and classification

Both CT & MRI are recommended to evaluate fractures, TAL & its bony attachments.

Assessing the integrity of the transverse ligament

1. disruption of the TAL may be inferred indirectly from
   a) Rule of Spence: on open-mouth odontoid x-ray, if the total overhang of both C1 lateral masses on C2 is ≥ 7 mm
   b) atlantodental interval (ADI) (p. 213): > 3 mm in adults, > 4 mm in peds
2. MRI may be able to image the TAL directly. Findings of disruption (axial MRI): high signal within TAL on gradient-echo MRI, loss of continuity of TAL, blood at insertion site[36]
3. CT demonstrates bony injuries in regions of TAL insertion on C1 tubercles

Classification of TAL disruption

See reference.[37]

- Type I. Anatomic disruption. Tear of TAL itself. Rare (the odontoid usually fractures before the TAL tears). Unlikely to heal. Requires surgical stabilization

 Type II. Physiologic disruption. Detachment of the tubercle of C1 to which TAL is attached (Fig. 1.12) as may occur in comminuted C1 lateral mass fractures. 74% chance of healing with immobilization (halo recommended[37])

Treatment

- For TAL disruption: One approach is to fuse all Type I TAL injuries, and those Type II TAL injuries that are still unstable after 3–4 months of immobilization.[37] Fusion is also recommended with irreducible subluxations. If C1 is intact, a C1–2 fusion is usually adequate. For situations involving C1 fractures, see below.
- For odontoid fractures with intact TAL managed as outlined (p.979)

64.4 Atlas (C1) fractures

64.4.1 General information

Acute C1 fractures account for 3–13% of cervical spine fractures.[38] 56% of 57 patients had isolated C1 fractures; 44% had combination C1–2 fractures; 9% had additional non-contiguous C-spine fractures. 21% had associated head injuries.[38]

64.4.2 Classification of C1 fractures

See reference.[39]

Type I: fractures involving a single arch (31–45% of C1 fractures)
Type II: burst fracture (37–51%): the classic Jefferson fracture (see below)
Type III: lateral mass fractures of the atlas (13–37%)

64.4.3 Jefferson fracture

Described by Sir Geoffrey Jefferson.[40] Classically a four-point (burst) fracture of the C1 ring,[41] but the term is now often used to include the more common three or two-point fractures,[42] the latter through the C1 arches (thinnest portion). Usually from axial load (a "blow-out" fracture). 41% chance of an associated C2 fracture.

In pediatrics, it is critical to differentiate a C1 fracture from the normal synchondroses (p.214) and from pseudospread of the atlas (p.933). A fracture may also occur through the unfused synchondroses.

64.4.4 Stability

To reiterate: stability of the occiptoatlantoaxial complex is primarily due to ligaments, with little contribution from bony articulations; see Occiptoatlantoaxial-complex anatomy (p.68). Integrity of transverse ligament (TAL) is the most important determinant of stability (see Assessing the integrity of the transverse ligament above).

Jefferson fractures are unstable, however, there is usually no neurologic deficit for isolated Jefferson fractures (due to the large canal diameter at this level, plus the tendency for fragments to be forced outwards away from spinal cord).

64.4.5 Clinical

Neurologic deficit is rare. 3 of 25 patients with Jefferson fractures sustained neurologic injuries (1 complete injury, 2 central cord syndromes) in one series.

64.4.6 Evaluation

Thin cut high-resolution CT is the diagnostic test of choice. It is critical to evaluate from C1 through C3 to delineate details of the C1 fracture and to assess for associated C2 injury.

MRI may be able to assess the integrity of the TAL, but images are often difficult to interpret for this.

64.4.7 Treatment

Treatment options depend heavily on the status of the TAL. Practice guidelines are shown here. Specifics are delineated in Table 64.6.[43] When external immobilization is employed, it is used for 8–16 weeks (mean = 12).

Practice guideline: Treatment of isolated atlas fractures

64

Level III[44]: for isolated atlas fractures:
- treatment is based on the fracture type and integrity of transverse atlantal ligament
- if the transverse ligament is intact: cervical immobilization alone
- if the transverse ligament is disrupted: either (note: disruption of the TAL may be anatomic or physiologic; see text for details)
  a) cervical immobilization alone
  b) or, surgical fixation and fusion

Fusion options when surgery is indicated[37]:
1. unilateral ring or anterior C1 arch fractures: C1–2 fusion
2. multiple ring fractures or posterior C1 arch fractures: occipital-cervical fusion

Surgical options: 1. fusion A. unilateral ring…, B. multiple ring… 2. Surgical options that do not involve arthrodesis include: posterior C1 screw placement, anterior transoral screw/plate placement.

64.4.8 Outcome

In many series,[38,45] treatment without surgery results in satisfactory outcome when the TAL is not disrupted.

64.5 Axis (C2) fractures

64.5.1 General information

Acute fractures of the axis represent ≈ 20% of cervical spine fractures. Neurological injury is uncommon, and occurs in < 10% of cases. Most injuries may be treated by rigid immobilization.

Steele's rule of thirds: each of the following occupies one–third of the area of the canal at the level of the atlas: dens, space, spinal cord.[46]

64.5.2 Types of C2 fractures

1. odontoid fractures (p.978): type II odontoid fracture is the most common injury of the axis
2. hangman's fracture: see below
3. miscellaneous C2 fractures (p.982)

Table 64.6 Treatment options for isolated C1 fractures

Fracture type	Treatment options
anterior or posterior arch	collar or SOMI
anterior AND posterior arch (burst)	
• stable (TAL[a] intact)	collar or SOMI, halo
• unstable (TAL disrupted)	halo, C1–2 stabilization & fusion
lateral mass fractures	
• comminuted fracture	collar or SOMI, halo
• transverse process fracture	collar or SOMI

[a]abbreviations: TAL = transverse atlantal ligament

64.5.3 Hangman's fracture

General information

Key concepts

- bilateral fracture through the pars interarticularis of C2 with traumatic subluxation of C2 on C3, most often due to hyperextension + axial loading
- most are stable with no neurologic deficit
- classification: Levine system (Table 64.7). Critical dividing line: disruption of C2–3 disc (Types II and higher) which may render the fracture unstable
- W/U: cervical CT with sagittal & coronal recons for all. Cervical MRI to assess C2–3 disc disruption (Levine II). CTA for dissection if fx passes thru foramen transversarium (consider for all C2 fractures – see Table 55.7)
- most do well with non-halo immobilization x 8–14 weeks. Esceptions: severe/unstable fractures (p. 975) or those that do not remain aligned in brace

AKA traumatic spondylolisthesis of the axis (a term first used in 1964[47]).

Description: bilateral fracture through the pars interarticularis (isthmus) of the pedicle of C2 (Fig. 64.2; the configuration of C2 is unique, and the distinction between the pars and the pedicle is ambiguous). There is often anterior subluxation of C2 on C3.

The term "hangman's fracture" (HF) was coined by Schneider et al.[48] although the mechanism of most modern HFs (hyperextension and axial loading, from MVAs or diving accidents) differs from that sustained in judicial hangings (where submental placement of the knot results in hyperextension and distraction[49]). Some cases may be due to forced flexion or compression of the neck while in extension.

Pediatrics: rare in children < 8 years old where the forces tend to fracture the incompletely fused odontoid, see epiphyseal fracture (p. 214). In pediatrics, consider pseudosubluxation in the differential diagnosis (p. 934).

Usually stable. Deficit is rare. Nonunion is rare. 90% heal with immobilization only. Operative fusion is rarely needed. Fractures of C2 that do not go through the isthmus are not true hangman's fractures and may require different management (p. 982).

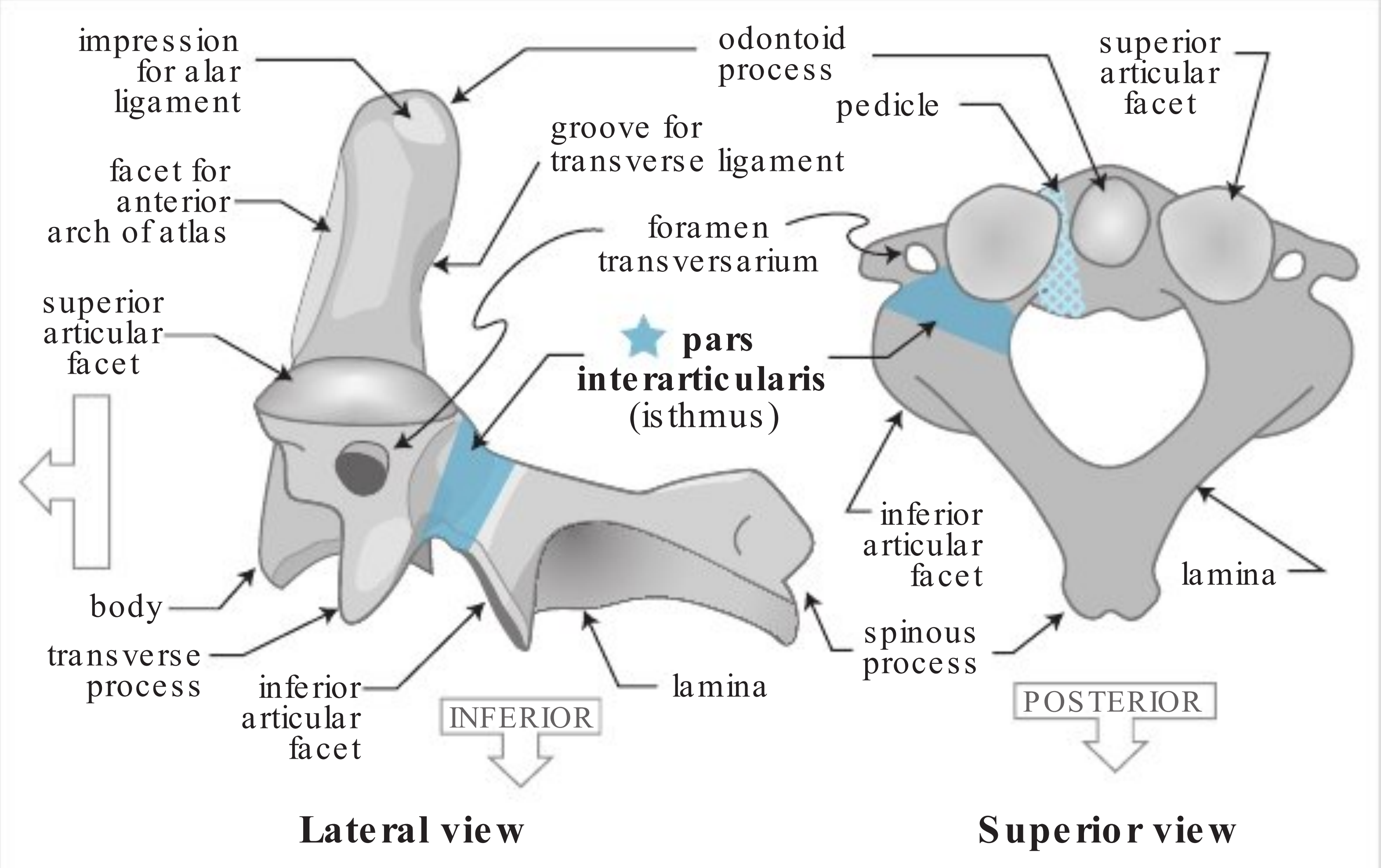


Fig. 64.2 Anatomy of axis (C2). The pars interarticularis is shown in dark blue

Classification

Levine/Effendi classification

The system of Effendi et al.[50] as modified by Levine[51] and others (Table 64.7) is widely used in grading adult HF (not applicable to peds). Angulation is measured as the angle between the inferior endplates of C2 and C3. Anterior subluxation of C2 on C3 > 3 mm (Type II) is a surrogate marker for C2–3 disc disruption which can be evaluated more directly with cervical MRI.

Grading system of Frances et al.

The grading system[54] is shown in Table 64.8.

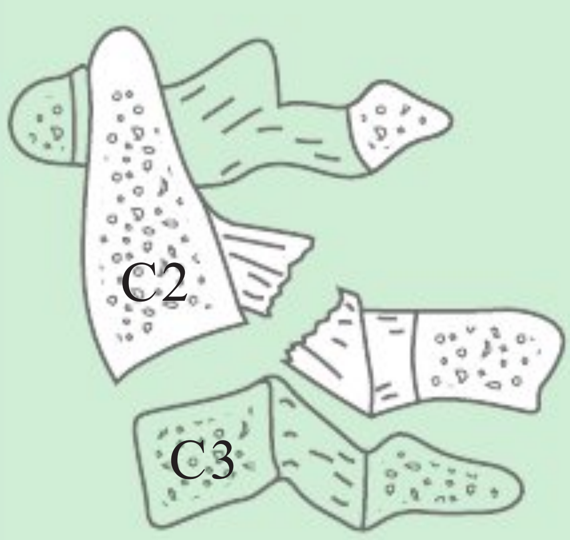
The methodology of measurements is depicted in Fig. 64.3.

Levine/Francis correlation

In a series of 340 axis fractures,[55] the most common fracture type was Type I in the Levine system (72%) and Grade I in the Francis system (65%); and there was a close correlation as follows:

Levine Type I ≈ Francis Grade I
Levine Type III ≈ Francis Grade IV

Table 64.7 Levine classification of hangman's fractures (modified Effendi system)[a]

Type	Description	Radiographic Findings	Mechanism	Comment
I	vertical pars fx just posterior to the VB	≤ 3 mm subluxation of C2 on C3 & no angulation	axial loading & extension	stable on flexion/extension x-rays. Neurologic deficit rare
I A	fx lines on each side are not parallel. Fx may pass thru foramen transversarium on one side	fx line may not be visible on x-ray. Anterior C2 VB may be subluxed 2–3 mm anteriorly on C3 & the C2 VB may appear elongated.	may be hyperextension + lateral bending	"atypical hangman's fracture".[52] Spinal canal may be narrowed. 33% incidence of paralysis
II	vertical fx thru pars. Disruption of C2–3 disc & posterior longitudinal ligament	subluxation of C2 on C3 > 3 mm and/or angulation[b]. Slight anterior compression of C3 possible	axial loading & extension with rebound flexion	may lead to early instability. Neurologic deficit rare. Usually reduces with traction
IIA	oblique fx (usually anterior-inferior to posterior superior)  little subluxation (usually ≤ 3 mm) but more angulation (can be > 15°)		flexion distraction (posterior arch fails in tension)	rare (< 10%). Unstable. Traction → increased angulation & widening of disc space do not use traction
III	Type II + bilateral C2–3 facet capsule disruption. C2 posterior arch is free floating. Anterior longitudinal ligament may be disrupted or stripped off C3	facets of C2/C3 may be subluxed or locked	unclear, may be flexion (capsule disruption) followed by compression (isthmus fracture)	rare. Neurologic deficit may occur & may be fatal. Facet dislocation usually cannot be reduced by closed reduction. Traction may be dangerous (see text)

[a]Effendi et al.,[50] Levine and Edwards,[51] Sonntag and Dickman[27] and Levine[53]
[b]amount of angulation was not specified in original article, but > 10° has been suggested by some

Other fracture types

Not all fractures fit into one or both of these classification systems.[56] Example: coronally oriented fracture extending through the posterior C2 vertebral body.

Presentation

Most (≈ 95%) are neurologically intact, those few with deficits are usually minor (paresthesias, monoparesis…) and many recover within one month.[54] Almost all conscious patients will have cervical pain usually in the upper posterior cervical region, and occipital neuralgia is not uncommon.[57] There is a high incidence of associated head injury and there will be other associated C-spine injuries – e.g. C1 fracture (see above) or clay shoveler's fracture (p.988) – in ≈ one third, with most occurring in the upper 3 cervical levels. There are usually external signs of injury to the face and head associated with the hyperextending and axial force.

Evaluation

Cervical CT: with sagittal & coronal reconstructions should be done to fully assess the fracture.

CTA: should be done to evaluate the vertebral arteries if fracture extends through foramen transversarium (especially Levine Type IA) and in patients with symptoms suggestive of stroke. Some

Table 64.8 Francis grading[a] system for hangman's fracture

Grade	Angulation θ	Displacement
I	<11°	d <3.5 mm
II	>11°	
III	<11°	d >3.5 mm and d/b <0.5
IV	>11°	
V		disc disruption

[a]see Fig. 64.3 for definitions

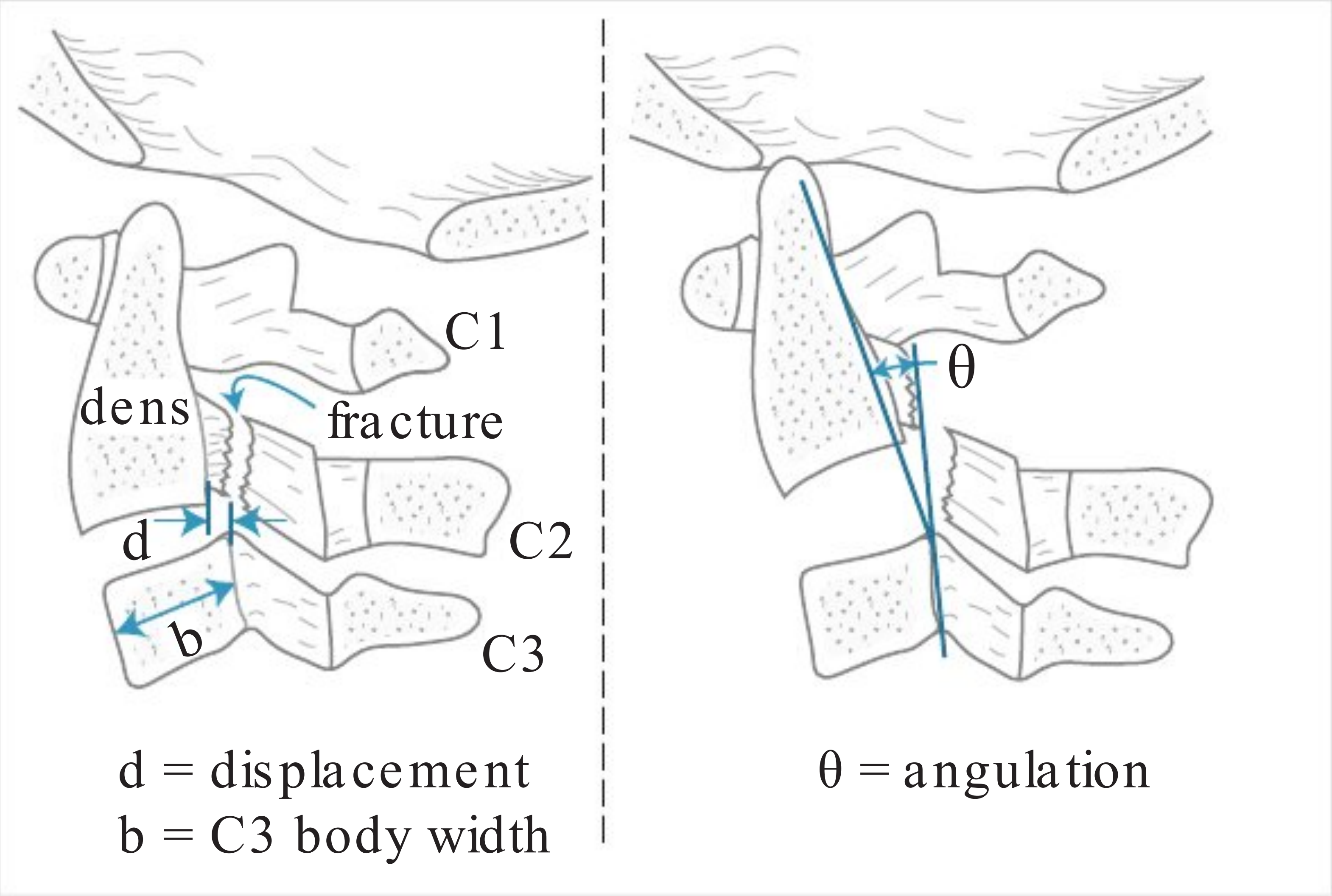


Fig. 64.3 Grading system of Francis

recommend CTA for all C2 fractures – Table 55.7). Angiography or MRA may be done as an alternative to CTA.

MRI: cervical MRI should be done to look for C2–3 disc disruption (a marker for instability (Levine grade II) which usually requires surgical stabilization). Findings may include abnormal increased signal intensity on MRI (best seen on sagittal FLAIR images or T2WI).

X-Rays: lateral C-spine x-rays show the fracture in 95% of cases. Also demonstrates C2 angulation and/or subluxation. Most fractures pass through the pars or the transverse foramen,[54] 7% go through the body of C2 (p.982). Instability can usually be identified as marked anterior displacement of C2 on C3 (guideline[54]: unstable if displacement exceeds 50% of the AP diameter of C3 vertebral body), excessive angulation of C2 on C3, or by excessive motion on flexion-extension films.

Patients suspected of having Levine Type I fractures and are neurologically intact should have physician-supervised flexion-extension x-rays to rule out a reduced type II fracture.

Treatment

General information

Nonsurgical management produces adequate reduction in 97–100% and results in a fusion rate of 93–100%[57,58,59] if the external immobilization is adequately maintained for 8–14 weeks[60] (average time for healing is ≈ 11.5 weeks[54]). Specific treatment depends on the reliability of the patient and the degree of stability as described below. Most cases do well with non-halo immobilization.[59] Practice guidelines are shown here, and details follow.

Practice guideline: Management of isolated hangman's fracture

Level III[61,62]

- hangman's fractures may initially be managed with external immobilization in most cases (halo or collar)
- surgical stabilization should be considered in cases of:
  a) severe angulation of C2 on C3 (Levine II, Francis II & IV)
  b) disruption of the C2–3 disc space (Levine II, Francis V)
  c) or inability to establish or maintain alignment with external immobilization

Stable fractures (Levine Types I or IA, or Francis Grades I or II)

Treat with immobilization (Aspen or Philadelphia collar[63 (p 2326)] or cervicothoracic orthosis (CTO) (e.g. SOMI) is usually adequate) × 3 months.[53] Halo-vest may be needed in unreliable patients or for combination C1-C2 fractures. Schneider reported 50 cases of Type I fracture treated with non-halo fixation, only 1 was taken to surgery and was found to already be fused.

Unstable fractures

Levine Type II

Reduce with gentle cervical traction (most reduce with ≤30 lbs[53]) with the head in slight extension (preferably in halo ring) under close x-ray monitoring to prevent "iatrogenic hanging" in cases with ligamentous instability.[54] Place in halo vest × 3 months. Follow patients with serial x-rays. Stabilize surgically if fracture moves.

Type II fractures with ≤5 mm of subluxation and angulation <10°

Once reduced, apply halo-vest and begin to mobilize (usually within 24 hrs of injury). Verify that immobilization is adequate in the halo with upright lateral C-spine x-ray, operate if inadequate. After 8–12 weeks, change to Philadelphia collar or CTO until fusion is definitely complete (usually 3–4 months).

Type II fractures with >5 mm subluxation or ≥10° of angulation

Surgical fusion in these patients is recommended because of the following concerns:

1. risk of settling if immediately mobilized in halo-vest
2. healing with significant angulation may result in chronic pain
3. if not reduced, the gap may be too large for bony bridging using traction alone

Alternatively, cervical traction can be maintained for ≈ 4 weeks and then reduction should be reassessed 1 hour after removing weight from traction, and if stable, again 24 hours after mobilizing in a halo vest. If unstable, return to traction and repeat trial at 5 & 6 weeks. If still unstable at 6 weeks, surgical fusion is recommended.[53]

Levine Type IIA

 Traction will accentuate the deformity.[53] Fractures should be reduced by immediate placement in halo vest (bypassing traction) with extension and compression applied. Halo-vest immobilization × 3 months produces ≈ 95%union rate.

Levine Type III

 Reduction with traction may be dangerous with locked facets. ORIF is recommended.[27] MRI prior to surgery is recommended to assess the C2–3 disc. Can follow ORIF with halo-vest for the fracture, or can fuse at the same time as ORIF.

Surgical treatment

Indications

Few patients have indications for surgical treatment of HF, and include those with:
1. inability to reduce the fracture (includes most Levine Type III & some Type II)
2. failure of external immobilization to prevent movement at fracture site
3. traumatic C2–3 disc herniation with compromise of the spinal cord[64]
4. established non-union: evidenced by movement on flexion-extension film (p.956) [54]; all failures of nonoperative treatment had displacement > 4 mm[27]

Hangman's fractures likely to need surgery[55]:
1. Levine Type II or III
2. or Francis grade II, IV or V
3. or if either:
   a) anterior displacement of C2 VB > 50% of the AP diameter of the C3 VB
   b) or if angulation produces widening of either the anterior or posterior borders of the C2–3 disc space > the height of the normal C3–4 disc below

Surgical options

1. fusion techniques:
   a) posterior approach: if the fracture is not transfixed (osteosynthesis – see below) then a C1–2 fusion is required. This depends on the integrity of the C2–3 disc and facet joint capsules, otherwise a C1–3 fusion is required. Occasionally the occiput is incorporated as well. Options for C1–2 fusion:
      - C1–2 wiring and fusion
      - C1–2 lateral mass screws/rods (p.1481)
   b) anterior C2–3 discectomy[54] with fusion. Optional anterior plating or zero-profile graft/plate. Performed via a transverse anterior cervical incision midway between the angle of the jaw and the thyroid cartilage[58,64]
      - preserves more motion by excluding C1
      - this approach is also recommended for established non-union[54]
      - not optimal for Levine Type III requiring ORIF for locked facets
      - also used when at least a partial reduction cannot be achieved
      - technique: for special considerations for approach to the C2–3 junction, see the section on operative techniques
2. osteosynthesis: screw placement from posterior approach through the C2 pedicle across the fracture fragment.[53 (p 443)] Reduction must be achieved before the screw holes are drilled.[65] The technique for C2 pedicle screws (p.1481) is used. The posterior fracture fragment may be overdrilled with a 3.5 mm drill. A "top hat" is placed in the hole and a 2.7 mm drill is used to drill the VB. Screw length: 30–35 mm for average adults. Alternatively, a lag screw may be used (with 20 mm unthreaded)

Treatment endpoint

Plain x-rays should show trabeculation across the fracture site or interbody fusion of C2 to C3. Flexion-extension lateral radiographs should show no movement at the fracture site.

64.5.4 Odontoid fractures

General information

Key concepts

- 10–15% of C-spine fx. Can occur in older patients with minor trauma (GLF), or in younger patients typically following MVA, falls from a height, skiing…
- may be fatal at time of injury, most survivors are intact. Neck pain is common
- classification: Anderson & D'Alonzo (Table 64.9). Type II (at base) is the most common
- Tx: surgery is considered for: Type II if age > 50 yrs, Type IIA, or Type II & III if displacement ≥ 5 mm or if alignment cannot be maintained with halo

Significant force is required to produce an odontoid fracture in a young individual, and is usually sustained in a motor vehicle accident (MVA), a fall from a height, a skiing accident, etc. In patients > 70 years age, simple ground level falls (GLF) with head trauma may produce the fracture. Odontoid fractures comprise ≈10–15% of all cervical spine fractures.[66] They are easily missed on initial evaluation, especially since significant associated injuries are frequent and may mask symptoms. Pathologic fractures can also occur, e.g. with metastatic involvement (p.1391).

Flexion is the most common mechanism of injury, with resultant anterior displacement of C1 on C2 (atlantoaxial subluxation). Extension only occasionally produces odontoid fractures, usually associated with posterior displacement.

Signs and symptoms

The frequency of fatalities at the time of the accident resulting directly from odontoid fractures is unknown, it has been estimated as being between 25–40%[67] 82% of patients with Type II fractures in a review of 7 reports in the literature were neurologically intact, 8% had minor deficits of scalp or limb sensation, and 10% had significant deficit (ranging from monoparesis to quadriplegia).[68] Type III fractures are rarely associated with neurologic injury.

Common symptoms are high posterior cervical pain, sometimes radiating in the distribution of the greater occipital nerve (occipital neuralgia). Almost all patients with high posterior cervical pain will also have paraspinal muscle spasm, reduced range of motion of the neck, and tenderness to palpation over the upper cervical spine. A very suggestive finding is the tendency to support the head with the hands when going between the upright and supine position. Paresthesias in the upper extremities and slight exaggeration of muscle stretch reflexes may also occur. Myelopathy may develop in patients with non-union (p.980).

Classification

The most widely used classification system of Anderson and D'Alonzo[69] is shown in Fig. 64.4 and Table 64.9.

Type I fractures are due to avulsion of the attachment of the alar ligament. They are very rare. Although long considered to be a stable injury, they may not occur as an isolated fracture and may

Table 64.9 Anderson and D'Alonzo classification of odontoid fractures

Type	Characteristics	Stability
I	through tip (above transverse ligament), rare	unstable[a]
II	through base of neck the most common dens fracture (may be best seen on AP x-ray)	usually unstable
IIA	similar to type II, but with large bone chips at fracture site,[70] comprise ≈ 3% of type II odontoid fractures. Diagnosed by plain radiographs and/or CT	usually unstable
III	through body of C2 (usually involves marrow space). May involve superior articular surface	usually stable

[a]controversial, see text

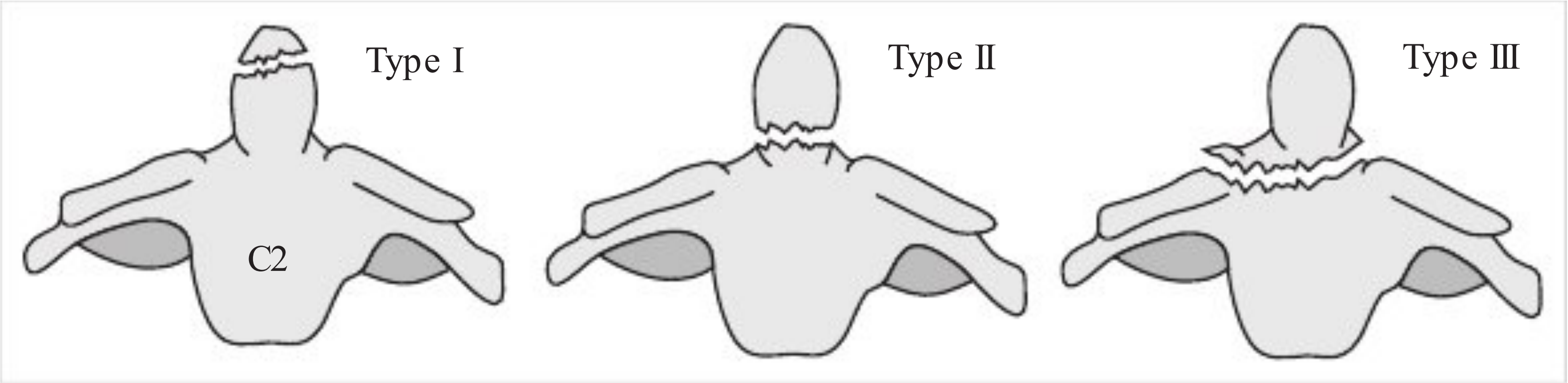


Fig. 64.4 Major types of odontoid fractures (AP view)

be a manifestation of atlanto-occipital dislocation.[71] Also, it may be a marker for possible disruption of the transverse ligament[72] which may result in atlanto-axial instability.

 Imaging pearl. A type III odontoid fracture may be misinterpreted as type II on sagittal CT reconstructions because the fracture appears to lie above the VB. Always check the coronal reconstruction which more readily demonstrates the relationship of the fracture to the VB.

Treatment

Practice guidelines

Practice guidelines are shown below. Details appear in the following sections.

Practice guideline: Management of isolated odontoid fractures

- Level II[62]: isolated Type II odontoid fractures in adults ≥ 50 years age should be considered for surgical stabilization & fusion
- Level III[62]
  - Nondisplaced type I, II & III fractures may be managed initially with external cervical immobilization, recognizing that type II odontoid fractures have a higher rate of nonunion
  - Type II & III: consider surgical fixation for:
    a) dens displacement ≥ 5 mm
    b) or Type IIA fracture (comminution of fracture)
    c) or inability to maintain or achieve alignment with external immobilization
  - For surgical intervention, either an anterior or posterior approach may be used

Immobilization

For those not meeting surgical indications, 10–12 weeks of immobilization as suggested in Table 64.10 is recommended. There is no Class I medical evidence comparing immobilization options.

Halo vest: fusion rate = 72%,[73] appears superior to a SOMI. If a halo is used, obtain supine and upright lateral C-spine x-rays in the halo. If there is movement at the fracture site, then surgical stabilization is recommended.

Rigid collar[73,74]: fusion rate = 53%

In patients who are poor surgical candidates, there is theoretical and anecdotal rationale to consider calcitonin therapy (p. 1010) in conjunction with a rigid cervical orthosis.[75]

Type I

So rare that meaningful analysis is difficult. If there is associated atlanto-axial instability, surgical fusion may at times be necessary.

Type II

General information

Treatment remains controversial. No agreement has been reached after many attempts to identify factors that will predict which type II fractures are most likely to heal with immobilization and

Table 64.10 Immobilization for odontoid fractures

Fracture Type	Option
Type I	collar, halo
Type II[a]	halo, collar[a]
Type IIA[a]	halo[a]
Type III[a]	collar, halo[a]

[a]consider surgery for these, use indicated brace when surgery not deemed appropriate

which will require operative fusion. Critical review of the literature reveals a paucity of well designed studies. A wide range of nonunion rates with immobilization alone (5–76%) is quoted: 30% is probably a reasonable estimate for overall nonunion rate, with 10% nonunion rate for those with displacement < 6 mm.[73] Possible key factors in predicting nonunion include:

1. degree of displacement: probably the most important factor
   a) some authors feel that displacement > 4 mm increases nonunion[69,76]
   b) some authors use ≥ 6 mm as the critical value, citing a 70% nonunion rate[60] in these regardless of age or direction of displacement
2. age:
   a) children < 7 yrs old almost always heal with immobilization alone
   b) some feel that there is a critical age above which the nonunion rate increases, and the following ages have been cited: age > 40 yrs (possibly ≈ doubling the nonunion rate),[76] age > 55 yrs,[77] age > 65 yrs,[78] yet others do not support increasing age as a factor[73]

Indications for surgery for odontoid Type II fractures

Given the above, there can be no hard and fast rules. The following is offered as a guideline (also, above).

 Surgical treatment (instead of external immobilization) is recommended for odontoid Type II fractures in patients ≥ 7 years age with any of the following:

1. displacement ≥ 5 mm
2. instability at the fracture site in the halo vest (see below)
3. age ≥ 50 years: increases nonunion rate (with halo) 21-fold[79]
4. nonunion (see Table 64.11 for radiographic criteria) including firm fibrous union,[80] especially if accompanied by myelopathy[57]
5. disruption of the transverse ligament: associated with delayed instability[37]

Surgical options

1. odontoid compression screw (p. 1480): appropriate for acute type II fractures with transverse ligament intact and attached
2. C1–2 arthrodesis (p. 1479): for options including wiring/fusion, transarticular screws, halifax clamps…

Type IIA

Early surgery is recommended for all type IIA fractures.[70]

Type III

≈ 90% heal with external immobilization (and analgesics) if adequately maintained for 8–14 weeks.[60] Halo-vest brace is probably best,[74] fusion rate ≈ 100% in 1 series.[73] Rigid collar: fusion rate = 50–70%; if used, monitor the patient with frequent C-spine x-rays to rule-out nonunion.

Surgical treatment options

See Atlantoaxial fusion (C1–2 arthrodesis) (p. 1479) and Anterior odontoid screw fixation (p. 1476) for surgical options and operative details.

Nonunion

The radiographic criteria for nonunion are shown in Table 64.11.

Table 64.11 Radiographic criteria of nonunion of odontoid fractures
• defect in the dens with contiguous sclerosis of both fragments (vascular pseudarthrosis) • defect in the dens with contiguous resorption of both fragments (rarefying osteitis or atrophic pseudarthrosis) • defect in the dens with definite loss of cortical continuity • movement of dens fragment demonstrated on flexion-extension x-rays

The most common symptom of nonunion is continued high posterior cervical pain beyond the time that the brace is removed. Late myelopathy can develop in as many as 77% of mobile nonunions[67,81] as a result of motion and soft tissue proliferation around the unstable fracture site.

Os odontoideum

General information

A separate bone ossicle of variable size with smooth cortical borders separated from a foreshortened odontoid peg, occasionally may fuse with the clivus. May mimic Type 1 or 2 odontoid fracture. Etiology is debated with evidence to support both of the following (diagnosis & treatment do not depend on which etiologic theory is correct):

1. congenital: developmental anomaly (nonunion of dens to body of axis). However, does not follow known ossification centers (Fig. 12.4) and has been demonstrated in 9 patients with previously normal odontoid processes[82]
2. acquired: postulated to represent an old nonunion fracture or injury to vascular supply of developing odontoid[82,83]

True os odontoideum is rare. Ossiculum terminale: nonunion of the apex at the secondary ossification center, is more common.

Two anatomic types:

1. orthotopic: ossicle moves with the anterior arch of C1
2. dystopic: ossicle is functionally fused to the basion. May sublux anterior to the C1 arch

Presentation

Main groups identified in the literature[84]:

1. occipitocervical/neck pain
2. myelopathy: further subdivided[82]
   a) transient myelopathy: common following trauma
   b) static myelopathy
   c) progressive myelopathy
3. intracranial signs or symptoms: from vertebrobasilar ischemia
4. incidental finding

Most patients are neurologically intact and present with atlantoaxial instability which may be discovered incidentally. Many symptomatic and asymptomatic patients have been reported with no new problems over many years of follow-up.[85] Conversely, cases of precipitous spinal cord injury after seemingly minor trauma have been reported.[86]

Σ

The natural history is variable, and predictive factors for deterioration, especially in asymptomatic patients, have not been identified.[87]

Evaluation

Practice guideline: Diagnosis of os odontoideum

Level III[88]

- recommended: the following plain C-spine x-rays: AP, open-mouth odontoid, lateral (static & flexion-extension) with or without tomography (CT or plain) and/or MRI of craniocervical junction

It is critical to R/O C1–2 instability. However, myelopathy does not correlate with the degree of C1–2 instability. An AP canal diameter < 13 mm does correlate with the presence of myelopathy.

Treatment

Regardless of whether os odontoideum is congenital or an old non-union fracture, immobilization is unlikely to result in fusion. Therefore, when treatment is elected, surgery – usually atlantoaxial arthrodesis (p. 1479) – is required.

Practice guideline: Management of os odontoideum

Level III[88]

- patients without neurologic signs or symptoms:
  - may be followed with clinical & radiographic surveillance
  - or posterior C1–2 fusion may be done
- patients with neurologic signs or symptoms or C1–2 instability: posterior C1–2 internal fixation and fusion
- if surgery is done: post-op halo immobilization is recommended (e.g. following posterior wiring & fusion) unless rigid internal instrumentation is used
- for patients with irreducible cervicomedullary compression and/or evidence of associated occipitoatlantal instability: occipital-cervical fusion ± C1 laminectomy
- for patients with irreducible cervicomedullary compression, consider ventral decompression

64.5.5 Miscellaneous C2 fractures

Comprise ≈ 20% of C2 fractures.[27] Includes fractures of spinous process, lamina, facets, lateral mass or C2 vertebral body. Fractures of spinous process or lamina may be treated with Philadelphia collar or cervicothoracic orthosis (CTO). Fractures which compromise the anterior or middle columns (i.e. fractures of facets, C2 body, or lateral mass) requires CTO or halo-vest if nondisplaced, or halo if displaced.

Practice guideline: Management of fractures of the axis (C2) body

Level III[61,62]:

- fractures may initially be managed with external immobilization in most cases (halo or collar)
- surgical stabilization should be considered in cases of:
  a) severe ligamentous instability
  b) or inability to establish or maintain alignment with external immobilization
- evaluate for vertebral artery injury in cases of comminuted fracture of the axis body

64.6 Combination C1–2 injuries

64.6.1 General information

Combination C1–2 injuries are relatively common and may imply more significant structural and mechanical injury than isolated C1 or C2 fractures. The frequency of C2 fractures in C1–2 combination injuries is shown in Table 64.12. 5–53% of patients with Type II or III odontoid fractures and 6–26% of hangman's fractures have an associated C1 fracture.[89]

Table 64.12 Accompanying C2 injuries

Injury	%
Type II dens fracture	40%
Type III dens fracture	20%
hangman's fracture	12%
other	28%

Table 64.13 Treatment options for combination C1-C2 injuries

Injury	Treatment options
C1 + hangman's	
• stable	collar, halo, surgery[a]
• unstable (C2–3 angulation ≥ 11°)	halo, surgery
C1 + Type II odontoid fracture	
• stable (ADI[a] < 5 mm)	collar, halo, surgery
• unstable (ADI ≥ 5 mm)	halo, surgery
C1 + Type III odontoid fracture	halo
C1 + miscellaneous C2	collar, halo

[a]abbreviations: ADI = atlantodental interval; surgery = surgical fixation & fusion

64.6.2 Treatment

Practice guideline: Treatment of combination atlas and axis fractures

Level III[89]

1. recommended: base treatment primarily on the type of C2 injury
2. recommended: external immobilization of most C1–2 fractures
3. consider surgical stabilization for these situations. Note: loss of integrity of the C1 ring may necessitate modification of the surgical technique; these injuries are potentially unstable: see Axis (C2) fractures (p. 972):
   a) C1-Type II odontoid combination fractures with an ADI ≥ 5 mm
   b) C1-hangman's combination fractures with C2–3 angulation ≥ 11°

Treatment options are summarized in Table 64.13.[89]

64.6.3 Outcome

Only 1 nonunion (C1 + Type II odontoid, treated initially with halo). No new neuro deficits.

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65 Subaxial (C3 through C7) Injuries / Fractures

65.1 Classification systems

65.1.1 General information

Various systems have been proposed to help assess stability and/or guide management. The Allen-Ferguson system (p.987) is based on the mechanism of injury. Attempts at quantifying biomechanical stability include the White and Panjabi system (p.987) and the more recent subaxial injury classification (SLIC) (see below). Measurements for spine injuries are based on methods outlined by Bono et al.[1]

Practice guideline: Subaxial cervical spine injury classification

Level I[2]
- Use the Subaxial Injury Classification (SLIC) and severity scale for SCI (see section 65.1.2)
- Classify the stability and fracture pattern using the Cervical Spine Injury Severity Score (CSISS): the CSISS is somewhat complicated and may be better suited to clinical trials than daily practice (see reference[2])

65.1.2 Spine Trauma Study Group subaxial cervical spine injury classification (SLIC)

 General information. The subaxial injury classification (SLIC)[3] is shown in (Table 65.1) and assesses injuries to the disco-ligamentous complex (DLC) in addition to neurologic and bony injuries. Inter-rater reliability intraclass correlation coefficient is 0.71.

Table 65.1 Subaxial injury classification (SLIC)[3]

Injury (rate the most severe injury at that level)	Points
Morphology	
No abnormality	0
Simple compression (compression fx, endplate disruption, sagittal or coronal plane VB fx.)	1
Burst fracture	2
Distraction (perched facet, posterior element fx.)	3
Rotation/translation (facet dislocation, teardrop fx., advanced compression injury, bilateral pedicle fx., floating lateral mass (p.994). Guidelines: relative axial rotation ≥ 11°[4] or any translation not related to degenerative causes	4
Discoligamentous complex (DLC)	
Intact	0
Indeterminate (isolated interspinous widening with < 11° relative angulation & no abnormal facet alignment, ↑ signal on T2WI MRI in ligaments…)	1
Disrupted (perched or dislocated facet, <50% articular apposition, facet diastasis > 2 mm, widened anterior disc space, ↑ signal on T2WI MRI through entire disc…)	2
Neurologic status	
Intact	0
Root injury	1
Complete spinal cord injury	2
Incomplete spinal cord injury	3
● Continuous cord compression with neuro deficit	+ 1

 DLC integrity[3] . The DLC includes: anterior longitudinal ligament (the strongest component of the anterior DLC), posterior longitudinal ligament, ligamentum flavum, facet capsule (the strongest component of the posterior DLC), interspinous and supraspinous ligaments. The DLC is the hardest SLIC parameter to evaluate. Largely inferred indirectly from MRI findings. Healing is less predictable than bone healing in the adult. More data needs to be accrued before this parameter can be reliably quantified.

 Management based on the total SLIC score is shown in Table 65.2.

 A given injury can be described using the SLIC as follows:
1. spinal level
2. SLIC morphology (from Table 65.1): use the most severe injury type at this level
3. description of bony injury: e.g. fracture or dislocation of transverse process, pedicle, endplate, superior or inferior articular process, lateral mass…
4. SLIC DLC status (from Table 65.1) with descriptors: e.g. herniated disc…
5. SLIC neurologic status (from Table 65.1)
6. confounders: e.g. presence of ankylosing spondylitis, DISH, osteoporosis, previous surgery, degenerative disease…

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65.1.3 Cervical spine injury classification on the basis of mechanism of trauma

A modification of the Allen-Ferguson system[5] divides cervical spine fracture/dislocations into 8 major groups based on the dominant loading force and neck position at the time of injury as shown in Table 65.3. Grades of severity within each group are described, and any of these fractures may also be associated with damage from rotatory loads.

Details on some of these fracture types are provided in following sections.

65.1.4 Stability model of White and Panjabi

Guidelines for determination of clinical instability (p. 930) of the subaxial cervical spine published by White and Panjabi[6 (p 314)] are shown in Table 65.4. In general, all else being equal, compromise of

Table 65.2 Management based on total SLIC score

SLIC score	Management
1–3	non surgical
4	not specified
≥ 5	surgical

Table 65.3 Examples of types of cervical spine injuries[a]

Major loading force	Acting alone	With compression	With distraction
Flexion (p. 989)	unilateral or bilateral facet dislocation (p. 992)	• anterior VB fx. with kyphosis • disruption of interspinous ligament • teardrop fx (p. 989)	• torn posterior ligaments (may be occult) • dislocated or locked facets (p. 992)
Extension[b] (p. 994)	fractured spinous process and possibly lamina[b]	fracture through lateral mass or facet[b], including horizontalization of facet (p. 994)	disruption of ALL with retrolisthesis of superior vertebrae on inferior one[b]
Neutral position		burst fracture (p. 989)	complete ligamentous disruption (very unstable)

[a]abbreviations: ALL = anterior longitudinal ligament; VB = vertebral body, fx = fracture; numbers in parentheses are page numbers for that topic
[b]any of the extension injuries may produce SCIWORA in young patients, or central cord syndrome in the presence of stenosis

Table 65.4 Guidelines for diagnosing clinical instability of the mid & lower C-spine[6]

Item	Points[a]
anterior elements[b] destroyed or unable to function	2
posterior elements[b] destroyed or unable to function	2
positive stretch test[c]	2
spinal cord damage	2
nerve root damage	1
abnormal disc narrowing	1
developmentally narrow spinal canal, either • sagittal diameter < 13 mm, OR • Pavlov ratio[d] < 0.8	1
dangerous loading anticipated[e]	1
Radiographic criteria	
neutral position x-rays	
• sagittal plane displacement > 3.5 mm or 20%	2
• relative sagittal plane angulation > 11°	2
OR	
flexion-extension x-rays	
• sagittal plane translation > 3.5 mm or 20%	2
• sagittal plane rotation > 20°	2
Unstable if total ≥ 5	

[a]if there is inadequate information for any item, add half of the value for that item to the total
[b]in the C-spine, posterior elements = anatomic components posterior to the posterior longitudinal ligament
[c]stretch test: apply incremental cervical traction loads of 10 lbs q 5 mins up to 33% body wt. (65 lbs max). Check X-ray and neuro exam after each ≈. Positive if ≈ in separation > 1.7 mm or ≈ angle > 7.5° on x-ray or change in neuro exam. This test is contraindicated if obvious instability
[d]Pavlov ratio = the ratio of (distance from the midlevel of the posterior VB to the closest point on the spinolaminar line): (the AP diameter of the middle of the VB)
[e]e.g. heavy laborers, contact sports athletes, motorcyclists

anterior elements produces more instability in extension, whereas compromise of the posterior elements produces more instability in flexion (important in patient transfers and immobilization). NB: certain conditions such as ankylosing spondylitis (p. 1123) may cause an otherwise stable injury to be unstable.

Stretch test: The cervical stretch test may be helpful in cases where stability is difficult to determine based on other factors. It may also be useful in detecting instability in cases such as an athlete with no obvious bony or ligamentous disruption. It is performed by applying graduated cervical traction with the patient lying supine on an x-ray table. Serial neurologic exams and lateral radiographs are performed as outlined in the footnote of Table 65.4.

65.2 Clay shoveler's fracture

Avulsion of spinous processes (usually C7) first described in Perth, Australia (pathomechanics: during the throwing phase of shoveling, clay may stick to the shovel jerking the trapezius and other muscles which are attached to cervical spinous processes).[7] Can also occur with: whiplash injury,[8] injuries that jerk the arms upwards (e.g. catching oneself in falling), neck hyperflexion, or a direct blow to the spinous process.

This fracture is stable, and by itself poses little risk. If the patient is intact, they should have further study (flexion-extension C-spine x-rays or CT scan through the affected level) to R/O other occult fractures. A rigid collar is used PRN pain.

65.3 Vertical compression injuries

In order to apply a purely compressive force to the cervical spine without flexion or extension, reversal of the normal cervical lordosis is required, as may occur in a slightly flexed posture. Burst fractures are the most common result, with the possibility of retropulsion of bone into the spinal canal with neurologic deficit.

65.4 Flexion injuries of the subaxial cervical spine

65.4.1 General information

Constitute up to 15% of cervical spine trauma. Common causes include: MVAs, falls from a height, and diving into shallow water.[9]

65.4.2 Compression flexion injuries

The classic diving injury is the prototypical example. Posterior element fractures occur in up to 50% of compression flexion injuries.[10] Although flexion-compression injuries do distract the posterior elements to some degree, most do not produce posterior ligamentous injuries. Subtypes of compression-flexion fractures include: teardrop fractures (see below), quadrangular fractures (p.991).

Treatment: mild cervical compression fractures without neurologic deficit or retropulsion of bone into the spinal canal are usually treated with a rigid orthosis until x-rays show healing has occurred (usually 6–12 wks). Stability is assessed with flexion-extension views (p.956) before completely discontinuing the brace. More severe compression fractures heal in a halo brace with ≈ 90% rate of ankylosing fusion.

65.4.3 Teardrop fractures

General information

Originally described by Schneider & Kahn.[11] Results from hyperflexion or axial loading at the vertex of the skull with the neck flexed (eliminating the normal cervical lordosis)[12] (often mistakenly attributed to hyperextension because of the retrolisthesis). Two forces are involved: 1) compression of the anterior column, and 2) tension on DLC. There are varying degrees of severity. In its most severe form, the injury consists of complete disruption of all of the ligaments, the facet joints and the intervertebral disk[13] and ≥ 3 mm of posterior displacement of the body into the canal. As originally described, an important feature is displacement of the inferior margin of the fractured vertebral body posteriorly into the spinal canal.[11] Usually unstable.

Seen in ≈ 5% of patients in a large series of patients with x-ray evidence of cervical spine trauma.[14] Patients are often quadriplegic, although some may be intact and some may have anterior cervical cord syndrome (p.947).

Findings

Possible associated injuries and radiographic findings include[13,15]:

1. a small chip of bone (the "teardrop") just beyond the anterior inferior edge of the involved vertebral body (VB) on lateral cervical spine film
2. often associated with a fracture through the sagittal plane of the VB (sagittal split) which can almost always be seen on AP view (may be midline or off-center). Thin cut CT scan is more sensitive
3. a large triangular fragment of the anterior inferior VB
4. other fractures through the vertebral body may also occur
5. the fractured vertebrae is usually displaced posteriorly on the vertebra below (easily appreciated on oblique x-rays, ▯ Fig. 65.1). However, cases without retrolisthesis are also described[10]
6. the fractured body is often wedged anteriorly (kyphosis), and may also be wedged possibly laterally
7. disruption of the facet joints which may be appreciated as separation of the joints on lateral x-ray, often unmasked by cervical traction

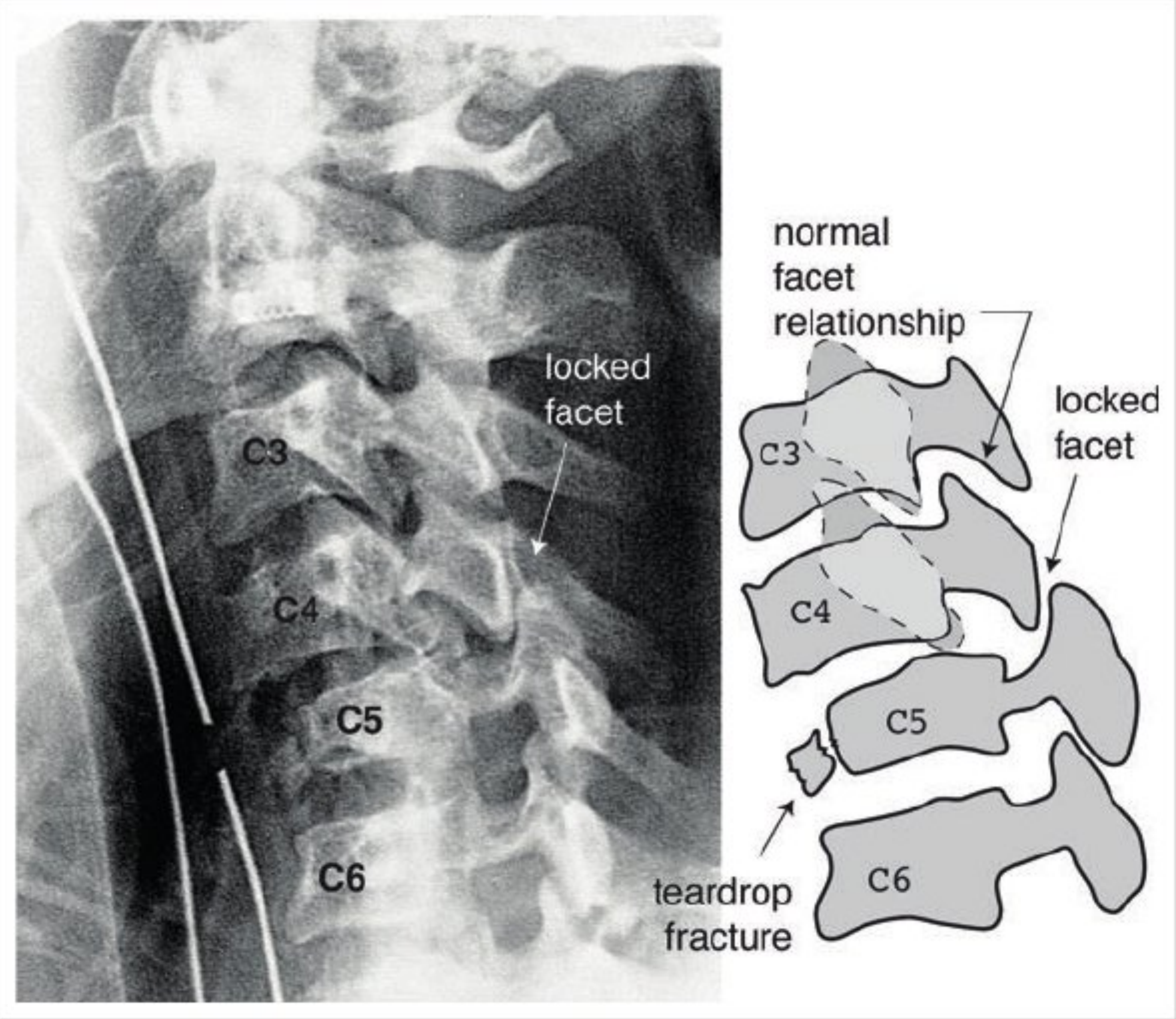


Fig. 65.1 Unilateral locked facets (left C4 on C5) & C5 teardrop fracture (p.989). 60° LAO C-spine x-ray on left, and schematic on right (sagittally oriented VB fracture through C5 seen on CT scan, not shown). Note the anterior subluxation of C4 on C5, and the slight retrolisthesis of C5 on C6

8. prevertebral soft-tissue swelling, see for measurements (p.214)
9. narrowing of the intervertebral disc below the fracture (indicating disruption)

Distinguishing between teardrop fracture and avulsion fracture

Rationale: Teardrop fractures must be distinguished from a simple avulsion fracture which may also result in a small chip of bone off the anterior inferior VB, usually pulled off by traction of the anterior longitudinal ligament (ALL) in hyperextension. Although there may be disruption of the ALL in these cases, it does not usually cause instability.

Methodology: In a patient with a small bone chip off of the inferior anterior VB, a "teardrop" fracture needs to be ruled-out. Determine if the following criteria are met:
- neurologically intact (because of the need for cooperation, this includes mental status, and excludes the inebriated or concussed patient)
- size of bone fragment is small
- no malalignment of vertebral bodies
- no evidence of VB fracture in sagittal plane on AP C-spine x-rays or on CT
- no posterior element fracture on x-ray or CT
- no prevertebral soft tissue swelling (p.214) at level of fragment
- and no loss of vertebral body height or disc space height

If the above criteria are met, obtain flexion-extension C-spine x-rays (p.956). If no abnormal movement, discharge patient in rigid collar (e.g. Philadelphia collar), and repeat the films in 4–7 days (i.e. after the pain has subsided to be certain that alignment is not being maintained by cervical muscle spasm from pain), D/C collar if 2nd set of films is normal.

If the patient does not meet the above criteria, treat them as an unstable fracture and obtain a CT scan through the fractured vertebra to evaluate for associated fractures (e.g. sagittal plane fracture that may not be apparent on plain x-ray).

MRI assesses the integrity of the disc and gives some information about the posterior ligaments.

65.4.4 Treatment of teardrop fracture

If the disc and ligaments are intact (determined by MRI) then an option is to employ a halo brace until the fragment is healed (perform flexion-extension x-rays after removing the halo to rule-out persistent instability). Alternatively, surgical stabilization may be performed, especially if ligamentous or disc injury is seen on MRI. When the injury is primarily posterior due to disruption of the posterior ligaments and facet joints, and if there is no anterior compromise of the spinal canal, then posterior fusion suffices (p.998). Severe injuries with canal compromise often require a combined anterior decompression and fusion (performed first) followed by posterior fusion using either a modified Bohlman triple-wire technique or lateral mass screws and rods.

65.4.5 Quadrangular fractures

See reference.[16]

Four features:

1. oblique vertebral body (VB) fracture passing from anterior-superior cortical margin to inferior end plate
2. posterior subluxation of superior VB on the inferior VB
3. angular kyphosis
4. disruption of disc and anterior and posterior ligaments

Treatment:

May require combined anterior and posterior fusion.

65.5 Distraction flexion injuries

65.5.1 General information

Ranges from hyperflexion sprain (mild, see below) to minor subluxation (moderate) to bilateral locked facets (severe, see below). Posterior ligaments are injured early and are usually evidenced by widening of the interspinous distance (p.214).

65.5.2 Hyperflexion sprain

A purely ligamentous injury that involves disruption of the posterior ligamentous complex without bony fracture. May be missed on plain lateral C-spine x-rays if they are obtained in normal alignment; requires flexion-extension views (p.956). Instability may be concealed when films are obtained shortly after the injury if spasm of the cervical paraspinal muscles splints the neck and prevents true flexion.[17] For patients with limited flexion, a rigid collar should be prescribed, and if the pain persists 1–2 weeks later the films should be repeated (including flexion-extension).

Radiographic signs of hyperflexion sprain[18] (x-rays may also be normal):

1. kyphotic angulation
2. anterior rotation and/or slight (1–3 mm) subluxation
3. anterior narrowing and posterior widening of the disc space
4. increased distance between the posterior cortex of the subluxed vertebral body and the anterior cortex of the articular masses of the subjacent vertebra
5. anterior and superior displacement of the superior facets (causing widening of the facet joint)
6. fanning (abnormal widening) of the interspinous space on lateral C-spine x-ray, or increased interspinous distance on AP; see Interspinous distances (p.214)

65.5.3 Subluxation

Cadaver studies have shown that horizontal subluxation > 3.5 mm of one vertebral body on another, or > 11° of angulation of one vertebral body relative to the next indicates ligamentous instability[19,20] (Table 65.4). Thus, if subluxation of ≤ 3.5 mm on plain films is seen, and there is no neuro deficit, obtain flexion-extension films; see Flexion-extension cervical spine x-rays (p.956). If no abnormal movement, remove cervical collar.

65.5.4 Locked facets

General information

Severe flexion injuries can result in locked facets (AKA "sprung" facets AKA "jumped" facets) with reversal of the normal "shingled" relationship between facets (normally the inferior facet of the level above is posterior to the superior facet of the level below). Involves disruption of facet capsule. Facets that have not completely locked but have had significant ligamentous disruption allowing distraction just short of the point of locking are known as "perched facets."

Flexion + rotation → unilateral locked facets. Hyperflexion → bilateral locked facets.

Unilateral locked facets

25% of patients are neurologically intact, 37% have root deficit, 22% have incomplete cord injuries, and 15% are complete quadriplegics.[21]

Bilateral locked facets

Occurs with disruption of ligaments of apophyseal joints, ligamentum flavum, longitudinal and interspinous ligaments, and the anulus. Rare. Most common at C5–6 or C6–7. 65–87% have complete quadriplegia, 13–25% incomplete, ≤ 10% are intact. Adjacent fractures (VB, facet, lamina, pedicle…) occur in 40–60%[5,22] Nerve root deficits may also occur.

Diagnosis

Sagittal CT: usually the optimal manner to identify locked facets.

C-spine x-rays: both unilateral (ULF) and bilateral locked facets (BLF) will produce subluxation (ULF → rotatory subluxation).

BLF: usually produces > 50% subluxation on lateral C-spine x-ray.

ULF:

1. AP: spinous processes above the subluxation rotate to the same side as the locked facet (with respect to those below)
2. lateral: "bow-tie sign" (visualization of left & right facets at the level of the injury instead of the normal superimposed position[21]). Subluxation may be seen. Disruption of the posterior ligamentous complex may produce widening of the interspace between spinous processes
3. oblique (Fig. 65.1): may demonstrate the locked facet which will be seen blocking the neural foramen (use ≈ 60° LAO for left locked facet, 60° RAO for right; LAO = left anterior oblique, RAO = right anterior oblique: e.g. with RAO the patient is angled with the right shoulder closer to the film)

Axial CT: "naked facet sign": the articular surface of the facet will be seen with the appropriate articulating mate either absent or on the wrong side of the facet (Fig. 65.2). With ULF, CT also demonstrates the rotation of the level above anteriorly on the level below on the side of the locked facet.

MRI: the best test to rule-out traumatic disc herniation (found in 80% of BLF).[23]

Treatment

Practice guidelines

See Practice guideline: Initial closed reduction in fracture/dislocation cervical SCI (p.958).

Closed reduction of locked facets

Contraindicated if traumatic disc herniation is demonstrated on MRI. Patients who cannot be assessed neurologically may be done using SSEP/MEP monitoring. Two methods of closed reduction:

1. traction: more commonly employed in the U.S.
   a) initial weight (in lbs) ≈ 3 × cervical vertebral level, increase in 5–10 lb increments usually at 10–15 minute intervals until desired alignment is attained (assess neurologic exam (or SSEP/MEP) and lateral C-spine x-ray or fluoroscopy after each ≈ to avoid overdistraction)
   b) end points (i.e. stop the procedure):

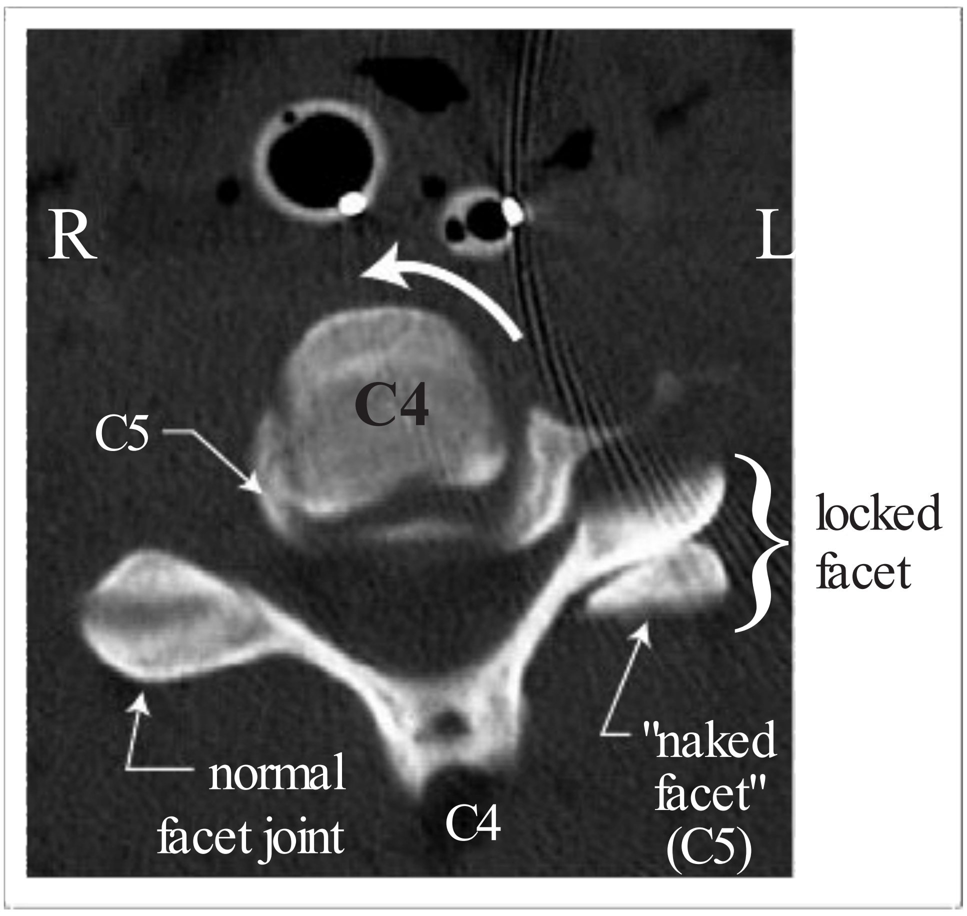


Fig. 65.2 Locked facet (left C4–5). (CT scan). Note the rotation of the C4 vertebral body on C5 (curved arrow)

- do not exceed 10 lbs per vertebral level (some say 5 lbs/level) under most circumstances. This is a guideline – you are trying to avoid overdistraction at the index level and at normal levels
- distraction of perched/locked facet or desired reduction is achieved
- if occipitocervical instability develops
- if any disc space height exceeds 10 mm (overdistraction)
- if any neurologic deterioration or deterioration of SSEP/MEP

c) with unilateral locked facets, one may add gentle manual torsion towards the side of the locked facets. With bilateral locked facets, one may add gentle manual posterior tension (e.g. with a rolled towel under the occiput)

d) once the facets are perched or distracted, gradual reduction of the weights will usually result in reduction – verify with x-ray (placing the neck in slight extension, e.g. with small shoulder roll, may help maintain the reduction)

2. manipulation (usually under anesthesia): less commonly employed,[21] more frequently used in Europe. Involves manually applying axial traction and sagittal angulation sometimes with rotation and direct pressure at the fracture level under fluoroscopy

Paraspinal muscle relaxation (but not enough to cause obtundation) may assist in reduction. Use IV diazepam (Valium®) and/or narcotic. General anesthesia may be used in difficult cases (with SSEP/MEP monitoring).

Once reduction is achieved, the patient is left in 5–10 lbs of traction for stabilization.

Disadvantage of closed reduction

1. fails to reduce ≈ 25% of cases of BLF
2. risks overdistraction at higher levels or worsening of other fractures
3. neurologic worsening following closed reduction may occur with traumatic disc herniation[22,24] and should be evaluated immediately with MRI and if confirmed treated with prompt discectomy
4. adds time and potentially pain to the patient's care, especially since many will go on to have surgical fusion anyway

Following closed reduction, the need for internal (operative) stabilization vs. external stabilization (i.e. bracing) may be addressed (below).

Open reduction and fixation is usually required if reduction is not achieved. Closed reduction is often more difficult with bilateral locked facets than with unilateral.

Open reduction of locked facets

1. posterior approach: the most common approach. Although rare, still subjects the patient to risk of deterioration from traumatically herniated disc. Therefore a pre-op MRI should be done if

possible. Often requires drilling of the superior aspect of the articular facet of the level below. A foraminotomy is recommended when there are root symptoms to visualize and decompress the root
2. anterior approach: by removing the disc at the subluxed level and exploring the anterior epidural space, the risk of worsening deficit due to a traumatic herniated disc is theoretically reduced. Reduction may be achieved by adding simultaneous manual traction
3. combined anterior/posterior (360°) approach: using anterior plate and posterior lateral mass screws/rods eliminates need for post-op external immobilization

Stabilization

Surgical fusion is commonly performed after successful closed reduction, failed closed reduction, or following open reduction.

If there are fracture fragments about the articular surfaces, there may be satisfactory healing with halo vest immobilization (for 3 months) once closed reduction is achieved.[25] Frequent x-rays are needed to rule-out redislocation.[26] Flexion-extension x-rays are obtained upon halo removal and surgery is required for continued instability. Up to 77% of patients with unilateral or bilateral facet dislocation (with or without facet fracture fragments) will have a poor anatomic result with halo vest alone (although late instability was uncommon), suggesting that surgery should be considered for all of these patients.[27] Surgical fusion is more clearly indicated in cases without facet fracture fragments (ligamentous instability alone may not heal) or if open reduction is required.

If surgery is indicated, an MRI should be done beforehand if possible. A posterior approach is preferred if there are no anterior masses (such as traumatic disc herniation or large osteophytic spurs), if subluxation of the bodies is > one–third the VB width (suggesting severe posterior ligamentous injury), or for fractures of the posterior elements. A posterior approach is mandatory if there is an unreducible dislocation. See also Options for posterior approach (p. 998).

65.6 Extension injuries of the subaxial cervical spine

65.6.1 Extension injury without bony injury

Extension injuries can produce spinal cord injury (SCI) without evidence of bony injury. Injury patterns include central cord syndrome (p. 944) usually in an older adult with cervical spondylosis, and SCIWORA (see below) usually in young children. Middle aged adults with hyperextension dislocations that reduce spontaneously immediately may present with SCI and no bony abnormality on x-ray, but there may be rupture of the anterior longitudinal ligament ALL and/or intervertebral disc on MRI or autopsy. Extension forces may also be associated with carotid artery dissections (p. 1324).

65.6.2 Minor extension injuries

Results from extension acting alone. Includes spinous process and lamina fractures. By themselves, are stable.

65.6.3 Extension compression injury

This is the most common mechanism of lateral mass/facet fractures (see below).

65.6.4 Lateral mass and facet fractures of the cervical spine

General information

Often results from extension combined with compression.

Classification of cervical lateral mass and facet fractures

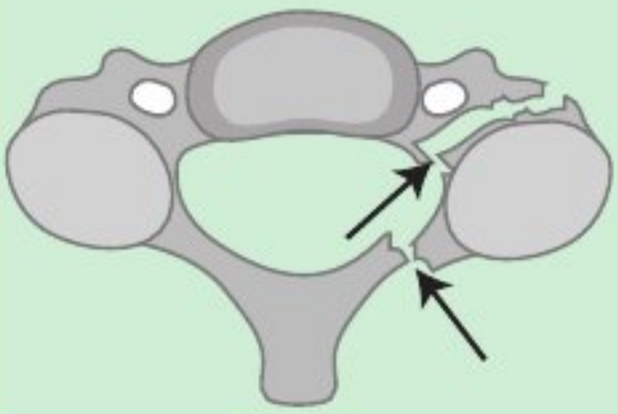
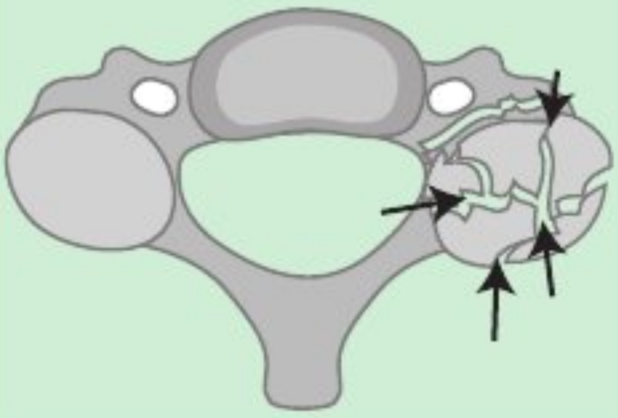
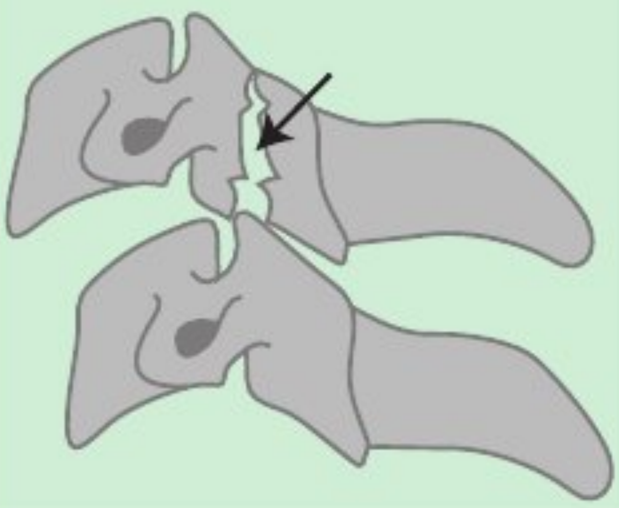
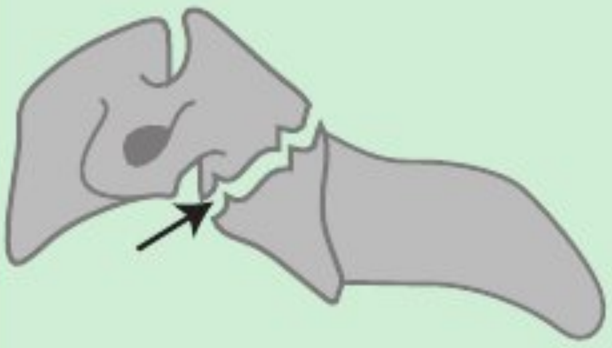
4 patterns identified in lateral mass and facet fractures[28] are shown in Table 65.5.

Anterior subluxation of the fractured vertebra was observed in 77% of whole lateral-mass fractures.[28]

Horizontal facet or separation fracture of the articular mass

Extension combined with compression and rotation may produce fracture of one pedicle and ipsilateral lamina which permits the detached articular mass ("floating" lateral mass) to rotate forward to

Table 65.5 Classification of cervical lateral mass & facet fractures[28]

Designation	Diagram	Description
separation fracture		fractures through lamina and ipsilateral pedicle. Permits horizontalization of facet[29] (see text)
comminuted fracture		multiple fractures. Often associated with lateral angulation deformity
split fracture		coronally oriented vertical fracture in 1 lateral mass, with invagination of the superior articular facet of the level below
traumatic spondylolysis		bilateral horizontal fractures through pars interarticularis, separating anterior spinal elements from posterior

a more horizontal orientation[29] (horizontalization of the facet) (Table 65.5). May be associated with rupture of the anterior longitudinal ligament (ALL) and fissure of the disc at one or two levels. Neuro deficit is common. Unstable.

Failure of nonoperative treatment

A study of CT scans of 26 unilateral cervical facet fractures[30] identified the risk factors shown below for failure of nonoperative treatment (Fig. 65.3 for illustration of the measurement definitions): where the fracture fragment (FF) height was defined as the maximum tip-to-tip cephalocaudal height on sequential sagittal reconstructions.

Nonoperative management is likely to fail if FF is:

1. >1 cm, or
2. >40% of LM (the height of the intact contralateral lateral mass at the same level, defined as the maximum tip-to-tip cephalocaudal height on sequential sagittal reconstructions)

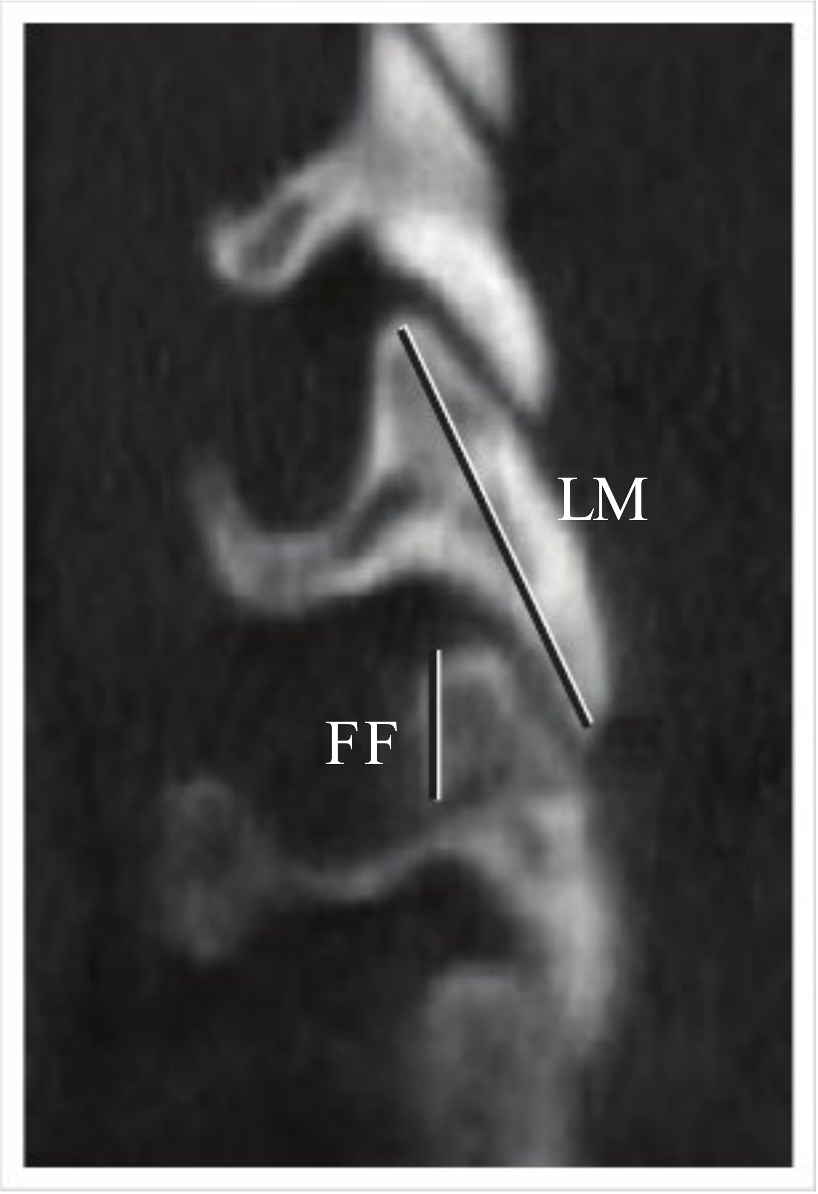


Fig. 65.3 Facet fracture fragment measurements. Sagittal reconstructed CT. FF = fracture fragment height, LM = lateral mass height (measured on the contralateral side at the same level as the fracture, (not as shown here which just illustrates the technique used to measure LM)

Surgical treatment of cervical lateral mass and facet fractures

Most cases can be treated with a posterior approach using fixation screws (lateral mass screws or pedicle screws[28]) and rods extending at least 1 level above and below the level of fracture (usually omitting a screw on the side of the fracture at the index level). Simultaneous neural decompression is performed when needed. Additional treatment with an anterior approach may be required for release of rigid deformity or for additional anterior column support.[28] Some separation fractures may be candidates for osteosynthesis (to preserve motion) using a cervical pedicle screw[28] that traverses the fracture.

An anterior approach is an alternative. Advantage: usually only 1 level needs to be fused. Disadvantages: decompression of compressing fragments cannot always be accomplished and requires disrupting an area that may not be compromised (if there is subluxation, the anterior column is probably compromised).

65.7 Treatment of subaxial cervical spine fractures

65.7.1 General information

Practice guideline: Treatment of subaxial cervical spine fractures or dislocations

Level III[31]
- closed or open reduction of subaxial fractures or dislocations with the goal of decompression of the spinal cord and restoration of the spinal canal
- stable immobilization either by internal fixation or by external immobilization to facilitate early patient mobilization and rehabilitation. If surgical treatment is employed, either anterior or posterior fixation is acceptable when a particular approach for decompression of the spinal cord is not required
- treatment with prolonged bed rest in traction if more contemporary treatment options are not available
- for patients with ankylosing spondylitis

- routine use of CT and MRI is recommended even after minor trauma
- when surgical stabilization is required, posterior long segment instrumentatin and fusion or a combined anterior/posterior procedure (360° fusion). Anterior standalone instrumentation and fusion procedures are associated with a failure rate of up to 50%in these patients

65.7.2 Management overview

Management of some specific types of C-spine fractures is covered in the corresponding preceding sections. For injuries not specifically addressed, general management principles are as follows[6]:

1. immobilize and reduce externally (if possible): may use traction × 0–7 days
2. determine if there is an indication for decompression as soon as practical (clinical conditions permitting), and decompress if needed. Although controversial, the following are generally accepted indications for acute decompression in patients without complete spinal cord injury:
   a) radiographic evidence of bone or foreign material in the spinal canal with associated spinal cord symptoms
   b) complete block on CT, myelogram or MRI
   c) clinical judgement: e.g. a progressive incomplete spinal cord injury where the surgeon believes that decompression would be beneficial
3. ascertain stability of the injury (see Table 65.4)
   a) stable fractures: treat in non-halo orthosis for 1–6 weeks (p.935)
   b) unstable fractures: all of the following choices are appropriate, with little evidence (based on long-term spinal stability) to recommend one scheme over another in most cases
      - traction × 7 weeks, followed by orthosis × 8 weeks
      - halo × 11 weeks, followed by orthosis × 4 weeks
      - surgical fusion, followed by orthosis × 15 weeks
      - surgical fusion with internal immobilization (lateral mass screws & rods…) ± orthosis for short period of time (≈ several weeks)

65.7.3 Surgical treatment

In patients with complete spinal cord lesions

Operating on a patient with a complete cord injury (ASIA A and not in spinal shock) does not result in significant recovery of neurologic function.[32] If there is ongoing spinal cord compression and the bulbocavernosus reflex is absent, the patient may be in spinal shock–operate at the earliest time that is safe to do so at your institution. However, aggressive non-surgical reduction of traumatic subluxation should be pursued.

The primary goal of surgery in this setting is spinal stabilization, allowing the patient to be placed in a sitting position for improved pulmonary function, for psychological benefit, and to allow initiation of rehabilitation. Although the spine will fuse spontaneously in many cases (taking ≈ 8–12 weeks), surgical stabilization expedites the mobilization process and reduces the risk of delayed kyphotic angulation deformity. Early surgery may lead to further neurological injury, and should be delayed until the patient has stabilized medically and neurologically. In most cases, performing surgery within 4–5 days (if the patient is otherwise stable) is probably early enough to help reduce pulmonary complications.

In patients with incomplete lesions

Patients with incomplete cord injuries who have compromise of the spinal canal (by bone, disc, unreducible subluxation or hematoma) and either do not improve with nonoperative therapy or deteriorate neurologically should undergo surgical decompression and stabilization.[32] This may facilitate some further return of spinal cord function. An exception may be the central cord syndrome (p.944).

Anterior or posterior?

The choice of technique depends to a large degree on the mechanism of injury, as the treatment should tend to counteract the instability, and ideally should not compromise structures that are still functioning. Instrumentation (wires/cables, lateral mass screws & rods, clamps…) immobilize the area of instability while bony fusion is occurring. In the absence of bony fusion, all mechanical devices will eventually fail, and so it becomes a "race" between fusion and instrument failure. Extensive injuries (including teardrop fractures (p.989) and compression burst fractures) may require a combined anterior and posterior approach (staged, or in a single sitting; anterior decompression precedes posterior fusion).

Posterior immobilization and fusion

Indications: The procedure of choice for most flexion injuries. Useful when there is minimal injury to the vertebral bodies and in the absence of anterior compression of the spinal cord and nerves. Including: posterior ligamentous instability, traumatic subluxation, unilateral or bilateral locked facets, simple wedge compression fractures.

The most common technique consists of open or closed reduction, followed by lateral mass screws & rods (p.972). Interlaminar Halifax clamps are an alternative.[33] Although successes have been reported using methylmethacrylate,[34] it does not bond to bone and weakens with age, and thus its use in the setting of traumatic injury is discouraged.[35]

Choice of posterior technique: If the anterior weight-bearing column is significantly damaged, or if there is absence or compromise of the lamina or spinous processes, then either a combined anterior-posterior approach is needed or posterior rigid instrumentation (e.g. lateral mass screw-plate or rod fixation) with fusion is recommended.[36]

Anterior approach

Does not depend on integrity of posterior elements to achieve stability.

Indications:

1. fractured vertebral body with bone retropulsed into spinal canal (burst fracture)
2. most extension injuries
3. severe fractures of posterior elements that preclude posterior stabilization and fusion
4. may be used for traumatic subluxation of the cervical spine

Usually consists of:

1. corpectomy: decompresses the neural elements (if necessary) and removes fractured and structurally compromised bone
   a) decompression usually requires wide corpectomy, at least ≈ 16 mm (palpate anterior surface of vertebral body to determine width; note position of vertebral arteries on pre-op CT). NB: it is suggested to take the corpectomy no wider than 3 mm lateral to the medial edge of the longus coli muscle, this leaves ≈ 5 mm margin of safety to the foramen transversarium[37]
   b) if decompression is not needed, ≈ 12 mm corpectomy suffices (i.e. about the width of a half-inch cottonoid)
2. AND
   a) strut graft fusion: replaces the involved body or bodies with either:
      - bone (usually iliac crest, rib or fibula, either homologous or cadaveric)
      - or synthetic cage (e.g. titanium or PEEK)
   b) usually accompanied with compression plates
   c) usually followed with external immobilization
   d) corpectomy of > 1 level, or presence of injury to posterior elements is usually an indication for augmentation with posterior instrumentation

Complications of surgical treatment

1. hardware problems
   a) anterior cage problems
      - cage displacement/extrusion
      - cage subsidence/telescoping into endplace
      - vertebral body fracture
   b) problems with plating
      - screw pull-out, loosening or breakage
      - fatigue fracture of plate
      - screw injury: nerve root, spinal cord or vertebral artery
2. inadequate post-operative immobilization
   a) improper brace selected
   b) poor patient compliance with immobilization device
3. failure of graft to take (nonunion)

4. judgmental error
   a) failure to incorporate all unstable levels
   b) improper surgical approach

65.8 Spinal cord injury without radiographic abnormality (SCIWORA)

65.8.1 General information

Although spinal cord injuries are uncommon in children, there is a subgroup of these in which no radiographic evidence of bony or ligamentous disruption can be demonstrated (including on dynamic flexion-extension x-rays). This is attributed to the normally increased elasticity of the spinous ligaments and paravertebral soft-tissue in the young population[38] and has been dubbed SCIWORA (an acronym for "Spinal Cord Injury Without Radiographic Abnormality"). The age range of children with SCIWORA is 1.5–16 years, it has a much higher incidence in age $\leq$9 yrs.[39] The spinal cord may undergo contusion, transection, infarction, stretch injuries, or meningeal rupture. Additional etiologies include: blunt abdominal trauma with disruption of blood flow from the aorta or segmental branches, traumatic disc herniation. There may be an increased risk of SCIWORA among young children with asymptomatic Chiari I malformation.[40]

54% of children with SCIWORA had a delay between injury (at which time some children experience transient numbness, paresthesias, Lhermitte's sign, or a feeling of total body weakness) and the onset of objective sensorimotor dysfunction ("latent period") ranging from 30 minutes to 4 days.

Practice guideline: Diagnosis of SCIWORA

Level III[41]
- MRI of the region of suspected injury
- radiographic screening of entire spine
- assess spinal stability with flexion-extension x-rays in the acute setting and at a late follow-up, even if MRI is negative for extraneural injury

not recommended: spinal angiography or myelography

65.8.2 Radiographic evaluation

In addition to plain films and flexion-extension films (to identify overt instability which would require surgical fusion), should include MRI which may show increased signal within the spinal cord parenchyma on T2WI. There were no intraspinal space occupying lesions in 13 patients studied with myelography/CT.[38]

65.8.3 Management

Practice guideline: Management of SCIWORA

Level III[41]
- external immobilization of the injured spinal segment for up to 12 weeks
- early discontinuation of external immobilization for patients who become asymptomatic and are confirmed to have no instability on flexion-extension x-rays
- avoidance of "high-risk" activities for up to 6 months after SCIWORA

Surgical intervention, including laminectomy, has shown no benefit in the few cases where it has been tried.[43]

Due to a 20% rate of repeat injury (some due to trivial trauma, and some without identifiable trauma) within 10 weeks of the original trauma when treated with only a rigid collar and restriction

Table 65.6 Treatment protocol for SCIWORA (modified[42])
• admit patient to hospital (helps emphasize seriousness of injury) • BR with rigid cervical collar until flexion-extension films are normal • MRI of cervical spine to document presence of spinal cord injury • detailed discussion with patient and family about seriousness of injury and rationale for treatment outlined here • immobilization in Guilford brace for 3 months[a] • prohibition of contact and noncontact sports • regular follow-up visits for monitoring condition and compliance • liberalize activities at 3 months if flexion-extension films are normal
[a]this represents an extremely conservative recommendation, a less restrictive recommendation is immobilization for 1–3 weeks[43]; see Practice guideline: SCIWORA (p. 999).

of contacts sports (both for 2 months), more aggressive measures were initially recommended (Table 65.6).

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66 Thoracic, Lumbar and Sacral Spine Fractures

66.1 Assessment and management of thoracolumbar fractures

66.1.1 General information

A widely used model for thoracolumbar spine stability is the 3-column model of Denis (see below). See also the more recently proposed TLICS system (p.1006).

66.1.2 Three column model

General information

Denis' 3 column model of the spine (Fig. 66.1) attempts to identify CT criteria of instability of thoracolumbar spine fractures.[1] This model has generally good predictive value, however, any attempt to create "rules" of instability will have some inherent inaccuracy.

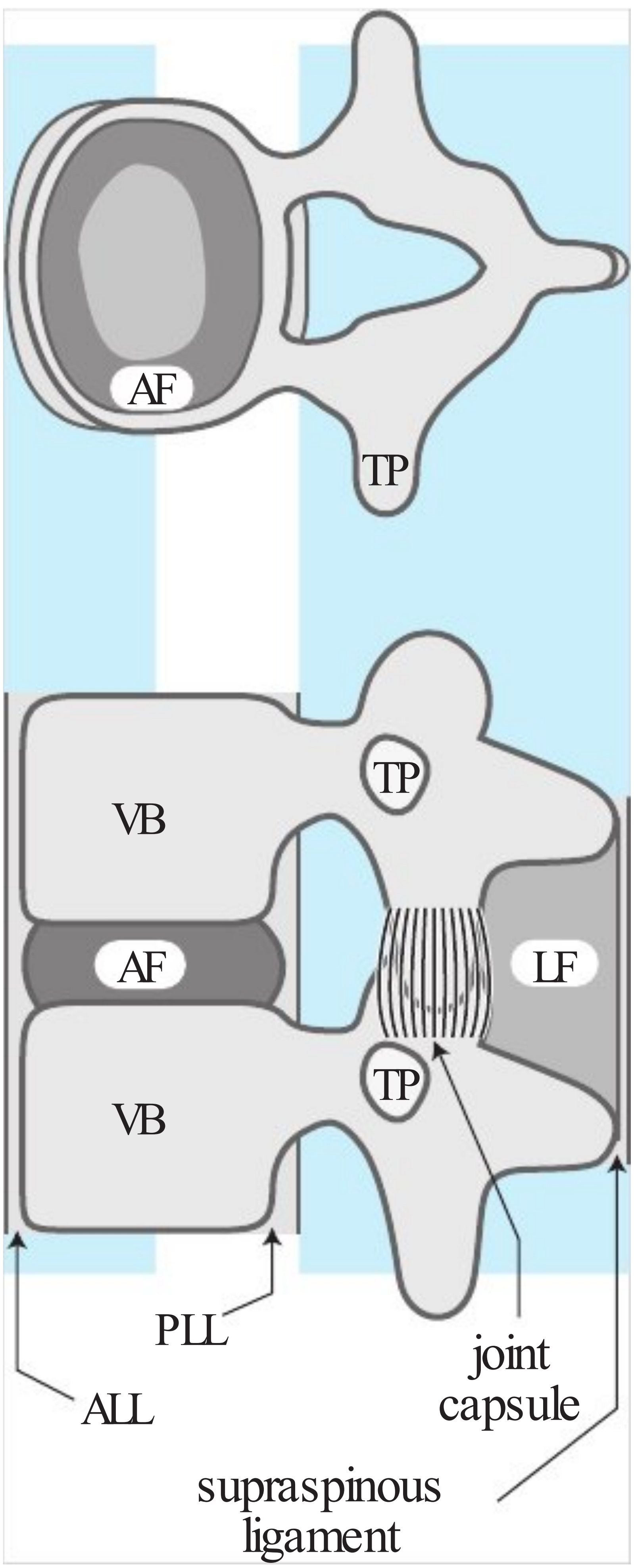


Fig. 66.1 Three column model of the spine (TP = transverse process, see text for other abbreviations) (Adapted from Spine, Denis F, Vol. 8, pp. 317–31, 1983, with permission)

Definitions

- anterior column: anterior half of disc and vertebral body (VB) (includes anterior anulus fibrosus (AF)) plus the anterior longitudinal ligament (ALL)
- middle column: posterior half of disc and vertebral body (includes posterior wall of vertebral body and posterior AF), posterior longitudinal ligament (PLL), & the pedicles
- posterior column: posterior bony complex (posterior arch) with interposed posterior ligamentous complex (supraspinous and interspinous ligament, facet joints and capsule, and ligamentum flavum (LF)). Injury to this column alone does not cause instability

Classification into major and minor injuries

Minor injuries

Involve only a part of a column and do not lead to acute instability (when not accompanied by major injures). Includes:

1. fracture of transverse process: usually neurologically intact except in two areas:
   a) L4–5 → lumbosacral plexus injuries (there may be associated renal injuries, check U/A for blood)
   b) T1–2 → brachial plexus injuries
2. fracture of articular process or pars interarticularis
3. isolated fractures of the spinous process: in the TL spine: these are usually due to direct trauma. Often difficult to detect on plain x-ray
4. isolated laminar fracture: rare. Should be stable

Major injuries

The McAfee classification describes 6 main types of fractures.[2] A simplified system with four categories follows (also see Table 66.1):

Type 1: Compression fracture: compression failure of anterior column. Middle column intact (unlike the 3 other major injuries below) acting as a fulcrum,

1. 2 subtypes:
   a) anterior: most common between T6-T8 and T12-L3
      - lateral x-ray: wedging of the VB anteriorly, no loss of height of posterior VB, no subluxation
      - CT: spinal canal intact. Disruption of anterior end-plate
   b) lateral (rare)
2. clinical: no neurologic deficit

Type 2: Burst fracture: pure axial load → compression of vertebral body → compression failure of anterior and middle columns. Occur mainly at TL junction, usually between T10 and L2

1. 5 subtypes; L5 burst fractures may constitute a rare subtype (p.1006)
   a) fracture of both end-plates: seen in lower lumbar region (where axial load → increased extension, unlike T-spine where axial load → flexion)
   b) fracture of superior end-plate: the most common burst fracture. Seen at TL junction. Mechanism = axial load + flexion
   c) fracture of inferior end-plate: rare

Table 66.1 Column failure in the four major types of thoracolumbar spine injuries

Fracture type	Column		
	Anterior	Middle	Posterior
compression	compression	intact	intact, or distraction if severe
burst	compression	compression	intact
seat-belt	intact or mild compression of 10–20% of anterior VB	distraction	
fracture-dislocation	compression, rotation, shear	distraction, rotation, shear	

[a]adapted[1] with permission

d) burst rotation: usually midlumbar. Mechanism = axial load + rotation
e) burst lateral flexion: mechanism = axial load + lateral flexion

2. radiographic evaluation
   a) lateral x-ray: cortical fracture of posterior VB wall, loss of posterior VB height, retropulsion of bone fragment from end plate(s) into canal
   b) AP x-ray: increase of interpediculate distance (IPD), vertical fracture of lamina, splaying of facet joints: ↑ IPD indicates failure of middle column
   c) CT: demonstrates break in posterior wall of VB with retropulsed bone in spinal canal (average: 50% obstruction of canal area), increase in IPD with splaying of posterior arch (including facets)
   d) MRI: compromise of anterior canal by bone fragment; possible cord compression usually with fragments occupying > 50% of the canal diameter
   e) MRI or myelogram: compression in spinal canal
3. clinical: depends on level (thoracic cord more sensitive and less room in canal than conus region), the impact at the time of disruption, and the extent of canal obstruction
   a) ≈ 50% intact at initial examination (half of these recalled leg numbness, tingling, and/or weakness initially after trauma that subsided)
   b) of patients with deficits, only 5% had complete paraplegia

Type 3: Seat-belt fracture (some call this a flexion-distraction fracture, but that term is also used for a subtype of fracture-dislocation): flexion across a fulcrum anterior to the anterior column (e.g. seat belt) → compression of anterior column & distraction failure of both middle and posterior columns. May be bony or ligamentous

1. 4 subtypes
   a) Chance fracture (named for G. Q. Chance, 1948): one level, totally through bone
   b) one level, through ligaments
   c) two level, through bone in middle column, through ligament in anterior and posterior columns
   d) two level, through ligament in all 3 columns
2. radiographic evaluation
   a) plain x-ray: ↑ interspinous distance, pars interarticularis fractures, and horizontal split of pedicles and transverse process. No subluxation
   b) CT: axial cuts are poor for this type (most of fracture is in plane of axial CT cuts). Sagittal and coronal reconstructions demonstrate well. May demonstrate pars fracture
3. clinical: no neurologic deficit

Type 4: Fracture-dislocation: failure of all 3 columns due to compression, tension, rotation or shear → subluxation or dislocation

1. x-ray: occasionally, may be reduced when imaged. Look for other markers of significant trauma (multiple rib fractures, unilateral articular process fractures, spinous process fractures, horizontal laminar fractures)
2. 3 subtypes
   a) flexion rotation: posterior and middle columns totally ruptured, anteriorly compressed → anterior wedging
      - lateral x-ray: subluxation or dislocation. Preserved posterior VB wall. Increased interspinous distance
      - CT: rotation and offset of VBs with → canal diameter. Jumped facets
      - clinical: 25% neurologically intact. 50% of those with deficits were complete paraplegics
   b) shear: all 3 columns disrupted (including ALL)
      - when trauma force directed posteriorly to anteriorly (more common) VB above shears forward fracturing the posterior arch (→ free floating lamina) and the superior facet of the inferior vertebra
      - clinical: all 7 cases were complete paraplegics
   c) flexion distraction
      - radiographically resemble seat-belt type with addition of subluxation, or with compression of anterior column > 10–20%
      - clinical: neurologic deficit (incomplete in 3 cases, complete in 1)

Associated injuries

In addition to the above, associated injuries include: vertebral end-plate avulsion, ligamentous injuries, and hip and pelvic fractures. Thoracolumbar fractures may be associated with hemodynamic instability as a result of hemothorax or aortic injury. Fractures of the transverse processes may be associated with abdominal trauma (e.g. renal injuries at L4–5).

Stability and treatment of thoracolumbar spine fractures

Minor injuries

Isolated thoracolumbar transverse process fractures (as demonstrated on spinal CT) do not require intervention or consultation of a spine service.[3,4]

Major spine injuries

Denis categorized the instability as:

- 1st degree: mechanical instability
- 2nd degree: neurological instability
- 3rd degree: both mechanical & neurological instability

Anterior column injury

Isolated anterior column injuries are usually stable and are treated as outlined in Table 66.2

The following exceptions may be unstable (1st degree) and often require surgery[1,5]:

Unstable compression fractures

1. a single compression fracture with:
   a) loss of > 50% of height with angulation (particularly if the anterior part of the wedge comes to a point)
   b) excessive kyphotic angulation at one segment (various criteria are used, none are absolute. Values quoted: > 30°, > 40°)
2. 3 or more contiguous compression fractures
3. neurologic deficit (generally does not occur with pure compression fracture)
4. disrupted posterior column or more than minimal middle column failure
5. progressive kyphosis: risk of progressive kyphosis is increased when loss of height of anterior vertebral body is > 75% Risk is higher for lumbar compression fractures than thoracic

Middle column failure

These are unstable (often requiring surgery) with the following exceptions which should be stable (stable injuries may be treated as outlined in Table 66.2.

Stable middle column fractures

- above T8 if the ribs and sternum are intact (provides anterior stabilization)
- below L4 if the posterior elements are intact
- Chance fracture (anterior column compression, middle column distraction)
- anterior column disruption with minimal middle column failure

Posterior column disruption

Not acutely unstable unless accompanied by failure of the middle column (posterior longitudinal ligament and posterior anulus fibrosus). However, chronic instability with kyphotic deformity may develop (especially in children).

Seat-belt type injuries without neurologic deficit

No immediate danger of neurologic injury. Treat most with external immobilization in extension (e.g. Jewett hyperextension brace or molded TLSO).

Table 66.2 Treatment of stable anterior or middle column thoracolumbar spine injuries

• treat initially with analgesics and recumbency (bed-rest) for comfort × 1–3 weeks • diminution of pain is a good indication to commence mobilization with or without external immobilization (corset or Boston brace or extension TLSO × ≈ 12 weeks) depending on the degree of kyphosis • vertebroplasty (± kyphoplasty) may be an option (p. 1011) • serial x-rays to rule-out progressive deformity

Fracture-dislocation

Unstable. Treatment options:

1. surgical decompression and stabilization: usually needed in cases with
   a) compression with > 50% loss of height with angulation
   b) or, kyphotic angulation > 40° (or > 25%)
   c) or, neurologic deficit
   d) or, desire to shorten length of time of bedrest
2. prolonged bedrest: an option if none of the above are present

When vertebral body resection (vertebral corpectomy) is performed, options to access: transthoracic or transabdominal approach (or combined), transpedicular (for thoracic spine), lateral (retroperitoneal/retropleural) approach. Fracture and compression usually occurs at the superior margin of vertebral body, thus start resection at the inferior disc interspace. Followed by strut graft (cage or bone: iliac crest or fibula or tibia). Posterior instrumentation is usually required; see Spinal instrumentation (p. 1007).

Burst fractures

Not all burst fractures are alike. Some burst fractures may eventually cause neurologic deficit (even if no deficit initially). Middle column fragments in canal endanger the neuro elements. Criteria have been proposed to differentiate mild burst fractures from severe ones. No system is uniformly accepted. Recommendations.[1,6]

Surgical indications for burst fractures: burst fracture with any of the following:

- anterior vertebral body height ≤ 50% of the posterior height
- residual canal diameter ≤ 50% of normal (note: retropulsed bone in the canal is often resorbed with either bracing or surgery and is therefore controversial as an isolated indication for surgery[7,8])
- kyphotic angulation ≥ 20°
- when the increased interpediculate distance usually present on the initial film widens further on AP x-ray when standing in brace/cast
- neurologic deficit (incomplete)
- progressive kyphosis

Common surgical options for burst or severe compression fractures:

1. If instrumentation alone is needed
   a) can place pedicle screws in 2 levels above and 2 levels below the fracture
   b) if the index level can be included (i.e. if the pedicles are intact enough to accept shorter screws), similar biomechanical stability can be achieved by placing screws at the index level (the fractured level) and then just 1 above and 1 below[9]
2. If decompression of the spinal canal and/or anterior support is needed, corpectomy and strut graft (e.g. with expandable cage) with percutaneous pedicle screws may be used. Approaches:
   a) from posterior approach e.g. laminectomy with transpedicular approach and impacting bone anteriorly out of canal with a mallet and reverse angled Scoville curette, or
   b) lateral corpectomy and removal of bone from canal

For those not undergoing surgery (i.e. when surgery is not required or is contraindicated), an option is to treat with recumbency from 1–6 weeks (the duration depending on pain and degree of deformity).[6] Avoid early ambulation → further axial loading (even in cast). When appropriate, begin ambulation in an orthosis (e.g. molded thoracolumbar sacral orthosis (TLSO) or a Jewett brace) and follow patient for 3–5 months with serial x-rays to detect progressive collapse or angulation which may need further intervention. L5 burst fractures may be an exception to the usual management (see below).

66.1.3 Thoracolumbar injury classification and severity score (TLICS)

The TLICS system has been proposed to simplify classification and discussion of thoracolumbar fractures.[10,11] Points are assigned as shown in Table 66.3. The scores are summed, and management guidelines are given in Table 66.4.

Neurologic deficit, especially when partial, favors surgery.

Table 66.3 Thoracolumbar injury classification & severity score (TLICS)

Category	Finding	Points
Radiographic findings	compression fx	1
	burst component or lateral angulation > 15°	1
	distraction injury	2
	translational/rotational injury	3
Neurologic status	intact	0
	root injury	2
	complete SCI	2
	incomplete SCI	3
	cauda equina syndrome	3
Integrity of posterior ligamantous complex	intact	0
	undetermined	2
	definite injury	3
TLICS = Total Points →		

Table 66.4 Management based on TLICS

TLICS	Management
≤ 3	nonoperative candidates
4	"grey zone" may be considered for operative or nonoperative management
≥ 5	surgical candidates

66.2 Surgical treatment

66.2.1 Ligamentotaxis

May work with fragments retropulsed into the anterior canal if the PLL is intact (may not be the case with middle column failure), distraction may be able to "pull" the fragments back into their normal position (ligamentotaxis) although this is not assured.[12] Ligamentotaxis has a better chance of succeeding if performed within 48 hours of injury. From a posterior approach with laminectomy: intraoperative ultrasound may demonstrate residual canal fragments,[13] and if needed the fragments may be impacted anteriorly out of the canal, e.g. using tamps such as Sypert spinal impactors. It is important not to overdistract to avoid neural injury.

66.2.2 Choice of surgical approach

The posterior approach is preferred when there is not a specific need to go from the front.

66.2.3 Spinal instrumentation

Anterior instrumentation of the lower lumbar spine is difficult, and is usually not recommended below ≈ L4.

66.2.4 Burst fractures

Choice of approach

Surgical considerations: a posterior approach is preferred if there is a dural tear, whereas a burst fracture with partial deficit and canal compromise may be treated more effectively from an anterior approach.[2] A small progression in angular deformity may occur when posterior stabilization is performed alone (since the injury to the anterior column is not corrected), but by itself usually does not require intervention.

For a posterior approach

In ideal situation (good bone quality, pedicle screw placement goes well (i.e. no fracture, no breach), and non-smoking patient) then one can fuse/rod one above and one below the fracture (using pedicle screws; longer constructs are needed with laminar hooks). With a short segment fusion like this, approximately 10° of lordosis be lost with time, therefore, one should try to overcorrect a little to accommodate the anticipated settling. If the patient does not meet the above criteria (e.g. poor bone quality), an option is to "rod long, fuse short" (e.g. rod 2 levels above and below the fracture but fuse only 1 level above and below) and then to remove the hardware when the fusion is solid (e.g. at 8–12 months) – this avoids fusing a nonpathologic segment just to get a better anchor. Junctional deterioration to the point that further surgery is needed often occurs at 3 years when 4 segments are fused, whereas it occurs at 8–9 years when only 3 levels are fused. Fusing across critical levels (i.e. thoracolumbar junction with T11 or L1 compression fractures) requires that the fusion incorporate 2–3 levels on each side of the junction (the forces of the long segment of the relatively immobile thoracic spine with the lumbar spine at the T-L junction increase the risk of nonunion).

For thoracic fractures that are not severe and do not require decompression, an option is to place pedicle screws and rods (which can be done percutaneously) without placing any graft. The concept is that the ribs anteriorly and the screws/rods posteriorly provide adequate stabilization while the fractured VB heals. The hardware can be removed electively once the fusion is solid (usually at 8–12 months). This is more commonly practiced in Europe than the U.S.

66.2.5 Wound infections

Postoperative wound infections with spinal instrumentation are usually due to Staph. aureus. With titanium hardware it may respond to debridement of devitalized tissue (e.g. onlay bone graft) and thorough washout (typically with 3 L of antibiotic irrigation flushed into the wound using a pulse lavage device – avoiding direct irrigation of any exposed dura) without removal of instrumentation, followed by antibiotics.[2] Persistent infection may respond to. If this inadequate, removal of instrumentation may occasionally be required.

66.3 Osteoporotic spine fractures

66.3.1 General information

Osteoporosis is defined as a condition of skeletal fragility as a result of low bone mass, microarchitectural deterioration of bone, or both.[14] It is found most commonly in post-menopausal white females, and is rare prior to menopause. Lifetime risk of symptomatic vertebral body (VB) osteoporotic compression fractures is 16% for women, and 5% for men. There are ≈ 700,000 VB compression fractures per year in the U.S.

These patients are often found to have significant VB compression fractures on plain films after presenting with back pain following a seemingly minor fall. CT often shows an impressive appearing amount of bone retropulsed into the canal.

66.3.2 Risk factors

Factors that increase the risk of osteoporosis include:

1. weight < 58 kg
2. cigarette smoking[15]
3. low-trauma VB fracture in the patient or a first degree relative
4. drugs
   a) heavy alcohol consumption
   b) AEDs (especially phenytoin)

c) warfarin
d) steroid use:
- bone changes can be seen with 7.5 mg/d of prednisone for >6 months
- VB fractures occur in 30–50% of patients on prolonged glucocorticoids

5. postmenopausal female
6. males undergoing androgen deprivation therapy (e.g. for prostate Ca). Orchiectomy or ≥9 doses of gonadotropin-releasing hormone agonists had a 1.5 fold increase in risk of all fractures[16]
7. physical inactivity
8. low calcium intake
9. low serum levels of vitamin D (which decreases calcium absorption – see below). Lab: serum 25-hydroxyvitamin D [25(OH)D], AKA calcidiol is the best indicator of vitamin D status ▶ Table 66.5

Factors that protect against osteoporosis include impact exercise and excess body fat.

66.3.3 Diagnostic considerations

To differentiate osteoporotic compression fractures from other pathologic fractures, see Pathologic fractures of the spine (p. 1391).

Pre-fracture diagnosis

1. measuring bone fragility is not possible
2. the best correlate with bone fragility is radiographic measurement of bone mineral density (BMD) using DEXA scan (see below)
3. patients with low-trauma fractures or fragility fractures are considered osteoporotic even if their BMD are greater than these cutoffs

DEXA scan (dual energy x-ray absorptiometry): the preferred way to measure BMD

1. proximal femur: BMD measurement in this location is the best predictor for future fractures
2. LS spine: best location to assess response to treatment (need AP and lateral views, since AP often overestimates BMD because of superimposition of overlying posterior elements and aortic calcifications)
3. forearm BMD may be used if hip or spine are unsuitable

Interpretation of DEXA scan results:

1. findings are reported as
   a) T-score: norms for healthy young adults
   b) Z-score: norms of subjects of same age and sex as the patient
2. diagnostic criteria: WHO definitions (with a normal distribution 1 SD below the mean is the lowest 25th percentile, 2 SD below is 2.5th percentile)
   a) normal: >–1 standard deviations (SD)
   b) osteopenia: from –1 to –2.5 SD
   c) osteoporosis: < than –2.5 SD[17]

Post-fracture considerations

1. other causes of pathologic fracture, especially neoplastic (e.g. multiple myeloma, metastatic breast cancer) should be ruled out

Table 66.5 Serum 25-hydroxyvitamin D levels

ng/ml[a]	nmol/L[a]	Interpretation
<10–11	<25–27.5	vit D deficiency → rickets (in peds) and osteomalacia (adults)
<10–15	<25–37.5	inadequate for bone and overall health
≥15	≥37.5	adequate for bone and overall health
consistently > 200	consistently > 500	potentially toxic → hypercalcemia & hyperphosphatemia

[a] 1 ng/ml = 2.5 nmol/L

2. younger patients with osteoporosis require evaluation for a remediable cause of the osteoporosis (hyperthyroidism, steroid abuse, hyperparathyroidism, osteomalacia, Cushing's syndrome)

66.3.4 Treatment

See references.[18,19,20,21]

Prevention of osteoporosis

High calcium intake during childhood may increase peak bone mass. Weight-bearing exercise in adulthood helps slow calcium loss from bones. Also effective: estrogen (see below), bisphosphonates (alendronate and risedronate), and raloxifene.

Treating established osteoporosis

Drugs that increase bone formation include:
1. intermittent low-dose parathyroid hormone: still experimental
2. sodium fluoride: 75 mg/d increases bone mass but did not significantly reduce the fracture rate. 25 mg PO BID of a delayed-release formulation (Slow Fluoride®) reduced fracture rate but may make bone more fragile and could increase risk of hip fractures. Fluoride increases demand for Ca^{++}, therefore supplement with 800 mg/d Ca^{++} and 400 IU/d vitamin D. Not recommended for use > 2 yrs

Drugs that reduce bone resorption are less effective on cancellous bone (found mainly in the spine and at the end of long bones[19]). Improvement in spine bone mineral density accounts for only a small part of the observed reduction in the risk of vertebral fracture.[22] Medications include:
1. estrogen: cannot be used in men. Studies of estrogen hormone replacement therapy (HRT) have shown increased vertebral bone mass by > 5%and decreased rate of vertebral fractures by 50% Also relieves post-menopausal symptoms and reduces risk of CAD. However, because HRT increases the risk of breast cancer[23] and of breast cancer recurrence[24] as well as DVT, its use has diminished substantially
2. calcium: current recommendation for postmenopausal women: 1,000-1,500 mg/d taken with meals[25]
3. vitamin D or analogues: promote calcium absorption from the GI tract. Typically administered with calcium therapy (either calcium or vitamin D alone are less effective). Vitamin D 400–800 IU/d is usually sufficient. If urinary Ca^{++} remains low, high dose vitamin D (50,000 IU q 7–10 d) may be tried. Since high-dose formulations have been discontinued in the U.S., analogues such as calcifediol (Calderol®) 50 mcg/d or calcitriol (Rocaltrol®) up to 0.25 mcg/d may be tried with Ca^{++} supplement. Serum levels of 25-hydroxyvitamin D [25(OH)D], AKA calcidiol is the best indicator of vitamin D status. The significance of vitamin D levels are shown in Table 66.5.[25] With high dose vitamin D or analogues, monitor serum and urinary Ca^{++}
4. calcitonin: a hormone synthesized by the thyroid gland which decreases bone resorption by osteoclasts. May be derived from a number of sources, salmon is one of the more common ones. The skeletal response is maximal during the first 18–24 months of therapy. Benefit in preventing fractures is less well established[21]
   a) parenteral salmon calcitonin (Calcimar®, Miacalcin®): indicated for patients for whom estrogen is contraindicated. Expensive ($1,500–3,000/yr) and must be given IM or sub-Q. 30–60% of patients develop antibodies to the drug which negates its effect. ℞: 0.5 ml (100 U) of calcitonin (given with calcium supplements to prevent hyperparathyroidism) SQ q d
   b) intranasal forms (Miacalcin nasal spray): less potent (works better in older women > 5 yrs post menopause). 200–400 IU/d given in one nostril (alternate nostrils daily) plus Ca^{++} 500 mg/d and vitamin D
5. bisphosphonates: carbon-substituted analogues of pyrophosphate have a high affinity for bone and inhibit bone resorption by destroying osteoclasts. Not metabolized. Remain bound to bone for several weeks
   a) etidronate (Didronel®), a 1st generation drug. Not FDA approved for osteoporosis. May reduce rate of VB fractures, not confirmed on F/U. Possible increased risk of hip fractures due to inhibition of bone mineralization may not occur with 2nd & 3rd generation drugs listed below. ℞ 400 mg PO daily × 2 wks followed by 11–13 weeks of Ca^{++} supplementation
   b) alendronate (Fosamax®): can cause esophageal ulcers. ℞ Prevention: 5 mg PO daily; treatment 10 mg PO daily; taken upright with water on an empty stomach at least 30 minutes before eating or drinking anything else. Once weekly dosing of 35 mg for prevention and 70 mg for treatment.[21,26] Taken concurrently with 1000–1500 mg/d Ca^{++} and 400/d IU of vitamin D

c) risedronate (Actonel®): □ Prevention or treatment: 5 mg PO daily, or 35 mg once/week[26] on an empty stomach (as for alendronate, see above)

6. estrogen analogues:
   a) tamoxifen (Nolvadex®), an estrogen antagonist for breast tissue but an estrogen agonist for bone, has a partial agonist effect on uterus associated with an increased incidence of endometrial cancer
   b) raloxifene (Evista®): similar to tamoxifen but is an estrogen antagonist for uterus.[27] Decreases the effect of warfarin (Coumadin®).
      □: 60 mg PO q d. Supplied: 60 mg tablets
7. RANK ligand (RANKL) inhibitors: RANKL binds to RANK receptors and stimulates precursor cells to mature into osteoclasts and inhibits their apoptosis.[28] Agents undergoing investigation include denosumab (Prolia®) 60 mg SQ q 6 months appears more effective than alendronate[29]

Treatment of osteoporotic vertebral compression fractures

Patients rarely have neurologic deficit. They are also usually fragile elderly women who usually do not tolerate large surgical procedures well, and the rest of their bones are also osteoporotic which are poor for internal fixation.

Management consists primarily of analgesics and bed rest followed by progressive mobilization, often in an external brace (often not tolerated well). Surgery is rarely employed. In cases where pain control is difficult to obtain or where neural compression causes deficit, limited bony decompression may be considered. Percutaneous vertebroplasty (see below) is a newer option.

Typical time course of conservative treatment:

1. initially, severe pain may require hospital or subacute care facility admission for adequate pain control utilizing
   a) sufficient pain medication
   b) bed rest for about 7–10 days (DVT prophylaxis recommended)
2. begin physical therapy (PT) after ≈ 7–10 days as patient tolerates (prolonged bed rest can promote "disuse osteoporosis")
   a) pain control as patient is mobilized may be enhanced by a lumbar brace which may work by reducing movement which causes repetitive "microfractures"
   b) discharge from the hospital with lumbar brace for outpatient PT
3. pain subsides on the average after 4–6 weeks (range 2–12 weeks)

Vertebral body augmentation

Percutaneous vertebroplasty (PVP)

Transpedicular injection of polymethylmethacrylate (PMMA) AKA "methylmethacrylate cement" into the compressed bone with the following goals (note: PMMA injection is FDA approved for treatment of compression fractures due to osteoporosis or tumor, but not for trauma as PMMA would prevent healing of the fracture):

1. to shorten the duration of pain (sometimes providing pain relief within minutes to hours). Remember: the natural history is that pain will eventually diminish in essentially all of these patients. Mechanism of pain relief may be due to stabilization of bone and/or due to disruption of nerve pain transmission by heat released during the exothermic curing of the cement
2. to try and stabilize the bone: may prevent progression of kyphosis

Randomized studies published in 2009 found no benefit in vertebroplasty over a sham procedure at 1 month[30] or at any time up to 6 months post-procedure.[31] NB: kyphoplasty (see below) was not studied; use with metastatic spine tumors was also not evaluated. Patient selection issues may make these results more or less applicable to a specific patient.

Kyphoplasty

Similar to PVP, except first, a balloon is inserted into the compressed VB through the pedicle. The balloon is inflated and then deflated and removed. PMMA is injected into the thusly created defect. Potential benefits of this over vertebroplasty: there may be some restoration of height, and there may be less tendency for PMMA extravasation/embolization (due to the cavity creation and the thicker PMMA used). In the (industry sponsored) randomized non-blinded FREE study[32] there was a significant positive difference in pain reduction and quality of life improvement in the kyphoplasty group compared to the nonoperated group at 1 month that diminished by 1-year post-op.

Indications

1. painful osteoporotic compression fractures:
   a) usually do not treat fractures producing < 5–10% loss of height
   b) severe pain that interferes with patient activity
   c) failure to adequately control pain with oral pain medication
   d) pain localized to fracture level
   e) acute fractures: procedure is not effective for healed fractures. In questionable cases, look for changes on STIR MRI (see below)
2. levels: FDA approved for use from T5 through L5, however has been used off-label (primarily for tumor, e.g. multiple myeloma) from T1 through sacrum, and has been described (for tumor) in the cervical spine from an anterior approach
3. vertebral hemangiomas that cause vertebral collapse or neurologic deficit as a result of extension into the spinal canal (not for incidental hemangiomas) (p. 795): the first indication for PVP[33]
4. osteolytic metastases and multiple myeloma[34]: pain relief and stabilization
5. pathologic compression fractures[35] from metastases: PVP does not give as rapid pain relief as with osteoporotic compression fractures (it may actually be necessary to increase pain meds for 7–10 days post PVP)
6. pedicle screw salvage when pedicle fractures or screws strip during pedicle screw placement

Contraindications

1. coagulopathy
2. completely healed fractures (no edema on MRI or cold on bone scan)
3. active infections: sepsis, osteomyelitis, discitis and epidural abscess
4. spinal instability
5. focal neurologic exam: may indicate herniated disc, retropulsed fragment in canal. Get CT or MRI to rule these out
6. relative contraindications:
   a) fractures > 80% loss of VB height (technically challenging)
   b) acute burst fractures
   c) significant canal compromise from tumor or retropulsed bone
   d) partial or total destruction of the posterior VB wall: not an absolute contraindication
7. iodine allergy: there is a small risk of a balloon rupturing with spill of the iodinated contrast used to fill the balloons prior to injecting the PMMA. Options include: iodine allergy prep (p. 221), use of gadolinium instead of iodinated contrast

Complications

Complication = rate: 1–9% Lowest when used to treat osteoporotic compression fractures, higher with vertebral hemangiomas, highest with pathological fractures

1. methacrylate leakage:
   a) into soft tissues: usually of little consequence
   b) into spinal canal: symptomatic spinal cord compression is very rare
   c) into neural foramen: may cause radiculopathy
   d) into disc space
   e) venous: can get into spinal venous plexus or vena cava with ≈ 0.3–1% risk of clinically significant methycrylate pulmonary embolism (PE)[36]
2. radiculopathy: 5–7% incidence. Some cases may be due to heat released during cement curing. Often treated conservatively: steroids, pain meds, nerve block…
3. pedicle fracture
4. rib fracture
5. transverse process fracture
6. anterior penetration with needle: puncture of great vessels, pneumothorax…
7. increased incidence of future VB compression fractures at adjacent levels

Management of some associated developments

1. chest pain
   a) get rib x-rays
   b) VQ scan if indicated
2. patient starts coughing during injection: fairly common. May be reaction to rib pain or to odor of PMMA, may also indicate solvent in lungs. Stop injecting
3. back pain: take x-ray to rule-out new fracture or PMMA in veins
4. neurologic symptoms: get CT scan

Pre-procedure evaluation

1. plain x-rays: minimum requirement, most practitioners get MRI or bone scan
2. CT: helps rule-out bony compromise of spinal canal which may indicate increased risk of leakage for PMMA into canal during procedure
3. MRI: not mandatory, may be helpful in some cases
   a) short tau inversion recovery (STIR) images demonstrate bone edema indicative of acute fractures (not as good for differentiating pathology)[37]
   b) MRI can also disclose neurologic compression by soft tissue (e.g. tumor)
4. patients with multiple compression fractures: consider getting bone scan and perform PVP in the VB near the level of pain that lights up the most (↑ activity on bone scan correlates strongly with good outcome from PVP)

Booking the case: Kyphoplasty

Also see default values (p. 27).

1. position: prone
2. anesthesia: may be done under general, or under MAC
3. equipment: 2 C-arms for bi-plane fluoro
4. implants:
   a) kyphoplasty set
   b) iodinated contrast from radiology to fill balloons
5. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: insertion of a needle into the fractured/abnormal bone, sometimes getting a biopsy as well, and then inflating a balloon in the bone to try and bring it back to a more normal size and then to inject a liquid cement which will then harden inside the bone to strengthen it
   b) alternatives: nonsurgical management, open surgery, in cases of tumor sometimes radiation therapy can be done
   c) complications: leakage of cement which can compress nerves and may need to be removed surgically if possible, rib fracture (from positioning), injury to large blood vessel or lung by the needle, failure to achieve the desired pain relief

Procedure

1. pain medication
   a) remember, this procedure is done with the patient lying on their stomach and is usually performed on frail, elderly females who smoke. Therefore use caution to avoid oversedation and respiratory compromise
   b) sedation and pain medication
   c) use of local anesthetic during needle placement
   d) additional pain medication just prior to injection
2. use bi-plane fluoro to pass needle through the pedicle to enter VB – see Percutaneous pedicle screws (p. 1495) – and place tip ≈ 1/2 to 2/3 of the way through the VB
3. test inject with contrast, e.g. iohexol (Omnipaque 300) (p. 219); do digital subtraction study if equipment is available. For kyphoplasty, the balloon is inflated at this time
   a) a little venous enhancement is acceptable
   b) if you visualize vena cava
      - do not pull needle back (the fistula has already been created)
      - push needle in a little further, or
      - push some gelfoam (soaked in contrast) through the needle, or
      - inject a very small amount of PMMA under visualization and allow it to set to block the fistula
4. inject PMMA (that has been opacified with tantalum or barium-sulfate) under fluoroscopic visualization until:
   a) 3–5 cc injected (minimal compression fractures accept more cement, sometimes up to ≈ 8 cc). No correlation between amount of PMMA injected and pain relief[34]
   b) PMMA approaches posterior VB wall. Stop if cement ever: enters disc space, vena cava, pedicle, or spinal canal

Post-procedure

1. PVP is often an outpatient procedure, but sometimes overnight admission is used
2. watch for
   a) chest or back pain (may indicate rib fracture)
   b) fever: may be reaction to cement
   c) neurologic symptoms
3. activity
   a) gradual mobilization after ≈ 2 hours
   b) ± physical therapy
   c) ± short term use of external brace (most centers do not use)
4. institute medical treatment for osteoporosis: remember the patient with fragility fractures by definition has osteoporosis with risk of future fractures

66

66.4 Sacral fractures

66.4.1 General information

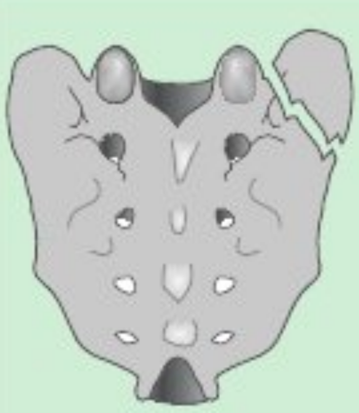
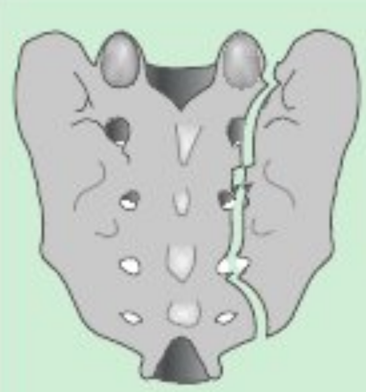
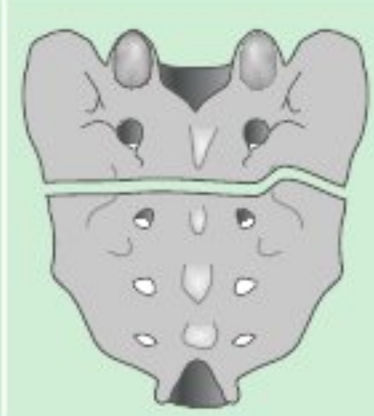
Uncommon. Usually caused by shear forces. Identified in 17% of patients with pelvic fractures[38] (□ keep in mind that neurologic deficits in patients with pelvic fractures may be due to associated sacral fractures). Neurologic injuries occur in 22–60%[38]

The sacrum below S2 is not essential to ambulation or support of the spinal column, but may still be unstable since pressure to the area may occur when supine or sitting.

66.4.2 Classification

Three characteristic clinical presentations based on zone of involvement[38,39] as shown in □ Table 66.6.

Table 66.6 Classification of sacral fractures

Zone I	Zone II	Zone III Vertical	Zone III Transverse
			
Zone I: Region of ala sparing the central canal and neural foramina. Occasionally associated with partial L5 root injury possibly as a result of entrapment of the L5 root between the upwardly migrated fracture fragment and the transverse process of the L5 vertebra	Zone II: Region of sacral foramina (sparing the central canal). A vertical fracture which may be associated with unilateral L5, S1 and/or S2 nerve root involvement (producing sciatica). Bladder dysfunction is rare	Zone III: Region of sacral canal. Frequently associated with sphincter dysfunction (occurs only with bilateral root injuries) and saddle anesthesia. Subdivided[38]:	
		Vertical: almost always associated with pelvic ring fracture	Transverse (horizontal): rare. Often due to a direct blow to the sacrum as in a fall from a great height. Marked displacement of fracture fragment can produce severe deficit[a] (bowel & bladder incontinence)

[a]significant deficit is rare in fractures at or below S4

66.4.3 Treatment

In one series,[40] all 35 fractures were treated without surgery, and only 1 patient with a complete cauda equina syndrome did not improve. Others feel that surgery may have a useful role[38]:
1. operative reduction and internal fixation of unstable fractures may aid in pain control and promote early ambulation
2. decompression and/or surgical reduction/fixation may possibly improve radicular or sphincter deficits

Some observations[38]:
1. reduction of the ala may promote L5 recovery with Zone I fractures
2. Zone II fractures with neurologic involvement may recover with or without surgical reduction and fixation
3. horizontal Zone III with severe deficit: controversial. Reduction & decompression does not ensure recovery, which may occur with nonoperative management

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67 Penetrating Spine Injuries and Long Term Management / Complications

67.1 Gunshot wounds to the spine

67.1.1 General information

Most are due to assaults with handguns. Distribution: cervical 19–37%, thoracic 48–64%, and lumbosacral 10–29% (roughly proportional to lengths of each segment). Spinal cord injury due to civilian GSWs are primarily due to direct injury from the bullet (unlike military weapons which may create injury from shock waves and cavitation). Steroids are not indicated (p.951).

67.1.2 Indications for surgery

1. injury to the cauda equina (whether complete or incomplete) if nerve root compression is demonstrated[1]
2. neurologic deterioration: suggesting possibility of spinal epidural hematoma
3. compression of a nerve root
4. CSF leak
5. spinal instability: very rare with isolated GSW to the spine
6. to remove a copper jacketed bullet: copper can cause intense local reaction[2]
7. incomplete lesions: very controversial. Some series show improvement with surgery,[3] others show no difference from unoperated patients
8. debridement to reduce the risk of infection: more important for military GSW where there is massive tissue injury, not an issue for most civilian GSW except in cases where the bullet has traversed GI or respiratory tract
9. vascular injuries
10. surgery for late complications:
    a) migrating bullet
    b) lead toxicity[4] (plumbism): absorption of lead from a bullet occurs only when it lodges in joints, bursae, or disc space. Findings include: anemia, encephalopathy, motor neuropathy, nephropathy, abdominal colic
    c) late spinal instability: especially after surgery

67.2 Penetrating trauma to the neck

67.2.1 General information

Most often, injuries to the soft-tissues of the neck fall into the purvey of general/trauma surgeons and/or vascular surgeons. However, depending on local practice patterns, neurosurgeons may participate in care of these injuries, or they may get involved by virtue of associated spinal injuries (p.1017).

The mortality rate for penetrating injury to the neck is ≈ 15%, with most early deaths due either to asphyxiation from airway compromise, or exsanguination externally or into the chest or upper airways. Late death is usually due to cerebral ischemia or complications from spinal cord injury.

67.2.2 Vascular injuries

Venous injuries occur in ≈ 18% of penetrating neck wounds, and arterial injuries in ≈ 12%. Of the cervical arteries, the common carotid is most usually involved, followed by the ICA, the ECA, and then the vertebral artery. Outcome probably correlates most closely with neurologic condition on admission, regardless of treatment.

Vertebral artery (VA): the majority of injuries are penetrating. Due to the proximity of other vessels, the spinal cord and nerve roots, injuries are rarely isolated to the VA. 72% of documented VA injuries had no related physical findings on exam.[5]

67.2.3 Classification

Trauma surgeons have traditionally divided penetrating injuries of the neck into 3 zones,[6] and although definitions vary, the following is a general scheme[7]:

Zone I: inferiorly from the head of the clavicle to include the thoracic outlet

Zone II: from the clavicle to the angle of the mandible

Zone III: from the angle of the mandible to the base of the skull

67.2.4 Evaluation

Neurologic examination: global deficits may be due to shock or hypoxemia due to asphyxiation. Cerebral neurologic deficits are usually due to vascular injury with cerebral ischemia. Local findings may be related to cranial nerve injury. Unilateral UE deficits may be due to nerve root or brachial plexus involvement. Median or ulnar nerve dysfunction can occur from compression by a pseudoaneurysm of the proximal axillary artery. Spinal cord involvement may present with complete injury, or with an incomplete spinal cord injury syndrome (p.944). Shock due to spinal cord injury is usually accompanied by bradycardia (p.931), as opposed to the tachycardia seen with hypovolemic shock.

Cervical spine x-rays: assesses trajectory of injury and integrity C-spine.

Angiography: indicated in most cases if the patient is stable (especially for zone I or III injuries, and for zone II patients with no other indication for exploration, or for patients with penetration of the posterior triangle or wounds near the transverse processes where the VA may be injured). Patients actively hemorrhaging need to be taken to the OR without pre-op angiography. Angiographic abnormalities include:

1. extravasation of blood
   a) expanding hematoma into soft tissues: may compromise airway
   b) pseudoaneurysm
   c) AV fistula
   d) bleeding into airways
   e) external bleeding
2. intimal dissection, with
   a) occlusion, or
   b) luminal narrowing (including possible “string sign”)
3. occlusion by soft tissue or bone

67.2.5 Treatment

Airway

Stable patients without airway compromise should not have “prophylactic” intubation to protect the airway. Immediate intubation is indicated for hemodynamically unstable patients or for airway compromise. Options:

- endotracheal: preferred
- cricothyroidotomy: if endotracheal intubation cannot be performed (e.g. due to tracheal deviation or patient agitation) or if there is evidence of cervical spine injury and manipulation of the neck is contraindicated, then cricothyroidotomy is performed with placement of a #6 or 7 cuffed endotracheal tube (followed by a standard tracheostomy in the OR once the patient is stabilized)
- awake nasotracheal: may be considered in the setting of possible spinal injury

Indications for surgical exploration

Surgical exploration has been advocated for all wounds that pierce the platysma and enter the anterior triangles of the neck,[8] however, 40–60%of these explorations will be negative. Although a selective approach may be based on angiography, false negatives have resulted in some authors recommending exploration of all zone II injuries.[9]

Surgical treatment for vascular injuries

Endovascular techniques may be suitable for select cases, especially for patients who are already in the endovascular suite for angiography. However, patients who are actively bleeding usually end up in the O.R. with an open procedure.

Carotid artery: choices are primary repair, interposition grafting, or ligation. Patients in coma or those with severe strokes caused by vascular occlusion of the carotid artery are poor surgical candidates for vascular reconstruction due to a high mortality rate ≥40%[7] however the outcome with ligation is worse. Repair of injuries is recommended in patients with no or only minor neurologic deficit. ICA ligation is recommended for bleeding that cannot be controlled and was used for extravasation of dye at the base of the skull in 1 patient.[10]

Vertebral artery: injuries are more often managed by ligation than by direct repair,[11] especially when bleeding occurs during exploration. Less urgent conditions (e.g. AV fistula) requires knowledge of the patency of the contralateral VA and the ability to fill the ipsilateral PICA from retrograde flow through the BA before ligation can considered (arteriographic anomalies contraindicate ligation in 15%of cases). Proximal occlusion may be accomplished with an anterior approach after the sternocleidomastoid is detached from the sternum. The VA is the normally the first branch of the subclavian artery. Alternatively, endovascular techniques may be used, e.g. detachable balloons for proximal occlusion, or thrombogenic coils for pseudoaneurysms. Distal interruption may also be required, and this necessitates surgical exposure and ligation. Optimal management of a thrombosed injured VA in a foramen transversarium is unknown, and may require arterial bypass if ligation is not a viable option.

67.3 Delayed cervical instability

67.3.1 General information

Definition (adapted[12]): cervical instability that is not recognized until beyond 20 days after the injury (p.930). The instability itself may be delayed, or the recognition may be delayed.

67.3.2 Etiologies

Reasons for delayed cervical instability:
1. inadequate radiologic evaluation[13]
   a) incomplete studies (e.g. must see all the way down to C7-T1 junction)
   b) suboptimal studies: motion artifact, incorrect positioning… Etiologies include: poor patient cooperation as a result of agitation/intoxication, portable films, poor technique…
2. abnormality missed on x-ray
   a) overlooked fracture, subluxation
   b) injury failed to be demonstrated despite sufficiently adequate x-rays[12]; see recommendations of extent of radiologic workup (p.1088)
      • type of fracture not demonstrated on the radiographs obtained
      • patient positioning (e.g. supine) may reduce some malalignment
      • spasm of cervical muscles may reduce and/or stabilize the injury
      • microfractures
3. inadequate models: some findings may be judged to be stable using certain models, but in the long-run may prove to be unstable (there is no perfect model for instability)

67.3.3 Indications for additional studies

Further studies or repeat x-rays several weeks after the trauma should be considered in patients with neurologic deficit, persistent pain, significant degenerative changes when the original films were suboptimal, subluxations <3 mm, or when surgery is contemplated.[14]

67.4 Delayed deterioration following spinal cord injuries

Etiologies include:
1. posttraumatic syringomyelia (p.1148). Latency to symptoms: 3 mos-34 yrs
2. subacute progressive ascending myelopathy (SPAM): rare. Median time of occurrence: 13 days post injury (range: 4–86 days).[15] Signal changes extending to ≥4 levels above the original injury
3. unrecognized spinal instability[16]: mean delay in diagnosis was 20 days
4. tethered spinal cord: may be due to scar tissue at site of injury
5. delayed spinal epidural hematoma (SEH): most symptomatic SEH occur within 72 hours of surgery, however longer delays have been reported[17]
6. apoptosis of neurons, oligodendrogliocytes, and astrocytes[18]: initiated during the acute phase, deterioration occurs during the chronic phase of SCI (months to years after SCI)
7. glial scar formation: mass effect as well as release of factors that may damage surviving neurons[19 (p 43–5)]

67.5 Chronic management issues with spinal cord injuries

67.5.1 Overview

Most of the following topics are treated elsewhere in this manual, but are pertinent to spinal cord injured (SCI) patients, and reference to the specific section is made.
- autonomic hyperreflexia: see below
- ectopic bone, includes para-articular heterotopic ossification: ossification of some joints that occurs in 15–20% of paralyzed patients
- osteoporosis and pathologic fracture (p.1009)
- spasticity (p.1528)
- syringomyelia (p.1144)
- deep vein thrombosis (p.952): see below
- shoulder-hand syndrome: possibly sympathetically maintained

67.5.2 Respiratory management problems in spinal cord injuries

In attempting to wean high level SCI patients from a ventilator, it may be helpful to change tube feedings to Pulmonaid® which lowers the CO_2 load.

Patients with cervical SCIs are more prone to pneumonia due to the fact that most of the effort in a normal cough originates in the abdominal muscles which are paralyzed.

67.5.3 Autonomic hyperreflexia

General information

Key concepts

- exaggerated autonomic response to normally innocuous stimuli
- in spinal cord injury, occurs only in patients with lesions above ≈ T6
- patients complain of pounding headache, flushing and diaphoresis above lesion
- can be life threatening, requires rapid control of hypertension and a search for an elimination of offending stimuli

AKA autonomic dysreflexia. Autonomic hyperreflexia[20,21] (AH) is an exaggerated autonomic response (sympathetic usually dominates) secondary to stimuli that would only be mildly noxious under normal circumstances. It occurs in ≈ 30% of quadriplegic and high paraplegic patients (reported range is as high as 66–85%), but does not occur in patients with lesions below T6 (only patients with lesions above the origin of the splanchnic outflow are prone to develop AH, and the origin is usually T6 or below). It is rare in first 12–16 weeks post-injury.

During attacks, norepinephrine (NE) (but not epinephrine) is released. Hypersensitivity to NE may be partially due to subnormal resting levels of catecholamines. Homeostatic responses include vasodilatation (above the level of the injury) and bradycardia (however, sympathetic stimulation may also cause tachycardia).

Stimuli sources

Stimulus sources causing episodes of autonomic hyperreflexia:
1. bladder: 76% (distension 73%, UTI 3%, bladder stones…)
2. colorectal: 19% (fecal impaction 12%, administering enema or suppository 4%)
3. decubitus ulcers/skin infection: 4%
4. DVT
5. miscellaneous: tight clothing or leg bag straps, procedures such as cystoscopy or debriding decubitus ulcers, case report of suprapubic tube

Presentation

1. paroxysmal HTN: 90%
2. anxiety
3. diaphoresis

4. piloerection
5. pounding H/A
6. ocular findings:
   a) mydriasis
   b) blurring of vision
   c) lid retraction or lid lag
7. erythema of face, neck and trunk: 25%
8. pallor of skin below the lesion (due to vasoconstriction)
9. pulse rate: tachycardia (38%) or mild elevation over baseline, bradycardia (10%)
10. "splotches" over face and neck: 3%
11. muscle fasciculations
12. increased spasticity
13. penile erection
14. Horner's syndrome
15. triad seen in 85%: cephalgia (H/A), hyperhidrosis, cutaneous vasodilatation

Evaluation

In the appropriate setting (e.g. a quadriplegic patient with an acutely distended bladder), the symptoms are fairly diagnostic.

Many features are also common to pheochromocytoma. Studies of catecholamine levels have been inconsistent, however they can be mildly elevated in AH. The distinguishing feature of AH is the presence of hyperhidrosis and flushing of the face in the presence of pallor and vasoconstriction elsewhere on the body (which would be unusual for a pheochromocytoma).

Treatment

1. immediately elevate HOB (to decrease ICP), check BP q 5 min
2. treatment of choice: identify and eliminate the offending stimulus
   a) make sure bladder is empty (if catheterized check for kinks or sediment plugs). Caution: irrigating bladder may exacerbate AH (consider suprapubic aspiration)
   b) check bowels (avoid rectal exam, may exacerbate). Palpate abdomen or check abdominal x-ray (AH from this usually resolves spontaneously without manual disimpaction)
   c) check skin and toenails for ulceration or infection
   d) remove tight apparel
3. HTN that is extreme or that does not respond quickly may require treatment to prevent seizures and/or cerebral hemorrhage/hypertensive encephalopathy. Caution must be used to prevent hypotension following the episode. Agents used include: sublingual nifedipine[22] 10 mg SL, IV phentolamine – alpha cholinergic blocker (p.655) – or nicardipine (p.126).
4. consider diazepam (Valium®) 2–5 mg IVP (@<5 mg/min). Relieves spasm of skeletal and smooth muscle (including bladder sphincter). Is also anxiolytic

Prevention

Good bowel/bladder and skin care are the best preventative measures.

Prophylaxis in patients with recurrent episodes:

- phenoxybenzamine (Dibenzyline®): an alpha blocker. Not helpful during the acute crisis. May not be as effective for alpha stimulation from sympathetic ganglia as with circulating catecholamines.[23] The patient may also develop hypotension after the sympathetic outflow subsides. Thus this is used only for resistant cases (note: will not affect sweating which is mediated by acetylcholine).
  Adult: wide range quoted in literature: average 20–30 mg PO BID
- beta-blockers: may be necessary in addition to α-blockers to avoid possible hypotension from β_2-receptor stimulation (a theoretical concern)
- phenazopyridine (Pyridium®): a topical anesthetic that is excreted in the urine. May decrease bladder wall irritation, however, the primary cause of irritation should be treated if possible.
  Adult: 200 mg PO TID after meals. Supplied : 100, 200 mg tabs.
- "radical measures" such as sympathectomy, pelvic or pudendal nerve section, cordectomy, or intrathecal alcohol injection have been advocated in the past, but are rarely necessary and may jeopardize reflex voiding
- prophylactic treatment prior to procedures may employ use of anesthetics even in regions rendered anesthetic by the cord injury. Nifedipine 10 mg SL has also been used effective for AH during cystoscopy and prophylactically[22]

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XVI

68 Low Back Pain and Radiculopathy

68.1 General information

Key concepts

See reference.[1]

- low back pain is common, and in ≈ 85% of cases no specific diagnosis can be made
- initial assessment is geared to detecting "red flags" (indicating potentially serious pathology), and in the absence of these, imaging studies and further testing of patients is usually not helpful during the first 4 weeks of low back symptoms
- relief of discomfort is usually best achieved with nonprescription pain meds and/or spinal manipulation
- while activities may need to be modified, bed rest beyond 4 days may be more harmful than helpful, and patients are encouraged to return to work or their normal daily activities as soon as possible
- 89–90% of patients with low back problems will improve within 1 month even without treatment (including patients with sciatica from disc herniation)

Low back pain (LBP) is extremely prevalent, and is the second most common reason for people to seek medical attention.[2] After the common cold, it is the number two cause for loss of time at work. LBP accounts for ≈ 15% of all sick leave from work, and is the most common cause of disability for persons < 45 yrs age.[3] Estimates of lifetime prevalence range from 60–90%, and the annual incidence is 5%[4] Only 1% of patients will have nerve-root symptoms, and only 1–3% have lumbar disc herniation. The prognosis for most cases of LBP is good, and improvement usually occurs with little or no medical intervention.

68.2 Intervertebral disc

68.2.1 General information

The function of the intervertebral disc is to permit stable motion of the spine while supporting and distributing loads under movement. The intervertebral disc has been characterized as the largest nonvascularized structure in the human body, which imparts some unique attributes to it.

68.2.2 Anatomy

Anulus fibrosus (anulus may alternatively be spelled annulus, but fibrosus is the only correct spelling and is distinct from fibrosis)[5]: the multilaminated ligament that encompasses the periphery of the disc space. Attaches to the end-plate cartilage and ring apophyseal bone. Blends centrally with the nucleus pulposus.

Nucleus pulposus: the central portion of the disc. A remnant of the notocord.

Capsule[5]: combined fibers of the anulus fibrosus and the posterior longitudinal ligament (this term is useful because these 2 structures may not be distinguishable on imaging studies).

68.3 Nomenclature for disc pathology

Historically, the terminology for lumbar disc pathology has been contentious and nonstandardized. A committee tasked to standardize the nomenclature has issued version 2.0 of their recommendations.[6] Some of these standardizations are useful primarily for consistency related to radiographic reports and for research, and may not be as useful for day-to-day clinical practice A subset of the recommendations is shown in Table 68.1.

Degenerated disc: (see Table 68.1 for definition) some reports indicate that these can cause radicular pain possibly by an inflammatory mechanism,[7] but this is not universally accepted.

Vacuum disc: gas in the disc space (empty space on imaging), usually indicates disc degeneration, not infection.

Table 68.1 Nomenclature for lumbar disc pathology[6]

Term	Description
anular tears AKA anular fissures	separations between anular fibers, avulsions of fibers from their VB insertions, or breaks through fibers that extend radially, transversely, or concentrically
degeneration	desiccation, fibrosis, narrowing of the disc space, diffuse bulging of the anulus beyond the disc space, extensive fissuring (numerous anular tears), mucinous degeneration of the anulus, defects & sclerosis of endplates, & osteophytes at the vertebral apophyses
degenerative disc disease	clinical syndrome of symptoms related to degenerative changes in the intervertebral disc (described above), also often considered to encompass degenerative changes outside the disc as well
bulging disc	generalized displacement of disc material (arbitrarily defined as > 50% or 180°) beyond the peripheral limits of the disc space[a]. Not considered a form of herniation. May be a normal finding, not usually symptomatic
herniation	localized displacement of disc material (< 50% or 180°) beyond the limits of the intervertebral disc space[a]
	focal: < 25% of the disc circumference
	broad-based: 25–50% of the disc circumference
	protrusion: the fragment does not have a “neck” that is narrower than the fragment in any dimension
	extrusion: the fragment has a “neck” that is narrower than the fragment in at least 1 dimension. 2 subtypes a) sequestration: the fragment has lost continuity with the disc of origin (AKA free fragment) b) migration: the fragment is displaced away from the site of extrusion, regardless of whether sequestered or not
	intravertebral herniation (AKA Schmorl’s node (p. 1060): disc herniates in the cranio-caudal direction through the cartilaginous end-plate into the VB

Table 68.2 Modic’s classification

Modic Type	Intensity changes		Description	
	T1WI	T2WI		
1[a]	↓	↑	bone marrow edema associated with acute or subacute inflammation	
2	↑	iso or ↑	chronic changes	replacement of bone marrow by fat
3	↓	↑		reactive osteosclerosis

[a]Type 1 Modic changes disappear on STIR images (the signal characteristics mimic CSF/water). Back pain in this group may respond to fusion (p. 1036).

68.4 Vertebral body marrow changes

Associated with degenerative or inflammatory changes. Modic’s classification[8] of MRI characteristics is shown in Table 68.2.

68.5 Clinical terms

Radiculopathy. Dysfunction of a nerve root; signs and symptoms may include: pain in the distribution of that nerve root, dermatomal sensory disturbances, weakness of muscles innervated by that nerve root, and hypoactive muscle stretch reflexes of the same muscles

 Mechanical low back pain (p.1024) . AKA "musculoskeletal" back pain (both non-specific terms). The most common form of low back pain. May result from strain of the paraspinal muscles and/or ligaments, irritation of facet joints… Excludes anatomically identifiable causes (e.g. tumor, disc herniation…)

 Sciatica. Pain along the course of the sciatic nerve, usually resulting from nerve root compromise (the sciatic nerve is comprised of nerve roots L1 through L5)

68.6 Disability, pain and outcome determinations

Disability scales for low back pain have been developed to assess outcomes for research purposes. Some widely used measures include:
1. visual analogue scale: used for any type of pain. The patient is asked to mark their pain level on a line divided into segments with sequential labels 0 (no pain) to 10 (the worst pain)
2. Oswestry disability index (ODI)[9]: a categorical ordinal scale that is used for low back pain. There are 4 English versions in wide use,[10] version 2.0[11] is recommended.[10]
   It consists 10 questions related to activities of daily living. Each item is scored 0–5 (5 being the most disability) and the total is multiplied by 2% to obtain the final score (range: 0–100%). The interpretation of the final score is shown in Table 68.3. A score > 45% is essentially completely disabled. A score in the teens is very functional
3. Roland–Morris disability questionnaire[12]
4. Short Form 36 (SF36)[13]

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68.7 Differential diagnosis of low back pain

The differential diagnosis of low back pain (p.1024) overlaps with that of myelopathy. In ≈ 85% of cases of LBP no specific diagnosis can be made,[14] however, serious and/or dangerous conditions can usually be reliably ruled out.

68.8 Initial assessment of the patient with back pain

68.8.1 Background

Initial assessment consists of a history and physical exam focused on identifying serious underlying conditions such as: fracture, tumor, infection or cauda equina syndrome (p.1050). Serious conditions presenting as low back problems are relatively rare.

68.8.2 History

The following information has been found to be helpful in identifying patients with serious underlying conditions such as cancer and spinal infection.[1] Table 68.4 shows the sensitivity and specificity of some features of the history for various conditions.
1. age
2. history of cancer (especially malignancies that are prone to skeletal metastases: prostate, breast, kidney, thyroid, lung, lymphoma/myeloma)
3. unexplained weight loss

Table 68.3 Oswestry disability index score

Score	Interpretation
0–20%	minimal disability: can cope with most daily activities
21–40%	moderate disability: pain and difficulty with sitting, lifting & standing. The patient may be disabled from work
41–60%	severe disability: pain is the main problem, but other areas are affected
61–80%	crippled: back pain impinges on all aspects of the patient's life
81–100%	these patients are either bed-bound or else are exaggerating their symptoms

Table 68.4 Sensitivity and specificity of historical findings in patients with low back problems[1]

Condition	History	Sensitivity	Specificity
cancer	age ≥ 50 yrs	0.77	0.71
	previous Ca	0.31	0.98
	unexplained weight loss	0.15	0.94
	failure to improve after conservative therapy × 1 month	0.31	0.90
	any of the above	1.00	0.60
	pain > 1 month	0.50	0.81
spinal osteomyelitis	IV drug abuse, UTI, or skin infection	0.40	NA
compression fracture	age ≥ 50 yrs	0.84	0.61
	age ≥ 70yrs	0.22	0.96
	trauma	0.30	0.85
	steroid use	0.06	0.995
HLD	sciatica	0.95	0.88
spinal stenosis	pseudoclaudication	0.60	NA
	age ≥ 50 yrs	0.90[a]	0.70
ankylosing spondylitis	positive response to 4 out of 5 of the following	0.23	0.82
	age at onset ≤ 40 yrs	1.00	0.07
	pain not relieved when supine	0.80	0.49
	AM back stiffness	0.64	0.59
	pain ≥ 3 mos duration	0.71	0.54

[a]estimate

4. immunosuppression: from steroids, organ transplant medication, or HIV
5. prolonged use of steroids
6. duration of symptoms
7. responsiveness to previous therapy
8. pain that is worse at rest
9. history of skin infection: especially furuncle
10. history of IV drug abuse
11. UTI or other infection
12. pain radiating below the knee
13. persistent numbness or weakness in the legs
14. history of significant trauma. In a young patient: usually involves MVA, a fall from a height, or a direct blow to the back. In an older patient: minor falls, heavy lifting or even a severe coughing episode can cause a fracture especially in the presence of with osteoporosis
15. findings consistent with cauda equina syndrome (p.1050):
    a) bladder dysfunction (usually urinary retention, or overflow incontinence) or fecal incontinence
    b) saddle anesthesia (p.1050)
    c) unilateral or bilateral leg weakness or pain
16. psychological and socioeconomic factors may influence the patient's report of symptoms (p.1033), and one should inquire about:
    a) work status
    b) typical job tasks
    c) educational level

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d) pending litigation
e) worker's compensation or disability issues
f) failed previous treatments
g) substance abuse
h) depression

68.8.3 Physical examination

Less helpful than the history in identifying patients who may be harboring conditions such as cancer, but may be more helpful in detecting spinal infections.

1. spinal infection (p. 349): findings that suggest this as a possibility (but are also common in patients without infection)
   a) fever: common in epidural abscess and vertebral osteomyelitis, less common in discitis
   b) vertebral tenderness
   c) very limited range of spinal motion
2. findings of possible neurologic compromise: the following physical findings will identify most cases of clinically significant nerve root compromise due to L4–5 or L5-S1 HLD which comprise > 90% of cases of radiculopathy due to HLD; limiting the exam to the following might not detect the much less common upper lumbar disc herniations, which may be difficult to detect on PE (p. 1057)
   a) dorsiflexion strength of ankle and great toe: weakness suggests L5 and some L4 dysfunction
   b) achilles reflex: diminished reflex suggests S1 root dysfunction
   c) light touch sensation of the foot:
      - diminished over medial malleolus and medial foot: suggests L4 nerve root involvement
      - diminished over dorsum of foot: suggests L5
      - diminished over lateral malleolus and lateral foot: suggests S1
   d) straight leg raising (SLR); also check for crossed SLR (p. 1048)

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68.8.4 "Red flags" in the history and physical exam for low back problems

Based upon the above history and physical exam, the findings in Table 68.5 would suggest the possibility of a serious underlying condition as the cause of the low back problem. Also, thoracic region pain is relatively uncommon and should raise the index of suspicion.

68.8.5 Special diagnostic tests

For patients without features suggesting a serious underlying condition, special diagnostic tests are not needed during the first month of symptoms. This covers approximately 95% of patients with low back problems.[1]

Table 68.5 "Red flags" for patients with low back problems

Condition	Red flags
cancer or infection	1. age > 50 or < 20 yrs 2. history of cancer 3. unexplained weight loss 4. immunosuppression (see text) 5. UTI, IV drug abuse, fever or chills 6. back pain not improved with rest
spinal fracture	1. history of significant trauma (see text) 2. prolonged use of steroids 3. age > 70 yrs
cauda equina syndrome or severe neurologic compromise	1. acute onset of urinary retention or overflow incontinence 2. fecal incontinence or loss of anal sphincter tone 3. saddle anesthesia 4. global or progressive weakness in the LEs

68.9 Radiographic evaluation

68.9.1 General information

Diagnosing lumbar spinal stenosis or herniated intervertebral disc is usually helpful only in potential surgical candidates.[15] This includes patients with appropriate clinical syndromes who have not responded satisfactorily to adequate non-surgical treatment over a sufficient period of time, and who have no medical contraindications to surgery. Radiologic confirmation of these diagnoses usually requires CT, myelography, MRI, or some combination (see below). NB: myelography,[16] CT,[17] or MRI[18] may also show bulging or herniated lumbar discs (HLD) or spinal stenosis in asymptomatic patients (e.g. 24% of asymptomatic patients have herniated discs on MRI and 4% have spinal stenosis; these numbers become 36% and 21% respectively in patients 60–80 years old).[19] Thus, these tests must be interpreted in light of clinical findings, and the anatomic level and side should correspond to the history, examination, and/or other physiologic data. Diagnostic radiology is of limited benefit as the initial evaluation in the majority of spinal disorders.[20]

In the absence of red flags for serious conditions, imaging studies are not recommended in the first month of symptoms.[1] For patients who have had previous back surgery, MRI with contrast is probably the best test. Myelography (with or without CT) is invasive and has increased risk of complications, and is therefore indicated only in situations where MRI cannot be done or is inadequate, and the possibility of surgery is anticipated.

Σ

Patients for whom radiographic imaging is recommended are those with:
1. suspected benign conditions with symptoms persisting > 4 weeks of great enough severity to consider surgery, including:
   a) back related leg symptoms and clinically specific signs of nerve root compromise
   b) a history of neurogenic claudication (p. 1100) or other finding suggestive of lumbar spinal stenosis
   c) symptoms related to spinal deformity/imbalance, especially positional back pain that increases with time spent upright
2. red flags: physical examination or other test results suggesting other serious conditions affecting the spine (e.g. cauda equina syndrome, fracture, infection, tumor, or other mass lesions or defects)

Recommendations for use of MRI and discography to select patients for fusion are shown in the Practice Guideline (p. 1029).

Practice guideline: MRI and discography for patient selection for lumbar fusion*

1. Level II[21]:
   a) MRI is recommended as the initial diagnostic test
   b) normal appearing discs on MRI should not be considered for discography or treatment
   c) lumbar discography should not be used as a stand-alone test
   d) to consider a disc level for treatment, if discography is used, there should be a concordant pain response[a] and associated abnormalities on MRI[b]
2. Level III[21]: discography should be reserved for equivocal MRI findings, especially at levels adjacent to unequivocally abnormal levels

Notes:

* see also recommendations on use of facet injections (p. 1036)

[a] concordant pain response: pain identical or very similar to the patient's usual pain complaints (NB: discography can produce severe LBP in patients with no prior complaints[22,23])

[b] abnormal disc morphology on MRI: loss of T2WI signal intensity ("black disc"), disc space collapse, Modic changes (see Table 68.2), and high-intensity zones (these findings also frequently occur in asymptomatic patients[24])

68.9.2 Plain lumbosacral x-rays

General information

Unexpected findings occurred in only 1 in 2500 adults < 50 years age.[25] Diagnosis of surgical conditions of disc herniation and spinal stenosis cannot be made from plain films (although they may be inferred, further study would be required). Various congenital abnormalities of uncertain significance may be identified (e.g. spina bifida occulta), and evidence of degenerative changes (including osteophytes) are as frequent in symptomatic as in asymptomatic patients. Gonadal radiation is significant. Seldom indicated during pregnancy.

Recommendation

Not recommended for routine evaluation of patients with acute low back problems during the first month of symptoms unless a "red flag" is present (see below). Reserve LS x-rays for patients with a likelihood of having spinal malignancy, infection, inflammatory spondylitis, or clinically significant fracture. In these cases, plain x-rays are often just a starting point, and further study (CT, MRI…) may be indicated even if the plain x-rays are normal. "Red flags" for these conditions include the following:

- age > 70 years, or < 20 yrs
- systemically ill patients
- temp > 100°F (or > 38° C)
- history of malignancy
- recent infection
- patients with neurologic deficits suggesting possible cauda equina syndrome (saddle anesthesia, urinary incontinence or retention, LE weakness) (p. 1050)
- heavy alcohol or IV drug abusers
- diabetics
- immunosuppressed patients (including prolonged treatment with corticosteroids)
- recent urinary tract or spinal surgery
- recent trauma: any age with significant trauma, or > 50 yrs old with mild trauma
- unrelenting pain at rest
- persistent pain for more than ≈ 4 weeks
- unexplained weight loss

When spine x-rays are indicated, AP and lateral views are usually adequate.[26] Obliques and coned-down L5-S1 views more than double the radiation exposure, and add information in only 4–8% of cases,[27] and can be obtained in specific instances where warranted (e.g. to diagnose spondylolysis when spondylolisthesis is found on the lateral film).

68.9.3 MRI

Unless contraindicated, noncontrast MRI is the initial diagnostic test of choice for diagnosing most cases of disc herniation and spinal stenosis. Specificity and sensitivity for HLD are on the same order as CT/myelography, which is better than myelography alone.[1,28,29]

Advantages:

- Provides the most information about soft tissues (intervertebral discs, spinal cord, inflammation…) of any available diagnostic test
- provides information regarding tissue outside of the spinal canal, e.g. extreme lateral disc herniation (p. 1058), tumors…
- non-invasive and does not utilize ionizing radiation

Disadvantages:

- patients in severe pain or with claustrophobia may have difficulty holding still
- dose not visualize bone well
- poor for studying blood early (e.g. spinal epidural hematoma)
- expensive
- interpretation with scoliosis is more difficult, may be partially compensated by contouring visualization plane through center of canal
- a number of contraindications: see Contraindications to MRI (p. 230)

Findings:

In addition to demonstrating herniated lumbar disc (HLD) outside of the disc interspace compressing nerve root or thecal sac, MRI can demonstrate signal changes within the interspace suggestive of disc degeneration[30] (loss of signal intensity on T2WI, loss of disc space height) and is useful in diagnosing infections and tumors.

68.9.4 Lumbosacral CT

If technically adequate images can be obtained (e.g. good quality scanner, images not obscured by artifact from patient movement or obesity), CT can demonstrate most spine pathology. For HLD, sensitivity is 80–95%, and specificity is 68–88%[31,32] However, even some large disc herniations will be missed with plain CT. CT studies for HLD tend to be less satisfactory in the elderly. When MRI is an option, the main utility of CT is for imaging bone to assess fractures or to demonstrate details of bony anatomy for surgery.

Disc material has density (Hounsfield units) ≈ twice that of the thecal sac. Associated findings with herniated disc include:
- loss of epidural fat (normally seen as low density in the anterolateral canal)
- loss of normal "convexity" of thecal sac (indentation by herniated disc)

Advantages:
- excellent bony detail
- non-invasive
- outpatient evaluation
- evaluates paraspinal soft tissue (e.g. to rule out tumor, paraspinal abscess…)
- advantages over MRI: faster scanning (significant in patients who have difficulty laying still for long time), less expensive, less claustrophobic, fewer contraindications, see Contraindications to MRI (p.230)

Disadvantages:
- involves ionizing radiation (x-rays)
- sensitivity is significantly lower than MRI or myelogram/CT

68.9.5 Myelography

With water soluble intrathecal contrast introduced via lumbar puncture, sensitivity (62–100%) and specificity (83–94%)[33,34,35,36] are similar to CT for detection of HLD. Usually combined with post-myelographic CT scan (myelogram/CT), which increases the sensitivity and especially the specificity.[37] A herniated disk in the large space between thecal sac and posterior border of vertebral bodies at L5-S1 (insensitive space) may not be seen on myelography alone (CT or MRI are usually better at demonstrating this).

Advantages:
1. evaluates cauda equina better than noncontrast CT
2. provides "functional" information about degree of stenosis (a high-degree block will allow flow of dye only after certain position changes)
3. when combined with CT, may demonstrate some anatomy obscured by metal artifact on MRI in patients with prior instrumentation

Disadvantages:
1. may miss pathology outside of the dura (including far laterally herniated disc), sensitivity is improved with post-myelographic CT
2. invasive
   a) drugs e.g. warfarin must be stopped, and sometimes bridged to heparin
   b) with occasional side effects (post LP H/A, N/V, rare seizures)
3. iodine allergic patients
   a) requires iodine allergy prep
   b) may still be risky (especially in severely iodine allergic patients)

Findings:

HLD produces extradural filling defect at the level of the intervertebral disc. Massive disc herniation or severe lumbar stenosis may produce a total or near total block. In some cases of HLD, the finding may be very subtle and may consist of a cut-off of the filling (with contrast) of the nerve root sleeve (compared to normal nerve(s) on contralateral side or at other levels). Another subtle finding may be a "dual shadow" on lateral view.

68.9.6 Bone scan

See Bone scan for low back problems (p.1032).

68.9.7 Discography

Injection of water-soluble contrast agent directly into the nucleus pulposus of the intervertebral disc being studied by percutaneous needle access through Kambin's trianglein an awake patient. Results of the test depend on volume of dye accepted into the disc, the pressure needed to inject the dye, the configuration of the dye (including leakage from the confines of the disc space) on radiographic imaging (plain x-rays produce the so-called "discogram", CT scan is often also be utilized), and reproduction of the patient's pain on injection. Some of the basis for performing a discogram is to identify levels that may produce "discogenic pain" or "painful disc syndrome" (p.1032), a controversial point. When the pain produced mimics the patient's presenting pain, the pain is said to be "concordant."

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Critique:

Invasive. Interpretation is equivocal, and complications may occur (disc space infection, disc herniation, and significant radiation exposure with CT-discography). May be abnormal in asymptomatic patients[22,23] (as any of the above tests may be) although the false positive rate may not be quite this high.[38] See Practice guideline: MRI and discography for patient selection for lumbar fusion (p.1029) for recommendations.

68.10 Electrodiagnostics for low back problems

If the diagnosis of radiculopathy seems likely on clinical grounds, electrophysiologic testing is not recommended.[1]

1. needle EMG (p.242): can assess acute and chronic nerve root dysfunction, myelopathy and myopathy, and may be useful for patients with suspicion of other conditions (e.g. neuropathy) or when a reliable strength exam is not possible. Reduced recruitment may be seen within the first several days of onset, however, spontaneous activity takes 10–21 days to develop (p.242) (less helpful in the first ≈ 3 weeks). Also, not usually helpful with normal muscle strength exam. Accuracy is highly operator dependent and improves with knowledge about imaging studies and clinical information.[39] See findings in radiculopathy (p.243).
2. H-reflex (p.243): measures sensory conduction through nerve roots. Use is limited to assessing S1 radiculopathy.[40] Correlates with achilles reflex.
3. SSEPs (p.239): assesses afferent fibers which travel in peripheral nerve and the posterior column of the spinal cord. May be abnormal in conditions affecting the dorsal columns with impaired joint position and proprioception (e.g. cervical spondylotic spinal myelopathy)
4. nerve conduction studies (including NCVs): helps identify acute and chronic entrapment neuropathies that may mimic radiculopathy
5. not recommended for assessing acute low back problems[1]
   a) F-wave response (p.243): measures motor conduction through nerve roots, used to assess proximal neuropathies
   b) surface EMG: assesses acute and chronic recruitment patterns during static or dynamic tasks using surface (instead of needle) electrodes

68.11 Bone scan for low back problems

Description: injection of a radiolabeled compound (usually technetium-99m) that is taken up by metabolically active bone. A gamma camera localizes regions of uptake. Total radiation dose is ≈ to a set of lumbar spine x-rays.[1] Contraindicated during pregnancy. Breast feeding must be briefly suspended following a bone scan due to presence of radiotracer in the breast milk.

A moderately sensitive test which may be used in evaluating low back pain when spinal tumor,[41] infection,[42] or occult fracture is suspected from "red flags" (see Table 68.5) on history or examination, or results of lab tests or plain x-rays. Not very specific, but may locate occult lesions and help differentiate these conditions from degenerative changes. A positive bone scan suggesting one of these conditions usually must be confirmed by other diagnostic tests or procedures (no studies have compared bone scans to CT or MRI).

Low yield in patients with longstanding low back problems and normal plain x-rays and laboratory tests (especially ESR or CRP).[41]

SPECT scans may provide additional information to a bone scan.

68.12 Thermography for low back problems

 Not recommended.[1] Did not accurately predict absence or presence of nerve root compression seen at surgery,[43] and may be positive in a significant percentage of asymptomatic patients.[44]

68.13 Psychosocial factors

Although some patients with chronic LPB (>3 months duration) may have started off with a diagnosable condition, psychological and socioeconomic factors (such as depression, secondary gain…) may come to play a significant role in perpetuating or amplifying pain. Psychological factors, especially elevated hysteria or hypochondriasis scales on the Minnesota Multiphasic Personality Inventory (MMPI) were found to be a better predictor of outcome than findings on radiographic imaging in one study.[39] A screening scale of 5 factors has been proposed[45] (positive findings in any 3 suggests psychological distress):

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1. These items are potentially reliable[46]
   a) pain on simulated axial loading: press on top of head
   b) inconsistent performance: e.g. difficulty tolerating straight leg raising (SLR) while supine, but no difficulty when sitting
   c) overreaction during the physical exam
2. These items may not be reliable[46]
   a) inappropriate tenderness that is superficial or widespread
   b) motor or sensory abnormalities not corresponding to anatomic boundaries (e.g. for sensation: dermatomes, peripheral nerve distribution…)

However, the usefulness of this information is limited, and no effective interventions have been identified to address these factors. Therefore the AHCPR panel was unable to recommend specific assessment tools or interventions.[1]

68.14 Treatment

68.14.1 General information

An initial period of nonsurgical management (below) is indicated except in the following circumstances where urgent surgery is indicated:

Situations where conservative treatment is not indicated:

- symptoms of cauda equina syndrome: urinary retention, saddle anesthesia... (p.1050)
- progressive neurologic deficit, or profound motor weakness
- a relative indication for proceeding to urgent surgery without conservative management is severe pain that cannot be sufficiently controlled with adequate pain medication (rare)

If specific diagnoses such as herniated intervertebral lumbar disc or symptomatic lumbar stenosis are made, surgical treatment for these conditions may be considered if the patient fails to improve satisfactorily. In cases where no specific diagnosis can be made, management consists of conservative treatment and following the patient to rule out the possible development of symptoms suggestive of a more serious diagnosis that may not have initially been evident.

68.14.2 “Conservative” treatment

This term has regrettably come to be used for non-surgical management. With minor modification, similar approaches can be used for mechanical low back pain, as well as for acute radiculopathy from disc herniation.

Recommendations (based on AHCPR findings[1] in the absence of “red flags”; note: some key literature citations are given here, primarily those from the better studies that support the Agency for

Health Care Policy and Research [AHCPR] panel recommendations. However, refer to Bigos et al.[1] for full analysis and list of references):

1. activity modifications: no studies were found that met the panels review criteria for adequate evidence. However, the following information was felt to be useful:
   a) bed rest: for 2–3 days maximum
      - the theoretical objective is to reduce symptoms by reducing pressure on the nerve roots and/or intradiscal pressures which is lowest in the supine semi-Fowler position,[47] and also to reduce movements which are experienced as painful by the patient
      - deactivation from prolonged bed rest (> 4 days) appears to be worse for patients (producing weakness, stiffness, and increased pain) than a gradual return to normal activities[48]
      - recommendations: the majority of patients with low back problems will not require bed rest. Bed rest for 2–4 days may be an option for those with severe initial radicular symptoms, however, this may be no better than watchful waiting[49] and may be harmful[50]
   b) activity modification
      - the goal is to achieve a tolerable level of discomfort while continuing sufficient physical activity to minimize disruption of daily activities
      - risk factors: although there is not agreement on their exact role, the following were identified as having an increased incidence of low back problems. Jobs requiring heavy or repetitive lifting, total body vibration (from vehicles or industrial machinery), asymmetric postures, or postures sustained for long periods (including prolonged sitting)
      - recommendations: temporarily limit heavy lifting, prolonged sitting, and bending or twisting of the back. Establish activity goals to help focus attention on expected return to full functional status
   c) exercise (may be part of a physical therapy program):
      - during the 1st month of symptoms, low-stress aerobic exercise can minimize debility due to inactivity. In the first 2 weeks, utilize exercises that minimally stress the back: walking, bicycling, or swimming
      - conditioning exercises for trunk muscles (especially back extensors, and possibly abdominal muscles) are helpful if symptoms persist (during the first 2 weeks, these exercises may aggravate symptoms)
      - there is no evidence to support stretching of back muscles, or to recommend back-specific exercise machines over traditional exercise
      - recommended exercise quotas that are gradually escalated results in better outcome than having patients simply stop when pain occurs[51]
2. analgesics:
   a) for the initial short-term period, acetaminophen (APAP) or NSAIDs (p. 137) may be used. In one study[52] of acute LBP, NSAIDs did not add any benefit to APAP + standard education (see below)
   b) stronger analgesics – mostly opioids (p. 138) – may be required for severe pain, usually severe radicular pain. For non-specific back pain, there was no earlier return to full activity than with NSAIDs or APAP.[1] Opioids should not be used > 2–3 weeks, at which time NSAIDs should be instituted unless contraindicated
3. muscle relaxants
   a) the therapeutic objective is to reduce pain by relieving muscle spasm. However, muscle spasms have not been proven to cause pain, and the most commonly used muscle relaxants have no peripheral effect on muscle spasm
   b) probably more effective than placebo, but have not been shown to be more effective than NSAIDs, and their use in combination with NSAIDs has not been shown to be more effective than use of NSAIDs alone
   c) potential for side effects: drowsiness (in up to 30%). Most manufacturers recommend use for < 2–3 weeks. Agents such as chlorzoxazone (Parafon Forte® and others) may be associated with risk of serious and potentially fatal hepatotoxicity[53]
4. education: (may be provided as part of a physical therapy program)
   a) explanation of the condition to the patient[54] in understandable terms, and positive reassurance that the condition will almost certainly subside[55] have been shown to be more effective than many other forms of treatment
   b) proper posture, sleeping positions, lifting techniques… should be conveyed to the patient. Formal "back school" seems to be marginally effective.[56] There may be some early benefit, but long-term efficacy could not be shown.[57] The quality and expense of such programs varies widely[1]
5. spinal manipulation therapy (SMT): defined as manual therapy in which loads are applied to the spine using long or short lever methods with the selected joint being taken to its end range of

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voluntary motion, followed by application of an impulse loading (may be part of a physical therapy program)
   a) may be helpful for patients with acute low back problems without radiculopathy when used in the first month of symptoms (efficacy after 1 month is unproven) for a period not to exceed 1 month. One study[52] found no added benefit to APA+standard education
   b) there is insufficient evidence to recommend SMT in the presence of radiculopathy
   c) SMT should not be used in the face of severe or progressive neurologic deficit until serious conditions have been ruled out
   d) reports of arterial dissection: especially vertebral artery (p. 1325) and stroke, myelopathy & subdural hematoma with cervical SMT and cauda equina syndrome with lumbar SMT[58,59,60] and the uncertainty of benefits have led to the questioning of the use of SMT[58] (especially cervical)
6. epidural injections:
   a) epidural (cortico)steroid injections (ESI): there is no evidence that this is effective in treating acute radiculopathy.[61] Most studies that show benefit are retrospective and noncontrolled. Prospective studies yield varied results.[62] Some improvement at 3 & 6 weeks may occur (but no functional benefit, and no change in the need for surgery), with no benefit at 3 months.[63] The response in chronic back pain is poor in comparison to acute pain. ESI may be an option for short-term relief of radicular pain when control on oral medications is inadequate or for patients who are not surgical candidates
   b) there is no evidence to support the use of epidural injections of steroids, local anesthetics and/or opioids for LBP without radiculopathy
   c) reports on efficacy with conditions such as lumbar spinal stenosis are conflicting,[62] relief is almost uniformly temporary (4–6 weeks with initial injection, shorter times with subsequent ones)

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 Not recommended by the AHCPR panel[1] for treatment of acute low back problems in the absence of "red flags" (Table 68.5):
1. medications
   a) oral steroids: no difference was found at one week and 1 year after randomization to receive 1 week therapy with oral dexamethasone or placebo[64]
   b) colchicine: conflicting evidence shows either some[65] or no[66] therapeutic benefit. Side effects of N/V and diarrhea were common[1]
   c) antidepressant medications: most studies of these medications were for chronic back pain. Some methodologically flawed studies failed to show benefits when compared to placebo for chronic (not acute) LBP[67]
2. physical treatments
   a) TENS (transcutaneous electrical nerve stimulation): not statistically significantly better than placebo, and added no benefit to exercise alone[68]
   b) traction (including pelvic traction): not demonstrated to be effective.[69] One possible explanation for lack of benefit is that due to the sizable paraspinal muscles and ligaments (as compared to the cervical spine) the amount of weight required to distract the intervertebral disc space is approximately ≥ 2/3 of the patient's body weight, which is painful and/or pulls the patient to the foot of the bed
   c) physical agents and modalities: including heat (including diathermy), ice, ultrasound. Benefit is insufficiently proven to justify their cost, however, self-administered home programs for application of heat or cold may be considered. Ultrasound and diathermy should not be used in pregnancy
   d) lumbar corsets and support belts: not proven beneficial for acute back problems. Prophylactic use has been advocated to reduce time lost from work by individuals doing frequent lifting as part of their job, but this is controversial[70]
   e) biofeedback: has not been studied for acute back problems. Primarily advocated for chronic LBP, where effectiveness is controversial[71]
3. injection therapy
   a) trigger point and ligamentous injections: the theory that trigger points cause or perpetuate LBP is controversial and disputed by many experts. Injections of local anesthetic are of equivocal efficacy (saline may be as effective[72]) and are mildly invasive
   b) (zygapophyseal) facet joint injections: theoretical basis is that there exists a "facet syndrome" producing LBP which is aggravated by spine extension, with no nerve root tension signs (p. 1047). No studies have adequately investigated injections for pain < 3 months duration. For chronic LBP, neither the agent nor the location (intrafacet or pericapsular) made a significant difference in outcomes[73,74]

c) epidural injections in the absence of radiculopathy: see above
d) acupuncture: no studies were found that evaluated the use in acute back problems. All randomized clinical trials found were for patients with chronic LBP, and even the best studies were felt to be mediocre and contradictory. Meta-analysis found acupuncture was more effective in relieving chronic LBP than sham or no treatment,[75] but there was no comparison to other therapies

Practice guideline: Injection therapy for low-back pain

Therapeutic recommendations

Level III[76]: lumbar epidural injections or trigger point injections are not recommended for long-term relief of chronic LBP. These techniques or facet injections may be used to provide temporary relief in select patients

Diagnostic recommendations

Level III[76]: lumbar facet injections
- may predict the response to radiofrequency facet ablation
- □ not recommended as a diagnostic tool to predict the response to lumbar fusion

68.14.3 Surgical treatment

Indications for surgery for herniated lumbar disc

See the section on herniated lumbar discs (p.1049).

Indications for fusion for chronic LBP without stenosis or spondylolisthesis

Very controversial.

Practice guideline: Lumbar fusion for LBP without stenosis or spondylolisthesis

Level I[77]: lumbar fusion is recommended for carefully selected patients with disabling LBP due to one- or two-level degenerative disease without stenosis or spondylolisthesis (in the primary quoted study[78] patients had chronic LBP for ≥ 2 years and had radiologic evidence of disc degeneration at L4-L5, L5-S1, or both, and had failed best medical management)

Level III[77,79]: an intensive course of PT and cognitive therapy is recommended as an option for patients with LBP in whom conventional medical management has failed

Practice guideline: Choice of fusion technique

Level II[80]: for ALIF or ALIF + instrumentation, the addition of a posterolateral fusion is not recommended (the demonstrated benefit does not outweigh the additional time and blood loss involved)

Level III[80]:
- either a posterolateral fusion or an interbody fusion (PLIF, TLIF or ALIF) are options for patients with LBP due to DDD at 1 or 2 levels
- an interbody graft is an option to improve fusion rates and functional outcome (caution: the improvement in fusion rate and outcome is marginal, and interbody fusion is associated with an increased complication rate, especially with combined approaches, e.g. 360° fusion)

□ the use of multiple approaches (anterior + posterior) is not recommended as a routine option for LBP without deformity

Surgical treatment options

The type of surgical procedure chosen is tailored to the specific condition identified. Examples are shown in Table 68.6. Discussion of some options is also provided below.

Lumbar spinal fusion

Although there is no consensus on the indications,[81] lumbar spinal fusion (LSF) is accepted treatment for fracture/dislocation or instability resulting from tumor or infection.

For degenerative spine disease, practice parameters have been developed and are included herein. Pain associated with Modic type 1 changes (see Table 68.2) may respond to stabilization procedures, the other Modic types do not exhibit this association.

Practice guideline: Lumbar fusion for disc herniation

Level III[82]:
1. lumbar fusion is not routinely recommended following disc excision in patients with HLD or 1st time recurrent HLD causing radiculopathy
2. lumbar fusion is a potential adjunct to disc excision in cases of a HLD or recurrent HLD:
   a) with evidence of preoperative lumbar spinal deformity or instability
   b) in patients with chronic axial LBP associated with radiculopathy

Instrumentation as an adjunct to fusion

Practice guideline: Pedicle screw fixation

Level III[83]: pedicle screw fixation is recommended as a treatment option for patients with LBP treated with posterolateral fusion who are at high risk for fusion failure (routine use of pedicle screws is discouraged because of conflicting evidence of benefit, together with considerable evidence of increased cost and complications)

The use of instrumentation increases the fusion rate.[84] Hardware used in the absence of fusion will eventually fatigue, especially in the region of the lumbar lordosis. Therefore, instrumentation must be viewed as a temporary internal stabilizing measure while awaiting the fusion process to complete.

68.15 Chronic low back pain

Rarely can an anatomic diagnosis be made in patients with chronic LBP ≥ 3 months duration.[85] Also, see Psychosocial factors (p. 1033). Patients with chronic pain syndromes (CPS) refer to their problems with affective or emotional terms with a higher frequency than those with acute pain.[86] The

Table 68.6 Surgical options for low back problems

Condition	Surgical treatment options
"routine" HLD or initial recurrence of HLD	• standard discectomy and microdiscectomy are of similar efficacy • intradiscal procedures: nucleotome, laser disc decompression are not recommended (p. 1052)
foraminal or far lateral HLD	• partial or total facetectomy (p. 1059) • extracanal approach (p. 1059) • endoscopic techniques
lumbar spinal stenosis	• simple decompressive laminectomy • laminectomy plus fusion: may be indicated for patients with degenerative spondylolisthesis, stenosis and radiculopathy, adult degenerative scoliosis (ADS), or instability

Table 68.7 Chances of patients going back to work	
Time out of work	Chances of getting back to work
<6 mos	50%
1 yr	20%
2 yrs	<5%

amount of time that a patient has been out of work due to low back problems is related to the chances of the patient getting back to work as shown in Table 68.7.

68.16 Coccydynia

68.16.1 General information

Pain and tenderness around the coccyx. A symptom, not a diagnosis. Typically, discomfort is experienced on sitting or on rising from sitting. More common in females, possibly due to a more prominent coccyx. The condition is unusual enough in males that in the absence of local trauma, strong consideration should be given to an underlying condition.

68.16.2 Etiologies

For differential diagnosis, see Acute low back pain (p.1416). Better accepted etiologies include[87]:

1. local trauma (may be associated with fracture or dislocation):
   a) 25%of patients give a history of a fall
   b) 12%had repetitive trauma (rowing machine, prolonged bicycle riding…)
   c) 12%started with parturition
   d) 5%started following a surgical procedure (half of which were in the lithotomy position)
2. idiopathic: excluding traumatic cases, no etiology can be identified in most cases
3. neoplasms
   a) chordoma
   b) giant cell tumor
   c) intradural schwannoma
   d) perineural cyst
   e) intra-osseous lipoma
   f) carcinoma of the rectum
   g) sacral hemangioma[88]
   h) pelvic metastases (e.g. from prostate cancer)
4. prostatitis

Controversial etiologies include[87,89]:

1. local pressure over a prominent coccyx
2. referred pain:
   a) spinal disease
      - herniated lumbosacral disc
      - cauda equina syndrome
      - arachnoiditis
   b) pelvic/visceral disease
      - pelvic inflammatory disease (PID)
      - perirectal abscess
      - perirectal fistula
      - pilonidal cyst
3. inflammation of the various ligaments attached to the coccyx
4. neurosis or frank hysteria

Histological evaluation of the coccyx has not helped delineate the cause, even though avascular necrosis has been suggested.[90]

68.16.3 Evaluation

MRI: effective for detecting soft tissue masses, including presacral masses

CT scan: no characteristic finding in coccydynia findings. Very sensitive for detecting bony pathology (fracture, destructive lesion…).

Sacrococcygeal films are often performed to rule-out a bony destructive lesion. Often, the question of a fracture will be raised, and many times cannot be definitely ruled-in or out based on this study. There may or may not be any significance of such a fracture.

Nuclear bone scans were not helpful in 50 patients with coccydynia.[87]

68.16.4 Treatment

Numerous treatments have been proposed, and some are offered here for historical purposes[87] (and to dissuade casual attempts to effect a "new" cure that in reality has already been tried):
1. plaster jackets
2. hot baths (sitz baths), heating pads
3. massage therapy
4. XRT
5. psychotherapy

Most cases resolve within ≈ 3 months of conservative management consisting of NSAIDs, mild analgesics, and measures to reduce pressure on the coccyx (e.g. a rubber ring ("doughnut") sitting cushion, lumbar supports to maintain sitting lumbar lordosis to shift weight from coccyx to posterior thighs).[91]

Management recommendations for refractory cases[87,91]
1. local injection: 60% respond to corticosteroid + local anesthetic (40 mg Depo-Medrol® in 10 cc of 0.25% bipuvicaine). Recommended as initial treatment; response should be achieved by 2 injections
2. manipulation of the coccyx: usually under general anesthesia. ≈ 85% successful when combined with local injection
3. ± physiotherapy (diathermy & ultrasound): found to be of benefit only in ≈ 16% (may be more effective with the addition of gentle manipulation of the coccyx without general anesthesia[92])
4. caudal epidural steroid injection
5. blockade or neurolysis (with chemicals or by cryoablation[93]) of the ganglion impar (AKA ganglion of Walther, the lowest ganglion of the paired paravertebral sympathetic chain, located just anterior to the sacrococcygeal junction): some success has been described with this technique (traditionally used for intractable sympathetic perineal pain of neoplastic etiology[94])
6. neurolytic techniques directed to S4, S5 and coccygeal nerves
7. coccygectomy (surgical removal of the mobile portion of the coccyx, followed by smoothening of the residual bony prominence on the sacrum): was required in ≈ 20% of patients in one series,[87] with a reported success rate of 90% However, many practitioners do not view this as a highly effective treatment and feel that great restraint should be used in considering this form of therapy

68.16.5 Recurrence

Occurs in ≈ 20% of conservatively treated cases, usually within the first year. Repeat therapy was often successful in providing permanent relief. More aggressive treatment may be considered for refractory cases.

68.17 Failed back surgery syndrome

68.17.1 General information

Definition: failure to satisfactorily improve low back pain or radiculopathy following back surgery. These patients often require analgesics and are unable to return to work. The failure rate for lumbar discectomy to provide satisfactory long-term pain relief is ≈ 8–25%[95] Pending legal or worker's compensation claims were the most frequent deterrents to a good outcome.[96]

68.17.2 Etiologies

Factors that may cause or contribute to the failed back syndrome:
1. incorrect initial diagnosis
   a) inadequate pre-op imaging
   b) clinical findings not correlated with abnormality demonstrated on imaging
   c) other causes of symptoms (sometimes in the presence of what was considered to be an appropriate lesion on imaging studies which may have been asymptomatic): e.g. trochanteric bursitis, diabetic amyotrophy…
2. continued nerve root or cauda equina compression caused by:
   a) residual compression (retained disc material, osteophytes…)
   b) recurrent pathology at same level: disc reherniation at the same level, usually have pain-free interval > 6 mos post-op (p.1061); or restenosis (over many years[97] – was more common with midline fusions)
   c) adjacent level pathology: disc herniation or stenosis[97]
   d) compression of nerve root by peridural scar (granulation) tissue (see below)
   e) pseudomeningocele
   f) epidural hematoma
   g) conjoined nerve roots with compression at another level or in atypical location
   h) segmental instability: 3 patterns,[98] 1) lateral rotational instability, 2) post-op spondylolisthesis, 3) post-op scoliosis
3. permanent nerve root injury from the original disc herniation or from surgery, includes deafferentation pain which is usually constant and burning or ice cold
4. adhesive arachnoiditis: responsible for 6–16%of persistent symptoms in post-op patients[99] (see below)
5. discitis (p.356): usually produces exquisite back pain 2–4 weeks post-op
6. spondylosis
7. other causes of back pain unrelated to the original condition: paraspinal muscle spasm, myofascial syndrome… Look for trigger points, evidence of spasm
8. post-op reflex sympathetic dystrophy, RSD (p.1054)
9. “non-anatomic factors”: poor patient motivation, secondary gains, drug addiction, psychological problems (p.1053)…

68.17.3 Arachnoiditis (AKA adhesive arachnoiditis)

General information

Inflammatory condition of the lumbar nerve roots. Actually a misnomer, since adhesive arachnoiditis is really an inflammatory process or fibrosis that involves all three meningeal layers (pia, arachnoid, and dura).

Etiologies/risk factors

Many putative “risk factors” have been described for the development of arachnoiditis, including[100]:
1. spinal anesthesia: either due to the anesthetic agents or to detergent contaminants on the syringes used for same
2. spinal meningitis: pyogenic, syphilitic, tuberculous
3. neoplasms
4. myelographic contrast agents: less common with currently available nonionic water soluble contrast agents
5. trauma
   a) post-surgical: especially after multiple operations
   b) external trauma
6. hemorrhage
7. idiopathic

Radiographic findings in arachnoiditis

NB: Radiographic evidence of arachnoiditis may also be found in asymptomatic patients.[100] Arachnoiditis must be differentiated from tumor: the central adhesive type (see below) may resemble CSF seeding of tumor, and myelographic block may mimic intrathecal tumor.

Table 68.8 Myelographic classification of arachnoiditis

Type	Description
1	unilateral focal filling defect centered on the nerve root sleeve adjacent to disc space
2	circumferential constriction around thecal sac
3	complete obstruction with "stalactites" or "candle guttering", "candle-dripping", or "paint-brush" filling defects
4	infundibular cul-de-sac with loss of radicular striations

MRI

3 patterns on MRI[101,102]:

1. central adhesion of the nerve roots into 1 or 2 central "cords"
2. "empty thecal sac" pattern: roots adhere to meninges around periphery, only CSF signal is visible intrathecally
3. thecal sac filled with inflammatory tissue, no CSF signal. Corresponds with myelographic block and candle-dripping appearance

Enhancement: acute arachnoiditis may enhance. Chronic arachnoiditis usually does not enhance with gadolinium as much as e.g. tumor.

Myelogram

May demonstrate complete block, or clumping of nerve roots. One of many myelographic classification systems[103] for arachnoiditis is shown in Table 68.8.

68.17.4 Peridural scar

General information

Although peridural scar tissue is frequently blamed for causing recurrent symptoms,[104,105] there has been no proof of correlation between the two.[106] Peridural fibrosis is an inevitable sequelae to lumbar disc surgery just as post-op fibrosis is a consequence of any surgical procedure. Even patients who are relieved of their pain following discectomy develop some scar tissue post-op.[107] Although it has been shown that if a patient has recurrent radicular pain following a lumbar discectomy there is a 70% chance that extensive peridural scar will be found on MRI,[106] this study also showed that on post-op MRIs at 6 months, 43% of patients will have extensive scar, but 84% of the time this will be asymptomatic.[108] Thus, one must use clinical grounds to determine if a patient with extensive scar on MRI is in the 16% minority of patients with radicular symptoms attributable to scar.[108]

See a discussion of measures to reduce peridural scarring (p. 1053).

Radiologic evaluation

General information

Patients with only persistent low back or hip pain without a strong radicular component, with a neurologic exam that is normal or unchanged from pre-op, should be treated symptomatically. Patients with signs or symptoms of recurrent radiculopathy (positive SLR is a sensitive test for nerve root compression), especially if these follow a period of apparent recovery, should undergo further evaluation.

It is critical to differentiate residual/recurrent disc herniation from scar tissue and adhesive arachnoiditis as surgical treatment has generally poor results with the latter two (below).

MRI without and with IV gadolinium

Diagnostic test of choice. The best exam for detecting residual or recurrent disc herniation, and to reliably differentiate disc from scar tissue. Pre-contrast studies with T1WI and T2WI yields an accuracy of ≈ 83%, comparable to IV enhanced CT.[109,110] With the addition of gadolinium, using the protocol below yields 100% sensitivity, 71% specificity, and 89% accuracy.[111] May also detect adhesive arachnoiditis (see above). As scar becomes more fibrotic and calcified with time, the differential enhancement with respect to disc material attenuates and may become undetectable at some point, ≈ 1–2 years post-op[110] (some scar continues to enhance for > 20 yrs).

Recommended MRI protocol

See reference.[111]

Get pre-contrast T1WI and T2WI. Give 0.1 mmol/kg gadolinium IV. Obtain T1WI images within 10 minutes (early post-contrast). No benefit from post-contrast T2WI.

Findings on unenhanced MRI

Signal from a HLD becomes more intense as the sequence is varied from T1WI → T2WI, whereas scar tissue becomes less intense with this transition. Indirect signs (also applicable to CT):

1. mass effect: a nerve root is displaced away from disc material, whereas it may be retracted toward scar tissue by adherence to it
2. location: disc material tends to be in contiguity with the disc interspace (best seen on sagittal MRI)

Findings on enhanced MRI

On early (≤ 10 mins post-contrast) T1WI images: scar enhances inhomogeneously, whereas disc does not enhance at all. A nonenhancing central area surrounded by irregular enhancing material probably represents disc wrapped in scar. Venous plexus also enhances, and may be more pronounced when it is distorted by disc material, but the morphology is easily differentiated from scar tissue in these cases.

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On late (> 30 mins post-contrast) T1WI: scar enhances homogeneously, disc had variable or no enhancement. Normal nerve roots do not enhance even on late images.

CT scan without and with IV (iodinated) contrast

Unenhanced CT scan density measurements are unreliable in the postoperative back.[112] Enhanced CT is only fairly good in differentiating scar (enhancing) from disc (unenhancing with possible rim enhancement). Accuracy is about equal to unenhanced MRI.

Myelography, with post-myelographic CT

Postoperative myelographic criteria alone are unreliable for distinguishing disc material from scar.[100,113] With the addition of CT scan, neural compression is clearly demonstrated, but scar still cannot be reliably distinguished from disc.

Myelography (especially with post-myelographic CT) is very capable of demonstrating arachnoiditis[113] (see above).

Plain LS x-rays

Generally helpful only in cases of instability, malalignment, or spondylosis.[113] Flexion/extension views are most helpful when trying to demonstrate instability.

68.17.5 Treatment of failed back surgery syndrome

Post-operative discitis

For treatment of intervertebral disc-space infection, see Discitis (p. 356).

Symptomatic treatment

Recommended for patients who do not have radicular signs and symptoms, or for most patients demonstrated to have scar tissue or adhesive arachnoiditis on imaging. As in other cases of non-specific LBP treatment includes: short-term bed rest, analgesics (non-narcotic in most cases), anti-inflammatory medication (non-steroidal, and occasionally a short course of steroids), and physical therapy.

Surgery

Generally reserved for those with recurrent or residual disc herniation, segmental instability, or patients with a pseudomeningocele. Patients with post-op spinal instability should be considered for spinal fusion (p. 1037).[98]

In most series with sufficient follow-up, success rates after reoperation are lower in patients with only epidural scar (as low as 1%) compared to those patients with disc and scar (still only ≈ 37%).[95]

An overall success rate (>50% pain relief for >2 yrs) of ≈ 34% was seen in one series,[105] with better results in patients that were young, female, with good results following previous surgery, a small number of previous operations, employment prior to surgery, predominantly radicular (cf axial) pain, and absence of scar requiring lysis.

In addition to the absence of disc material, factors associated with poor outcome were: sensory loss involving more than one dermatome, and patients with past or pending compensation claims.[95,114]

Arachnoiditis:

Surgery for carefully selected patients with arachnoiditis (those with mild radiographic involvement (Types 1 & 2 in Table 68.8), and <3 previous back operations)[103] has met with moderate success (although in this series, no patient returned to work). Approximate success rate in other series[115,116]: 50% failure, 20% able to work but with symptoms, 10–19% with no symptoms. Surgery consists of removal of extradural scar enveloping the thecal sac, removing any herniated disc fragments, and performing foraminotomies when indicated. Intradural lysis of adhesions is not indicated since no means for preventing reformation of scar has been identified.[116]

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69 Lumbar and Thoracic Intervertebral Disk Herniation / Radiculopathy

69.1 Lumbar disc herniation and lumbar radiculopathy

69.1.1 General information

Key concepts

- Radiculopathy: pain and/or subjective sensory changes (numbness, tingling…) in the distribution of a nerve root dermatome, possibly accompanied by weakness and reflex changes of muscles innervated by that nerve root
- typical disc herniation → radiculopathy in the nerve exiting at the level below
- massive disc herniations can → cauda equina syndrome (a medical emergency). Typical symptoms: saddle anesthesia, urinary retention, LE weakness (p. 1050).
- most patients do as well with conservative treatment as with surgery, initial nonsurgical (conservative) treatment should be attempted for the vast majority
- surgery indications: cauda equina syndrome, progressive symptoms or neurologic deficits despite conservative treatment, or severe radicular pain >≈ 6 weeks

69.1.2 Pathophysiology

Intervertebral discs may undergo degenerative changes (p. 1096); see Table 68.1 for description: this includes desiccation and fibrosis which further leads to fissuring and tearing which in turn increases the risk of herniation of disc material outside the normal confines of the disc space (see same table for definitions).

Disc herniations may compress one or more nerve roots, producing lumbar radiculopathy or less frequently cauda equina syndrome.

69.1.3 Herniation zones

Central and paramedial disc herniations

The posterior longitudinal ligament is strongest in the midline, and the posterolateral annulus may bear a disproportionate portion of the load exerted by the weight from above. This may explain why most herniated lumbar discs (HLD) occur posteriorly, slightly off to one side within the central canal zone or in the subarticular zone as illustrated in Fig. 69.1. In the lumbar spine this characteristically compresses the nerve root en passage (that is, the nerve entering the lateral recess just before it exits through the neural foramen of the level below).

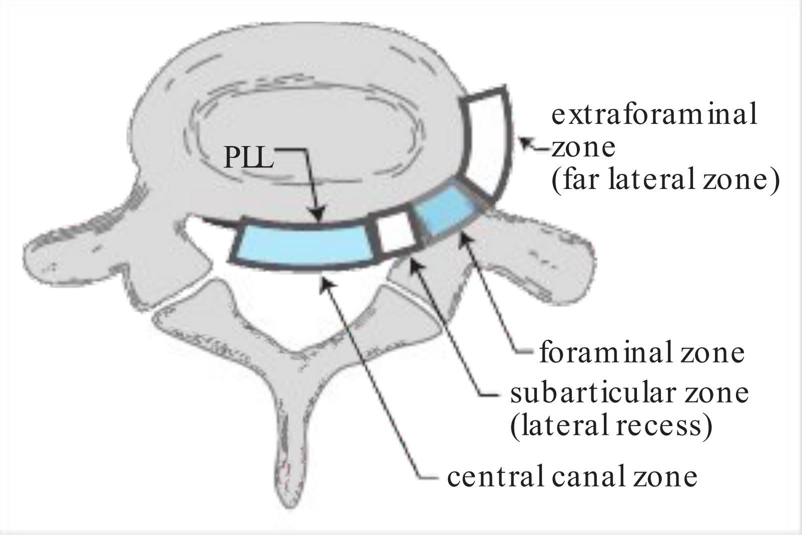

Fig. 69.1 Zones of lumbar disc herniation

Extreme lateral disc herniation

Disc herniations may also occur in the foraminal zone, which typically involves the nerve root exiting at that level.

Disc herniations in the extraforaminal zone occasionally involve the nerve root exiting at that level, however disc herniation here and those herniating anterior to the spine may not result in any nerve root involvement.

69.1.4 Other disc herniation variants

1. intravertebral disc herniation: AKA Schmorl's node. See Herniation through the endplate into the vertebral body (p.1060)
2. intradural disc herniation (p.1060)
3. limbus fracture: traumatic separation of a segment of bone from the edge of the vertebral ring apophysis at the site of anular attachment. May accompany HLD

69.1.5 Characteristic findings on the history

- symptoms may start off with back pain, which after days or weeks gradually or sometimes suddenly yields to radicular pain often with reduction of the back pain
- precipitating factors: various factors are often blamed, but are rarely identified[1] with certainty
- pain relief upon flexing the knee and thigh (e.g. lying supine with a pillow under the knees)
- patients generally avoid excessive movements, however, remaining in any one position (sitting, standing, or lying) too long may also exacerbate the pain, sometimes necessitating position changes at intervals that range from every few minutes to 10–20 minutes. This is distinct from constant writhing in pain e.g. with ureteral obstruction
- "cough effect": ↑ pain with coughing, sneezing, or straining at the stool. Occurred in 87% of patients with HLD in one series[2]
- bladder symptoms: the incidence of voiding dysfunction is 1–18%[3 (p 966)] Most common: difficulty voiding, straining, or urinary retention. Reduced bladder sensation may be the earliest finding. The possible causes of this are sensory loss, or incomplete interruption of the preganglionic parasympathetic fibers. Later it is not unusual to see "irritative" symptoms including urinary urgency, frequency (including nocturia), increased post-void residual. Less common: enuresis, and dribbling incontinence[4]; NB: frank urinary retention may indicate cauda equina syndrome (p.1050). Occasionally a HLD may present only with bladder symptoms which may improve after surgery.[5] Discectomy may improve bladder function, but this cannot be assured

Back pain per se is usually a minor component (only 1% of patients with acute low back pain have sciatica[6]), and when it is the only presenting symptom, other causes should be sought; see Low back pain (p.1024). Sciatica has such a high sensitivity for disc herniation, that the likelihood of a clinically significant disc herniation[7] in the absence of sciatica is ≈ 1 in 1000. Exceptions include a central disc herniation which may cause symptoms of lumbar stenosis (i.e. neurogenic claudication) or a cauda equina syndrome.

69.1.6 Physical findings in radiculopathy

General information

Nerve root impingement gives rise to a set of signs and symptoms present to variable degrees. Characteristic syndromes are described for the most common nerve roots involved; see Nerve root syndromes (p.1049).

In a series of patients referred to neurosurgical outpatient clinics for radiating leg pain, 28% had motor loss (yet only 12% listed motor weakness as a presenting complaint), 45% had sensory disturbance, and 51% had reflex changes.[8]

Findings suggestive of nerve root impingement include the following. Table 69.1 shows the sensitivity and specificity of some findings on the exam among patients with sciatica.

1. signs/symptoms of radiculopathy (Table 69.1)
   a) pain radiating down LE
   b) motor weakness
   c) dermatomal sensory changes
   d) reflex changes: mental factors may influence symmetry[9]
2. positive nerve root tension sign(s): including Lasègue's sign (see below)
3. tenderness over the sciatic notch

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Table 69.1 Sensitivity and specificity of physical findings for HLD in patients with sciatica[10]

Test	Comment	Sensitivity	Specificity
ipsilateral SLR	positive result: pain at < 60° elevation	0.80	0.40
crossed SLR	reproduction of contralateral pain	0.25	0.90
↓ ankle jerk	HLD usually at L5-S1 (total absence increases specificity)	0.50	0.60
sensory loss	area of loss is poor in localizing level of HLD	0.50	0.50
↓ patellar reflex	suggests upper HLD	0.50	NA
Weakness			
knee extension (quadriceps)	HLD usually at L3–4	< 0.01	0.99
ankle dorsiflexion (anterior tibialis)	HLD usually at L4–5	0.35	0.70
ankle plantarflexion (gastrocs)	HLD usually at L5-S1	0.06	0.95
great toe extension (EHL)	HLD at L5-S1 in 60%, at L4–5 in 30%	0.50	0.70

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Nerve root tension signs

Includes[11]:

1. Lasègue's sign: AKA straight leg raising (SLR) test. Helps differentiate sciatica from pain due to hip pathology. Test: with patient supine, raise afflicted limb by the ankle until pain is elicited[12] (should occur at < 60°, tension in nerve increases little above this angle). A positive test consists of leg pain or paresthesias in the distribution of pain (back pain alone does not qualify). The patient may also extend the hip (by lifting it off table) to reduce the angle. Although not part of Lasègue's sign, ankle dorsiflexion with SLR usually augments pain due to nerve root compression. SLR primarily tenses L5 and S1, L4 less so, and more proximal roots very little. Nerve-root compression produces a positive Lasègue's sign in ≈ 83% of cases[2] (more likely to be positive in patients < 30 yrs age with HLD[13]). May be positive in lumbosacral plexopathy (p.544). Note: flexing both thighs with the knees extended ("long-sitting" or sitting knee extension) may be tolerated further than flexing the single symptomatic side alone
2. Cram test: with patient supine, raise the symptomatic leg with the knee slightly flexed. Then, extend the knee. Results similar to SLR
3. crossed straight leg-raising test AKA Fajersztajn's sign: SLR on the painless leg causes contralateral limb pain (a greater degree of elevation is usually required than the painful side). More specific but less sensitive than SLR (97% of patients undergoing surgery with this sign have confirmed HLD[14]). May correlate with a more central disc herniation
4. femoral stretch test,[15] AKA reverse straight leg raising: patient prone, examiner's palm at popliteal fossa, knee is maximally dorsiflexed. Often positive with L2, L3, or L4 nerve root compression (e.g. in upper lumbar disc herniation), or with extreme lateral lumbar disc herniation (may also be positive in diabetic femoral neuropathy or psoas hematoma); in these situations SLR (Lasègue's sign) is frequently negative (since L5 & S1 not involved)
5. "bowstring sign": once pain occurs with SLR, lower the foot to the bed by flexing knee, keeping the hip flexed. Sciatic pain ceases with this maneuver, but hip pain persists
6. sitting knee extension test: with patient seated and both hips and knees flexed 90°, slowly extend one knee. Stretches nerve roots as much as a moderate degree of SLR

Other signs useful in evaluation for lumbar radiculopathy

1. FABER: an acronym for Flexion ABduction External-Rotation, AKA FABERE test (the trailing "e" is for extension), AKA Patrick's-test (after Hugh Talbot Patrick). A test of hip motion. Method: the hip and knee are flexed and the lateral malleolus is placed on the contralateral knee. The ipsilateral knee is gently displaced downward towards the exam table. This stresses the hip joint and does not usually exacerbate true nerve-root compression. Often markedly positive in the presence of hip joint disease – e.g. trochanteric bursitis (p.1101) – sacroiliitis or mechanical low-back pain

2. Trendelenburg sign: examiner observes pelvis from behind while patient raises one leg while standing. Normally the pelvis remains horizontal. A positive sign occurs when the pelvis tilts down toward the side of the lifted leg indicating weakness of the contralateral thigh adductors (primarily L5 innervated)
3. crossed adductors: in eliciting patellar reflex (knee jerk (KJ)), the contralateral thigh adductors contract. In the presence of a hyperactive ipsilateral KJ it may indicate an upper motor neuron lesion, in the presence of a hypoactive ipsilateral KJ it may be a form of pathological spread, indicating nerve root irritability
4. Hoover sign[16]: to distinguish unilateral functional weakness of iliopsoas from organic weakness using synergistic contraction of the contralateral gluteus medius. The supine patient is asked to lift one leg off the bed against resistance from the examiner's hand. The examiner simultaneously places the palm of his/her other hand under the heel of the unlifted leg and gently lifts. Test 1: when the patient lifts the normal leg, if the paretic leg pushes down with more force than was exhibited on manual testing of the limb beforehand, the weakness is judged functional, if the force is equally weak the weakness is judged organic. Test 1 cannot be used if the hip extensor was normal beforehand. Test 2: (the better known test) the patient is asked to lift the weak leg. If the heel on the normal side lifts passively by the examiner, it suggests the weakness is functional (i.e. the patient is not trying). Not totally reliable[17,18]
5. abductor sign: an alternative to the Hoover test, to differentiate functional from organic weakness in the thigh abductors using synergistic contraction of the contralateral thigh abductors.[18] With the patient supine, the examiner places a hand on the lateral aspect of both legs. The patient is asked to abduct one leg, and then the other while the examiner applies resistance with his/her hand. The examiner mentally notes the response of the non-abducting LE. The results are as noted in Table 69.2

Nerve root syndromes

Due to the facts listed below, a herniated lumbar disc (HLD) usually spares the nerve root exiting at that interspace, and impinges on the nerve exiting from the neural foramen one level below (e.g. a L5-S1 HLD usually causes S1 radiculopathy). This gives rise to the characteristic lumbar nerve root syndromes shown in Table 69.3.

Important applied anatomy in lumbar disc disease:

1. in the lumbar region, the nerve root exits below and in close proximity to the pedicle of its like-numbered vertebra
2. the intervertebral disc space is located well below the pedicle
3. not all patients have 5 lumbar vertebrae; see Localizing levels in spine surgery (p. 1436)

69.1.7 Radiographic evaluation

See Radiographic evaluation under Low back pain (p. 1416). 70% of herniated discs that migrate do so inferiorly.

69.1.8 Nonsurgical treatment

See nonsurgical treatment measures (p. 1033).

69.1.9 Surgical treatment

Indications for surgery

No predictive factors have been identified that can determine which patients are likely to improve on their own and which would be better served with surgery.

Table 69.2 Abductor sign

Abducting LE	Contralateral (nonabducting) LE	
	Organic weakness	Functional weakness
weak LE	maintains position	hyperadducts
normal LE	hyperadducts	maintains position

Table 69.3 Lumbar disc syndromes

Syndrome	Level of herniated lumbar disc		
	L3–4	L4–5	L5-S1
root usually compressed	L4	L5	S1
% of lumbar discs	3–10% (5% average)	40–45%	45–50%
reflex diminished	knee jerk[a] (Westphal's sign)	medial hamstring[b]	Achilles[a] (ankle jerk)
motor weakness	quadriceps femoris (knee extension)	tibialis anterior (foot drop) & EHL[c]	gastrocnemius (plantarflexion), ± EHL[c]
decreased sensation[d]	medial malleolus & medial foot	large toe web & dorsum of foot	lateral malleolus & lateral foot
pain distribution	anterior thigh	posterior LE	posterior LE, often to ankle

[a] Jendrassik maneuver may reinforce (see Table 29.2)
[b]medial hamstring reflex is unreliable (not always pure L5), may also stimulate adductors when eliciting
[c]see WEAKNESS in Table 69.1 for breakdown
[d]sensory impairment is most common in the distal extremes of the dermatome[19]

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Surgical indications in patients with a radiographically identified herniated disc that correlates with findings on the history and physical exam:

1. failure of non-surgical management to control pain after 5–8 weeks: over 85% of patients with acute disc herniation will improve without surgical intervention in an average of 6 weeks[20] (70% within 4 weeks[21]). Most clinicians advocate waiting somewhere between 5–8 weeks from the onset of radiculopathy before considering surgery (assuming none of the items listed below applies)
2. " EMERGENT SURGERY": (i.e. before the 5–8 weeks of symptoms have lapsed). Indications:
   a) cauda equina syndrome (CES): (see below)
   b) progressive motor deficit (e.g. foot drop). NB: paresis of unknown duration is a doubtful indication for surgery[1,22,23] (no study has documented that there is less motor deficit in surgically treated patients with this finding[24]). However, the acute development or progression of motor weakness is considered an indication for rapid surgical decompression
   c) "urgent" surgery may be indicated for patients whose pain remains intolerable in spite of adequate narcotic pain medication
3. ± patients who do not want to invest the time in a trial of non-surgical treatment if it is possible that they will still require surgery at the end of the trial

Cauda equina syndrome

The clinical condition arising from dysfunction of multiple lumbar and sacral nerve roots within the lumbar spinal canal. Usually due to compression of the cauda equina (the bundle of nerve roots below the conus medullaris arising from the lumbar enlargement and conus). See Table 53.2 for features to help differentiate CES from a conus lesion.

Possible findings in CES:

1. sphincter disturbance:
   a) urinary retention: the most consistent finding. Sensitivity ≈ 90% (at some point in time during course).[25,26] To evaluate acutely: have patient empty bladder and check post-void residual (by catheterization or with bladder ultrasound). In a patient without retention, only 1 in 1000 will have a CES. Cystometrogram (when done) shows a hypotonic bladder with decreased sensation and increased capacity
   b) urinary and/or fecal incontinence[27]: some patients with urinary retention will present with overflow incontinence
   c) anal sphincter tone: diminished in 60–80%
2. "saddle anesthesia": the most common sensory deficit. Distribution: region of the anus, lower genitals, perineum, over the buttocks, posterior-superior thighs. Sensitivity ≈ 75% Once total perineal anesthesia develops, patients tend to have permanent bladder paralysis[28]

3. significant motor weakness: usually involves more than a single nerve root (if untreated, may progress to paraplegia)
4. low back pain and/or sciatica (sciatica is usually bilateral, but may be unilateral or entirely absent, prognosis may be worse when absent or bilateral[26])
5. bilateral absence of Achilles reflex has been noted[29]
6. sexual dysfunction (usually not detected until a later time)

Etiologies of CES includes:
1. compression of cauda equina
   a) massive herniated lumbar disc: see below
   b) tumor
      - from compression: e.g. with metastatic disease to the spine with epidural extension
      - intravascular lymphomatosis (B-cell lymphoma) (p.711): a circulating lymphoma without solid mass. Often presents with CNS findings: dementia, enhancing meninges on MRI, lymphoma cells in CSF, and CES
   c) free fat graft following discectomy[30]
   d) trauma: fracture fragments compressing cauda equina
   e) spinal epidural hematoma
2. infection: may cause neurologic deficit from
   a) compression: typically from spinal epidural abscess complicating discitis or vertebral osteomyelitis
   b) a significant number of cases of CES from infection may be due to vascular compromise resulting from local septic thrombophlebitis. This may carry a worse prognosis as surgical decompression cannot correct this mechanism
3. neuropathy:
   a) ischemic
   b) inflammatory
4. ankylosing spondylitis: etiology is often obscure (p.1123)

CES from HLD: May be due to massive herniated disc, usually midline, most common at L4–5, often superimposed on a preexisting condition (spinal stenosis, tethered cord…).[27]

Prevalence of CES
1. 0.0004 in all patients with LBP[7]
2. only ≈ 1–2%of HLD that come to surgery[7]

Time course: CES tends to develop either acutely, or (less typically) slowly (prognosis is worse in the acute onset group, especially for return of bladder function, which occurred in only ≈ 50%).[25] 3 patterns[31]:
- Group I – sudden onset of CES symptoms with no previous low back symptoms
- Group II – previous history of recurrent backache and sciatica, the latest episode combined with CES
- Group III – presentation with backache & bilateral sciatica that later develop CES

Surgical issues: some advise a bilateral laminectomy[27] (but this is not mandatory). Occasionally, when it is difficult to remove a very tense midline disc, transdural removal may be helpful.[29]

Timing of discectomy in CES: controversial, and the point of contention in numerous lawsuits. In spite of early reports emphasizing rapid decompression,[29] other reports found no correlation between the time to surgery after presentation and the return of function.[25,26] Some evidence supports the goal of performing surgery within 48 hours (although performing surgery within 24 hours is desirable if possible, there is no statistically significant proof that delaying up to 48 hrs is detrimental).[32,33]

Booking the case: Lumbar discectomy

Also see defaults & disclaimers (p.27).
1. position: prone
2. equipment: microscope (if used), minimally invasive retractors (if used)
3. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: through the back to go between the bones and remove the piece of disc that is pressing on the nerve(s)

b) alternatives: nonsurgical management
c) complications: usual spine surgery complications (p. 1018) plus the disc can herniate again in the same place in ≈ 6% of cases, it is possible that a fragment of disc can be missed at the time of surgery, there might not be the amount of pain relief desired (back pain does not respond as well to surgery as nerve-root pain)

Surgical options for lumbar radiculopathy

Once it is decided to treat surgically, options include:
1. trans-canal approaches
   a) standard open lumbar laminectomy and discectomy: 65–85% reported no sciatica one year post-op compared to 36% for conservative treatment.[34] Long-term results (> 1 year) were similar. 10% of patients underwent further back surgery during the first year[34]
   b) "microdiscectomy"[35,36]: similar to standard procedure, however smaller incision is utilized. Advantages may be cosmetic, shortened hospital stay, lower blood loss. May be more difficult to retrieve some fragments.[37 (p 1319),38] Overall efficacy is similar to standard discectomy[39]
   c) sequestrectomy: removal of only the herniated portion of the disc without entering the disc space to remove disc material from there
2. intradiscal procedures (see below): a number of procedures have been devised over the years to percutaneously treat HLD by creating a cavity within the disc. Some have been abandoned for various reasons, not the least of which is controversy regarding the validity of the underlying premise that this can work
   a) chemonucleolysis: using chymopapain to enzymatically dissolve the disc (no longer used)
   b) automated percutaneous lumbar discectomy: utilizes a nucleotome
   c) percutaneous endoscopic intradiscal discectomy: see below
   d) intradiscal endothermal therapy (IDET or IDTA): see below
   e) laser disc decompression

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Intradiscal surgical procedures (ISP)

ISPs (see below for specific procedures) are among the most controversial procedures for lumbar spine surgery. The theoretical advantage is that epidural scarring is avoided, and that a smaller incision or even just a puncture site is used. This is also purported to reduce postoperative pain and hospital stay (often performed as an outpatient procedure). The conceptual problem with ISPs is that they are directed at removing disc material from the center of the disc space (which is not producing symptoms) and rely on the reduced intradiscal pressure to decompress the herniated portion of the disc from the nerve root. Only ≈ 10–15% of patients considered for surgical treatment of disc disease are candidates for an ISP. ISPs are usually done under local anesthetic in order to permit the patient to report nerve root pain to identify impingement on a nerve root by the surgical instrument or needle. Overall, ISPs are not recommended until rigorous controlled trials prove the efficacy.[10]

Indications utilized by proponents of intradiscal procedures:
1. type of disc herniation: appropriate only for "contained" disc herniation (i.e. outer margin of anulus fibrosus intact)
2. appropriate level: best for L4–5 HLD. May also be used at L3–4. Difficult but often workable (utilizing angled instruments or other techniques) at L5-S1 because of the angle required and interference by iliac crest
3. not recommended in presence of severe neurologic deficit[40]

Results:

"Success" rate (≈ pain free and return to work when appropriate) reported ranges from 37–75%[41,42,43]

Automated percutaneous lumbar discectomy: AKA nucleoplasty. Utilizes a nucleotome[44] to remove disc material from the center of the intervertebral disc space. 1 year success rate of 37% Complications include cauda equina syndrome from improper nucleotome placement.[45] In another study, nucleoplasty (with or without IDET (see below)) for HLD showed only modest reduction in pain at 9 months.[46]

Laser disc decompression: Insertion of a needle into the disc, and introduction of a laser fiberoptic cable through the needle to allow a laser to burn a hole in the center of the disc[47,48] (with or without endoscopic visualization).

The 2104 North American Spine Society Coverage Committee[49] position statement:

"Laser spine surgery in the cervical or lumbar spine is NOT indicated at this time. Due to lack of high quality clinical trials concerning laser spine surgery with the cervical or lumbar spine, it cannot be endorsed as an adjunct to open, minimally invasive, or percutaneous surgical techniques."

Percutaneous endoscopic lumbar discectomy (PELD) : This term refers to an essentially intradiscal procedure indicated primarily for contained disc herniations, although some small "noncontained" fragments may be treatable.[50] No large randomized study has been done to compare the technique to the accepted standard, open discectomy (with or without microscope). In one report[51] of 326 patients with L4–5 HLD, only 8 (2.4%) met study criteria (no previous operation, failure of conservative treatment, imaging study proving disc protrusion followed by discography to R/O "disc perforation") for PELD. Of these 8, only 3 were reported as having a good result. This study is not adequate for evaluating the technique.

Intradiscal endothermal therapy (IDET): AKA intradiscal (electro)thermal anuloplasty (IDTA). Efficacy: 23–60% at 1 year for treating "internal disc disruption"[52] (radial fissures in the nucleus pulposus extending into the anulus fibrosus) which is purported to account for 40% of patients with chronic low back pain of unknown etiology.[53]

Adjunctive treatment in lumbar laminectomy

Epidural steroids following discectomy

Perioperative epidural steroids after routine surgery for lumbar degenerative disease may result in a small reduction of post-op pain, length of stay, and the risk of not returning to work at 1 year, but most of the evidence originates from studies not using validated outcome assessment that favor positive results, and further study is recommended (various agents, dosages, co-administered drugs, and delivery methods were reported).[54] However, the combination of systemic steroids at the start of the case (Depo-Medrol® 160 mg IM and methylprednisolone sodium succinate (Solu-Medrol®) 250 mg IV) combined with infiltration of 30 ml of 0.25% bupivacaine (Marcaine®) into the paraspinal muscles at incision and closure, may reduce hospital stay and post-op narcotic requirements.[55]

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Methods to reduce scar formation

Epidural free fat graft

The use of an autogenous free fat graft in the epidural space has been employed in an attempt to reduce post-op epidural scar formation. Opinion varies widely as to the effectiveness, some feel it is helpful, others feel it actually exacerbates scarring.[56] In some patients, no evidence of the graft will be found on reoperation years later. The fat graft can very rarely be a cause of nerve root compression[57] or cauda equina syndrome[30] within the first few days post-op, and there is a case report of compression 6 years following surgery.[58]

Other measures

Other measures include the placement of barrier films or gels. There are numerous products available, none has been shown to have reproducible benefit.

Risks of lumbar laminectomy

General information

Overall risk of mortality in large series[59,60]: 6 per 10,000 (i.e. 0.06%), most often due to septicemia, MI, or PE. Complication rates are very difficult to determine accurately,[34] but the following is included as a guideline.

Common complications

1. infection:
   a) superficial wound infection: 0.9–5%[61] (risk is increased with age, long term steroids, obesity, ? DM): most are caused by S. aureus; see Laminectomy wound infection (p.345) for management
   b) deep infection: < 1% (see below under Uncommon complications)
2. increased motor deficit: 1–8% (some transient)
3. unintended "incidental" durotomy (the term "unintended durotomy" has been recommended in preference to "dural tear", see below): incidence is 0.3–13% (risk increases to ≈ 18% in re-do operations).[62] Possible sequelae include those listed in Table 69.4

Table 69.4 Possible sequelae of dural opening
Well documented
1. CSF leak a) contained: pseudomeningocele b) external: CSF fistula 2. herniation of nerve roots thorough opening 3. associated nerve root contusion, laceration or injury to the cauda equina 4. CSF leak collapses the thecal sac and may increase blood loss from epidural bleeding
Less well documented
1. arachnoiditis 2. chronic pain 3. bladder, bowel and/or sexual dysfunction

a) CSF fistula (external CSF leak): the risk of a CSF fistula requiring operative repair is ≈ 10 per 10,000[59]
b) pseudomeningocele: 0.7–2%[62] (may appear similar radiographically to spinal epidural abscess (SEA), but post-op SEA often enhances, is more irregular, and is associated with muscle edema)
4. recurrent herniated lumbar disc (same level either side) (p.1061): 4%(with 10 year follow-up)[63]
5. Post-operative urinary retention (POUR): usually temporary, but may delay hospital discharge

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Uncommon complications

1. direct injury to neural structures. For large disc herniations, consider a bilateral exposure to reduce risk
2. injury to structures anterior to the vertebral bodies (VB): injured by breaching the anterior longitudinal ligament (ALL) through the disc space, e.g. with pituitary rongeur. The depth of disc space penetration with instruments should be kept ≤ 3 cm, since 5%of lumbar discs had diameters as small as 3.3 cm.[64] Asymptomatic perforations of the ALL occur in up to 12%of discectomies. Breach of the ALL risks potential injuries to:
   a) great vessels[65]: risks include potentially fatal hemorrhage, and arteriovenous fistula which may present years later. Most such injuries occur with L4–5 discectomies. Only ≈ 50%bleed into the disc space intraoperatively, the rest bleed into the retroperitoneum. Emergent laparotomy or endovascular treatment[66] is indicated, preferably by a surgeon with vascular surgical experience, if available. Mortality rate is 37–67%
      - aorta: the aortic bifurcation is on the left side of the lower part of the L4 VB, and so the aorta may be injured above this level
      - below L4, the common iliac arteries may be injured
      - veins (more common than arterial injuries): vena cava at and above L4, common iliac veins below L4
   b) ureters
   c) bowel: at L5-S1 the ileum is the most likely viscus to be injured
   d) sympathetic trunk
3. wrong site surgery: incidence in self-reporting survey was 4.5 occurrences per 10,000 lumbar spine operations.[67] Factors identified as potential contributors to the error: unusual patient anatomy, not performing localizing radiograph. 32%of responding neurosurgeons indicated that they removed disc material from the wrong level at some time in their career
4. rare infections:
   a) meningitis
   b) deep infection: < 1% Including:
      - discitis (p.356): 0.5%
      - spinal epidural abscess (SEA) (p.349): 0.67%
5. cauda equina syndrome: may be caused by post-op spinal epidural hematoma (see below). Incidence was 0.21%in one series of 2842 lumbar discectomies[68] and 0.14%in a series of 12,000 spine operations.[69] Red flags: urinary retention, anesthesia that may be saddle or bilateral LE
6. postoperative visual loss (POVL)[70]: (see below)
7. complications of positioning:
   a) compression neuropathies: ulnar, peroneal nerves. Use padding over elbows and avoid pressure on posterior popliteal fossa

b) anterior tibial compartment syndrome: due to pressure on anterior compartment of leg (reported with Andrew's frame). An orthopedic emergency that may require emergent fasciotomy
c) pressure on the eye: corneal abrasions, damage to the anterior chamber
d) cervical spine injuries during positioning due to relaxed muscles under anesthesia

8. post-op arachnoiditis (p.1040): risk factors include epidural hematoma, patients who tend to develop hypertrophic scar, post-op discitis, and intrathecal injection anesthetic agents or steroids. Surgical treatment for this is disappointing. Intrathecal depo-medrol may provide short-term relief (in spite of the fact that steroids are a risk factor for the development of arachnoiditis).
9. thrombophlebitis and deep-vein thrombosis with risk of pulmonary embolism (PE)[59]: 0.1%; see Thromboembolism in neurosurgery (p.167)
10. complex regional pain syndrome AKA reflex sympathetic dystrophy (RSD) (p.497): reported in up to 1.2% of cases, usually after posterior decompression with fusion, often following reoperations[71] with onset 4 days to 20 weeks post-op. See also critique of RSD (p.497). Treatment includes some or all of: PT, sympathetic blocks, oral methylprednisolone, removal of hardware if any
11. very rare: Ogilvie's syndrome (pseudo-obstruction ("ileus") of the colon). Usually seen in hospitalized/debilitated patients. May be related to narcotics, electrolyte deficiencies, possibly from chronic constipation. Also reported following spinal surgery/trauma, spinal/epidural anesthesia, spinal metastases, & myelography[72]

Unintended durotomy

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Unintentional opening of the dura during spinal surgery has an incidence of 0–14%[73]

Terminology: The terms "unintended durotomy", "incidental durotomy",[73] or even just "dural opening", have been recommended in preference to "dural tear" which may imply carelessness[62] when none was present. Dural openings have been associated with one or more alleged complications or sequelae in medical malpractice suits involving surgery on the lumbar spine.

The injury: By itself, opening the dura intentionally or otherwise is not expected to have a deleterious effect on the patient.[62,74] In fact, dural opening is often a standard part of the operation for intradural disc herniation,[75] tumors, etc. Although not frequent, (for incidence, see above) unintended durotomy is not an unusual occurrence, and alone, is not considered an act of malpractice. However, it may result from an event or events that produce more serious injuries. These events and injuries should be dealt with on their own merits.

In the SPORT study, there was a 9% incidence of unintended durotomy in patients undergoing first-time open laminectomy.[76] There were no long-term differences in nerve root injuries, mortality, additional operations, or outcome measures. Short-term differences included longer inpatient stay, increased blood loss, and duration of surgery.[76]

Possible sequelae include those listed in Table 69.4. A CSF leak may produce "spinal headache" (p.1508) with its associated symptoms and if it breaches the skin it may be a risk factor for meningitis. Pain or sensory/motor deficits may be associated with injuries to nerve roots or delayed herniation of nerve roots through the dural opening.

Etiologies: Potential causes are many, and include[62]: unanticipated anatomic variations, adhesion of the dura to removed bone, slippage of an instrument, an obscured fold of dura caught in a rongeur or curette, thinning of the dura in cases of longstanding stenosis, and the possibility of a delayed CSF leak caused by perforation of the dura when it expands onto a surgically created spicule of bone.[77] The risk may be increased with anterior decompression for OPLL, with revision surgery, and with the use of high-speed drills.[73]

Treatment: If the opening is recognized at the time of surgery, watertight primary closure (with or without patch graft) should be attempted with nonabsorbable suture if at all possible to prevent pseudomeningocele and/or CSF fistula. A cottonoid placed over the opening prevents aspiration of nerve roots.[78] Care must be taken to avoid incorporating a nerve root into the closure. Most repairs will be accomplished with no complication or sequelae to the patient. When the opening is in the far (anterior) side of the dura, consideration may be given to intradural repair accessed through a posterior durotomy which is subsequently closed (this may risk additional injury to the nerve roots). Biocompatible fixatives (e.g. fibrin glue[73]) may be used to supplement primary closure.

Primary repair may be impossible in some situations (e.g. when the opening cannot be found or accessed, as is sometimes the case when it occurs on the nerve root sleeve) and alternatives here include placement of a fat or muscle graft over the suspected leak site, use of a the patient's own blood for a "blood patch" (one technique is to have the anesthesiologist draw ≈ 5–10 ml of the patient's blood from an arm vein, keeping it in the syringe for several minutes until it starts to coagulate, and then to have the anesthesiologist inject the blood onto the dura), use of gelfoam, fibrin glue… Some recommend that the wound not be drained post-op, with a water-tight closure of fascia, fat, and skin to add to the barrier. Others use a subcutaneous drain or epidural catheter. CSF diversionary procedures (e.g. through a drain inserted 1 or more levels away) may also be used.

Although bed rest × 4–7 days is often advocated to reduce symptoms and facilitate healing, when watertight closure has been achieved, normal post-op mobilization is not associated with a high failure rate (bed rest is recommended if symptoms develop).[73]

In one report of 8 patients with leaks that appeared post-op, re-operation was avoided when treated by resuturing the skin under local anesthesia, followed by bed rest in slight Trendelenburg position (to reduce pressure on the leakage site), broad spectrum antibiotics and antibiotic ointment over the skin incision, and daily puncture and drainage of the subcutaneous collection.[79]

See other treatment measures for H/A associated with CSF leak (p. 1049).

Postoperative visual loss

1. ischemic optic neuropathy[80]: the most common cause of the very uncommon post-operative loss of vision. Often bilateral. Usually associated with significant blood loss (median: 2 L), and/or prolonged operative time (≥ 6 hrs). All cases had anesthetic time > 5 hrs or blood loss > 1 L. Blood loss can cause hypotension (may cause release of endogenous vasoconstrictors in addition to reduced blood flow due to low hemodynamic pressure) and increased platelet aggregation. Is not due to direct pressure on the globe in most cases, and can occur at any age and even in otherwise healthy patients. No association with age, HTN, atherosclerosis, smoking or DM.
   The blindness can be extensive and is often permanent. Prevention is critical since there is no known effective treatment.[81]
   a) posterior ischemic optic neuropathy (PION)[80]: may follow surgery (surgical PION). Risk factors as above, plus:
      - surgery in the prone position (can cause periorbital edema, and rarely, direct pressure on the orbit)
      - lack of tight glycemic control
      - use of Trendelenburg position
      - hemodilution or overuse of crystalloid vs. colloid (blood) fluid replacement
      - prolonged hypotension
      - cellular hypoxia
      - decreased renal perfusion
   b) 6 independent risk factors for POVL[81]
      - male gender: odds ratio (OR) = 2.53
      - obesity: by clinical assessment or BMI ≥ 30 OR = 2.83
      - use of Wilson frame: OR = 4.30
      - length of anesthesia: OR = 1.39 per hour
      - EBL: OR = 1.34 per Liter
      - use of colloid as a percentage of nonblood replacement: less certain (small difference). OR = 0.67 per 5%colloid
   c) anterior ischemic optic neuropathy (AION): divided into arteritic (as with GCA) and nonarteritic (common with DM)
2. central retinal artery occlusion
3. cortical blindness: from occipital lobe infarction possibly due to embolis

Post-op care

Post-op orders

The following are guidelines for post-operative orders for a lumbar laminectomy without intra-operative complications; variations between surgeons and institutions must be taken into consideration:

1. admit post-anesthesia care unit (PACU)
2. vital signs on the nursing unit: q 2° × 4 hrs, q 4° × 24°, then q 8°
3. activity: up with assist, advance as tolerated
4. nursing care
   - I's & O's
   - intermittent catheterization q 4–6° PRN no void
   - optional: TED hose (may reduce risk of DVT) or PCB
   - optional (if drain used): empty drain q 8° and PRN
5. diet: clear liquids, advance as tolerated
6. IV: D5 1/2 NS + 20 mEq KCl/l @ 75 ml/hr, D/C when tolerating PO well (after antibiotics D/C'd if prophylactic antibiotics are used)

7. meds
   - laxative of choice (LOC) PRN
   - sodium docusate (e.g. Colace®) 100 mg PO BID when tolerating PO (stool softener, does not substitute for LOC)
   - optional: prophylactic antibiotics if used at your institution
   - acetaminophen (Tylenol®) 325 mg 1–2 PO or PR q 6° PRN
   - narcotic analgesic
   - optional: steroids are used by some surgeons to reduce nerve-root irritation from surgical manipulation
8. labs
   - optional (if significant blood loss during surgery): CBC

Post-op check

In addition to routine, the following should be checked:
1. strength of lower extremities, especially muscles relevant to nerve root, e.g. gastrocnemius for L5-S1 surgery, EHL for L4–5 surgery…
2. appearance of dressing: look for signs of excessive bleeding, CSF leak…
3. signs of cauda equina syndrome (p.1050), e.g. by post-op spinal epidural hematoma
   a) loss of perineal sensation ("saddle anesthesia")
   b) inability to void: may not be not unusual after lumbar laminectomy, more concerning if accompanied by loss of perineal sensation
   c) pain out of the ordinary for the post-op period
   d) weakness of multiple muscle groups

Any new neurologic deficit should prompt rapid evaluation for spinal epidural hematoma[69] (EDH). Delayed deficits may be due to EDH or epidural abscess. Post-op films in the recovery room can rule out graft or hardware malposition for fusions or instrumentation procedures, or changes in alignment. The diagnostic test of choice is MRI. If contraindicated or not available, CT/myelography may be indicated. An extradural defect immediately post-op suggests EDH.

Outcome of surgical treatment

In a series of 100 patients undergoing discectomy, at 1 year post-op 73% had complete relief of leg pain and 63% had complete relief of back pain; at 5–10 years the numbers were 62% for each category.[2] At 5–10 years post-op, only 14% felt that the pain was the same or worse than pre-op (i.e. 86% felt improved), and 5% qualified as having a failed back surgery syndrome (a heterogeneous not-precisely defined term, here meaning not returned to work, requiring analgesics, receiving worker's compensation, see Failed back surgery syndrome (p.1039)).

Attempts to measure relative merits of conservative treatment vs. surgery have failed. The recent SPORT study[82,83] suffered from significant selection bias since patients were allowed to crossover to the other arm of the study and thus more nearly approximated the current methodology of surgical selection than an actual RCT.[84] Earlier attempts at randomized trials also suffered from methodological flaws.[1] Conclusions that can be drawn from these studies[84]: most patients with manageable or improving pain and less disability typically choose conservative treatment and most have improvement in symptoms, whereas patients with severe, persistent or worsening pain and/or neurologic deficit are more likely to choose surgery with a resultant excellent outcome.

In patients with a diminished knee-jerk or ankle-jerk pre-op, 35% and 43% (respectively) still had reduced reflexes 1 year post-op[8]; reflexes were lost post-op in 3% and 10% respectively. The same study found that motor loss was improved in 80%, aggravated in 3%, and was newly present in 5% post-op; and that sensory loss was improved in 69% and was worsened in 15% post-op.

Foot drop: severe or complete paralysis of ankle dorsiflexion occurs in 5–10% of HLD, and about 50% of cases recover with or without treatment. Discectomy does not improve the outcome, especially in cases of painless foot drop.[24]

See Recurrent disc herniation (p.1061).

69.1.10 Herniated upper lumbar discs (levels L1–2, L2–3, and L3–4)

General information

L4–5 & L5-S1 herniated lumbar discs (HLD) account for most cases of HLD (realistically ≈ 90%, possibly as high as 98%[7]). 24% of patients with HLD at L3–4 have a past history of a HLD at L4–5 or L5-S1, suggesting a generalized tendency towards disc herniation. In a series of 1,395 HLDs, there were 4 at L1–2 (0.28% incidence), 18 at L2–3 (1.3%), and 51 at L3–4 (3.6%).[85]

Presentation

Typically presents with LBP, onset following trauma or strain in 51%. With progression, paresthesias and pain in the anterior thigh occur, with complaints of leg weakness (especially on ascending stairs).

Signs

Quadriceps femoris was the most common muscle involved, demonstrating weakness and sometimes atrophy.

Straight leg raising was positive in only 40%. Psoas stretch test was positive in 27%. Femoral stretch test may be positive (p.1048).

50% had reduced or absent knee jerk; 18% had ankle jerk abnormalities; reflex changes were more common with L3–4 HLD (81%) than L1–2 (none) or L2–3 (44%).

69.1.11 Extreme lateral lumbar disc herniations

General information

Definition: herniation of a disc at (foraminal disc herniation) or distal to (extraforaminal disc herniation) the facet (some authors do not consider foraminal disc herniation to be “extreme lateral”). See Fig. 69.1.

Incidence (Table 69.5): 3–10% of herniated lumbar discs (HLD) (series with higher numbers[86] include some HLD that are not truly extreme lateral).

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Occurs most commonly at L4–5 and next at L3–4 (see Table 69.5), thus L4 is the most common nerve involved and L3 is next. With a clinical picture of an upper lumbar nerve root compression (i.e. radiculopathy with negative SLR), chances are ≈ 3 to 1 that it is an extremely lateral HLD rather than an upper lumbar disc herniation.

Differs from the more common central and subarticular HLD in that:
- the nerve root involved is usually the one exiting at that level (c.f. the root exiting at the level below)
- straight leg raising (SLR) (p.1048) is negative in 85–90% of cases ≥ 1 week after onset (excluding double herniations; ≈ 65% will be negative if double herniations are included); may have positive femoral stretch test
- pain is reproduced by lateral bending to the side of herniation in 75%
- higher incidence of extruded fragments (60%)
- higher incidence of double herniations on the same side at the same level (15%)
- pain tends to be more severe (may be due to fact that the dorsal root ganglion may be compressed directly) and often has more of a burning dysesthetic quality

Presentation

Quadriceps weakness, reduction of patellar reflex, and diminished sensation in the L3 or L4 dermatome are the most common findings.

Differential diagnosis

1. lateral recess stenosis or superior articular facet hypertrophy
2. retroperitoneal hematoma or tumor

Table 69.5 Incidence of extreme lateral HLD by level[a]

Disc level	No.	%
L1–2	1	1%
L2–3	11	8%
L3–4	35	24%
L4–5	82	60%
L5-S1	9	7%

[a]series of 138 cases[86]

3. diabetic neuropathy (amyotrophy) (p.545)
4. spinal tumor
   a) benign (schwannoma or neurofibroma)
   b) malignant tumors
   c) lymphoma
5. infection
   a) localized (spinal epidural abscess)
   b) psoas muscle abscess
   c) granulomatous disease
6. spondylolisthesis (with pars defect)
7. compression of conjoined nerve root
8. on MRI, enlarged foraminal veins may mimic extreme lateral disc herniation

Radiographic diagnosis

NB: if actively sought, many asymptomatic far-lateral disc herniations may be demonstrated on MRI or CT. Careful clinical correlation is usually required.

MRI: diagnostic test of choice. Sagittal views through the neural foramen may help demonstrate the disc herniation.[87] MRI may have ≈ 8% false positive rate due to presence of enlarged foraminal veins that mimic extreme lateral HLD.[88]

Myelography: myelography alone is rarely diagnostic (usually requires post myelo CT.[89,90] Fails to disclose the pathology in 87% of cases due to the fact that the nerve root compression occurs distal to the nerve root sleeve (and therefore beyond the reach of the dye).[91]

CT scan[90]: reveals a mass displacing epidural fat and encroaching on the intervertebral foramen or lateral recess, compromising the emerging root. Or, may be lateral to foramen. Sensitivity is ≈ 50% and is similar with post-myelographic CT.[91] Post-discography CT[91,92] has also been able to demonstrate.

Surgical treatment

NB: compression of the dorsal root ganglion may result in a slower recovery from discectomy and overall less satisfying outcome than with the more commonplace paramedian disc herniation.

Foraminal discs

Usually requires mesial facetectomy to gain access to the region lateral to the dural sac without undue retraction on nerve root or cauda equina. Caution: total facetectomy combined with discectomy may result in a high incidence of instability (total facetectomy alone causes ≈ 10% rate of slippage), although other series found this risk to be lower (≈ 1 in 33[93,94]). An alternative technique is to remove just the lateral portion of the superior articular facet below.[95] Endoscopic techniques may be well suited for herniated discs in this location.[96]

Discs herniated beyond (lateral to) the foramen

Numerous approaches are used, including:

1. traditional midline hemilaminectomy: the ipsilateral facet must be partially or completely removed. The safest way to find the exiting nerve root is take the laminectomy of the inferior portion of the upper vertebral level (e.g. L4 for a L4–5 HLD) high enough to expose the nerve root axilla, and then follow the nerve laterally through the neural foramen by removing facet until the HLD is identified
2. lateral approach (i.e. extra-canal) through a paramedian incision.[97] Advantages: the facet joint is preserved (facet removal combined with discectomy may lead to instability), muscle retraction is easier. Disadvantages: unfamiliar approach for most surgeons and the nerve cannot be followed medial to lateral. A localizing x-ray is taken with a spinal needle. A 4–5 cm vertical skin incision is made 3–4 cm lateral to the midline on the side of the disc herniation. The incision is taken down to the thoracolumbar fascia and the subcutaneous tissue is dissected off the fascia. Above L4, one may palpate the groove between multifidus (medial) and longissimus (lateral), where the fascia is incised. The facet joint is palpated, and blunt dissection is used to gain access to the lateral facet joint and transverse processes above and below the level of the disc herniation. The correct level is confirmed on x-ray using a probe as a marker. The intertransversarius muscle and fascia are divided. Care must be taken to avoid mechanical and electrocautery injury to the nerve and dorsal root ganglion (which lies immediately beneath the intertransverse ligament). The radicular artery, vein, and nerve root are located just beneath the transverse processes, usually slightly

medial to this position. The nerve root hugs the pedicle of the level above as it exits the neural foramen (palpating this e.g. with a dental dissector helps locate it), and it may be splayed over the herniated disc fragment. If more medial exposure is necessary, the lateral facet joint may be resected. The HLD is removed. Additional removal of disc material from the disc space may be performed with down-biting pituitary rongeurs. Extracanal approach to L5-S1 requires removal of part of the sacral ala in order to access the space caudal to the L5 transverse process

69.1.12 Lumbar disc herniations in pediatrics

Less than one percent of surgery for herniated lumbar disc is performed on patients between the ages of 10 and 20 yrs (one series at Mayo found 0.4% of operated HLD in patients < 17 yrs age[98]). These patients often have few neurologic findings except for a consistently positive straight leg raising test.[99] Herniated disc material in youths tends to be firm, fibrous and strongly attached to the cartilaginous end-plate unlike the degenerated material usually extruded in adult disc herniation. Plain radiographs disclosed an unusually high frequency of congenital spine anomalies (transitional vertebra, hyperlordosis, spondylolisthesis, spina bifida…). 78% did well after their first operation.[98]

69.1.13 Intradural disc herniation

Herniation of a fragment of disc into the thecal sac, or into the nerve root sleeve (the latter sometimes referred to as "intraradicular" disc herniation) has been recognized with a reported incidence of 0.04–1.1% of disc herniations.[75,100] Although it may be suspected on the basis of pre-op MRI or myelography, the diagnosis is rarely made preoperatively.[100] Intraoperatively, it may be suggested by the appreciation of a tense firm mass within the nerve root sleeve or by the negative exploration of a level with obvious clinical signs and clear cut radiographic abnormalities (after verifying that the correct level is exposed).

Surgical treatment:

Although a surgical dural opening may be utilized,[75] others have found this to be necessary in a minority of cases.[101]

69.1.14 Intravertebral disc herniation

General information

AKA Schmorl's node or nodule. Named for German pathologist Christian Georg Schmorl (1861–1932) the term was first used in 1971.[102] AKA Schmor's (no "l") nodule AKA Geipel hernia.[103] Disc herniation through the cartilaginous end plate into the cancellous bone of the vertebral body (VB) (AKA intraspongious disc herniation). Often an incidental finding on x-ray or MRI. Clinical significance is controversial. May produce low back pain initially that lasts ≈ 3–4 months after onset. Diffuse displacement (as may be seen in osteoporosis) is sometimes referred to as a balloon disc.[104]

Clinical findings

During the acute (symptomatic) phase, patients may exhibit LBP that is aggravated by weight bearing and movement. There may be tenderness to percussion or manual compression over the involved segment.

Radiographic findings

MRI: the extrusion of disc material into the VB is easily appreciated on sagittal images. It has been suggested[105] that acute (symptomatic) lesions may appear differentiated from chronic (asymptomatic) lesions by the presence of MRI findings of inflammation in the bone marrow immediately surrounding the node as outlined in Table 69.6.

CT: demonstrates defect in endplate and vertebral body since disc material has a significantly lower density than bone.

Plain x-ray: ≤ 33% may be seen on plain x-rays.[106] They may not be detectable acutely until sclerotic osseous bone casting develops.

Treatment

Conservative treatment is indicated, usually consisting of non-steroidal anti-inflammatory drugs (NSAIDs). Occasionally stronger pain medication and/or lumbar bracing may be required. Surgery is rarely indicated.

Table 69.6 MRI signal intensity in Schmorl's nodes[a]

Lesion	T1WI	T2WI
symptomatic (acute)	low	high
asymptomatic (chronic)	high[b]	low[b]

[a]signal intensity in surrounding marrow
[b]the same as normal marrow

Outcome

With conservative treatment, symptoms generally resolve within 3–4 months of onset (as with most vertebral body fractures).

69.1.15 Recurrent herniated lumbar disc

General information

Rates quoted in the literature range from 3–19% with the higher rates usually in series with longer follow-up.[107] In an individual series with 10 year mean F/U, the rate of recurrent disc herniation was 4% (same level, either side), one third of which occurred during the 1st year post-op (mean: 4.3 yrs).[63] A second recurrence at the same site occurred in 1% in another series[107] with mean F/U of 4.5 yrs. In this series,[107] patients presenting for a second time with disc herniation had a recurrence at the same level in 74%, but 26% had a HLD at another level. Recurrent HLD occurred at L4–5 more than twice as often as L5-S1.[107]

It is often possible for a smaller amount of recurrent herniated disc to cause symptoms than in a "virgin back", due to the fact that the nerve root is often fixated by scar tissue and has little ability to deviate away from the fragment.[56]

Treatment

Initial recommended treatment is as with a first time HLD. Nonsurgical treatment should be utilized in the absence of progressive neurologic deficit, cauda equina syndrome (CES) or intractable pain.

69.1.16 Surgical treatment

Disagreement occurs regarding optimal treatment. See Practice guideline: Lumbar fusion for disc herniation (p.1037).

Surgical outcome:

As with first time HLD, the outcome from surgical treatment is worse in worker's compensation cases and in patients undertaking litigation, only ≈ 40% of these patients benefit.[107,108] A worse prognosis is also associated with: patients with < 6 mos relief after their first operation, cases where fibrosis without recurrent HLD is found at operation.

69.1.17 Spinal cord stimulation

One study actually showed a better response rate to spinal cord stimulation than to reoperation.[109] Since surgery for recurrent HLD carries a higher risk of dural and nerve root injury, and a lower success rate than first time operations, this may be a viable option for some patients.

69.2 Thoracic disc herniation

69.2.1 General information

Key concepts

- comprise only 0.25% of herniated discs, and < 4% of operations for herniated disc
- usually occur at or below T8 (the more mobile portion of the thoracic spine)
- frequently calcified → get CT through disc (may affect choice of surgical approach)

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- primary indications for surgery: refractory pain, progressive myelopathy
- surgical treatment: laminectomy is usually not appropriate

Account for 0.25–0.75% of all protruded discs.[110] 80% occur between the 3rd and 5th decades. 75% are below T8 (the more mobile portion of the thoracic spine), with a peak of 26% at T11–12. 94% were centrolateral and 6% were lateral.[111] A history of trauma may be elicited in 25% of cases.

Most common symptoms: pain (60%), sensory changes (23%), motor changes (18%). With thoracic radiculopathy, pain and sensory disturbance is in a band-like distribution radiating anteriorly and inferiorly along the involved root's dermatome. Motor involvement is difficult to document.

69.2.2 Evaluation

MRI is the mainstay of diagnosis. However, it is almost mandatory to also obtain a CT scan to determine if the disc is soft or calcified ("hard disc") which can have a profound effect on the approach. The CT is also useful to demonstrate bony detail if instrumentation is needed.

69.2.3 Indications for surgery

Herniated thoracic discs requiring surgery are rare.[111] Indications: refractory pain (usually radicular, bandlike) or progressive myelopathy. Uncommon: symptomatic syringomyelia originating at level of disc herniation.

69.2.4 Surgical approaches

Surgery for thoracic disc disease is problematic because of: the difficulty of anterior approaches, the proportionately tighter space between cord and canal compared to the cervical and lumbar regions, and the watershed blood supply which creates a significant risk of cord injury with attempts to manipulate the cord when trying to work anteriorly to it from a posterior approach. Herniated thoracic discs are calcified in 65% of patients considered for surgery[111] (more difficult to remove from a posterior or lateral approach than non-calcified discs).

Open surgical approaches[111,112]:

1. posterior (midline laminectomy): primary indication is for decompression of posteriorly situated intracanalicular pathology (e.g. metastatic tumor) especially over multiple levels. There is a high failure and complication rate when used for single-level anterior pathology (e.g. midline disc herniation)
2. posterolateral
   a) lateral gutter: laminectomy plus removal of pedicle.
   b) transpedicular approach[113]
   c) costotransversectomy (see below)
   d) transfacet pedicle sparing
3. anterolateral (transthoracic): usually through the pleural space
4. lateral extracavitary (retrocolic)[114] staying external to the pleural space

Thoracoscopic surgery is an alternative to open surgery.

69.2.5 Choosing the approach

General information

See anterior approaches to the thoracic spine (p. 1489).

Intraoperative SSEPs and MEPs may be helpful for patients with myelopathy.

For a laterally herniated thoracic disc without myelopathy: posterolateral approach with medial facetectomy is technically simple, and has generally good results. For a central disc herniation, or when myelopathy is present: transthoracic approach has the lowest incidence of cord injury with the best operative results (Table 69.7). For anterior access, unless pathology is predominantly left-sided, a right-sided thoracotomy is preferred because the heart does not impede access.

Costotransversectomy

Indications: in the past this was often used to drain tuberculous spine abscess. It may be used for lateral disc herniation, biopsy of VB or pedicle, limited unilateral decompression of spinal cord from

Table 69.7 Results with various approaches for thoracic spine pathology[115]

Approach	Indication	Total no.	Outcome			
			Normal	Improved	Same	Worse
laminectomy	posteriorly located tumor	129	15%	42%	11%	32%
posterolateral (transpedicular)	radicular pain with lateral disc herniation; biopsy of tumor	27	37%	45%	11%	7%
lateral (costotransversectomy)	fair for midline disc; good ipsilateral access, poor access to opposite side	43	35%	53%	12%	0
transthoracic	best for midline lesions, especially for reaching both sides of cord	12	67%	33%	0	0

tumor or bone fragments, or sympathectomy. Can be used at ≈ any T-spine level. Limitations: difficult to visualize anterior canal to access midline anterior pathology. Better for soft disc than for calcified central disc.

Involves resection of the transverse process and at least ≈ 4–5 cm of the posterior rib. A serious risk of this approach is interruption of a significant radicular artery which may compromise spinal cord blood supply; see Spinal cord vasculature (p.87). There is also a risk of pneumothorax which is less grave.

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Booking the case: Costotransversectomy

Also see defaults & disclaimers (p.27).
1. position: prone, usually on chest rolls
2. equipment:
   a) microscope (not used for all cases)
   b) C-arm
3. implants: if post-op instability is anticipated, thoracic pedicle screws and possibly a cage (e.g. for fracture or tumor, not typically for disc herniation)
4. neuromonitoring: SSEP/MEP
5. blood availability: type and cross 2 U PRBC
6. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the back of the chest to remove a small piece of rib to permit removal of the herniated/calcified disc
   b) alternatives: nonsurgical management, surgery from the side through the chest
   c) complications: spinal cord injury with paralysis, lung complications including pneumothorax or hemothorax (blood or air outside lungs), possible seizures with MEPs

69.2.6 Surgical technique

The approach can be somewhat difficult due to the infrequent encounter with the anatomy by most neurosurgeons. Be prepared for a "deep, red hole, where everything initially looks the same and the bony anatomy is not easy to define." With patience and persistence and the help of an anatomic model in the O.R., the surgeon can get his/her bearings. One of the most helpful landmarks is following the NVB (or just the nerve root) medially to the neural foramen.

In the O.R., before the prep and skin incision, localizing x-rays are obtained; a spinal needle inserted between 2 spinous processes may be used as a marker.

Patient position: the approach is from the side of the pathology/symptoms; for central disc herniations a right-sided approach reduces risk of injury to artery of Adamkiewicz (located on the left in 80%(p.87)). Options:
1. lateral oblique, ≈ 30° elevated from straight prone, a "bean-bag" is good for stabilization. For a thin patient, the surgeon may stand in front of the patient (gives more horizontal angle of view – does not work as well with heavier patients due to mass of skin/muscle in the way laterally)

2. prone on chest rolls: the chest roll on the side of the pathology should be more medial to allow the shoulder and scapula to fall forward out of the way

Skin incision options:
1. curved paramedian skin incision: apex oriented away from the midline along the slight depression demarcating the junction of the lateral border of the paraspinal muscles with the ribs (≈ 6–7 cm lateral to midline) centered over the interspace of interest extending ≈ 3 vertebral bodies (VB) above and below. The incision is carried through the skin, subcutaneous fat, trapezius, and (for lower 6 thoracic levels, where most thoracic disc herniations occur) the latissimus dorsi, down to the ribs, and this musculocutaneous flap can be reflected medially as a unit
2. midline incision: need to extend 3–4 levels above and below the level of pathology to get an angle low enough to visualize posterior to the facet in order to access the posterior vertebral body. The inferior aspect can be curved laterally towards the side of pathology. Advantage: a laminectomy can more easily be performed if needed (if the angle does not provide adequate visualization, as a "bail-out" contingency, a facetectomy may be performed, and pedicle may even be removed to access inferior to the disc space. This usually permits easy decompression of the entire thecal sac. In the thoracic spine, stabilization is optional, and if chosen, unilateral pedicle screws and fusion are usually adequate)

Rib removal and thoracic exposure: for a simple biopsy or drainage of a small abscess, removal of only 1 rib may suffice. The rib to be removed is from the level inferior to the disc space to be accessed[116] (e.g. remove the T5 rib to access T4–5 disc space). For most other pathologies, 2 or 3 ribs are often removed.[117] To access a VB, the like-numbered rib and the rib below are removed.

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There are a number of ligaments attached to the rib: the intercostal neurovascular bundle (NVB) courses medial to the superior costotransverse ligament which extends from the superior aspect of the rib to the transverse process of the level above. This ligament and the lateral costotransverse ligament are divided and the transverse process is rongeured off (the base of which lies on the lamina directly posterior to the pedicle). This exposes the rib anterior to the transverse process. The periosteum is incised on the rib from the angle of the rib to the costovertebral articulation, and by subperiosteal dissection around its circumference the pleura is dissected off the anterior surface of the rib. The NVB is dissected from the deep-inferior surface along with the periosteum. The rib is then transected laterally at the angle (≈ 5 cm lateral to the rib head) with rib shears, it is gripped with a clamp, and is rotated while the ligaments (including the radiate ligaments which attach the rib to the both the VB above and the VB below the disc space at the superior and inferior costal facet, respectively, except T1, 11 & 12 which only articulate with their like-numbered VB) are sharply dissected off the rib which is then removed. The removed rib material may be used for fusion substrate except in cases of tumor or infection. The pleura is then dissected from the deep surface of the adjacent ribs and VB (taking care not to injure the segmental vessels and to dissect the sympathetic trunk off the VB with the pleura). The pleura is then retracted laterally with a malleable ribbon or Deaver retractor.

The intervertebral foramen of interest may be located by following the NVB of the rib above proximally, the intercostal nerve (the ventral ramus of the nerve root at that level) enters between the two pedicles. The dura may then be exposed by enlarging the neural foramen by removing part of the pedicles with a high-speed drill and Kerrison rongeurs.

Instrumentation/fusion are rarely required for simple discectomy. Instability due to fracture, tumor, or extensive resection (e.g. with total facet takedown) necessitates surgical stabilization, typically with pedicle screws/rods extending 2 levels above and 2 levels below. Prior to closure, check for air leak by filling the opening with saline and having the anesthesiologist apply a valsalva maneuver. If an air leak is identified, a Cook catheter may be placed into the pleural space through the surgical exposure, or alternatively a chest tube is placed through a separate intercostal incision after the laminectomy wound is closed. A post-op CXR is obtained regardless of whether an air leak is identified.

Transpedicular approach

Drilling down the pedicle and removing a small amount of bone from the vertebral body, then pushing material from the epidural space into the defect created and removing it. Requires removal of just the rib head. Advantages: minimal risk of pneumothorax, more familiar anatomy. Disadvantages: requires instrumentation especially if done bilaterally, the angle is not very obliques so visualization of epidural space is minimal, may need to be done bilaterally if there are extensive bilateral components to the pathology.

Booking the case: Transpedicular approach

Same as for costotransversectomy (p. 1062).

Transthoracic approach

Indications: thoracic disc disease with central fragment or calcified disc, burst fractures of the thoracic spine, etc.

Advantages[118]:

- excellent anterior exposure (especially advantageous for multiple levels)
- little compromise of stability (due to supporting effect of rib cage)
- low risk of mechanical cord injury

Disadvantages:

- requires thoracic surgeon (or familiarity with thoracic surgery)
- some risk of vascular cord injury (due to sacrifice of intercostal arteries)
- definitive diagnosis may not be possible if it is uncertain prior to procedure

Possible complications:

- pulmonary complications: pleural effusion, atelectasis, pneumonia, empyema, hypoventilation
- CSF-pleural fistula

Booking the case: Transthoracic spine surgery

Also see defaults & disclaimers (p. 27).

1. position: typically on the side, often on a beanbag
2. equipment:
   a) microscope (not used for all cases)
   b) C-arm
3. anesthesia: double lumen tube
4. implants: if post-op instability is anticipated, thoracic pedicle screws and possibly a cage (e.g. for fracture or tumor, not typically for disc herniation)
5. neuromonitoring: SSEP/MEP
6. blood availability: type and cross 2 U PRBC
7. some surgeons use chest surgeon for the approach, closure and for follow-up
8. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the chest with removal of a small piece of rib to permit removal of the herniated/calcified disc
   b) alternatives: nonsurgical management, surgery from the side or through the back
   c) complications: spinal cord injury with paralysis, pneumothorax, possible seizures with MEPs

Key technical points

1. the services of an experienced thoracic surgeon are usually engaged
2. position: true lateral (facilitates intra-op localizing x-rays); approached from the more involved side. For the upper thoracic midline region, some prefer right-side-up to eliminate thoracic aorta from obstructing exposure and to reduce the possibility of encountering the artery of Adamkiewicz,[119] others prefer left-side-up to use the aorta as a landmark[118] (for levels below the cardiophrenic angle, a left-sided approach is preferred because the inferior vena cava is difficult to mobilize)
3. usually one rib is resected; most often the rib of the vertebra immediately above the disc space desired (facilitates exposure). Multiple ribs may be resected to increase exposure
4. when removing the vertebral body (VB) (corpectomy, e.g. for osteomyelitis, especially Pott's disease or for kyphoscoliosis)

   a) the posterior cortex of the VB must be pulled anteriorly (e.g. with angled curettes) to avoid mechanical cord trauma
   b) anterior fusion may be performed using the removed rib. If inadequate, fibula or iliac crest may be used
5. sizeable radicular arteries are spared. The intercostal nerve is used as a guide to the intervertebral foramen (nerve enters foramen superiorly and posteriorly)
6. the disc space is situated off the caudal aspect of the intervertebral foramen for most thoracic levels
7. one or two intervertebral arteries and veins usually have to be sacrificed; to minimize the risk of ischemic cord injury, cut them as close to the midline of the spine as possible (collaterals tend to lie on the lateral aspect of the spine)
8. the sympathetic chain is dissected off the VBs and is pushed posteriorly

Lateral (retrocolic) approach)

See reference.[114]

The same instrumentation used for lateral lumbar interbody fusion (p.1498) may be used to access the lateral thoracic bodies for thoracic disc herniation.

Check the pre-op MRI for the location of the aorta and to rule-out aortic aneurysm. Above T11, enter on the right side. The access retractor is “reversed” so that the center blade is positioned anteriorly and the shim is place so that as the retractor is expanded in the AP direction, the lateral blades move posteriorly, giving more access to the posterior disc space. In general, do not penetrate the contralateral anulus because of proximity of aorta. At or just below L12-L1, the diaphragm attaches to the VB. A dual lumen endotracheal tube is not required. A chest tube is mandatory if there is an air leak, otherwise it is optional (a pigtail catheter may suffice).

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70 Cervical Disc Herniation

70.1 General information

Important applied anatomy in herniated cervical disc (HCD):

1. in the cervical region, the nerve root exits above the pedicle of its like-numbered vertebra (opposite to the situation in the lumbar spine, due to the fact that there are 8 cervical nerve roots and only 7 cervical vertebrae)
2. each root exits passes through its neural foramen in close relation to the undersurface of the pedicle
3. the intervertebral disc space is located close to the inferior portion of the pedicle (unlike the lumbar region)

70.2 Cervical nerve root syndromes (cervical radiculopathy)

70.2.1 General information

Due to the facts listed above, a HCD usually impinges on the nerve exiting from the neural foramen at the level of the herniation (e.g. a C6–7 HCD usually causes C7 radiculopathy). This gives rise to the characteristic cervical nerve root syndromes shown in Table 70.1.

70.2.2 Miscellaneous clinical facts

C4 radiculopathy is not common, and may produce nonradiating axial neck pain.

Left C6 radiculopathy (e.g. from C5–6 HCD) occasionally presents with pain simulating an MI (pseudo-angina).

C8 and T1 nerve root involvement may produce a partial Horner's syndrome.

The most common scenario for patients with herniated cervical disc is that the symptoms were present upon awakening in the morning, without identifiable trauma or stress.[1]

70.3 Cervical myelopathy and SCI due to cervical disc herniation

Acute cord compression presenting with myelopathy or spinal cord injury (SCI) (including complete SCI and incomplete syndromes, especially central cord syndrome (p.944) and sometimes Brown-Sequard syndrome (p.947) [2] is well described in association with traumatic cervical disc herniation.[3] Less commonly, these findings may occur in non-traumatic cervical disc herniation.

Table 70.1 Cervical disc syndromes

Syndrome	Cervical disc syndromes			
	C4–5	C5–6	C6–7	C7-T1
% of cervical discs	2%	19%	69%	10%
compressed root	C5	C6	C7	C8
reflex diminished	deltoid & pectoralis	biceps & brachioradialis	triceps	finger-jerk[a]
motor weakness	deltoid	forearm flexion	forearm ext (wrist drop)	hand intrinsics
paresthesia & hypesthesia	shoulder	upper arm, thumb, radial forearm	fingers 2 & 3, all fingertips	fingers 4 & 5

[a]not everyone has a finger flexor reflex. Description: gently lift the fingertips of the patient's pronated hand and tap the underside of the fingers with a reflex hammer. When present, fingers flex in response

70.4 Differential diagnosis

See Differential diagnosis (p.1420).

70.5 Physical exam for cervical disc herniation

70.5.1 Overview

1. evaluation for radiculopathy
   a) lower motor neuron findings
      - weakness usually in one myotome group on one side
      - muscle bulk and tone: atrophy and fasciculations, may be present
   b) sensation: with nerve root compression, sensory loss will follow a dermatomal pattern and will be in the same nerve root distribution as the weakness
   c) muscle stretch reflexes
   d) mechanical signs: reproduction of radicular symptoms with axial loading of the head
2. evidence of spinal cord involvement (myelopathy)
   a) upper motor neuron findings, usually in the lower extremities
      - weakness may occur without atrophy or fasciculations
      - spasticity: poor control of the Legs when walking, scissoring of the legs
   b) sensation: any loss below the level of involvement will follow spinal cord patterns
      - complete loss
      - Brown-Sequard pattern: unilateral loss of pinprick with contralateral vibratory and position sense loss
      - Central cord syndrome: suspended sensory loss in upper extremities, less impaired below
      - Pathologic reflexes: Hoffmann's reflex, Babinski sign, ankle clonus

70.5.2 Signs useful in evaluating cervical radiculopathy

General information

Almost all herniated cervical discs cause painful limitation of neck motion. Neck extension usually aggravates pain when cervical disc disease is present (a minority of patients instead exhibit pain with flexion). Some patients find relief in elevating the arm and cupping the back or the top of the head with the hand (abduction relief sign, see below for shoulder abduction test). Lhermitte's sign (electrical shock-like sensation radiating down the spine) may be present; see DDx (p.1421).

Miscellaneous

The following tests were found to be specific, but not particularly sensitive in detecting cervical root compression[4]:
1. Spurling's sign[5]: radicular pain reproduced when the examiner exerts downward pressure on vertex while tilting head towards symptomatic side (sometimes adding neck extension). Causes narrowing of the intervertebral foramen and possibly increases disc bulge. Used as a "mechanical sign" analogous to SLR for lumbar disc herniation
2. axial manual traction: 10–15 kg of axial traction is applied to a supine patient with radicular symptoms (pull up on patient's mandible and occiput). The reduction or disappearance of radicular symptoms is a positive finding
3. shoulder abduction test[6]: a sitting patient with radicular symptoms lifts their hand above their head. The reduction or disappearance of radicular symptoms is a positive finding. Moderately sensitive, fairly specific[7]

70.6 Radiologic evaluation

70.6.1 MRI

The study of choice for initial evaluation for herniated cervical disc (HCD). Accuracy is less than water soluble contrast myelogram/CT (≈ 85–90% accuracy for MRI because of only fair to good imaging of neural foramen), but is non-invasive. For myelopathy, MRI is > 95% effective in diagnosing.

Protocol:
1. sagittal T1WI
2. multiple echo cardiac gated sagittal images (Tr = 1560, Te = 25, 4th echo)

3. GRASS image: axial partial flip-angle fast scan (Tr = 25, Te = 13, angle = 8°). Dark material adjacent to disc space is bone, disc is higher signal, CSF and flowing blood are high signal.

70.6.2 CT and myelogram/CT

Indications: when MRI cannot be done or when more bony detail than what MRI provides is required. Evaluates for ossification of the posterior longitudinal ligament (OPLL) when suspected.

Plain CT: is usually good at C5–6, is variable at C6–7 (due to artifact from patient's shoulders, depending on body habitus), and is usually poor at C7-T1.

Myelogram/CT (water soluble intrathecal contrast): invasive, on rare occasions requires overnight hospitalization. Accuracy is ≈ 98% for cervical disc disease.

70.6.3 Electrodiagnostics (EMG and NCV)

Compression may occur at the level of the dorsal (pre-ganglionic) sensory root (which, if occurs alone, produces a sensory-only radiculopathy) and/or at the ventral (motor) root. When motor exam is normal, EMG is unlikely to show abnormality. The AANEM practice parameter for cervical radiculopathy[8,9,10] reports sensitivity of 50-71% for the needle EMG examination and correlation between positive needle EMG and radiologic findings of 65-85%

EMG can also be normal in sensory only radiculopathy, which occasionally occurs in cervical spine, but not in lumbar spine. Since most muscles have at least dual innervation this poses a particular challenge for proximal cervical radiculopathies in which many muscles have the same shared innervation, e.g. biceps, deltoid, brachioradialis, infraspinatus and supraspinatus are all innervated by C5-C6.

For both cervical and lumbosacral radiculopathy, screening 6 muscles representing all root levels to include paraspinal muscles yield consistently high identification rates.[11]

For muscles to demonstrate fibrillations and positive waves there must be axonal loss in the motor nerve axons which innervates a muscle. Muscle demonstrates fibs and positive waves within 1 to 2 weeks following loss of innervation depending on the distance from the nerve to the muscle.

NCV is helpful to assess for peripheral neuropathies which may have symptoms similar to radiculopathy (e.g. carpal tunnel syndrome vs. C6 radiculopathy; ulnar neuropathy vs. C8 radiculopathy). A good physical exam can differentiate these entities in most cases.

Practice guideline: EDX guidelines for cervical radiculopathy

See reference.[10]

1. Guideline: EMG needle examination:
   a) needle examination of at least 1 muscle innervated by C5, C6, C7, C8 and T1 spinal roots in a symptomatic limb
   b) cervical paraspinal muscles at 1 or more levels (except in patients with prior posterior approach cervical surgery
   c) if abnormalities are identified, perform studies of 1 or 2 additional muscles innervated by the suspected root and different peripheral nerve
2. Guideline: At least 1 motor and 1 sensory nerve conduction study (NCS) in the clinically involved limb to determine if there is concomitant polyneuropathy or nerve entrapment. Motor and sensory NCS of median and ulnar nerves if symptoms and signs suggest CTS or ulnar neuropathy. If 1 or more NCS are abnormal or if clinical features suggest polyneuropathy, further evaluation may include NCS of other nerves in the ipsilateral and contralateral limb

70.7 Treatment

70.7.1 General information

Over 90% of patients with acute cervical radiculopathy due to cervical disc herniation can improve without surgery,[12] and regression of an extruded cervical disc has been demonstrated radiographically by CT and MRI.[13,14,15] The recovery period may be made more tolerable by adequate pain medication, anti-inflammatory medication (NSAIDs or short-course tapering steroids) and intermittent cervical traction (e.g. gradually escalating up to 10–15 lbs for 10–15 minutes, 2–3 × daily).

Surgery is indicated for those that fail to improve or those with progressive neurologic deficit while undergoing non-surgical management.

Management of myelopathy/central cord syndrome associated with acute cervical disc herniation is controversial, since the natural history is favorable in most cases. However, some patients have poor recovery and experience permanent deficits even with emergency surgery.[16]

70.7.2 Conservative management

Modalities include:
1. physical therapy, which may also include cervical traction.
2. Interventional pain management
   a) Trigger point injections
   b) Facet blocks
   c) Epidural steroid injection: not used as often and with lumbar spine

70.7.3 Surgery

Surgical options

1. anterior cervical discectomy: see below
   a) without any prosthesis or fusion: rarely used today
   b) combined with interbody fusion: the most common approach
      - without anterior cervical plating
      - with anterior cervical plating or with zero profile
   c) with artificial disc AKA cervical disc arthroplasty
2. posterior approaches
   a) cervical laminectomy: not typically used for a herniated cervical disc, more common for cervical spinal stenosis, OPLL
      - without posterior fusion
      - with lateral mass fusion
   b) keyhole laminotomy: sometimes permits removal of disc fragment

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For practice guidelines regarding intra-op electrophysiologic, see monitoring for surgery for cervical radiculopathy (p.1090).

Anterior cervical discectomy with fusion (ACDF)

Without special modifications, a routine anterior approach is usually able to access levels C3–7. In patients with short thick necks, access may be even more limited. In some cases, with long thin necks, up to C2–3 or as low as C7-T1 can be approached anteriorly.

Advantages over posterior (nonfused) approach:
1. safe removal of anterior osteophytes
2. fusion of disc space affords immobility (up to 10%incidence of subluxation with extensive posterior approach)
3. only viable means of directly dealing with centrally herniated disc

Disadvantages over posterior approach: immobility at fused level may increase stress on adjacent disc spaces. If a fusion is performed, some surgeons prescribe a rigid collar (e.g. Philadelphia collar) for 6–12 weeks. Multiple level ACDF can devascularize the vertebral body (or bodies) between discectomies.

Booking the case: ACDF

Also see defaults & disclaimers (p.27).
1. position: supine, some use halter traction with this
2. equipment:
   a) microscope (not used by all surgeons)
   b) C-arm
3. implants: graft (e.g. PEEK, cadaver bone, titanium cage…) and anterior cervical plate (optional, especially on single level ACDF)

4. neuromonitoring: (optional) some surgeons used SSEP/MEP
5. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the front of the neck to remove the degenerated disc and bone spurs, and to place a graft where the disc was, and possibly place a metal plate on the front of the spine. Some surgeons take bone from the hip to replace the removed disc
   b) alternatives: nonsurgical management, surgery from the back of the neck, artificial disc (in some cases)
   c) complications: swallowing difficulties are common but usually resolve, hoarseness of the voice (<4%chance of it being permanent), injury to: foodpipe (esophagus), windpipe (trachea), arteries to the brain (carotid), spinal cord with paralysis, nerve root with paralysis, possible seizures with MEPs

Technique

A summary of the steps involved is included here. For C5–6, the skin incision is made at level of cricoid cartilage, for other levels, appropriate adjustments up or down may be made, sometimes with the assistance of fluoroscopy. The incision is approximately 4–5 cm horizontally, centered on the SCM. Many right handed surgeons prefer operating from the right side of the neck, although the risk to the recurrent laryngeal nerve (RLN) is lower with a left sided approach (the RLN lies in a groove between the esophagus and trachea). The skin may be undermined off the platysma to permit a vertical incision in the platysma in the same orientation as its muscle fibers. Alternatively, some incise the platysma horizontally with scissors horizontally.

Dissect in tissue plane medial to SCM. For the C5–6 interspace, angle slightly cranially during dissection. For the C6–7 disc, proceed almost straight down to spine. Sweep omohyoid medially (to stay out of it and to protect the RLN). The trachea + esophagus are retracted medially. The carotid sheath + SCM are retracted laterally.

After verification of level with lateral C-spine x-ray with spinal needle in the interspace, bipolar the prevertebral fascia and medial edges of the longus coli muscles longitudinally in the midline. Self-retaining retractor blades are inserted underneath the fascia to retract the longus coli muscles laterally. The anesthesiologist is asked to deflate the cuff on the endotracheal tube and then to re-inflate it using minimal leak technique to reduce the risk of compression injury from the retractor. The disc space is incised with a 15 scalpel blade. The discectomy is performed with curettes and pituitary rongeurs; a vertebral body spreader aids the exposure. The posterior longitudinal ligament is incised, one technique is to elevate it with a sharp nerve hook and then incise it with a # 11 scalpel. The subligamentous space is probed with a blunt nerve hook. The posterior lip of the VB above and below are removed with a Kerrison rongeur with a small foot-plate. Decompression of the roots is verified with the blunt nerve hook. Fusion is performed at this time if desired by placing the graft in the interspace.

For redo operations (same or different levels): approach is usually from the same side as previous operation(s) since many patients have swallowing issues post-op, and some may be due to partial recurrent laryngeal nerve injury (which can be subclinical) and which could result in a permanent need for a feeding tube if a contralateral injury occurs. If for some reason it is desired to go to the opposite side, an evaluation by an ENT physician is recommended, and should include scoping the patient to rule-out subclinical problems that could turn into major difficulties if bilateral.

Choice of graft material

Autologous bone (usually from iliac crest), non-autologous bone (cadaveric), bone substitutes (e.g. hydroxylapatite[17]) or synthetics (e.g. PEEK or titanium cage) filled with osteogenic material. Substitutes for autologous bone eliminate problems with the donor site (p.1075), but may have a higher rate of absorption. There were also cases of HIV transmission from cadaveric bone grafts in 1985, however, as a result of the heightened awareness of AIDS since that time together with significant improvements in antibody testing and careful screening of donors, no further cases have been reported.

Anterior cervical plating

Recommendations for plating following ACDF are shown in Practice guideline: Anterior cervical plating (p.1074).

Practice guideline: Anterior cervical plating

1 level ADCF: The addition of an anterior plate to an ACDF is recommended to reduce the pseudarthrosis rate and graft problems (Level D Class III) and to maintain lordosis (Level C Class II) but it does not improve clinical outcome alone (Level B Class II)[18]

2 level ADCF: Plating is recommended to improve arm pain. Plating does not improve other outcome parameters (Level C Class II)[18]

Use of bone morphogenic proteins (BMP)

Practice guideline: Use of BMP in cervical interbody grafting

Current evidence does not support the routine use of rhBMP-2 for cervical arthrodesis (Level C Class II)[19] (note: italics added. Use with precautions (see text) may be indicated in cases with high risk of nonunion).

Use of BMP in anterior cervical discectomies is not FDA approved, but has been used off-label. Complication rates as high as 23–27% have been reported (including post-op swallowing or respiratory difficulties as a result of edema which is usually temporary) compared to 3% without BMP.[19] If used, it is recommended that a smaller dose be employed than in the lumbar spine (25% has been advocated) and to avoid contact of BMP with soft tissues in the neck.

Post-op check

In addition to routine, the following should be checked
1. evidence of airway obstruction – post-op wound hematoma: should be first consideration. Wound may need to be emergently opened at bedside (before getting to the OR) if airway is compromised, see Carotid endarterectomy, disruption of arteriotomy closure, management (p.1294). Also consider swelling from IJV thrombosis (rare) in differential diagnosis (see below)
   a) respiratory distress
   b) extreme difficulty swallowing: alternatively may indicate anterior extrusion of bone graft impinging upon esophagus (check lateral C-spine x-ray)
   c) tracheal deviation: may be visible or may be seen on AP C-spine x-ray
2. weakness of nerve root of level operated: e.g. biceps for C5–6, triceps for C6–7
3. long tract signs (Babinski sign…) which may indicate cord compression by spinal epidural hematoma
4. hoarseness: may indicate vocal cord paresis from recurrent laryngeal nerve injury: hold oral feeding until this can be further assessed

ACDF complications

General information

Common ones listed below, see references[20,21] for more details. The most common complication following ACDFs: swallowing difficulties (may be multifactorial).
1. exposure injuries
   a) perforation of viscus: minimize risk by blunt retraction until longus colli is separated from its attachment to vertebrae
      - pharynx
      - esophagus: difficult to manage, and require multidisciplinary effort including ENT involvement.[22] The incidence may be higher with use of anterior cervical plate, and injury may not manifest until years after the fusion (may be due to repetitive motion of esophagus over the plate). Treatment of esophageal perforation is usually facilitated by plate removal
      - trachea
   b) vocal cord paresis: due to injury of the recurrent laryngeal nerve (RLN) or vagus. Incidence: 11% temporary, 4% permanent paresis. Symptoms include: hoarseness, breathiness, cough, aspiration, mass sensation, dysphagia, and vocal cord fatigue.[23] Avoid sharp dissection in paratracheal muscles. Some cases may be due to prolonged retraction against trachea and not to nerve division; to reduce this risk, after the self-retaining retractor is placed, have the

anesthesiologist deflate the cuff on the ET tube and then inflate it to minimal leak pressure. More common with right sided approaches, primarily in the lower cervical spine (C5–6 and below) where the RLN is more vulnerable[23]

c) vertebral artery injury: thrombosis or laceration. 0.3% incidence.[21] Treatment alternatives include: packing, direct repair by temporary clipping with aneurysms clips and repair with 8–0 prolene[24] and endovascular trapping. Risks of treating hemorrhagic complications with packing include: recurrent bleeding, AV fistula, pseudoaneurysm, arterial thrombosis,[21] distal embolic stroke (primarily in cerebellum)
d) carotid injury: thrombosis, occlusion, or laceration (usually by retraction)
e) CSF fistula: usually difficult to repair directly. Place fascial graft beneath bone plug. Keep HOB elevated post-op. Consider: dural sealant (fibrin glue, DuraSeal®…), lumbar drain
f) Horner's syndrome: sympathetic plexus lies within longus coli, thus do not extend dissection far laterally into these muscles
g) thoracic duct injury: in exposing lower cervical spine, primarily on left
h) thrombosis of internal jugular vein[25]: rare. Carries 2–3% risk of PE.[26] Treatment options: anticoagulation (oral or IV) may lower the mortality,[27] SVC filter if anticoagulation is contraindicated,[28] percutaneous thrombectomy[29]

2. spinal cord or nerve root injuries
   a) spinal cord injury: especially risky in myelopathy due to narrowed canal. Minimize risk by penetrating the osteophyte at the lateral margin of interspace (however, this increases risk to nerve root)
   b) avoid hyperextension during intubation: anesthesiologist may need to determine patient's tolerance pre-op. Consider fiberoptic guided or awake nasotracheal intubation in extreme stenosis
   c) bone graft must be shorter than interspace depth. Exercise caution in tapping graft into position
   d) sleep induced apnea: rare but serious complications of C3–4 level operations.[30] May be associated with bradycardia & cardiorespiratory instability. Possibly due to disruption of the afferent component of the central respiratory control mechanism
3. bone fusion problems
   a) failure of fusion (pseudarthrosis): see below
   b) anterior (kyphotic) angulation deformity: may be as high as 60% with Cloward technique (may be reduced by collar immobilization). May develop in Hirsch technique with excessive bone removal
   c) graft extrusion: 2% incidence (rarely requires re-operation unless compression of cord posteriorly, or esophagus or trachea anteriorly occurs)
   d) donor site complications: hematoma/seroma, infection, fracture of ilium, injury to lateral femoral cutaneous nerve, persistent pain due to scar, bowel perforation
4. miscellaneous
   a) wound infection: incidence < 1%
   b) post-op hematoma: see above. Placing cervical collar in O.R. may delay recognition
   c) dysphagia and hoarseness: common. Usually transient (see below)
   d) adjacent level degeneration: controversial whether this represents a sequelae to altered biomechanics from surgery, or a predisposition to cervical spondylosis.[31] Many (≈ 70%) are asymptomatic[32]
   e) postoperative discomfort:
      - globus: the sensation of a lump in throat (see below)
      - nagging discomfort in neck, shoulder, and very commonly in interscapular regions (may last several months). May correlate with amount of distraction of the disc space
   f) complex regional pain syndrome (CRPS) AKA reflex sympathetic dystrophy (RSD): rarely described in the literature,[33] possibly due to stellate ganglion injury; see discussion of RSD (p. 497)
   g) angioedema: massive edema of the tongue and neck.[34] A dramatic hypersensitivity reaction (not really a direct complication of ACDF, but superficially can mimic some findings of post-op hematoma). If limited to the tongue, the airway is not compromised. See treatment (p. 222)
   h) pneumothorax or hemothorax[35]: accessing C7-T1 or lower may expose the pleural apex

Dysphagia following ACDF

Symptoms: Include: difficulty swallowing (solids, liquids including saliva), pain with swallowing (odynophagia), globus (sensation of a lump in throat) and compromise of ability to protect against aspiration. Food may stick in the throat (or feel as if it is) and there may be coughing or choking.

Incidence: Early dysphagia is common. Incidence: 60%[36] in retrospective survey after noninstrumented fusion (dysphagia occurred in 23% in a control group undergoing unrelated lumbar spine surgery[36]), 50% in prospective study.[37] At 6 months, only ≈ 5% reported moderate or severe dysphagia.[37] Surgery at multiple levels increased the risk at 1 & 2 months.[37] Decreases significantly in most cases by 6 months.[37]

Etiologies: Etiologies of post-op dysphagia include:

1. post-op hematoma. If severe, may cause tracheal obstruction (see above)
2. post-op edema, due in part to retraction of esophagus
3. effects of general anesthesia: e.g. irritation from ET tube. Accounts for up to 23% of early symptoms (dysphagia occurred in 23% in a control group undergoing unrelated lumbar spine surgery[36]). Usually subsides within ≈ 24–72 hours
4. recurrent laryngeal nerve dysfunction:
   a) temporary: usually due to traction on the nerve
   b) permanent: 1.3% at 12 months[37]
5. esophageal injury
   a) at time of surgery
   b) delayed: possibly from repetitive abrasion on surgical site/hardware or from unrecognized esophageal injury at time of surgery[22]
6. cervical collar
   a) prevents patient from lowering jaw during swallow phase, which compromises effective glottic closure of airway
   b) may be too tight thereby directly squeezing the throat
7. protrusion of graft/hardware anterior to the vertebral bodies
   a) some protrusion is present with most anterior hardware. This may be minimized with “zero profile” instrumentation
   b) hardware failure (screw-pullout/backout/breakage, plate pullout)
   c) interbody graft migration: without anterior plate, or in conjunction with anterior plate displacement
8. excessive adhesions[38]
9. denervation of the pharyngeal plexus[38]
10. rare conditions: swelling from IJV thrombosis, angioedema

Management:

1. initial management: rule-out emergent/serious conditions (severe edema, hematoma with airway compromise, risk of aspiration)
   a) if there is significant stridor or dysphonia, especially if tracheal deviation is obvious, someone must stay with the patient as efforts are made to emergently take the patient to the O.R. for wound exploration evacuation. Consider opening the wound at the bedside if delays occur or if symptoms are severe; see Carotid endarterectomy, disruption of arteriotomy closure, management (p. 1294). Emergent anesthesia consultation for airway protection – alert them to the likelihood of deviated trachea which challenges even the most expert at intubating
2. once emergent conditions are ruled out, early management is geared towards amelioration of symptoms
   a) advise patient to eat softer foods (temporarily avoiding steak or bread), to chew food well, to wash down dry foods with a drink. Reassure patient that most cases largely resolve with 6 months[37]
   b) if significant symptoms persist > 2 weeks
      - refer patient to ENT for laryngoscopy to rule out vocal cord paralysis (from RLN injury) or other etiologies. For RLN, see below
      - modified barium swallow
3. persistent symptoms may be amenable to surgical intervention, including hardware removal and lysing of adhesions,[38] management of esophageal perforation usually requires consultation with ENT

Management of esophageal perforation:

There is no consensus on optimal management. A multidisciplinary approach with head and neck surgeons together with the spine surgeon is suggested with the following[22]:

- Closure with a sternocleidomastoid muscle flap, or possibly a pedicle flap
- Removal of all anterior hardware. If there is evidence that the fusion is not solid, posterior instrumentation may be necessary

Recurrent laryngeal nerve paralysis

Heralded by breathy voice, hoarseness, or aspiration. Refer to ENT to have patient scoped to determine if cord is paralyzed and its position. 4 possible positions: 1) median, 2) paramedian, 3) intermediate, 4) lateral (cadaveric). Many patients can compensate for median or paramedian position. Patients requiring intervention usually are treated with medialization technique, either 1) injection, or 2) medialization thyroplasty using an implant. For injections, different materials may be selected for desired duration of effect (Teflon used to be the only available agent, and was essentially permanent) and so early intervention may be employed with temporary materials (instead of waiting 1 year, which was the previous paradigm).

Pseudarthrosis (or pseudoarthrosis) following ACDF

Pseudarthrosis may occur with or without supplemental anterior cervical plating.

Practice guideline: Assessment of subaxial fusion

>2 mm movement between spinous processes on dynamic (flexion-extension) cervical spine x-rays is recommended as a criteria for pseudarthrosis (Level B Class II), this measurement is unreliable when performed by the treating surgeon (Level C Class II).[39]

Visualization of bone trabeculation across the fusion on static films is a less reliable marker for fusion (Level D Class III) (2D reformatted CT increases the accuracy (Level D Class III))[39]

Incidence: Difficult to assess because of lack of validated criteria. Estimate: 2–20% Higher with dowel technique (Cloward) than with keystone technique of Bailey & Badgley or with interbody method of Smith-Robinson (10%) or with non-fusion advocated by Hirsch. One criteria: motion >2 mm between the tips of the spinous processes on lateral flexion/extension x-rays.[40,41] Other criteria: lucencies around the screws of an anterior plate, toggling of the screws on flexion/extension x-rays.

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Presentation: Not uniformly associated with symptoms or problems.[40,42] Some patients may have chronic or recurrent neck pain, some may present with radicular symptoms. (NB: when DePalma's data is analyzed with patients reclassified as failures if neck and/or arm symptoms persist, the success rate of surgery is lower with pseudarthrosis[43]).

Management: Guidelines are shown in Practice guideline: Management of anterior cervical pseudarthrosis (p. 1077). No treatment is required for asymptomatic pseudarthrosis. Options for symptomatic patients include re-resection of the bone graft with repeat fusion[44] (some recommend using autologous bone if allograft was used, a plate may be considered if one was not used previously), cervical corpectomy with fusion,[44] or posterior cervical fusion.

Practice guideline: Management of anterior cervical pseudarthrosis

Revision of symptomatic pseudarthrosis should be considered (Level D Class III).[45] Posterior approaches may be associated with higher fusion rates on revision than anterior approaches (Level D Class III)[45]

Cervical disc arthroplasty

An alternative to fusion. Uses an artificial disc to preserve motion at the level of the discectomy. Some of the available cervical disc replacement (CDR) models are shown in Table 70.2.[46]

Contraindications described by the FDA have included: isolated axial neck pain, ankylosing spondylitis or pregnancy, rheumatoid arthritis, autoimmune disease, diffuse idiopathic skeletal hyperostosis, severe spondylosis with bridging osteophytes or ossification of the posterior longitudinal ligament, disc height loss >50%, spinal infection, metal allergy to components of the prosthesis, severe osteoporosis/osteopenia, active malignancy, metabolic bone disease, trauma, segmental instability, 3 or more levels requiring treatment, insulin dependent diabetes mellitus, human immunodeficiency virus, hepatitis B/C, morbid obesity, absence of motion (<2 degrees), and posterior facet arthrosis.

Table 70.2 Artificial cervical discs

Trade name	Manufacturer	Material	IAR[a]	Comment
Prestige®	Medtronic	MOM[a] (chrome cobalt stainless steel)	variable ball in trough	1st FDA approved CDR; lots of MRI artifact
Bryan®	Medtronic	lubricated elastic nucleus sealed in a flexible membrane	variable in center of disc space	
Advent®	Blackstone (Orthofix)	flexible elastomer core		withdrawn from market
ProDisc-C	Synthes	metal-on-polyethylene	posterior part of inferior VB	midline keel inserts into VB; lots of MRI artifact
Mobi-C®	LDR Spine	metal-on-polyethylene	variable in center of disc space	approved for 1 or 2 levels
PCM®	Cervitech	metal-on-polyethylene	gliding motion	contoured to endplates

[a]Abbreviations: IAR = instantaneous axis of rotation, MOM = metal-on-metal, PCM = Porous-coated Motion

Practice guideline: Cervical disc arthroplasty

Cervical arthroplasty is a recommended alternative to ACDF in selected patients for control of arm and neck pain (Level B Class II)[18]

Booking the case: Cervical disc arthroplasty

Also see defaults & disclaimers and/or more significant (p. 27).

1. position: supine, some use halter traction with this
2. equipment:
   a) microscope (not used by all surgeons)
   b) C-arm
3. implants: schedule vendor to provide desired artificial disc
4. neuromonitoring: (optional) some surgeons used SSEP/MEP
5. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the front of the neck to remove the degenerated disc and bone spurs, and to place an artificial disc
   b) alternatives: nonsurgical management, surgical fusion (from the front or the back of the neck)
   c) complications: swallowing difficulties are common but usually resolve, hoarseness of the voice (< 4% chance of it being permanent), injury to: foodpipe (esophagus), windpipe (trachea), arteries to the brain (carotid) with stroke, spinal cord with paralysis, nerve root with paralysis, possible seizures with MEPs (if used). The disc may eventually wear out and further surgery may be needed

Post-op orders:

1. no cervical collar (the goal is to preserve motion at the operated level)
2. NSAIDs around the clock for ≈ 2 weeks (this inhibits bone growth which theoretically helps avoid undesirable fusion at the operated level)

Posterior cervical decompression (cervical laminectomy)

Not necessary for unilateral radiculopathy (use either ACD or keyhole laminotomy). Consists of removal of cervical lamina (laminectomy) and spinous processes in order to convert the spinal canal from a “tube” to a “trough.”

Usually reserved for the following conditions:
1. multiple cervical discs or osteophytes (anterior cervical discectomy (ACD) is usually used to treat only 2, or possibly 3, levels without) with myelopathy
2. where the anterior pathology is superimposed on cervical stenosis, and the latter is more diffuse and/or more significant (p.1083)
3. in professional speakers or singers where the 4%risk of permanent voice change due to recurrent laryngeal nerve injury with ACD may be unacceptable

Booking the case: Cervical laminectomy

Also see defaults & disclaimers (p.27).
1. position: prone, some use pin headholder
2. equipment:
   a) C-arm
   b) high-speed drill
3. implants: cervical lateral mass screws and rods if fusion is being done
4. neuromonitoring: some surgeons used SSEP/MEP
5. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the back of the neck to remove the bone over the compressed spinal cord and nerves and possibly to place screws and rods to fuse the boned together
   b) alternatives: nonsurgical management, surgery from the front of the neck, posterior surgery without fusion, laminoplasty
   c) complications: nerve root injury (C5 nerve root is the most common), may not relieve symptoms necessitating further surgery, possible seizures with MEPs. If fusion is not done, risk of progressive bone slippage which would require further surgery

Posterior keyhole laminotomy

AKA "keyhole foraminotomy." First described in 1951.[47] A technique to decompress only individual nerve roots (but not the spinal cord) by creating a small "keyhole" in the lamina to access the nerve root.

Practice guideline: Cervical laminoforaminotomy

Cervical laminoforaminotomy is recommended as a surgical treatment option for symptomatic cervical radiculopathy caused by disc herniation or lateral recess narrowing (Level D Class III)[48]

Indications for keyhole approach (as opposed to anterior discectomy):
1. monoradiculopathy with posterolateral soft disc sequestration (small lateral osteophytic spurs may also be addressed). This approach does not provide adequate decompression with central or broad-based disc herniation or with stenosis of the spinal canal
2. radiculopathy in patients who are professional speakers or singers where the risk of recurrent laryngeal nerve injury is untenable (see above)
3. for lower (e.g. C7, C8 or T1) or upper (e.g. C3 or C4) cervical nerve root compression, especially in a patient with a short thick neck, making an anterior approach more difficult
4. in patients with a herniated disc when it is desired to avoid a fusion (as would be done with an anterior approach)

Booking the case: Cervical keyhole laminectomy

Also see defaults & disclaimers (p.27).
1. position: prone, some use pin headholder
2. equipment:
   a) microscope (not used by all surgeons)
   b) C-arm

3. instrumentation: some surgeons use a tube retractor system
4. neuromonitoring: some surgeons used SSEP/MEP
5. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the back of the neck to remove the bone over the compressed nerve root and possibly remove fragment of herniated disc
   b) alternatives: nonsurgical management, surgery from the front of the neck, posterior surgery with fusion
   c) complications: nerve root injury, may not relieve symptoms necessitating further surgery, possible seizures with MEPs

Technique

See references.[49,50,51]

Position:

a) prone, on chest rolls. Adhesive tape is used to retract shoulders down for any level below about C4–5. The head is stabilized on a horse-shoe headrest or in a Mayfield head holder.
b) sitting position: generally abandoned. However, may be used with proper precautions (p.1445)

"Open" keyhole foraminotomy

The desired level is localized with intra-op fluoroscopy before making the skin incision, a 2–3 cm midline incision is adequate. A unilateral exposure suffices. Periosteal elevators are used to dissect muscles off the lamina and facet joint in the sub-periosteal plane. A Kocher clamp may be placed on the spinous process to permit confirmation of the correct level on intra-operative x-ray. A Scoville retractor or equivalent is employed.

A high-speed drill (e.g. with diamond burr) is used to make an opening in the medial one-third to one-half of the inferior facet of the vertebra above the desired disc space, extending slightly medially into the junction with the lamina. Once the inferior facet is penetrated, the superior facet of the inferior vertebral level will be visualized. This is also thinned with the drill (it is critical to remove the bone of the superior facet of the level below caudally to where it meets the pedicle). A small Kerrison rongeur may be used to slightly enlarge the laminectomy. An opening is made in the ligamentum flavum overlying the lateral aspect of the spinal cord dura. The nerve root can be identified as it exits from the thecal sac, and can be followed as it travels between the pedicles of the vertebrae above and below. Soft tissues (including ligamentum flavum) form fibrous bands across the dorsum of the nerve, and are removed to further expose the dura of the nerve root. The venous plexus around the nerve root is coagulated with bipolar cautery and then divided to mobilize the nerve. The nerve may then be gently moved a few millimeters rostrally using a micro nerve hook. The dura overlying the spinal cord should not be manipulated, and the disc space need not be entered. Inspection for free disc fragments should begin in the nerve root axilla using a probe (e.g. blunt nerve hook). Next, the space anterior to the root (the region of the disc) may be palpated. Any disc fragments that are dislodged are removed with a small pituitary rongeur. If the disc fragment is contained anterior to the posterior longitudinal ligament (PLL), the PLL may be incised in the region of the nerve root axilla with a #11 scalpel blade in a motion that is directed downward and laterally, away from the nerve root and spinal cord. The foraminotomy may be extended slightly laterally if the foramen still feels tight when probed. Small osteophytes can potentially be reduced using a small reversed-angled curette, although some surgeons believe that the need for this is obviated by the decompression provided by the keyhole opening. In some cases, simple posterior decompression of the nerve root (without removing a disc fragment) may be adequate to relieve compression. Spinal stability is usually preserved if less than half the facet joint is removed.

MIS keyhole foraminotomy

Positioning as described above.

1. skin incision
   a) use fluoro to locate the correct level for the incision
   b) incision 1 cm off midline on the side of the pathology at the level of the disc space
   c) remove adhesive plastic barrier (e.g. Ioban®) from around the opening to prevent pieces from being dragged into the incision

2. avoid using a guidewire to reduce the risk of penetrating the interlaminar space. STAY LATERAL and insert the thinnest dilator. Dock the dilator on the lateral mass and insert progressively sized dilators
3. use Bovie to expose lateral lamina and medial facet joint. Start laterally where bone is more easily felt and there is little danger of penetrating the interlaminar space and injuring the spinal cord
4. use a straight curette to expose the inferior edge of the superior lateral lamina and the medial facet joint
5. drill off the medial inferior facet, to expose the superior facet of the level below
6. drill the medial superior facet until you are flush with the superior aspect of the pedicle below
7. this completes the bony work, the soft tissue work proceeds as above under open keyhole foraminotomy

Outcome

A number of large series have reported good or excellent outcome in the range of 90–96%[50]

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71 Degenerative Cervical Disc Disease and Cervical Myelopathy

71.1 General information

Cervical degenerative disc disease is generally discussed in terms of “cervical spondylosis”, a term which is sometimes used synonymously with “cervical spinal stenosis.” Spondylosis usually implies a more widespread age-related degenerative condition of the cervical spine including various combinations of the following:

1. congenital spinal stenosis (the “shallow cervical canal”[1])
2. degeneration of the intervertebral disc producing a focal stenosis due to a “cervical bar” which is usually a combination of:
   a) osteophytic spurs (“hard disc” in neurosurgical jargon)
   b) and/or protrusion of intervertebral disc material (“soft disc”)
3. hypertrophy of any of the following (which also contributes to canal stenosis):
   a) lamina
   b) dura
   c) articular facets
   d) ligaments, including
      - increased stenosis in extension is more common than with flexion (based on MRI studies[2] and cadaver studies), largely due to posterior inbuckling of ligamentum flavum[3]
      - posterior longitudinal ligament: may include ossification of the posterior longitudinal ligament (OPLL) (p. 1127).[4] May be segmental or diffuse. Often adherent to dura
      - ossification of the ligamentum flavum[5] (yellow ligament)
4. subluxation: due to disc and facet joint degeneration
5. altered mobility: severely spondylotic levels may be fused and are usually stable, however there is often hypermobility at adjacent or other segments
6. telescoping of the spine due to loss of height of VBs → “shingling” of laminae
7. alteration of the normal lordotic curvature[6] (NB: the amount of abnormal curvature did not correlate with the degree of myelopathy)
   a) reduction of lordosis: including
      - straightening
      - reversal of the curvature (kyphosis): may cause “bowstringing” of the spinal cord across osteophytes
   b) exaggerated lordosis (hyperlordosis): the least common variant (may also cause bowstringing)

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Although the majority of individuals > 50 yrs old have radiologic evidence of significant degenerative disease of the cervical spine, only a small percentage will experience neurologic symptoms.[7]

71.2 Pathophysiology

Pathogenesis is controversial. Theories include the following alone or in combination:

1. direct cord compression between osteophytic bars and hypertrophy or infolding of the ligamentum flavum, especially if superimposed on congenital narrowing or cervical subluxations
2. ischemia due to compression of vascular structures[8] (arterial deprivation[9] and/or venous stasis[10])
3. repeated local cord trauma by normal movements in the presence of protruded discs and/or osteophytic (spondylotic) bars (cord and root injuries[11])
   a) cephalad/caudad movement with flexion extension[12]
   b) anterior/posterior traction on the cord by dentate ligaments[13] & nerve roots
   c) diameter of spinal canal varies during flexion and extension
      - increased stenosis is more common in extension (see above)
      - unstable segments may sublux (so-called pincer mechanism)[14]

Histologically,[15] there is degeneration of the central grey matter at the level of compression, degeneration of the posterior columns above the lesion (particularly in the anteromedial portion), and demyelination in the lateral columns (especially the corticospinal tracts) below the lesion. Anterior spinal tracts are relatively spared. There may be atrophic changes in the ventral and dorsal roots and neurophagia of anterior horn cells.

71.3 Clinical

71.3.1 General information

Cervical spondylosis condition may produce several types of clinical problems[16]:

1. Myeloradiculopathy: some combination of
   a) Radiculopathy: nerve root compression may cause nerve-root (radicular) complaints
   b) spinal cord compression may cause myelopathy. Some stereotypical syndromes are presented below (see Cervical spondylotic myelopathy (CSM) below)
2. pain and paresthesias in the head, neck and shoulders with little or no suggestion of radiculopathy nor abnormal physical findings. This group is the most difficult to diagnose and treat, and often requires a good physician-patient relationship to decide if surgical treatment should be undertaken in an attempt to provide relief

Cervical spondylosis is the most common cause of myelopathy in patients > 55 yrs of age.[17] CSM is rare in patients < 40 years of age.

Cervical spondylotic myelopathy (CSM) develops in almost all patients with ≥ 30% narrowing of the cross-sectional area of the cervical spinal canal[18] (although some patients with more severe cord compression do not have myelopathy[19,20]).

Gait disturbance, often with LE weakness or stiffness, is a common early finding in CSM.[21] Ataxia may result from spinocerebellar tract compression. Early on, patients may experience difficulty running. Cervical pain and mechanical signs are uncommon in cases of pure myelopathy. See Table 71.1 for the frequency of symptoms in CSM in one series. In most cases the disability is mild, and the prognosis for these is good.

71.3.2 Motor

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Findings can be due to cord (UMN) and/or root (LMN) compression. The earliest motor findings are typically weakness in the triceps and hand intrinsics.[23] There may be wasting of the hand muscles.[24] Slow, stiff opening and closing of the fists may occur.[25] Clumsiness with fine motor skills (writing, buttoning buttons…) is common.

There is often proximal weakness of the lower extremities (mild to moderate iliopsoas weakness occurs in 54%) and spasticity of the LEs.

71.3.3 Sensory

Sensory disturbance may be minimal, and when present are often not radicular in distribution. There may be a glove-distribution sensory loss in the hands.[26] A sensory level may occur a number of levels below the area of cord compression.

LEs often exhibit loss of vibratory sense (in as many as 82%), and occasionally have reduced pinprick sensation (9%) (almost always restricted to below the ankle). Compression of the spinocerebellar tract may cause difficulty running. Lhermitte's sign was present in only 2 of 37 cases. Some patients may present with a prominence of posterior column dysfunction (impaired joint position sense and 2 point discrimination).[27]

71.3.4 Reflexes

In 72–87%, reflexes are hyperactive at a varying distance below the level of stenosis. Clonus, Babinski's sign (p.90) or Hoffman's sign (p.90) may also be present. Dynamic Hoffman's sign[28] may be more sensitive: test for Hoffman's sign during multiple cervical flexion and extension movements as tolerated by the patient. 94% of asymptomatic individuals with Hoffman's reflex will have significant spinal cord compression on MRI.[29] Inverted radial reflex: flexion of the fingers in response to eliciting the brachioradialis reflex, said to be pathognomonic of CSM.[30]

A hyperactive jaw jerk indicates upper motor neuron lesion above the midpons, and distinguishes long tract findings due to pathology above the foramen magnum from those below (e.g. cervical myelopathy): not helpful if absent (a normal variant). Primitive reflexes (grasp, snout, rooting) are not reliable localizing signs (except perhaps the grasp reflex) of frontal lobe pathology.

71.3.5 Sphincter

Urinary urgency and frequency are common in CSM, however these complaints are also protean in the aging population. Urinary incontinence is rare. Anal sphincter disturbances are uncommon.

Table 71.1 Frequency of symptoms in CSM (37 cases[22])

Finding	%
pure myelopathy	59%
myelopathy + radiculopathy	41%
reflexes	
• hyperreflexia	87%
• Babinski	54%
• Hoffman	13%
sensory deficits	
• sensory level	41%
• posterior column	39%
• dermatomal arm	33%
• paresthesias	21%
• positive Romberg	15%
motor deficits	
• arm weakness	31%
• paraparesis	21%
• hemiparesis	18%
• quadriparesis	10%
• Brown-Séquard	10%
• muscle atrophy	13%
• fasciculations	13%
pain	
• radicular arm	41%
• radicular leg	13%
• cervical	8%
spasticity	54%
sphincter disturbance	49%
cervical mechanical signs	26%

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71.3.6 Syndromes

Clustering of CSM into these 5 clinical syndromes has been described[25]:

1. transverse lesion syndrome: involvement of corticospinal and spinothalamic tracts and posterior columns, with anterior horn cells segmentally involved. Most frequent syndrome, possibly an "end-stage" of the disease process
2. motor system syndrome: primarily corticospinal tract and anterior horn involvement with minimal or no sensory deficit. This creates a mixture of lower motor neuron findings in the upper extremities and upper motor neuron findings (myelopathy) in the lower extremities which can mimic ALS (see below). Reflexes may be hyperactive below the area of maximal stenosis (including the upper extremities), occasionally beginning several levels below the stenosis
3. central cord syndrome: motor and sensory deficit affecting the UEs more than the LEs. This syndrome is characterized by dysfunction of the watershed areas located centrally within the cord,

which may be responsible for prominence of hand symptoms[31] (results in "numb-clumsy hands"[32]). Lhermitte's sign may be more common in this group
4. Brown-Séquard syndrome: often with asymmetric narrowing of the canal with the side of greater narrowing producing ipsilateral corticospinal tract (upper motor neuron weakness) and posterior column dysfunction with contralateral loss of pain and temperature sensation
5. brachialgia and cord syndrome: radicular UE pain with lower motor neuron weakness, and some associated long tract involvement (motor and/or sensory)

71.3.7 Grading

1. the modified Japanese Orthopaedic Association scale (mJOA) (▶ Table 71.2) is a valid and reliable grading system, although it is non-specific
2. Neck Disability Index[33]: a 10 question survey similar to the Oswestry Disability Index for the lumbar spine (see ▶ Table 68.3). Mild disability is defined as a score of 10–28%, moderate = 30–48%, severe = 50–68%, complete ≥ 72%
3. other commonly used scales (not tested for validity or reliability):
   a) Nurick[34] (see ▶ Table 74.2)
   b) Harsh

71.3.8 Natural history

The time course of symptoms is highly variable and unpredictable. In ≈ 75% of cases of CSM, there is progression either in a stepwise fashion (in one–third) or gradually progressive (two–thirds).[35] In some series, the most common pattern was that of an initial phase of deterioration followed by a stabilization that typically lasts for years and may not change thereafter.[36,37] In these cases, the degree of disability may be established early in the course of CSM. Others disagree with such a "benign" outlook and cite that over 50% of cases continue to deteriorate with conservative treatment.[7] Sustained spontaneous improvement is probably rare.[17]

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In patients < 75 years age and mJOA score > 12 (mild myelopathy), the clinical condition remained stable in 3-years of follow-up (Class I)[38] (however, these patients can still have significant disability that can respond to surgery). Patients with stenosis without myelopathy who have electrodiagnostic abnormalities or clinical radiculopathy are at risk of developing myelopathy (Class I).[38] Longstanding severe stenosis over many years may cause irreversible deficit due to necrosis in gray and white matter (Class III).[38]

71.4 Differential diagnosis

71.4.1 General information

See Myelopathy (p. 1407) for other possible causes. Some of these (e.g. spinal cord tumor, OPLL) may be demonstrated radiographically. Asymptomatic cervical spondylosis is very common, and ≈ 12% of cases of cervical myelopathy attributed to spondylosis are later found to be due to another disease process including:
1. ALS: see below
2. multiple sclerosis (MS): spinal cord demyelination may mimic CSM. With MS, remissions and exacerbations are common, and patients tend to be younger
3. herniated cervical disc (soft disc): patients tend to be younger than with CSM. Course is more rapid
4. subacute combined system disease: abnormal vitamin B12 level and possibly macrocytic anemia (p. 1409)
5. hereditary spastic paraplegia: family history is key. Diagnosis of exclusion[39]
6. (spontaneous) intracranial hypotension (p. 178)

71.4.2 Amyotrophic lateral sclerosis (ALS)

AKA (anterior horn) motor neuron disease; also see Amyotrophic lateral sclerosis (p. 183). Can mimic the motor system syndrome of CSM (see above), and spinal cord compression may be seen on MRI in > 60% of patients with ALS.[40]

"Triad" of ALS:
1. atrophic weakness of hands and forearms (early) – LMN finding
2. mild LE spasticity – UMN finding
3. diffuse hyperreflexia – UMN finding

Table 71.2 Modified JOA score for cervical myelopathy[23a]

Score	Description	
Upper extremity (UE) motor dysfunction		
0	unable to feed self	
1	unable to use knife & fork; can eat with spoon	
2	can use knife & fork with much difficulty	
3	can use knife & fork with slight difficulty	
4	none (normal)	
Lower extremity (LE) motor dysfunction		
0	unable to walk	
1	can walk on flat surface with walking aid	
2	can walk up and/or down stairs with handrail	
3	lack of smooth and stable gait	
4	none (normal)	
Sensory deficit		
0	UE	severe sensory loss or pain
1		mild sensory loss
2		none (normal)
0	LE	severe sensory loss or pain
1		mild sensory loss
2		none (normal)
0	trunk	severe sensory loss or pain
1		mild sensory loss
2		none (normal)
Sphincter dysfunction		
0	unable to void	
1	marked voiding difficulty (retention)	
2	some voiding difficulty (urgency or hesitation)	
3	none (normal)	
[a]total score ranges from 0 to 17 (normal)		

Inevitably, some cases of demyelinating disease will be misdiagnosed initially as CSM until some features suggestive of ALS occur (in one series of 1500 ALS patients, 4%underwent spine surgery (56% cervical, 42%lumbar, 2%thoracic)[40] before ALS was correctly diagnosed).

Features that may help differentiate ALS from CSM:

1. ALS: sensory changes are conspicuously absent
2. ALS: bulbar symptoms (dysarthria, hyperactive jaw-jerk…)[41]

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3. ALS: extensive weakness/muscle atrophy of hands, usually with fasciculations[42]
4. ALS: lower-motor neuron (LMN) findings in the tongue (visible fasciculations, or positive sharp waves on EMG) or in the LEs (e.g. fasciculations and atrophy) favors the diagnosis of ALS over CSM (however, LMN findings in the LEs may occur in CSM if there is coincidental lumbar radiculopathy)
5. CSM or herniated cervical disc: usually includes neck and shoulder pain, limitation of neck movement, sensory changes, and LMN findings restricted to 1 or 2 spinal cord segments

71.5 Evaluation

71.5.1 Plain x-rays

General information

Minimum evaluation consists of AP, lateral (neutral position), and open-mouth odontoid views. If desired, flexion-extension views and/or oblique views may be obtained but requires specific orders.

When MRI is available, the additional information provided by plain cervical spine x-rays in patients with CSM is limited. In this setting, x-rays may be best for:
1. Demonstrating dynamic instability with flexion-extension views (see below)
2. Sagittal balance measured on standing lateral cervical spine x-rays may provide prognostic information[43]
3. X-rays may be able to compensate for the following MRI deficiencies, but cervical CT is much better
   a) Differentiating calcified discs or bone spurs from “soft discs”
   b) Differentiating OPLL form a thickened posterior longitudinal ligament
   c) Bone abnormalities: fractures, bony lytic lesions

Cervical spinal stenosis

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Cervical spinal stenosis can be inferred from plain x-rays. NB: Canal diameters measured on x-rays are actually surrogate markers for the entity of interest: viz. spinal canal narrowing sufficient to produce spinal cord compression and thereby spinal cord symptoms. This can be directly demonstrated on MRI or CT/myelo, and MRI can also detect intrinsic spinal cord signal abnormalities.

See Canal diameter (p.214) for normal dimensions and measurement techniques. Patients with CSM have an average minimal AP canal diameter of 11.8 mm,[44] and values ≤ 10 mm were likely to be associated with myelopathy.[45] Patients with an AP diameter < 14 mm may be at increased risk,[46] and CSM is rare in patients with a diameter > 16 mm, even with significant spurs.[17]

Cervical spinal stenosis is also suggested on plain films when the spinolaminar line is close to the posterior margin of the lateral masses.

Pavlov ratio (AKA Torg ratio[47,48]): the ratio of the AP diameter of the spinal canal at the mid VB level to the VB at the same location. A ratio < 0.8 is sensitive for transient neuropraxia, but has been shown to have poor positive predictive value for CSM.

Oblique views

Oblique views can delineate foraminal compromise caused by osteophytic spurs.

Flexion-extension views

Lateral flexion/extension x-rays may provide valuable information by detecting dynamic instability (abnormalities that manifest with movement) that cannot be appreciated on (static) CT or MRI, including widening of the atlanto-dental interval on flexion (p.213).

71.5.2 MRI

MRI provides information about the spinal canal, and can also show intrinsic cord abnormalities (demyelination, syringomyelia, spinal cord atrophy, edema…). MRI also rules out other diagnostic possibilities (Chiari malformation, spinal cord tumor…).

Bony structures and calcified ligaments are poorly imaged. These shortcoming and the difficulties in differentiating osteophytes from herniated discs on MRI are overcome with the addition of plain cervical spine films[49] or to better advantage with thin-section CT bone windows.

Findings that correlate with poor outcome (Class III)[50]:
1. multilevel T2WI hyperintensity within the spinal cord parenchyma

2. single level T2WI hyperintensity with corresponding T1WI hypointensity (single level T2WI hyperintensity without T1WI changes are of uncertain prognostic significance)
3. spinal cord atrophy (transverse area < 45 mm^2)

Other MRI findings seen with CSM:
1. reduced transverse area of the spinal cord (TASC) at the level of maximum compression. A "banana" shaped cord on axial images has a high correlation with the presence of CSM.[46] There is conflicting evidence whether the degree of canal stenosis predicts outcome.[50] Sagittal T2WIs tend to exaggerate the magnitude of spinal cord compression by osteophytes and/or discs, and therefore axial images and T1WIs also need to be considered in the evaluation. Narrowing is not specific for CSM: ≈ 26% of asymptomatic individuals > 64 years of age have spinal cord compression on MRI[51]
2. "snake eyes" (AKA "owl's eyes") within the spinal cord on axial T2WI (Fig. 71.1) may be related to cystic necrosis of the cord[52] and may correlate with poor outcome (Class III)[50]

71.5.3 CT and CT/myelogram

Plain CT scans may demonstrate a narrow canal, but do not provide adequate information regarding soft tissues (discs, ligaments, spinal cord and nerve roots). However, bony detail may prove invaluable in the surgical treatment of CSM.

Cervical myelography followed by high-resolution CT scanning provides sagittal and axial information (including spinal cord atrophy), and delineates bony detail better than MRI.[49] Unlike MRI, CT/myelogram is invasive (requires LP) and involves ionizing radiation and does not provide information about changes within the spinal cord parenchyma.

71.5.4 EMG

Not routinely useful in CSM. EMG has poor sensitivity in cervical radiculopathy and is not reliable in predicting outcome from surgery for CSM (Class III).[50] EMG is most helpful in suspicious cases to eliminate etiologies such as peripheral neuropathy or ALS.

71.5.5 Sensory evoked potentials (SEPs)

SSEPs are of limited usefulness, although a normal pre-op SEP or normalization of SEPs in the early post-op period are associated with better outcome.[53]

Practice guideline: Pre-op SEPs in CSM

Pre-op SEPs should be considered if the additional prognostic information would help treatment decisions (Level B Class II)[53]

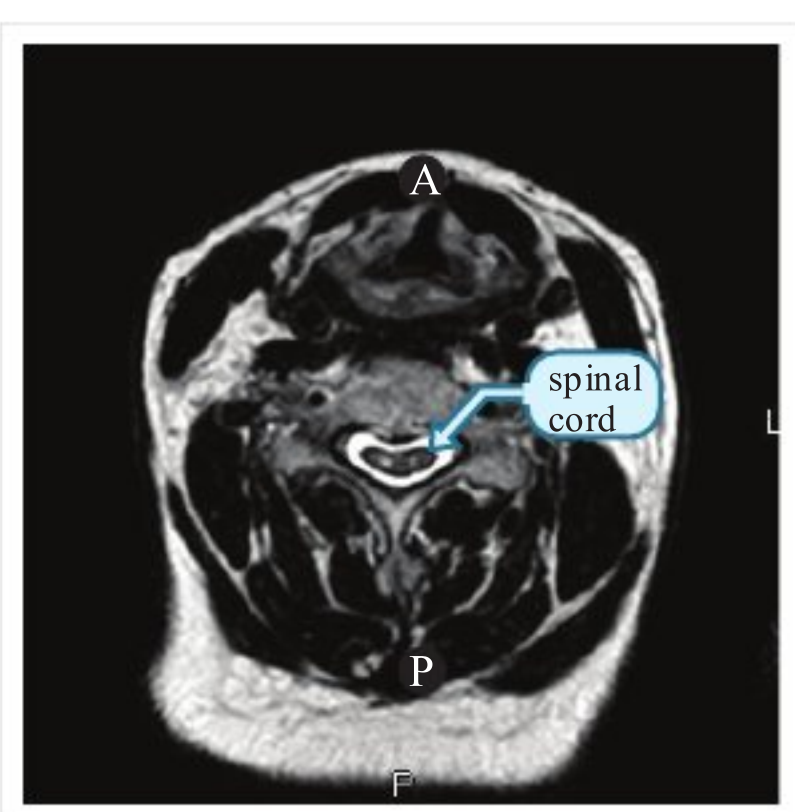

Fig. 71.1 Snake eyes (two foci of high signal) within a slightly flattened and mildly atrophic spinal cord on axial T2WI MRI

71.6 Treatment

71.6.1 Nonoperative management

Measures include: prolonged immobilization with rigid cervical bracing in an attempt to reduce motion and hence the cumulative effects of trauma on the spinal cord, modified activity to eliminate "high-risk" activities or bed rest, and anti-inflammatory medications.[54]

71.6.2 Surgical treatment

Indications for surgery

See Practice guideline: Surgical vs. nonsurgical management (p.1090).

Practice guideline: Surgical vs. nonsurgical management

Mild myelopathy (mJOA score > 12): in the short-term (3 years) patients may be offered the option of surgical decompression or nonoperative management (prolonged immobilization in a rigid cervical collar, anti-inflammatory medications, and "low-risk" activities or bed rest (Level C Class II)).[55] Note: patients with mJOA scores > 12 (see Table 71.2) may not always be mildly impaired, they may derive significant improvement from surgery, and deterioration from this point may be ominous.

More severe myelopathy: should be treated with surgical decompression with benefits maintained at 5 and 15 years post-op (Level D Class III)[55]

Level B Class I[56] Degenerative cervical radiculopathy: patients do better with anterior decompression ± fusion (compared to conservative management) for

- rapid relief (within 3–4 months) of arm & neck pain and sensory loss
- relief of longer term (≥ 12 months) symptoms of weakness of wrist extension, elbow extension, shoulder abduction and internal rotation

Intraoperative electrophysiologic monitoring

Practice guideline: Intra-operative electrophysiologic monitoring during surgery for CSM or radiculopathy

Use of intra-op EP monitoring during routine surgery for CSM or cervical radiculopathy is not recommended as an indication to alter the surgical plan or administer steroids since this paradigm has not been observed to reduce the incidence of neurologic injury (Level D Class III)[57]

Choice of approach

General information

The debate between anterior approaches (anterior cervical discectomy or corpectomy) and posterior approaches (decompressive cervical laminectomy or laminoplasty) dates back to the time that both became widely practiced.[16] General sentiment is to treat anterior disease at the disc level (e.g. osteophytic bar, herniated disc...) usually limited to ≤ 3 levels (or occasionally 4) with an anterior approach, and to use a posterior approach as the initial procedure in the situations outlined below. Considerations of spinal curvature may need to enter into the decision process.

Practice guideline: Choice of surgical approach for CSM

There was not enough evidence to recommend any of the following techniques over the other (in terms of short-term success in treating CSM): ACDF, anterior corpectomy and fusion, laminectomy (with or without fusion) and laminoplasty (Level D Class III)[58]

Laminectomy without fusion, however, is associated with a higher incidence of late kyphotic deformity (Level D Class III; incidence 14–47%, not all cases are symptomatic, not all cases need treatment: see text)[58]

Posterior approach

Options include:

1. laminectomy alone laminectomy/arthrodesis (i.e. laminectomy + lateral mass fusion): Class III (this procedure was found to be effective, the class shows the strength of the evidence)[59])
2. laminoplasty (Class III; this procedure was found to be effective, the class shows the strength of the evidence[60]): methods include unilateral (“open door”) and midline enlargement (“French door”)
3. multilevel foraminotomies: usually not adequate for central canal stenosis

Situations where a posterior approach would generally be the initial approach:

1. congenital cervical stenosis where removing osteophytes will still not provide at least ≈ 12 mm of AP canal diameter
2. disease over ≥ 3 levels (although up to 4 may occasionally be dealt with anteriorly)
3. primary posterior pathology (e.g. infolding of ligamentum flavum)
4. some cases of OPLL (anterior approach has higher risk of dural tear)

Disadvantages of the posterior approach:

1. laminectomy without fusion
   a) degeneration and osteophytes continue to progress following surgery
   b) risk of subsequent subluxation or progressive kyphotic angulation (“swan neck” deformity) (facetiously dubbed “spina bifida neurosurgica”)
      - quoted incidence: 14–47%[61,62,63] (risk may be minimized by careful preservation of facet joints)
      - not all cases need to be treated: in one series, 31% (18/58) developed post-op kyphosis, and 16% of these (3/18) required surgical stabilization[64]
      - the development of kyphotic deformity does not appear to diminish the clinical outcome[63] and does not correlate with neurologic deterioration when deterioration occurs[65]
2. more painful initially post-op and sometimes more prolonged rehabilitation
3. long-term complaints of a heaviness of the head possibly associated with atrophy of the paraspinal muscles
4. contraindicated with pre-existing swan neck deformity, and not recommended in the presence of reversal of the normal cervical lordosis (i.e. kyphotic curve)[66] where the spinal cord won’t tend to move away from the anterior compression or in the presence of ≥ 3.5 mm subluxation or > 20° rotation in the sagittal plane[46] and caution must be exercised in hyperlordosis (see below)

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Anterior approach

Also shown to be effective (Class III[55]).

Instrumentation options: in terms of fusion rates for 2-level anterior operations (i.e. 2 disc spaces) (Class III)[58]:

2-level ACDF with anterior plate ¼ 1-level corpectomy with plate > 1-level corpectomy without plate* > 2-level ACDF without plate

* however; the graft extrusion rate is higher for corpectomy than ACDF

Worsening of myelopathy has been reported in 2–5% of patients after anterior decompression[67,68] (intraoperative SSEP monitoring may reduce this rate[68]) and C5 radiculopathy may occur (see below).

Anterior cervical plating

Many systems are available, with more similarities than differences. All include some method of preventing screw back out. Some general pointers:

1. for single level fusion, typical plate length is 22–24 mm
2. screw length: rule of thumb is 12 mm for females, 14 mm for males
3. do not completely tighten a single screw (to avoid kicking up plate) until the diagonally opposite screws are placed and loosely tightened
4. most systems have fixed and variable angle screws. Variable angle screws allow for load sharing with the graft (here is where a derivative of Wolff’s law is often invoked: the weight sharing helps stimulate fusion). Avoid over-angling screws which may prevent the locking mechanism from properly engaging

5. optimal plate placement allows for contact of the plate with the VB at the screw locations. This may require
   a) contouring of plate to follow the lordosis of the c-spine
   b) reduction of anterior osteophytes

Posterior approach

For decompression, some recommend cervical laminectomy extending one or two levels beyond the stenosis above and below.[69,70] A C3–6 laminectomy is often considered a "standard" laminectomy. An "extended laminectomy" includes C7 and/or C2.

Curvature considerations: extending the laminectomy to include C2 and sometimes C1 has been recommended for patients with straightening of the cervical curvature.[6] In cases of hyperlordosis, posterior migration of the spinal cord following an extensive laminectomy may put increased tension on the nerve roots and blood vessels (with possible neurologic worsening), and a limited laminectomy just where the cord is compressed is often recommended (below).

"Keyhole foraminotomies" or medial facetectomy with undercutting of the facets may be performed at levels involved with radiculopathy.

Position: choices are primarily: prone, lateral oblique, or sitting. The prone position has a major disadvantage of difficulty elevating the head above the heart, resulting in venous engorgement with significant operative bleeding. The sitting position has a number of inherent risks (p.1445), including cord hypoperfusion[68] and air embolism. The lateral oblique position may introduce some distortion to the anatomy due to asymmetrical positioning.

The reported rate of post-op spinal deformity is 25–42%. Neurologic worsening has been reported in 2% in some series, higher in others. C5 radiculopathy may occur (see below).

To avoid significant destabilization of the cervical spine:

1. during the dissection, do not remove soft tissue overlying the facet joints (to preserve their blood supply)
2. take the laminectomy only as far lateral as the extent of the spinal canal, carefully preserving the facet joints[7] (use keyhole laminotomies where necessary)
3. avoid removing a total of one facet at any given level

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Outcome

General information

Even excluding cases that are later proven to have demyelinating disease, the outcome from surgery for CSM is often disappointing. Once CSM is clinically apparent, complete remission almost never occurs. The prognosis with surgery is worse with increasing severity of involvement at the time of presentation[69] and with longer duration of symptoms (48% showed clinical improvement or cure if operated within 1 yr of onset, whereas only 16% responded after 1 yr[7]). The success of surgery is also lower in patients with other degenerative diseases of the CNS (ALS, MS…).

Progression of myelopathy may be arrested by surgical decompression. This is not always borne out, and some early series[34,37] showed similar results with conservative treatment as with laminectomy which yielded improvement in 56%, no change in 25%, and worsening in 19%. Also as discussed earlier some cases of CSM develop most of the deficit early and then stabilize (p.1086).

Some series show good results, with ≈ 64–75% patients having improvement in CSM post-op.[22] However, other authors remain less enthusiastic. Utilizing a questionnaire in 32 post-op patients operated anteriorly, 66% had relief from radicular pain, while only 33% had improvement in sensory or motor complaints.[22] In one series, half the patients had improvement in fine motor function of the hands, but the other half worsened postoperatively.[71] Spinal cord atrophy as a result of continued pressure or ischemia may be partly responsible for poor recovery. Bedridden patients with severe myelopathy rarely recover useful function.

Post-op C5 palsy

Criteria: weakness of deltoid and/or biceps with no worsening of myelopathy. Follows ≈ 3–5% of extensive anterior or posterior decompression (including laminoplasty).[67,72] 50% have motor involvement only (deltoid > biceps), 50% also have C5 dermatomal sensory loss and/or C5 dermatomal pain (shoulder). Most occur < 1 week post-op.[72] 92% are unilateral.[72] No pre-op risk factors have been identified.[73] Etiology: unproven, may be related to traction on the nerve root from posterior migration of the cord after decompression or to bone graft displacement. Prognosis for spontaneous recovery is generally good; more severe deficits take longer to recover.[72]

Late developments

Some patients who show early improvement will develop late deterioration (7–12 yrs after reaching a plateau),[46] with no radiographically apparent explanation in up to 20%of these cases.[74] In others, degeneration at levels adjacent to the operated segments may be demonstrated.

Adjacent segment disease (ASD): degeneration that develops at a motion segment adjacent to a previous fusion. Findings include: disc degeneration, stenosis, facet hypertrophy, scoliosis, listhesis and instability. After ACDF, ASD occurred at a rate of 2.9%per year over 10 years observation.[75] Estimate: 25%of patients will develop symptomatic adjacent level changes within 10 years of surgery.[75] This rate was higher with single level fusion at C5–6 or C6–7 than it was with multilevel fusion, and natural progression of the disease was felt to be a significant contributor[75] (i.e. it was not all attributable to the fusion). Most cases of ASD observed radiographically are asymptomatic.

71.7 Coincident cervical and lumbar spinal stenosis

In 5%, lumbar and cervical stenoses are symptomatic simultaneously.[76]

Coincident symptomatic lumbar and cervical spinal stenosis is usually managed by first decompressing the cervical region, and later operating on the lumbar region (unless severe neurogenic claudication dominates the picture). It is also possible, in selected cases, to operate on both in a single sitting.[76,77]

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72 Thoracic and Lumbar Degenerative Disc Disease

72.1 General information about degenerative disc disease (DDD)

Since structures outside of the disc are usually also involved, the term degenerative spine disease (DSD) may be preferable to degenerative disc disease. Spondylosis is a non-specific term which may include degenerative spine disease. “Cervical spondylosis” is occasionally used synonymously with cervical stenosis (p. 1083).

Symptomatic spinal stenosis in the thoracic region is rare,[1] and is generally seen in the setting of a calcified disc. The majority of this chapter deals with lumbar DSD. For the cervical region, see that chapter (p. 1083).

72.2 Anatomic substrate

72.2.1 General information

DSD is a progressive deterioration of the structures of the spine including:

1. disc abnormalities:
   a) the proteoglycan content of the disc nucleus decreases with age
   b) disc desiccation (loss of hydration) occurs
   c) tears develop in the disc annulus and progress to internal disruption of the lamellar architecture. Herniation of the nucleus may occur from increased nuclear pressure under mechanical loads
   d) mucoid degeneration and ingrowth of fibrous tissue ensues (disc fibrosis)
   e) subsequently disc resorption occurs
   f) there is a loss of disc space height and increased susceptibility to injury
2. facet joint abnormalities: hypertrophy and capsular laxity
3. osteophytes often form on the edges of the VB bordering the degenerated disc
4. spondylolisthesis: subluxation of one VB on another (see Spondylolisthesis below)
5. hypertrophy of the ligamentum flavum

Neurologic involvement in lumbar spinal stenosis may occur in one or more of the following 3 sites:

1. central canal stenosis: narrowing of the AP dimension of the spinal canal below a critical value. The reduction in canal size may cause local neural compression and/or compromise of the blood supply to the spinal cord (cervical) or the cauda equina (lumbar)
   a) Congenital (as in the achondroplastic dwarf)
   b) Acquired: as with hypertrophy of facets and ligamentum flavum
   c) Most commonly – acquired superimposed on congenital narrowing.
2. foraminal stenosis: narrowing of the neural foramen. May be the result of any combination of: foraminal disc protrusion, spondylolisthesis, facet hypertrophy, disc space collapse, hypertrophy of uncovertebral joints (cervical), synovial cyst
3. lateral recess stenosis; lumbar spine only (p. 1097)

72.2.2 Lumbar spinal stenosis

Key concepts

- caused by hypertrophy of facets and ligamentum flavum, may be exacerbated by disc bulging or spondylolisthesis, may be superimposed on congenital narrowing
- most common at L4–5 and then at L3–4
- symptomatic stenosis produces gradually progressive back and leg pain with standing and walking that is relieved by sitting or lying (neurogenic claudication)
- symptoms differentiated from vascular claudication which is usually relieved at rest regardless of position
- usually responds to decompressive surgery (sometimes with fusion) or interspinous spacer

Symptomatic lumbar stenosis is most common at L4–5, then L3–4, L2–3 and lastly L5-S1.[2] It is rare at L1–2. Generally occurs in patients with congenitally shallow lumbar canal – see Normal LS spine measurements (p.1102) – with superimposed acquired degeneration in the form of some combination of facet hypertrophy, hypertrophy of the ligamentum flavum, protruding (and often calcified) intervertebral discs, and spondylolisthesis. First recognized as a distinct clinical entity producing characteristic symptoms in the 1950's and 60's.[3,4]

May be classified as[5]:

1. stable form of lumbar spinal stenosis: hypertrophy of facets and ligamentum flavum accompanied by disc degeneration and collapse
2. unstable: have the above with superimposed
   a) degenerative spondylolisthesis (p.1098), the unisegmental form
   b) degenerative scoliosis: the multisegmental form

72.2.3 Central canal stenosis

The central canal may be narrowed by a combination of any of the following:
- Hypertrophy of the ligamentum flavum
- Hypertrophy of the facet joints
- Congenitally shot pedicles
- Bulging intervertebral discs
- Osteophytes arising from the posterior VB
- Juxtafacet cysts
- Spondylolisthesis

72.2.4 Lateral recess syndrome

The lateral recess is the "gutter" alongside the pedicle which the nerve root enters just proximal to its exit through the neural foramen (Fig. 72.1). It is bordered anteriorly by the vertebral body, laterally by the pedicle, and posteriorly by the superior articular facet of the inferior vertebral body. Hypertrophy of this superior articular facet compresses the nerve root. Narrowing of the lateral recess is present in essentially all cases of central canal stenosis, but it can be symptomatic by itself.[6] L4–5 is the most commonly involved facet.

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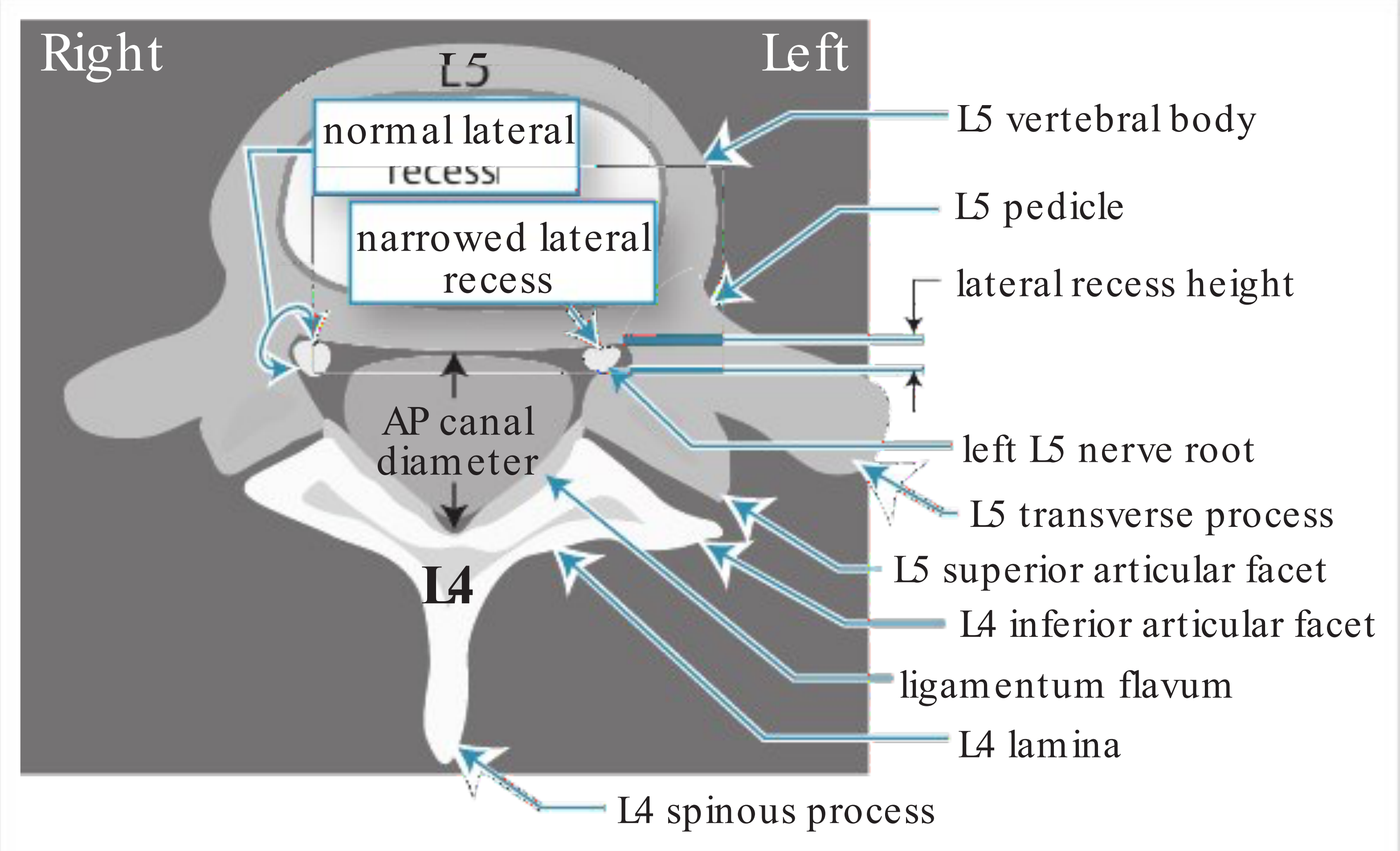

Fig. 72.1 Schematic axial CT through the L4-5 facet joint showing the lateral recesses (normal on patient's right, stenotic on left)

72.2.5 Spondylolisthesis

General information

Anterior subluxation of one vertebral body (VB) on another, most commonly the upper VB is anterior to the inferior one. Usually L5 on S1, the next most common is L4 on L5.

Disc herniation and nerve root compression with spondylolisthesis: It is rare for a herniated lumbar disc to occur at the level of the listhesis, however the disc may "roll" out as it is uncovered and produce findings on MRI that may resemble a herniated disc which has been termed a "pseudodisc." It is more common to see a herniated disc at the level above the listhesis.

If the listhesis does cause nerve root compression, it tends to involve the nerve exiting below the pedicle of the anteriorly subluxed upper vertebra (e.g. if an L4–5 spondylolisthesis causes nerve root compression, it will generally involve the L4 root). The compression is usually due to upward displacement of the superior articular facet of the level below together with disc material, and symptoms typically resemble neurogenic claudication, although true radiculopathy may sometimes occur. There also may be a contribution from a fibrous/inflammatory mass from the nonunion.

Isthmic spondylolisthesis rarely produces central canal stenosis since only the anterior part of the vertebral body shifts forward. May present with radiculopathy or neurogenic claudication from compression in the neural foramen, with the nerve exiting under the pedicle at that level being the most vulnerable. May also present with low back pain. Many cases are asymptomatic.

Adolescent spondylolisthesis

In adolescents and teens, spondylolisthesis usually occurs in athletes subject to repetitive hyperextension of the lumbar spine. In girls, this is frequently encountered in gymnasts and softball pitching. In boys, football is common.

In these youngsters, cessation of sports for several months usually produces resolution.

Surgery is sometimes performed for patients who are unwilling to discontinue athletics.

Grading spondylolisthesis

The Meyerding[7,8] grading of subluxation in the sagittal plane is shown in Table 72.1.

Types of spondylolisthesis

1. Type 1: dysplastic: congenital. Upper sacrum or arch of L5 permits the spondylolisthesis. No pars defect. 94%are associated with spinal bifida occulta. Some of these may progress (no way to accurately identify those that will progress)
2. Type 2: isthmic spondylolisthesis AKA spondylolysis: a failure of the neural arch as a result of a defect in the pars interarticularis (identifiable as a discontinuity in the neck of the "Scotty dog" on oblique LS-spine x-ray). May be seen in 5–20%of spine x-rays.[9] Rarely produces central canal stenosis since only the anterior part of the vertebral body shifts forward. May cause narrowing of the neural foramen. Three subtypes:
   a) lytic: fatigue fracture or insufficiency fracture of pars. In the pediatric age group may occur in athletes (especially gymnasts or football players); in some this may be an exacerbation of a pre-existing defect, in others it may be a result of repetitive trauma
   b) elongated but intact pars: possibly due to repetitive fractures and healing
   c) acute fracture of pars

Table 72.1 Spondylolisthesis grading

Grade	% subluxation[a]
I	<25%
II	25–50%
III	50–75%
IV	75%-complete
spondyloptosis	>100%

[a]% of the AP diameter of the VB

3. Type 3: degenerative: due to long-standing intersegmental instability. Usually at L4–5. No break in the pars. Found in 5.8% of men and 9.1% of women (many of whom are asymptomatic)[9]
4. Type 4: traumatic: due to fractures usually in areas other than the pars
5. Type 5: pathologic: generalized or local bone disease, e.g. osteogenesis imperfecta

Natural history

Progression of spondylolisthesis may occur without surgical intervention, but is more common following surgery.[10]

72.2.6 Degenerative scoliosis

One of the main differentiating features from juvenile scoliosis is that in degenerative scoliosis, the disc spaces are asymmetrically narrowed in the coronal plane and the vertebral bodies tend to maintain a more normal configuration.

72.3 Risk factors

1. The risk of developing DSD is multifactorial and includes:
2. The most powerful determinant in developing DSD in a study of twins was genetic influence, and possibly other unidentified factors.[11] Environmental factors studied (including sedentary vs. active lifestyles, occupation, cigarette smoking…) exerted only a modest influence, which may explain why conflicting findings for these have been reported
3. cumulative effects of microtrauma and macrotrauma to the spine
4. osteoporosis
5. cigarette smoking: several epidemiologic studies have shown that the incidence of back pain, sciatica and spinal degenerative disease is higher among cigarette smokers than among non-smokers[12,13]
6. in the lumbar spine:
   a) stresses on the spine including effects of excess body weight
   b) loss of muscle tone (primarily abdominals and paraspinals) resulting in increased dependence on the bony spine for structural support

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72.4 Associated conditions

1. congenital:
   a) achondroplasia
   b) congenitally narrowed canal
2. acquired:
   a) spondylolisthesis
   b) acromegaly
   c) post-traumatic
   d) Paget’s disease (p.1120)
   e) ankylosing spondylitis (p.1123)
   f) ossification of the ligamentum flavum: more common in East Asians, rare in Caucasians.[14] Often, but not always, associated with OPLL[15]

72.5 Clinical presentation

72.5.1 General information

1. the degenerative abnormalities may produce spinal stenosis which can lead to neural compromise producing the following symptoms
   a) radicular symptoms (more common in cervical spine than lumbar)
   b) neurogenic claudication (lumbar) or spinal myelopathy (cervical)
2. Sagittal imbalance and scoliosis as a result of degenerative changes can place focal stress on specific spine structures which may be painful. Also, muscles used to compensate for the imbalance can cause pain from overuse fatigue
3. discogenic pain (controversial) may be less prevalent in the late stages of DSD. May contribute to “musculoskeletal low back pain” but the actual pain generators here are not definitively identified

4. Many cases of DSD (including spinal stenosis and spondylolisthesis) are asymptomatic, and the degenerative changes are discovered incidentally

72.5.2 Neurogenic claudication

Lumbar spinal stenosis often presents as neurogenic claudication (NC), (claudicate: from Latin, claudico, to limp) AKA pseudoclaudication. To be differentiated from vascular claudication (AKA intermittent claudication) which results from ischemia of exercising muscles (see Table 72.2 for distinguishing characteristics).

NC characteristics: unilateral or bilateral buttock, hip, thigh or leg discomfort that is precipitated by standing or walking and characteristically relieved by a change in posture (usually sitting with the waist flexed, squatting, or lying in the fetal position). Painful burning paresthesias of the lower extremities are also described. Valsalva maneuvers usually do not exacerbate the pain. Many patients report increased pain first thing in the morning that improves once they have been out of bed for varying periods (usually an hour more or less).

The time course is usually gradually progressive over many months to years. As the condition progresses, the ability to get relief from position changes tends to decrease. However, presenting with acute, unrelenting pain is not characteristic and other causes should be sought.

In comparison, a HLD usually causes increased pain on sitting, has a more abrupt onset, has pain on straight leg raising, and is worsened by Valsalva maneuvers.

NC is thought to arise from ischemia of lumbosacral nerve roots, as a result of increased metabolic demand from exercise together with vascular compromise of the nerve root due to pressure from surrounding structures. NC is only moderately sensitive (≈ 60%) but is highly specific for spinal stenosis.[17] Pain may not be the major complaint, instead, some patients may develop paresthesias or LE weakness with walking. Some may complain of muscle cramping, especially in the calves.

Relief from symptoms: occurs with positions that decrease the lumbar lordosis which increases the diameter of the central canal (by reducing inward buckling of the ligamentum flavum) and distracts the facet joints (which enlarges the neural foramina. Favored positions include sitting, squatting and recumbency. Patients may develop “anthropoid posture” (exaggerated waist flexion). “Shopping cart sign” patients often can walk farther if they can lean forward e.g. as on a grocery cart. Riding a bicycle is also often well tolerated.

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Table 72.2 Clinical features distinguishing neurogenic from vascular claudication[16]

Feature	Neurogenic claudication	Vascular claudication
distribution of pain	in distribution of nerve (dermatomal)	in distribution of muscle group with common vascular supply (sclerotomal)
sensory loss	dermatomal distribution	stocking distribution
inciting factors	variable amounts of exercise, also with prolonged maintenance of a given posture (65% have pain on standing at rest); coughing produces pain in 38%	reliably reproduced with fixed amount of exercise (e.g. distance ambulated) that decreases as disease progresses; rare at rest (27% have pain on standing at rest)
relief with rest	slow (often > 30 min), variable, usually positional (stooped posture or sitting often required, standing and resting is usually not sufficient)	almost immediate; not dependent on posture (relief of walking induced symptoms with standing is a key differentiating feature)
claudicating distance	variable day-to-day in 62%	constant day-to-day in 88%
discomfort on lifting or bending	common (67%)	infrequent (15%)
foot pallor on elevation	none	marked
peripheral pulses	normal; or if ↓ usually reduced only unilaterally	↓ or absent; femoral bruits are common
skin temp of feet	normal	decreased

72.5.3 Neurologic exam

The neurologic exam is normal in ≈ 18% of cases (including normal muscle stretch reflexes and negative straight leg raising). Weakness in the anterior tibialis and/or extensor hallicus longus may occur in some cases of central canal stenosis at L4–5, or with foraminal stenosis of L5-S1. Absent or reduced ankle jerks and diminished knee jerks are common,[17] however this is also prevalent in the aged population. Pain may be reproduced by lumbar extension.

72.6 Differential diagnosis

72.6.1 General considerations

1. vascular insufficiency: (AKA vascular or intermittent claudication) see above
2. hip disease: trochanteric bursitis (see below), degenerative joint disease
3. disc herniation (lumbar or thoracic)
4. facet joint pain (controversial): may respond to medial branch block (therapeutic & diagnostic)
5. Baastrup's syndrome[18]: AKA arthrosis interspinosa. Radiographically: contact of adjacent spinous processes ("kissing spines") with enlargement, flattening and reactive sclerosis of apposing interspinous surfaces. Produces localized midline lumbar pain & tenderness on back extension relieved by flexion, local anesthetic injection or partial excision of the involved spinous processes
6. juxtafacet cyst: (p.1143)
7. arachnoiditis
8. intraspinal tumor
9. Type I spinal AVM (spinal dural AVM) (p.1140)
10. diabetic neuritis: with this, the sole of the foot is usually very tender to pressure from the examiner's thumb
11. delayed onset muscle soreness (DOMS): onset usually 12–48 hours after beginning a new activity or changing activities (NC occurs during the activity). Symptoms typically peak within 2 days and subside over several days
12. inguinal hernia: typically produces groin pain
13. functional etiologies

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Degenerative hip disease

Trochanteric bursitis (TBS) and degenerative arthritis of the hip are also included in the differential diagnosis of NC.[19,20] Although TBS may be primary, it can also be secondary to other conditions including lumbar stenosis, degenerative arthritis of the lumbar spine or knee, and leg length discrepancy. TBS produces intermittent aching pain over the lateral aspect of the hip. Although usually chronic, it occasionally may have acute or subacute onset. Pain radiates to lateral aspect of thigh in 20–40% (so called "pseudoradiculopathy"), but rarely extends to the posterior thigh or as far distally as the knee. There may be numbness and paresthesia-like symptoms in the upper thigh which are usually not dermatomal in distribution. Like NC, the pain may be triggered by prolonged standing, walking and climbing, but unlike NC it is also painful to lie on the affected side. Localized tenderness over the greater trochanter can be elicited in virtually all patients, with maximal tenderness at the junction of the upper thigh and greater trochanter. Pain increases with weight bearing (and is often present from the very first step, unlike NC) and with certain hip movements, especially external rotation (over half the patients have a positive Patrick-FABERE test (p.1048), and rarely with hip flexion/extension. Treatment includes NSAIDs, local injection of glucocorticoid (usually with local anesthetic), physical therapy (with stretching and muscle strengthening exercises) and local application of ice. No controlled studies have compared these modalities.

72.7 Diagnostic evaluation

72.7.1 Radiographic evaluation

Comparison of modalities

MRI: demonstrates impingement on neural structures and loss of CSF signal on T2WI due to central canal stenosis, lateral recess stenosis, foraminal stenosis as well as juxtafacet cysts. Increased fluid in the facet joint and vacuum disc MRI is poor for visualizing bone which contributes significantly to the pathology. Asymptomatic abnormalities are demonstrated in up to 33% of patients 50–70 years old without back-related symptoms.[2]

Lumbosacral spine x-rays: may disclose spondylolisthesis. AP diameter of canal is usually narrowed (congenitally or acquired) (below) whereas the interpediculate distance (IPD) may be normal.[16] Oblique films may demonstrate pars defects. Adding flexion/extension views can assess "dynamic" instability.

Standing scoliosis x-rays: provide information about scoliosis and sagittal balance. See Adult degenerative scoliosis for technique and measurements.

CT scan (either routine, or following water-soluble myelography): classically shows "trefoil" canal (cloverleaf shaped, with 3 leaflets). CT also demonstrates AP canal diameter, hypertrophied ligaments, facet arthropathy, pars fractures and occasionally may show bulging annulus or herniated disc

Myelogram: lateral films often show "washboard pattern" (multiple anterior defects), AP films often show "wasp-waisting" (narrowing of dye column), may also show partial or complete (especially in prone position) block. May be difficult to perform LP if stenosis is severe (poor CSF flow and difficulty avoiding nerve roots the LP needle).

Normal LS spine measurements

Normal dimensions of the lumbar spine are shown in Table 72.3 for plain film and Table 72.4 for CT.

72.7.2 Adjuncts to radiographic evaluation

"Bicycle test": patients with NC can usually tolerate longer periods of exercise on a bicycle than patients with intermittent (vascular) claudication because the position in bicycling flexes the waist.

Noninvasive studies to rule-out vascular insufficiency: Ratio of ankle to brachial blood pressure (A:B ratio): > 1.0 is normal; mean of 0.59 in patients with intermittent claudication; 0.26 in patients with rest pain; < 0.05 indicates impending gangrene.

EMG with NCV may show multiple nerve-root abnormalities bilaterally.

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Table 72.3 Normal AP diameter of lumbar spinal canal on lateral plain film (from spinolaminar line to posterior vertebral body)[21]

average (normal)	22–25 mm
lower limits of normal	15 mm
severe lumbar stenosis	< 11 mm

Table 72.4 Normal lumbar spine measurements on CT[22]

AP diameter	≥ 11.5 mm
interpediculate distance (IPD)	≥ 16 mm
canal cross-sectional area	≥ 1.45 cm^2
ligamentum flavum thickness[23]	≤ 4–5 mm
height of lateral recess (see below)	≥ 3 mm

Table 72.5 Dimensions of lateral recess on CT (bone windows)

Lateral recess height	Degree of lateral recess stenosis
3–4 mm	borderline (symptomatic if other lesion co-exists, e.g. disc bulging)
< 3 mm	suggestive of lateral recess syndrome
< 2 mm	diagnostic of lateral recess syndrome

72.8 Treatment

72.8.1 General information

In one study of 27 unoperated patients, 19 remain unchanged, 4 improved, and 4 worsened (mean follow-up: 49 months; range: 10–103 months).[24]

NSAIDs (acetaminophen may be as effective) and physical therapy are usually the initial measures of nonsurgical management. Unlike the cervical spine, traction is not usually helpful.

Brace therapy, e.g. with an LSO may be attempted. Guidelines are shown below.

Practice guideline: Brace therapy

Level II[25]:
- short-term use (1–3 weeks) of a rigid lumbar support is recommended for treatment of LBP of relatively short duration (<6 months)
- bracing in patients with LBP > 6 months duration is not recommended because it has not been shown to have long-term benefit

Level III[25]:
- lumbar braces may reduce the number of sick days due to LBP among workers with a previous lumbar injury. Braces are not recommended for LBP in the general working population
- the use of pre-op bracing or transpedicular external fixation as tools to predict outcome for lumbar fusion is not recommended

Interventional pain management s an option for persistent pain. Epidural steroids may provide temporary relief (usually days to weeks at most). Facet blocks and, if helpful, rhizotomies (for longer relief), may be employed.

72.8.2 Management of isthmic spondylolisthesis

See reference.[9]

Some special management considerations for isthmic spondylolisthesis as a subset of spinal stenosis.

1. lesions with sclerotic borders are usually well established with little chance of healing.
2. Surgery is reserved for patients with neurologic deficit or incapacitating symptoms or progression of spondylolisthesis
3. lesions without sclerosis that show increased uptake on bone scan (indicating active lesion with potential for healing) or MRI high signal changes on T2WI[26] or STIR may heal in a rigid orthosis such as the Boston brace for ≥ 3 months
4. management of symptoms:
   a) LBP only: treat with NSAIDs, PT
   b) LBP with myelopathy, radiculopathy, or neurogenic claudication: surgical treatment[27] (see Table 72.6 for surgical options)

Table 72.6 Surgical recommendations for spondylolisthesis

Nature of spondylolisthesis	Nature of problem	Type of procedure needed
degenerative	nerve root compression within confines of spinal canal	decompression (preserving facets)
	spinal stenosis at the level of spondylolisthesis	decompression; some advocate with intertransverse-process fusion[29]
	nerve root compression far lateral, outside confines of spinal canal	radical decompression (Gill procedure, see below) plus fusion
traumatic	(does not matter)	decompression plus fusion

5. in pediatrics: may be managed with TLSO and long course of PT (e.g. 6–9 months) for symptoms. Resumption of sports may be considered when symptoms subside, but recurrence should prompt elimination of athletics or consideration of surgery

72.8.3 Indications for surgery

Surgical intervention is undertaken when symptoms become severe in spite of conservative management. The goals of surgery are pain relief, halting progression of symptoms, and possibly reversal of some existing neurologic deficit. Most authors do not consider surgery unless the symptoms have been present > 3 months, and most patients who have surgery for this have symptoms of > 1 year duration.

72.8.4 Surgery

Surgical options

1. laminectomy: posterior (direct) decompression of central canal and neural foramina without or with fusion. Fusion options:
   a) posterolateral fusion ± pedicle screw/rod fixation
   b) interbody fusion: generally not done as a “stand-alone” (i.e. usually requires additional stabilization, options here include: pedicle screws, facet screws, facet dowels, spinous process clamp…)
      • posterior lumbar interbody fusion (PLIF) (p.1497): usually bilateral graft placement
      • transforaminal lumbar interbody fusion (TLIF) (p.1497): unilateral graft placement though a facet take-down on that side
2. procedures to increase disc space height and thereby indirectly decompress neural foramina without direct decompression
   a) anterior lumbar interbody fusion (ALIF) (p.1493): through laparotomy
   b) lateral lumbar interbody fusion (p.1498): some techniques trademarked as extreme lateral interbody fusion (XLIF™) or direct-lateral (DLIF™)
   c) axial lumbar interbody fusion (Ax-LIF): L5-S1 only
3. limitation of extension by interspinous spacer: e.g. X-Stop® (see below)

Choosing which procedure to use

Items that factor into consideration when choosing which procedure to use include:

1. consider indirect decompression (lateral interbody fusion (e.g. XLIF® or DLIF®), ALIF, interspinous decompression (e.g. X-Stop):
   a) when foraminal stenosis appears to be the dominant problem (e.g. with loss of disc space height, facet hypertrophy, on the concave side of a scoliotic curve)
   b) previous spine surgery that might make exposure of the nerves more difficult or risky
   c) when the disc space is compressed (more difficult to distract an already tall disc space with normal disc)
2. consider direct decompression (e.g. laminectomy)
   a) “pinpoint” central canal stenosis especially when disc height and neural foramina are well preserved
   b) where a significant contributor to the compression is a focal, correctable lesion, e.g. herniated disc, synovial cyst, intraspinal tumor
   c) to avoid a fusion (in select cases)
3. consider motion-preservation surgery when a fusion is undertaken at a level and the adjacent level is already starting to show some degenerative changes that have not yet reached a surgical magnitude. Motion preservation at this adjacent segment theoretically shields it from some of the transmitted stresses from the fused level
4. situations where a fusion should be considered in addition to direct or indirect decompression of the nerves:
   a) spondylolisthesis (especially > Grade I)
   b) symptomatic sagittal imbalance or degenerative scoliosis
   c) dynamic instability on flexion/extension lateral lumbar spine x-rays
   d) expectation that the decompression will destabilize the spine (e.g. facet takedown for a TLIF)
   e) multiply recurrent herniated disc (when this is the third or more operation for the same disc)
   f) controversial: “black disc” on MRI with positive concordant discogram at this level: fusion without decompression has been advocated when there is no neural compression

When spondylolisthesis is present

May occur without decompression, but is more common following surgery.[10] However, lumbar instability following decompressive laminectomy is rare (only ≈ 1% of all laminectomies for stenosis will develop progressive subluxation). Fusion is rarely required to prevent progression of subluxation with degenerative stenosis.[28]

For Grade I and low Grade II spondylolisthesis, laminectomy without fusion may be considered. Stability (without need for instrumentation) is thought to be maintained if > 50–66% of the facets are preserved during surgery and the disc space is not violated (maintains integrity of anterior and middle column). Younger or more active patients are at higher risk of subluxing. Patients with a tall (normal) disc space are at higher risk of subluxing than those with collapsed disc space.

One approach is to obtain flexion/extension x-rays pre-op, and follow patients after decompression. Those who develop symptomatic slippage post-op are treated by fusion, possibly in conjunction with spinal instrumentation.

When surgery is indicated, Table 72.6 serves as a guide to the type of procedure.

Laminectomy/laminotomy – surgical technique

Posterior approach with removal of the spines and lamina of affected levels (surgical "unroofing"), along with the associated ligamentum flavum. Individual nerve roots are palpated for compression within their neural foramen, with foraminotomies performed at appropriate levels. Doing a total L4 laminectomy for stenosis allows access to the L4–5 foramen, and the upper part of the L5-S1 foramen. If, in addition, the lower part of L3 is also removed, access is gained to the inferior pedicle of L3 and thus the L3–4 neural foramen. Undercutting the superior articular facet is often necessary to decompress the nerves in the lateral recess and neural foramen (p. 1097). Treatment of moderate stenosis at adjacent levels appears warranted as these levels have been shown to have a significant likelihood of becoming symptomatic later.[30]

Alternatively, laminotomies (as opposed to laminectomies) may be performed in cases where the central canal has a normal AP diameter, but the lateral canal gutter is stenotic.[31,32] Multilevel subarticular fenestrations are another slight variation on this theme.[33]

Position: either of the following is acceptable

1. prone: on a frame or chest rolls or knee-chest position to decompress the abdomen to decrease venous pressure and thus reduce bleeding
2. lateral decubitus position: if there is no laterality to symptoms, right lateral decubitus (left-side-up) is easier for most right-handed surgeons to use angled Kerrison rongeur parallel to nerve roots

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Booking the case: Lumbar laminectomy

Also see defaults & disclaimers (p. 27).

1. position: prone
2. implants: for fusions, schedule with the vendor for the desired implants and associated instrumentation
3. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: through the back to remove bone, ligament and any other tissue that is pressing on the nerve(s). If a fusion is to be done, then typically this will be accomplished using screws, rods and small cages, as required
   b) alternatives: nonsurgical management
   c) complications: usual spine surgery complications (p. 28), plus there might not be the amount of pain relief desired (back pain does not respond as well to surgery as nerve-root pain. The surgery slightly weaken the spine, and as a result, about 15% people will need a fusion at a later date

Minimally invasive spine surgery (MISS) decompression

Usually using ≈ 1" incisions and expandable retractors.

1. options include bilateral laminotomies (see above)
2. bilateral decompression through a unilateral laminotomy
   a) entry site: 3.5–4 cm off the midline to permit the needed angle

b) when using a retractor with an "open side" orient the retractor with the open side facing laterally (e.g. with the Nuvasive Maxcess® place the handles medially) to permit the angle needed for contralateral decompression
c) the laminectomy and facet takedown (usually for a TLIF) are done
d) open the ligamentum flavum on the side you're working on, to visualize the posterior extent of the spinal canal, to permit finding the plane between the posterior part of the ligamentum flavum and the undersurface of the bone
e) the ligamentum flavum is left in place on the contralateral side to protect the dura during drilling
f) complete the decompression and disc removal on the side you're working on
g) the undersurface of the bone (spinous process and contralateral lamina) are then drilled to decompress the contralateral side
h) once the undersurface of the contralateral posterior canal has been drilled, the ligamentum flavum is removed with pituitary rongeurs. It is possible to even do a contralateral foraminotomy at this point (curved Kerrison rongeurs are very helpful for this)
i) pedicle screws are placed through the open side, and then percutaneously through the contralateral side
j) this is generally followed by a transforaminal lumber interbody fusion (TLIF)

Interspinous process decompression/stabilization/fusion

Interspinous spacers (e.g. X-Stop™ (Medtronic)) limit extension at 1 or 2 levels (without fusion), preventing narrowing of the associated neural foramen, and may also off-load the facet joints and even the disc. "Success rate": 63%at 2 years. This device may be used as a standalone.

Interspinous plates (e.g. Aspen® (Lanx), Affix™ (Nuvasive), Spire® (Medtronic)) clamp across two spinous processes to fixate them (unlike X-Stop™ which just limits extension). The Aspen® clamps have a space for a graft which may optionally be used to promote fusion between the spinous processes. Interspinous plates may be used to augment other constructs e.g. lateral interbody fusion,[34] but are not intended for standalone use. Biomechanical stability is reported to be similar to bilateral pedicle screws in flexion, and unilateral pedicle screws in lateral bending.[35]

Contraindications (includes exclusionary criteria from the IDE study):

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1. instability at level considered for procedure: spondylolisthesis > Grade 1 or scoliosis with Cobb angle ≥ 25°
2. cauda equina syndrome
3. acute fracture of the spinous process
4. bilateral pars defects (disconnects spinous process from the anterior elements)
5. osteoporosis. Contraindications per the IDE: DEXA scan (p. 1009) with spine or hip T-score < –2.5 (i.e. more than 2.5 SD below the mean for normal adults) in the presence of ≥ 1 fragility fractures. Concerns: spinous process fracture at the time of insertion, or late subsidence due to microfractures. However, Kondrashov[36] interprets a T-score < –2.5 anywhere as indicative of osteoporosis (even without fragility fractures). Options here include:
   a) augmenting the spinous processes by injecting ≈ 0.5–1 cc of PMMA into each spinous process (SP) with a 13 Ga needle inserted ≈ halfway into the SP on lateral fluoro[36] prior to dilating the interspace or placing the X-Stop. Verify central position within SP on AP fluoro, and monitor injection on fluoro
   b) X-Stop^{PK}® made of titanium and PEEK (the modulus of elasticity of PEEK is closer to bone than titanium is)
6. ankylosed level (i.e. already fused)
7. L5-S1 level: the spinous process of S1 is usually too small (not usually an issue since symptomatic stenosis at L5-S1 is rare)
8. age < 50 years: not studied in IDE investigation

Surgical pointers:
1. it is critical that the spacer sit in the anterior third of the spinous process
2. results may be better with the patient awake, under local anesthesia, lying on their side in a position that they feel is relieving their pain (thus opening up the critical levels). This may reduce the risk of undersizing the prosthesis

Post-op (based on manufacturer's recommendations):
1. to avoid spinous process stress fracture: build-up physical activity gradually
2. 1st 6 weeks post-op: no spine hyperextension, no heavy lifting. Minimize stair climbing
3. initially, walking (for < 1 hour) is recommended as long as it is comfortable
4. at 2 weeks post-op: cycling (stationary or bicycle) may be added
5. 6 months post-op: may add sports such as swimming, golf, racquetball, tennis, running or jogging

Gill procedure

This procedure, and its modifications,[37] consist of radical decompression of nerve roots including removal of the loose posterior elements and total facetectomy. This is most usually followed by fusion (posterolateral or interbody). Fusion rate may be enhanced with the use of internal fixation (e.g. transpedicular screw-rod fixation).[38]

Reduction of spondylolisthesis

Reduction of spondylolistheseis can be accomplished with instrumentation, and requires a fusion.

The risk of nerve root injury with reduction of grade I or II spondylolisthesis is low.

Reduction of high-grade (grade III or IV) spondylolisthesis carries a risk of radiculopathy (e.g. L5 radiculopathy in cases of L5-S1 spondylolisthesis) in 50% of cases (some permanent) and may produce a cauda equina syndrome, probably from stretching nerve roots by distraction. Some have recommended intraoperative stimulation of nerve while performing EMG recording as the listhesis is gradually reduced, and stopping if the current required for stimulation increases 50% above baseline.

Isthmic spondylolisthesis (spondylolysis) – pars interarticularis defect

Instrumentation and/or fusion

Practice guideline: Fusion in patients with lumbar stenosis without spondylolisthesis

Level III[39]:
- in situ posterolateral fusion is not recommended following decompression in patients with lumbar stenosis in whom there is no evidence of preexisting spinal instability or likely iatrogenic instability due to facetectomy
- in situ posterolateral fusion is recommended in patients with lumbar stenosis in whom there is evidence of spinal instability
- the addition of pedicle-screw instrumentation is not recommended in conjunction with posterolateral fusion following decompression

Practice guideline: Fusion in patients with lumbar stenosis and spondylolisthesis

Level II[40]: posterolateral fusion is recommended for patients with stenosis and associated degenerative spondylolisthesis who require decompression

Level III[40]: pedicle screw fixation as an adjunct to posterolateral fusion should be considered in patients with stenosis and spondylolisthesis in cases where there is pre-op evidence of spinal instability or kyphosis at the level of the spondylolisthesis or when iatrogenic instability is anticipated (note: the definition of "instability" and "kyphosis" varies, and has not been standardized).

Fusion may accelerate degenerative changes at adjacent levels. Some surgeons recommend fusion at levels of spondylolisthetic stenosis.[5,30] Patients with combined degenerative spondylolisthesis, stenosis, and radiculopathy may be reasonable candidates for fusion.[41]

Booking the case: Lumbar lami ± fusion for stenosis

Also see defaults & disclaimers (p. 27).
1. position: prone
2. implants: for fusions, schedule with the vendor for the desired implants and associated instrumentation
3. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: through the back to remove bone, ligament and any other tissue that is pressing on the nerve(s). If a fusion is to be done, then typically this will be accomplished using screws, rods and small cages, as required
   b) alternatives: nonsurgical management
   c) complications: usual spine surgery complications (p. 28), plus there might not be the amount of pain relief desired (back pain does not respond as well to surgery as nerve-root pain). There can be problems with the implants, including breakage, migration (slippage), or undesirable positioning which may require further surgery.

72.9 Outcome

72.9.1 Morbidity/mortality

Risk of in-hospital mortality is 0.32%[17] Other risks include: unintended durotomy (p. 1055) (0.32%[17] to ≈ 13%[28,42]), deep infection (5.9%), superficial infection (2.3%), and DVT (2.8%); see also Risks of lumbar laminectomy (p. 1053).

72.9.2 Nonunion

Risk factors for nonunion in fusion operations (does not necessarily correlate with success of operation):
1. cigarette smoking delays bone healing and increases the risk of pseudoarthrosis following spinal fusion procedures, especially in the lumbar spine[12]
2. number of levels: in lumbar fusions, fusing 2 levels resulted in increased nonunion rates compared to fusing 1 level[43]
3. NSAIDs: controversial
   a) short-term (≤ 5 days) post-op use: high-dose ketorolac (120–240 mg/d) was associated with increased risk of nonunion, but low-dose ketorolac (≤ 110 mg/d), and celecoxib (200–600 mg/d) or rofecoxib (50 mg/d) were not[43]
   b) some feel that long-term NSAID use does lower fusion rate[44]

72.9.3 Success of operation

General information

Patients with a postural component to their pain had much better results (96% good result) than those without a postural component (50% good results), and the relief of leg pain was much more successful than relief of back pain.[45] Surgery is most likely to reduce LE pain and improve walking tolerance.[41]

Outcome studies

SPORT study

There have been many attempts to ascertain the benefit of surgery, including the $13.5 million SPORT study. Shortcomings of the study include: patients were allowed to decline randomization and were then entered into an observational cohort which may introduce bias into the groups, crossovers were allowed between patients randomized to surgery and those randomized to nonsurgical treatment (degrading the "intention to treat" analysis), no standardized surgical or nonsurgical technique, relatively low long-term follow-up (52% at 8 years), change in paradigm from analyzing intention-to-treat to an as-treated analysis.

Results indicated a strong benefit of surgery at 4-years follow-up[46] that appeared to diminish by 8-years in the randomized cohort but persisted in the observational cohort.[47]

Other long-term outcome studies

Literature review[17] with long-term follow-up found good or excellent outcome after surgery with a mean of 64% (range: 26–100%). A patient satisfaction survey indicated that 37% were much improved and 29% somewhat improved (total: 66%) post-op.[48] A prospective study found a success rate of 78–88% at 6 wks and 6 months, which dropped to ≈ 70% at 1 year and 5 yrs.[49] Success rates were slightly lower for lateral recess syndrome.

Reasons for surgical failure

Surgical failure may be divided into two groups:

1. patients with initial improvement who develop recurrent difficulties. Although short-term improvement after surgery is common,[46] many patients progressively deteriorate over time.[47,50] One study found a 27% recurrence of symptoms after 5 years follow-up[30] (30% due to restenosis at the operated level, 30% due to stenosis at a new level ("adjacent segment failure"); 75% of these patients respond to further surgery). Other etiologies include: development of herniated lumbar disc, development of late instability including kyphosis ("proximal junction kyphosis" – PJK), coexisting medical conditions
2. patients who fail to have any post-op pain relief (early treatment failures). In one series of 45 such patients[51]:
   a) the most common finding was a lack of solid clinical and radiographic indications for surgery (e.g. non-radicular LBP coupled with modest stenosis)
   b) technical factors of surgery had less influence on outcome, with the most common finding being failure to decompress the lateral recess (which, in non-fusion cases, requires judicious medial facet resection or undercutting the superior articular facet)
   c) other diagnoses (e.g. arachnoiditis), missed diagnosis (e.g. spinal AVM)

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73 Adult Spinal Deformity and Degenerative Scoliosis

73.1 General information

Key concepts

- Adult spinal deformity (ASD) encompasses scoliosis and sagittal imbalance
- Sagittal balance correlates with quality of life measures
- Basic spine measurements needed: LL, PI, PT, ± SVA
- Major alignment objectives: LL = PI ± 9°, PT < 20°, SVA < 5 cm

Adult spinal deformity (ASD) is a broad term that refers to a wide spectrum of structural abnormalities of a mature spine. ASD encompass abnormalities in the coronal plane (scoliosis) as well as abnormalities in the sagittal plane.

The term "adult degenerative scoliosis" (ADS) (as distinguished from idiopathic juvenile scoliosis (IJS)) is often used interchangeably with ASD. Definition of adult degenerative scoliosis: spinal deformity with Cobb angle[1] > 10° in a skeletally mature individual.[2] ADS may be the result of childhood idiopathic scoliosis persisting into adulthood, or may be de novo.

Deformity in ASD can be primarily due to asymmetric disc degeneration or secondary to hip pathology, osteoporosis and asymmetric loads.[3] It subsequently involves posterior elements (including facet joints) and thereafter axial rotation, lateral olisthesis and ligamentous laxity.[2,4] Progressive facet and discogenic degeneration may lead to segmental instability and subsequent central/foraminal stenosis secondary to ligamentum flavum hypertrophy and osteophyte formation[5] and well as spondylolisthesis.

While treatment goals include reduction of pain, symptomatic neural compression and disability due to deformity, the methodology and biomechanics of ADS treatment differ greatly from treating IJS in the adolescent.

ADS tends to progress at an average rate of 3° per year (range: 1–6°).[4] Factors associated with higher rates of progression: Cobb angle > 30°, apical rotation > Grade II (on the Nash-Moe system,[6] which is falling into disuse), lateral listhesis > 6 mm, and an intercrest line through L5.[4] Factors not correlated with rate of progression: age and gender. Controversial associations: osteopenia.

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73.2 Epidemiology

ASD is more prevalent in patients age > 60 years, however true prevalence is not well defined. > 50% of adults hospitalized with spinal deformity are > 65 years.[7] Incidence of asymptomatic scoliosis ranges from 1.4%-32% and up to 68% in patients > 60.[8]

73.3 Clinical evaluation

Location, time and duration pain (leg vs. axial back) are important factors in evaluation of a patient with ASD. These patients may also have symptoms of spinal stenosis (central or radicular), which may require concomitant decompression. The patient's ability to perform activities of daily living and medical co-morbidities (e.g. cardiac, osteoporosis etc.) need be taken into consideration for treatment planning.

Some patients present with obvious spinal deformity (scoliosis, forward flexion at the waist, walking with knees bent).

As with neurogenic claudication, patients tend to be more symptomatic when up on their feet. A significant amount of pain may be generated by attempting to correct for spinal imbalance by using paraspinal muscles as well as retroverting the pelvis (rotating it backward at the hips) and not fully extending the knees. All this extra muscle activity is fatiguing and begins to produce muscle pain in the back and thighs. Patients with ASD tends to be better in the morning when they are rested.

Unlike lumbar spinal stenosis in the absence of scoliosis, symptoms may not relieved by flexion.[2] There may be some relief when supporting the trunk with the arms.

73.4 Diagnostic testing

 CT / MRI. Both are recommended for the evaluation of symptomatic spondylosis and ASD to determine extent of neural compression.

 DEXA (dual-energy x-ray absorptiometry). Patients should be evaluated for osteopenia/osteoporosis prior to surgical planning. Medical treatment may be beneficial in the peri-operative period.

Some surgeons use Forteo for 3 months (off-label use – controversial) in an effort to try and quickly increase osteoporotic bone strength for surgery.

 Standing scoliosis x-rays. Recommended for the evaluation of global and regional spinal balance. Pre and postoperative plain films help confirm that alignment objectives are achieved.

Measurements related to sagittal balance are taken from standing scoliosis x-rays (CT & MRI are obtained supine and are not equivalent). Technical requirements for the lateral image:
- x-ray must image from C7 down to the femoral heads
- the patient needs to try to keep the knees straight (extended)
- arms should be folded in front of the chest (and they should not lean or hold on to anything)

Dynamic scoliosis x-rays (“lateral bending films”) help determine the degree of curve rigidity preoperatively.

73.5 Pertinent spine measurements

73.5.1 General information

Quantification of severity of spinal deformity and classification helps guide appropriate treatment paradigms.[9,10]

73.5.2 Scoliosis nomenclature

Scoliosis is measured using Cobb angles. On an AP x-ray, the “end vertebrae” are identified at the top and bottom of the scoliotic curve and are defined as the vertebrae with the greatest angle relative to the horizontal plane. One horizontal line is drawn through the superior endplate of the superior “end vertebra”, and a second is drawn through the inferior endplate of the inferior “end vertebra.” The Cobb angle is the angle between these 2 lines. Curves are named for the convex side (dextroscoliosis = convex to right, levoscoliosis = convex to left).

A non-structural curve can correct on side bending. A structural curve is not flexible.

The major curve is the largest structural curve. A fractional curveis the curve below the major curve.

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73.5.3 Spino-pelvic parameters

Measurement methodology and pertinent information are shown in Table 73.1 and illustrated in Fig. 73.1 and Fig. 73.2. Basic measurements that can be correlated with pain reduction and quality of life measures:
- LL (lumbar lordosis)
- PI (pelvic incidence)
- PT (pelvic tilt)
- ± SVA (sagittal vertical alignment): while this can be helpful at times, it also appears to be subject to variability depending on how much pain the patient has with standing up straight

With the exception of CSVL, the measurements shown in Table 73.1 are all taken from a lateral standing x-ray (Fig. 73.1 and Fig. 73.2).

73.6 SRS-Schwab classification of adult spinal deformity

Adult scoliosis has been classified by the Scoliosis Research Society (SRS)[16] based on its regional/global radiographic features (a modification of previously established adolescent King/Moe and Lenke classifications) and most recently by spino-pelvic parameters as it relates to health-related quality of life[17,18,19] is shown here.
- Coronal Curve types
  - T: Thoracic only (with lumbar curve < 30°)
  - L: Thoracolumbar/Lumbar only (with thoracic curve < 30°)
  - D: Double curve (both T and T/L curves > 30°)
  - N: No major coronal deformity (all coronal curves < 30°)
- Sagittal Modifiers
  - Pelvic harmony (PI minus LL)
    - 0: non pathologic (PI–LL < 10°)
    - +: moderate deformity (10° < PI–LL < 20°)
    - ++: marked deformity (PI–LL > 20°)

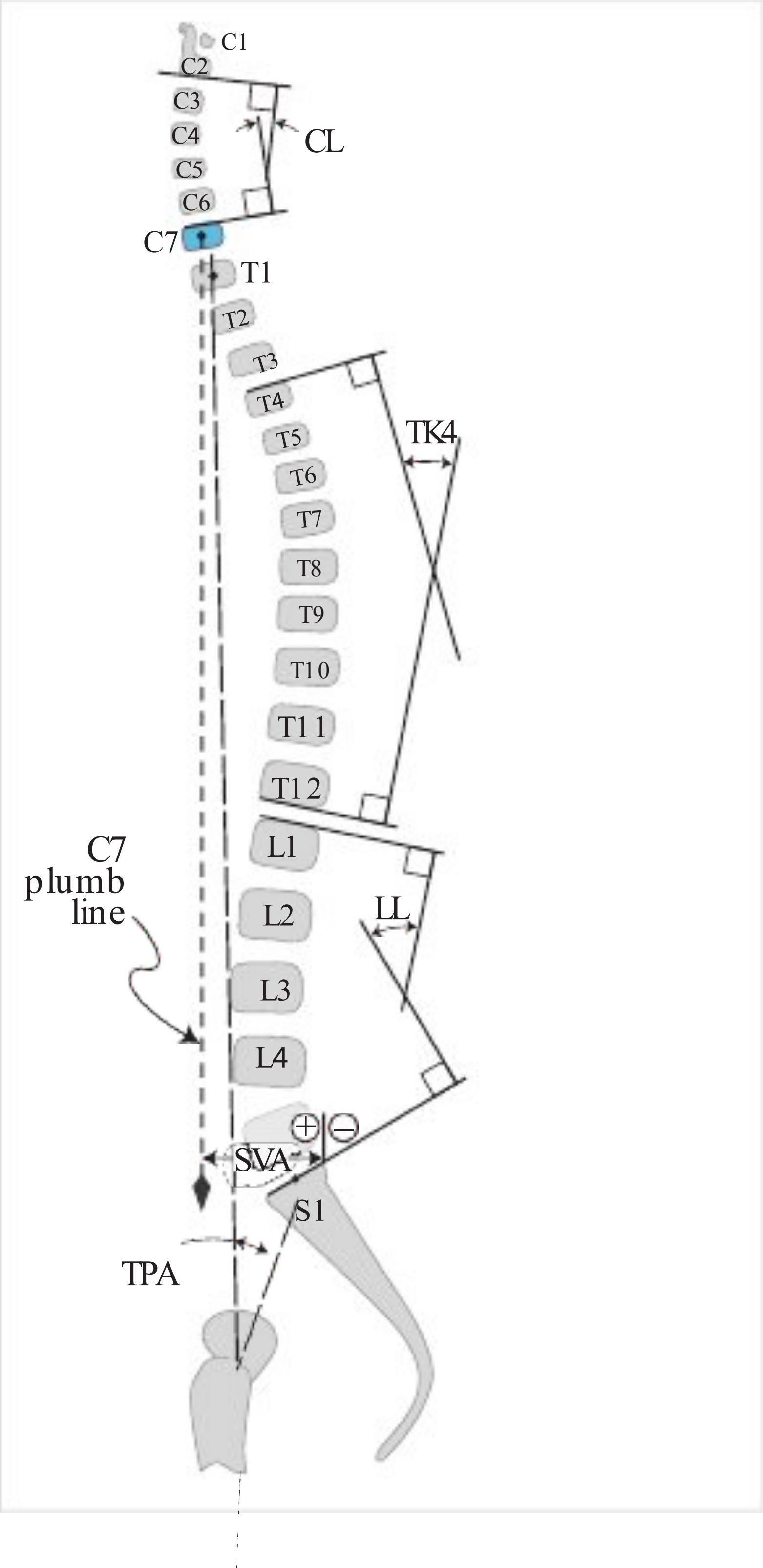

Fig. 73.1 Schematic lateral spine diagram showing method for measuring CL, TK, LL SVA and TPA.

- Global alignment (SVA)
  - 0: non pathologic (SVA<4 cm)
  - +: moderate deformity (4 cm <SVA<9.5 cm)
  - ++: marked deformity (SVA>9.5 cm)
- Pelvic Tilt (PT)
  - 0: non pathologic (PT<20°)
  - +: moderate deformity (20°<PT<30°)
  - ++: marked deformity (PT>30°)

73.7 Treatment/management

73.7.1 Options

1. observation
2. focal decompression
3. surgical correction of deformity
   a) MIS (minimally invasive (spine) surgery)
   b) Hybrid (MIS+open)
   c) Traditional open surgery (transforaminal lumbar interbody fusion (TLIF), posterior lumbar interbody fusion (PLIF)…)

Treatment options are based on clinical symptoms (axial back pain ± radiculopathy vs. radiculopathy alone) and degree of abnormalities in sagittal (balance need for open osteotomies or anterior column release ACR). Neuropathic symptoms most often originate from foraminal compromised on the

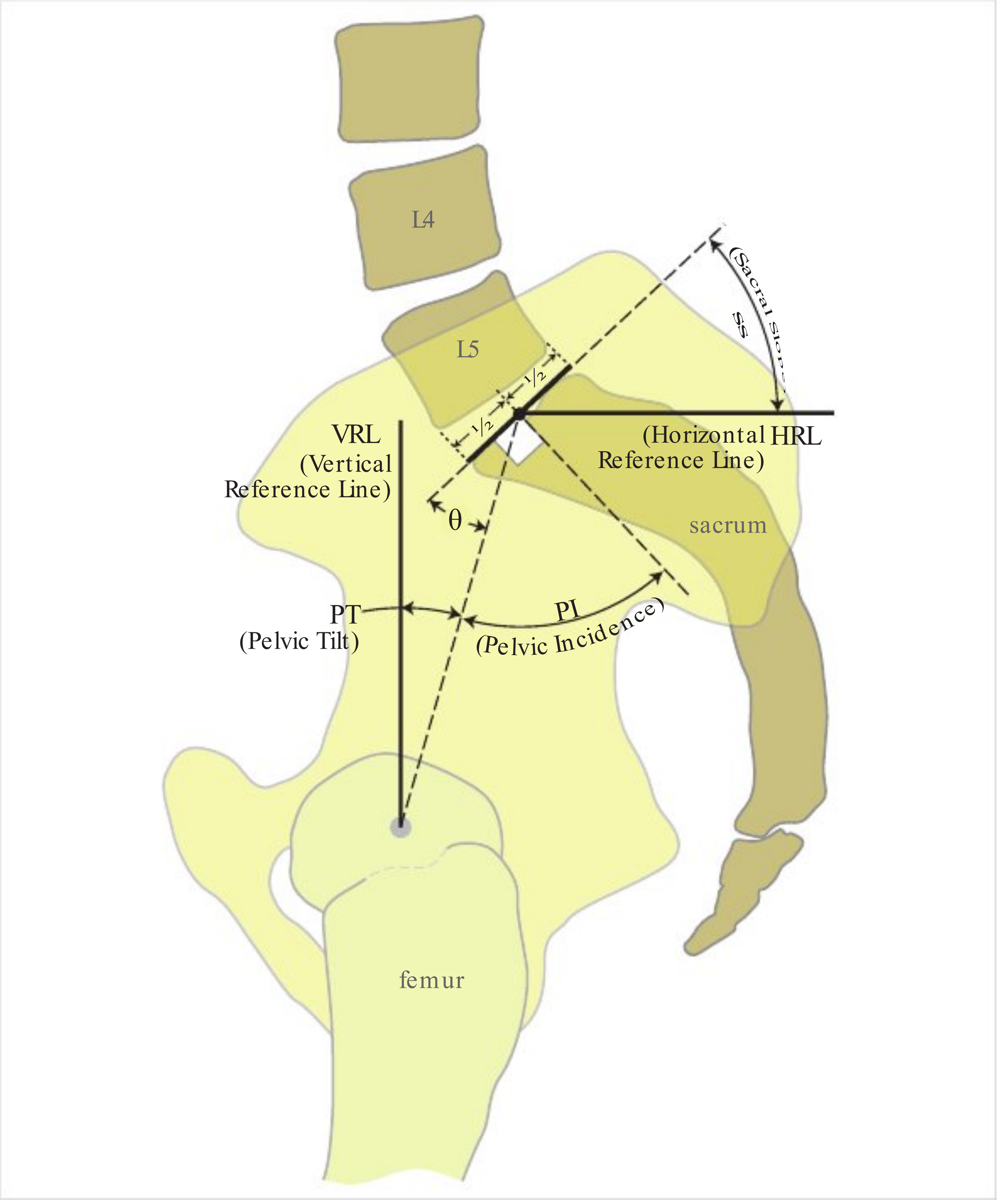

Fig. 73.2 Schematic lateral spine diagram showing method for measuring PT, PI and SS.

concavity of the curve, but can be seen on the convexity in the setting of facet hypertrophy and may improve with indirect decompression and correction in the coronal plane. Significant central stenosis (neurogenic claudication) may require concomitant direct decompression in addition of deformity correction.

Goal of surgery: to improve quality of life measures, neuropathic and axial pain. A surgeon's armamentarium includes traditional open, MIS and hybrid procedures, which is patient and deformity specific. More recently, the surgical decision making paradigms have included MIS techniques to limit approach related morbidity. MIS techniques include lateral interbody fusion, ALLR, MIS-TLIF and percutaneous pedicle screw fixation, which can be used with posterior osteotomies to enhance corrective power.

73.7.2 Correction of global spinal balance

Indications for surgery
- Axial back pain ± neuropathic symptoms (deleterious to ADLs)
  - Abnormal SVA
  - ± CSVL (central sacral vertical line) abnormality
  - or derangement of spino-pelvic parameters.
- Patient age and co-morbidities must be taken in to consideration (osteopenia, anesthetic risk and medical co-morbidities can limit correction goal and amount of surgery that is safe)

Summary of spino-pelvic objectives
- LL = PI ± 9°
- PT < 20°
- SVA < 5 cm

In most instances, correction of sagittal imbalance is due to a shortfall of lumbar lordosis (LL) relative to the pelvic incidence (PI) – which can be considered flat back syndrome. That is, LL is usually more than 9° below PI.

Pelvic tilt > 20° suggests the patient is trying to compensate by retroverting the pelvis (some authors accept up to 25° as normal).

The minimum amount of correction that the surgeon tries to achieve is therefore that amount that LL needs to increase to bring it within 9° of PI, and typically also adds in the amount that the patient is compensating (i.e. the amount that PT is greater than 20°) which yields the following approximation (applies when LL is more than 9° lower than PI, and PT is greater than 20°); see Eq (73.1):

$$\text{Increase in LL needed} \approx (PI - LL - 9°) + (PT - 20°) \qquad (73.1)$$

Coronal balance. Measured on AP standing scoliosis x-ray. A plumb line is dropped from the center of the C7 VB. If it falls > 4 cm from the midline of the sacrum (where the CSVL is located), there is coronal imbalance (if the plumb line falls to the right of the CSVL it is positive, to the left it is negative).

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73.7.3 Surgical options for increasing lumbar lordosis

Various surgical techniques may be used to increase the lumbar lordosis to bring it within the desired specifications. If necessary, these techniques can be combined with procedures to decompress the neural elements (e.g. laminectomy). A comparison of the approximate amount of lordosis than can be achieved with different techniques is shown in Table 73.2.

Transforaminal lumbar interbody fusion (TLIF) and posterior lumbar interbody fusion (PLIF). Traditional operation. May be done open or MIS.

Lateral lumbar interbody fusion (LLIF). E.g. XLIF™, DLIF™, OLIF™. Approach through psoas muscle (XLIF, DLIF) or anterior to psoas muscle (OLIF) through a lateral or anterolateral approach. Can distract the vertebral bodies by increasing the height of the disc space and thereby indirectly decompressing the neural elements. If bone quality is good, and there is no instability nor spondylolisthesis > Grade I, a standalone procedure (i.e. without screw instrumentation) may be an option if cage width of at least 22 mm (or preferably 26 mm) in the AP dimension is used.

Table 73.2 Comparison of the amount of lumbar lordosis that can be obtained from various surgical techniques.

Technique	TLIF/PLIF	LLIF	ALIF	SPO + ACR	PSO
Degrees of lumbar lordosis	< 0 (i.e. kyphosis) up to 2° [20]	1° [21]	6° [20]	16° [22]	30–40° [22,23]

Abbreviations: TLIF = transforaminal lumbar interbody fusion; PLIF = posterior lumbar interbody fusion; LLIF = lateral lumbar interbody fusion; ALIF = anterior lumbar interbody fusion; SPO = Smith-Petersen osteotomy; ACR = anterior column release; PSO = pedicle subtraction osteotomy

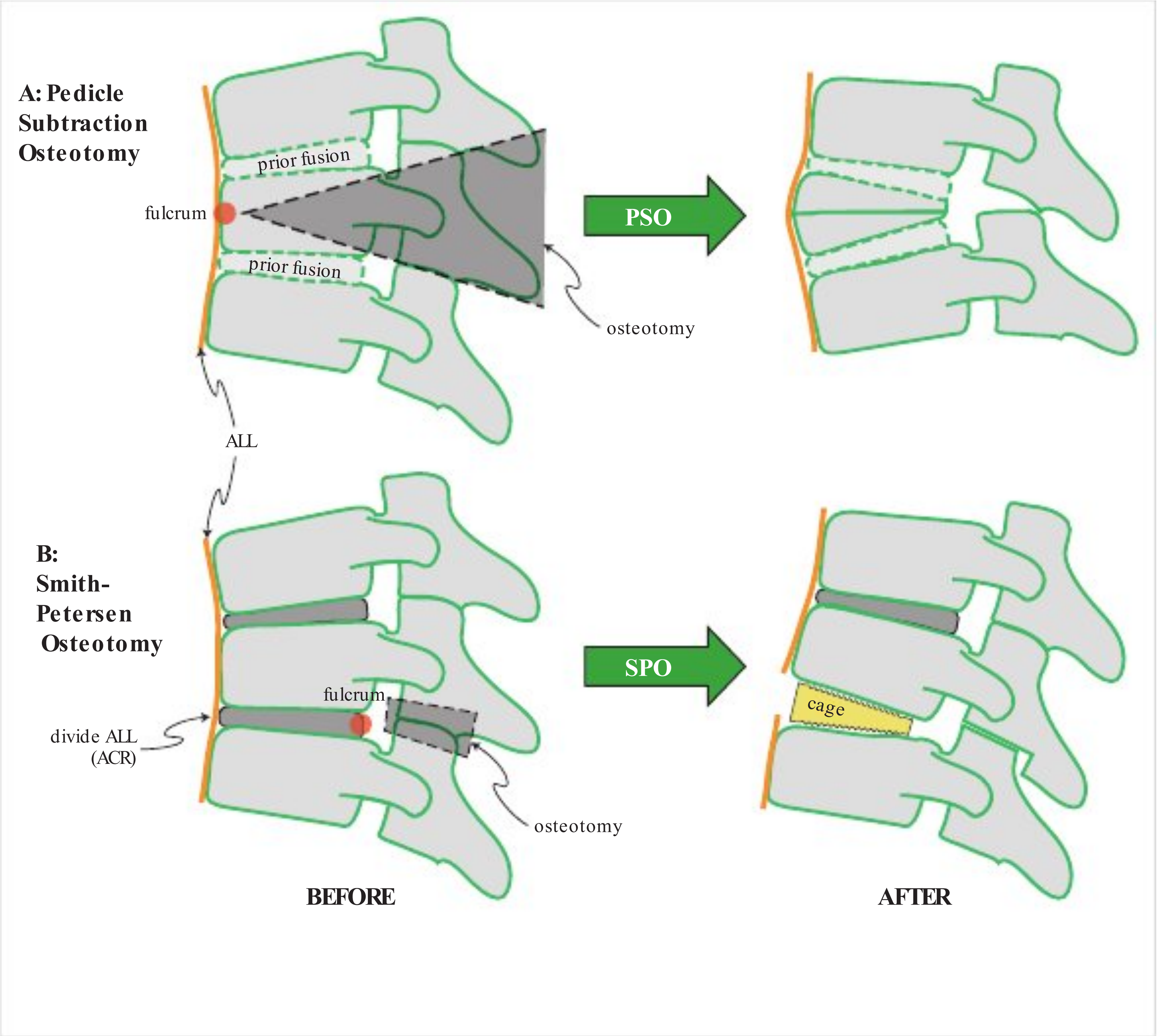

Fig. 73.3 Comparison of A: Pedicle subtraction osteotomy and B: Smith Petersen osteotomy plus ACR. Abbreviations: ALL=anterior longitudinal ligament, ACR=anterior column release.

Anterior column release (ACR). Involves division of the anterior longitudinal ligament (ALL release (ALLR)) typically with placement of a “hyperlordotic” cage (20–30° lordosis) from an anterior approach (e.g. lateral interbody technique). This is followed by posterior fixation, often with a Smith-Petersen osteotomy (especially for 30° cages) and compression. It can increase LL up to 12° per ACR level and improvement of SVA up to 3 cm (depending on the level at which it is performed).[22,24]

Risk of injury to great vessels, either directly (when cutting the ALL) or indirectly by elongating the anterior column. It is critical to evaluate the great vessels on axial MRI or CT or on angiogram and to not do the procedure if the vessels appear tightly approximated to the bodies or to osteophytes, at that level.

Smith-Petersen osteotomy (SPO). AKA “chevron or extension osteotomy” can increase lordosis up to 10° per level. Approximately one degree for each millimeter of bone resected.[25,26] Involves removal of bilateral superior and inferior facets along with the ligamentum flavum and portion of the lamina above and below. The created gap is then closed with compression of the posterior elements to give lordosis (resecting posterior elements and using the middle column as a fulcrum to elongate the anterior column, Fig. 73.3 A).[26,27] LL is increased approximately 1° for every 1 mm of bone removed.

Table 73.3 MIS management recommendations based on severity of ASD[9] with approximate SRS-Schwab Class[17] equivalence.

	Mild (balanced)		Moderate (compensated)		Severe (uncompensated)	
	Deukmedjian at al.[9]	SRS-Schwab[17]	Deukmedjian at al.[9]	SRS-Schwab[17]	Deukmedjian et al.[9]	SRS-Schwab[17]
CCA	<30°	N	>30°	T, L or D	>30°	T, L or D
PI – LL	<20°	0 or +	20–30°	++	>30°	++
SVA	<5 cm	0	5–9 cm	+	>10 cm	++
PT[a]	<25°	0	25–30°	+	>30°	++
Recommendations for anterior procedure	MIS LLIF		MIS-LLIF to neutral vertebrae + ACR		MIS-LLIF to neutral vertebrae ± ACR	
Recommendations for posterior procedure	If PT<20° consider standalone[b], otherwise percutaneous fixation		Percutaneous fixation to S1 ± facetectomy(ies)		Open fixation to S2 or iliac + osteotomy(ies)	

To use this table, determine which category the patient fits into (Mild, Moderate or Severe) using either the Deukmedjian parameters (on which this reference is based) or the roughly equivalent SRS-Schwab parameters shown

Abbreviations: CCA = Cobb coronal angle; LLIF = lateral lumbar interbody fusion (e.g. XLIF, DLIF, OLIF…); ACR = anterior column release
[a] in the SRS-Schwab classification, PT<20° is considered normal
[b] standalone meaning no posterior fixation, assumes that bone quality is not osteoporotic and that a cage width of at least 22 mm is used (to reduce the risk of subsidence)

The term "Ponte osteotomy" is often used interchangeably with SPO, but the Ponte was originally described for treating Scheuermann's kyphoscoliosis.

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● Pedicle subtraction osteotomy (PSO). Involves removal of the posterior elements including the ligmanetum flavum, lamina and facets widely, followed by isolation and resection of pedicles bilaterally and wedge shaped removal of the vertebral body just barely up to the ventral cortex. The created gap is then closed by compression of the posterior elements and subsequent greenstick fracture of the isolated ventral cortex (Fig. 73.3 B).[23] Can increase LL by 30°- 40° per level, improvement of SVA 5.5 – 13 cm per level.[22,23]

This procedure is technically challenging and is associated with high blood loss (3 L in 1 series[28]) and increased risk of complications (including proximal junctional kyphosis (PJK) in 23%[28]) compared to SPO. Generally reserved for spines that have previously been fused where it is not possible to get the amount of lordosis needed from the unfused levels. The PSO is a spine "shortening" procedure.

Uses the anterior column as a fulcrum. Usually limited to levels below the conus medullaris (i.e. L1–2) due to inward buckling of the dura (L3 is the most common level). Intraoperative electrophysiologic monitoring is required. Relative contraindication: poor bone quality.

● Anterior lumbar interbody fusion (ALIF). Best for L5-S1 (where the great vessels tend not to interfere with the access, and where every degree of correction produces a more significant amount of improvement in SVA than at other levels as a result of being at the lowest point in the spine).

73.7.4 Guidelines for MIS treatment for ASD

A simple algorithm for MIS management of sagittal imbalance based on spino-pelvic parameters and the approximate SRS-Schwab class is shown in Table 73.3.[9] See references[10,29] for a more detailed protocol.

Currently under investigation by the SRS: when, and to what degree, does improvement in sagittal balance occur with simple decompression (possibly with a minimal fusion) as a result of pain relief permitting the patient to stand up straighter with less pain.

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74 Special Conditions Affecting the Spine

74.1 Paget's disease of the spine

74.1.1 Pathophysiology

Paget's disease (PD) (AKA osteitis deformans) is a disorder of osteoclasts (possibly virally induced) causing increased rate of bone resorption with reactive osteoblastic overproduction of new, weaker, woven bone, producing characteristic "mosaic pattern."

Initially there is a "hot" phase with elevated osteoclastic activity and increased intraosseous vascularity. Osteoblasts lay down a soft, nonlamellar bone. Later a "cool" phase occurs with disappearance of the vascular stroma and osteoblastic activity leaving sclerotic, radiodense, brittle bone[1] ("ivory bone").

74.1.2 Malignant degeneration

A misnomer, since the malignant changes actually occur in the reactive osteoblastic cells. About 1% (reported range: 1–14%) degenerate into sarcoma (osteogenic sarcoma, fibrous sarcoma, or chondrosarcoma),[2 (p 2642)] with the possibility of systemic (e.g. pulmonary) metastases. Malignant degeneration is much less common in the spine than in the skull or femur.

74.1.3 Epidemiology

Prevalence: ≈ 3% of population > 55 years old in the U.S. and Europe, much lower in Asia.[3] Slight male predominance. Family history of Paget's disease is found in 15–30% of cases (accuracy is poor since most are asymptomatic).

74.1.4 Common sites of involvement

Affinity for axial skeleton, long bones and skull. In approximate descending order of frequency: pelvis, thoracic and lumbar spine, skull, femur, tibia, fibula, and clavicles.

74.1.5 Neurosurgical involvement

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PD may present to the neurosurgeon as a result of:
1. back pain: usually not as a direct result of vertebral bone involvement (see below)
2. spinal cord and/or nerve root symptoms
   a) compression of the spinal cord or cauda equina (relatively rare)
   b) spinal nerve-root compression
   c) vascular steal due to reactive vasodilatation adjacent to involved areas
3. with skull involvement:
   a) compression of cranial nerves as they exit through bony foramina: 8th nerve is most common, producing deafness or ataxia (p.1400)
   b) skull base involvement → basilar invagination
4. to ascertain diagnosis in unclear bone lesions of the spine or skull

74.1.6 Presentation

General information

Only ≈ 30% of pagetic sites are symptomatic,[4] the rest are discovered incidentally. The overproduction of weak bone may produce bone pain (the most common symptom), predilection for fractures and compressive syndromes: cranial nerve (p.1400), spinal nerve root… Painless bowing of a long bone may be the first manifestation. A number of patients present due to pain from joint dysfunction related to PD.

The overwhelming majority of pagetic lesions are asymptomatic[5 (p 1413)] with lesions detected on radiographs or bone scan obtained for other reasons or as part of a work-up for an elevated alkaline phosphatase. Although the most common complaint in patients with Paget's disease is of back pain, this is attributable to pagetic involvement alone in only ≈ 12%;[6] in the remainder it is secondary to other factors, some of which are described below.

Symptoms that may be related to the Paget's disease itself

Symptoms from the following are slowly progressive (usually present for > 12 months; rarely < 6 mos):

1. neural compression
   a) causes of compression
      - due to expansion of woven bone
      - due to osteoid tissue
      - pagetic extension into ligamentum flavum and epidural fat[7]
   b) sites of compression
      - spinal cord (see below)
      - nerve root in neural foramen
2. osteoarthritis of facet joints (Paget's disease may precipitate osteoarthritis[6])

Symptoms from the following tend to progress more rapidly:

1. malignant (sarcomatous) change of involved bone (rare, see above)
2. pathologic fracture (pain usually sudden in onset)
3. neurovascular (compromise of vascular supply to nerves or spinal cord) by
   a) compression of blood vessels (arterial or venous)
   b) pagetic vascular steal (see below)

Spinal cord symptoms

Myelopathy or cauda equina syndrome may be due to spinal cord compression or from vascular effects (occlusion, or "steal" due to reactive vasodilatation of nearby blood vessels[5 (p 1415)]). Only ≈ 100 cases had been described as of 1981.[8] Characteristically, 3–5 adjacent vertebrae are involved,[9 (p 2307)] whereas monostotic involvement is usually asymptomatic.[10] In case reports in the literature, progressive quadri- or paraparesis was the most common presentation.[11] Sensory changes are usually the first manifestation, progressing to weakness and sphincter disturbance. Pain was the only symptom in a neurologically intact patient in only 5.5%

A rapid course (averaging 6 wks) with a sudden increase in pain is more suggestive of malignant degeneration.

74.1.7 Evaluation

1. lab work (serum markers may be normal in monostotic involvement):
   a) serum alkaline phosphatase: usually elevated (this enzyme is involved in bone synthesis and so may not be elevated in purely lytic Paget's disease[5 (p 1416)]); mean 380 ± 318 IU/L (normal range: 9–44).[6] Bone-specific alkaline phosphatase may be more sensitive and may be useful in monostotic involvement[3]
   b) calcium: usually normal (if elevated, one should R/O hyperparathyroidism)
   c) urinary hydroxyproline: hydroxyproline is found almost exclusively in cartilage. Due to the high turnover of bone, urinary hydroxyproline is often increased in PD with a mean of 280 ± 262 mg/24 hrs (normal range 18–38)[6]
2. bone scan: lights up in areas of involvement in most, but not all[6] cases
3. plain x-rays:
   a) localized enlargement of bone: a finding unique to PD (not seen in other osteoclastic diseases, such as prostatic bone mets)
   b) cortical thickening
   c) sclerotic changes
   d) osteolytic areas (in skull → osteoporosis circumscripta; in long bones → "V" shaped lesions)
   e) spinal Paget's disease often involves several contiguous levels. Pedicles and lamina are thickened, vertebral bodies are usually dense and compressed with increased width. Intervening discs are replaced by bone
4. CT: hypertrophic changes at the facet joints with coarse trabeculations

74.1.8 Treatment

Medical treatment for Paget's disease

General information

There is no cure for Paget's disease. Medical treatment is indicated for cases that are not rapidly progressive where the diagnosis is certain, for patients who are poor surgical candidates, and pre-op if excessive bleeding cannot be tolerated. Medical therapy reverses some neurologic deficit in 50%of cases,[12] but generally requires prolonged treatment (≈ 6–8 months) before improvement occurs, and may need to be continued indefinitely due to propensity for relapses. Medications used include the following.

Calcitonin derivatives

Parenteral salmon calcitonin (Calcimar®)[12]: reduces osteoclastic activity directly, osteoblastic hyperactivity subsides secondarily. Relapse may occur even while on calcitonin. Side effects include nausea, facial flushing, and the development of antibodies to salmon calcitonin (these patients may benefit from a more expensive synthetic human preparation (Cibacalcin®) starting at 0.5 mg SQ q d[13]).

□ 50–100 IU (medical research council units) SQ q d × 1 month, then 3 injections per week for several months.[3] If used pre-op to help decrease bony vascularity, ≈ 6 months of treatment is ideal. Doses as low as ≈ 50 IU units 3 × per week may be used indefinitely post-op or as a sole treatment (alkaline phosphatase and urinary hydroxyproline decline by 30–50% in > half of patients in 3–6 months, but they rarely normalize).

Bisphosphonates

These drugs are pyrophosphate analogues that bind to hydroxyapatite crystals and inhibit reabsorption. They also alter osteoclastic metabolism, inhibit their activity, and reduce their numbers. They are retained in bone until it is resorbed. Oral absorption of all is poor (especially in the presence of food). Bone formed during treatment is lamellar rather than woven.

Etidronate (Didronel®) (AKA EHDP): reduces normal bone mineralization (especially at doses ≥ 20 mg/kg/d) producing mineralization defects (osteomalacia) which may increase the risk of fracture but which tend to heal between courses.[14] Contraindicated in patients with renal failure, osteomalacia, or severe lytic lesions of a LE. □ 5–10 mg/kg PO daily (average dose: 400 mg/d, or 200–300 mg/d in frail elderly patients) for 6 months, may be repeated after a 3–6 month hiatus if biochemical markers indicate relapse.

Tiludronate (Skelid®): unlike etidronate, does not appear to interfere with bone mineralization at recommended doses. Side effects: abdominal pain, diarrhea, N/V. □ 400 mg PO qd with 6–8 ounces of plain water > 2 hrs before or after eating × 3 months. Available: 200 mg tablets.

Pamidronate (Aredia®): much more potent than etidronate. May cause a transient acute flu-like syndrome. Oral dosing is hindered by GI intolerance, and IV forms may be required. Mineralization defects do not occur in doses < 180 mg/course. □ 90 mg/d IV × 3 days, or as weekly or monthly infusions.

Alendronate (Fosamax®): does not produce mineralization defects (p.1010).

Clodronate (Ostac®, Bonefos®): □ 400–1600 mg/d PO × 3–6 months. 300 mg/d IV × 5 days (may be available outside the U.S.).

Risedronate (Actonel®): does not interfere with bone mineralization in recommended doses.[15] □: 30 mg PO q d with 6–8 oz. of water at least 30 minutes before the first meal of the day.

Surgical treatment

General information

In general, conservative treatment of fractures in PD are associated with a high rate of delayed union.

Surgical indications for spinal Paget's disease

1. rapid progression: indicating possible malignant change or spinal instability
2. spinal instability: severe kyphosis or compromise of canal by bone fragments from pathologic fracture. Although the collapse is usually gradual, sudden compression may occur
3. uncertain diagnosis: especially to R/O metastatic disease (osteoblastic lesions)
4. failure to improve with medications

Surgical considerations

1. profuse bleeding is common: if significant bleeding would present an unusual problem, treat for as long as feasible pre-op with a bisphosphonate or calcitonin (see above)
   a) use bone wax to help control bleeding
   b) hemostasis may be difficult

2. to treat resultant spinal stenosis: decompressive laminectomy is the standard procedure in the thoracic region.[11] However, if most of the pathology is anterior, consideration should be given to anterior approach
3. bone is often thickened, and may be fused with obliteration of interspace landmarks. A high-speed drill is usually helpful
4. post-op medical treatment may be necessary to prevent recurrences[12]
5. osteogenic sarcoma
   a) surgery and chemotherapy are used, cure is less likely than in primary osteosarcoma of non-pagetic origin
   b) biopsy proven of the scalp requires en-bloc excision of scalp and tumor

Surgical outcome

See reference.[11]

In 65 patients treated with decompressive laminectomy, 55 (85%) had definite but variable degrees of improvement. Patients who had only minimal improvement were often ones with malignant changes. One patient was worse after surgery, and the operative mortality was 7 patients (10%). Survival with malignant degeneration is < 5.5 mos after admission.

74.2 Ankylosing spondylitis

74.2.1 General information

Key concepts

- Prototypical spondyloarthritidy, now referred to as radiographic axial spondyloarthritis
- Seronegative (absence of rheumatoid factor), associated with HLA-B27
- Begins in SI joints (sine qua non of involvement) progressing rostrally
- Clinical: morning back stiffness, kyphotic deformity limits chest expansion
- X-ray findings: "bamboo spine", Andersson lesions, progressive thoracic kyphosis
- Fragile rigid spine highly susceptible to fracture and SCI even after low energy trauma
- Severe deformity, neurologic involvement or unstable fracture warrants surgical intervention

Historically known as Marie-Strümpell disease, Ankylosing spondylitis (AS) is an HLA-B27 associated inflammatory disease and the prototype of a group of diseases known as the spondyloarthritides (SpA). Common features are inflammatory back pain, negative serology for rheumatoid factor and absence of rheumatoid nodules, and asymmetric oligoarthritis predominantly of lower extremities. Currently referred to in the literature as Radiographic axial Spondyloarthritis (RaxSpA), AS is believed to be the late stage of a single disease entity, axial Spondyloarthritis (axSpA) distinguished by definitive evidence of sacroilitis on plain radiographs. Other conditions under this rubric include: psoriatic arthropathy, Reiter's disease, juvenile spondyloarthropathy… AS has in the past also been known as rheumatoid spondylitis, or rheumatoid arthritis of the spine, but use of these terms is discouraged because of the lack of rheumatoid factor. The spine is the primary skeletal site involved, usually starting in the sacroiliac joints and lumbar spine and progressing rostrally.

Enthesopathy: nongranulomatous inflammatory changes at the entheses (attachment points of ligaments, tendons or capsules on bones; the locus of involvement in AS) stimulates re-placement of ligaments by bone ultimately resulting in osteoporotic VBs, calcified intervertebral discs (sparing the nucleus pulposus), and ossified ligaments, producing square appearing VBs with bridging syndesmophytes, the so-called "bamboo spine" or "poker spine". Extra-articular manifestations (EAMs) include: anterior uveitis, inflammatory bowel disease (IBD) and psoriasis.

Neurosurgical involvement usually results from the following:

1. cauda equina syndrome (CES-AS) : the etiology of CES in AS is frequently unclear, but is usually not due to stenosis or compressive lesion. Onset is slow and insidious and there is a high incidence of dural ectasia.[16] Any patient with AS and neurologic deficit should be assumed to have CES until proven otherwise. Without treatment most patients' neurological status continues to deteriorate[16]
2. rotatory subluxation: at occipitoatlantal and atlantoaxial joints. May occur as these are typically the last mobile segments of the spine. Incidence is much less than with rheumatoid arthritis. Lesions that might be stable in otherwise normal spines are often not stable in AS
3. myelopathy secondary to bow-stringing of the cord: laminectomy may aggravate

4. acute spinal cord injury (SCI): risk of SCI or CES due to fracture is increased in AS, and may occur following minimal trauma. Injuries are more common in the lower cervical spine. The akylosed spine of AS when fractured creates long lever arms, limiting the ability to absorb impact and rendering even minor fractures very unstable.[17] Delayed deterioration may be due to spinal epidural hematoma.[18]
5. Andersson lesion: discovertebral lesion that results from inflammation or fracture, mechanical stresses prevent lesion from fusion resulting in pseudarthrosis.[19]
6. spinal deformity
7. spinal stenosis: rare
8. basilar impression

74.2.2 Epidemiology

Incidence in the general population is ≈ 0.44-7.3 cases per 100,000.[20] Traditionally reported male: female ratio 3:1, however, likely stems from underdiagnosis in women, and more rapid progression of spinal ankylosis in men.[20] Peak incidence: 17-35 yrs age. > 90% of patients with AS are HLA-B27 positive (only 8% of people without AS have this antigen), but only 2% of people with HLA-B27 develop clinical AS. Although AS is not hereditary, first degree relatives are at increased risk.

74.2.3 Clinical

■ Symptoms. Typical initial presentation is with nonradiating low back pain, morning back stiffness, hip pain and swelling (due to large joint arthritis), exacerbated by inactivity and improved with exercise.[21]

■ Signs. Patrick's test (p. 1048) usually positive. Compressing the pelvis with the patient in the lateral decubitus position produces pain.

Schober test (measure distraction between skin marks on the back before and after forward flexion to detect reduced mobility of the spine due to fusion). Is not specific for inflammatory spondylopathies[22] but may be helpful for monitoring ongoing physical therapy.

74.2.4 Diagnosis

Diagnosis by experienced rheumatologist is closest thing to a gold standard.[23] The Assessment of SpondyloArthritis international Society (ASAS) recently presented its recommendations for a modified Berlin Algorithm[23] as a potentially useful tool for rheumatologists in diagnosing AS. SI joint involvement is the sine qua non for definite diagnosis. Diagnosis is very involved, and includes: chronic low back pain, buttock pain, sacroiliitis, family history, psoriasis, an inflammatory bowel disease or an arthritis followed in ≤ 1 month with urethritis, cervicitis or acute diarrhea, an enthesopathy, and a family history, and positive x-rays. The (obsolete) New York Criteria (■ Table 74.1) are shown here to provide a small insight into early attempts to establish diagnostic criteria, but should no longer be used for definitive diagnosis.

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74.2.5 Radiographic evaluation

■ Plain x-rays. Vital for diagnosis and follow-up. Sacroiliac (SI) joint involvement (on AP pelvic x-rays or on oblique views through the plane of the SI joints) is one of the earliest findings, and the often symmetric osteoporosis followed by sclerosis is characteristic. "Bamboo spine" (see above) is also classic. X-ray of the entire spine is recommended since multiple, non-contiguous (and often unsuspected) fractures are not unusual. See ■ Fig. 74.1.

■ CT. Useful for diagnosis of cervical fracture not apparent on plain radiographs, and in preoperative assessment of bony anatomy.

■ MRI. Can rule out spinal epidural hematoma and the occasional herniated disc. May demonstrate dural ectasia in cases of CES-AS syndrome. Andersson lesions: pathologic changes at ligament insertion sites (MRI signal abnormalities at front and back of the endplates) are characteristic. Erosive changes due to pseudarthrosis at the disc space can mimic discitis (high signal on T1WI & T2WI with enhancement).

■ Bone scan. Ratio of uptake of SI joint to sacrum > 1.3:1 is suggestive of AS.

Table 74.1 Modified New York Criteria for AS[24] – not to be used for diagnosis (obsolete)
Diagnosis (see criteria below)
Definite AS: radiologic criterion + ≥ 1 clinical criteria
Probable AS: radiological criterion without clinical criteria, or 3 clinical criteria without radiological criterion
Clinical criteria
low back pain > 3 months, improved by exercise, not relieved by rest
limitation of lumbar spine motion in both sagittal and frontal planes
limitation of chest expansion relative to normal values for age and sex
Radiological criterion
sacroiliitis

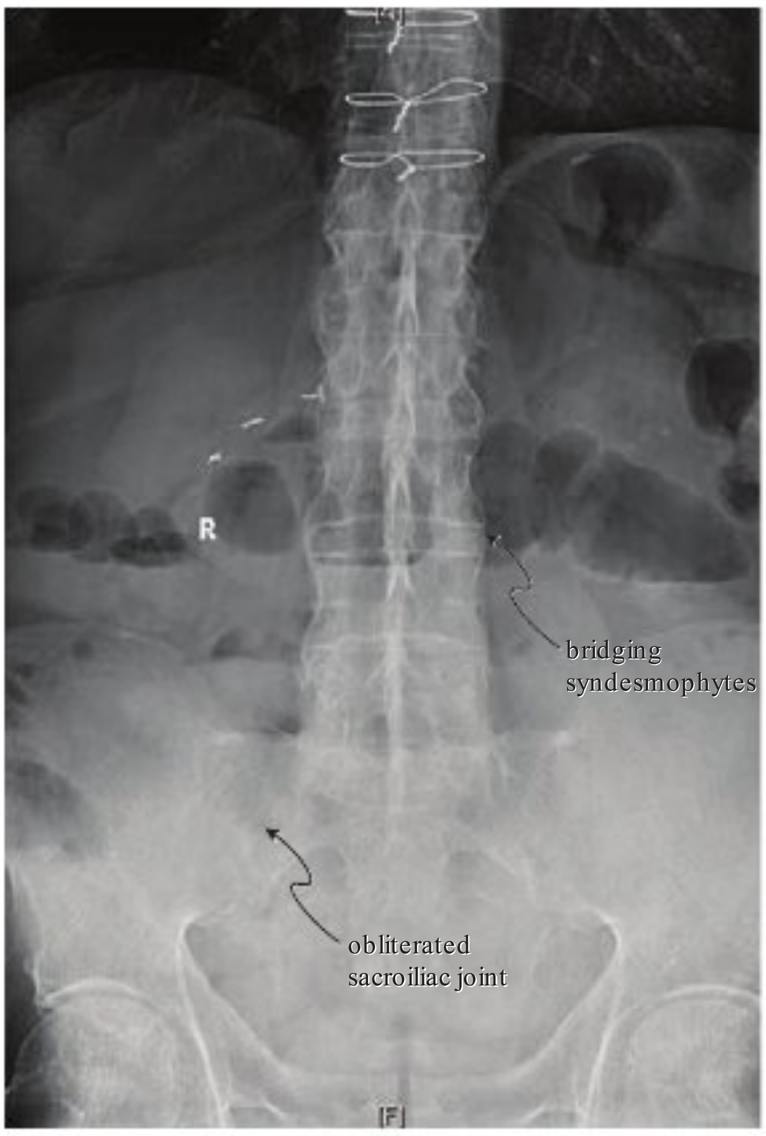

Fig. 74.1 AP lumbar/pelvic x-ray demonstrating "Bamboo spine" and sclerosis of sacroiliac joints

74.2.6 Differential diagnosis

1. early on, AS may resemble rheumatoid arthritis. However, in AS nodules do not form in joints, and rheumatoid factor is absent in the serum
2. metastatic prostate Ca in elderly male patients with sacroiliac pain and blastic changes compatible with sacroiliitis
3. Forestier's disease (p. 1130) and DISH (p. 1129): these overlapping conditions produce exuberant bony overgrowth anterior and lateral to the disc without degeneration and ossification of the disc as in AS. Both spare the facets and SI joints, do not produce flexion deformity, and tend to occur in men > 50 yrs old (older than typical AS)[25]
4. Psoriasis, Reactive Arthritis (Reiter's syndrome), Enteropathic (IBD-related) Arthritis: the spondylitis with these tends to be milder and less uniform, and SI joint involvement is asymmetrical. Cutaneous findings (erythema nodosum, and pyoderma gangrenosum) are absent in AS.[26]

74.2.7 Natural history

Progression is slow, and patients usually remain functionally active. Thoracic kyphosis with compensatory increase in cervical and lumbar lordosis is common. The shift in center of gravity together with spine stiffness and fragility predisposes to frequent falls and further spine injuries. Eventually can progress to involvement of costovertebral joints resulting in restrictive lung disease pattern. AS patients are also predisposed to development of fibrotic lung disease in late stages.

74.2.8 Treatment

General information

Combined ASAS/EULAR recommendations for management of AS are the most comprehensive currently available (copy available at ASAS website). Multidisciplinary, coordinated by a rheumatologist.[27] Goal of treatment is long term quality of life through symptomatic control and prevention of progressive structural damage. NSAIDs are first line pharmacological management.[27] Management of the disease itself may involve TNF inhibitors in patients with persistently high disease activity.[27]

Surgical treatment

General information

The most common surgical intervention is orthopedic total hip arthroplasty.[27]

Cervical fracture

The cervical spine is the most frequent fracture site in AS patients.[28] Patients often cannot distinguish acute fracture pain from chronic inflammatory pain; therefore there should be a low threshold to obtain imaging. It is imperative to determine pre-injury alignment as application of C-collar may cause hyperextension injury.[29] Gentle, low-weight traction with the force vector directed anteriorly and superiorly may be used for initial stabilization.[30,31] Halo vest or surgery for unstable fracture pattern.

Surgical indications:

- irreducible deformity
- deteriorating neurological status in the presence of epidural hematoma or other source of compression[32]
- unstable fracture (e.g. most 3-column fractures which are very unstable as a result of long lever arms of fused segments above and below): halo-vest immobilization is becoming a less frequently employed option for this

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Procedures: Decompressive laminectomy if evidence of spinal cord compression.[32] Given poor bone quality and extended lever arms, good fusion bed and a construct extending multiple levels above and below the fracture are critical.[33] Proximally, lateral mass screws up to C3, possible pedicle screw in C2, and pedicle screws in thoracic spine for distal fixation.[33] 360° fusion may provide optimal stabilization in some cases (where feasible).

Thoracolumbar fracture

The majority occur at thoracolumbar junction.[33] Can be divided into 3 types[34]:

1. shearing injury: typically acute. Resembles chance fracture, highly unstable 3 column injury[34]
2. wedge compression: typically chronic
3. pseudarthrosis: typically subacute previously missed fracture

Wedge compression or pseudarthrosis; rule out posterior element involvement to determine if fracture is unstable.[33] Stable fracture can be treated with external orthosis. Unstable fractures, consider thicker rods or more rigid rod material to account for increased forces across fracture, and PMMA augmentation to prevent screw pullout.

Kyphotic deformity

ASAS/EULAR recommendations include corrective osteotomy for severe disabling deformity.[27] Can be accomplished via open wedge osteotomy, polysegmental wedge oseotomy, or closed wedge osteotomy (lowest complication rate).[35] Cervical deformity is most commonly addressed with wedge osteotomy at C7 and T1 given absence of vertebral artery in foramen transversarium at these levels. Current trend in literature is to address deformity simultaneously with acute fracture fixation.[31,32]

Cauda equina syndrome

Although evidence is limited, in the absence of demonstrable neural compression, a lumbo-peritno-neal (LP) shunt may provide the best chance of improving neurologic dysfunction or halting progression of neurologic deficit.[36]

Surgical considerations

- Anesthesia team should be aware of Kyphotic deformity and fracture location: nasotracheal or fiberoptic intubation to prevent hyperextension of the neck and exacerbation of neurologic injury.[33]
- Extensive pre-op evaluation: fracture pattern, posterior ligamentous restraint, neurologic compression and function, preexisting bone quality.[33]
- Surgical positioning modified to account for preexisting deformity; support in all regions to prevent hyperextension and exacerbation of neurologic injury.[33]
- ICBG is gold standard; however, often a significant pain source potentially limiting mobilization and increasing likelihood of stasis sequelae (e.g. DVT), consider allograft.[32]
- Extensive knowledge of lateral mass and pedicle anatomy to ensure adequate hardware placement despite distorted bony anatomy and obscuring of typical landmarks.[32]
- Post-op immobilization via halo-vest or TLSO.[33]
- Expedited mobilization out-of-bed as AS patients are predisposed to pulmonary complications.[33]
- Plastic surgery to help manage skin necrosis and wound closure.[33]

74.3 Ossification of the posterior longitudinal ligament (OPLL)

74.3.1 General information

Key concepts

- fibrosis followed by calcification and then ossification of the posterior longitudinal ligament. The process may involve the dura
- more common in Asian population
- most patients have only mild subjective complaints
- 50% of patients have impaired glucose tolerance, respiratory compromise may result from ossification of the costotransverse and costovertebral ligaments
- surgery is best for moderate neuro involvement (Nurick grade 3 & 4)

The age of patients with OPLL ranges from 32–81 years (mean = 53), with a slight male predominance. The prevalence increases with age. Duration of symptoms averages ≈ 13 months. It is more prevalent in the Japanese population (2–3.5%).[37,38]

74.3.2 Pathophysiology

The pathologic basis of OPLL is unknown, but there is an increased incidence of ankylosing hyperostosis which suggests a hereditary basis.

OPLL begins with hypervascular fibrosis in the PLL which is followed by focal areas of calcification, proliferation of periosteal cartilaginous cells and finally ossification.[39] The process frequently extends into the dura. Eventually active bone marrow production may occur. The process progresses at varying rates among patients, with an average annual growth rate of 0.67 mm in the AP direction and 4.1 mm longitudinally.[40]

When hypertrophied or ossified, the posterior longitudinal ligament may cause myelopathy (due to direct spinal cord compression, or ischemia) and/or radiculopathy (by nerve root compression or stretching).

Changes within the spinal cord involve the postero-lateral gray matter more than white matter, suggesting an ischemic basis for the neurologic involvement.

74.3.3 Distribution

Average involvement: 2.7–4 levels. Frequency of involvement:
1. cervical: 70–75% of cases of OPLL. Typically begins at C3–4 and proceeds distally, often involving C4–5 and C5–6 but usually sparing C6–7
2. thoracic: 15–20% (usually upper, ≈ T4–6)
3. lumbar: 10–15% (also usually upper, ≈ L1–3)

74.3.4 Pathologic classification

See reference.[41]
1. segmental: confined to space behind vertebral bodies, does not cross disc spaces
2. continuous: extends from VB to VB, spanning disc space(s)
3. mixed: combines elements of both of the above with skip areas
4. other variants: includes a rare type of OPLL that is contiguous with the endplates and is confined to the disc space (involves focal hypertrophy of the PLL with punctate calcification)

74.3.5 Clinical

Most patients are asymptomatic, or have only mild subjective complaints. This is probably explained by the protective effect of the fusion resulting from OPLL and the very gradual compression.

Natural history: 17% of patients without myelopathy developed myelopathy in one study[42] over 1.6 years mean follow-up. Statistically, the myelopathy-free rate in patients without initially presenting with myelopathy was 71% after 30 years.[42]

74.3.6 Evaluation

Plain x-rays

Often fail to demonstrate OPLL. Flexion/extension views may be helpful in assessing stability.

MRI

OPLL appears as a hypointense area and is difficult to appreciate until it reaches ≈ 5 mm thickness. On T1WI it blends in with the hypointensity of the ventral subarachnoid space; on T2WI it remains hypointense while the CSF becomes bright. Sagittal images may be very helpful in providing an overview of the extent of involvement, and T2WI may demonstrate intrinsic spinal cord abnormalities which may be associated with a worse outcome.

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Myelography/CT

Myelography with post-myelographic CT (especially with 3D reconstructions) is probably best at demonstrating and accurately diagnosing OPLL.

74.3.7 Treatment

Treatment decisions

Based on clinical grade[41] as follows:
1. Class I: radiographic evidence without clinical signs or symptoms. Most patients with OPLL are asymptomatic.[38] Conservative management unless severe
2. Class II: patients with myelopathy or radiculopathy. Minimal or stable deficit may be followed expectantly. Significant deficit or evidence of progression warrants surgical intervention
3. Class IIIA: moderate to severe myelopathy. Usually requires surgical intervention
4. Class IIIB: severe to complete quadriplegia. Surgery is considered for incomplete quadriplegics showing progressive slow worsening. Rapid deterioration or complete quadriplegia, advanced age or poor medical condition are all associated with worse outcome

In moderate grade patients (Nurick grades 3 & 4,[43] (see Table 74.2), surgery provided a statistically significant reduction in deterioration. There was no difference between surgery and conservative treatment in mild grade (Nurick 1 or 2), and surgery was ineffective in severe grade (Nurick 5).[42]

Table 74.2 Nurick grade of disability from cervical spondylosis[43]	
Grade	Description
0	signs or symptoms of root involvement without myelopathy
1	myelopathy, but no difficulty in walking
2	slight difficulty in walking, able to work
3	difficulty in walking but not needing assistance, unable to work full-time
4	able to walk only with assistance or walker
5	chairbound or bedridden

Pre-op assessment

Appropriate cardiorespiratory assessment should be made knowing that:

1. respiratory compromise may result from ossification of the costotransverse and costovertebral ligaments
2. 50% of patients have impaired glucose tolerance with the attendant risks associated with diabetes

Technical considerations for surgery

Severe OPLL increases the risk of spinal cord injury during neck positioning for intubation, and strong consideration should be given to awake nasotracheal intubation.

An anterior approach is generally favored, although laminectomy may be acceptable. SSEP monitoring has been recommended by some.[39] Distraction should be avoided until the spinal cord has been decompressed from the OPLL.

Some authors advocate complete removal of bone from the dura, while others feel it is permissible to leave a thin rim of bone adherent to the dura. Care must be taken in removing bone because it tends to blend imperceptibly with dura and the next thing one may see is bare spinal cord.

Depending on the distance of vertical involvement, vertebral corpectomy with strut grafting may be required. Internal plate fixation is often used as an adjunct. Post-operative immobilization for at least 3 months is employed with rigid collars for single level ACDF or 1–2 level corpectomies, or halo-vest traction for corpectomies > 2 levels.

74.3.8 Results with surgery

The incidence of pseudarthrosis after vertebral corpectomy and strut graft ranges from 5–10% and increases with the number of levels fused.

In one series there was a 10% incidence of transient worsening of neurologic function following anterior surgery[40] which may have been related to distraction.

The risk of dural tear with CSF leak following an anterior approach depends on the aggressiveness with which bone is removed from the dura, and ranges ≈ 16–25%

Other risks of anterior approaches, e.g. esophageal injury (p. 1074), also pertain.

74.4 Ossification of the anterior longitudinal ligament (OALL)

OALL of the cervical spine and/or hypertrophic anterior cervical osteophytes may produce dramatic radiographic findings and minimal clinical symptoms. Distinct from Forestier's disease (see below). Cervical involvement may produce dysphagia.[44]

74.5 Diffuse idiopathic skeletal hyperostosis (DISH)

Key concepts

- usually asymptomatic, but may present with globus
- W/U: ☐ speech therapy consult for dysphagia evaluation (usually includes ☐ modified barium swallow), ☐ CT of cervical spine, ± ☐ digital video esophagoscopy

AKA “DISH”, AKA spondylitis ossificans ligamentosa, AKA ankylosing hyperostosis, among others. A condition characterized by flowing osteophytic formation of the spine in the absence of degenerative, traumatic, or post-infectious changes. Affects Caucasians and males more commonly, and usually seen in patients in their mid-60s.

97% of cases occur in the thoracic spine, also in the lumbar spine in 90%, cervical spine in 78%, and all three segments in 70%. Sacroiliac joints are spared, unlike ankylosing spondylitis (AS) (p.1123). As with AS, unfused levels may be very unstable.

Risk factors for DISH include: elevated body mass index,[45] elevated serum uric acid,[45] diabetes mellitus,[45] elevated growth hormone or insulin levels.[46]

Usually does not produce clinical symptoms. Patients may have early morning stiffness and mild limitations of activities. Cervical involvement may present with dysphagia or globus pharyngis (the subjective sensation of a lump in the throat, not to be confused with globus hystericus which refers to the sensation of a lump in the throat where there is no identifiable pathology) due to compression of the esophagus between the osteophytes and the rigid laryngeal structures[47] (part of Forestier’s disease[48]).

Plain x-rays and CT scan demonstrate the pathology. In cases of dysphagia, evaluation should include speech therapy consult for dysphagia evaluation, barium swallow to help localize the site of obstruction, and DVE (digital video esophagoscopy) to rule-out intrinsic esophageal disease.

In cases of dysphagia, evaluation should include speech therapy consult for dysphagia evaluation, barium swallow to help localize the site of obstruction, and DVE (digital video esophagoscopy) to rule-out intrinsic esophageal disease. Plain x-rays and CT scan helps demonstrate the pathology. Cases that do not respond satisfactorily to dietary modifications in patients who are losing weight or are having recurrent episodes of choking or pneumonia should be considered for surgery. An anterior cervical approach, and utilization of a high-speed drill with careful protection of soft-tissue structures (esophagus, carotid sheath) without need for discectomy nor spine stabilization has been recommended.[47] Patients need to be made aware that post-op they are likely to be worse initially (from manipulation of esophagus and possibly disruption of some of the autonomic innervation of the esophagus) and will probably need a gastrostomy feeding tube. By 1 year post-op there may be some improvement.

74.6 Scheuermann’s kyphosis

74.6.1 General information

AKA Scheuermann juvenile kyphosis AKA Scheuermann disease AKA juvenile osteochondrosis of the spine.

Definition: anterior wedging of at least 5° of ≥ 3 adjacent thoracic vertebral bodies.

Other findings include: Schmorl nodes (p.1060) and endplate narrowing.

74.6.2 Presentation

Adolescents: often present as a result of the cosmetic deformity associated with progressive kyphosis which may be mistaken for “slouching.”

Adults: often present with pain.

74.6.3 Evaluation

Patients require standing scoliosis x-rays and a careful neurologic exam.

The role of MRI is controversial, and not all practitioners obtain this.

74.6.4 Radiographic findings

Anterior wedge deformities at multiple levels. End plate irregularities and Schmorl’s nodes.

74.6.5 Treatment

Bracing may be used in adolescents, unless the patient meets surgical criteria listed below.

Adults presenting with pain often respond to nonsurgical treatment including: physical therapy and NSAIDs.

Surgery is generally indicated for:
1. Curves exceeding 70–75°
2. Unacceptable cosmetic appearance
3. Refractory pain
4. Progressive kyphosis
5. Neurologic deficit

74.7 Spinal epidural hematoma

74.7.1 General information

Rare. Over 200 cases of varying etiology have been reported,[49] although one–third of recent cases have been associated with anticoagulation therapy.[50] NSAIDs may also be a risk factor.[51] Etiologies include:
1. traumatic: including following LP or epidural anesthesia,[49,52,53,54] fracture (see below), spinal surgery[55] or chiropractic manipulation.[56] Occurs predominantly in patient who is: anticoagulated,[57] thrombocytopenic, or has bleeding diathesis or a vascular lesion
2. spontaneous[58]: rare. Etiologies: hemorrhage from spinal cord AVM (p.1140), from vertebral hemangioma (p.794) or tumor

May occur at any level of the spine, however, thoracic is most common. Most often located posterior to spinal cord (except for hematomas following anterior cervical procedures), facilitating removal via laminectomy.[50]

74.7.2 Traumatic spinal epidural hematoma (TSEH) associated with spine fracture

In one series,[59] among 74 trauma patients who underwent emergent spinal MRI, ≈ half of the patients with spine fractures also had TSEH. Treatment was based solely on the fracture, and the outcome in patients with neurologic deficits was no worse in the group with TSEH than in the group without.

74.7.3 Presentation

The clinical picture of spontaneous spinal epidural hematoma is fairly consistent but nonspecific. Usually starts with severe back pain with radicular component. It may occasionally follow minor straining, and is less commonly preceded by major straining or back trauma. Spinal neurologic deficits follow, usually progressing over hours, occasionally over days. Motor weakness may go unnoticed when patients are bedridden with pain.

74.7.4 Treatment

Recovery of neurologic deficit without surgery is rare (only a handful of case reports in the literature[51]), therefore optimal treatment is immediate decompressive laminectomy in those patients who can tolerate surgery.[50] In one series, most patients who recovered underwent decompression within 72 hrs of onset of symptoms.[60] In another, decompression within 6 hours was associated with better outcome.[55]

High-risk patients: for medically high-risk patients (e.g. acute MI) on anticoagulation, surgical mortality and morbidity is extremely high, and this must be considered when making the decision of whether or not to operate. In patients not operated, anticoagulants should be stopped, and reversed if possible; see Correction of coagulopathies or reversal of anticoagulants (p.166). Consider use of high dose methylprednisolone to minimize cord injury; see Methylprednisolone under spinal cord injury (p.951). Percutaneous needle aspiration may be a consideration in high-risk patients.

74.8 Spinal subdural hematoma

Rare. May be posttraumatic (including iatrogenic causes) or may occur spontaneously. Spinal subdural hematomas (SSH) that occur spontaneously or following lumbar puncture usually occur in patients with coagulopathies (primary or iatrogenic).[61]

Conservative treatment is possible in nontraumatic SSHs with minimal neurologic impairment.[61]

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74

75 Other Non-Spine Conditions with Spine Implications

75.1 Rheumatoid arthritis

75.1.1 General information

More than 85% of patients with moderate or severe rheumatoid arthritis (RA) have radiographic evidence of C-spine involvement.[1]

The grading system of Ranawat et al.[1] for myelopathy is shown in Table 75.1 is used in RA as well as some other etiologies of myelopathy.

75.1.2 Cervical spine involvement in RA

Common involvement

1. upper cervical spine: involved in 44–88% of RA cases[2] (often found together):
   a) anterior atlantoaxial subluxation: the most common manifestation of RA in the cervical spine, found in up to 25% of patients with RA (see below)
   b) basilar impression (BI): upward translocation of the odontoid process, found in ≈ 8% of patients with RA (p.1138)
   c) pannus of granulation tissue: forms around the odontoid
2. subaxial C-spine (i.e. below C2) subluxation (p.1138)

Less common involvement of the cervical spine in RA

1. posterior subluxation of the atlantoaxial joint: must have either associated fracture of or near total arthritic erosion of odontoid
2. vertebral artery insufficiency secondary to changes at the cranio-cervical junction[3]

75.1.3 Atlantoaxial subluxation (AAS) in RA

General information

Inflammatory involvement of the atlantoaxial synovial joints causes erosive changes in the odontoid process (anteriorly at the synovial joint with the C1 arch, and posteriorly at the synovial joint with the transverse ligament) and decalcification and loosening of the insertion of the transverse ligament on the atlas. These changes lead to instability allowing a scissoring effect with anterior subluxation of C1 on C2. AAS occurs in ≈ 25% of patients with RA.[3] Mean time between onset of RA symptoms to the diagnosis of AAS in 15 patients: 14 years.[4]

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Clinical

Signs and symptoms of AAS are shown in Table 75.2.

AAS is usually slowly progressive. Mean age at onset of AAS symptoms: 57 years.

Pain is experienced locally (upper cervical and suboccipital regions, often from compression of C2 nerve root) or is referred (to mastoid, occipital, temporal, or frontal regions).

VBI may occur from VA involvement (p.1305).

Table 75.1 Ranawat classification of myelopathy

Class	Description
I	no neural deficit
II	subjective weakness + hyperreflexia + dysesthesia
III	objective weakness + long tract signs III A = ambulatory III B = quadriparetic & non ambulatory

Table 75.2 Signs and symptoms of AAS[a] (15 patients with AAS[4])

Finding	%
pain	
• local	67%
• referred	27%
hyperreflexia	67%
spasticity	27%
paresis	27%
sensory disturbance	20%
[a]other possible findings not reported in this series: clumsiness, neurogenic bladder, Babinski sign	

Radiographic evaluation

General information

The magnitude of AAS is usually increased with neck flexion.

Lateral C-spine x-ray

Both the ADI and PADI below are surrogate markers for instability and for spinal cord compression. With the availability of MRI, the ability to directly assess spinal cord compression has diminished the usefulness of these measures, especially for that aspect.

Anterior atlantodental interval (ADI)

The ADI (p.213) only gives information about the stability of the C1–2 joint. The normal ADI in adults is < 3–4 mm.[5,6] Widening of the ADI suggests possible incompetence of the transverse ligament. However, the ADI does not correlate with the risk of neurologic injury[7,8] and is not predictive of progression from asymptomatic AAS to symptomatic AAS.

Posterior atlantodental interval (PADI)

The amount of room available for the spinal cord can vary for any given ADI depending on the AP diameter of the spinal canal and the thickness of any pannus. The PADI (p.213) and the AP diameter of the subaxial canal measured on a lateral C-spine x-ray correlates with the presence and severity of paralysis.[7]

The PADI also predicts neurologic recovery following surgery. Patients with paralysis from AAS showed no recovery if the pre-op PADI was < 10 mm.[7]

PADI ≤ 14 mm has been proposed as an indication for surgical stabilization.

MRI

The optimal test to evaluate the source and magnitude of upper cord or medulla compression. Demonstrates location of odontoid process, extent of pannus, and effects of subluxation (may need to be performed with head flexed to evaluate this).

Treatment

General information

Requires knowledge of the following information:

1. natural history: AAS in most patients progresses, with a small percentage either stabilizing or fusing spontaneously. In one series[9] with 4.5 years mean follow-up, 45% of patients with 3.5–5 mm subluxation progressed to 5–8 mm, and 10% of these progressed to > 8 mm
2. once myelopathy occurs, it may be irreversible
3. the worse the myelopathy, the higher the risk for sudden death
4. the chances of finding myelopathy are significantly increased once the subluxation reaches ≥ 9 mm[10]

5. associated cranial settling further decreases the tolerance for AAS
6. the life expectancy of patients with RA is 10 years less than the general population[9]
7. the morbidity and mortality of surgical treatment (see below)
8. pannus may regress some after medical treatment

When to treat?

1. symptomatic patients with AAS: almost all require surgical treatment (C1–2 fusion in most cases). For management, see below. Some surgeons do not operate if the maximal dens-C1 distance is < 6 mm
2. asymptomatic patients: controversial
   a) some authors feel surgical fusion is not necessary in asymptomatic patient if the dens-C1 distance is below a certain cutoff. Recommendations for this cutoff have ranged from 6 to 10 mm,[11] with 8 mm commonly cited (an unvalidated delineation)
   b) these patients are often placed in a rigid cervical collar, e.g. while outside the home, even though it is generally acknowledged that a collar probably does not provide significant support or protection
   c) NB: some cases of sudden death in previously asymptomatic RA patients may be due to AAS and may then be erroneously attributed to cardiac arrhythmias, etc.[12]

Surgical management

It is necessary to either reduce the subluxation or to decompress the upper cord before doing a C1-C2 or occipital-C1-C2 fusion.

Menezes assesses all subluxed patients for reducibility using MRI compatible Halo cervical traction as follows: start with 5 lbs, and gradually increase over a period of a week. Most cases reduce within 2–3 days. If not reduced after 7 days then it is probably not reducible. Only ≈ 20% of cases are not reducible (most of these have odontoid > 15 mm above foramen magnum).

Most require stabilization via posterior wiring and fusion, either of C1 to C2, or of occiput to C2. The latter is used when fusion is combined with decompression (posterior laminectomy of C1 with posterior enlargement of the foramen magnum). See Atlantoaxial fusion (C1–2 arthrodesis) (p. 1479).

Posterior fusion alone does not provide adequate relief if the subluxation is irreducible, or if pannus causes significant compression (however, there may be some reduction of pannus after fusion). In these cases, transoral odontoidectomy may be indicated. Performing the posterior stabilization and decompression first allows some patients to avoid a second operation, and permits the remainder to undergo the anterior approach without becoming destabilized. Still, some surgeons do the odontoidectomy first[11] (requires the patient to remain in traction until the fusion).

Reminder: the patient must be able to open the mouth greater than ≈ 25 mm in order to perform transoral odontoidectomy without splitting the mandible.

Posterior fusion

See Atlantoaxial fusion (C1–2 arthrodesis) (p. 1479) for technique. In RA, erosion and osteoporosis weakens the C1 arch, and extra care is needed to avoid fracturing it.

75.1.4 Surgical morbidity and mortality

Because of the frequency of simultaneous involvement of other systems in RA including pulmonary, cardiac, and endocrine, operative mortality ranges from 5–15%[11]

The non-fusion rate for C1–2 wiring and fusion has been reported as high as 50%,[13] typical rates are lower (with 18% of patients in one series developing a fibrous union[11]). The most common site of failure of osseous fusion is the interface between the bone graft and the posterior arch of C1.[14]

75.1.5 Post-operative care

The patient is usually mobilized almost immediately post-op in halo vest traction (some use an optional period of maintained traction before mobilization). Impaired healing in RA dictates that the Halo be worn until fusion is well established, as seen on x-ray (usually 8–12 weeks). Sonntag evaluates the patient with flexion-extension lateral C-spine x-rays by disconnecting the halo ring from the vest.

75.1.6 Basilar impression in rheumatoid arthritis

General information

AKA atlantoaxial impaction. Erosive changes in the lateral masses of C1 → telescoping of the atlas onto the body of C2 causing ventral migration of C1 with resultant ↓ in AP diameter of the spinal canal. There is concomitant upward displacement of the dens. The posterior arch of C1 often protrudes superiorly through the foramen magnum. All of these factors lead to compression of the pons and medulla. Rheumatoid granulation tissue behind the odontoid also contributes to the brainstem compression. Vertebral artery and/or anterior spinal artery compression may also cause neurologic dysfunction.

The degree of erosion of C1 correlates with the extent of odontoid invagination.

Clinical

See Table 75.3 for signs & symptoms.

Pain may occur as a result of compression of the C1 and/or C2 nerve roots. Compression of the medulla can cause cranial nerve dysfunction.

Motor exam usually difficult because of severe polyarticular degeneration and associated pain. Sensory findings (all non-localizing): diminished vibratory, position, and light touch.

Radiographic evaluation

See Basilar invagination & basilar impression (BI) (p.217) for radiographic criteria of BI. Erosion of the tip of the odontoid, commonly seen in RA, obviates use of any measurement that is based on the location of the tip of the odontoid.[15] For this reason, other measures have been developed, including the Clark station,[14] Redlund-Johnell criteria,[16] and Ranawat criteria.[1] Since even these methods will miss up to 6%of cases of BI in RA,[15] it is recommended that suspicious cases be investigated further (e.g. with CT and/or MRI).

MRI: optimal for demonstrating brain stem impingement, poor for showing bone. Cervicomedullary angle: the angle between a line drawn through the long axis of the medulla on a sagittal MRI

Table 75.3 Symptoms and signs of BI (45 patients with RA[2])

Finding	%
headache	100%
progressive difficulty ambulating	80%
hyperreflexia + Babinski	80%
limb paresthesias	71%
neurogenic bladder	31%
cranial nerve dysfunction	22%
• trigeminal nerve anesthesia	20%
• glossopharyngeal	
• vagus	
• hypoglossal	
miscellaneous findings	
• internuclear ophthalmoplegia	
• vertigo	
• diplopia	
• downbeat nystagmus	
• sleep apnea	
• spastic quadriparesis	

and a line drawn through the cervical spinal cord. The normal CMA is 135–170°. A CMA < 135° correlates with signs of cervicomedullary compression, myelopathy or C2 radiculopathy.[17]

CT: primarily done to assess bony anatomy (erosion, fractures…).

CTA should be performed when surgery is contemplated, to show detail of VA anatomy.

Myelography (water soluble) with CT: also good for delineating bony pathology.

Treatment

See also Craniocervical junction and upper cervical spine abnormalities (p.1151).

Cervical traction

May attempt with Gardner-Wells tongs. Begin with ≈ 7 lbs, and slowly increase up to 15 lbs. Some may require several weeks of traction to reduce.

Surgery

Reducible cases: posterior occipitocervical fusion ± C1 decompressive laminectomy.

Irreducible cases: requires transoral resection of odontoid. May perform before posterior fusion (but then must be kept in traction while waiting for posterior fusion).

75.1.7 Subaxial subluxation in rheumatoid arthritis

The direct effects of RA on the subaxial spine involves the facet joints posteriorly. Degenerative disc disease, which is generally a late manifestation in RA, is not the result of synovitis.[18] Involvement is most common at C2–3 and C3–4.

75.2 Down syndrome

75.2.1 General information

Down syndrome is associated with ligamentous laxity of the spine. This has implications whenever a fusion is contemplated, as adjacent segment failure with kyphosis is very common. Ligamentous laxity may also result in atlantoaxial subluxation (AAS).

75.2.2 Atlantoaxial subluxation (AAS) in Down syndrome

General information

Not all cases of AAS are unstable (an unstable spine, by definition, needs treatment).

Incidence of AAS in Down syndrome (DS) is 20%,[19] but only 1–2% of DS patients have symptomatic AAS.[20] AAS in DS appears to be due to laxity of the transverse atlantal ligament (TAL). This laxity may decrease with age as the TAL stiffens.

Management

Controversial. There have been position statements[21] and rebuttals.[20,22]

Recommendations (modified[23]); ADI = atlantodental interval (p.213), PADI = posterior atlantodental interval (p.213):

1. children who have been screened and do not have AAS: no further screening after age 10 years (since AAS does not develop later; the cutoff age is controversial)
2. os odontoideum: surgical fusion
3. symptomatic AAS
   a) symptoms may include: gait difficulties, neck pain, limited neck motion, torticollis, clumsiness, sensory deficits, and other symptoms of myelopathy
   b) for ADI > 4.5 mm or PADI < 14 mm or spinal cord damage on cervical MRI: surgical fusion
4. asymptomatic AAS seen on lateral C-spine x-ray:
   a) for ADI ≤ 4.5 mm and PADI ≥ 14 mm: no need for further testing
   b) for ADI > 4.5 mm or PADI < 14 mm: cervical MRI
      - if the MRI shows spinal cord damage: surgical fusion
      - if MRI shows no spinal cord damage: surgical fusion is optional. If fusion is not done, prohibit high-risk activities and restudy in 1 year

75.3 Morbid obesity

Morbid obesity, defined as a body mass index (BMI) > 40. Approximately doubles risks (13.6% vs. 6.9%) of complications of all types (cardiac, renal, pulmonary, wound complications...)[24] with spine surgery. Mortality is tripled (but is still low, 0.41 in morbid obese vs. 0.13).[24] Hospital costs and length of stay is also longer.

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76 Special Conditions Affecting the Spinal Cord

76.1 Spinal vascular malformations

76.1.1 General information

Often also referred to by the term spinal AVMs which technically refers to a subset of spinal vascular malformations (SVMs). Incidence of SVM is about 4% of primary intraspinal masses. 80% occur between age 20 and 60 years.[1 (p 1850–3)]

76.1.2 Classification

For a review of the history of classification systems, see the excellent review by Black[2]).

There are three current era classification systems.

The "American/English/French Connection" classification

References for classification: include[2,3,4,5,6,7,8,9,10]

1. Type I: dural AVM AKA AV-fistula (AVF). The most common type (80%) of SVM in the adult.[11] Fed by radicular artery which forms an AV shunt (fistula) at the dural root sleeve (located in the intervertebral foramen),[8] drains into an engorged spinal vein on posterior cord. Usually in lumbar or lower thoracic spine. Slow flow. High pressure in draining vein may cause venous congestion of the cord. Cord involvement may be distant to the fistula. Symptoms: LBP and progressive myeloradiculopathy or cauda equina syndrome (due to venous congestion) with urinary retention usually in middle-aged patients, 90%males. Up to 35%have pain. 15–20%are associated with other AVMs (cutaneous or other). Rarely bleed
   a) Type IA: single arterial feeder
   b) Type IB: two or more arterial feeders
2. intradural AVMs (high flow): 75%present with acute onset of symptoms, usually from hemorrhage (SAH or intramedullary)
   a) Type II: AKA spinal glomus AVM. Intramedullary. True AVM of the spinal cord. 15–20%of all SVMs. Compact nidus fed by medullary arteries with the AV shunt contained at least partially within the spinal cord or pia. May be associated with feeding artery aneurysms. Worse prognosis than dural AVM.[8] Fed by 1, or at most 2–3 feeders 80%of the time
   b) Type III: AKA juvenile spinal AVM. Essentially an enlarged glomus AVM which occupies the entire cross-section of the cord and invades the vertebral body which may cause scoliosis
   c) Type IV[7]: intradural perimedullary AVM (also called arteriovenous fistulae (AVF)). Direct fistula between artery supplying spinal cord (usually anterior spinal artery, often artery of Adamkiewicz) and draining veins. Typically occur in younger patients than Type I, and may present catastrophically with hemorrhage into the subarachnoid space.[12] Table 76.1 shows the 3 subtypes.[9]
3. miscellaneous spinal vascular lesions:
   a) spinal cord cavernomas
   b) spinal cord venous angiomas: extremely rare. Difficult to visualize angiographically
   c) vertebral body hemangiomas (p.794)

Hôpital Bicêtre classification

See reference.[13]

Table 76.1 Merland's subclassification of Type IV (perimedullary) AV fistulas[a]

Subtype	Arterial supply	AVF	Venous drainage
I	single (thin ASA)	single, small	slowly ascending perimedullary venous system
II	multiple (dilated ASA & PSA)	multiple, medium	slowly ascending perimedullary venous system
III	multiple (dilated ASA & PSA)	single, giant	giant venous ectasia, rapid metameric venous drainage

[a]AVF = arteriovenous fistula; ASA = anterior spinal artery; PSA = posterior spinal artery

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1. AVMs
2. fistulae: micro- or macrofistulae
3. genetic classification of spinal cord

AV shunts
a) genetic hereditary lesions: macrofistulae and hereditary hemorrhagic telangiectasias
b) genetic nonhereditary lesions: multiple lesions with metameric or myelomeric associations
c) single lesions: incomplete associations of categories a or b

Spetzler et al. classification

See reference.[14]

This system reincorporated vascular spinal neoplasms.

1. Neoplastic vascular lesions
   a) hemangioblastoma
   b) cavernous malformation
2. spinal aneurysms (rare)
3. arteriovenous lesions
   a) AVFs
      - extradural
      - intradural: dorsal or ventral
   b) AVMs
      - extradural-intradural
      - intradural
      - intramedullary
      - intramedullary-extramedullary
      - conus medullaris

76.1.3 Presentation

85% present as progressive neuro deficit (back pain associated with progressive sensory loss and LE weakness over months to years). Yet, SVMs account for < 5% of lesions presenting as spinal cord "tumors." 10–20% of SVMs present as sudden onset of myelopathy usually in patients < 30 yrs age,[15, 16] secondary to hemorrhage (causing SAH, hematomyelia, epidural hematoma, or watershed infarction). Coup de poignard of Michon = sudden excruciating back pain with SAH (clinical evidence of SVM).

Foix-Alajouanine syndrome (subacute necrotic myelopathy): acute or subacute neurologic deterioration in a patient with a SVM without evidence of hemorrhage. Presents as spastic → flaccid paraplegia, with ascending sensory level and loss of sphincter control. Initially thought to be due to spontaneous thrombosis of the AVM causing subacute necrotizing myelopathy[17] which would be irreversible. However, more recent evidence suggests that the myelopathy may be due to venous hypertension with secondary ischemia, and there may be improvement with treatment.[18]

Clinical. Auscultation over spine reveals a bruit in 2–3% of cases. Cutaneous angioma over back is present in 3–25%; valsalva maneuver may enhance the redness of the angioma.[16]

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76.1.4 Evaluation

Spinal angiography: necessary for treatment planning. Best performed at centers that do this study regularly. For Type I dural AVMs, angiography must encompass all dural feeders of the neuraxis, which includes:

1. ICAs: because of the artery of Bernasconi & Cassinari (p. 79)
2. every radicular artery including the artery of Adamkiewicz (p. 87)
3. internal iliac arteries: for sacral feeders

MRI: detects some SVMs with greater sensitivity and safety than angiography,[19] but is inadequate for treatment planning. 82% show extramedullary flow voids. Variable degree of cord enhancement (from venous congestion or venous infarction). Negative MRI does not rule out diagnosis.

Myelography: classically shows serpiginous intradural filling defects. Generally superseded by MRI. If done, patient should be imaged prone and supine (to avoid missing a dorsal AVM) Risk of bleeding from puncture of a dilated artery/vein with myelography needle.

76.1.5 Treatment

Type I (dural AVMs): usually require treatment. Usually amenable to endovascular techniques using glue, in which case the proximal vein must be taken as well. If you don't completely eliminate a dural fistula (spinal or intracranial) it will come back!

Type II (spinal glomus AVMs): may be amenable to interventional neuroradiologic procedures including embolization,[20] especially type IIA (single feeder). However recurrence may be higher with endovascular treatment than surgery, and surgery is often preferred for Type IIB (≥2 feeders). Surgical strategy: similar to intracranial AVMs except that the parenchyma cannot be retracted, bleeding is rarely life threatening, and arteries of passage must be preserved to avoid devastating deficits. Intraoperative ICG angio is often helpful. The nidus is compact, and the hemosiderin ring around the nidus on MRI often represents a plane that can be exploited.

Type III (juvenile spinal AVMS): the natural history is probably better than the prognosis with any type of treatment.

Type IV (perimedullary fistulae): suggested management[10] is shown in Table 76.2.

76.2 Spinal meningeal cysts

76.2.1 General information

Spinal meningeal cysts (SMC): diverticula of the meningeal sac, nerve root sheath or arachnoid. May have familial tendency.

Terminology in literature is confusing. One classification system is shown in Table 76.3. Previously AKA Tarlov's perineural cysts, spinal arachnoid cysts, and extradural diverticula, pouches or cysts. Only congenital lesions are considered here.

1. Type I SMCs above the sacrum usually have a pedicle adjacent to entrance of dorsal nerve root
2. Type II SMCs: formerly called Tarlov's cysts and were differentiated from nerve root diverticula because the former were defined as communicating with subarachnoid space, and the latter not. However, intrathecal contrast CT (ICCT) shows both communicate. Often multiple, occur on dorsal roots anywhere, but are most prominent and symptomatic in sacrum
3. Type III SMCs: may also be multiple and asymptomatic. More common along posterior subarachnoid space. Attributed to proliferation of arachnoid trabeculae

Table 76.2 Suggested management for Type IV arteriovenous fistulae[10]

Subtype	Diagnosis	Embolization	Surgery
Subtype I	difficult; ? reliability of MRI[a]; tomomyelography; angiotomomyelography	difficult	easy on filum terminale; difficult on conus medullaris
Subtype II	easy: MRI or myelography	incomplete occlusion	on posterolateral AVFs
Subtype III		effective	difficult, dangerous

[a]due to inaccuracy, do not delay angiogram to get MRA, etc.

Table 76.3 Types of spinal meningeal cysts[21]

Type	Description
Type I	extradural meningeal cysts without spinal nerve root fibers
IA	"extradural meningeal/arachnoid cyst"
IB	(occult) "sacral meningocele"
Type II	extradural meningeal cysts with spinal nerve root fibers ("Tarlov's perineural cyst", "spinal nerve root diverticulum")
Type III	spinal intradural meningeal cysts ("intradural arachnoid cyst")

76.2.2 Presentation

May be asymptomatic (i.e. incidental finding). May cause radiculopathy by pressure on adjacent nerve root (may or may not cause symptoms of nerve root from which it actually arises). Symptom complex depends on size of SMC, and proximity to spinal cord and nerve roots.

1. Type I SMCs: in thoracic and cervical region, may present with acute myelopathy (spasticity and sensory level); lumbar region → LBP and radiculopathy; sacral region → sphincter disturbance
2. Type II SMCs: often asymptomatic, but sacral lesions may → sciatica and/or sphincter disturbance
3. Type III SMCs: may also be multiple and asymptomatic; more common along posterior subarachnoid space

76.2.3 Evaluation

MRI to identify the mass, then water-soluble ICCT scan to evaluate communication of cyst with subarachnoid space.

1. Type II SMCs: all 18 cases had bony erosion (demonstrated by canal widening, pedicle erosion, foraminal enlargement, or vertebral body scalloping)
2. Type III SMCs: may also cause bony erosion; appear on myelogram as intradural defect, may not appear on ICCT if they communicate with subarachnoid space which causes them to blend with adjacent subarachnoid space

76.2.4 Treatment

1. Type I SMCs: close ostium between cyst and subarachnoid space. Above sacrum, can usually be dissected from dura; occasionally fibrous adhesions prevent this
2. Type II SMCs: no pedicle, thus either partially resect and oversew cyst wall, or excise cyst and involved nerve root. Simple aspiration is not recommended
3. Type III SMCs: excise completely unless dense fibrous adhesions prevent this, in which case marsupialize cyst. Tend to recur if incompletely excised

76.3 Juxtafacet cysts of the lumbar spine

76.3.1 General information

The term juxtafacet cyst (JFC) was originated by Kao et al.[22] in 1974 and includes both synovial cysts (those having a synovial lining membrane) and ganglion cysts (those lacking synovial lining) adjacent to a spinal facet joint or arising from the ligamentum flavum. Distinction between these two types of cysts may be difficult without histology (see below) and is clinically unimportant.[23]

JFC occur primarily in the lumbar spine (although cysts in the cervical[24,25,26] and thoracic[27] spine have been described). They were first reported in 1880 by von Gruker during an autopsy,[28] and were first diagnosed clinically in 1968.[29] The etiology is unknown (possibilities include: synovial fluid extrusion from the joint capsule, latent growth of a developmental rest, myxoid degeneration and cyst formation in collagenous connective tissue…), increased motion seems to have a role in many cysts, and the role of trauma in the pathogenesis is debated[25,30] but probably plays a role in a small number (≈ 14%).[31] JFC are relatively rare, only 3 cases were identified in a series of 1,500 spinal CT exams,[32] but the frequency of diagnosis may be on the rise due to the widespread use of MRI and an increasing awareness of the condition.

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76.3.2 Pathology

Cyst walls are composed of fibrous connective tissue of varying thickness and cellularity. There is usually no signs of infection or inflammation. There may be a synovial lining[33] (synovial cyst) or it may be absent[34] (ganglion cyst). The distinction between the two may be difficult,[23] possibly owing in part to the fact that fibroblasts in ganglion cysts may form an incomplete synovial-like lining.[35] Proliferation of small venules is seen in the connective tissue. Hemosiderin staining may be present, and may or may not be associated with a history of trauma.[31]

76.3.3 Clinical

The average age was 63 years in one series[31] and 58 years in a review of 54 cases in the literature[33] (range: 33–87) with a slight female preponderance in both series. Most occur in patients with severe spondylosis and facet joint degeneration,[34] 25% had degenerative spondylolisthesis.[31] L4–5 is the most common level.[31,36] They may be bilateral. Pain is the most common symptom, and is usually radicular. Some JFC may contribute to canal stenosis and can produce neurogenic claudication (p.1100) [37] or on occasion a cauda equina syndrome. Symptoms may be more intermittent in nature than with firm compressive lesions, such as HLD. A sudden exacerbation in pain may be due to hemorrhage within the cyst. Some JFC may be asymptomatic.[38]

76.3.4 Differential diagnosis

Also see Differential diagnosis, Sciatica (p.1410). Differentiating JFC from other masses relies largely on the appearance and location. Other distinguishing features include:

1. neurofibroma: unlikely to be calcified
2. free fragment of HLD: not cystic in appearance
3. epidural or nerve root metastases: not cystic
4. dural subarachnoid root sleeve dilatation (p.1142)
5. arachnoid cyst (from arachnoid herniation through a dural defect): not associated with facet joint, margins thinner than JFC[39]
6. perineurial cysts (Tarlov's cyst): arise in space between perineurium and endoneurium, usually on sacral roots,[40] occasionally show delayed filling on myelography. Usually associated with remodelling of adjacent bone

76.3.5 Evaluation

Identifying a JFC pre-op helps the surgeon, as the approach differs slightly from that for HLD, and the cyst might otherwise be missed or unknowingly deflated and unnecessary time wasted afterwards trying to find a compressive lesion. Or, the unwitting surgeon may misinterpret the cyst as a "transdural disc extrusion" and needlessly open the dura. Pre-op diagnoses were incorrect in 30% of operated cases of JFC.[31]

Myelography: posterolateral filling defect (whereas most discs are situated anteriorly, an occasional fragment may migrate posterolaterally, whereas a JFC will always be posterolateral), often with a round extradural appearance.

CT scan: shows a low density epidural cystic lesion typically with a posterolateral juxtaarticular location. Some have calcified rim,[38] and some may have gas within.[41] Erosion of bony lamina is occasionally seen.[36,42]

MRI: variable findings (may be due to differing composition of cyst fluid: serous vs. proteinaceous[43]). Unenhanced signal characteristics of non-hemorrhagic JFC are very similar to CSF. Hemorrhagic JFC are hyperintense. MRI usually misses bony erosion.

76.3.6 Treatment

Optimal treatment is not known. There is one case report of a cyst that resolved spontaneously.[32] If symptoms persist with conservative treatment, some promote cyst aspiration or facet injection with steroids,[44] while most advocate surgical excision of the cyst.

Surgical treatment considerations: The cyst may be adherent to the dura. The cyst may also collapse during the surgical approach and may be missed. A JFC may serve as a marker for possible instability and should prompt an evaluation for the same. Some argue for performing a fusion since JFC may arise from instability, however, it appears that fusion is not required for a good result in many cases.[44] Therefore it is suggested that consideration for fusion be made on the basis of any instability and not merely on the basis of the presence of a JFC.

Minimally invasive spine surgery (MISS) has also been used for removal,[45] long-term follow-up is lacking. A 15 mm entry incision is made 1.5 cm lateral to midline.

Following surgical treatment, symptomatic JFCs may recur or may develop on the contralateral side.[31]

76.4 Syringomyelia

76.4.1 General information

Key concepts

- AKA syrinx. Cystic cavitation of the spinal cord
- 70% are associated with Chiari I malformation, 10% with basilar invagination. May also be posttraumatic or associated with tumor, infection…

- symptoms: progressive neurologic deterioration over months to years, usually affecting UE first
- diameter > 5 mm + associated edema predict a more rapid deterioration
- preferred treatment is directed at correcting the causative pathophysiology

AKA syrinx. Cystic cavitation of the spinal cord. Other terms not precisely defined include: hydrosyringomyelia, communicating or noncommunicating syringomyelia.

Syringobulbia: Rostral extension into brainstem (usually medulla). May present with (bilateral) peri-oral tingling and numbness, due to compression of the spinal trigeminal tracts as the fibers decussate.

76.4.2 Etiologies

Primary syringomyelia

This term is used differently by different authors.[46] Herein, refers to syrinx in the absence of identifiable cause

Secondary syringomyelia

Most cases are thought to be secondary to partial obstruction of the spinal subarachnoid space.[46] Unanswered question: Why then do patients with varying degrees of degenerative cervical spinal stenosis generally not get syringomyelia? Etiologies include:

1. Chiari I malformation: the most common cause ofsyrinx (p.277)
2. postinflammatory
   a) postinfectious
      - granulomatous meningitides (TB and fungal)
      - postoperative meningitis, especially after intradural procedure
   b) chemical or other sterile inflammations
      - rarely after SAH
      - after myelography: especially with older oil-based agents no longer in use
3. posttraumatic: also see below
   a) with severe posttraumatic kyphotic deformity: e.g. with retropulsed bone, scarring…
   b) arachnoid scarring without recognized trauma
   c) severe injury to spinal cord and/or its coverings. Blood may be a contributing factor

 the older concept of syrinx developing as a coalescence of foci of traumatic hematomyelia has not been borne out

1. postsurgical: has been identified many years after uncomplicated intradural neoplasm removal (e.g. neurofibromas)
2. basilar arachnoiditis:
   a) idiopathic
   b) postinfectious: see above
3. basilar impression, with constriction of the foramen magnum (p.217)
4. associated with spinal tumors: this is distinct from a tumor cyst
5. associated with disc protrusion:
6. cerebellar ectopia
7. Dandy Walker syndrome

76.4.3 Epidemiology

See reference.[47]

Prevalence of non-posttraumatic syringomyelia: 8.4 cases/100,000 population. Usually presents between ages 20–50.

Associated clinical syndromes are shown in Table 76.4.

76.4.4 Pathophysiology

Major theories of formation of the cyst:

1. hydrodynamic (“water-hammer”) theory of Gardner: systolic pulsations are transmitted with each heartbeat from the intracranial cavity to the central canal. Has been essentially disproven using MRI[48]

Table 76.4 Conditions associated with syringomyelia	
Condition	%[a]
Chiari type 1 malformation	70
basilar invagination	10
intramedullary spinal cord tumors	4
[a]percent of cases of syringomyelia	

2. Williams' ("craniospinal dissociation") theory: maneuvers that raise CSF pressure (valsalva, coughing...) cause "hydrodissection" through the spinal cord tissue. May be more common in noncommunicating syringomyelia
3. Heiss-Oldfield theory: occlusion at the foramen magnum causes CSF pulsations during cardiac systole to be transmitted through the Virchow-Robin spaces which increases the extracellular fluid which coalesce to form a syrinx[47] (i.e. through the cord parenchyma)

76.4.5 Clinical

Presentation: highly variable. Usually progresses over months to years, with a more rapid deterioration early that gradually slows.[47] Initially, pain, weakness, atrophy and loss of pain & temperature sensation in the upper extremities (with cervical syrinx) is common. Myelopathy that progresses slowly over years ensues.

76.4.6 Characteristic syndrome

(nonspecific for intramedullary spinal cord pathology):
1. sensory loss (similar to central cord syndrome) with a suspended ("cape") dissociated sensory loss (loss of pain and temperature sensation with preserved touch and joint position sense → painless ulcerations from unperceived injuries and/or burns)
2. pain: commonly cervical and occipital. Dysesthetic pain often occurs in the distribution of the sensory loss[47]
3. weakness: lower motor neuron weakness of the hand and arm
4. painless (neurogenic) arthropathies (Charcot's joints) especially in the shoulder & neck due to loss of pain & temperature sensation: seen in < 5%

76.4.7 Evaluation

Prior to the CT/MRI era, diagnosis relied on myelography or on autopsy.

MRI: defines anatomy in sagittal as well as axial plane. Test of choice. Cervical & thoracic spine and brain MRI (without & with contrast, to include craniocervical junction) should be obtained. Syringomyelic cavities may be complex, with noncommunicating channels (more common with posttraumatic syrinx).

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CT: low attenuation area within cord seen on either plain CT or myelogram/CT (with water soluble contrast).

Myelogram: rarely used alone (usually performed in conjunction with CT). When used alone: often normal (false negative), some → complete block at level of syrinx; iodine contrast studies may show fusiform widening of spinal cord, whereas air contrast studies may show collapse of the cord.[49] Dye may slowly leach into the cyst.

EMG: no characteristic findings, but may be useful to R/O other conditions that may be responsible for symptoms (e.g. peripheral neuropathy causing paresthesias).

76.4.8 Distinguishing from similar entities

1. tumor cyst:
   a) especially with intramedullary spinal cord gliomas. Tumors may secrete fluid, or may cause microcysts that eventually coalesce. Most (but not all) intramedullary tumors will enhance with IV contrast on MRI
   b) tumor cyst fluid is usually highly proteinaceous, syrinx fluid usually has the same MRI characteristics as CSF (NB: true syrinx can occur with tumor)

2. central spinal canal:
   a) residual central spinal canal: the central canal is present within the spinal cord at birth and normally gradually involutes with age.[50] Persistence of the canal is a normal variant. Characteristic imaging features:
      - linear or fusiform on sagittal MRI
      - ≤2–4 mm in maximal width
      - may be singular, or there may be several discontinuous regions in the rostral-caudal direction
      - perfectly round in cross-section and centrally located on axial MRI
      - if IV contrast is given, there should be no enhancement
   b) simple dilatation of the central canal with ependymal cell lining has sometimes been called hydromyelia, but this usage is ambiguous

76.4.9 Management

General information

For an incidentally discovered syrinx (i.e. asymptomatic and no neurologic deficit) with no identified etiology, if the size remains stable over 2–3 years of observation, F/U studies at 2–3 year intervals can be done if there are no changes in symptoms.

Surgical treatment

Intervention is considered for symptomatic lesions (not all are symptomatic). If an underlying cause cannot be determined, it may be very difficult to treat a very small syrinx directly (however, these are unlikely to be causing reversible symptoms).

Options include:

1. current philosophy is to treat the underlying pathophysiology (and to use syrinx draining procedures as second choice when this is not feasible)
   a) posterior decompression: procedure of choice when posterior anomalies (e.g. Chiari malformation) are present
   b) decompression if a different site of compression is identified
2. shunts:
   a) disadvantages:
      - complication rate: 16%
      - clinical stabilization rate: 54% at 10 yrs
      - may produce traction on spinal cord with potential for further injury
      - prone to obstruction: 50% at 4 years
      - does not correct underlying pathophysiology and so syrinx may recur
   b) indications: cases of diffuse arachnoiditis (e.g. following tuberculous or chemical meningitis) where the obstruction extends over many levels, and syrinx diameter > 3–4 mm
   c) K or T tube drainage. Choice of distal sites includes:
      - peritoneum[51] (difficult in cervical region): e.g. Edwards-Barbaro syringoperitoneal shunt (distributed by Integra)
      - pleural cavity
      - subarachnoid space (e.g. Heyer-Schulte-Pudenz system): requires normal CSF flow in subarachnoid space, therefore cannot use in arachnoiditis
3. percutaneous aspiration of the cyst[52] (may be used repeatedly)
4. no longer recommended:
   a) plugging the obex with muscle, teflon or other material
   b) opening the subarachnoid space & removing inferior tonsils
   c) syringostomy: usually fails to remain patent, therefore using a stent or a shunt (syringosubarachnoid or syringoperitoneal) is recommended

Technical considerations:

1. intraoperative ultrasound is often helpful for:
   a) localizing the cyst
   b) assessing for septations (to avoid shunting only part of cyst)
2. if Chiari malformation is not present, consider syringosubarachnoid shunt as the initial procedure. If this fails, then syringoperitoneal shunt may be inserted
3. Rhoton suggests performing the myelotomy in the dorsal root entry zone (DREZ), between the lateral and posterior columns (instead of the midline as with a tumor) because this is consistently

the thinnest part and there is usually already an upper extremity proprioceptive deficit from the syrinx.[53 (p 1317)] There is ≈ 10%incidence of posterior column dysfunction with shunting
4. with syringosubarachnoid shunts, be sure the distal shunt tip is subarachnoid (and not just subdural) or else it will not function
5. with syringopleural shunt, the pleural opening can be made posteriorly, adjacent to one of the ribs as described for ventriculopleural shunt (p.1515)
6. for peritoneal terminus of shunt: the proximal shunt may be placed in the syrinx with the patient in the prone position, and the peritoneal end of the catheter may be tunneled to an intermediate position in the flank in the midaxillary line where it can be coiled into a small subcutaneous pouch which can be temporarily closed with staples and dressed with Tegaderm. After the spinal incision is closed, the patient can be undraped and rotated to the supine position, the Tegaderm is removed, and the abdomen and the flank incision are prepped, the staples are removed and the catheter is retrieved from the sub-Q pocket and tunneled to the peritoneal opening that is made at this time. A small bump (rolled sheet…) to elevate the flank may be employed in both parts of the operation to facilitate access to the pouch

76.4.10 Outcome

Assessing treatment results is difficult due to rarity of the condition, variability of natural history (which may arrest spontaneously), and too short follow-up.[54] Enthusiasm for direct treatment (shunts, fenestration…) is low among neurosurgeons because of the perceived poor response and risk of iatrogenic neurologic worsening. However, this may remain as the only option for a deteriorating patient and positive outcomes do occur.[55]

76.5 Posttraumatic syringomyelia

76.5.1 General information

Posttraumatic syringomyelia (PTSx) may follow significant spinal trauma (with or without clinical spinal cord injury). Includes penetrating injury or non-penetrating "violent" trauma to the spinal cord (injuries such as post-spinal anesthesia or following thoracic disc herniation are not included).

76.5.2 Epidemiology

Often a late presentation following spinal cord injury, therefore incidence is higher in series with longer follow-up. Incidence increasing with increasing survival following spinal cord injury and with increasing use of MRI. Range: ≈ 0.3–3%of cord injured patients (see Table 76.5).

In a large number of patients followed via multicenter cooperative data bank, there were fewer cases of syrinx following cervical injuries than following thoracic injuries[57] (may be artifactual since patients with lower lesions may be more aware of ascending levels).

Latency following spinal cord injury:
1. latency to symptoms: 3 mos to 34 yrs (mean 9 yrs) (earlier in complete cord lesions than incomplete: mean 7.5 vs. 9.9 yrs)
2. latency to diagnosis: up to 12 yrs (mean 2.8 yrs) after onset of new symptoms

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Table 76.5 Incidence of posttraumatic syringomyelia

Type of injury	No./risk[a]	Incidence
all spinal cord injury patients	30/951	3.2%
complete quadriplegics	14/177	7.9%
incomplete quadriplegics	4/181	4.5%
complete paraplegics	4/282	1.7%
incomplete paraplegics	4/181	2.2%

[a]number occurring over number at risk in 951 patients followed for 11 years[56]

Table 76.6 Presentation (in 30 SCI patients with syrinx[56])

Symptom	Initial	At time of diagnosis
pain[a]	57%	70%
numbness	27%	40%
increased motor deficit	23%	40%
increased spasticity	10%	23%
increased sweating (hyperhidrosis)	3%	13%
autonomic dysreflexia	3%	3%
no symptoms	7%	7%
Signs	**Frequency**	
ascending sensory level	93%	
depressed tendon reflexes	77%	
increased motor deficits	40%	

[a]pain is often quite severe, and unrelieved with analgesics[59]

76.5.3 Clinical

The presentation of patients with PTSx is shown in Table 76.6. The late appearance of upper extremity symptoms in a paraplegic patient should raise a high index of suspicion of posttraumatic syringomyelia.[58]

Hyperhidrosis may be the only feature of descending syringomyelia in patients with complete cord lesions.[60] For the differential diagnosis, see Delayed deterioration following spinal cord injuries (p.1019).

76.5.4 Evaluation

One end of the cavity is often found at a site of spinal column fracture or abnormal angulation.

76.5.5 Management

General information

Many authors advocate early surgical drainage of cyst as a means of reducing increased delayed deficit.[61] Some authors feel that aside from disturbing sensory symptoms, that motor loss was infrequent and therefore conservative management is indicated in most cases.[62]

Medical

Managed non-surgically: 31%remained stable, 68%progressed over yrs (longer F/U in latter).

Surgical

There is probably no benefit in operating on a patient with a small syrinx.[56]

Surgical options

Same as in Communicating syringomyelia, with the following differences:

1. cord transection (cordectomy)[63]: an option in complete injuries only
2. plugging the obex is probably not indicated (controversial in congenital syrinx)

Outcome

In 9 PTSx patients treated with syringosubarachnoid shunt[56]: pain relieved in all 9 (1 only slightly), motor recovery in 5/8, improved tendon reflex in 1/10. Some post-op complications in 9 patients included: 1 incomplete lesion became complete, 1 sensorimotor deterioration, transient pain in 3.

Most results are good for radicular symptoms, with dubious efficacy for autonomic symptoms or spasticity.

76.6 Spinal cord herniation (idiopathic)

76.6.1 General information

Rare. Spinal cord herniates though a defect in the dura usually located anteriorly or anterolaterally between T2–8.[64] Bone erosion anterior to the dural defect may occasionally be seen. Frequently associated with a calcified disc fragment which theoretically may have gradually eroded through the dura.

76.6.2 Differential diagnosis

Main DDx is with dorsal arachnoid cyst (p.1407). Both result in increased subarachnoid space posterior to the cord, and a ventral kinking of the spinal cord. Contiguous CSF pulsation artifact on MRI can be seen with cord herniation, whereas an arachnoid cyst tends to interrupt this.

76.6.3 Presentation

Commonly presents as an incomplete Brown-Séquard syndrome (with relative sparing of posterior columns). Symptoms may be due to distortion of the spinal cord, but vascular injury may also play a role.

76.6.4 Surgery

Requires a lateral or anterolateral approach to minimize spinal cord manipulation (p.1061). The dural defect is widened which usually results in reduction of the spinal cord herniation. A sling of dural substitute can then be slid anterior to the cord to prevent reherniation.

76.7 Spinal epidural lipomatosis (SEL)

76.7.1 General information

Hypertrophy of epidural fat. Most commonly seen with prolonged exogenous steroid therapy (in 75%of cases[65]) usually moderate to high dosage for years,[66] but may also be associated with: Cushing's disease, Cushing's syndrome, obesity,[67] hypothyroidism or may be idiopathic.[68] Male: female = 3:1.[66]

Back pain usually precedes all other symptoms. Progressive LE weakness and sensory changes are common. Sphincter disturbance occurs but is rare. SEL is most common in the thoracic spine (≈ 60% of cases), the rest are in the lumbar spine (no cases reported in cervical spine).

It is often very difficult to differentiate the patient with increased epidural fat (even to the point that CSF may be obliterated at the levels of involvement) that is not causing symptoms, from those cases where the exuberant fat is responsible for the findings.

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76.7.2 Evaluation

CT: density of adipose tissue is extremely low (-80 to -120 Hounsfield units)[69] which distinguishes SEL from most other lesions (except lipoma).

MRI: signal follows fat (high signal on T1WI, intermediate on T2WI). Suggested diagnostic criteria: epidural adipose should be > 7 mm thick to be considered SEL.[67,70]

76.7.3 Treatment

In those patients who can be weaned off steroids and lose weight, surgery may be avoided in some cases.[71] If SEL is related to obesity, weight loss alone can be successful.

Surgery is indicated for symptomatic patients in whom the above interventions are unsuccessful or not feasible. An effort to normalize cortisol levels in those with endogenous hypercortisolism (Cushing's disease…) should be made before laminectomy is performed. Due to potential complications and slow growth of the tissue, the decision to operate should be made with caution.

Surgery usually consists of laminectomy with removal of adipose tissue. Occasionally repeat surgery is needed for reaccumulation of adipose tissue.

76.7.4 Outcome

Surgery usually results in significant improvement.[70] Idiopathic cases may fare better than those due to steroid excess. Cauda equina compression responds better than thoracic myelopathy.

Complications rates may be higher than expected in part due to medical comorbidities. Fessler et al.[72] reported 22% 1-year mortality.

76.8 Craniocervical junction and upper cervical spine abnormalities

76.8.1 Associated conditions

Also see Axis (C2) vertebra lesions (p.1391).

Abnormalities in this region are seen in a number of conditions including:

1. rheumatoid arthritis (p.1134)
2. traumatic & post-traumatic: including fractures of odontoid, occipital condyles…
3. ankylosing spondylitis (p.1123): may result in fusion of the entire spine which spares the occipitoatlantal and/or atlantoaxial joints which can lead to instability there
4. congenital conditions:
   a) Chiari malformations (p.277)
   b) Klippel-Feil syndrome (p.271)
   c) Down syndrome
   d) atlantoaxial dislocation (AAD)
   e) occipitalization of the atlas: seen in 40% of congenital AAD[73]
   f) Morquio syndrome (a mucopolysaccharidosis): atlantoaxial subluxation occurs due to hypoplasia of the odontoid process and joint laxity
5. neoplasms: metastatic (p.815) or primary
6. infection
7. following surgical procedures of the skull base or cervical spine: e.g. transoral resection of the odontoid

76.8.2 Types of abnormalities

Abnormalities include:

1. basilar impression/invagination: as with Paget’s disease
2. atlanto-occipital dislocation
3. atlantoaxial dislocation
4. occipitalization of the atlas, or thin or deficient posterior arch of atlas[74]

76.8.3 Treatment

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Fractures of the occipital condyles, atlas or axis are usually adequately treated with external immobilization; also see Occipital condyle fractures (p.966). Because traumatic occipitocervical dislocations are usually fatal, optimal treatment is not well defined. Occipitalization of the atlas may be treated by creating an “artificial atlas” from the base of the occiput and wiring to that.[74]

Indications and techniques are outlined in Atlantoaxial fusion (C1–2 arthrodesis) (p.1479).

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SAH and Aneurysms

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XVII

77 Introduction and General Information, Grading, Medical Management, Special Conditions

77.1 Introduction and overview

77.1.1 Definition

Blood in the subarachnoid space, i.e. between the arachnoid membrane and the pia mater.

77.1.2 Miscellaneous facts about SAH

1. May be post-traumatic or spontaneous. Trauma is the most common cause
2. Most cases of spontaneous SAH are due to aneurysmal rupture
3. peak age for aneurysmal SAH is 55–60 years, ≈ 20% of cases occur between ages 15–45 yrs[1]
4. 30% of aneurysmal SAHs occurs during sleep
5. sentinel headaches that precede the SAH-associated ictus have been reported by 10-50% of patients and most commonly occur within 2-8 weeks before overt SAH.[2,3,4]
6. headache is lateralized in 30%, most to the side of the aneurysm
7. SAH is complicated by intracerebral hemorrhage in 20–40%, by intraventricular hemorrhage in 13–28% (p. 1192) and by subdural blood in 2–5%, usually due to p-comm aneurysm when over convexity, or distal anterior intracerebral artery (DACA) aneurysm with interhemispheric subdural (p. 1211)
8. soft evidence suggests that rupture incidence is higher in spring and autumn
9. patients ≥ 70 yrs age have a higher proportion with a severe neurologic grade[5]
10. seizures may occur in up to 20% of patients after SAH, most commonly in the first 24 hours and are associated with ICH, HTN and aneurysm location (MCA & acomm)[6,7]

77.1.3 Outcome of aneurysmal SAH

1. 10–15% of patients die before reaching medical care
2. mortality is 10% within first few days
3. 30-day mortality rate was 46% in one series,[8] and in others over half the patients died within 2 weeks of their SAH[9]
4. median mortality rate in epidemiologicval studies from U.S. has been 32% vs. 44% in Europe and 27% in Japan (may be an underestimate based on underreported prehospital death)[10]
5. causes of mortality
   a) 25% die as a result of medical complications of SAH[11]
      - neurogenic pulmonary edema (p. 1178)
      - neurogenic stunned myocardium (p. 1177)
   b) about 8% die from progressive deterioration from the initial hemorrhage[12 (p 27)]
6. among patients surviving the initial hemorrhage treated without surgery, rebleeding is the major cause of morbidity and mortality (p. 1167), the risk is ≈ 15–20% within 2 weeks. The goal of early surgery is to reduce this risk (p. 1200)
7. of those reaching neurosurgical care, vasospasm (p. 1178) kills 7%, and causes severe deficit in another 7%[13]
8. about ≈ 30% of survivors have moderate to severe disability,[14] with rates of persistent dependence estimated between 8-20% in population based studies[10]
9. ≈ 66% of those who have successful aneurysm clipping never return to the same quality of life as before the SAH[14,15]
10. patients ≥ 70 yrs age fare worse for each neurologic grade.[5] A multivariate analysis revealed age and WFNS grade to be most predictive of long-term outcome, regardless of treatment modality[16]
11. the severity of clinical presentation is the strongest prognostic indicator

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77.2 Etiologies of SAH

Etiologies of subarachnoid hemorrhage (SAH) include[17]:

1. trauma: the most common cause of SAH.[18,19] In all of the following discussion, only non-traumatic (i.e. “spontaneous”) SAH will be considered

2. “spontaneous SAH”
   a) ruptured intracranial aneurysms: 75–80% of spontaneous SAHs (p.1191)
   b) cerebral arteriovenous malformation (AVM): 4–5% of cases; AVMs more commonly cause ICH & IVH than SAH (p.1239)
   c) certain vasculitides that involve the CNS, see Vasculitis and vasculopathy (p.195)
   d) rarely due to tumor (many case reports[20,21,22,23,24,25,26,27,28,29,30,31])
   e) cerebral artery dissection (may also be post-traumatic)
      - carotid artery (p.1324)
      - vertebral artery: may cause intraventricular blood (especially 4th and third ventricle) (p.1325)
   f) rupture of a small superficial artery
   g) rupture of an infundibulum (p.1161)
   h) coagulation disorders:
      - iatrogenic or bleeding dyscrasias
      - thrombocytopenia
   i) dural sinus thrombosis
   j) spinal AVM: usually cervical or upper thoracic (p.1140)
   k) cortical subarachnoid hemorrhage
   l) pretruncal nonaneurysmal SAH (p.1231)
   m) rarely reported with some drugs: e.g. cocaine (p.207)
   n) sickle cell anemia
   o) pituitary apoplexy (p.720)
   p) no cause can be determined in 14–22% (p.1230)

77.3 Incidence

Estimated annual rate of aneurysmal SAH in the United States: 9.7-14.5 per 100,000 population.[32,33] Reported rates are lower in South and Central America,[34] and higher in Japan and Finland.[35] Incident of SAH increases with age (avg. age of onset > 50[33,36,37,38]; tends to be higher in women (1.24 times higher than men),[34] and appears to be higher in African Americans and Hispanics (compared to Caucasians).[32,39,40]

77.4 Risk factors for SAH

See references.[17,41]

1. behavioral
   - hypertension
   - cigarette smoking[42]
   - alcohol abuse
   - sympathomimetic drugs (p.207)
2. gender and race (see above)
3. history of cerebral aneurysm
   - ruptured aneurysm
   - unruptured aneurysm (esp. those that are symptomatic, larger in size and located in posterior circulation)
   - morphology: bottleneck shape[43] and increased ratio of size of aneurysm to parent vessel have been associated with increased risk of rupture[44,45]
4. family history of aneurysms (at least 1 first-degree family member and especially if ≥ 2 are affected)
5. genetic syndromes
   - autosomal dominant polycystic kidney disease
   - type IV Ehlers-Danlos syndrome
6. pregnancy – there does not appear to be an increased risk of aneurysmal SAH in pregnancy, delivery, and puerperium[46,47]

77.5 Clinical features

77.5.1 Symptoms of SAH

Sudden onset of severe H/A (see below), usually with vomiting, syncope (apoplexy), neck pain (meningismus), and photophobia. If there is LOC, patient may subsequently recover consciousness.[48] Focal cranial nerve deficits may occur (e.g. third nerve palsy from aneurysmal compression of the third cranial nerve, causing diplopia and/or ptosis). Low back pain may develop due to irritation of lumbar nerve roots by dependent blood.

77.5.2 Headache

The most common symptom, present in up to 97% of cases. Usually severe (classic description: "the worst headache of my life") and sudden in onset (paroxysmal). The H/A may clear and the patient may not seek medical attention (referred to as a sentinel hemorrhage or headache, or warning headache; they occur in 30–60% of patients presenting with SAH). If severe or accompanied by reduced level of consciousness, most patients present for medical evaluation. Patients with H/A due to minor hemorrhages will have blood on CT or LP. However, warning headaches may also occur without SAH and may be due to aneurysmal enlargement or to hemorrhage confined within the aneurysmal wall.[49] Warning H/A are usually sudden in onset, milder than that associated with a major rupture, and may last a few days.

Differential diagnosis of severe, acute, paroxysmal headache (25% will have SAH[50]):

1. subarachnoid hemorrhage, AKA "warning headache" or sentinel H/A (see above)
2. benign "thunderclap headaches" (BTH) or crash migraine.[51] Severe global headaches of abrupt onset that reach maximal intensity in < 1 minute, accompanied by vomiting in ≈ 50% They may recur, and are presumably a form of vascular headache. Some may have transient focal symptoms. There are no clinical criteria that can reliably differentiate these from SAH[52] (although seizures and diplopia, when they occurred, were always associated with SAH). There is no subarachnoid blood on CT and LP, which should probably be performed on at least the first presentation to R/O SAH. Earlier recommendations to angiogram these individuals[53] have since been tempered by experience[54,55]
3. reversible cerebral vasoconstrictive syndrome (RCVS)[56] (AKA benign cerebral angiopathy or vasculitis[57]): severe H/A with paroxysmal onset, ± neurologic deficit, and string of beads appearance on angiography of cerebral vessels that usually clears in 1–3 months. > 50% report prior use of vasoconstrictive substances (cocaine, marijuana, nasal decongestants, ergot derivatives, SSRIs, interferon, nicotine patches) sometimes combined with binge drinking. May also occur post-partum. Complications occurred in 24% including:
   a) usually during the 1st week: SAH, ICH, seizures, RPLS
   b) usually during the 2nd week: ischemic events (TIA, stroke)
4. benign orgasmic cephalgia: a severe, throbbing, sometimes "explosive" H/A with onset just before or at the time of orgasm (distinct from pre-orgasmic headaches which intensify with sexual arousal[58]). In a series of 21 patients[59] neurologic exam was normal in all, and angiography done in 9 was normal. 9 had a history of migraine in the patient or a family member. No other symptoms developed in 18 patients followed for 2–7 yrs. Recommendations for evaluation are similar to that for thunderclap headaches above

77.5.3 Signs

Meningismus (see below), hypertension, focal neurologic deficit (e.g. oculomotor palsy, hemiparesis), obtundation or coma (see below), ocular hemorrhage (see below).

Meningismus

Nuchal rigidity (especially to flexion) often ensues in 6 to 24 hrs. Patients may have a positive Kernig sign (flex thigh to 90° with knee bent, then straighten knee, positive sign if this causes pain in hamstrings) or Brudzinski sign (flex the supine patient's neck, involuntary hip flexion is a positive sign).

Coma following SAH

Coma may follow SAH because of any one or a combination of the following[60]:

1. increased ICP
2. damage to brain tissue from intraparenchymal hemorrhage (may also contribute to increased ICP)
3. hydrocephalus
4. diffuse ischemia (may be secondary to increased ICP)
5. seizure
6. low blood flow (reduced CBF) due to reduced cardiac output (p.1177)

Ocular hemorrhage

Three types of ocular hemorrhage (OH) may be associated with SAH. They occur alone or in various combinations in 20–40% of patients with SAH.[61]

1. subhyaloid (preretinal) hemorrhage: seen funduscopically in 11–33% of cases as bright red blood near the optic disc that obscures the underlying retinal vessels. May be associated with a higher mortality rate[62]
2. (intra)retinal hemorrhage: may surround the fovea
3. hemorrhage within the vitreous humor (Terson syndrome). First described by the French ophthalmologist Albert Terson. Occurs in 4–27% of cases of aneurysmal SAH,[63,64,65] usually bilateral. May occur with other causes of increased ICP including ruptured AVMs. Funduscopy reveals vitreous opacity. The location of the origin of the vitreous hemorrhage differs in various reports (subhyaloid, epiretinal, subinternal limiting membrane).[66] May be more common with anterior circulation aneurysms (especially ACoA), although 1 study found no correlation with location.[64] Also rarely reported with SDH and traumatic SAH. Often missed on initial examination. When sought, usually present on initial exam, however it may develop as late as 12 days post SAH, and may be associated with rebleeding.[64] The mortality rate may be higher in SAH patients with vitreous hemorrhage than in those without. Patients should be followed for complications of OH (elevated intraocular pressure, retinal membrane formation → retinal detachment, retinal folds[67]). Most cases clear spontaneously in 6–12 mos. Vitrectomy should be considered in patients whose vision fails to improve[65] or if more rapid improvement is desired.[68] The long-term prognosis for vision is good in ≈ 80% of cases with or without vitrectomy[68]

The pathomechanics of OH are controversial. OH was originally attributed to extension of the blood from the subarachnoid space into the vitreous, but no communication exists between these two spaces. In actuality may be due to compression of the central retinal vein and the retinochoroidal anastomoses by elevated CSF pressure[65] causing venous hypertension and disruption of retinal veins.

77.6 Work-up of suspected SAH

77.6.1 Overview

1. tests to diagnose SAH
   a) non-contrast high-resolution CT scan: see below
   b) if CT is negative: LP in suspicious cases (for findings, see below)
2. test to identify source of SAH. Options: CTA, MRA, or catheter angiography. The choice needs to take into account the patient's age, renal function, and even best guess of where an aneurysm might be located
   a) MRA: no radiation, and 2D-TOF MRA does not use contrast (p.232). Poor sensitivity for aneurysm detection early after SAH (see below)
   b) CTA vs. angiogram: one needs to balance the risk of the procedure and ease of obtaining it against the information expected to be obtained
      - total iodine load in a healthy adult should be < 90 gm in 24 hours. In older patients and/or possible compromised renal function, this volume should be less. CTA typically uses 65–75 cc of contrast with ≈ 300 mg iodine/ml, or ≈ 21 gm iodine. The amount of contrast with a cerebral arteriogram varies. However if an angiogram is needed after a CTA, in most cases you do not have to wait 24 hours
      - if there is concern about renal function (e.g. serum creatinine > 100 mcmol/L) hydrate the patient and optionally give Mucomyst® (p.221)
      - catheter angiography may be necessary after a positive CTA to better delineate the anatomy, or to determine dominant filling and cross flow, or in highly suspicious cases with a negative CTA (see below). While CTA permits reliable assessment of feasibility of endovascular treatment in most cases,[69] DSA is still necessary in some
3. if CTA/angiogram is negative: see SAH of unknown etiology (p.1230)

77.6.2 Laboratory/radiographic findings

CT scan

A good quality (e.g. no motion artifact) non-contrast high-resolution CT will detect SAH in ≥ 95% of cases if scanned within 48 hrs of SAH. Blood appears as high density (white) within subarachnoid

spaces. For subtle SAH, look in the occipital horns of the lateral ventricles and the dependent portions of the sylvian fissures. CT also assesses:
1. ventricular size: hydrocephalus occurs acutely in 21% of aneurysmal ruptures (p.1170) [70]
2. hematoma: intracerebral hemorrhage or large amount of subdural blood with mass effect may need emergent evacuation
3. infarct: not sensitive in first 24 hours after infarct (p.1280)
4. amount of blood in cisterns and fissures: important prognosticator for vasospasm (p.1180) and can identify pretruncal nonaneurysmal hemorrhage (p.1231)
5. CT can predict aneurysm location based on the pattern of blood in ≈ 78% of cases (but mostly for MCA and A-comm aneurysms)[71]
   a) blood predominantly in anterior interhemispheric fissure (± blood in lateral ventricles) or within the gyrus rectus suggests a-comm aneurysm
   b) blood predominantly in 1 sylvian fissure is compatible with p-comm or MCA aneurysm on that side
   c) blood predominantly in the prepontine or peduncular cistern suggests a basilar apex or SCA aneurysm
   d) blood predominantly within ventricles (p.1192)
      - blood primarily in 4th and third ventricle: suggests lower posterior fossa source, such as PICA aneurysm or VA dissection
      - blood primarily in the 3rd ventricle suggests a basilar apex aneurysm
6. with multiple aneurysms, CT may help identify which one bled by the location of blood (see above). See also other “clues” (p.1226)

Differential diagnosis of SAH on CT

Things that can mimic the appearance of SAH on CT include:
1. pus
2. following contrast administration: sometimes IV, and especially intrathecal
3. occasionally the pachymeningeal thickening seen in spontaneous intracranial hypotension (p.389)

Lumbar puncture

The most sensitive test for SAH. However, false positives – e.g. with traumatic taps; see Differentiating SAH from traumatic tap (p.1506) – occur with enough frequency that this test is falling out of favor for diagnosis of SAH.

Caution: lowering the CSF pressure may possibly precipitate rebleeding by increasing the transmural pressure (p.1170). Therefore remove only a small amount of CSF (several ml) and use a small (≤20 Ga) spinal needle.

Findings (also, see Table 23.4):
1. opening pressure: elevated
2. appearance:
   a) non-clotting bloody fluid that does not clear with sequential tubes
   b) xanthochromia: yellow coloration of CSF supernatant (specimen must be centrifuged in the lab) due to heme pigments released by the breakdown of RBCs. The most reliable means of differentiating traumatic tap from SAH. In patients with a negative head CT, the minimal amount of time required for bilirubin to become detectable in the CSF, as well as the minimal amount of blood the needs to enter the CSF to give a positive xanthochromia remains unknown. However, xanthochromia is usually not apparent until 2–4 hours after the SAH. Is present in almost 100% by 12 hours after the bleed, and remains in 70% at 3 weeks, and is still detectable in 40% at 4 weeks. Spectrophotometry is more sensitive than visual inspection, but may lack sufficient specificity to warrant widespread use.[72,73] False positives: xanthochromia may occur with jaundice or high protein levels in the CSF
3. cell count: RBC count usually > 100,000 RBCs/mm^3. Compare RBC count in first to last tube (should not drop significantly)
4. protein: elevated due to blood breakdown products
5. glucose: normal, or reduced (RBCs may metabolize some glucose with time)

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MRI

Not sensitive for SAH acutely within the first 24–48 hrs[74] (too little met-Hb) especially with thin layers of blood. Better after ≈ 4–7 days (excellent for subacute to remote SAH, > 10–20 days). FLAIR

MRI is the most sensitive imaging study for detecting blood in the subarachnoid space. May be helpful in determining which of multiple aneurysms bled (p.1226).[75]

Magnetic resonance angiography (MRA)

Based on a systematic review, sensitivity is 87% and specificity is 92% for detecting intracranial aneurysms (IAs) (compared to catheter DSA) with significantly poorer sensitivity for aneurysms < 3 mm diameter.[76,77,78]

MRA's ability to detect IAs depends on aneurysm size, rate and direction of blood flow in the aneurysm relative to the magnetic field, and aneurysmal thrombosis and calcification. MRA may be most useful as a screening test in high-risk patients including patient's with two first degree relatives with IAs, especially those who are also smokers or hypertensive themselves.[79]

CT angiography (CTA)

Many centers have shown good results with CTA (p.227), with a prospective study detecting 97% of aneurysms and demonstrating CTA as safe and effective when used as the initial and sole imaging study for ruptured and unruptured cerebral aneurysms.[80] CTA shows a 3-dimensional image (as can modern catheter angiography) which can help differentiate adherent vessels from those arising from the aneurysm. CTA also demonstrates the relation to nearby bony structures which can be important in surgical planning. CTA use is increasing for evaluation of vasospasm.[81]

Catheter angiogram

General information

Injection of radio-opaque (iodinated) contrast ("dye") into selective vessels using a catheter typically inserted into the femoral artery at the upper thigh, while taking serial x-rays to obtain a "video-like" representation of the vasculature.

The gold standard for evaluation of cerebral aneurysms. Current state of the art uses digital subtraction angiography (DSA). Demonstrates source (usually aneurysm) in ≈ 80–85%; remainder are so-called "SAH of unknown etiology" (p.1230). Shows if radiographic vasospasm is present – clinical vasospasm almost never occurs < 3 days following SAH (p.1178) – and assesses primary feeding arteries, collateral flow in case of a need for arterial sacrifice.

General principles:

1. study the vessel of highest suspicion first (in case patient's condition should change, necessitating discontinuation of procedure)
2. continue to do complete 4 vessel angiogram (even if aneurysm(s) have been demonstrated) to rule out additional aneurysms and assess collateral circulation
3. if there is an aneurysm or suspicion of one, obtain additional views to help delineate the neck and orientation of the aneurysm (see index for specific aneurysm)
4. if no aneurysm is seen, before an arteriogram can be considered negative, must:
   a) visualize both PICA origins: 1–2% of aneurysms occur at PICA origin. Both PICAs can usually be visualized with one VA injection if there is enough flow to reflux down the contralateral VA. Occasionally it is necessary to see more of the contralateral VA than what refluxes to PICA and selective catheterization may be required
   b) flow contrast through the ACoA: if both ACAs fill from one side, this is usually satisfactory. It may be necessary to perform a cross compression AP study with carotid injection (first, rule-out plaque in the carotid to be compressed), or use a higher injection rate to facilitate flow through the ACoA
   c) if an infundibulum (see below) colocalizes to the SAH, it may be unwise to label the case as angiogram-negative and exploration is recommended by some[82]

Infundibulum

A funnel shaped initial segment of an artery, to be distinguished from an aneurysm. Found in 7–13% of otherwise normal arteriograms,[83,84] with a higher incidence in cases of multiple or familial aneurysms. Bilateral in 25%[84] Most commonly found at the origin of the p-comms, but they rarely occur at other sites. Criteria for differentiating infindibula from aneurysms are shown in Table 77.1. Infundibula may represent incomplete remnants of previous fetal vessels[85 (p 272)]

Although they may bleed,[82,87,88,89] there is less risk of rupture than with a saccular aneurysm (no infundibulum < 3 mm in size bled[90] in the cooperative study). However, infundibula have been documented to progress to an aneurysm (i.e. they are preaneurysmal) which may bleed (13 case reports in the literature as of 2009). Recommended treatment: at the time of surgery for another reason,

Table 77.1 Criteria of an infundibulum
1. triangular in shape 2. mouth (widest portion) < 3 mm[a] [86] 3. vessel at apex
[a]a widely accepted but probably arbitrary dimension

Table 77.2 Hunt and Hess classification[a] of SAH[94]

Grade	Description
1	asymptomatic, or mild H/A and slight nuchal rigidity
2	Cr. N. palsy (e.g. III, VI), moderate to severe H/A, nuchal rigidity
3	mild focal deficit, lethargy, or confusion
4	stupor, moderate to severe hemiparesis, early decerebrate rigidity
5	deep coma, decerebrate rigidity, moribund appearance
Add one grade for serious systemic disease (e.g. HTN, DM, severe atherosclerosis, COPD) or severe vasospasm on arteriography.	
[a]original paper did not consider patient's age, site of aneurysm, or time since bleed; patients were graded on admission and pre-op	

consider treating an infundibulum with wrapping, or placing in an encircling clip, or sacrificing the artery if it can be done safely (infundibula lack a true neck).

Angiographic findings

1. general features to take note of when analyzing an aneurysm on angiogram (special considerations for specific aneurysms are covered in designated sections)
   a) size of aneurysm dome:
      - MRI or CT helps with this since the aneurysm may be partially thrombosed and the portion that is patent and fills with contrast and is therefore visualized on angiogram may be much smaller than the actual size
      - large aneurysms (≥ 15 mm dia.) are associated with lower rates of complete occlusion by endovascular coiling[91,92]
   b) neck size
      - narrow necks < 5 mm are ideal for coiling[93]
      - broad necks ≥ 5 mm are associated with increased risk of incomplete occlusion and recanalization with coiling[92]
      - stent or balloon-assisted coiling may be needed for wide necked aneurysms. Stents should be avoided if possible (p. 1586).
   c) dome:neck ratio ≥ 2 is associated with higher rate of successful coil occlusion[93]
2. for basilar bifurcation aneurysms (p. 1218)

77.7 Grading SAH

77.7.1 General information

Four grading scales are in common use. The 2 most widely quoted grading scales are presented here.

77.7.2 Hunt and Hess grade

See Table 77.2 and Table 77.3 for grading system. Grades 1 and 2 were operated upon as soon as an aneurysm was diagnosed. Grade ≥ 3 managed until the condition improved to Grade 2 or 1. Exception: life threatening hematoma or multiple bleeds (which were operated on regardless of grade).

Analysis of data from the International Cooperative Aneurysm Study revealed that with normal consciousness, Hunt and Hess (H&H) grades 1 and 2 had identical outcome, and that hemiparesis and/or aphasia had no effect on mortality.

Table 77.3 Modified classification[95] adds the following

Grade	Description
0	unruptured aneurysm
1 a	no acute meningeal/brain reaction, but with fixed neuro deficit

Table 77.4 WFNS SAH grade[96]

WFNS grade	GCS score[a]	Major focal deficit[b]
0[c]		
1	15	–
2	13–14	–
3	13–14	+
4	7–12	+ or –
5	3–6	+ or –

[a]GCS = Glasgow Coma Scale, see Table 18.1
[b]aphasia, hemiparesis or hemiplegia (+ = present, – = absent)
[c]intact aneurysm

Mortality:
Admission Hunt and Hess Grade 1 or 2: 20%
Patients taken to O.R. (for any procedure) at H&H Grade 1 or 2: 14%
Major cause of death in Grade 1 or 2 is rebleed.
Signs of meningeal irritation increases surgical risk.

77.7.3 World Federation of Neurosurgical Societies / World Federation of Neurological Surgeons (WFNS) grading of SAH

Due to lack of data on the significance of features such as headache, nuchal rigidity, and major focal neurologic deficit, the WFNS Committee on a Universal SAH Grading Scale[96,97] developed a grading system which is shown in Table 77.4. It uses the Glasgow Coma Scale (GCS) (see Table 18.1) to evaluate level of consciousness, and uses the presence or absence of major focal neurologic deficit to distinguish grade 2 from grade 3.

77.8 Initial management of SAH

77.8.1 General information

Practice guideline

Practice guideline: Initial management of aneurysmal SAH

Level I[41]:
- administer oral nimodipine to all patients with aneurysmal SAH. The value of other calcium channel blockers is uncertain
- maintain euvolemia and normal circulating blood volume

Level II[41]:
- control of HTN: the ideal BP to reduce the risk of rebleeding has not been established. A reasonable target is to maintain SBP < 160 mm Hg

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Initial management concerns

1. rebleeding: the major concern during the initial stabilization
2. hydrocephalus: precipitous development acute hydrocephalus may be obstructive (due to blockage of CSF flow by blood clot), but presence of ventriculomegaly early after SAH as well as at later stages is often due to communicating hydrocephalus (p.1170) (due to toxic effect of blood breakdown products on arachnoid granulations)
3. delayed ischemic neurologic deficit (DIND), usually attributed to vasospasm. Begins to be of concern several days following the SAH
4. hyponatremia with hypovolemia (p.1166)
5. DVT and pulmonary embolism (p.167)
6. seizures (p.1167)
7. determining source of bleeding: should be investigated early with CTA or catheter angiography. The timing and choice of study takes into consideration the patient's condition (unstable or premorbid patients are not candidates), the feasibility of early treatment (ideal), and the likelihood of endovascular therapy (based on patient's age and predicted aneurysm location as well as availability)

Goals of medical management related to neurologic injury

In addition to prevention of hyponatremia, hypovolemia, seizures, etc. (see above), the goals of initial medical management include:
1. augmenting CBF: the main device for accomplishing this is hyperdynamic therapy (p.1186). Goals are:
   a) increasing cerebral perfusion pressure (CPP)
   b) improving blood rheology: RBC aggregability increases after SAH[98]
   c) maintaining euvolemia: the majority of patients become hypovolemic in the first 24 hrs after SAH. Also, avoid prophylactic hypervolemia
   d) maintaining normal ICP
2. neuroprotection: there are currently no medications shown to be effective or approved for use as neuroprotective agents for this or any other type of brain injury. Animal studies have shown time and again that the concept may someday be translated into clinical practice[99]

77.8.2 Monitors/tubes

Also see below.
1. arterial-line: for patients who are hemodynamically unstable, stuporous or comatose, those with difficult to control hypertension, or those requiring frequent labs (e.g. ventilator patients)
2. intubate patients who are comatose or unable to protect airway (e.g. stridorous)
3. pulmonary-artery catheter (PA-catheter, AKA Swann-Ganz catheter): the safety and efficacy of this device has been debated in the critical care literature for over a decade now, with some calling for a moratorium on PA catheter use.[100] It is possible that newer technologies may supplant the need for this invasive procedure while allowing close hemodynamic monitoring.[101] Nevertheless, a PA catheter can be considered for:
   a) Hunt and Hess (H&H) grade ≥3 (except good grade 3 patients)
   b) patients with possible CSW or SIADH
   c) hemodynamically unstable patients
4. cardiac rhythm monitor: arrhythmias may occur following SAH (p.1177)
5. intraventricular catheter (IVC) AKA ventriculostomy. Possible indications:
   a) patients developing acute hydrocephalus following SAH or in those with significant intraventricular blood (allows measurement of ICP as well as drainage of blood laden CSF). IVC causes symptomatic improvement in almost two-thirds.[70] May increase the risk of rebleeding (p.1170), however, the risk of untreated hydrocephalus is probably higher[102]
   b) H&H grade ≥3 (except good grade 3 patients). If a high grade patient improves with an IVC, the prognosis may be more favorable. If ICP is elevated, management includes the use of mannitol; see Treatment measures for elevated ICP (p.866).

77.8.3 Admitting orders

1. admit to ICU (monitored bed)
2. VS with neuro checks q 1 hr

3. activity: BR with HOB at 30°. SAH precautions (i.e. low level of external stimulation, restricted visitation, no loud noises)
4. nursing
   a) strict I's & O's
   b) daily weights
   c) knee high TED hose and pneumatic compression boots (PCB)
   d) indwelling Foley catheter if patient lethargic, incontinent, or unable to void in urinal or bed-pan. Consider temperature sensing catheter for strict fever control
5. diet: NPO (in preparation for surgery or endovascular intervention)
6. IV fluids: early aggressive fluid therapy to head off cerebral salt wasting
   a) NS+20 mEq KCl/L at ≈ 2 ml/kg/hr (typically 140–150 ml/hr) (below)
   b) if Hct < 40%[103] give 500 ml of 5%albumin over 4 hrs upon admission
7. medications (avoid IM medications to reduce pain)
   a) prophylactic anticonvulsants: see post-SAH seizures below
   b) sedation (not oversedation): e.g. with propofol
   c) analgesics: fentanyl (unlike morphine, does not cause histamine release. Lowers ICP) 25–100 mcg (0.5–2 ml) IVP, q 1–2 hrs PRN (avoid Demerol® because it may lower seizure threshold)
   d) dexamethasone (Decadron®): may help with H/A and neck pain. Effect on edema controversial. Usually given pre-op prior to craniotomy
   e) stool softener in patients able to take PO (docussate 100 mg PO BID)
   f) anti-emetics: avoid phenothiazines (especially in patients who seize) which may lower seizure threshold. Use e.g. Zofran® (ondansetron) 4 mg IV over 2–5 minutes, may repeat in 4 & 8 hours, and then q 8 hours for 1–2 days
   g) calcium channel blockers (p.1183): nimodipine (Nimotop®) 60 mg PO/NG q 4 hrs initiated within 96 hrs of SAH (some use 30 mg q 2 hrs to avoid periodic dips in BP). IV administration is equally as effective[104] where available. Oral nimodipine should be administered to all patients with aSAH.
   h) H2 blockers (e.g. ranitidine) or proton pump inhibitors (e.g. Prevacid® (lansoprazole) 30 mg p.o. or IV q d): to reduce risk of stress ulceration
   i) these agents impair coagulation and are used with caution: ASA, dextran,[105] heparin, and repeated administration of hetastarch (Hespan®)[106,107] over a period of days
   j) statins: several clinical trials have investigated the utility of statins, with variable results. More recently, a meta-analysis reported no evidence for clinical benefit.[108] In addition, a multicenter randomized phase 3 trial did not detect any short- or long-term benefit with the use of simvastatin[109]
8. oxygenation
   a) in non-intubated patient: O_2 2 L per NC PRN (based on ABG) and if tolerated
   b) in ventilated patient: strive for normocarbia and $pO_2 > 100$ mm Hg
9. temperature (normothermia): medications (Tylenol) and cooling measures (e.g. ice packs, Arctic Sun external cooling device) to reduce and prevent fever are encouraged, as fever has been shown to be independently associated with worse cognitive and functional outcome in survivors of aSAH[110,111,112]
10. HTN: SBP 120–160 mm Hg by cuff is a guideline with unclipped aneurysm (below)
11. labs
   a) ABG, electrolytes, CBC, PT/PTT on admission
   b) ABG, electrolytes, CBC q day (ABG q 6 hrs if patient unstable, electrolytes q 12 hrs if hyponatremia develops, see Hyponatremia following SAH below)
   c) serum and urine osmolality if urine output high or low; see Syndrome of inappropriate anti-diuretic hormone secretion (SIADH) (p.112)
   d) hemoglobin and hematocrit: some studies suggest that higher hemoglobin values are associated with improved outcomes after aSAH.[113,114] However, liberal RBC transfusion has been associated with worse outcomes in aSAH.[115,116] The optimal hemoglobin goal after aSAH in not yet known, and may depend on the presence or absence of vasospasm
   e) serum glucose: effective glucose control after aSAH can significantly reduce the risk of poor outcome[117]
   f) CXR daily until stable: patients undergoing triple-H therapy can develop dangerous pulmonary edema as they "fall off" the Starling curve with volume expansion. Patients with SAH are also rarely at risk for neurogenic pulmonary edema (p.1178) [118]
   g) if available, transcranial doppler to monitor MCA, ACA, ICA, VA and BA velocities and Lindegaard ratio (p.1182) q Mon, Weds & Fri

77.8.4 Blood pressure and volume management

General information

With an unsecured (unclipped or uncoiled) aneurysm, gentle volume expansion with slight hemodilution and mild elevation of blood pressure may help prevent or minimize the effects of vasospasm[119] and cerebral salt wasting. However, extreme hypertension must be avoided (to reduce risk of re-bleeding). Hypervolemia is to be avoided since it does not mitigate vasospasm and increases complications.[120]

Initial blood pressure

Ideal blood pressure is controversial, and must take patient's baseline into consideration. The magnitude of blood pressure control to reduce the risk of rebleeding has not been established, but a decrease in systolic blood pressure to < 160 mm Hg is reasonable.

If blood pressure is labile, labetalol or nicardipine should be used in conjunction with an arterial-line. Avoid hypotension as it may exacerbate ischemia.

Long acting drugs (e.g. ACE inhibitors should be started in patients requiring continued therapy. In patients who were normotensive prior to SAH with easily controlled hypertension, ACE inhibitors may be used PRN in conjunction with a beta blocker, e.g. labetalol (p. 126).

77.8.5 Hyponatremia following SAH

Background

Hypovolemia and hyponatremia frequently follow SAH as a result of natriuresis and diuresis. The reported incidence of hyponatremia in aSAH ranges from 10-30%[41] Although hyponatremia had been attributed to a rise in ADH[121] (thought to produce SIADH with hypervolemia), the ADH increment is usually transient, lasting only ≈ 4 days and hypervolemia did not occur. Another theory is based on the fact that there is often a delayed peak in atrial natriuretic factor (ANF) (a 28-amino acid polypeptide) after an initial smaller rise[122] that was frequently followed by urinary loss of sodium (cerebral salt wasting (CSW) (p. 118) that mimics SIADH, and volume depletion. Although CSW has clearly been shown to be the cause of hyponatremia in the majority of these patients,[123] there are still doubts that ANF is the operative natriuretic factor in SAH.[124] A rise in ANP and brain natriuretic peptide (BNP) after SAH is associated with the development of a negative fluid balance.[125]

Routine labs are identical in SIADH and CSW,[126] but the extracellular fluid volume (which is more difficult to measure) is low in CSW and is normal or elevated in SIADH (see Table 5.5 or a comparison of the two conditions). The neurologic effects of hyponatremia (p. 112) may mimic delayed ischemic neurologic deficit from vasospasm, and hyponatremic patients have about 3 times the incidence of delayed cerebral infarction after SAH than normonatremic patients.[127] Hyponatremia has been chronologically associated with the onset of sonographic and clinical vasospasm[128,129]

Factors that may increase the risk of hyponatremia after SAH include: history of diabetes, CHF, cirrhosis, adrenal insufficiency, or the use of any of the following drugs: NSAIDs, acetaminophen, narcotics, thiazide diuretics.[130]

Treatment

Caution! Restricting fluids which is the treatment for SIADH may be hazardous in the case of CSW (which is more likely to occur after SAH than is SIADH) since dehydration increases blood viscosity which exacerbates ischemia from vasospasm.[127]

- reat hypovolemia aggressively with infusions of crystalloid (e.g. NS), PRBCs, or colloids
- hypertonic saline (3%) has been shown to be effective in correcting hyponatremia[131] and appears to increase regional cerebral blood flow, brain tissue oxygen, and pH in patients with high-grade aSAH[132]
- fludrocortisone has been shown to help correct hyponatremia, with a reduced need for fluids.[133,134] Similarly hydrocortisone administration has been associated with reduced natriuresis and a lower rate of hyponatremia[135]

77.8.6 Post-SAH seizures

General information

No RCT has been performed to help guide decisions on prophylaxis or treatment of seizures. There is also conflicting evidence on whether onset seizures are predictive of late seizures or post-SAH epilepsy.[136,137] As such, there is no consensus amongst practitioners regarding the need for AEDs, the best AED to use, which patients should receive prophylactic AEDs, nor the optimal dose or duration of treatment.

Epidemiology

Incidence. The incidence of seizure-like episodes varies widely between observational studies. One literature review[138] reported 4-26%of SAH patients had onset seizures, 1-28%had early seizures (w/in first 2 weeks), and 1-35%had late seizures (after 2 weeks).[139] Additionally, non-convulsive status epilepticus has been reported in 3-18%of SAH patients, and should be suspected in patients with a poor neurological exam or in the setting of neurological deterioration.[140,141]

Risk factors for post-SAH seizures. [6,41,138,139,140,142,143,144]

- increasing age
- MCA aneurysm
- volume of subarachnoid blood/thickness of clot
- associated intracerebral or subdural hematoma
- poor neurological grade
- Rebleeding
- cerebral infarction
- vasospasm
- hyponatremia
- hydrocephalus
- hypertension
- treatment modality, see coiling vs. clipping (p.1195)

Outcome

The association between seizures and functional outcome remain unclear. One study[140] showed that an in-hospital seizure was independently predictive of one year mortality (65%with seizures vs. 23% without seizures), but others have shown no association with a poorer prognosis.[138,143,145] Two large, retrospective, single-institution studies of patients with aSAH found that nonconvulsive status epilepticus is a very strong predictor of poor outcome.[41,141,146]

AEDs

Studies have assessed neurological outcome following short- and long-term phenytoin use, with higher doses and longer duration associated with poorer outcomes.[147,148] When Keppra is compared to phenytoin, keppra is associated with a higher rate of short-term seizure recurrence,[149] but improved long-term outcomes and fewer side effects.[139,150] Although use of prophylactic AED for aSAH is controversial, a generalized seizure may be devastating in the presence of a tenuous aneurysm. As such, AEDs are given by many authorities in the acute setting, at least until the aneurysm is secured. One paradigm: Keppra® (levetiracetam) 1 g IV q 12 hours until the aneurysm is secured.

Practice guideline: Post-SAH seizures

- Level II[41]: prophylactic anticonvulsants may be considered in the immediate posthemorrhagic period
- Level III[41]: the routine long-term use of anticonvulsants is not recommended
- Level II[41]: long-term anticonvulsants may be considered with known risk factors for delayed seizure disorder (e.g. prior seizure, intracerebral hematoma, intractable hypertension, infarction, or MCA aneurysm)

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77.9 Rebleeding

77.9.1 General information

Approximately 3000 North Americans die each year from rebleeding of ruptured cerebral aneurysms.[151] For untreated ruptured aneurysms, the maximal frequency of rebleeding is in the 1st day (between 4% and 13.6%),[152,153,154,155] with more than 1/3 of rebleeds occurring within 3 hours and 1/2 within 6 hours of symptoms onset.[156] After the first day, the subsequent risk is 1.5% daily for 13 d. Overall, 15–20% rebleed within 14 d, 50% will rebleed within 6 months, thereafter the risk is ≈ 3%/yr with a mortality rate of 2%/yr.[157] (Note: to understand the calculation of long-term cumulative risk for aneurysmal rupture, see Annual and lifetime risk of hemorrhage and recurrent hemorrhage (p.1240); that discussion is related to AVMs but the same concepts pertain to aneurysms). 50% of deaths occur in the 1st month.

There is a risk of rebleeding during any period that the aneurysm is untreated. As such, early treatment of the ruptured aneurysm can reduce risk of rebleeding[158] (see Timing of aneurysm intervention). In addition, higher Hunt and Hess grades,[159] larger aneurysm size, and poorly controlled blood pressure (> 160 mmHg) have also been associated with an increased risk of rebleeding.[154,155,160]

Pre-operative ventriculostomy – e.g. for acute post-SAH hydrocephalus (p.1170) – and possibly lumbar spinal drainage (p.1202) increase the risk of rebleeding.

The risk of rebleeding in SAH of unknown etiology and with AVMs, as well as the risk of bleeding with incidental multiple unruptured aneurysms, are all similar at ≈ 1%/yr; may actually be less in SAH of unknown etiology (p.1230).[161]

77.9.2 Prevention of rebleeding

The optimal method of preventing rebleeding is early coiling or surgical clipping. Bed rest and hyperdynamic therapy do not prevent rebleeding.[162]

77.9.3 Antifibrinolytic therapy

The role of clot lysis in early rebleeding is uncertain.

Practice guideline: Antifibrinolytic therapy

Level II[41]: for patients with aneurysmal SAH where there is an unavoidable delay in treatment of the aneurysm, in whom there is a significant risk of rebleeding and no compelling medical contraindications, up to 72 hours of therapy with tranexamic acid or aminocaproic acid is reasonable

Drug info: Tranexamic Acid (Cyklokapron®)

Reduces the risk of early rebleeding.[153]

□: 1 gm IV as soon as diagnosis of SAH is verified (if patient is to be transported to another facility for definitive care, the dose is given before the patient is transported), followed by 1 gm q 6 hours until the aneurysm is occluded; this treatment did not exceed 72 hours.

Drug info: Epsilon-aminocaproic acid (Amicar®)

Epsilon-aminocaproic acid (EACA) an antifibrinolytic agent, competitively inhibits activation of plasminogen to plasmin. Existing plasmin is neutralized by endogenous antiplasmins. EACA does reduce the risk of rebleeding. However, the incidence of hydrocephalus and delayed ischemic deficits (vasospasm) have been shown to be higher with prolonged use[163] with prolonged use. There may also be a lag of 24–48 hrs before effectiveness occurs.[164]

Because of the increased rate of cerebral infarction, EACA was found not to reduce early mortality, and its use was discouraged.

Reevaluation in a non-randomized study[165] excluding grade IV and V patients, suggests that the problems with EACA may be minimized by use of an IV loading dose (to eliminate the lag-period to effectiveness) and by limiting the length of time of use to that time until the patient can undergo early surgery. A more recent investigation[166] showed a significant decrease in rebleeding in EACA-treated patients versus non-EACA patients (2.7 vs. 11.4%). There was a 76% reduction in mortality attributable to rebleeding, a 13% increase in favorable outcome in good-grade (Hunt Hess I-III) EACA treated patients, and a 6.8% increase in poor grade patients (Hunt Hess IV/V), but these results did not reach statistical significance. Although there was an 8-fold increase in DVT in the EACA group, there was no increase in pulmonary embolism. Additionally, there was no difference in ischemic complications between groups.

□ [166]: EACA 4 g IV loading dose, followed by 1 g/h with cessation of infusion 4 hours before angiography for a maximum duration of 72 hours after SAH.

77.10 Pregnancy and intracranial hemorrhage

77.10.1 General information

Intracranial hemorrhage (subarachnoid or intraparenchymal) is a rare occurrence during pregnancy (estimated range of incidence: 0.01–0.05% of all pregnancies[167]) and yet is responsible for 5–12% of maternal deaths during pregnancy.

Intracranial hemorrhage of pregnancy (ICHOP) commonly occurs in the setting of eclampsia, and is more commonly intraparenchymal[168] and may be associated with loss of cerebrovascular autoregulation PRES (p. 194).[169] Symptoms of eclampsia with or without ICHOP include H/A, mental status changes, and seizures.

A literature review of 154 reported cases of ICHOP-related SAH revealed 77% were aneurysmal and 23% were from ruptured AVM (other series show the percentage of AVMs range from 21–48%). Mortality is ≈ 35% for aneurysmal and ≈ 28% for AVM hemorrhage (the latter being higher than in non-gravid patients). There is an increasing tendency for bleeding with advancing gestational age for both aneurysms and AVMs (earlier it had been asserted that this held true for aneurysms only[170]).

Patients with ICHOP having AVMs tend to be younger than those with aneurysm, paralleling the occurrence in the general population. One major oft-quoted study showed an increased risk of hemorrhage from AVMs during pregnancy[171] (citing an 87% hemorrhage rate), however another investigation disputes this assertion,[172] and found the risk of hemorrhage to be 3.5% during the pregnancy in patients with no history of hemorrhage, or 5.8% in those with previous hemorrhage. Another study evaluated risk of aneurysm rupture during pregnancy and delivery form the Nationwide Inpatient data and calculated the rupture risk during pregnancy and delivery to be 1.4% and 0.05%, respectively.[173] Literature review[167] found a risk of recurrent hemorrhage following ICHOP from aneurysm or AVM during the remainder of the pregnancy was 33–50%

77.10.2 Management modifications for pregnant patients

Modifications of evaluation and treatment techniques may be necessary for the pregnant patient.

1. neuroradiologic studies
   a) CAT scan: with shielding of the fetus, CAT scanning of the brain produces minimal radiation exposure to the child
   b) MRI:
      - generally felt to have low potential for complications, however, many centers will not do MRI during first trimester.
      - gadolinium based contrast agents (GBCAs) are teratogenic in animals in high repeated doses. It has not been studied in human pregnancy. A cohort of 26 women who received GBCAs during the first trimester showed no evidence of teratogenicity or mutagenicity.[174] There have also been no reported issues related to nephrogenic systemic fibrosis. GBCAs are FDA Class C drugs – not recommended for use during pregnancy, but may be used if benefits outweigh potential risks.
   c) angiography: with shielding of the fetus, radiation exposure is minimal. Iodinated contrast agents pose little risk to the fetus, The mother should be well hydrated during and after the study[167]
2. antiepileptic drugs: see Pregnancy and antiepileptic drugs (p. 458)
3. diuretics: the use of mannitol in pregnancy should be avoided to prevent fetal dehydration and maternal hypovolemia with uterine hypoperfusion
4. antihypertensives: nitroprusside should not be used in pregnancy
5. nimodipine is potentially teratogenic in animals, the effect on humans is unknown. It should be used only when the potential benefit justifies the risk

77.10.3 Neurosurgical management

The currently recommended treatment of a ruptured aneurysm in the pregnant patient is immediate surgical treatment to avoid rebleeding and ischemic complications due to vasospasm. A meta-analysis has demonstrated that mother and fetus both benefit from surgical treatment – with maternal mortality decreasing from 63% to 11% and fetal mortality decreasing from 27% to 5%[167,175] Successful endovascular treatment for aSAH has been reported, but fetal exposure to radiation is a concern. The absorbed fetal dose has been estimated to range from 0.17 to 2.8 mGy, corresponding to a fetal risk of a hereditary disease at birth and a cumulative risk for a fatal cancer by age 15 which are both substantially lower than that which naturally occur.[176] Because endovascular treatment requires heparin for systemic anticoagulation, it carries the risk of hemorrhagic implications when labor spontaneously begins during or around the time of embolization.

77.10.4 Obstetric management following ICHOP

Several reports have indicated that the fetal and maternal outcome is no different for vaginal delivery vs. C-section, and is probably more dependent on whether the offending lesion has been treated. However, there are no formal studies to help guide the optimal treatment of pregnant women with aSAH. One strategy[175] is to perform an emergent c-section, followed by aneurysm treatment, if the fetus is mature enough for survival outside the uterus. If the fetus is < 24 weeks, treat the aneurysm and maintain the pregnancy. If the fetus is between 24-28 weeks, a strategy should be tailored according to the maternal and fetal status. C-section may be used for fetal salvage for a moribund mother in the third trimester. During vaginal delivery, the risk of rebleeding may be reduced by the use of caudal or epidural anesthesia, shortening the 2nd stage of labor, and low forceps delivery if necessary.

77.11 Hydrocephalus after SAH

77.11.1 Hydrocephalus after traumatic SAH

See also posttraumatic hydrocephalus (p. 920).

77.11.2 Acute hydrocephalus

General information

The frequency of hydrocephalus (HCP) on the initial CT after SAH depends on the defining criteria used, with a reported range of 9–67%[177] A realistic range is ≈ 15–20% of SAH patients, with 30–60% of these showing no impairment of consciousness.[177,178] 3% of those without HCP on initial CT develop HCP within 1 week.[177]

Factors felt to contribute to acute HCP include: blood interfering with CSF flow through the Sylvian aqueduct, fourth ventricle outlet, or subarachnoid space, and/or with reabsorption at the arachnoid granulations.

Findings associated with acute HCP include[178]:

1. increasing age
2. admission CT findings: intraventricular blood, diffuse subarachnoid blood, and thick focal accumulation of subarachnoid blood (intraparenchymal blood did not correlate with chronic HCP, and patients with a normal CT had a low incidence)
3. hypertension: on admission, prior to admission (by history), or post-op
4. by location:
   a) posterior circulation aneurysms have a higher incidence of HCP
   b) MCA aneurysms correlate with low incidence of HCP
5. miscellaneous: hyponatremia, patients who were not alert on admission, use of preoperative antifibrinolytic agents, and low Glasgow outcome score

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Treatment

About half the patients with acute HCP and impaired consciousness improved spontaneously.[177] Patients in poor grade (H&H IV-V) with large ventricles may be symptomatic from the HCP and consideration should be given to ventriculostomy which caused improvement in ≈ 80% of patients in whom it was used.[177] There may be an increased risk of aneurysmal rebleeding in patients undergoing ventriculostomy shortly after SAH[177,179,180] especially if performed early and if ICP is lowered precipitously. The risk of aneurysm rebleeding with EVD has been studied in retrospective case series with mixed results.[181,182,183] The mechanism is controversial, but may be due to an increase in the transmural pressure (the pressure across the aneurysm wall which equals the difference between arterial pressure and ICP).

When a ventriculostomy is used, it is recommended to keep ICP in the range of 15–25 mm Hg[184] and to avoid rapid pressure reduction (unless absolutely necessary) to decrease the risk of IVC induced aneurysmal rebleeding. One paradigm is to keep the EVD open with the drip chamber nozzle 15-20 cm above the tragus.

Practice guideline: Acute hydrocephalus associated with aSAH

Level B[41]: CSF diversion (EVD or lumbar drain) for acute symptomatic hydrocephalus associated with aSAH.

77.11.3 Chronic HCP

Practice guideline: Chronic hydrocephalus associated with aSAH

- Level B[41]: permanent CSF diversion (shunt) for symptomatic chronic hydrocephalus following aSAH.
- Level C[41]: Weaning an EVD over > 24 hours does not appear to reduce the need for permanent CSF diversion.
- Level C[41]: Routine fenestration of the lamina terminalis is not recommended as it does not reduce the need for permanent CSF diversion.

Chronic HCP is due to pia-arachnoid adhesions or permanent impairment of the arachnoid granulations. Acute HCP does not inevitably lead to chronic HCP. 8–45%(reported range in literature[185]) of all ruptured aneurysm patients, and ≈ 50%of those with acute HCP following SAH need permanent CSF diversion. A number of studies have attempted to identify factors predictive of aSAH-associated shunt dependent chronic hydrocephalus. Intraventricular blood increases this risk.[185] There is controversy as to whether the use of ventriculostomy for acute HCP increases[186] or possibly even decreases[185] the incidence of shunt dependency. There may be a positive association between Fisher grade and likelihood of requiring CSF diversion for chronic hydrocephalus.[187] In addition, Hoh et al.[188] found age (2% increase/ year), comorbidity score (presence of DM, HTN, or alcohol abuse), admission type, insurance type (increased with medicaid and private payer) and hospital aneurysm volume (high > low) to be predictive of shunt placement in ruptured aneurysm patients. Treatment type (clip versus coil) has also been studied with no clear advantage for one modality over the other (p.1195).

The method of determining which patients require shunt placement has also been studied in a single center RCT.[189] There was no difference in the rate of shunt placement between those who underwent rapid weaning (< 24 hrs) versus gradual weaning (96 hrs) of the EVD (63.4%rapid versus 62.5%gradual).

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78 Critical Care of Aneurysm Patients

78.1 Neurogenic stress cardiomyopathy (NSC)

78.1.1 General information

Key concepts

- impaired cardiac function (reduced ejection fraction) not attributable to underlying coronary artery disease or myocardial abnormalities. May be reversible
- cardiac enzymes (troponin) tend to be lower than expected for the degree of myocardial impairment, distinguishes NSC from acute MI
- putative mechanism: catecholamine surge (possibly in myocardial sympathetic nerves) as a result of hypothalamic stimulation or injury from the SAH
- possible sequalae: hypotension, CHF, arrhythmias…all of which may further exacerbate cerebral ischemia
- peak incidence: 2 days to 2 weeks post SAH
- risk factors: higher Hunt and Hess grade
- treatment: may include dobutamine (for SBP < 90 and low SVR) and/or milrinone (for SBP > 90 and increased SVR)

Older terms: reversible postischemic myocardial dysfunction,[1] neurogenic stunned myocardium. Classically seen in patients following cardiac surgery, and attributed to a defect in troponin-I (TnI).[2] Some patients may develop myocardial hypokinesis following SAH.[3] May appear compatible with an MI on echocardiography, yet, troponin levels are typically lower (often < 2.8 ng/ml) than would be predicted given the level of myocardial impairment.[4] Peak incidence: 2 days to 2 weeks post SAH. The condition reverses completely in most cases within about 5 days as normal myocardial cells replace those with defective TnI. However, ≈ 10% of patients may progress on to an actual MI.

Stroke volume and cardiac output are reduced. Risk factors include higher Hunt Hess grade (> 3),[5,6,7,8] female gender,[8,9] smoking status and age.[5] Hypotension does not always occur since the reduced cardiac output (CO) may be offset by an increase in SVR. However, the reduced CO may impair the ability to tolerate barbiturates administered for cerebral protection during early surgery due to their myocardial suppressant effect. Intraoperative TEE monitoring may be a useful guide for titrating pressors. The reduced CO may also impede the use of hyperdynamic therapy for vasospasm.

78.1.2 Arrhythmias and EKG changes

EKG changes in over 50% of cases of SAH and include: broad or inverted T-waves, Q-T prolongation, S-T segment elevation or depression, U-waves, premature atrial or ventricular contraction, SVT, V-flutter or V-fib,[10] bradycardia. In some cases EKG abnormalities may be indistinguishable from an acute MI.[11,12]

78.1.3 Possible mechanism

Elevation of intracranial pressure secondary to aSAH is thought to cause sympathetic activation resulting in hypercontraction of cardiac myocytes and subsequent myocardial injury.[5] A related theory invokes hypothalamic ischemia resulting in increased sympathetic tone, whereby the hypothalamic ischemia results in increased sympathetic tone and the resultant catecholamine surge may produce subendocardial ischemia[13] or coronary artery vasospasm.[3] The catecholamine surge appears to be more focal (i.e. in the heart) than systemic.

78.1.4 Treatment

Interventions that have been studied for increasing cardiac output in NSC[14,15]:

1. milrinone: used when SBP > 90 mm Hg and normal SVR, or when the patient is on chronic beta blockers
2. dobutamine: more effective with hypotension (SBP < 90 mm Hg) and low SVR
3. other options: stellate ganglion block, magnesium

78.2 Neurogenic pulmonary edema

78.2.1 General information

A rare condition associated with a variety of intracranial pathologies, including:
- subarachnoid hemorrhage
- generalized seizures
- and head injury

78.2.2 Pathophysiology

Two possibly synergistic mechanisms. Sudden increased ICP or hypothalamic injury may produce a salvo of sympathetic discharge causing redistribution of blood to the pulmonary circulation, resulting in elevation of pulmonary capillary wedge pressures (PCWP) and increased permeability. Secondly, the associated surge of catecholamines directly disrupts the capillary endothelium which increases alveolar permeability.

78.2.3 Treatment

Supportive, using measures such as positive pressure ventilation with low levels of PEEP (p.871) and treatment to normalize ICP.

A PA-catheter is usually helpful.

There may be some efficacy in using a dobutamine infusion[16] supplemented with furosemide as needed. The theoretical advantage of dobutamine over previously attempted alpha- and beta-blockers is that dobutamine does not reduce cerebral perfusion.

78.3 Vasospasm

78.3.1 General information

Key concepts

- delayed cerebral ischemic symptoms and/or cerebral arterial narrowing on angiography that follows some cases of SAH (usually), trauma, or other insults
- time course: almost never before day 3 post SAH, peak incidence 6–8 days post SAH, rarely starts after day 17. Main time of risk: 3–14 days post SAH
- risk factors: higher SAH grade, more blood on CT
- results in pathologic changes within the vessel walls (not just vasoconstriction)
- diagnosis: may be clinical, angiographic, or with transcranial Doppler
- treatment: none are curative. Mainstays of treatment:
  - euvolemia and hemodynamic augmentation (formerly “triple H” therapy)
  - neuroendovascular intervention: angioplasty or intraarterial verapamil

Cerebral vasospasm is a condition that is most commonly seen following some cases of aneurysmal subarachnoid hemorrhage (SAH), but may also follow other intracranial hemorrhages (e.g. intraventricular hemorrhage from AVM,[17] and SAH of unknown etiology), head trauma (with or without SAH),[18] brain surgery, lumbar puncture, hypothalamic injury, infection, and may be associated with preeclampsia (p.194). The concept of vasospasm was originated in 1951 by Ecker.[19] Vasospasm has two not-necessarily reconcilable definitions (below):

1. clinical vasospasm: see below
2. radiographic vasospasm: see below

78.3.2 Definitions

Delayed cerebral ischemia (DCI) and early brain injury (EBI)

There has been a move away from thinking of the deleterious effects of SAH in terms of vasospasm, and the concepts of DCI and EBI are coming to the fore.[20]

DCI: Delayed development of a neurological deficit, decline in Glasgow coma scale of at least 2 points, and/or cerebral infarction unrelated to aneurysm treatment or other causes. DCI is an umbrella term that encompasses a number of clinical entities including symptomatic vasospasm, delayed ischemic neurological deficit (DIND), and asymptomatic delayed cerebral infarction.[21]

EBI: In addition to direct mechanical damage from the SAH, EBI also refers to a number of other factors including the transient increase in ICP, reduction of CBF, apoptosis and edema formation.

Clinical vasospasm

Sometimes referred to as delayed ischemic neurologic deficit (DIND), or symptomatic vasospasm. A delayed ischemic neurologic deficit following SAH. Clinically characterized by confusion or decreased level of consciousness sometimes with focal neurologic deficit (speech or motor). The diagnosis is one of exclusion, and sometimes cannot be made with certainty.

See clinical findings (p.1179).

Radiographic vasospasm (AKA angiographic vasospasm)

Arterial narrowing demonstrated on cerebral angiography, often with slowing of contrast filling. The diagnosis is solidified by previous or subsequent angiograms showing the same vessel(s) with normal caliber. In some cases a DIND corresponds to a region of vasospasm seen angiographically. The incidence of angiographic vasospasm following SAH is around 50%(range: 20-100%).[22]

78.3.3 Characteristics of cerebral vasospasm

Clinical findings

Findings usually develop gradually, and may progress or fluctuate. May include:

1. non-localizing findings
   a) new or increasing H/A
   b) alterations in level of consciousness (lethargy…)
   c) disorientation
   d) meningismus
2. focal neurological signs may occur including cranial nerve palsies[23,24] and focal motor deficits. Also, symptoms may cluster into one of the following "syndromes" (vasospasm incidence is higher in the distribution of the ACA than in that of the MCA)
   a) anterior cerebral artery (ACA) syndrome: frontal lobe findings predominate (abulia, grasp/suck reflex, urinary incontinence, drowsiness, slowness, delayed responses, confusion, whispering). Bilateral anterior cerebral artery distribution infarcts are usually due to vasospasm following an AComm aneurysm rupture
   b) middle cerebral artery (MCA) syndrome: hemiparesis, monoparesis, aphasia (or apractagnosia of non-dominant hemisphere – inability to use objects or perform skilled motor activities, due to lesions in the lower occipital or parietal lobes; subtypes: ideomotor apraxia and sensory apraxia)

Incidence

1. Radiographic cerebral vasospasm (CVS) is identified in 20–100%of arteriograms performed around the 7th day following SAH, whereas clinical vasospasm associated with radiographic CVS occurs in only ≈ 30%of patients with SAH[25]
2. radiographic CVS may occur in the absence of clinical deficit, and vice-versa

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Severity

1. CVS is a significant cause of morbidity and mortality in patients surviving SAH long enough to reach medical care, exceeded only by the direct effects of aneurysmal rupture as well as re-bleeding[26,27]
2. CVS ranges in severity from mild reversible dysfunction, to severe permanent deficits secondary to ischemic infarction in up to 60%of SAH patients,[28] extensive enough to be fatal in 7% of SAHs[25,28]
3. earlier onset of CVS is associated with greater deficit

Time course of vasospasm

1. onset: almost never before day 3 post-SAH[29]
2. maximal frequency of onset during days 6–8 post-SAH (however, rarely can occur as late as day 17). Typical at-risk period is quoted as days 3–14[30]
3. clinical CVS is almost always resolved by day 12 post-SAH. Once radiographic CVS is demonstrated, it usually resolves slowly over 3–4 weeks
4. onset is usually insidious, but ≈ 10%have an abrupt and severe deterioration

Correlated findings

1. Risk is higher in conditions where arterial blood at high pressure contacts the vessels at the base of the brain. CVS rarely occurs in the setting of intraparenchymal or pure intraventricular hemorrhage (e.g. from AVM) or in SAH with distribution limited to the cerebral convexity
2. blood clots are especially spasmogenic when in direct contact with the proximal 9 cm of the ACA and the MCA
3. not all patients with SAH develop CVS, and CVS can follow other insults besides SAH, such as mass resection,[31] meningitis,[32] amygdalohippocampectomy.[33] Can even be associated with sexual intercourse[34]and over-consumption of black licorice[35]
4. the Hunt and Hess grade on admission correlates with the risk of CVS (Table 78.1)
5. the amount of blood on CT correlates with the severity of CVS[36,37] (Table 78.2; also holds true for traumatic SAH[38])
6. higher incidence with increasing age of patient
7. a history of active cigarette smoking is an independent risk factor[39]
8. history of preexisting hypertension
9. there is good but not perfect correlation between the site of major blood clots on CT, the focality of delayed ischemic neurological deficits, and the visualization of angiographic CVS in corresponding arteries
10. pial enhancement on CT ≈ 3 days after SAH (with IV contrast administration) may correlate with higher risk of CVS (indicates increased permeability of BBB),[40] but this is controversial[41]

Table 78.1 Correlation of DIND with Hunt and Hess grade

Hunt and Hess grade	% DIND (clinical vasospasm)
1	22%
2	33%
3	52%
4	53%
5	74%

Table 78.2 Modified[43] grading system of Fisher[36] (correlation between the amount of blood on CT and the risk of vasospasm)

Modified Fisher scale group	Blood on CT[a]	Symptomatic vasospasm
	No SAH or IVH	
1	focal or diffuse thin SAH, no IVH	24%
2	focal or diffuse thin SAH, with IVH	33%
3	focal or diffuse thick SAH, no IVH	33%
4	focal or diffuse thick SAH, with IVH	40%

[a]measurements made in the greatest longitudinal & transverse dimension on a printed EMI CT scan (no scaling to actual thickness) performed within 5 d of SAH in 47 patients; falx never contributed more than 1 mm thickness to interhemispheric blood

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11. for patients undergoing early surgery, if there is little SAH left on a CT done 24 hours post-op, there is little risk of vasospasm
12. antifibrinolytic therapy reduces rebleeding, but increases the risk of hydrocephalus and vasospasm (p.1168) [42]
13. angiographic dye can exacerbate CVS
14. hypovolemia

78.3.4 Pathogenesis

Pathogenesis is still poorly understood.

In humans, CVS is a chronic condition with definite long-term changes in the morphology of the involved vessels. Much of CVS is poorly understood because of lack of a good animal model (humans show a mild acute phase, and most animal studies fail to demonstrate a chronic phase).

Pathological changes observed within the vessel wall are outlined in Table 78.3.

Direct mediators. Vasospasm is caused by smooth muscle contraction, due to either impaired vasodilatory mediators, overactive vasoconstrictive mediators, or more likely, both.

- The formed components of blood have each been shown to contribute to vasospasm
  - Oxyhemoglobin in pure form can cause contraction of cerebral arteries when contacting the abluminal surface of the vessel
  - Hemoglobin scavenges Nitric Oxide, a powerful vasorelaxant[46]
  - Platelet-derived growth factor induces vascular proliferation → vascular stiffening and impaired ability to dilate[47]
- Endothelial dysfunction: theories include decreased production of Nitrous Oxide and prostacyclins, and overproduction of Endothelin-1
- Vessel innervation by the sympathetic nervous system. Interruption of sympathetic innervation prevents vasospasm in rats[48]

Proposed mechanisms of vasospasm include

- contraction of the smooth muscle in the media of the vessel wall, as a result of:
  - vasoconstrictors within the hemorrhagic arterial blood[49] (see below)
  - vasoactive substances released into the CSF[50,51]
  - neuronal mechanisms via nervi vasorum (nerves in the vessel wall)
    - increased vasoconstrictor tone (possibly due to denervation supersensitivity)
    - loss of vasodilator tone
    - time dependent relative imbalance favoring vasoconstrictor over vasodilator innervation[52]
    - sympathetic hyperactivity: e.g. due to hypothalamic injury from elevated ICP[53]
  - impairment of endothelial derived relaxant factor (EDRF): vascular endothelium plays an obligatory role in vasodilatation caused by several pharmacologic agents by releasing a relaxant substance called EDRF[54]
- proliferative vasculopathy
- immunoreactive process
- inflammatory process
- mechanical phenomenon
  - stretching of arachnoid fibers
  - direct compression by blood clot
  - platelet aggregation[49]

Table 78.3 Pathological changes in vasospasm

Time	Vessel layer	Pathologic change
day 1–8	adventitia	↑ inflammatory cells (lymphocytes, plasma cells, mast cells) and connective tissue
	media	muscle necrosis and corrugation of elastica
	intima	thickening with endothelial swelling and vacuolization, opening of interendothelial tight junctions[44,45]
day 9–60	intima	proliferation of smooth muscle cells → progressive intimal thickening

78.3.5 Diagnosis of cerebral vasospasm

General information

Diagnosis requires appropriate clinical criteria, and ruling-out other conditions that can produce delayed neurologic deterioration, as shown in Table 78.4.

Ancillary tests for vasospasm

In addition to angiographically demonstrating vasospasm:
- transcranial doppler (TCD): see below
- alterations in intracranial pulse wave[56]
- CTA: specific for vasospasm, but may overestimate the degree of stenosis[57]
- MRA: may be useful for management of vasospasm (not a practical alternative to conventional angiography)[58]
- continuous quantitatively analyzed EEG monitoring in the ICU:
  - a decline of the percent of alpha activity (defined here as 6–14 Hz) called "relative alpha" (RA) from a mean of 0.45 to 0.17 predicted the onset of vasospasm earlier than TCD or angiographic changes[59]
  - a decline of total EEG power (amplitude) was 91%sensitive for predicting vasospasm[60]
- alterations in cerebral blood flow (CBF):
  - MRI: DWI and PWI may detect early ischemia (p.232)
  - CT perfusion study (p.227)
  - Xenon CT: may detect large global changes in CBF, but too insensitive to detect focal blood flow changes[61,62] and does not correlate with increase TCD velocities positron emission tomography (PET)[63] or SPECT scans (nonquantitative, and takes longer than xenon studies)

Transcranial doppler (TCD)

A noninvasive method of semiquantitatively measuring velocity of blood flow in a specific artery through the skull (in regions of thinner bone – insonation windows) utilizing ultrasound phase shift.

Narrowing of the arterial lumen as occurs in vasospasm elevates the blood flow velocity which may be detected with TCD.[64,65,66] Detectable changes may precede clinical symptoms by up to 24–48 hrs. Findings are often more helpful when baseline studies performed before vasospasm is likely to have begun are available.

Typical values are shown for the MCA in Table 78.5. Also, daily increases of > 50 cm/sec may suggest vasospasm. There is less correlation between velocities and vasospasm in the anterior cerebral

Table 78.4 Diagnosis of clinical vasospasm[55]

- delayed onset or persisting neuro deficit
- onset 4–20 days post-SAH
- deficit appropriate to involved arteries
- rule-out other causes of deterioration
  - rebleeding
  - hydrocephalus
  - cerebral edema
  - seizure
  - metabolic disturbances: hyponatremia…
  - hypoxia
  - sepsis
- ancillary tests (see text)
  - transcranial Doppler
  - CBF studies

Table 78.5 Interpretation of transcranial doppler for vasospasm

Mean MCA velocity (cm/sec)	MCA:ICA (Lindegaard) ratio	Interpretation
<120	<3	normal
120–200[a]	3–6	mild vasospasm[a]
>200	>6	severe vasospasm

[a]velocities in this range are specific for vasospasm but are only ≈ 60% sensitive

arteries (ACA). Distinguishing vasospasm from hyperemia (which increases blood flow velocities in both the MCA and the ICA) is facilitated by using the ratio of these velocities (the so-called Lindegaard ratio) also shown in Table 78.5.

Once values become elevated, it often takes several weeks to go back down.

Comparison of diagnostic modalities

Determining the sensitivities and specificities for the tests shown in Table 78.6 have allowed calculation of the positive predictive value (PPV) and negative predictive values (NPV) as listed[21]

78.3.6 Treatment for vasospasm

General information

See management protocol (p. 1185).

Numerous treatments for cerebral arterial vasospasm (CVS) have been evaluated.[67,68] Vasospasm in humans does not respond to the large variety of drugs that reverse experimental vasospasm in animal models.

Prevention of vasospasm

To date, there is no effective prophylactic intervention for CVS.[69] Vasospasm can often be mitigated by preventing post-SAH hypovolemia and anemia by employing hydration and blood transfusion. Although early aneurysm treatment (clipping or coiling) does not prevent CVS (in fact, manipulation of vessels may increase the risk), it facilitates treatment of CVS by eliminating the risk of rebleeding (permitting safe use of hypertension as needed) and surgical removal of clot (see below) may reduce the incidence of CVS; see Timing of aneurysm surgery (p. 1200) for discussion of early surgery. Prophylactic (i.e. before vasospasm has been diagnosed) hyperdynamic therapy – triple H therapy (p. 1186) – is not indicated (it may cause complications and does not provide any benefit).[70]

Vasospasm treatment options

Treatment options fall into the following categories:

1. direct pharmacological arterial dilatation
   a) smooth muscle relaxants:
      - calcium channel blockers (generally accepted for standard usage): nimodipine does not counteract vasospasm, but does improve neurologic outcomes (p. 1186)
      - endothelin receptor antagonists (experimental or research technique with potential for future application): ET_A antagonists (clazosentan) and $ET_{A/B}$ antagonists[71,72]
      - Ryanodine receptor blocker: Dantrolene. Mediates intracellular calcium release from the sarcoplasmic reticulum. One of the few drugs shown to both prevent and reverse vasospasm[73,74]
      - Magnesium: MASH-2 study showed no improvement in clinical outcome[75]

   b) sympatholytics (technique that is accepted for use but not necessarily standard or available at all centers)

Table 78.6 PPV & NPV of various tests for cerebral vasospasm

Test		PPV (%)	NPV (%)
TCD	MCA	83-100	29-98
	ACA	41-100	37-80
	ICA	73	56
	PCA	37	78
	BA	63	88
	VA	54	82
CTA		43-100	37-100
CTP		71-100	27-99

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c) intra-arterial papaverine[76,77]: short-lived (see below)
d) αICAM-1 inhibition (antibody to intracellular adhesion molecule; technique that is accepted for use but not necessarily standard or available at all centers)
2. direct mechanical arterial dilatation: balloon angioplasty (see below)
3. indirect arterial dilatation: utilizing hyperdynamic therapy (generally accepted for standard usage; see below)
4. surgical treatment to dilate arteries: cervical sympathectomy (technique not generally used or no longer accepted)[78]
5. removal of potential vasospasmogenic agents
   a) removal of blood clot: does not completely prevent vasospasm
      - mechanical removal at the time of aneurysm surgery[79,80]
      - subarachnoid irrigation with thrombolytic agents at the time of surgery or post-op through cisternal catheters[81,82,83,84] (must be initiated within ≈ 48 hrs of clipping) or intrathecally.[85] Hazardous with incompletely clipped aneurysm[84]
   b) CSF drainage: via serial lumbar punctures, continuous ventricular drainage, or postoperative cisternal drainage[86]
6. protection of the CNS from ischemic injury: calcium channel blockers (p.1183) – generally accepted for standard usage
7. improvement of the rheologic properties of intravascular blood to enhance perfusion of ischemic zones; also an endpoint of hyperdynamic therapy (p.1185); generally accepted for standard usage
   a) includes: plasma, albumin, low molecular weight dextran (technique not generally used or no longer accepted), perfluorocarbons (experimental or research technique with potential for future application), mannitol (p.1202)
   b) the optimal hematocrit is controversial, but ≈ 30–35%is a good compromise between lowered viscosity without overly reducing O_2 carrying capacity (hemodilution is used to lower Hct; phlebotomy is not used)
8. statins: no benefit was detected with simvastatin[87]
9. extracranial-intracranial bypass around zone of vasospasm (technique not generally used or no longer accepted)[88,89]

Vasodilatation by angioplasty

Catheter directed balloon angioplasty of vessels demonstrated to be in vasospasm[90,91]: available only in centers with interventional neuroradiologists. Risks of the procedure: arterial occlusion, arterial rupture, displacement of aneurysm clip,[92,93] arterial dissection. Only feasible in large cerebral vessels (distal arteries not accessible). Clinical improvement occurs in ≈ 60–80% Improvements in vessel diameter and neuro deficits have been observed in most studies.[94]

Prophylactic TBA: phase II prospective trial failed to show primary end point benefit (Glasgow Outcome Score), but fewer patients developed vasospasm.[95]

Criteria for transluminal balloon angioplasty (TBA):
1. failure of hyperdynamic therapy
2. ruptured aneurysm is repaired
3. optimal results when performed within 12 hours of onset of symptoms
4. may be done immediately post-clipping for vasospasm that was observed pre-op
5. controversial: asymptomatic vasospasm seen on the contralateral side during angioplasty for ipsilateral vasospasm. Some would balloon the asymptomatic side, but others cite the complication rate and would observe
6. recent cerebral infarction (stroke): a contraindication to TBA. Prior to TBA, perform CT or MRI to rule-out

Vasodilatation by intra-arterial drug injection

Vasodilatation by intra-arterial drug (IAD) injection produces effects that are shorter-lived and less profound at their peak than with angioplasty. While IAD can be repeated, this requires multiple arterial catheterizations. IAD is also of value to help open up vessels to allow placement of the angioplasty balloon, and for vessels inaccessible to angioplasty balloons.

Agents currently used for chemical spasmolysis (p.1587):
1. verapamil: the primary drug employed
2. nicardipine: a dihydropiridine calcium channel blocker which acts preferentially on vascular smooth muscle more than cardiac smooth muscle. Restores vessels to at least 60%of normal diameter. 70%of those treated had no stroke on CT. May cause a drop in SBP, but not >30%[96] intra-arterial therapy: 10–40 mg per procedure. Three retrospective case series have reported vessel dilation and transient improvement in neuro deficits.[94]

3. papaverine
4. nitroglycerine

78.3.7 Vasospasm management

Pertinent guidelines

Practice guideline: Management of cerebral vasospasm/DCI after aneurysmal SAH

- Level I[69]: maintain euvolemia and normal circulating blood volume
- Level I[69]:Induce hypertension unless BP is elevated at base-line or if precluded by cardiac stents
- Level II[69]: Endovascular angioplasty and/or selective intra-arterial vasodilator therapy is reasonable for patients not responding rapidly to or candidates for hypertensive therapy

Specific measures for Vasospasm/DCI after aSAH

Patients with clinical suspicion of vasospasm (DIND), or with transcranial doppler (TCD) increases of > 50 cm/sec or with absolute velocities > 200:

1. general care measures
   a) serial neuro exams: while important, sensitivity for CVS/DCI is limited in poor grade patients[69]
   b) activity: bed rest, HOB elevated to ≈ 30°
   c) TED hose and/or sequential compression boots
   d) strict I & O measurements
2. diagnostic measures (primarily to rule-out other causes of deficit)
   a) STAT non-contrast CT to rule-out hydrocephalus, edema, infarct or rebleed
   b) option: perfusion CT or MRI (if available)
   c) STAT bloodwork
      - electrolytes to rule-out hyponatremia[97]
      - CBC to assess rheology and rule-out sepsis or anemia
      - ABG to rule out hypoxemia
   d) repeat TCD if available to detect changes indicative of vasospasm
3. monitors
   a) A-line to monitor BP
   b) PA catheter to monitor PCWP and cardiac output when possible (central line to monitor CVP when PA catheter cannot be placed)
   c) insert ICP monitor if ICP felt to be problematic, treat elevated ICP with mannitol or CSF drainage before institution hemodynamic augmentation (caution: the diuresis from mannitol in treating ICP may produce hypovolemia; also, exercise caution in lowering ICP with unsecured aneurysm)
4. treatment measures
   a) continue nimodipine therapy. Give via NG tube if pt unable to swallow
   b) administer O_2 to keep pO_2 > 70 mm Hg
5. ensure euvolemia: patients with SAH often develop hypovolemia early in their course.[98,99,100]
   a) primary IV fluid is crystalloid, usually isotonic (e.g. NS)
   b) blood (whole or PRBC) when Hct drops < 40%
   c) colloid: plasma fraction or 5% albumin (at 100 ml/hr) to maintain 40% Hct (if Hct is > 40%, use crystalloids[101])
   d) mannitol 20% at 0.25 gm/kg/hr as a drip may improve rheologic properties of blood in the microcirculation (avoid hypovolemia from resultant diuresis)
   e) replace urinary output (U.O.) with crystalloid (if Hct < 40%, then use 5% albumin, usually @ ≈ 20-25 ml/hr)
   f) avoid hetastarch (Hespan®) (p.1165) and dextran which impair coagulation
6. monitoring labs
   a) ABG and H/H daily
   b) serum and urine electrolytes and osmolalities q 12 hr (creatinine elevations may indicate peripheral ischemia from vasopressors)

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   c) CXR daily
   d) frequent EKG
7. initiate hyperdynamic (triple-H) therapy (see below) unless BP is elevated at baseline or cardiac stents preclude it) for 6 hours
8. if no response to 6 hrs of triple-H therapy, or if doppler or perfusion CT or MRI suggests vasospasm, patient is taken to angiography to confirm presence of vasospasm and for interventional neuroradiologic treatment (intraarterial verapamil, angioplasty…)
1. move patient to the ICU and placed on triple-H therapy for 6 hours if this is not already instituted
2. option: perfusion CT or MRI (if available)
3. if no response to 6 hrs of triple-H therapy, or if perfusion CT suggests vasospasm, patient is taken to angiography to confirm presence of vasospasm and for interventional neuroradiologic treatment (intraarterial verapamil, angioplasty…)

Hemodynamic augmentation (formerly "TRIPLE-H" therapy). Many older treatment schemes for CVS included so-called "triple-H" therapy (for: Hypervolemia, Hypertension and Hemodilution).[102] This has given way to "hemodynamic augmentation" consisting of maintenance of euvolemia and induced arterial hypertension.[103] While potentially confusing, this has now sometimes been referred to as triple-H therapy.[69]

Inducing HTN may be risky with an unclipped ruptured aneurysm. Once the aneurysm is treated, initiating therapy before CVS is apparent may minimize morbidity from CVS.[104,105]

Use fluids to maintain euvolemia.

Administer pressors to increase SBP in 15% increments until neurologically improved or SBP of 220 mm Hg is reached. Agents include:
- dopamine (p.128)
  - start at 2.5 mcg/kg/min (renal dose)
  - titrate up to 15-20 mcg/kg/min
- levophed
  - start at 1-2 mcg/min
  - titrate every 2-5 minutes: double the rate up to 64 mcg/min, then increase by 10mcg/min
- neosynephrine (phenylephrine): does not exacerbate tachycardia
  - start at 5 mcg/min
  - titrate every 2-5 minutes: double the rate up to 64 mcg/min, then increase by 10mcg/min up to a max of 10 mcg/kg
- dobutamine: positive inotrope
  - start at 5 mcg/kg/min
  - increase dose by 2.5 mcg/kg/min up to a maximum of 20 mcg/kg/min

Complications of hemodynamic augmentation:
- intracranial complications[106]
  - may exacerbate cerebral edema and increase ICP
  - may produce hemorrhagic infarction in an area of previous ischemia
- extracranial complications
  - pulmonary edema in 17%
  - 3 rebleeds (1 fatal)
  - MI in 2%
  - complications of PA catheter[107]:
    - catheter related sepsis: 13%
    - subclavian vein thrombosis: 1.3%
    - pneumothorax: 1%
    - hemothorax: may be promoted by coagulopathy from dextran[106]

78.4 Post-op orders for aneurysm clipping

- admit PACU, transfer to ICU (neuro unit if available) when stable
- VS: q 15 min × 4 hrs, then q 1 hr. Temperature q 4 hrs × 3 d, then q 8 hrs. Neuro check q 1 hr
- activity: bed rest (BR) with HOB elevated 20–30°
- knee high TED hose and pneumatic compression boots
- I & O q 1 hr (if no Foley: straight cath q 4 hrs PRN bladder distension)
- incentive spirometry q 2 hrs while awake (do not use following transsphenoidal surgery)
- IVF: NS + 20 mEq KCl/L @ 90 ml/hr

For extubated patients:
- diet: NPO except minimal ice chips and meds as ordered
- O_2: 2 L per NC

For intubated patients:
- diet: NPO. NG tube to intermittent suction. May clamp for 1 hour after meds given
- ventilator orders

For all patients:
- meds:
  a) H2 antagonist, e.g. ranitidine 50 mg IVPB q 8 hrs
  b) Keppra® (levetiracetam): 500 mg PO or IV q 12 hours. Maintain therapeutic AED levels for 2–3 months post-op for most supratentorial craniotomies
  c) Cardene® drip: titrate to keep SBP < 160 mm Hg and/or DBP < 100 mm Hg (use cuff pressures, may use A-line pressures if they correlate with cuff pressures)
  d) analgesics: fentanyl (unlike morphine, does not cause histamine release. Lowers ICP) 25–100 mcg (0.5–2 ml) IVP, q 1–2 hrs PRN
  e) acetaminophen (Tylenol®) 650 mg PO/PR q 4 hrs PRN temperature T > 100.5° F (38 C)
  f) mini-dose heparin or enoxaparin (for DVT prophylaxis; no difference in heparin-induced thrombocytopenia with these 2 agents[108])
  g) calcium channel blockers; see admitting orders (p. 1165): nimodipine (Nimotop®) 60 mg PO/NG q 4 hrs or 30 mg q 2 hrs to avoid dips in BP. May be given IV where available
  h) continue prophylactic antibiotics if used: (e.g. cefazolin (Kefzol®) 500–1000 mg IVPB q 6 hrs x 24 hrs, then D/C)
- if available, transcranial doppler (p. 1182) to monitor MCA, ACA, ICA, VA and BA velocities and Lindegaard ratio (typical protocol is 3 × per week)
- labs:
  a) CBC once stabilized in ICU and q d thereafter
  b) renal profile once stabilized in ICU and q 12 hrs thereafter
  c) ABG once stabilized in ICU and q 12 hrs × 2 days, then D/C (also check ABG after any ventilator change if patient on ventilator)
- call M.D. if any deterioration in crani checks, for T > 101° (38.5 C), sudden increase in SBP, SBP < 120, U.O. < 60 ml/2-hrs

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79 SAH from Cerebral Aneurysm Rupture

79.1 Epidemiology of cerebral aneurysms

The incidence of cerebral aneurysms is difficult to estimate. Range of autopsy prevalence of aneurysms: 0.2–7.9% (variability depends on use of dissecting microscope, hospital referral and autopsy pattern, overall interest). Estimated prevalence of incidental aneurysms range from 1-5% of the population,[1,2,3,4,5] and they are becoming increasingly detected in clinical practice as use of CT and MRI is becoming more common.[6] Ratio of ruptured:unruptured (incidental) aneurysm is 5:3 to 5:6 (rough estimate is 1:1, i.e. 50% of these aneurysms rupture).[7] Unruptured intracranial aneurysms are more common in women (≈3:1 ratio)[8,9] and in the elderly,[10] and only 2% of aneurysms present during childhood.[11] When present in children, they tend to occur more frequently in males (2:1) and to a greater degree in the posterior circulation (40-45%).[12,13]

79.2 Etiology of cerebral aneurysms

The exact pathophysiology of the development of aneurysms is still controversial. In contrast to extracranial blood vessels, there is less elastic in the tunica media and adventitia of cerebral blood vessels, the media has less muscle, the adventitia is thinner, and the internal elastic lamina is more prominent.[14,15] This, together with the fact that large cerebral blood vessels lie within the subarachnoid space with little supporting connective tissue[16 (p 1644)] may predispose these vessels to the development of saccular aneurysms. Aneurysms tend to arise in areas where there is a curve in the parent artery, in the angle between it and a significant branching artery, and point in the direction that the parent artery would have continued had the curve not been present.[17]

The etiology of aneurysms may be:
1. congenital predisposition (e.g. defect in the muscular layer of the arterial wall, referred to as a medial gap)
2. “atherosclerotic” or hypertensive: presumed etiology of most saccular aneurysms, probably interacts with congenital predisposition described above
3. embolic: as in atrial myxoma
4. infectious – so called “mycotic aneurysms” (p. 1228)
5. traumatic; see Traumatic aneurysms (p. 1227)
6. associated with other conditions (see below)

79.3 Location of cerebral aneurysms

Saccular aneurysms, AKA berry aneurysms are usually located on major named cerebral arteries at the apex of branch points which is the site of maximum hemodynamic stress in a vessel.[18] More peripheral aneurysms do occur, but tend to be associated with infection (mycotic aneurysms) or trauma. Fusiform aneurysms are more common in the vertebrobasilar system. Dissecting aneurysms should be categorized with arterial dissection (p. 1323).

Saccular aneurysms location:
1. 85–95% in carotid system, with the following 3 most common locations:
   a) ACoA (single most common): 30% (ACoA & ACA more common in males)
   b) p-comm: 25%
   c) middle cerebral artery (MCA): 20%
2. 5–15% in posterior circulation (vertebro-basilar)
   a) ≈ 10% on basilar artery: basilar bifurcation, AKA basilar tip, is the most common, followed by BA-SCA, BA-VA junction, AICA
   b) ≈ 5% on vertebral artery: VA-PICA junction is the most common
3. 20–30% of aneurysm patients have multiple aneurysms (p. 1226) [19]

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79.4 Presentation of cerebral aneurysms

79.4.1 Major rupture

The most frequent presentation
1. most commonly produces SAH (p. 1167), which may be accompanied by:
2. intracerebral hemorrhage: occurs in 20–40% (more common with aneurysms distal to the Circle of Willis, e.g. MCA aneurysms)

3. intraventricular hemorrhage: occurs in 13–28%[20] (see below)
4. subdural blood occurs in 2–5%

79.4.2 Intraventricular hemorrhage

See also other etiologies of intraventricular hemorrhage (IVH) (p.1386).

IVH occurs in 13–28% of ruptured aneurysms in clinical series (higher in autopsy series)[20] and appears to carry a worse prognosis (64% mortality).[20] The size of the ventricles on admission was the most important prognosticator (large vents being worse). Patterns that may occur:

1. distal PICA aneurysms: may rupture directly into 4th ventricle through the foramen of Luschka[21]
2. a-comm aneurysm: it has been asserted that IVH occurs from rupture through the lamina terminalis into the anterior 3rd or lateral ventricles, however, this is not always borne out at the time of surgery
3. distal basilar artery or carotid terminus aneurysms: may rupture through the floor of the 3rd ventricle (rare)

79.4.3 Presentation other than major rupture

General information

May be thought of as possible “warning signs.”

1. mass effect
   a) giant aneurysms: including brain stem compression producing hemiparesis and cranial neuropathies
   b) cranial neuropathy (average latency from symptom to SAH was 110 days; note: the average latency quoted for some of these symptoms comes from a retrospective study of patients presenting with SAH who were identified as having a warning symptom[22]). See below:
   c) intra- or suprasellar aneurysm producing endocrine disturbance[23] due to pituitary gland or stalk compression
2. minor hemorrhage: warning or sentinel hemorrhage; see Headache (p.1158). This group had the shortest latency (10 days) between symptom and SAH (note: the average latency quoted for some of these symptoms comes from a retrospective study of patients presenting with SAH who were identified as having a warning symptom[22])
3. small infarcts or transient ischemia due to distal embolization (including amaurosis fugax, homonymous hemianopsia…)[24]: average latency from symptom to SAH was 21 days A
4. seizures: at surgery, an adjacent area of encephalomalacia may be found.[24] The seizures may arise as a result of localized gliosis and do not necessarily represent aneurysmal expansion as there is no data to indicate an increased risk of hemorrhage in this group
5. headache[24] without hemorrhage: abates after treatment in most cases
   a) acute: may be severe and “thunderclap” in nature,[25] some describe as “worst headache of my life.” Has been attributed to aneurysmal expansion, thrombosis, or intramural bleeding,[26] all without rupture
   b) present for ≥ 2 weeks: unilateral in about half (often retro-orbital or periorbital), possibly due to irritation of overlying dura. Diffuse or bilateral in the other half, possibly due to mass effect → increased ICP
6. incidentally discovered (i.e. asymptomatic, e.g. those found on angiography, CT or MRI obtained for other reasons)

Cranial neuropathies from aneurysmal compression

1. oculomotor (3rd nerve) palsy (ONP): occurs in ≈ 9% of p-comm aneurysms[27] (ruptured and unruptured), less common with basilar apex aneurysm. Symptoms of ONP may include:
   a) extraocular muscle palsy (eye deviates “down and out” → diplopia
   b) Ptosis
   c) dilated unreactive pupil (□ non-pupil-sparing third nerve palsy is the classic finding of 3rd nerve compression (p.562))
2. visual loss due to[24]
   a) compressive optic neuropathy with ophthalmic artery aneurysms: characteristically produces nasal quadrantanopsia
   b) chiasmal syndromes due to ophthalmic, a-comm, or basilar apex aneurysms
3. facial pain syndromes in the ophthalmic or maxillary nerve distribution that may mimic trigeminal neuralgia can occur with intracavernous or supraclinoid aneurysms[24,28]

 Note. The development of a third nerve palsy in a patient with an unruptured aneurysm is a medical emergency as it probably results from aneurysmal expansion and may portend impending rupture.

79.5 Conditions associated with aneurysms

79.5.1 Overview

1. autosomal dominant polycystic kidney disease: (see below)
2. fibromuscular dysplasia (FMD): prevalence of aneurysms in renal FMD is 7%, in aortocranial FMD 21%
3. arteriovenous malformations (AVM) including moyamoya disease; see AVMs and aneurysms (p.1241)
4. connective tissue disorders[29]:
   a) Ehlers-Danlos, especially type IV (deficient collagen type III) which also has a high rate of arterial dissection including with angiography or coiling
   b) Marfan syndrome (p.1322)
   c) pseudoxanthoma elasticum
5. multiple other family members with intracranial aneurysms. Familial intracranial aneurysm syndrome (FIA): 2 or more relatives, third degree or closer, harbor radiographically proven intracranial aneurysms. Also, see Familial aneurysms (p.1226)
6. coarctation of the aorta[30]
7. Osler-Weber-Rendu syndrome
8. atherosclerosis[31]
9. bacterial endocarditis
10. multiple Endocrine Neoplasia type I[32]
11. hereditary hemorrhagic telangiectasia[33]
12. neurofibromatosis type I[34]

79.5.2 Autosomal dominant polycystic kidney disease

General information

Adult polycystic kidney disease is seen in 1 of every 500 autopsies, and approximately 500,000 people in the U.S. carry the mutant gene for autosomal dominant polycystic kidney disease (ADPKD; there is also an autosomal recessive polycystic kidney disease). Renal function is usually normal during the first few decades of life, with progressive chronic renal failure ensuing. HTN is a common sequelae. Transmission is autosomal dominant, with 100% penetrance by 80 yrs of age.[35] Cystic disease of other organs may occur (viz.: liver in ≈ 33%, and occasionally lung, pancreas).[36]

The first association between ADPKD and cerebral aneurysms is attributed to Dunger in 1904. Reported prevalence of intracranial aneurysms with ADPKD: 10–30%,[37] with 15% being a reasonable estimate.[38] Most were located on the MCA, with multiple aneurysms present in 31%.[39] In addition to the increased incidence of aneurysms, there appears to be an increased risk of rupture,[40] with 64% occurring before age 50. As a result, patients with ADPKD carry a 10–20 fold increased risk of SAH compared to the general population.[41] Aneurysms are rarely detectable before age 20 years. The average rate of rupture of incidental aneurysms is ≈ 2%/yr (p.1167).

Recommendations

Using the above statistics, together with the life expectancy of patients with ADPKD, and other estimations (of operative morbidity and mortality, etc.), results of decision analysis is that arteriography not be routinely employed in patients older than 25 yrs.[37] However, patients with symptoms possibly due to unruptured aneurysms, and those with SAH, should undergo angiography and subsequent treatment of any aneurysms discovered (especially those > 1 cm diameter). A decision analysis study[38] determined that screening with MRA was beneficial compared to treating patients once they became symptomatic. Repeat MRA screening may be effectively repeated as follows:

1. every ≈ 2-3 years for a young patient with ADPKD with either:
   a) a history of aneurysms, or
   b) a kindred of ADPKD with aneurysms
2. every 5-20 years for a patient with ADPKD in a kindred of ADPKD with no history of aneurysms.[38]

79.6 Treatment options for aneurysms

79.6.1 General information

The optimal treatment for an aneurysm depends on the age and condition of the patient, the anatomy of the aneurysm and associated vasculature, the ability of the surgeon and the availability of endovascular treatment options, and must be weighed against the natural history of the condition. Also, treatment of the aneurysm facilitates treatment of vasospasm, should it occur.

Natural history:
1. risk of bleeding into subarachnoid space
   a) for ruptured aneurysms: this is the risk of rebleeding (p.1170)
   b) for unruptured aneurysms (p.1222)
   c) for cavernous carotid artery aneurysms: this risk is low (p.1225)
2. spontaneous thrombosis of an aneurysm is a rare occurrence[42,43,44] (estimates in autopsy series is 9–13%[44]). However they may reappear,[45,46] and delayed rupture may occur sometimes even years later
3. enlargement of aneurysm to cause mass effect: some aneurysms go on to become giant aneurysms and can cause mass effect with or without rupture

Although still controversial, endovascular treatment should be initially considered in treating amenable ruptured aneurysms. See also unruptured aneurysms (p.1225).

79.6.2 Therapies that do not directly address the aneurysm

The hope here is that the aneurysm will not bleed and that it will thrombose (see above).
1. continue medical management initiated on admission: i.e. control of HTN, continue calcium-channel blockers, stool softeners, activity restrictions…
2. treatment options generally not used
   a) antifibrinolytic therapy (e.g. ε-aminocaproic acid (EACA)): NB: NOT USED. Reduces rebleeding, but increases the incidence of arterial vasospasm and hydrocephalus[47]
   b) serial LPs: an historical treatment,[48] may increase the risk of aneurysmal re-rupture

79.6.3 Endovascular techniques to treat the aneurysm

1. thrombosing the aneurysm:
   a) “coiling” with Guglielmi electrolytically detachable coils (see below)
   b) Onyx HD 500 (p.1586) has been used for wide-necked or giant ICA aneurysms.[49] Out of 22 patients, there was 1 parent ICA stenosis and 2 ICA occlusions caused by Onyx migration
   c) “flow diversion” with “covered stents” (tightly woven stents) which promotes thrombosis of the aneurysm (p.1586)
2. trapping: effective treatment requires distal AND proximal arterial interruption, usually by endovascular techniques,[50] occasionally by direct surgical means (ligation or clip occlusion), or some combination. May also incorporate vascular bypass (e.g. EC-IC bypass) to maintain flow distal to trapped segment[51]
3. proximal ligation (so-called hunterian ligation after Hunter ligated the popliteal artery proximal to a peripheral aneurysm in 1784[52]): useful for giant aneurysms.[53,54] For non-giant aneurysms provides little benefit and adds the risk of thromboembolism (which may be reduced by occluding the CCA rather than the ICA[54]). May also elevate the risk of developing aneurysms in the contralateral circulation[55]

79.6.4 Surgical treatment options for aneurysms

1. clipping: the surgical gold standard. Surgical placement of a clip across the neck of the aneurysm to exclude the aneurysm from the circulation (see below) without occluding normal vessels
2. wrapping or coating the aneurysm: although this should never be the goal of surgery, situations may arise in which there is little else that can be done (e.g. fusiform basilar trunk aneurysms, aneurysms with significant branches arising from the dome, or part of the neck within the cavernous sinus)
   a) with muscle: the first method used to surgically treat an aneurysm[56] (the patient described died from rebleeding)

b) with cotton or muslin: popularized by Gillingham.[57] An analysis of 60 patients showed that 8.5% rebled in ≤6 mos, and the annual rebleeding rate was 1.5% thereafter[58] (similar to the natural history)

c) with plastic resin or other polymer: may be slightly better than muscle or gauze.[59] One study with long follow-up found no protection from rebleeding during the first month, but thereafter the risk was slightly lower than the natural history.[59] Other studies show no difference from natural course[60]

d) teflon and fibrin glue[61]

79.6.5 Treatment decisions: coiling vs. clipping

General information

The use of endovascular treatment for aneurysms has increased, with coil embolization being the most common endovascular modality (see above for other options). From 2002-2008, the rate of aneurysm coiling in the U.S. and U.K. increased from 17 and 35% to 58 and 68%, respectively.[62,63] There is considerable controversy and debate as to the best therapeutic approach for aneurysms (ruptured and unruptured). Impediments to resolving the debate include methodological shortcomings of published studies, the fact that endovascular methods are still rapidly evolving which renders many studies obsolete before completion, and the critical need for long-term follow-up of endovascular results.

This section reviews some of the available information comparing surgical treatment to coil embolization.

Ruptured intracranial aneurysms

To date, four randomized controlled trials have been published comparing functional outcome after coil embolization versus surgical clip ligation for ruptured intracranial aneurysms: the “Finnish Study”,[64] ISAT 2002,[65] the “Chinese Study”,[66] and BRAT 2012.[67] Table 79.1 summarizes the treatment data from the 4 RCTs.

ISAT. The largest trial, the International Subarachnoid Hemorrhage Aneurysm Trial (ISAT), enrolled 2143 patients and ran from 1997 to 2002, and was stopped prematurely because of a significant outcome difference between the 2 groups favoring endovascular coil embolization. Despite the limitations of ISAT (see below), the findings have often been generalized to all patients with aneurysms, resulting in a dramatic change in management.

Table 79.1 Summary of rebleeding, complete occlusion and retreatment rates as a function of treatment modality (clip vs. coil) for the 4 randomized controlled trials.

	Rebleed[a]: Clip	Rebleed[a]: Coil	Complete occlusion: Clip	Complete occlusion: Coil	Retreatment: Clip	Retreatment: Coil
Finnish	0%	0%	73.7%[b]	50%[b]	7%	23.1%
ISAT	1.0%	2.6%	82%	66%	4.2%	15.1%
$ISAT_5$[c]	0.3%*	0.9%*	n/a	n/a	———	———
$ISAT_{10}$[c]	0.4%	1.6%	n/a	n/a	———	———
Chinese	3.3%	3.2%	83.7%*	64.9%*	———	———
BRAT[d]	0.8%[e]	0%	85%	58%	4.5%*	10.6%*
$BRAT_3$[d]	0%	0%	87%	52%	5%*	13%*

* statistically significant difference ($p < 0.05$)
[a] Rebleeding from target aneurysm after first procedure
[b] Result achieved after treatment during first hospitalization
[c] $ISAT_5$ & $ISAT_{10}$ refer to the 5 and 10 year follow-up studies. Rebleeding results for these studies refer to recurrent SAH after the 1st year of follow-up
[d] $BRAT_3$ refers to the 3-year follow-up study. BRAT & $BRAT_3$ are “as-treated” results
[e] Both rebleeding events occurred during the initial hospitalization

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Table 79.2 Methodological shortcomings of ISAT

1. only 20% of 9559 patients presenting with SAH were randomized[a]
   a) selection could introduce bias
   b) more nonrandomized patients underwent MS than EDC
   c) guidelines not provided for which patients to consider for EDC
2. most study centers were located in Europe, Australia & Canada
3. the expertise of the surgeons and the interventionalists were not reported and were not necessarily comparable
4. the following features are not entirely representative of SAH patients at large
   a) 80% of patients were in good clinical condition (H&H grade 1 or 2)
   b) 93% of aneurysms were ≤ 10 mm diameter
   c) 97% were in the anterior circulation
5. rebleeding rate: after EDC (2.4%) or MS (1.0%) was high for both groups, and the difference could be more significant beyond the 1 year follow-up provided

[a]most SAH patients were referred specifically for MS or EDC. The only patients that were randomized were those for whom a panel decided it was not clear which procedure would be superior. Outcomes were not provided for non-randomized patients

Results: At 1 year, there was an absolute reduction of risk of having a poor outcome (i.e. Modified Rankin score > 2) by 7% with coiling (24%) compared to open surgery (31%; p=0.0019). Although not statistically significant, rebleeding in the first year after treatment was higher for coiling (2.6%) than clipping (1.0%). As such, the durability of coil embolization and its ability to prevent subsequent rebleeding of the treated aneurysm was questioned. Additionally, ISAT had many important shortcomings, as detailed in Table 79.2.

Following the initial report, medium-term follow-up results have been published.[68] In the endovascular cohort, there were 10 episodes of rebleeding from the treated aneurysm after 1 year, in 8447 person-years of follow-up. In the surgical cohort, 3 patients rebled from the treated aneurysm after 1 year, in 8177 person-years of follow-up (one of these patients had declined surgery after randomization and underwent coiling instead). There was a non-significant increased risk of rebleeding from the treated aneurysm in the endovascular cohort (p=0.06), by an intention-to-treat analysis, but a significant difference when analysis was by actual treatment (p=0.02). The probability of death at 5 years was significantly lower in the coiled group (11%) than in the clipping group (14%; p=0.03). However, when patients who died before treatment are excluded from this analysis, the statistical difference is no longer present (p=0.1).[69] The probability of independent survival for those patients alive at 5 years was no different between groups (83% coil; 82% clip).

The 10-year results have been reported for the UK cohort from the initial trial.[70] Similar to the 5-year results, the proportion of patients with a good outcome did not differ between the two groups, but the probability of being alive with a good outcome compared with death or dependence was significantly better for the endovascular group. Thirteen patients in the endovascular group rebled from the target aneurysm (1 per 641 patient-years) compared to 4 in the surgical group (1 per 2041 patient-years). Although the risk of rebleeding was higher in the endovascular group, the overall risk was small and the risk of death or dependency from a rebleed did not differ between the groups.

A follow up study, ISAT II (multicenter RCT), is currently being conducted to help elucidate differences in outcomes between treatment modalities.[71]

Chinese study[66] . 192 patients with aSAH randomized to coiling or clipping. Surgical clipping increased the risk of symptomatic vasospasm (OR 1.24), and there were significantly more new cerebral infarctions in the clipping group (21.7 vs. 12.8%). Incidence of complete aneurysm occlusion was significantly lower in the coiling group (64.9 vs. 83.7%). Rebleeding rates were similar in both groups (≈3%). At 1year, there was no significant difference in probability of mortality (coiling: 10.6%, clipping: 15.2%). Furthermore, there was no significant difference in probability of a good outcome (coiling: 75%, clipping: 67.9%).

BRAT[67] . Initiated at Barrow Neurologic Institute in 2002. Designed to reflect "real-world" practices of ruptured aneurysm treatment in North America. Randomly assigned in an alternate fashion every patient with SAH who agreed to participate. A large number of patients allocated to endovascular treatment crossed over to the surgical arm because patients could be enrolled regardless of whether the aneurysm was amenable to both treatment modalities (75 crossed over from coil to surgery; 4 crossed over from surgery to coil). Proportion of patients with a poor outcome (i.e. mRS > 2) was 33.7% in the surgical group versus 23.2% in the endovascular group (p=0.02, intention-to-treat analsysis). An "as-treated" analysis yielded similar results (33.9 vs 20.4%, p=0.01). There were 2 episodes of rebleeding following treatment – one assigned to and treated with clipping and the other assigned to coil, but treated with surgical clipping. Twelve patients (2.9%) required retreatment during the initial hospitalization (9 surgical and 3 coil patients). Overall, during the 1st year, there was a significant increased probability of retreatment in patients actually treated with coiling compared with those actually treated by clipping (10.6% of coil vs. 4.49% of surgical patients, p=0.03).

At 3 years,[72] there was no significant difference in poor-outcome between coiling (30%) and clipping (35.8%). Subgroup analysis: there were no differences in mRS scores between treatment groups at any time point among patients with anterior circulation aneurysms (83%). However, among posterior circulation aneurysms (17%), mRS scores were significantly better after endovascular management than after surgical treatment at every time point. Of note, with the exception of basilar tip aneurysms, the randomization of the posterior circulation aneurysms was unexpectedly skewed (large majority of SCA and PICA were clipped, whereas majority of PCA, vertebral and basilar where coiled). The lack of anatomical parity between the treatment groups makes it difficult to draw strong conclusions. In addition, the degree of aneurysm obliteration (87 vs. 52%), rate of aneurysm recurrence and rate of retreatment (5 vs. 13%) were significantly better in the group treated with clipping compared with coiling. However, no rebleeding occurrences were documented in the 2nd or 3rd year of BRAT.

Meta-analyses. Lanzino et al.[73] conducted a meta-analsysis on the 3 prospective controlled studies (Finnish, ISAT, BRAT). Pooled data showed poor outcome at 1 year to be lower in the embolization group; no difference in mortality between groups; and rebleeding rates within the first month were higher in coiled patients. However, the results were largely skewed by ISAT data.

Li et al.[74] conducted a meta-analysis on the 4 RCTs (see above) and 23 observational studies. The result of the RCT analysis with regard to poor outcome at 1 year paralleled that by Lanzino et al. However, there was no difference in poor outcome between groups in the nonrandomized controlled trial analysis. Additional subgroup analysis showed a higher incidence of rebleeding after coiling (≈2-3 vs. 1%), corresponding to a better complete occlusion rate of clipping (84 vs. 66.5%). Procedural complications rates and 1-year mortality did not differ significantly between the groups.

Vasospasm. Whether coiling or clipping has an independent correlation with symptomatic vasospasm is debatable. One meta-analysis[75] suggested a trend toward less symptomatic vasospasm after coiling compared with clipping. However, the analysis had multiple limitations – the two treatment groups were not comparable (age, clinical grade, aneurysm location); there were differences in study design and definitions of vasospasm; and there was a lack of angiographic diagnosis of vasospasm. In the Chinese RCT (above) symptomatic vasospasm and consequent cerebral infarction was more common in the coiling group. Li et al.[74] found vasospasm was more common after clipping (48.8 vs. 43.1%), however, ischemic infarction did not differ significantly. Treatment choice may also alter the spasm pattern: on one study,[76] patients who underwent clipping developed localized vasospasm around the rupture site, whereas those treated with coiling demonstrated progressive distal vasospasm over time (possibly related to treatment-specific effects on CSF circulation).

Shunt-dependent hydrocephalus. One study showed a lower incidence of shunt-dependent hydrocephalus in the surgical treatment group (19.9 vs. 47.1%),[77] however many others have failed to show this relationship.[74,78,79,80,81,82,83,84,85,86] A suggestion that fenestration of the lamina terminalis at the time of surgery may decrease shunt-dependent chronic hydrocephalus was refuted by a meta-analysis[87] of 11 non-randomized studies (hydrocephalus rates of 10% with fenestration compared to 15% without).

Seizures. A literature review[88] of seizures after aSAH reported a rate of ≈2% following either neurosurgical clipping or endovascular coiling. In contrast, ISAT showed endovascular intervention had lower seizure rates (13.3% to 3.3%) compared to surgical clipping (2.2-5.2%) in the first year. As such, there is no consensus as to whether treatment modality independently impacts seizure and/or epilepsy occurrence.

Factors to consider (clipping vs. coiling)
- health care environment / equipment available
- skill set and experience of the neurosurgeon and interventionalist
  - greater annual numbers of aneurysms treated by individual practitioners were significantly related to decrease in morbidity[89]
- anatomy and location of the aneurysm
  - favorable dome/neck ratio versus wide neck aneurysms
  - MCA aneurysms may be difficult to coil because of a branch near the neck
  - Basilar apex: favors coiling
  - associated IPH/SDH: surgery allows both evacuation of hemorrhage and treatment of aneurysm

- symptoms due to mass effect: clipping[90,91] may be better than coiling. In 13 patients with p-comm aneurysms and oculomotor nerve (3rd nerve) palsy (ONP), 6 of 7 patients clipped vs. 2 of 6 with coiling recovered completely.[27] Partial ONP improved with either treatment, but complete ONP recovered in 3 of 4 patients clipped vs. 0 of 3 coiled[27]
- patient age
  - younger age: lower risk of surgery, and lower lifetime risk of recurrence than with coiling
- clinical state / comorbidities
  - good outcome seen in 63% clip vs. 46% coil in poor grade (WFNS IV/V) pts (contrary to findings in practice guidelines[92]) therefore microsurgery and endovascular treatment, when selected primarily according to angiographic features, were equally likely to achieve good outcome[93]
  - patients on anticoagulation (e.g. Plavix) favor endovascular treatment

Practice guideline: Aneurysm treatment decisions

Level C[92]: Treatment decisions should be multidisciplinary (made by experienced cerebrovascular and endovascular specialists) based on characteristics of the patient and aneurysm.

Level C[92]: Microsurgical clipping may receive increased consideration in patients presenting with large (> 50 ml) intraparenchymal hematomas and middle cerebral artery aneurysms.

Level C[92]: Endovascular coiling may receive increased consideration in the elderly (> 70 yo), in those presenting with poor-grade WFNS classification (IV/V) aSAH, and in those with aneurysms of the basilar apex

Level B[92]: For patients with ruptured aneurysms judged to be technically amenable to both endovascular coiling and neurosurgical clipping, endovascular coiling should be considered

Unruptured intracranial aneurysms

As with ruptured aneurysms, there is controversy over the best treatment method for unruptured intracranial aneurysms (IUAs), and additional uncertainty surrounds the question of which IUAs need to be treated (vs. observed). Darsaut et al.[94] found that practitioners do not agree regarding management of IUAs, even when they share a background in the same specialty, similar capabilities in aneurysm management, or years of practice.

There are no prospective randomized studies of treatment interventions vs. conservative management,[95] or comparing treatment options to each other. Most data are either from personal series or are retrospective.

Surgical clipping. One summary of 260 patients (including a retrospective multicenter analysis) show no surgical mortality, and morbidity of 0-10.3% (6.5% major and 8% minor morbidity in the multicenter study).[5] Findings from one meta-analysis of 733 patients who underwent surgical clipping showed a mortality rate of 1% and major morbidity rate of 4%[96] A larger meta-analysis of 2460 patients revealed mortality and morbidity rates of 2.6% and 10.9%, respectively.[97]

The ISUIA investigators found surgical mortality to be 2.3% at 30 days, and 3.8% at 1 year.[98] Additionally, they found a combined morbidity and mortality at 1 year of 12.6% for those without previous hemorrhage and 10.1% for those with previous subarachnoid hemorrhage from another aneurysm. For patients treated with surgical clipping, morbidity and mortality was greatest in those with aneurysms that were large or in the posterior circulation, and in patients older than age 50. In comparison, the combined morbidity and mortality for the 451 patients treated with an endovascular procedure was 9.1% at 30 days and 9.5% at 1 year. Predictors of adverse outcome included aneurysm size and posterior circulation aneurysms. Additionally, the presence of calcification (independent of aneurysm size) has been shown to increase the likelihood of poor outcome.[99]

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Comparison of clipping to coil embolization. Early retrospective studies have shown a lower incidence of in-hospital death and discharge to skilled nursing facilities with endovascular therapy compared to those treated surgically.[100,101] A recent retrospective study conducted at a single center showed early outcome and lower complication rate favoring clipping, but the results did not remain significant long-term.[102] A meta-analysis[103] showed that clipping resulted in significantly higher disability compared with coiling (OR 2.38-2.83). However, subgroup analysis by outcome-measurement time, revealed clipping to be associated with greater risk of disability in the short-term (< 6 months), but not in the long-term (> 6 months). In addition, mortality (in-hospital and overall), hemorrhage, and infarction were no different between groups. Despite inclusion of a large number of studies and patients, it is very challenging to draw any conclusion from the meta-analysis, as all the studies were observational (i.e. low levels of evidence), and the analysis did not stratify outcomes based on size and/or location of aneurysms.

Lawson et al.[104] compared natural history rupture risk to national treatment risk for coiling and clipping (taken from the Nationwide Inpatient Sample from 2002-2008). Overall mortality rate for clipping and coiling was 2.66% and 2.17%, respectively. Poor outcome was significantly greater for clipping (4.75%) versus coiling (2.16%). Data regarding the homogeneity of the two groups regarding aneurysm size or location was not available. Treatment risk curves were generated and compared against natural history actuarial risk curves calculated from four prominent studies.[105,106,9,107] Overall, the analysis demonstrated rationale for clipping small, unruptured aneurysms in patients < 61-70 years, and coiling small unruptured aneurysms in patients < 70-80 years.

Additional studies have focused on the effect of age on outcomes. Mahaney et al.[108] showed that procedural and in-hospital morbidity and mortality increased with age in patients treated with surgery, but remained relatively constant with endovascular treatment. Poor neurological outcome from aneurysm or procedure-related morbidity and mortality did not differ between management groups for patients 65 years old and younger, but was significantly higher in the surgical group for patients older than 65 years. Surgery appeared to show a surgical benefit in patients < 50 years old at 1 year. Others have suggested an overall benefit of endovascular treatment over surgical clipping, which becomes more pronounced with age.[109]

Cost. Several studies have compared total hospital costs for treatment of unruptured aneurysms with mixed results. Halkes et al.[110] and Hoh et al.[111] found endovascular treatment to be associated with higher total hospital costs. A later study by Hoh et al.[112] found that on a national level, surgical clipping was associated with higher costs. A long-term outcome study[113] showed that clipping was associated with higher initial costs, but overall costs at 2 and 5 years were similar to coiling (due to higher number of follow-up angiograms and outpatient costs). More recently, total hospital cost was shown to be lower for clipping, despite higher fixed-direct and fixed-indirect costs.[114] This is a function of much higher variable costs (i.e. the cost of coils and devices) overcoming any substantial cost reduction due to shorter length of stay in patients treated endovascularly.

Miscellaneous. Oculomotor nerve palsy: Complete recovery of oculomotor nerve palsy associated with p-comm aneurysms is more common with surgical clipping than with endovascular treatment (87 % vs. 44%).[115]

Pregnancy: No studies have directly compared clipping versus coiling. Clipping may be preferred by some[116]; see pregnancy and SAH (p. 1169).

79.7 Timing of aneurysm surgery

79.7.1 Background

Historically, there was controversy between so-called "early surgery" (generally, but not precisely defined as ≤ 48–96 hrs post SAH) and "late surgery" (usually ≥ 10–14 days post SAH). The current consensus is that there should be intervention for a ruptured aneurysm (clipping or coiling) as promptly as possible to secure the aneurysm and prevent rebleeding. In a review of all patients with SAH treated by clipping or coiling in the Nationwide Inpatient Sample between 2002-2010, treatment at non-teaching hospitals and older age (> 80) were associated with delays in time to aneurysm clipping, but these associations were not seen when endovascular treatment was performed.[117] Increased time to procedure (> 3 days) was significantly associated with an increased likelihood of moderate to severe neurological deficit. Ultra-early (< 24 h after SAH) coiling of ruptured aneurysms has also been associated with improved clinical outcomes (mRS 0-2) compared to coiling at > 24 hours in poor grade SAH patients (Hunt-Hess IV/V).[118] This does not rule out a selection bias, however. Additionally, the increased morbidity and mortality associated with surgical intervention on patients who present subacutely with evidence of vasospasm on imaging, may be better suited for endovascular intervention.

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Early surgery advocated for the following reasons:

1. if successful, virtually eliminates the risk of rebleeding which occurs most frequently in the period immediately following SAH (p. 1167)
2. facilitates treatment of vasospasm which peaks in incidence between days 6–8 post SAH (never seen before day 3) by allowing induction of arterial hypertension and volume expansion without danger of aneurysmal rupture
3. allows lavage to remove potentially vasospasmogenic agents from contact with vessels, including use of thrombolytic agents (p. 1183)
4. although operative mortality is higher, overall patient mortality is lower[119]

Arguments against early surgery, in favor of late surgery include:
1. inflammation and brain edema are most severe immediately following SAH
   a) this necessitates more brain retraction
   b) at the same time this softens the brain making retraction more difficult (retractors have more tendency to lacerate the more friable brain)
2. the presence of solid clot that has not had time to lyse impedes surgery
3. the risk of intra-operative rupture is higher with early surgery
4. possible increased incidence of vasospasm following early surgery from mechanotrauma to vessels

Factors that favor choosing early surgery include:
1. good medical condition of patient
2. good neurologic condition of patient (Hunt and Hess (H&H) grade ≤ 3)
3. large amounts of subarachnoid blood, increasing the likelihood and severity of subsequent vasospasm (p. 1186), Table 78.2. Having the aneurysm clipped permits use of hyperdynamic therapy for vasospasm
4. conditions that complicate management in face of unclipped aneurysm: e.g. unstable blood pressure; frequent and/or intractable seizures
5. large clot with mass effect associated with SAH
6. early rebleeding, especially multiple rebleeds
7. indications of imminent rebleeding: (see below)

Factors that favor choosing delayed surgery (10–14 days post SAH) include:
1. poor medical condition and/or advanced age of patient (age may not be a separate factor related to outcome, when patients are stratified by H&H grade[120])
2. poor neurologic condition of patient (H&H grade ≥ 4): controversial. Some say the risk of rebleeding and its mortality argues for early surgery even in bad grade patients[121] since denying surgery on clinical grounds may result in withholding treatment in some patients who would do well (54% of H&H grade IV and 24% of H&H grade V patients had favorable outcome in one series[120]). Some data show no difference in surgical complications in good and bad grade patients with anterior circulation aneurysms[122]
3. aneurysms difficult to clip because of large size, or difficult location necessitating a lax brain during surgery (e.g. difficult basilar bifurcation or mid-basilar artery aneurysms, giant aneurysms)
4. significant cerebral edema seen on CT
5. the presence of active vasospasm

79.7.2 Conclusions

Practice guideline: Timing of intervention for ruptured aneurysm

Level B[92]: Surgical clipping or endovascular coiling of a ruptured aneursym causing aSAH should be performed as early as feasible in the majority of patients to reduce the risk of rebleeding.

79.7.3 Imminent aneurysm rupture

Findings that may herald impending aneurysm rupture and may therefore increase the need for expedient intervention include:
1. progressing cranial nerve palsy e.g. development of 3rd nerve palsy with p-comm aneurysm; traditionally regarded as an indication for urgent treatment (p. 1192)
2. increase in aneurysm size on repeat angiography
3. beating aneurysm sign[123]: pulsatile changes in aneurysm size between cuts or slices on imaging (may be seen on angiography, MRA, or CTA)

79.8 General technical considerations of aneurysm surgery

79.8.1 General information

The goal of aneurysm surgery is to prevent rupture or further enlargement of the aneurysm, while at the same time preserving all normal vessels and minimizing injury to brain tissue and cranial nerves. This is usually accomplished by excluding the aneurysm from the circulation with a clip across its neck. Placing the clip too low on the aneurysm neck may occlude the parent vessel, while too distal placement may leave a so-called "aneurysmal rest" which is not benign since it may enlarge (see below).

See Intraoperative aneurysm rupture (p.1204) for general measures to reduce the risk of this complication during surgery.

79.8.2 Aneurysmal rest

When a portion of the aneurysm neck is not occluded by a surgical clip, it is referred to as an aneurysmal rest. A "dog-ear" occurs when a clip is angled to leave part of the neck at one end, and obliterates the neck at the other. Rests are not innocuous, even if only 1–2 mm, because they may later expand and possibly rupture years later, especially in younger patients.[124] The incidence of rebleeding was 3.7% in one study, with an annual risk of 0.4–0.8% during the observation period of 4–13 yrs.[125] Patients should be followed with serial angiography, and any increase in size should be treated by reoperation or endovascular techniques if possible.

Booking the case: Craniotomy for aneurysm

Also see defaults & disclaimers (p.27).
1. position: (depends on location of aneurysm), radiolucent head-holder
2. intraoperative angiography (optional)
3. equipment: microscope (with ICG capability if used)
4. blood: type and cross 2 U PRBC
5. post-op: ICU
6. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery through the skull to place a permanent clip on the base of the aneurysm to prevent future bleeding, intraoperative angiogram, possible placement of external (ventricular) drain, possible lumbar drain
   b) alternatives: nonsurgical management, endovascular treatment only for aneurysms that are candidates
   c) complications: usual craniotomy complications (p.28) plus (the following are not really complications of surgery but are possible developments) post-op vasospasm, hydrocephalus, formation of new aneurysms

79.8.3 Surgical exposure

General information

To avoid excessive brain retraction, surgical exposure requires sufficient bony removal and adequate brain relaxation (see below).

Brain relaxation

More critical for ACoA and basilar tip than for easier to reach aneurysms such as p-comm or MCA.... Techniques include:
1. hyperventilation
2. CSF drainage: provides brain relaxation and a field dry of CSF, and removes blood & blood breakdown products along with the CSF. CSF drainage before opening the dura is associated with an increased risk of aneurysmal rebleeding (p.1167)
   a) ventriculostomy: risks include seizures, bleeding from catheter insertion, infection (ventriculitis, meningitis), possible increased risk of vasospasm
      • placed pre-op in cases of acute post-SAH hydrocephalus (p.1170)
      • placed intra-op
   b) lumbar spinal drainage (see below)
   c) intra-operative drainage of CSF from cisterns
3. diuretics: mannitol and/or furosemide. Although proof is lacking, lowering ICP by this or any means may theoretically increase the risk of rebleeding[126]

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Lumbar spinal drainage

May be inserted with Tuohy needle following induction of anesthesia (to minimize BP elevation), prior to final positioning. CSF is gradually withdrawn by the anesthesiologist only after the dura is opened (to minimize chances of intraoperative aneurysmal bleeding), usually a total of 30–50 cc are removed in ≈ 10 cc aliquots.

Risks include[127]: aneurysmal rebleeding (≤ 0.3%), back pain (10%, may be chronic in 0.6%), catheter malfunction preventing CSF drainage (< 5%), catheter fracture or laceration resulting in retained catheter tip in the spinal subarachnoid space, post-op CSF fistula, spinal H/A (may be difficult to distinguish from post-craniotomy H/A), infection, neuropathy (from nerve root impingement with needle), epidural hematoma (spinal and/or intracranial).

Cerebral protection during surgery

Pathophysiology of cerebral ischemia

The cerebral metabolic rate of oxygen consumption ($CMRO_2$) (p.1265) arises from neurons utilizing energy for two functions: 1) maintenance of cell integrity (homeostasis) which normally accounts for ≈ 40% of energy consumption, and 2) conduction of electrical impulses. Occlusion of an artery produces a central core of ischemic tissue where the $CMRO_2$ is not met. The oxygen deficiency precludes aerobic glycolysis and oxidative phosphorylation. ATP production declines and cell homeostasis cannot be maintained, and within minutes irreversible cell death occurs; a so-called cerebral infarction. Surrounding this central core is the penumbra, where collateral flow (usually through leptomeningeal vessels) provides marginal oxygenation which may impair cellular function without immediate irreversible damage. Cells in the penumbra may remain viable for hours.

Cerebral protection by increasing the ischemic tolerance of the CNS

1. drugs that mitigate the toxic effects of ischemia without reducing $CMRO_2$
   a) calcium channel blockers: nimodipine, nicardipine, flunarizine
   b) free radical scavengers: superoxide dismutase, dimethylthiourea, lazaroids, barbiturates, Vitamin C
   c) mannitol: although not a cerebral protectant per se, it may help re-establish blood flow to compromised parenchyma by improving the microvascular perfusion by transiently increasing CBV and decreasing blood viscosity
2. reduction of $CMRO_2$
   a) by reducing the electrical activity of neurons: titrating these agents to a isoelectric EEG reduces $CMRO_2$ by up to a maximum of ≈ 50%
      - barbiturates: in addition to reducing $CMRO_2$, they also redistribute blood flow to ischemic cortex, quench free radicals, and stabilize cell membranes. For dosing of thiopental, see below
      - isoflurane (p.105): shorter acting and less myocardial depression than with barbiturates
   b) by reducing the maintenance energy of neurons: no drugs developed to date can accomplish this, only hypothermia has any effect on this. Below mild hypothermia, extracerebral effects must be monitored (p.871)
      - mild hypothermia (core temperatures down to 33° C): in a multicenter RCT[128] mild hypothermia was demonstrated to be safe, but did not improve the neurological outcome after craniotomy among good-grade (Hunt Hess I-III) patients with aSAH
      - moderate hypothermia: 32.5–33° C has been used for head injury
      - deep hypothermia to 18° C permits the brain to tolerate up to 1 hour of circulatory arrest
      - profound hypothermia to < 10° C allows several hours of complete ischemia (the clinical usefulness of this has not been substantiated)

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Adjunctive cerebral protection techniques used in aneurysm surgery

1. systemic hypotension
   a) usually used during final approach to aneurysm and during manipulation of aneurysm for clip application
   b) theoretical goals
      - to reduce turgor of aneurysm facilitating clip closure, especially with atherosclerotic neck
      - to decrease transmural pressure (p.1170) to reduce the risk of intraoperative rupture
   c) One retrospective study[129] suggests that a decrease in MAP > 50% is associated with poor outcome. However, after adjustment for age, this association was no longer statistically significant. Because of the potential danger of hypoxic injury to brain and other organs (including areas of impaired autoregulation as well as normal areas), some surgeons avoid this method.

2. "focal" hypotension: using temporary aneurysm clips (specially designed with low closing force to avoid intimal injury) placed on parent vessel (small perforators will not tolerate temporary clips without injury)
   a) used in conjunction with methods of cerebral protection against ischemia
   b) may be combined with systemic hypertension to increase collateral flow
   c) the proximal ICA can tolerate an hour or more of occlusion in some cases, whereas the perforator bearing segments of the MCA and the basilar apex may tolerate clipping for only a few minutes
   d) in addition to the risk of ischemia, there is the risk of intravascular thrombosis and subsequent release of emboli upon removal of the clip
3. circulatory arrest, utilized in conjunction with deep hypothermia
   a) candidates include patients with large aneurysms that contain significant atherosclerosis and/or thrombosis that impedes clip closure and a dome that is adherent to vital neural structures
4. blood glucose: intraoperative hyperglycemia has been associated with long-term decline in cognition and gross neurologic function[130] and should be avoided

Systematic approach to cerebral protection

See reference.[131]

The following factors may mandate the use of temporary clips (and associated techniques of cerebral protection): giant aneurysm, calcified neck, thin/fragile dome, adherence of dome to critical structures, vital arterial branches near the aneurysm neck, intraoperative rupture. Aside from giant aneurysms, most of these factors may be difficult to identify pre-op. Therefore, Solomon provides some degree of cerebral protection to all patients undergoing aneurysm surgery.

1. spontaneous cooling is permitted during surgery, which usually results in a body temperature of 34° C by the time that dissection around the aneurysm begins
2. if temporary clipping is utilized
   a) if a long segment of the ICA is being trapped, administer 5000 U IV heparin to prevent thrombosis and subsequent emboli
   b) <5 mins temporary clip occlusion: no further intervention
   c) up to 10 or 15 mins occlusion: administer IV brain protection anesthesia (e.g. thiopental, propofol, and/or etomidate) and titrate to burst suppression on EEG
      - administration of IV brain protection anesthesia to burst suppression has been shown to significantly decrease infarction rate with temporary clipping within this time range[132]
      - intermittent reperfusion has been shown to be advantageous in some studies,[132] while in others findings have been contradictory[133,134]
   d) >20 mins occlusion: not tolerated (except possibly ICA proximal to p-comm), terminate operation if possible and plan repeat operation utilizing
      - deep hypothermic circulatory arrest (see above)
      - endovascular techniques
      - bypass grafting around the segment to be occluded

79.8.4 Postoperative angiography

Due to the fact that unexpected findings (aneurysmal rest, unclipped aneurysm, or major vessel occlusion) were seen on 19% of post-op angiograms (the only predictive factor identified was a new post-op deficit, which signaled major vessel occlusion) the use of routine post-op angiography has been recommended.[135]

79.8.5 Some drugs useful in aneurysm surgery

Drug info: Propofol (Diprivan®)

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May be used to achieve burst suppression[136] with shorter duration of action than other barbiturates. Results are preliminary, further investigation is needed to demonstrate the degree of neuroprotection. Has been reported at doses 170 mcg/kg/min for neuroprotection[137] (if tolerated) but this may be risky. May also be used as a continuous drip for sedation (p. 133), and for ICP management (p. 869). Reverses rapidly upon discontinuation (usually within 5–10 minutes).

Side effects: possible anaphylactic reaction with angioneurotic edema (angioedema) of the airways,[138] Propofol Infusion syndrome (p. 133).

79.8.6 Intraoperative aneurysm rupture

Epidemiology

Reported rates of intraoperative aneurysm rupture (IAR) range from ≈ 18% in the cooperative study (1963–1978)[139] to ≈ 36% in a pre-microscope series[140] (NB: this series had an unexplained high IAR rate of 61% with the microscope) and 40% in a more recent series.[141] Although rupture rate may be higher in early surgery than with late surgery,[141] other series found no difference.[142]

Morbidity and mortality for patients experiencing significant IAR is ≈ 30–35% (vs. ≈ 10% in the absence of this complication), although IAR may primarily affects outcome when it occurs during induction of anesthesia or opening of dura.[141]

See aneurysm rupture during coiling (p. 1587).

Prevention of intraoperative rupture

Presented as a list here to be incorporated into general operative techniques.

1. prevent hypertension from catecholamine response to pain:
   a) insure deep anesthesia during headholder pin placement and skin incision
   b) consider local anesthetic (without epinephrine) in headholder pin-sites and along incision line
2. minimize increases in transmural pressure: reduce MAP to slightly below baseline just prior to dural opening
3. reduce shearing forces on aneurysm during dissection by minimizing brain retraction:
   a) radical removal of sphenoid wing for circle of Willis aneurysms
   b) reduce brain volume by a number of mechanisms: diuretics (mannitol, furosemide), CSF drainage through lumbar subarachnoid drain placed pre-operatively and opened by the anesthesiologist at the time of dural incision, hyperventilation
4. reduce risk of large tear in aneurysm fundus or neck:
   a) utilize sharp dissection in exposing aneurysm and in removing clot from around aneurysm
   b) whenever possible, completely mobilize and inspect aneurysm before attempting clip application

Details of intraoperative rupture

Rupture can occur during any of the three following stages of aneurysm surgery[143]:

1. initial exposure (predissection)
   a) rare. Brain can become surprisingly tight even when bleeding seems to be into open subarachnoid space. Usually carries poor prognosis
   b) possible causes:
      - vibration from bone work: dubious
      - increasing transmural pressure upon opening the dura
      - hypertension from catecholamine response to pain (see above)
   c) management tactics:
      - have anesthesiologist radically drop BP
      - control bleeding (with anterior circulation aneurysms) by placing temporary clip across ICA as it exits from cavernous sinus, or if not possible then compress ICA in patient's neck through drapes
      - if necessary to gain control, resect portions of frontal or temporal lobe
2. dissection of the aneurysm: accounts for the majority of IARs, two basic types:
   a) tears caused by blunt dissection
      - tends to be profuse, proximal to the neck, and difficult to control
      - do not attempt definitive clipping unless adequate exposure has been achieved (which is usually not the case with these tears)
      - temporary clipping: this step is often necessary in this situation, after the temporary clip is in place return the MAP to normal and administer neuroprotective agent (e.g. propofol)
      - once the temporary clip is in place, it is better to take a few extra moments to improve the exposure and apply a well placed permanent clip instead of hastily clipping and trying to restore circulation
      - microsutures may need to be placed to close any portion of the tear that extends onto the parent vessel
   b) laceration by sharp dissection
      - tend to be small, often distally on fundus, and usually easily controlled by a single suction
      - may respond to gentle tamponade with a small cottonoid

Table 79.3 Follow-up schedule for treated aneurysms

Perform indicated study at the following times after treatment	
Coiled aneurysms	Clipped aneurysms
Study: CTA or gad-MRA[a]	Study: CTA
6 mos	1 year
1.5 years	5 years
3.5 years	every 10 years thereafter
? every 5–10 years (as with clipped aneurysms)	

[a]gad-MRA indicates gadolinium MRA which is more sensitive here than TOF-MRA (p. 232). Use the same modality for each follow-up to facilitate accurate comparison

- may shrink down with repeated low current strokes with the bipolar (avoid the temptation to use continuous high current)

3. clip application: bleeding at this point is usually due to either
   a) inadequate exposure of aneurysm: clip blade may penetrate unseen lobe of aneurysm. Similar to tears caused by blunt dissection (see above). Bleeding worsens as clip blades become approximated
      - prompt opening and removal of clip at the first hint of bleeding may minimize the extent of the tear
      - utilize 2 suckers to determine if definitive clipping can be done, or what is more common, to allow temporary clipping (see above)
   b) poor technical clip application: tends to abate as clip blades become approximated. Inspect the blade tips for the following:
      - to be certain that they span the breadth of the neck. If not, a second longer clip is usually applied parallel to the first, which may then be advanced
      - to verify that they are closely approximated. If not, tandem clips may be necessary, and sometimes multiple clips are needed

79.8.7 Aneurysm recurrence after treatment

Incompletely treated aneurysms may increase in size and/or bleed. This includes aneurysms that are clipped or coiled where there is still aneurysm filling, as well as a persistent aneurysm rest or a neck (p. 1201). While most aneurysm rests appear to be stable, there is a small subset that may enlarge or rupture.[144]

Additionally, even an aneurysm that has been completely obliterated may recur, and therefore one has to consider the durability of treatment. The risk of recurrence of a completely clipped aneurysm is ≈ 1.5%at 4.4 years.[144]

79.8.8 Follow-up after aneurysm treatment

Based on the above, together with the small risk of de novo aneurysm formation,[144] there is a trend to indefinitely follow patients with known aneurysms. One suggested follow-up schedule is shown in Table 79.3.

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80 Aneurysm Type by Location

80.1 Anterior communicating artery aneurysms

80.1.1 General information

The single most common site of aneurysms presenting with SAH.[1] May also present with diabetes insipidus (DI) or other hypothalamic dysfunction.

80.1.2 CT scan

SAH in these aneurysms results in blood in the anterior interhemispheric fissure in essentially all cases, and is associated with intracerebral hematoma in 63% of cases.[2] Intraventricular hematoma is seen in 79% of cases, with the blood entering the ventricles from the intracerebral hematoma in about one–third of these. Acute hydrocephalus was present in 25% of patients (late hydrocephalus, a common sequelae of SAH, was not studied).

Frontal lobe infarcts occur in 20%, usually several days following SAH.[2] One of the few causes of the rare finding of bilateral ACA distribution infarcts is vasospasm following hemorrhage from rupture of an ACoA aneurysm. This results in prefrontal lobotomy-like findings of apathy and abulia.

80.1.3 Angiographic considerations

Also see Table 102.2 in Endovascular section. Essential to evaluate contralateral carotid, to determine if both ACAs fill the aneurysm. If the aneurysm fills with one side only, it is desirable to inject the other side while cross compressing the side that fills the aneurysm to see if collateral flow is present. Also, determine if either carotid fills both ACAs, or if each ACA fills from the ipsilateral carotid injection (may permit trapping, see below).

If additional views are needed to better demonstrate aneurysm. Try oblique 25° away from injection side, center beam 3–4 cm above lateral aspect of ipsilateral orbital rim, orient x-ray tube in Towne's view. A submental vertex view may also visualize the area but the image may be degraded by the large amount of interposed bone.

80.1.4 Surgical treatment

Approaches

General information

1. pterional approach: the usual approach (see below)
2. subfrontal approach: especially useful for aneurysms pointing superiorly when there is a large amount of frontal blood clot (allows clot removal during approach)
3. anterior interhemispheric approach[3]: contraindicated for anteriorly pointing aneurysms as the dome is approached first and proximal control cannot be obtained (see below)
4. transcallosal approach

Pterional approach

Side of craniotomy:

A right pterional craniotomy is used with the following exceptions (for which a left pterional crani is used):

1. large ACoA aneurysm pointing to right: left crani exposes neck before dome
2. dominant left A1 feeder to aneurysm (with no filling from right A1): left crani provides proximal control
3. additional left sided aneurysm

See Pterional craniotomy (p. 1453) for positioning, etc. (use shoulder roll, rotate head 60° from vertical; see Fig. 94.5). Craniotomy is as shown in Fig. 94.7 (slightly more frontal lobe needs to be exposed than, e.g. for a p-comm aneurysm).

Lumbar drain (if IVC not already inserted) assists with brain relaxation.

Microsurgical dissection

Dissect down sylvian fissure with gentle retraction of frontal lobe away from base of skull. Olfactory nerve visualized first, then optic nerve. Open arachnoid over carotid and optic cistern and drain CSF. Elevate temporal tip, coagulate any bridging temporal tip veins that are present, and expose ICA.

Follow the ICA distally, looking for A1 (exposure of this allows temporary clipping in event of rupture). If the A1 take-off is too high, it may be hidden and would require excessive retraction to expose. Options to increase exposure include

1. gyrus rectus resection: a 1 cm long gyrus rectus corticectomy is performed[4] just medial to the olfactory tract. Helps find the ipsilateral A1 and often ACoA and A2. This is also helpful for down-pointing aneurysms because it permits visualization of the contralateral A1 before exposing the dome of the aneurysm (for proximal control). May lead to neuropsychiatric deficits. A subpial resection is performed with preservation of the small arterial branch that is consistently located here
2. fronto-temporal-orbital-zygoma removal
3. splitting the sylvian fissure: about 50%of experts do this routinely
4. ventricular drainage

Once found, A1 is followed until the ipsilateral A2 is identified. Then the contralateral A2 is identified and is followed proximally until the contralateral A1 is exposed. The a-comm is usually encountered in the process.

Critical branches to preserve: recurrent artery of Heubner; small ACoA perforators (may be adherent to aneurysm dome). If the aneurysm cannot be clipped, it may be trapped by clipping both ends of the ACoA only if each ACA fills from the carotid on its own side.

Post clipping, some authors recommend fenestrating the lamina terminalis in an effort to reduce the need for post-op shunting.

Anterior interhemispheric approach

See reference.[3]

Involves minimal brain retraction.

More suitable for an aneurysm that points straight up, but even with this proximal control is poor.

Position: supine with the neck extended ≈ 15°. A transverse skin incision is made in a skin crease in the lower forehead. The authors[3] describe using a 1.5 inch trephine craniotomy in the midline just superior to the glabella. Alternatively, better advantage of the dural opening may be possible with a more rectangular opening. The dural flap is hinged on the superior sagittal sinus. The depth of the aneurysm is ≈ 6 cm from the dura. Proximal control of the A1 branch of the ACA is difficult with this approach.

80.2 Distal anterior cerebral artery aneurysms

80.2.1 General information

Aneurysms of the distal anterior cerebral artery (DACA) (i.e. the ACA distal to the ACoA) are usually located at the origin of the frontopolar artery, or at the bifurcation of the pericallosal and callosomarginal arteries at the genu of the corpus callosum. Aneurysms located more distally are usually posttraumatic, infectious (mycotic), or due to tumor embolus.[5] DACA aneurysms are often associated with intracerebral hematoma or interhemispheric subdural hematoma[6] since the subarachnoid space is limited here. Conservative treatment of DACA aneurysms is often associated with poor results. Unruptured DACA aneurysms have a higher incidence of bleeding than unruptured aneurysms in other locations. These aneurysms are fragile and adherent to the brain, which predisposes to frequent premature intraoperative rupture.

On arteriography, if both ACAs fill from a single sided carotid injection, it may be difficult to make the important determination as to which ACA feeds the aneurysm. Multiple aneurysms are commonly associated with DACA aneurysms.

80.2.2 Treatment

Mycotic aneurysms should be treated as outlined (p. 1228).

Aneurysms up to 1 cm from the ACoA may be approached through a standard pterional craniotomy with partial gyrus rectus resection.

Aneurysms > 1 cm distal to the ACoA up to the genu of the corpus callosum, including those of the pericallosal/callosomarginal bifurcation, may be approached surgically by a basal frontal interhemispheric approach[7] via a frontal craniotomy using a bicoronal skin incision. The patient is positioned supine with the neck slightly extended, positioned vertically or just a few degrees to the left. A right sided craniotomy is preferred in most instances (exception: aneurysm dome buried in the right cerebral hemisphere making retraction hazardous), but should cross to the contralateral side by a couple centimeters. It must be taken all the way to the floor of the frontal fossa to permit exposure of the anterior cerebral artery for proximal control. The craniotomy extends ≈ 8 cm above the supraorbital ridge in order to provide leeway in circumnavigating veins bridging to the superior sagittal sinus. The dural flap is based on the superior sagittal sinus. If the sinus needs to be mobilized, it may be divided low anteriorly.

ACA aneurysms distal to the genu of the corpus callosum may also be approached by an interhemispheric approach using a unilateral skin incision. For these, the patient's neck is not extended, and a parasagittal craniotomy is used that doesn't need to be as low on the frontal fossa. The cingulate gyri may be difficult to separate, and care must be taken because excessive retraction may pull the cingulate gyrus off the dome of the aneurysm and produce premature rupture.

Ideally, A2 proximal to the aneurysm should be identified initially for proximal control and then followed distally to the aneurysm. When this is not possible, dissection should follow distal ACA branches proximally, towards the aneurysm, taking care not to disturb the aneurysm. Often, a portion of the cingulate gyrus may need to be removed and sometimes up to 1–2 cm of the anterior corpus callosum may need to be divided.

Surgical complications: Prolonged retraction on the cingulate gyrus may produce akinetic mutism that is usually temporary. The pericallosal arteries are small in caliber and may be atherosclerotic, which together increases the risk of occlusion of the parent artery with the aneurysm clip.

80.3 Posterior communicating artery aneurysms

80.3.1 General information

May occur at either end of p-comm; that is at the junction with the PCA, or more commonly at the junction with carotid (typically points laterally, posteriorly, and inferiorly). May impinge on the third nerve in either case and cause third nerve palsy (ptosis, mydriasis, “down and out” deviation) that, is not pupil sparing in 99% of cases. Surgical clipping may be more advantageous than endovascular coiling to treat oculomotor nerve palsies caused by pcomm aneurysms.[8,9]

80.3.2 Angiographic considerations

Also see Table 102.2 in Endovascular section. Vertebral artery (VA) injection is necessary to help evaluate the p-comm artery:

1. if the p-comm is patent: determine if there is a “fetal circulation” where the posterior circulation is fed only through the p-comm
2. determine if the aneurysm fills from VA injection

If additional views are needed to better demonstrate aneurysm

Try paraorbital oblique 55° away from injection side, center beam 1 cm posterior to inferior portion of lateral rim of ipsilateral orbit, orient x-ray tube 12° cephalad.

80.3.3 Surgical treatment

Pterional approach

See Pterional craniotomy (p. 1453) for positioning, etc. For the more common aneurysm at the ICA-p-comm junction, rotate head 15–30° from vertical (Fig. 94.5). Craniotomy is as shown in Fig. 94.7 (less frontal lobe needs to be exposed than for an ACoA aneurysm).

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Microsurgical dissection

Ultimately, the major vector of retraction will be on tip of temporal lobe (less on frontal lobe than in ACoA aneurysm), but the initial approach will be more anterior to reduce risk of intra-operative rupture.

1. dissect down sylvian fissure, retract frontal lobe and come down on optic nerve

2. cautiously elevate temporal tip (aneurysm may be adherent to temporal tip and/or to tentorium), coagulate bridging temporal tip veins if necessary
3. incise arachnoid membrane along the optic nerve from anterior to posterior
4. open arachnoid and drain CSF to gain relaxation
5. start to dissect carotid at anterior margin (at junction with optic nerve) and work towards the posterior margin of carotid where the aneurysm is located (isolating the carotid gives proximal control)

The aneurysm dome usually points laterally, posteriorly and inferiorly, and is encountered before and usually blocks visualization of the p-comm. The aneurysm frequently projects behind the tentorial edge which then obscures the dome.

Critical branches to preserve: anterior choroidal artery, posterior communicating artery (p-comm). If necessary, the p-comm may be sacrificed (e.g. included in clip) without deleterious effect in most cases if there is not a fetal circulation.

80.4 Carotid terminus (bifurcation) aneurysms

80.4.1 Angiographic considerations

See Table 102.2 in Endovascular section.

If additional views are needed to better demonstrate aneurysm. Try oblique 25° away from injection side, center beam 3–4 cm above lateral aspect of ipsilateral orbital rim, orient x-ray tube in Towne's view. Also may try submentovertex view.

80.4.2 Surgical considerations

See Pterional craniotomy (p.1453) for positioning, etc. (rotate head 30° from vertical, see Fig. 94.5. Craniotomy is as shown in Fig. 94.7.

80.5 Middle cerebral artery (MCA) aneurysms

80.5.1 General information

The following considers MCA aneurysms of the M1-M2 junction (referred to as "trifurcation" region, although this is not a true trifurcation (p.76)).

80.5.2 Surgical treatment

Approaches

1. trans-sylvian approach through a pterional craniotomy: this is the most commonly used approach
2. superior temporal gyrus approach[10]:
   a) advantages: minimizes brain retraction, possible reduced vasospasm from manipulation of proximal vessels
   b) disadvantages: proximal control difficult, slightly larger bone flap, possible increased risk of seizures

Craniotomy vs. craniectomy

Primary decompressive craniectomy (vs. craniotomy) for poor-grade MCA aneurysm SAH (WFNS IV/V) with associated IPH (>30cc) has not shown to provide any survival benefit and is not associated with improved outcome.[11]

Pterional approach

See Pterional craniotomy (p.1453) for positioning, etc. (rotate head 45° from vertical, Fig. 94.5).

Craniotomy

Craniotomy is as shown in Fig. 94.7. Less frontal lobe needs to be exposed than for, e.g. an ACoA aneurysm (distance "B" in Fig. 94.7 only needs to be ≈ 1 cm). The height "H" of the bony opening should be ≈ 5–6 cm (larger than for circle of Willis aneurysms).

Microsurgical dissection

Dissect down sylvian fissure with major vector of retraction on tip of temporal lobe (less on frontal lobe than in ACoA aneurysm). Open arachnoid and drain CSF. Elevate temporal tip, coagulate bridging temporal tip veins, and expose the ICA for proximal control in the event of rupture.

Follow the ICA distally by splitting the sylvian fissure to expose the M1 (again, for proximal control). Although exposure for proximal control is helpful to have as a contingency, one may be able to avoid temporary clipping of the MCA in the event of intraoperative rupture by controlling bleeding with a large suction, and subsequent clip placement (since the blood flow through the MCA is not as voluminous as through the ICA, and the surgical access to these aneurysms is usually fairly unrestricted).

Critical branches to preserve: distal MCA branches, recurrent perforators from the origin of the major MCA branches.

80.6 Supraclinoid aneurysms

See reference.[12]

80.6.1 Applied anatomy

The carotid artery exits the cavernous sinus and enters the subarachnoid space at the dural constriction known as the carotid ring (AKA clinoidal ring). The supraclinoid portion of the carotid artery may be divided into the following segments[13]:

1. ophthalmic segment: the largest portion of the supraclinoid ICA. Lies between the take-off of the ophthalmic artery and the posterior communicating artery (PCoA) origin. The proximal portion of this (including the origin of the ophthalmic artery) is often obscured by the anterior clinoid process. Branches include:
   a) ophthalmic artery: usually originates from the supracavernous ICA just after the ICA enters the subarachnoid space; see variants (p. 79). Enters the optic canal positioned inferolateral to the optic nerve
   b) superior hypophyseal artery: the largest of several perforators supplying the dura of the cavernous sinus and the superior pituitary gland and stalk
2. communicating segment: from the PCoA origin to the origin of the anterior choroidal artery (AChA)
3. choroidal segment: from AChA origin to the terminal bifurcation of the ICA

80.6.2 Ophthalmic segment aneurysms (OSAs)

See reference.[14]

General information

Ophthalmic segment aneurysms (OSAs) OSAs include (NB: nomenclature varies among authors):

1. ophthalmic artery aneurysms:
2. superior hypophyseal artery aneurysms:
   a) paraclinoid variant: usually does not produce visual symptoms
   b) suprasellar variant: when giant, may mimic pituitary tumor on CT

Presentation (excluding incidental discovery)

Ophthalmic artery aneurysms

Arise from the ICA just distal to the origin of ophthalmic artery. They project dorsally or dorsomedially towards the lateral portion of the optic nerve.

Presentation:

1. ≈ 45% present as SAH
2. ≈ 45% present as visual field defect:
   a) as the aneurysm enlarges it impinges on the lateral portion of the optic nerve → inferior temporal fiber compression → ipsilateral monocular superior nasal quadrantanopsia
   b) continued enlargement → upward displacement of the nerve against the falciform ligament (or fold) → superior temporal fiber compression → monocular inferior nasal quadrantanopsia
   c) in addition to near-complete loss of vision in the involved eye, compression of the optic nerve near the chiasm may also produce a superior temporal quadrant defect in the contralateral

eye (junctional scotoma AKA "pie in the sky" defect) from injury to the anterior knee of Wilbrand (nasal retinal fibers that course anteriorly for a short distance after they decussate in the contralateral optic nerve[15])

3. ≈ 10%present as both

Superior hypophyseal artery aneurysms

Originate in the small subarachnoid pocket medial to the ICA near the lateral aspect of the sella. The direction of enlargement is dictated by the size of this pocket and the height of the lateral sellar wall, resulting in two variants: paraclinoid & suprasellar.

Suprasellar variant may actually grow to a size large enough to compress the pituitary stalk and cause hypopituitarism and "classic" chiasmal visual symptoms (bilateral temporal hemianopsia).

Angiographic considerations

Also see Table 102.2 in Endovascular section. A notch can often be observed in the in the anterior, superior, medial aspect of giant ophthalmic artery aneurysms due to the optic nerve.[16]

 If additional views are needed to better demonstrate aneurysm. Try oblique 25° away from injection side, center beam 3–4 cm above lateral aspect of ipsilateral orbital rim, orient x-ray tube in Towne's view. Try submentovertex view.

80.6.3 Surgical treatment

See reference.[12]

Ophthalmic artery aneurysms

If necessary, the ophthalmic artery may be sacrificed without worsening of vision in the vast majority. Clipping a contralateral ophthalmic artery aneurysm is not technically difficult, and is not uncommonly required as OSAs are often multiple.

The aneurysm arises from the superomedial aspect of the ICA just distal to the ophthalmic artery origin, and projects superiorly.

Cutting the falciform fold early decompresses the nerve, and helps minimize worsening of visual deficit from surgical manipulation.

For unruptured aneurysms, drill off anterior clinoid via an extradural approach before opening dura to approach neck; for ruptured aneurysms, this may not be as safe.

In most cases, a side-angled clip can be placed parallel to the parent artery along the neck of the aneurysm.

Superior hypophyseal artery aneurysms

If necessary, the superior hypophyseal artery on one side may be clipped without demonstrable deleterious effect (due to bilateral supply to stalk and pituitary). Clipping a contralateral superior hypophyseal aneurysms is not really feasible.

With a usual pterional approach, the carotid artery is usually encountered first, and with large aneurysms is usually bowed laterally towards the surgeon. Clinoidal removal is usually required. The entire ICA wall may appear to be involved, and it may necessitate temporary ICA clipping (with cerebral protection) to reconstitute the ICA using encircling clips parallel to the parent vessel.

80.7 Posterior circulation aneurysms

80.7.1 General information

See also basilar tip aneurysms (p.1218). Clinical syndrome of SAH in the posterior fossa is indistinguishable from that due to anterior circulation aneurysms except for possible increased tendency towards respiratory arrest and subsequent neurogenic pulmonary edema (p.1178).[17] Vasospasm following posterior fossa SAH may be more likely to cause midbrain symptoms than vasospasm due to SAH elsewhere.

80.7.2 Hydrocephalus

In Yamaura's series,[18] 12%of patients required external ventricular drainage (EVD) following posterior fossa SAH to remove bloody CSF causing hydrocephalus, and 20%eventually required permanent ventricular shunt.

80.7.3 Vertebral artery aneurysms

General information

Traumatic vertebral artery aneurysms (VAA) (AKA dissecting aneurysms) are more common than non-traumatic VAAs. The following discussion concerns non-traumatic VAA.

Most VAAs arise at the VA-PICA junction. Other sites: VA-AICA, VA-BA.

Angiographic considerations

Also see Table 102.2 in Endovascular section. Angiography of VAA should assess the contralateral VA for patency in case of the need to trap the aneurysm. Allcock test (vertebral artery injection with carotid compression) may be used to assess patency of circle of Willis. Test occlusion with a balloon catheter can determine if patient will tolerate occlusion (a double lumen balloon will even allow measurement of distal back pressure).

PICA aneurysms

General information

For PICA anatomy, see Fig. 2.6. For arteriogram, see Fig. 2.7.

Comprise ≈ 3% of cerebral aneurysms. 3 common sites:

1. VA at the VA-PICA junction[19]:
   a) saccular aneurysms: most commonly at the distal (superior) angle. An aneurysm in this location should be suspected with a CT showing blood predominantly in the 4th ventricle[20] (aneurysmal dome may adhere to foramen of Luschka; rupture fills the ventricles with little subarachnoid blood visible on CT). The level is as varied as the PICA origin, and ranges from as low as the foramen magnum to as high as the ponto-medullary junction. Most VA-PICA aneurysms lie in the anterolateral portion of the medullary cistern,[21] anterior to the first dentate ligament.[22] However, the PICA origin may sometimes lie in the midline or across it
   b) fusiform aneurysms: usually the result of prior arterial dissection (p.1325)
2. PICA aneurysms distal to the VA-PICA junction: tend to be fragile and often develop multiple hemorrhages in a relatively short period, should be treated promptly, even when discovered incidentally
3. fusiform VA aneurysms involving PICA

Angiographic considerations

See Table 102.2 in Endovascular section.

Treatment

Options:

1. direct aneurysmal clipping is the preferred treatment
2. endovascular coil embolization: not as effective as clipping for relief of symptoms due to brainstem or cranial nerve compression
3. choices for unclippable and uncoilable aneurysms (e.g. fusiform, giant, or dissecting aneurysms) include:
   a) proximal (hunterian) VA ligation[23] which must be distal to the PICA origin to prevent severe morbidity or mortality[24]
   b) occlusion of the VA distal to the PICA origin (usually done endovascularly)
   c) midcervical VA occlusion (allows collateral flow through suboccipital muscular branches) e.g. endovascular Amplatzer plug

Surgical clipping of VA-PICA junction saccular aneurysms

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One approach to the VA-PICA junction is via a low extreme-lateral p-fossa approach. However, if the aneurysm is too far anterior to the brain stem, it may be totally out of vision or reach. Also, since these aneurysms usually project posteriorly and superiorly, the critical PICA will be directly in harms' way. Direct lateral approach more directly exposes the aneurysm[25] through a lateral suboccipital transcondylar approach.

Position: options include sitting position – less frequently used, see Sitting position (p.1445) – or lateral oblique ("park bench").

Lateral oblique position

Position: side of involved PICA is up, thorax elevated ≈ 15°. Head in-line with the thorax, neck slightly flexed, and slightly rotated 20° toward the floor (away from the side of the aneurysm). Upper shoulder depressed with adhesive tape. Lumbar spinal subarachnoid catheter placed, allows CSF drainage once dura is opened.

Options for skin incision:

Avoid opening too far laterally, otherwise the muscle mass impedes surgeon's vision.[26 (p 1747)]

1. From just above superior nuchal line to C2 vertebra[21]
   a) paramedian vertical incision
   b) midline vertical incision (hockey stick)
2. "sigmoid" incision starting 2 cm medial to mastoid notch, and curving to midline at level of C1 arch[27]

Craniectomy: lateral exposure of bone to the base of the mastoid, medially crossing the midline. Need not be quite as high as the transverse sinus. The foramen magnum is removed to its lateral margin. Removing the posterior arch of C1 from midline to the sulcus arteriosus (under VA) may help with proximal VA exposure[27] but is not usually necessary.[28]

Dural opening: K-shaped dural opening with a linear incision across the band at the foramen magnum (some patients have a sinus known as the arcuate sinus here that may require vascular clips).

Approach: first, gain proximal control of the VA where it first becomes intradural (in case of aneurysmal rupture). Retract cerebellum superiorly (caution: aneurysm dome may be adherent). Follow VA up from point where it enters dura; PICA origin then encountered usually just at neck of aneurysm (PICA origin may be confused for continuation of VA). Dissection must spare branches of pharyngeal filaments of spinal accessory nerve and lower filaments of vagus. May place temporary clip on VA proximal to PICA. Permanent clip usually placed between the fibers of IX & X above and XI below. It is better to leave a small residual aneurysm than to risk compromising PICA.[28]

Postoperative care: when neuropraxia of the lower cranial nerves is likely (in cases of difficult dissection or traction applied during clipping) the patient is kept intubated overnight. Patients who do not tolerate extubation at this point are immediately reintubated and elective tracheostomy is scheduled. Tracheostomy is maintained until the neuropraxia resolves.

Surgical clipping of distal PICA aneurysms

Aneurysms distal to the lateral medullary segment are approached through a craniectomy that extends across the midline.

80.7.4 Vertebrobasilar junction aneurysms

General information

Saccular aneurysms located where the two vertebral arteries join often form at the location of a basilar artery fenestration (basilar fenestration aneurysm).

Angiographic considerations

See ▯ Table 102.2 in Endovascular section.

CT-angiogram may be helpful as an adjunct because it can opacify both vertebral arteries simultaneously (not generally feasible with catheter angiogram).

Surgical approaches

1. suboccipital approach: for most; performed in lateral oblique position
2. subtemporal-transtentorial approach if the vertebrobasilar junction is high; performed in supine position

Suboccipital approach in lateral oblique position

NB: the side of approach must be chosen based on angiogram, as the extreme tortuosity of the VAs may cause the aneurysm of one VA to lie on the contralateral side of the brain stem.

Position: thorax elevated ≈ 15°. Head in-line with the thorax, neck slightly flexed, and slightly rotated away from side of aneurysm. Upper shoulder depressed with adhesive tape. Spinal subarachnoid catheter placed for CSF drainage, opened only once dura is opened.

80.7.5 AICA aneurysms

Angiographic considerations

See Table 102.2 in Endovascular section.

80.7.6 Basilar bifurcation aneurysms

General information

AKA basilar tip aneurysms. The most common posterior circulation aneurysm. Comprise ≈ 5% of intracranial aneurysms. Considered inoperable until Drake reported 4 cases in 1961,[29] with larger series reported later.[30]

Presentation

Most present with SAH indistinguishable from SAH due to anterior circulation aneurysmal rupture. Enlargement of the aneurysm prior to rupture may rarely compress the optic chiasm → bitemporal field cut (mimicking pituitary tumor), or occasionally may compress the third nerve as it exits from the interpeduncular fossa → oculomotor nerve palsy.[17]

CT/MRI scan

May occasionally be seen on CT or MRI as round mass in region of suprasellar cistern. With SAH, tend to see blood in interpeduncular cistern with some reflux into 4th (and to a lesser extent, third and lateral) ventricle. Occasionally may mimic pretruncal nonaneurysmal SAH (p.1231).

Angiography

Also see Table 102.2 in Endovascular section. Dome usually points superiorly. Should evaluate flow through posterior communicating arteries (may require Allcock test) in case trapping is required. Need to assess the height of the basilar bifurcation in relation to the dorsum sella (below).

Critical angiographic features to assess: On angiogram or CTA:
1. general features (p.1162)
2. orientation: determines whether surgery is an option. Posteriorly pointing aneurysms obscure perforators which may be adherent to the aneurysm, making surgery more difficult
3. patency of PCAs & SCAs
4. patency and size of p-comms
   a) diameter of p-comm > 1 mm is needed to support collateral flow (expert opinion)
   b) to determine if the P1's can be sacrificed
   c) P-comm patency and size is important for endovascular treatment as a potential route for deployment of horizontally oriented stent extending from P1 to contralateral P1[31,32]
   d) which can facilitate temporary clipping, or sacrifice, or placement of stents.
5. height of the aneurysm relative to the posterior clinoid process which will affect the selection of surgical approach[33,34] (the range of height of the posterior clinoid is 4–14 mm[34])
   a) supraclinoidal: aneurysm neck > 5 mm superior to posterior clinoid process
   b) clinoidal: aneurysm neck within 5 mm of posterior clinoid process
   c) infraclinoidal: aneurysm neck > 5 mm inferior to posterior clinoid process

Surgical treatment

Timing

Initial experience tended to favor allowing basilar tip aneurysms to "cool-down" for ≈ 10–14 days after SAH before attempting surgery to permit cerebral edema to subside. More recently, early surgery for these aneurysms has been advocated as for anterior circulation aneurysms (p.1199).[35] However, some surgeons still recommend waiting ≈ 1 week,[36] and most would agree that if there are obvious technical difficulties because of size, configuration or location of the aneurysm, that early surgery may not be appropriate. Also, if during the craniotomy it becomes apparent that cerebral edema is impairing the exposure, the operation should be aborted and attempted again at a later date.

Approaches

1. right subtemporal craniotomy (classical approach of Drake): approached through the incisura or division of the tentorium. Most basilar tip aneurysms are probably best approached via pterional approach (see below) except for posteriorly pointing aneurysms

a) advantage:
   - less distance to basilar tip
   - may be better than pterional approach for aneurysms projecting posteriorly or posteroinferiorly[36]

b) disadvantages:
   - requires temporal lobe retraction (minimized with lumbar drainage, mannitol, and possibly zygomatic arch section[37])
   - poor visualization of contralateral P1 segment and thalamoperforators

2. pterional approach (described by Yasargil): trans-Sylvian (see below)
   a) advantages:
      - little or no retraction on temporal lobe (unlike subtemporal approach)
      - better visualization of both P1 segments and thalamoperforators
      - other aneurysms, e.g. of the anterior circulation, can be dealt with at the same sitting
   b) disadvantages:
      - increases reach to aneurysm by ≈ 1 cm compared to subtemporal
      - requires wide splitting of the sylvian fissure
      - operating field is narrower than subtemporal approach
      - perforators arising from the posterior aspect of P1 may not be visible
3. modified pterional craniotomy: may allow trans-sylvian or subtemporal approach.[36] The craniotomy is taken further posteriorly than a standard pterional craniotomy
4. orbitozygomatic approach: allows access to portions of the basilar artery below the bifurcation. May be augmented by removal of the top of the clivus

Optional resection of the temporal tip will increase exposure of either approach. Unlike most anterior circulation aneurysms, securing proximal control is very difficult.

If the basilar bifurcation is high above the dorsum sella, then more retraction is required on a subtemporal approach than for a normal bifurcation height (near the dorsum sella). A high bifurcation is dealt with on a trans-sylvian approach by opening the sylvian fissure more widely, or by a subfrontal approach through the third ventricle via the lamina terminalis.[38] A low bifurcation may require splitting the tentorium behind the 4th nerve.

Pterional approach

See reference.[39]

Risks include: oculomotor palsy in ≈ 30%(most are minimal and temporary).

Approach is from the right unless:

1. additional left sided aneurysm (e.g. p-comm aneurysm) which could be treated simultaneously by a left sided approach
2. aneurysm points to the right
3. aneurysm is located to the left of midline (the operation is more difficult when the aneurysm is even just 2–3 mm contralateral to the craniotomy)[36]
4. patient has right hemiparesis or left oculomotor palsy

See Pterional craniotomy (p. 1453) for general information. Rotate the head ≈ 30° off the vertical so that the malar eminence points directly upward (Fig. 94.5). Slight neck flexion is used for low-lying aneurysms, slight extension for high ones. Craniotomy is as shown in Fig. 94.7, with aggressive removal of the sphenoid wing. The sphenoid wing and the orbital roof may be reduced with a drill. The posterior clinoid can be removed to improve exposure.

Approach

The sylvian fissure is split until the take-off of the proximal M1 from the carotid terminus is identified. The approach is medial to the ICA (between the ICA and optic nerve) when this space is ≥ 5–10 mm. If the ICA is close to the optic nerve, an approach lateral to the ICA may be used, aided by medial retraction of the ICA/M1 segment (Fig. 94.8). Here, the exposure is limited by the height of the M1 branch above the skull base, and if the basilar tip height above the skull base greatly exceeds this, clipping via this approach is not feasible.[18]

The 3rd nerve is identified. Also the p-comm and the anterior choroidal artery (AChA) are located as they arise from the posterior surface of the ICA (to differentiate between them: the p-comm origin is proximal to that of the AChA, p-comm courses perpendicular to Liliequist's membrane whereas AChA courses obliquely into the crural cistern). The p-comm is followed posteriorly through Liliequist's membrane which is opened revealing the prepontine cistern. The p-comm is followed until it joins the PCA at the P1/P2 junction. If p-comm is absent, follow the third nerve back to find where it emerges between PCA and SCA. P1 is followed proximally to the basilar bifurcation region where the contralateral P1 and both SCAs are identified. Caudal dissection of Liliequist's membrane exposes the interpeduncular cistern with proximal BA (this exposure is critical for proximal control of BA in the event of aneurysmal rupture).

Thalamoperforating arteries (ThPAs) arise from the distal p-comm and proximal PCA, and often compromise the access. Early poor results with clipping of basilar tip aneurysms has been attributed to sacrificing these vessels, which produces lacunar infarcts in the thalamus, midbrain, subthalamic, and pretectal regions. If hypoplastic, the p-comm may be divided between clips to improve exposure (preserving the ThPAs which will then arise from the stumps). Similarly, a hypoplastic P1 may be divided if the PCA fills from the p-comm. If the ThPAs make it impossible to clip the aneurysm, some may have to be sacrificed, which is best done at their origin. Fortunately, there are some anastamoses[40] and thus they are not entirely end-arteries as originally thought.

Outcome

If the aneurysm cannot be treated with endovascular technique, then the surgical option can be considered. Overall mortality is 5%, and morbidity is 12%(mostly due to injury to perforating vessels).[41]

80.7.7 Basilar trunk aneurysms

Most aneurysms of the basilar trunk are fusiform in morphology. Surgical access for these is extremely difficult.

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81

81 Special Aneurysms and Non-Aneurysmal SAH

81.1 Unruptured aneurysms

81.1.1 General information

Unruptured intracranial aneurysms (UIA) includes incidental aneurysms (those that do not produce any symptoms and are discovered incidentally) and aneurysms that produce symptoms other than those due to hemorrhage (e.g. pupillary dilatation due to third nerve compression). UIA merit consideration for treatment since the outcome from SAH with or without surgery is poor even under the best of circumstances. About 65% of patients die from the first SAH,[1] and even in patients with no neurologic deficit after aneurysm rupture, only 46% fully recover, and only 44% return to their former jobs.[2] However, the risk of aneurysmal rupture without intervention should be weighed against the risks of surgical clipping or endovascular treatment. Estimated prevalence of incidental aneurysms is 5–10% of the population.[2]

81.1.2 Presentation

See items other than those listed under “rupture” in Presentation of aneurysms (p.1191).

81.1.3 Natural history

Risk of bleeding from UIA differs from aneurysms that have ruptured. True risk is not known with certainty. Early studies found annual bleeding rate of 6.25%, whereas later reports estimate lifetime risk for a 20-year-old with an UIA to be 16%, which drops off to 5% for a 60-year-old.[2] A more recent study[3] estimates the annual rupture rate to be ≈ 1%. The International Study of Unruptured aneurysms (ISUIA)[4] was the first large-scale, prospective study evaluating the natural history of unruptured aneurysms as well as the risks of treatment of unruptured aneurysms. The authors concluded that rupture rate was related to size and location of the aneurysm, and that risk is increased with previous aSAH from a separate aneurysm (see below). However, there were important limitations of ISUIA (see Table 81.1).

Σ

There appears to be 2 distinct types of aneurysms: those that rupture, and those that tend to remain stable. Most UIAs seen in the clinic fall into the latter group

Spontaneous thrombosis of unruptured aneurysms may occur rarely (p.1194).

Additional retrospective and prospective studies have been conducted to assess the natural history of unruptured aneurysms. Overall, several variables have been identified as risk factors for rupture:

1. patient factors
   a) history of previous aSAH from a separate aneurysm[4,5]
   b) multiple aneurysms[6,7]
   c) age: there is conflicting evidence, as some studies have found an inverse relationship between age and rupture risk,[7,8] while others have found an increased risk of rupture for those aged 40 or older,[9] or no effect of age on rupture risk[10]
   d) medical conditions:
      - hypertension,[7]
      - smoking[8]
   e) geographic location: North America/Europe < Japan < Finland[11]

Table 81.1 Main methodological limitations of ISUIA study

• Patients were not randomized to surgery (vs. no surgery), and there were substantial differences between treated and untreated groups
• Follow-up was < 5 years in of 50% of patients
• Selection bias: low recruitment numbers from each center

f) gender?: Risk of rupture was greater amongst women compared to men in one study, but only approached statistical significance[9]
g) family history?: In the Familial Intracranial Aneurysm study[12] rupture rate in patients with unruptured aneurysm and first-degree relative with intracranial aneurysm was 17x higher than that for patients with unruptured intracranial aneurysm in ISUIA (after matching for aneurysm size and location) – although conclusions are limited secondary to small number of ruptures in the study. Other studies have failed to demonstrate an increased risk in this subgroup

2. aneurysm characteristics
   a) size: risk of rupture appears critically dependent on aneurysm diameter. ISUIA estimated the annual risk of rupture of aneurysms < 10 mm to be 0.05%/year, however a number of other studies have demonstrated a rupture risk closer to ≈1%/year for aneurysms < 10 mm.[5,8,13,14,15] Furthermore, the Small Unruptured Intracranial Aneurysm Verification Study[7] demonstrated that the rupture risk of smaller aneurysms (< 5 mm) is not trivial, and estimated to be ≈0.5%/year. A more recent retrospective review demonstrated the majority (62%) of ruptured aneurysms to be < 7 mm, with the majority of these being anterior communicating aneurysms.[16] Some speculate that this may be due to a shrinkage of aneurysms following rupture. Larger aneurysms (10-25 mm) are estimated to have ≈ 3-18%/year risk, while giant aneurysms (≥25 mm) have a risk of ≈8-50%/year
   b) location: ISUIA showed an increase rupture risk for pcomm and posterior circulation aneurysms.[16] Ishibashi et al.[5] also demonstrated increased risk amongst posterior circulation aneurysms. Conversely, some studies found an increased risk of rupture with anterior communicating aneurysms[8,16,17]
   c) morphology: presence of a daughter sac,[15] bottleneck shape[18] and increased ratio of size of aneurysm to parent vessel have all been associated with increased risk of rupture[19,20]

Estimation of absolute risk of aneurysm rupture in a patient based on a combination of risk factors is complex. Recently, a scoring system (PHASES) was developed by pooling patient data from six prospective studies,[5,7,8,15,21,22] to help estimate 5-year rupture risk by risk factor status.[11] The predictors comprising the PHASES aneurysm rupture risk score and the predicted 5-year cumulative risk of aneurysm rupture based on score is summarized below (Table 81.2). Further studies are needed however to externally validate the score.

81.1.4 Management

Cumulative rupture risk

To understand the calculation of cumulative risk for aneurysmal rupture, see discussion of this issue related to AVMs which is also relevant to aneurysms (p.1240).

Decision analysis

Decision analysis is a means of mathematically modeling outcomes of various decision options using probabilities and assigning “desirability factors” to the outcomes. This analysis requires data about the natural history (see above), life expectancy, and morbidity and mortality of SAH and aneurysm surgery. Although it is only a model, it does yield some insights in some complicated decisions.

In one such study,[23] using the values shown in Table 81.3, the result obtained was that a life expectancy of 12 more years is the break-even point, i.e. if the patient is not expected to live for 12 more years, then non-surgical management is a better choice than surgery (this result involves numerous assumptions and estimations; e.g. 5% “risk aversiveness” (intermediate) relates to patient’s fears of immediate surgical risk vs. risk of rupture spread over many years). Another analysis of various scenarios for a 50 year old female found that treatment was cost effective for UIAs that were symptomatic, ≥10 mm diameter, or with a previous history of SAH.[24]

Management recommendations

Decisions are based on natural history data compared with morbidity and mortality of intervention (surgery/endovascular), with recommendations being based mainly on expert opinion as high level evidence is lacking. Size, patient age and location appear to be the most important factors in determining whether to treat and by which means to treat an unruptured aneurysm (in a patient without prior SAH). In addition, treatment should be recommended for patients with a history of aSAH, strong family history, symptomatic aneurysms, and for enlargement or change in configuration of

Table 81.2 Predictors comprising the PHASES aneurysm rupture risk score; predicted 5-year cumulative risk of aneurysm rupture based on score[11]

Predictor	Points
(P) Population	
North American, European (other than Finnish)	0
Japanes	3
Finnish	5
(H) Hypertension	
No	0
Yes	1
(A) Age	
<70 years	0
≥70 years	1
(S) Size	
<7 mm	0
7–9 mm	3
10–19.9 mm	6
20 mm	10
(E) Earlier rupture from another aneurysm	
No	0
Yes	1
PHASES risk score	5-year risk of aneurysm rupture
2	0.4%
3	0.7%
4	0.9%
5	1.3%
6	1.7%
7	2.4%
8	3.2%
9	4.3%
10	5.3%
11	7.2%
12	17.8%

the aneurysm.[25] Numerous recommendations have been made for a critical size above which an unruptured aneurysm should be considered for surgery, and have included 3 mm,[26] 5 mm,[27] 7 mm,[28] and 9 mm.[29] The most recent American Heart Association guidelines did not advocate for repair of small incidental aneurysms (<10 mm) in patients with no history of subarachnoid hemorrhage,[25] but this report was prior to more recent prospective trials. In addition, the patient's expected longevity must also be taken into account, and therefore special consideration for treatment should be

Table 81.3 Data used in decision analysis of management of unruptured aneurysms[23]

	Typical value	Range
annual risk of rupture[a]	1%	0.5–2%
3 month mortality of SAH	55%	50–60%
serious morbidity after SAH	15%	10–20%
surgical morbidity & mortality	2% & 6%	4–10%

[a]this is an intermediate risk for aneurysms 6–10 mm diameter (NB: size may change; small aneurysms may grow)

given to young patients in this group. In all treatment decisions, coexisting medical conditions must also be taken into account.

One recent proposed strategy[30] in management is summarized here:

1. Large and/or symptomatic aneurysms (esp. in young pts) → intervention
2. Patients < 60 years old:
   a) < 7 mm
      - anterior circulation, NO risk factors → medical management or intervention
      - pcomm/posterior circulation, symptomatic aneurysm, strong family hx → intervention

   b) > 7 mm → intervention (surgery or endovascular based on size, location)
3. Patients > 60 years old:
   a) < 7 mm
      - no family hx and asymptomatic → medical management
      - +risk factors → intervention

   b) 7-12 mm
      - anterior circulation → medical management or intervention
      - pcomm/posterior circulation → intervention

   c) > 12 mm → Intervention

Recommended follow-up for UIAs treated conservatively

Σ

Annual follow-up with MRA/CTA is recommended for most incidental aneurysms that are not treated. Intervention is indicated for any documented growth. If no growth, may consider repeat imaging at a reduced frequency.

Background: The morbidity from catheter arteriograms is probably too high to recommend them for this purpose. CTA is more accurate than MRA, but involves iodine contrast and radiation. A TOF-MRA (not gadolinium-MRA) has no known risks and does not involve radiation, but has lower spatial resolution.

Unfortunately, most aneurysms rupture without demonstrable enlargement on follow-up. Aneurysms do not grow at a constant rate, and it may take several years to appreciate a millimeter of increased size on MRA.

Studies have identified risk factors for growth, including size,[31,32,33,34] location (MCA, basilar bifurcation),[33,34] > 1 aneurysm,[32,33] family history of SAH[33] and smoking.[31]

Risk of rupture in setting of growth is hard to estimate, given most aneurysms that increase in size are subsequently treated. In one study, the rupture rate was 2.4%/year for aneurysms showing growth (vs. 0.2%/year in those without growth).[31] In another study of 18 Japanese patients, rupture risk after growth was 18.5%/year.[35]

Unruptured cavernous carotid artery aneurysms (CCAAs)

Cavernous carotid artery aneurysms (CCAAs) have a unique risk profile among intracranial aneurysms. Most develop on the horizontal segment of the artery.

Presentation:

1. CCAAs may be discovered incidentally
   a) on arteriography for other reason

   b) on MRI
   c) occasionally on CT
2. when symptomatic:
   a) usually present with:
      - headache
      - cavernous sinus syndrome (p.1401): primarily produces diplopia (due to ophthalmoplegia). Classically the third nerve palsy from enlarging CCAA will not produce a dilated pupil because the sympathetics which dilate the pupil are also paralyzed[36 (p 1492)]
      - those that expand through the carotid ring into the subarachnoid space may cause monocular blindness from strangulation of the optic nerve[37]
   b) rarely, pain (retro-orbital or pain mimicking trigeminal neuralgia[38,39]) or a carotid-cavernous fistula (CCF) are the sole manifestation
   c) when CCAAs rupture, they usually produce a CCF
   d) life threatening complications are rare, but may be more common with giant intracavernous aneurysms.[40] Manifestations include:
      - SAH[40,41]: primarily with CCAAs that straddle the carotid ring (subarachnoid extension of CCAAs may be indicated by "waisting" of the aneurysm on angiography[42])
      - arterial epistaxis from rupture into sphenoid sinus; usually with traumatic aneurysms (p.1227); subarachnoid extension of CCAAs may be indicated by "waisting" of the aneurysm on angiography[42]
      - emboli

Indications for treatment:
1. unruptured CCAAs: the natural history is not precisely known
   a) symptomatic: patients with intolerable pain or visual problems[43]
   b) giant aneurysms: especially those that straddle the clinoidal ring (subarachnoid extension of CCAAs may be indicated by "waisting" of the aneurysm on angiography[42])
   c) aneurysms that enlarge on serial imaging
   d) controversial: incidental aneurysms in the distribution of a stenotic carotid artery for which carotid endarterectomy is indicated. There has been no evidence that doing the endarterectomy increases the risk of rupture, and, as indicated above, most ruptures are not life threatening and so the carotid disease should be treated according to its own merits
2. ruptured CCAAs:
   a) emergent treatment for cases with epistaxis or SAH
   b) urgent treatment for CCFs with severe eye pain or threat to vision

Treatment options for CCAAs:

Treatment of small incidental intracavernous CCAAs is not generally indicated.[25]

For other unruptured CCAAs, options include detachable coils in an attempt to thrombose the aneurysm (p.1194). This results in reduction of mass effect in ≈ 50% Open surgical treatment is rarely appropriate. Aneurysms that rupture and produce a carotid-cavernous fistula may be treated by endovascular occlusion (p.1256).

81.2 Multiple aneurysms

Multiple aneurysms are present in 15–33.5%of cases of SAH.[2] In one study of multiple factors, hypertension was found to be the most important one associated with multiplicity.[44]

When a patient presents with SAH and is found to have multiple aneurysms, the following may be clues as to which aneurysm has bled:
1. epicenter (center of greatest concentration) of blood on CT or MRI[45,46]
2. area of focal vasospasm on angiogram
3. irregularities in the shape of the aneurysm (so-called "Murphy's teat")
4. if none of the above help, then suspect the largest aneurysm
5. NB: in one series, the most common cause of post-op bleeding in 93 patients with multiple aneurysms was felt to be from rebleeding of the original aneurysm that ruptured that was actually missed on initial angiogram[47]

81.3 Familial aneurysms

81.3.1 General information

The role of inheritance in the development of intracranial aneurysms (IA) is well established for disorders such as polycystic kidney disease, and connective tissue disorders such as Ehlers-Danlos type IV, Marfan syndrome, and pseudoxanthoma elasticum (p.1193). Overall, it is not uncommon for patients with aSAH to have a family history. In one study of patients with subarachnoid hemorrhage,[48] 9.4%had a first-degree relative with aSAH or intracranial aneurysm and 14%had a second-degree relative with these diagnosis In families with two or more affected members, the age-adjusted prevalence of intracranial aneurysm among first-degree relatives was 9.2%in those aged 30 years or older.[49,50]

Additional cases of IAs in identical twins[51,52] as well as familial aggregations of IAs without a recognized inherited disorder have also been reported but are felt to be rare (it has been estimated that < 2%of IAs are familial[53]). Most reported cases consist of only 2 family members with IAs, and these are most commonly siblings.[54] In fact, siblings of an affected aSAH patient have a higher risk of aneurysm, than do children of an affected patient.[30] Analysis of case reports reveals that when IAs occur in siblings they tend to occur at identical or mirror image sites, and in comparison to sporadic IAs, familial IAs tend to rupture at a smaller size and at a younger age, and that the incidence of anterior communicating artery aneurysms is lower.[55] It has been postulated that IAs occurring in siblings may represent a distinct population of IAs.[56]

81.3.2 Screening recommendations

The indications and best method for investigation of asymptomatic relatives of a patient found to harbor an intracranial aneurysm are controversial. Negative studies do not guarantee that at a later date an aneurysm will not be discovered that either subsequently developed or expanded, or was simply not detected on the initial study.[57,58,59] Cerebral angiography is the most sensitive study, however, the risk and expense may not justify its use as a screening test in many cases. Furthermore, there is some evidence that aneurysms that rupture tend to do so shortly after their formation[29] which would reduce the value of screening.

81.3.3 Genetics

A large meta-analysis identified 19 single nucleotide polymorphisms associated with sporadic intracranial aneurysms.[60] The strongest associations were found on chromosomes 9 (CDKN2B; antisense inhibitor gene), 8 (SOX17; transcription regulator gene), and 4 (EDNRA gene).

Screening with either MRA or CTA is typically recommended for first-degree relatives (especially siblings) of affected family members when two or more members of the family have an intracranial aneurysm or aSAH.[30] The overall need for screening second-degree relatives is less clear. Screening of first-degree relatives is not typically recommended if only one family member is affected.[25,58] In patients with coarctation of the aorta, screening is typically recommended. Finally, patients with ADPKD who have a family history of intracranial aneurysm or aSAH should be screened. Findings suspicious for intracranial aneurysms should be followed-up with four vessel arteriography to confirm suspected lesions (MRA has a high false-positive rate of ≈ 16%[49]) and to rule-out additional aneurysms.

81.4 Traumatic aneurysms

81.4.1 General information

Traumatic aneurysms (TAs) comprise < 1% of intracranial aneurysms.[61,62] Most are actually false aneurysms, AKA pseudoaneurysms (a rupture of all the vessel wall layers with the "wall" of the aneurysm being formed by surrounding cerebral structures[63]). They may occur rarely in childhood. The mechanism of injury usually falls into one of the following groups[64]:

- Those arising from penetrating trauma. Usually from gunshot wounds, although penetration with a sharp object (which is less common) may be more prone to cause traumatic aneurysms.[65]

- Those arising from closed head injury. More common. Theories of pathogenesis include traction injury to the vessel wall or entrapment within a fracture. Tend to occur either:

1. peripherally
   a) distal anterior cerebral artery aneurysms: secondary to impact against the falcine edge
   b) distal cortical artery aneurysms: often associated with an overlying skull fracture, sometimes a growing skull fracture
2. at the skull base, usually involving the ICA in one of the following sites:
   a) petrous portion (virtually always associated with basal skull fractures):
   b) cavernous carotid artery (virtually always associated with basal skull fractures):

- aneurysm enlargement may cause a progressive cavernous sinus syndrome
- rupture may lead to a posttraumatic carotid-cavernous fistula (p.1256) or to massive epistaxis in the presence of a sphenoid sinus fracture[66,67,68]

c) supraclinoid carotid artery

Iatrogenic. Following surgery in or around the skull base, the sinuses, or orbits (including following transsphenoidal surgery[69]). The first such case was described in 1950

81.4.2 Presentation

1. delayed intracranial hemorrhage (subdural, subarachnoid, intraventricular, or intraparenchymal): the most common presentation. TAs tend to have a high rate of rupture
2. recurrent epistaxis
3. progressive cranial nerve palsy
4. enlarging skull fracture
5. may be incidental finding on CT scan
6. severe headache

81.4.3 Treatment

Although there are case reports of spontaneous resolution, treatment is usually recommended. ICA aneurysms at the skull base should undergo trapping or endovascular embolization. Peripheral lesions should be treated surgically with clipping of aneurysm neck, excision of the aneurysm, coiling, or wrapping if no other method is feasible.

81.5 Mycotic aneurysms

81.5.1 General information

The name “mycotic” originated with Osler in whose time the term referred to any infectious process[70] rather than the current usage which infers a fungal etiology. Currently accepted terminology favors infectious aneurysm (or bacterial aneurysm). Infectious aneurysms can, however, also occur with fungal infections.[71] Tend to form in distal (often unnamed) vessels.

81.5.2 Epidemiology and pathophysiology

1. comprise ≈ 4% of intracranial aneurysms
2. occurs in 3–15% of patients with subacute bacterial endocarditis (SBE)
3. most common location: distal MCA branches (75–80%)
4. at least 20% have or develop multiple aneurysms
5. increased frequency in immunocompromised patients (e.g. AIDS) and drug users
6. most probably start in the adventitia (outer layer) and spread inward

81.5.3 Evaluation

Blood cultures and LP may identify the infectious organism. Table 81.4 shows typical pathogens recovered. Patients with suspected infectious aneurysm(s) should undergo echo-cardiography to look for signs of endocarditis.

81.5.4 Treatment

These aneurysms usually have fusiform morphology and are usually very friable, therefore surgical treatment is difficult and/or risky. Most cases are treated acutely with antibiotics which are continued 4–6 weeks. Serial angiography (at 7–10 days and 1.5, 3, 6 and 12 months, even if aneurysms seem to be getting smaller, they may subsequently increase[73] and new ones may form) helps document effectiveness of medical therapy (serial MRA may be a viable alternative in some cases). Aneurysms may continue to shrink following completion of antibiotic therapy.[74] Delayed clipping may be more feasible; indications include:

1. patients with SAH
2. increasing size of aneurysm while on antibiotics[75] (controversial, some say not mandatory[74])
3. failure of aneurysm to reduce in size after 4–6 weeks of antibiotics[75]

Table 81.4 Pathogens implicated in mycotic aneurysms[72 (p 933–40)]

Organism	%	Comment
streptococcus	44%	S. viridans (classic cause of SBE)
staphylococcus	18%	S. aureus (cause of acute bacterial endocarditis)
miscellaneous	6%	(pseudomonas, enterococcus, corynebacteria…)
multiple	5%	
no growth	12%	
no info	14%	
total	99%	

Patients with SBE requiring valve replacement should have bioprosthetic (i.e. tissue) valves instead of mechanical valves to eliminate the need for risky anticoagulation.

81.6 Giant aneurysms

81.6.1 General information

Definition: >2.5 cm (≈ 1 inch) diameter. Two types: saccular (probably an enlarged "berry" aneurysm) and fusiform. Comprise 3–5% of intracranial aneurysms; peak age of presentation 30–60 years; female:male ratio = 3:1.

Drake's series of 174 giant aneurysms[76]: 35% presented as hemorrhage, with 10% showing some evidence of remote bleeding. The bleeding rate is unknown, but is probably less than the ≈ 2%/year for non-giant aneurysms.

May also present as TIAs (by reducing flow or by emboli) or as a mass. About one third have a neck amenable to clipping.

81.6.2 Evaluation

General information

Drake contended that even after thorough radiographic evaluation, actual operative visualization is the only way to definitively assess the aneurysm and its branches. 3D-CTA can add substantial information that rivals and may exceed direct visualization.

Angiogram

Often underestimates the size of the lesion secondary to thrombosed regions of the aneurysm that do not fill with contrast. CT or MRI is required to visualize the thrombosed portion.

CT scan

Frequently have a significant amount of edema surrounding the aneurysm. May see contrast enhancement of the brain surrounding the aneurysm; probably due to increased vascularity secondary to inflammatory reaction to the aneurysm.

MRI scan

Turbulence within → complicated signal on T1WI. Pulsation artifact (linear distortion radiation through aneurysm) on MRI helps differentiate giant aneurysms from solid or cystic lesions.

81.6.3 Treatment

Options include:
1. direct surgical clipping: usually possible in only ≈ 50% of cases
2. vascular bypass of aneurysm with subsequent clipping

3. trapping
4. proximal arterial ligation (hunterian ligation)
   a) for vertebral-basilar aneurysms[77]: results in improvement of cranial nerve deficit in ≈ 95% of patients. A reasonable alternative in the presence of an adequately sized contralateral VA that unites with the VA to be ligated
5. wrapping (p.1194)
6. endovascular treatment

81.7 Cortical subarachnoid hemorrhage

Cortical SAH (cSAH) appears as SAH over the convexity. Trauma is the most common cause. Nontraumatic etiologies are shown below.

Etiologies of nontraumatic cSAH[78]:

- pial AVMs
- dural AV fistulas
- cerebrovascular arterial dissection
- dural or cortical venous thrombosis
- vasculitis
- Reversible cerebral vasoconstriction syndrome (RCVS), AKA Call-Fleming syndrome,[79] a group of disorders sharing the cardinal clinical and angiographic features of reversible segmental multifocal cerebral vasoconstriction with severe headaches, focal ischemia, and/or seizures. May present as a hemorrhage restricted to a cortical sulcus
- PRES
- Cerebral amyloid angiopathy (CAA)
- Coagulopathies
- Brain tumors (primary or metastatic)

81.8 SAH of unknown etiology

81.8.1 General information

Incidence: traditionally quoted as 20–28% of all SAH, but this includes data from older series (some did not perform true pan-angiography, and/or CT was not available to R/O intracerebral hemorrhage). Recent estimates of incidence: 7–10% This is a heterogeneous category, and a better term might be "angiogram-negative SAH"; see requirements to be met before considering an arteriogram to be negative (p.1161). The quantity of blood on CT may predict the chances of an arteriogram disclosing a cerebral aneurysm.[80,81,82,83]

Patients with angiogram-negative SAH tend to be younger, less hypertensive, and more commonly male than those with positive angiography.[81]

Possible causes of SAH with a negative angiogram include:

□ Aneurysm that fails to be demonstrated in initial angiogram:
1. inadequate angiography, causes include:
   a) incomplete angio: (p.1161)
      - must see both PICA origins (1–2% of aneurysms occur here)
      - need to cross-fill through the ACoA (p.1161)
   b) degradation of images due to
      - poor patient cooperation (e.g. from agitation). Either sedate patient (use caution in non-intubated patients) or repeat the study at a later time when patient more cooperative
      - poor quality equipment providing substandard images
2. obliteration of aneurysm by the hemorrhage
3. thrombosis of the aneurysm after SAH (p.1194)
4. aneurysm too small to be visualized[84]: although "microaneurysms" may be a source of SAH, their natural history and optimal treatment are unknown
5. lack of filling of aneurysm due to vasospasm (of parent artery or of aneurysmal orifice)

□ Nonaneurysmal SAH from source that fails to show up on angiography. See for etiologies of SAH other than aneurysm (p.1156) (many of which may not be demonstrated on angiography), including:
1. angiographically occult (or cryptic) vascular malformation (p.1246)
2. pretruncal nonaneurysmal SAH: see below

81.8.2 Risk of rebleeding

Overall rebleed rate is 0.5%/yr, which is lower than with aneurysmal SAH or rebleeding from AVMs. There is also a smaller risk of delayed cerebral ischemia (vasospasm). Neurological outcome is likewise better.

81.8.3 Management

General measures

These patients are still at risk for the same complications of SAH as with aneurysmal SAH: vasospasm, hydrocephalus, hyponatremia, rebleeding, etc. (p. 1164) and (should be managed as any SAH (p. 1163). Some subgroups may be at lower risk for complications and may be managed accordingly (e.g. below).

Repeat angiography

Yield of positive second angiogram after technically adequate negative study: 1.8–9.8%[85] in early (pre-CT) studies, 2–24% quoted more recently.[84,86,87] CT scan findings are helpful in the decision to repeat angiography.[88] 70% of cases with diffuse SAH and thick layering of blood in the anterior interhemispheric fissure were associated with an ACoA aneurysm that showed up on repeat angiography.[82] The absence of blood on CT (performed within 4 days of SAH), or thick blood in the perimesencephalic cisterns alone (see below) were unlikely to be associated with a missed aneurysm.

Recommendations regarding repeat angio:

1. repeat angio after ≈ 10–14 days (allows vasospasm & some clot to resolve; note: between 5–10 days there is decreased chance of seeing an aneurysm because of vasospasm; angiography at ≈ day 10 permits surgery to be done if needed ≈ at day 14 which is about the earliest time after the "no-op" window of day 3–12)
   a) technically adequate 4 vessel angiogram is negative, and evidence for SAH is strong
   b) original angio was incomplete or if there are suspicious findings
2. if CT localizes blood clot to particular area, place special attention to this area on repeat angio
3. do not repeat angio for classic pretruncal SAH (see below) or if no blood on CT
4. patients are usually kept in the hospital 10–14 days while waiting for repeat angio (to watch for and manage complication of SAH or rebleeding)

Third arteriogram:

If the 1st 2 arteriograms are negative, and the history is suggestive of aneurysmal SAH, a 3rd arteriogram 3–6 months after SAH has ≈ 1% chance of showing a source of bleeding.

Other studies

1. imaging studies of the brain: MRI (with MRA if available) or CT (with angio-CT if available). This may visualize an aneurysm that fails to show up on angiography, and may identify other sources of SAH such as angiographically occult vascular malformation (p. 1246), tumor…
2. tests to rule-out spinal AVM: a rare cause of intracerebral SAH (p. 1140)
   a) spinal MRI: cervical, thoracic and lumbar
   b) spinal angiography: too difficult and risky to be justified in most cases of angio negative SAH. Consider in cases with high suspicion of spinal source

Surgical exploration

Advocated by some for cases of SAH with CT findings compatible with an aneurysmal source in which a suspicious area is demonstrated angiographically[84] with careful explanation to the patient and family of the possibility of negative operative findings.

81.9 Pretruncal nonaneurysmal SAH (PNSAH)

81.9.1 General information

Née perimesencephalic nonaneurysmal SAH.[89] The suggestion to change the name to pretruncal nonaneurysmal SAH was proposed because neuroimaging has shown the true anatomic localization of the blood to be in front of the brain stem (truncus cerebri) centered in front of the pons rather than perimesencephalic.[90] The existing literature on PNSAH is somewhat limited by the lack of a rigorous anatomic definition, with criteria of blood pattern differing amongst studies. Blood often extends into the interpeduncular or premedullary cisterns. It has also sometimes been referred to as the "Dutch disease" due to the initial profusion of information in that literature.

A distinct entity considered to be a benign condition with good outcome and less risk of rebleeding and vasospasm than other patients with SAH of unknown etiology[91] (no rebleeding occurred in 37 patients with PNSAH and 45 months mean follow-up,[92] nor in 169 patients with 8–51 months follow-up[87]; vasospasm has been reported in only 3 patients and may have been related to cerebral angiography rather than the PNSAH, and although it is low, the incidence of angiographic vasospasm may be higher than originally thought[93]).

The actual etiology has yet to be determined (there are 3 case reports of patients explored surgically with no abnormal findings[87] and one case where a pontine abnormality resembling a capillary telangiectasia was demonstrated on MRI[94]), but it may be secondary to rupture of a small perimesencephalic vein or capillary.[93] Studies have shown an association with abnormal venous anatomy, including primitive variants of the basal vein of Rosenthal,[95,96] which some authors have hypothesized results in hemorrhage secondary to central cerebral venous hypertension.[97,98] Other proposed etiologies include a ruptured perforating artery, cavernous malformation, intraluminal basilar dissection, and capillary telangiectasia.[99]

81.9.2 Presentation

Patients may present with severe paroxysmal H/A, meningismus, photophobia, and nausea. Loss of consciousness is rare. These patients are usually not critically ill (all were grade 1 or 2 (H&H or WFNS grading scale)), however, complications such as hyponatremia or cardiac abnormalities may still occur. Preretinal hemorrhages and sentinel H/A have not occurred. CT and/or MRI demonstrate characteristic findings (see below) although it may initially be missed on CT,[93] and LP may yield bloody CSF. By definition, all have negative angiography.

81.9.3 Epidemiology

PNSAH has been reported to comprise 20–68% of cases of angiogram-negative SAH[91,100] (depending on the timing of CT, adequacy of angiography, and the definition of PNSAH). However, the true incidence is probably more in the range of 50–75%[87]

The reported age range is 3–70 years (mean: 50 yrs),[87] 52–59% are male, and pre-existing HTN was present in 3–20% of patients.

81.9.4 Relevant anatomy

Posterior fossa cisterns:

The perimesencephalic cisterns include: interpeduncular, crural, ambient and quadrigeminal cisterns. The prepontine cistern lies immediately anterior to the pons.

Liliequist's membrane (LM)[101]:

Basically considered to separate the interpeduncular cistern from the chiasmatic cistern[102] (forming a competent barrier in only 10–30% of the population). In further detail, the superior leaflet of LM (diencephalic membrane) separates the interpeduncular cistern from the chiasmatic cistern medially and from the carotid cisterns laterally.[103,104] The inferior leaflet (the mesencephalic membrane) separates the interpeduncular from the prepontine cistern.

The diencephalic membrane is thicker and is more often competent, effectively isolating the chiasmatic cistern. However, the carotid cisterns often communicate with the crural cisterns and in turn with the interpeduncular cistern.[104]

Thus, blood in the carotid or prepontine cistern is compatible with a low-pressure pretruncal source of bleeding, however, blood in the chiasmatic cistern should raise concern about aneurysmal rupture.

81.9.5 Diagnostic criteria

Without knowledge of the actual substrate of PNSAH, the following suggested diagnostic criteria must be viewed as empiric (adapted[87]):

1. CT or MRI scan performed ≤ 2 days from ictus meeting the criteria shown in Table 81.5 (later scans render the diagnosis unreliable, e.g. washout could cause an aneurysmal SAH to fit the criteria). This criteria implies that blood should be contained inferior to Liliequist's membrane (LM) (i.e. perimesencephalic and/or prepontine cisterns). Extension into the suprasellar cistern is

Table 81.5 CT or MRI criteria for PNSAH[93,107]

1.	epicenter of hemorrhage immediately anterior to brain stem (interpeduncular or prepontine cistern)
2.	there may be extension into anterior part of ambient cistern or basal part of the sylvian fissure
3.	absence of complete filling of anterior interhemispheric fissure
4.	no more than minute amounts of blood in lateral portion of sylvian fissure
5.	absence of frank intraventricular hemorrhage (small amounts of blood sedimenting in the occipital horns of the lateral ventricles is permissible)

Table 81.6 Alternative anatomic CT criteria for PNSAH[108]

- epicenter of bleeding located immediately anterior to and in contact with the brainstem in the prepontine, interpeduncular, or posterior suprasellar cistern
- blood limited to the prepontine, interpeduncular, suprasellar, crural, ambient, and/or quadrigeminal cisterns and/or cisterna magna
- NO extension of blood into the Sylvian or interhemisheric fissures
- intraventricular blood limited to incomplete filling of the fourth ventricle and occipital horns of the lateral ventricles
- no intraparenchymal blood

common. Significant amounts of blood penetrating LM to the chiasmatic, sylvian, or interhemispheric cisterns should be viewed with suspicion

2. a negative high-quality 4-vessel cerebral angiogram[105] (radiographic vasospasm is common, and does not preclude the diagnosis nor does it mandate repeat angiography). NB: ≈ 3% of patients with a ruptured basilar bifurcation aneurysm meet the criteria of Table 81.5,[106] therefore an initial arteriogram is mandatory
3. appropriate clinical picture: no loss of consciousness, no sentinel H/A, SAH grade 1 or 2 (Hunt and Hess or WFNS grading scale) (p.1162), and absence of drug use. Variance from this should raise suspicion of alternate pathogenesis

Recently, a more stringent set of anatomic criteria (see Table 81.6) was tested and found to result in excellent inter-observer agreement (97.2%) in PNSAH. Additionally, no aneurysms were identified on formal angiography when the anatomical criteria were met.[108]

81.9.6 Repeat angiography

Controversial. Angiography carries ≈ 0.2–0.5% risk of permanent neurologic deficit in this population.[87] Most experts agree that repeat angiography is not indicated in patients meeting the criteria of PNSAH[86,105] (although others recommend repeat angiography in all surgical candidates[84,109]). One should probably repeat the study if any uncertainty exists or if there is a history of a condition associated with increased risk of cerebral aneurysms.[93]

81.9.7 Treatment

Optimal treatment is not known with certainty. The low risk of rebleeding and delayed ischemia suggests that extreme measures are not indicated. The following recommendations are made[87,93] (period not specified):

1. symptomatic treatment
2. cardiac monitoring
3. electrolyte monitoring for hyponatremia
4. follow patient clinically (and if appropriate, with repeat imaging studies) to rule-out hydrocephalus (transient ventricular enlargement is common, however, hydrocephalus requiring shunting is rare (only ≈ 1%)[87])
5. not recommended
   a) hyperdynamic therapy
   b) calcium channel blockers: use has not been investigated in PNSAH, but is probably not warranted due to low incidence of vasospasm and should be discontinued when normal angiographic findings are documented[93]
   c) activity restrictions (except in cases of increasing H/A with mobilization)
   d) anticonvulsants
   e) reduction of blood pressure below normal
   f) surgical exploration

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Part XVIII

Vascular Malformations

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XVIII

82 Vascular Malformations

82.1 General information and classification

This designation encompasses a number of non-neoplastic vascular lesions of the CNS. The four types originally described by McCormick in 1966 are shown in Table 82.1.[1]

Possible additional categories:

1. direct fistula AKA arteriovenous fistula (AV- fistula, not AVM). Single or multiple dilated arterioles that connect directly to a vein without a nidus. These are high-flow, high-pressure. Low incidence of hemorrhage. Usually amenable to interventional neuroradiological procedures. Examples include:
   a) vein of Galen malformation (aneurysm) (p. 1255)
   b) dural AVM (p. 1251)
   c) carotid-cavernous fistula (p. 1256)
2. mixed or unclassified angiomas: 11% of AOVM[2]

82.2 Arteriovenous malformation (AVM)

82.2.1 General information

Key concepts

- dilated arteries and veins with dysplastic vessels. Arterial blood flows directly between them with no capillary bed and no intervening neural parenchyma in nidus
- AVMs are medium-to-high pressure and high-flow
- usually presents with hemorrhage, less often with seizures
- these usually are congenital lesions with a lifelong risk of bleeding of ≈ 2–4% per year
- demonstrable on angiography, MRI, or CT (especially with contrast)
- main treatment options: stereotactic radiosurgery (usually for deep lesions < 3 cm dia) or surgical excision

82.2.2 Description

An abnormal collection of blood vessels wherein arterial blood flows directly into draining veins without the normal interposed capillary beds. There is no brain parenchyma contained within the nidus. AVMs are usually congenital lesions, that tend to enlarge somewhat with age and often progress from low flow juvenile lesions at birth to medium-to-high-flow high-pressure lesions in adulthood. AVMs appear grossly as a "tangle" of vessels, often with a fairly well-circumscribed center (nidus), and draining "red veins" (veins containing oxygenated blood). May be classified as:

1. parenchymal AVMs (discussed below). Subclassified as:
   a) pial
   b) subcortical
   c) paraventricular
   d) combined

Table 82.1 4 classic types of vascular malformation

Type	Prevalence %
Arteriovenous malformation (AVM)[a]	44–60%
Cavernous malformation (p. 1247)	19–31%
Capillary telangiectasia (p. 1247)	4–12%
Developmental venous anomaly (DVA) – formerly venous angioma	9–10%

[a]Sometimes referred to as "pial AVM" to distinguish it e.g. from dural AVM

2. pure dural AVM (p.1251)
3. mixed parenchymal and dural (rare)

82.2.3 Epidemiology

Prevalence: probably slightly greater than the usually quoted 0.14%

Slight male preponderance. Congenital (therefore risk of hemorrhage is lifelong).

A few types are known to be hereditary. Other types are part of known hereditary syndromes: 15–20%of patients with Osler-Weber-Rendu syndrome (hereditary hemorrhagic telangiectasia) have cerebral AVMs.

82.2.4 Comparison to aneurysms

See reference.[3]

AVM:aneurysm ratio in U.S. is 1:5.3 (pre-CT era data). The average age of patients diagnosed with AVMs is ≈ 33 yrs, which is ≈ 10 yrs younger than aneurysms.[4] 64%of AVMs are diagnosed before age 40 (c.f. 26%for aneurysms).

82.2.5 Presentation

General information

1. hemorrhage (most common)[5]: 50%(61%quoted elsewhere,[3] compared to 92%for aneurysms) (see below)
2. seizures
3. mass effect: e.g. trigeminal neuralgia due to CPA AVM
4. ischemia: by steal
5. H/A: rare. AVMs may occasionally be associated with migraines. Occipital AVMs may present with visual disturbance (typically hemianopsia or quadrantanopsia) and H/A that are indistinguishable from migraine[6]
6. bruit: especially with dural AVMs (p.1251)
7. increased ICP
8. findings limited almost exclusively to peds, usually with large midline AVMs that drain into an enlarged vein of Galen ("vein of Galen malformation" (p.1255)):
   a) hydrocephalus with macrocephaly: due to compression of Sylvian aqueduct by enlarged vein of Galen or to increased venous pressure
   b) congestive heart failure with cardiomegaly
   c) prominence of forehead veins (due to increased venous pressure)

Hemorrhage

General information

Peak age for hemorrhage is between 15–20 yrs.[3] The reported morbidity and mortality from AVM hemorrhage varies widely. An estimate is 10% mortality, 30–50% morbidity[7] (neurological deficit) from each bleed. For a discussion of hemorrhage during pregnancy, also see Pregnancy & intracranial hemorrhage (p.1169).

Hemorrhage location with AVMs

1. intraparenchymal (CH): 82%(the most common site of bleeding)[8]
2. intraventricular hemorrhage:
   a) usually accompanied by ICH as the result of rupture of the ICH into the ventricle
   b) pure IVH (with no ICH) may indicate an intraventricular AVM
3. subarachnoid: SAH may also be due to rupture of an aneurysm on a feeding artery; common with AVMs (p.1156)
4. subdural: uncommon. May be the sourced of a spontaneous SDH (p.904)

Hemorrhage rate related to AVM size

Small AVMs tend to present more often as hemorrhage than do large ones.[9,10] It was postulated that larger AVMs presented as seizure more often simply because their size made them more likely to involve the cortex. However, small AVMs are now thought to have much higher pressure in the feeding arteries[10] (see ▶ Table 82.2). Conclusion: small AVMs are more lethal than larger ones.

Table 82.2 Risk of bleeding by size of AVM[10]

Characteristic	Size of AVM		
	Small (<3 cm)	Medium (3–6 cm)	Large (>6 cm)
number of patients	44	31	17
% that bled	82%	29%	12%
size of hematoma (ave)	4.9 cm	2.7 cm	2.0 cm
feeding artery pressure (mm Hg)	66	47	35

Table 82.3 Annual average hemorrhage rates for various AVM subgroups[12]

<table>
<tr><th>Venous drainage</th><th>No Prior hemorrhage</th><th>Prior hemorrhage</th><th>Nidus location</th></tr>
<tr><td rowspan="2">No deep venous drainage</td><td>0.9%</td><td>4.5%</td><td>Not deep</td></tr>
<tr><td>3.1%</td><td>14.8%</td><td rowspan="2">Deep</td></tr>
<tr><td rowspan="2">Deep venous drainage</td><td>8.0%</td><td>34.4%</td></tr>
<tr><td>2.4%</td><td>11.4%</td><td>Not deep</td></tr>
</table>

Hemorrhage rate related to Spetzler-Martin grade

Controversial. Some studies show increased risk with Spetzler-Martin (S-M) grade 4–5 AVMs (p.1242) [11] (high grade), others show the opposite effect as:

S-M grade 1–3: annual risk of hemorrhage is 3.5%

S-M grade 4–5: annual risk of hemorrhage is 2.5%

Hemorrhage rate based on nidus depth, venous drainage and prior bleed

Breakdown of the risk of bleeding based on history of prior bleeding, venous drainage pattern, and nidus location using data from Stapf et al. is shown in Table 82.3.

Annual and lifetime risk of hemorrhage and recurrent hemorrhage

The average risk of hemorrhage from an AVM is ≈ 2–4% per year[13] (reminder: risk varies by AVM size, see above). The risk of bleeding over the remainder of one's life is given by Eq (82.1); this analysis includes a number of assumptions, among them: a constant risk of rebleeding even early after an initial bleed, no change in risk during the lifetime (which may not be true in pregnancy), no difference in risk for various AVM locations or age groups.

$$\text{risk of bleeding (at least once)} = 1 - (\text{annual risk of not bleeding})^{\text{expected years of remaining life}} \qquad (82.1)$$

where the annual risk of not bleeding is equal to 1 – the annual risk of bleeding. For example, if a 3% annual risk of bleeding is used as an average, and the remaining life expectancy is 25 years, the result is as illustrated in Eq (82.2).

$$\text{risk of bleeding (at least once in 25 years)}^{*} = 1 - 0.97^{25} = 0.53 = 53\% \qquad (82.2)$$

A simple to apply first approximation to Eq (82.1) is shown in Eq (82.3).

$$\text{risk of bleeding (at least once)}^{*} \approx 105 - \text{age in years} \qquad (82.3)$$

* Note: Eq (82.3) assumes a 3% per year bleeding rate.

Table 82.4 shows the risk for various ages using Eq (82.1) (longevity is taken from insurance life-tables).

Table 82.4 Lifetime risk of hemorrhage[a]

Age at presentation	Estimated years to live[b]	Lifetime risk of hemorrhage		
		For 1% annual risk[c]	For 2% annual risk	For 3% annual risk
0	76	53%	78%	90%
15	62	46%	71%	85%
25	52	41%	65%	79%
35	43	35%	58%	73%
45	34	29%	50%	64%
55	25	22%	40%	53%
65	18	16%	30%	42%
75	11	10%	20%	28%
85	6	5.8%	11%	17%

[a]modified from reference[13]
[b]based on 1992 Preliminary Life tables prepared by Metropolitan Life Insurance Company
[c]1% annual risk is also presented because it may be appropriate for incidental aneurysms (p. 1222)

A study of 166 symptomatic AVMS with long average follow-up (mean: 23.7 yrs)[4] found the risk of major bleeding was constant at 4% per year, and that this was independent of whether the AVM presented with or without hemorrhage. The mean time between presentation and hemorrhage was 7.7 yrs. The mortality rate was 1% per year, and the combined major morbidity and mortality rate was 2.7% per year.

Older studies may suffer from smaller numbers[9] or short follow-up (mean: 6.5 yrs).[3,5] These studies suggested a higher risk of (re)bleeding depending on whether the initial presentation was hemorrhage (≈ 3.7% per year) vs. seizure (1–2% per year). Crawford found that none of 8 incidental (asymptomatic) AVMs bled in 20 yrs follow-up; all died of unrelated causes.[9]

The hemorrhage risk (but not the rate) may be higher in pediatrics or with posterior-fossa AVMs.[13]

Rebleeding

The literature contains conflicting information. Reported rebleeding rate in the first year after hemorrhage was 6% in one series,[14] 18% in another series[15] which declined to 2% per year after 10 years, and in another large series[4] the annual rate was 4% and did not vary regardless of presentation as discussed above.

Seizures

The younger the patient at the time of diagnosis, the higher the risk of developing convulsions. 20 yr risk: diagnosis at age 10–19 → 44% risk; age 20–29 → 31%; age 30–60 → 6%. Patients presenting with hemorrhage have 22% risk of developing epilepsy in 20 yrs. No AVM found incidentally or presenting with neuro deficit developed seizures.[9]

AVMs and aneurysms

7% of patients with AVMs have aneurysms. 75% of these are located on major feeding artery (probably from increased flow).[9] These aneurysms may be classified into 1 of 5 types shown in Table 82.5. Aneurysms also may form within the nidus or on draining veins. When treating tandem AVMs and aneurysms, the symptomatic one is usually treated first (when feasible, both may be treated at the same operation).[16] If it is not clear which bled, the odds are that it was the aneurysm. Although a significant number (≈ 66%) of related aneurysms will regress following removal of the AVM, this does not always occur. In one series, none of the 9 associated aneurysms ruptured or enlarged following AVM removal[16]

Table 82.5 Categories of aneurysms associated with AVMs[a] [16]

Type	Aneurysm location
I	proximal on ipsilateral major artery feeding AVM
IA	proximal on major artery related but contralateral to AVM
II	distal on superficial feeding artery
III	proximal or distal on deep feeding artery (“bizarre”)
IV	on artery unrelated to AVM

[a]excludes intra-nidal and venous aneurysms

82.2.6 Evaluation

CT

Unenhanced brain CT is the best study to rule-out acute hemorrhage. Can also demonstrate calcifications within the lesion. Adding a contrast CT will show enhancement within the vessels, and can delineate the nidus (dense central area of an AVM). See below for CTA.

MRI

Characteristics of AVM on MRI:
1. flow void on T1WI or T2WI within the AVM
2. feeding arteries
3. draining veins
4. increased intensity on partial flip-angle (to differentiate signal dropout on T1WI or T2WI from calcium)
5. significant edema around the lesion may indicate a tumor that has bled rather than an AVM
6. gradient echo sequences (GRASS…) help demonstrate surrounding hemosiderin which suggests a previous significant hemorrhage
7. a complete ring of low density (due to hemosiderin) surrounding the lesion suggests AVM over neoplasm

82.2.7 CT angiography (CTA)

MR angiography (MRA)

Angiography

Characteristics of AVM on angiography:
1. tangle of vessels
2. large feeding artery
3. large draining veins
4. draining veins are visualized in the same images as arteries (arterial phase)

Most, but not all AVMs show up on angiography; see Angiographically occult vascular malformations (p.1246). Fewer cavernous malformations and venous angiomas do.

Grading

Spetzler-Martin grade of AVMs

Grade = sum of points from Table 82.6, ranges from 1 to 5. A separate grade 6 is reserved for untreatable lesions (by any means: surgery, SRS…), resection of these would almost unavoidably be associated with disabling deficit or death. This scale has been shown to have good prognostic predictability.[17] May not be applicable to pediatrics (AVMs are immature and change with time; AVMs mature at ≈ age 18 yrs and tend to become more compact).

Outcome based on Spetzler-Martin grade: 100 consecutive cases operated by an expert (Spetzler) had the outcomes shown in Table 82.7 (no deaths).

Spetzler has since published a 3-tiered management recommendation scheme[19] as follows:
- Class A (S-M Grade I & II): surgical resection

Table 82.6 Spetzler-Martin AVM grading system[18]

Graded feature	Points
Size[a]	
small (<3 cm)	1
medium (3–6 cm)	2
large (>6 cm)	3
Eloquence of adjacent brain	
non-eloquent[b]	0
eloquent[b]	1
Pattern of venous drainage[c]	
superficial only	0
deep	1

[a]largest diameter of nidus on non-magnified angiogram (is related to and therefore implicitly includes other factors relating to difficulty of AVM excision, e.g. number of feeding arteries, degree of steal, etc.)
[b]eloquent brain: sensorimotor, language and visual cortex; hypothalamus and thalamus; internal capsule; brain stem; cerebellar peduncles; deep cerebellar nuclei
[c]considered superficial if all drainage is through cortical venous system; considered deep if any or all is through deep veins (e.g. internal cerebral vein, basal vein, or pre-central cerebellar vein)

Table 82.7 Surgical outcome by Spetzler-Martin grade operated on by Spetzler

Grade	No.	No deficit		Minor deficit[a]		Major deficit[b]	
1	23	23	(100%)	0		0	
2	21	20	(95%)	1	(5%)	0	
3	25	21	(84%)	3	(12%)	1	(4%)
4	15	11	(73%)	3	(20%)	1	(7%)
5	16	11	(69%)	3	(19%)	2	(12%)

[a]minor deficit: mild brainstem deficit, mild aphasia, mild ataxia
[b]major deficit: hemiparesis, increased aphasia, homonymous hemianopsia

- Class B (S-M Grade III): multimodality treatment
- Class C (S-M Grade IV & V): follow clinically and repeat angiogram every 5 years. Treatment only for progressive neurologic deficit, steal-related symptoms, or aneurysms identified on surveillance angiograms

82.2.8 Treatment

General information

Options and some pros and cons of each include:

1. surgery: the treatment of choice for AVMs. When surgical risk is unacceptably high, alternative procedures (e.g. SRS) may be an option
   a) pros: eliminates risk of bleeding almost immediately. Seizure control improves
   b) cons: invasive, risk of surgery, cost (high initial cost of treatment may be offset by effectiveness or may be increased by complications)
2. radiation treatment
   a) conventional radiation: effective in ≈ 20%or less of cases.[20,21] Therefore not considered effective therapy
   b) stereotactic radiosurgery (SRS): accepted for some small (≤2.5–3 cm nidus), deep AVMs; see Stereotactic radiosurgery & radiotherapy (p.1564)

   - pros: done as an outpatient, non-invasive, gradual reduction of AVM flow, no recovery period
   - cons: takes 1–3 years to work (latency period). During that time there is a risk of bleeding, controversial whether it is increased or decreased); limited to lesions with nidus ≤ 3 cm
3. endovascular techniques: e.g. embolization (see below)
   a) pros: facilitates surgery
   b) cons: sometimes inadequate by itself to permanently obliterate AVMs, induces acute hemodynamic changes, may require multiple procedures, embolization prior to SRS reduces the obliteration rate from 70%(without embolization) to 47%(with embolization)[22]
4. combination techniques: e.g. embolization to shrink nidus then stereotactic radiosurgery

Considerations to take into account in managing AVMs:
1. associated aneurysms: on feeding vessels, draining veins or intra-nidal
2. flow: high or low
3. age of patient
4. history of previous hemorrhage
5. size and compactness of nidus
6. availability of interventional neuroradiologist
7. general medical condition of the patient

Embolization

Used as an initial procedure, embolization facilitates surgery[23] and possibly SRS. It is usually inadequate by itself to treat conventional AVMs (may recanalize later). See AVM embolization (p. 1589) in Endovascular neurosurgery section.

Surgical treatment

Pre-op medical management

Before direct surgical treatment, patient should ideally be pre-treated with propranolol 20 mg PO QID for 3 days to minimize post-op normal perfusion pressure breakthrough (postulated cause of postoperative bleeding and edema,[24] see below). Labetalol has also been used perioperatively to keep MAP 70–80 mm Hg.[25]

Booking the case: Craniotomy for AVM

Also see defaults & disclaimers (p. 27).
1. position: (depends on location of AVM), radiolucent headholder
2. pre-op embolization (by neuroendovascular interventionalist): typically 24–48 hours pre-op
3. intraoperative angiography (optional)
4. equipment
   a) microscope (with ICG capability if used)
   b) image-guided navigation: primarily for the bone flap placement
5. blood availability: type and cross 2 U PRBC
6. post-op: ICU
7. consent (in lay terms for the patient – not all-inclusive):
   a) procedure: surgery to open the skull and remove the abnormal tangle of blood vessels in the brain, intraoperative angiography
   b) alternatives: stereotactic radiosurgery, endovascular techniques (not considered definitive treatment for most AVMs, but often used as an adjunct)
   c) complications: usual craniotomy complications (p. 28) plus stroke (the main concern), bleeding intra-op (requiring transfusion) and post-op, neurologic deficit related to the area of AVM location, failure to be able to remove entire AVM, recurrence in future

Basic tenets of AVM surgery

1. wide exposure
2. occlude feeding (terminal) arteries before draining veins (lesions with a single draining vein can become impossible to deal with if premature blockage of the draining vein occurs, e.g. by kinking, coagulation)

3. excision of whole nidus is necessary to protect against rebleeding (occluding feeding arteries is not adequate)
4. identify and spare vessels of passage and adjacent (uninvolved) arteries
5. dissect directly on nidus of AVM, work in sulci and fissures whenever possible
6. in lesions that are high-flow on angiography, consider preoperative embolization
7. lesions with supplies from multiple vascular territories may require staging
8. clip accessible aneurysms on feeding arteries

Delayed postoperative deterioration

May be due to any of the following:
1. normal perfusion pressure breakthrough[24]: characterized by post-op swelling or hemorrhage. Thought to be due to loss of autoregulation, although this theory has been challenged.[26] Risk may be reduced by pre-op medication (see above)
2. occlusive hyperemia[27]: in the immediate post-op period probably due to obstruction of venous outflow from adjacent normal brain, in a delayed presentation may be due to delayed thrombosis of draining vein or dural sinus.[28] Risk may be elevated by keeping the patient "dry" post-op
3. rebleeding from a retained nidus of AVM
4. seizures

82.2.9 Follow-up of treated AVMs

When satisfactory complete angiographic obliteration of an AVM has been accomplished, recommended follow-up is with catheter angiogram (not CTA or MRA) at 1 & 5 years post treatment.

82.3 Venous angiomas

82.3.1 General information

Key concepts

- a vascular malformation that is part of the venous drainage of the involved area with intervening brain present. Therefore direct treatment is rarely indicated
- low-flow, low-pressure
- usually demonstrable on angiography as a starburst pattern
- rarely symptomatic: seizures rare, hemorrhage even more uncommon. Venous infarcts may occur (controversial)
- may have an associated cavernous malformation (p. 1247) which is more likely to be symptomatic

AKA venous malformation or (developmental) venous anomaly (DVA). A tuft of medullary veins that converge into an enlarged central trunk that drains either to the deep or superficial venous system. The veins lack large amounts of smooth muscle and elastic. No abnormal arteries are found. There is neural parenchyma between the vessels. Most common in regions supplied by the MCA[29] or in the region of the vein of Galen. They may be associated with a cavernous malformation (p.1247). Non hereditary. These are low-flow and low-pressure.

Most are clinically silent, but rarely seizures and even less frequently hemorrhage may occur. Venous infarcts have been described, but may be coincidental. If symptoms are present, look for an associated cavernous malformation (GRASS MRI images may reveal some cavernous malformations that might otherwise be occult).

82.3.2 Imaging

MRI

There may be some T2 hyperintensity on FLAIR.

Angiogram

Occasionally may be angiographically occult, however, they classically produce a distinct caput medusae (other descriptive terms include: a hydra, spokes of a wheel, a spider, an umbrella, a mushroom, or a sunburst or starburst).[30 (p 1471)] Other angiographic characteristics: appears as a long draining vein (longer than a normal vein) draining an excessive amount of brain tissue (it is theorized that venous restrictive disease occurs because of the length), arterial phase should show no AV shunting (characteristic of AVM).

82.3.3 Treatment

In general, these should not be treated as they are the venous drainage of the brain in that vicinity. If surgery is indicated for associated cavernous malformations, the angioma should be left alone. Surgery for the angioma itself is reserved only for documented bleeding or for intractable seizures that can be definitely attributed to the lesion.

82.4 Angiographically occult vascular malformations

82.4.1 General information

Terminology is controversial. The term "cryptic cerebrovascular malformations" was originally applied to angiographically and clinically silent lesions regardless of size.

Recommendation: use the term " angiographically occult (or cryptic) vascular malformations" (AOVM) to refer to cerebrovascular malformations that are not demonstrable on technically satisfactory catheter cerebral angiography (i.e. good quality films, with subtraction views, and the following as appropriate: magnification, angiotomography, rapid serial angiograms or delayed films).[31] Many lesions have large patent vessels at surgery in spite of negative angiography.[32] Other imaging modalities (i.e. CT, MRI) may be able to reveal these lesions. Although often used interchangeably, the term "occult malformation" (omitting the word "angiographically") is suggested for use with lesions that also do not appear on these other imaging modalities.

The reasons for a vascular lesion being angiographically cryptic include:

1. lesion that have hemorrhaged
   a) the bleeding may obliterate the lesion: difficult to substantiate[33]
   b) the clot may temporarily compress the lesion[33] which may re-open after weeks to months as clot dissolves
2. sluggish flow
3. small size of the abnormal vessels
4. may require very late angiographic films (i.e. delayed films) to visualize due to late filling

82.4.2 Epidemiology

Incidence of AOVM has been estimated as ≈ 10% of cerebrovascular malformations.[29] AOVMs were found at necropsy in 21 (4.5%) of 461 patients with spontaneous intracranial hemorrhage (ICH),[34] but refinements in angiography have occurred since this 1954 report.

The average age at diagnosis in one literature review[31] was 28 yrs.

82.4.3 Presentation

AOVM most often present with seizures or H/A. Less commonly they may present with progressive neurologic symptoms (usually as a result of spontaneous ICH).[35] They may also be discovered incidentally.

The natural history of this group of lesions is not accurately known.

82.5 Osler-Weber-Rendu syndrome

82.5.1 General information

AKA hereditary hemorrhagic telangiectasia (HHT), AKA capillary telangiectasia: slightly enlarged capillaries with low flow. Cannot be imaged on any radiographic study. Usually incidentally found at necropsy without clinical significance (risk of hemorrhage is very low, except possibly in brain stem). Has intervening neural tissue[29] (unlike cavernous malformations). Usually solitary, but may be multiple when seen as a part of a syndrome: Osler-Weber-Rendu (see below), Louis -Barr (ataxia telangiectasia), Myburn-Mason, Sturge-Weber.

Associated cerebrovascular malformations (CVM) include: telangiectasias, AVMs (the most common CVM, seen in 5–13% of HHT patients[36]), venous angiomas and aneurysms. Patients are also